CHAPTER

27

FLIGHT CONTROLS

CHAPTER 27
FLIGHT CONTROLS

Subject/Page	Date	COC
27-EFFECTIVE PAGES		
1 thru 34	Sep 15/2026	
27-CONTENTS		
1	May 15/2025	
2	May 15/2025	
3	May 15/2026	
4	May 15/2024	
O 5	Sep 15/2026	
O 6	Sep 15/2026	
7	May 15/2026	
8	May 15/2024	
9	May 15/2024	
10	May 15/2024	
11	May 15/2025	
O 12	Sep 15/2026	
13	May 15/2024	
14	Sep 15/2024	
15	May 15/2024	
16	May 15/2024	
O 17	Sep 15/2026	
O 18	Sep 15/2026	
19	May 15/2026	
20	Sep 15/2025	
21	May 15/2024	
22	May 15/2024	
23	May 15/2025	
O 24	Sep 15/2026	
O 25	Sep 15/2026	
26	May 15/2024	
O 27	Sep 15/2026	
28	Sep 15/2025	
29	Sep 15/2025	
30	Sep 15/2025	
31	May 15/2024	
32	May 15/2024	
33	May 15/2024	
34	May 15/2024	
27-CONTENTS (cont)		
35	May 15/2024	
36	May 15/2024	
O 37	Sep 15/2026	
O 38	Sep 15/2026	
39	Jan 15/2025	
40	May 15/2024	
41	May 15/2024	
42	May 15/2024	
43	May 15/2026	
O 44	Sep 15/2026	
O 45	Sep 15/2026	
46	May 15/2024	
47	May 15/2024	
48	May 15/2024	
49	Sep 15/2025	
50	May 15/2025	
51	May 15/2024	
52	Jan 15/2026	
53	Sep 15/2025	
54	Sep 15/2025	
55	Sep 15/2025	
56	Jan 15/2025	
57	May 15/2024	
58	May 15/2024	
O 59	Sep 15/2026	
60	BLANK	
27-00-00		
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902	May 15/2024	
903	May 15/2024	
904	May 15/2024	
905	Sep 15/2024	
R 906	Sep 15/2026	
907	May 15/2025	
908	May 15/2025	
909	May 15/2025	
27-00-00 (cont)		
910	May 15/2026	
911	May 15/2026	
912	May 15/2026	
913	May 15/2026	
914	May 15/2026	
915	May 15/2026	
916	May 15/2026	
917	May 15/2025	
918	May 15/2025	
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926	May 15/2025	
927	May 15/2025	
928	May 15/2025	
929	May 15/2025	
930	May 15/2025	
R 931	Sep 15/2026	
O 932	Sep 15/2026	
933	May 15/2025	
934	May 15/2025	
R 935	Sep 15/2026	
936	BLANK	
27-09-14		
201	Jan 15/2026	
202	Jan 15/2026	
203	Jan 15/2026	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	Sep 15/2025	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-09-14 (cont)		
R 209	Sep 15/2026	
210	May 15/2024	
211	May 15/2024	
212	May 15/2024	
213	May 15/2024	
R 214	Sep 15/2026	
215	May 15/2024	
216	May 15/2024	
217	May 15/2024	
218	May 15/2024	
R 219	Sep 15/2026	
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223	May 15/2024	
224	May 15/2024	
225	May 15/2024	
226	May 15/2024	
227	May 15/2024	
228	May 15/2024	
R 229	Sep 15/2026	
230	May 15/2024	
231	May 15/2024	
232	Jan 15/2026	
233	Jan 15/2026	
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240	Jan 15/2026	
241	Jan 15/2026	
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27-09-17		
201	May 15/2024	
202	May 15/2024	
27-09-18		
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404	May 15/2024	
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406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
27-09-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2025	
408	May 15/2024	
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27-09-21		
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508	BLANK	
27-09-31		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
27-09-31 (cont)		
406	Sep 15/2024	
407	May 15/2026	
R 408	Sep 15/2026	
409	May 15/2026	
R 410	Sep 15/2026	
411	May 15/2026	
412	BLANK	
27-09-31		
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502	May 15/2024	
27-09-45		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
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407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
27-09-61		
R 401	Sep 15/2026	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
R 406	Sep 15/2026	
O 407	Sep 15/2026	
408	BLANK	
27-09-61		
501	May 15/2024	
502	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

Subject/Page	Date	COC
27-09-71		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	Jan 15/2026	
507	Jan 15/2026	
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532	May 15/2024	
533	May 15/2024	
534	May 15/2024	
535	May 15/2024	
536	May 15/2024	

Subject/Page	Date	COC
27-09-91		
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603	May 15/2024	
604	May 15/2024	
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623	May 15/2024	
624	May 15/2024	
625	May 15/2024	
626	May 15/2024	
627	May 15/2024	
628	May 15/2024	
629	Jan 15/2026	
630	May 15/2024	
631	May 15/2024	
632	May 15/2024	
633	May 15/2024	
634	BLANK	

Subject/Page	Date	COC
27-11-00		
201	Sep 15/2025	
202	May 15/2025	
203	May 15/2025	
204	May 15/2024	
205	May 15/2024	
206	May 15/2025	
207	May 15/2024	
208	May 15/2024	
209	Sep 15/2025	
210	Jan 15/2025	
27-11-00		
R 501	Sep 15/2026	
R 502	Sep 15/2026	
R 503	Sep 15/2026	
R 504	Sep 15/2026	
R 505	Sep 15/2026	
R 506	Sep 15/2026	
R 507	Sep 15/2026	
R 508	Sep 15/2026	
R 509	Sep 15/2026	
510	May 15/2024	
511	May 15/2024	
R 512	Sep 15/2026	
513	May 15/2024	
R 514	Sep 15/2026	
R 515	Sep 15/2026	
R 516	Sep 15/2026	
R 517	Sep 15/2026	
R 518	Sep 15/2026	
R 519	Sep 15/2026	
R 520	Sep 15/2026	
R 521	Sep 15/2026	
R 522	Sep 15/2026	
O 523	Sep 15/2026	
O 524	Sep 15/2026	
O 525	Sep 15/2026	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-11-00 (cont)		
O 526	Sep 15/2026	
O 527	Sep 15/2026	
O 528	Sep 15/2026	
O 529	Sep 15/2026	
O 530	Sep 15/2026	
O 531	Sep 15/2026	
O 532	Sep 15/2026	
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O 536	Sep 15/2026	
O 537	Sep 15/2026	
O 538	Sep 15/2026	
R 539	Sep 15/2026	
R 540	Sep 15/2026	
O 541	Sep 15/2026	
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O 543	Sep 15/2026	
O 544	Sep 15/2026	
O 545	Sep 15/2026	
R 546	Sep 15/2026	
O 547	Sep 15/2026	
R 548	Sep 15/2026	
R 549	Sep 15/2026	
O 550	Sep 15/2026	
O 551	Sep 15/2026	
O 552	Sep 15/2026	
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O 554	Sep 15/2026	
O 555	Sep 15/2026	
O 556	Sep 15/2026	
O 557	Sep 15/2026	
O 558	Sep 15/2026	
O 559	Sep 15/2026	
O 560	Sep 15/2026	
O 561	Sep 15/2026	
O 562	Sep 15/2026	
O 563	Sep 15/2026	
O 564	Sep 15/2026	
O 565	Sep 15/2026	
O 566	Sep 15/2026	
O 567	Sep 15/2026	
O 568	Sep 15/2026	
O 569	Sep 15/2026	
O 570	Sep 15/2026	
O 571	Sep 15/2026	
O 572	Sep 15/2026	
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27-11-11		
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R 419	Sep 15/2026	
O 420	Sep 15/2026	
O 421	Sep 15/2026	
O 422	Sep 15/2026	
O 423	Sep 15/2026	
O 424	Sep 15/2026	
O 425	Sep 15/2026	
O 426	Sep 15/2026	
O 427	Sep 15/2026	
O 428	Sep 15/2026	
27-11-11		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
27-11-14		
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405	May 15/2024	
406	May 15/2024	
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408	May 15/2026	
27-11-21		
401	Jan 15/2026	
402	Jan 15/2026	
403	Sep 15/2025	
404	Sep 15/2025	
405	Sep 15/2025	
406	Sep 15/2025	
R 407	Sep 15/2026	
O 408	Sep 15/2026	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-11-21 (cont)		
O 409	Sep 15/2026	
O 410	Sep 15/2026	
411	Sep 15/2025	
412	BLANK	
27-11-21		
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602	May 15/2024	
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604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
27-11-24		
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403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
R 409	Sep 15/2026	
O 410	Sep 15/2026	
O 411	Sep 15/2026	
412	BLANK	
27-11-31		
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402	May 15/2026	
403	May 15/2024	
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405	May 15/2024	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
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Subject/Page	Date	COC
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602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2026	
606	May 15/2026	
607	May 15/2024	
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609	May 15/2024	
610	BLANK	
27-11-34		
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403	May 15/2024	
404	May 15/2024	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	BLANK	
27-11-34		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
27-11-41		
401	May 15/2024	
402	May 15/2024	
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404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2026	
408	Jan 15/2026	
409	Jan 15/2026	
410	May 15/2024	

Subject/Page	Date	COC
27-11-51		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
R 404	Sep 15/2026	
R 405	Sep 15/2026	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
27-11-54		
401	Sep 15/2025	
402	Sep 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Sep 15/2025	
27-11-61		
401	May 15/2024	
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405	May 15/2024	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
408	May 15/2026	
R 409	Sep 15/2026	
O 410	Sep 15/2026	
D 411	Sep 15/2026	
D 412	BLANK	
27-11-61		
601	May 15/2024	
R 602	Sep 15/2026	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-11-71		
O 401	Sep 15/2026	
R 402	Sep 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Jan 15/2026	
408	Jan 15/2026	
R 409	Sep 15/2026	
410	BLANK	
27-11-81		
401	May 15/2026	
402	May 15/2026	
403	Jan 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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415	May 15/2024	
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27-11-81		
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603	May 15/2024	
604	BLANK	

Subject/Page	Date	COC
27-11-94		
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402	Sep 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
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407	May 15/2024	
408	May 15/2024	
27-21-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
R 205	Sep 15/2026	
O 206	Sep 15/2026	
O 207	Sep 15/2026	
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R 210	Sep 15/2026	
211	May 15/2024	
R 212	Sep 15/2026	
213	Jan 15/2026	
214	Jan 15/2026	
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507	May 15/2024	
508	Sep 15/2025	
509	May 15/2024	
510	May 15/2024	
511	May 15/2024	
R 512	Sep 15/2026	
R 513	Sep 15/2026	

Subject/Page	Date	COC
27-21-00 (cont)		
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515	Sep 15/2024	
516	Sep 15/2025	
517	May 15/2024	
518	Sep 15/2024	
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547	May 15/2024	
548	May 15/2024	
549	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-21-00 (cont)		
550	Jan 15/2025	
551	May 15/2024	
552	May 15/2024	
553	May 15/2024	
554	May 15/2024	
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27-21-11		
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O 404	Sep 15/2026	
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27-21-17		
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27-21-24		
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403	May 15/2024	
404	May 15/2024	
405	Jan 15/2026	
27-21-24 (cont)		
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407	May 15/2024	
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27-21-24		
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27-21-31		
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402	Jan 15/2025	
403	Jan 15/2025	
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407	Jan 15/2025	
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27-21-41		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2025	
408	May 15/2025	
409	May 15/2025	
410	BLANK	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-21-51		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
R 404	Sep 15/2026	
405	May 15/2024	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
O 417	Sep 15/2026	
R 418	Sep 15/2026	
R 419	Sep 15/2026	
R 420	Sep 15/2026	
R 421	Sep 15/2026	
R 422	Sep 15/2026	
A 423	Sep 15/2026	
A 424	BLANK	
27-21-51		
601	May 15/2024	
602	May 15/2024	
27-21-61		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
R 409	Sep 15/2026	

Subject/Page	Date	COC
27-21-61 (cont)		
O 410	Sep 15/2026	
411	May 15/2026	
412	BLANK	
27-21-61		
R 601	Sep 15/2026	
O 602	Sep 15/2026	
603	May 15/2024	
604	May 15/2024	
605	Sep 15/2025	
R 606	Sep 15/2026	
O 607	Sep 15/2026	
608	BLANK	
27-21-71		
401	Sep 15/2024	
402	Sep 15/2024	
403	Sep 15/2024	
404	Sep 15/2024	
405	May 15/2024	
406	Sep 15/2024	
407	Sep 15/2024	
408	Sep 15/2024	
27-21-82		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	Jan 15/2025	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	

Subject/Page	Date	COC
27-21-82 (cont)		
416	May 15/2024	
417	May 15/2024	
418	BLANK	
27-21-82		
601	May 15/2024	
602	BLANK	
27-21-87		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
27-21-91		
401	Jan 15/2026	
402	May 15/2025	
403	Jan 15/2026	
404	May 15/2025	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	Jan 15/2026	
411	May 15/2025	
412	May 15/2025	
413	Jan 15/2026	
414	May 15/2025	
415	May 15/2025	
416	May 15/2025	
417	May 15/2024	
418	May 15/2024	
419	May 15/2025	
420	May 15/2025	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-21-94		
201	Sep 15/2024	
202	Sep 15/2024	
203	May 15/2024	
204	May 15/2024	
205	Sep 15/2024	
206	Sep 15/2024	
207	Sep 15/2024	
208	Sep 15/2024	
209	Sep 15/2024	
210	Sep 15/2024	
211	Sep 15/2024	
212	Sep 15/2024	
27-21-95		
401	Jan 15/2025	
402	May 15/2024	
403	May 15/2024	
404	Jan 15/2025	
405	Jan 15/2025	
406	BLANK	
27-31-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	Sep 15/2025	
206	May 15/2025	
207	May 15/2024	
208	May 15/2024	
209	May 15/2025	
210	Sep 15/2025	
211	Jan 15/2025	
212	Jan 15/2025	
213	Jan 15/2025	
R 214	Sep 15/2026	
O 215	Sep 15/2026	
216	May 15/2024	
217	May 15/2024	
27-31-00 (cont)		
218	May 15/2024	
219	May 15/2024	
220	May 15/2024	
R 221	Sep 15/2026	
R 222	Sep 15/2026	
R 223	Sep 15/2026	
224	May 15/2024	
225	May 15/2024	
226	May 15/2024	
227	Sep 15/2025	
228	May 15/2024	
229	May 15/2024	
230	Sep 15/2025	
231	Sep 15/2025	
232	May 15/2024	
233	May 15/2024	
234	Sep 15/2025	
235	May 15/2025	
236	May 15/2024	
237	May 15/2024	
238	May 15/2025	
239	Sep 15/2025	
240	Sep 15/2025	
27-31-00		
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504	May 15/2024	
505	May 15/2024	
506	Jan 15/2026	
507	Jan 15/2026	
508	Jan 15/2026	
509	Jan 15/2026	
510	Jan 15/2026	
R 511	Sep 15/2026	
O 512	Sep 15/2026	
27-31-00 (cont)		
R 513	Sep 15/2026	
O 514	Sep 15/2026	
O 515	Sep 15/2026	
516	Jan 15/2026	
R 517	Sep 15/2026	
518	Jan 15/2026	
519	Jan 15/2026	
520	Jan 15/2026	
R 521	Sep 15/2026	
522	May 15/2024	
523	May 15/2024	
524	Jan 15/2026	
525	Jan 15/2025	
526	Jan 15/2026	
R 527	Sep 15/2026	
528	Jan 15/2026	
529	Jan 15/2026	
530	Jan 15/2026	
531	Jan 15/2026	
532	Jan 15/2026	
533	Jan 15/2026	
534	Jan 15/2026	
535	Jan 15/2026	
536	Jan 15/2026	
537	Jan 15/2026	
538	Jan 15/2026	
539	May 15/2024	
540	Jan 15/2026	
541	Jan 15/2026	
542	Jan 15/2026	
R 543	Sep 15/2026	
544	Jan 15/2026	
545	Jan 15/2026	
546	Jan 15/2026	
547	Jan 15/2026	
548	Jan 15/2026	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-31-00 (cont)		
549	Jan 15/2026	
550	Jan 15/2026	
551	Jan 15/2026	
552	Jan 15/2026	
553	Jan 15/2026	
554	Jan 15/2026	
555	Jan 15/2026	
556	Jan 15/2026	
R 557	Sep 15/2026	
O 558	Sep 15/2026	
559	May 15/2024	
560	Jan 15/2026	
561	Jan 15/2026	
562	Jan 15/2025	
R 563	Sep 15/2026	
564	Jan 15/2026	
565	Jan 15/2026	
566	Jan 15/2026	
567	Jan 15/2026	
568	Jan 15/2025	
569	Jan 15/2026	
570	Jan 15/2026	
R 571	Sep 15/2026	
R 572	Sep 15/2026	
R 573	Sep 15/2026	
574	Sep 15/2025	
575	Jan 15/2026	
576	Jan 15/2026	
577	Jan 15/2026	
578	Jan 15/2026	
579	Sep 15/2025	
580	Jan 15/2026	
R 581	Sep 15/2026	
R 582	Sep 15/2026	
R 583	Sep 15/2026	
R 584	Sep 15/2026	
O 585	Sep 15/2026	
586	Jan 15/2025	
587	Jan 15/2025	
588	May 15/2024	
589	May 15/2024	
590	Jan 15/2026	
R 591	Sep 15/2026	
O 592	Sep 15/2026	
O 593	Sep 15/2026	
594	Jan 15/2026	
595	Jan 15/2026	
596	Jan 15/2026	
R 597	Sep 15/2026	
598	Sep 15/2025	
598.1	Jan 15/2026	
598.2	Jan 15/2026	
598.3	Jan 15/2026	
598.4	Sep 15/2025	
598.5	Jan 15/2026	
598.6	Jan 15/2026	
598.7	May 15/2025	
598.8	Jan 15/2026	
598.9	Jan 15/2026	
598.10	Sep 15/2025	
598.11	Sep 15/2025	
598.12	Jan 15/2026	
598.13	Jan 15/2026	
598.14	Jan 15/2026	
R 598.15	Sep 15/2026	
O 598.16	Sep 15/2026	
O 598.17	Sep 15/2026	
O 598.18	Sep 15/2026	
O 598.19	Sep 15/2026	
O 598.20	Sep 15/2026	
O 598.21	Sep 15/2026	
O 598.22	Sep 15/2026	
O 598.23	Sep 15/2026	
O 598.24	Sep 15/2026	
O 598.25	Sep 15/2026	
O 598.26	Sep 15/2026	
O 598.27	Sep 15/2026	
O 598.28	Sep 15/2026	
O 598.29	Sep 15/2026	
O 598.30	Sep 15/2026	
O 598.31	Sep 15/2026	
O 598.32	Sep 15/2026	
R 598.33	Sep 15/2026	
O 598.34	Sep 15/2026	
D 598.35	Sep 15/2026	
D 598.36	BLANK	
27-31-11		
401	Jan 15/2026	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
R 404	Sep 15/2026	
O 405	Sep 15/2026	
O 406	Sep 15/2026	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	Sep 15/2025	
413	Sep 15/2025	
414	May 15/2025	
415	Jan 15/2026	
416	May 15/2025	
417	May 15/2025	
418	May 15/2025	
419	May 15/2025	
420	May 15/2025	
421	May 15/2025	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-31-11 (cont)		
422	May 15/2025	
423	May 15/2025	
424	May 15/2025	
425	May 15/2025	
R 426	Sep 15/2026	
427	Jan 15/2026	
428	Jan 15/2026	
R 429	Sep 15/2026	
O 430	Sep 15/2026	
R 431	Sep 15/2026	
R 432	Sep 15/2026	
R 433	Sep 15/2026	
R 434	Sep 15/2026	
O 435	Sep 15/2026	
O 436	Sep 15/2026	
O 437	Sep 15/2026	
O 438	Sep 15/2026	
439	Jan 15/2026	
440	Jan 15/2026	
441	Jan 15/2026	
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443	May 15/2026	
444	Jan 15/2026	
445	Jan 15/2026	
446	Jan 15/2026	
447	Jan 15/2026	
448	BLANK	
27-31-14		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	Jan 15/2025	
407	Jan 15/2025	
408	Jan 15/2025	

Subject/Page	Date	COC
27-31-14 (cont)		
409	Jan 15/2025	
410	Jan 15/2025	
411	Jan 15/2025	
412	Jan 15/2025	
413	Jan 15/2025	
414	Sep 15/2025	
415	Sep 15/2025	
416	BLANK	
27-31-14		
601	May 15/2026	
602	May 15/2026	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	
610	May 15/2024	
27-31-17		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
R 204	Sep 15/2026	
O 205	Sep 15/2026	
O 206	Sep 15/2026	
O 207	Sep 15/2026	
O 208	Sep 15/2026	
R 209	Sep 15/2026	
O 210	Sep 15/2026	
O 211	Sep 15/2026	
O 212	Sep 15/2026	
R 213	Sep 15/2026	
O 214	Sep 15/2026	
O 215	Sep 15/2026	
O 216	Sep 15/2026	
R 217	Sep 15/2026	

Subject/Page	Date	COC
27-31-17 (cont)		
O 218	Sep 15/2026	
A 219	Sep 15/2026	
A 220	Sep 15/2026	
27-31-20		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2026	
407	May 15/2024	
408	May 15/2024	
27-31-20		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
27-31-31		
401	Jan 15/2025	
402	May 15/2026	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
R 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
O 414	Sep 15/2026	
O 415	Sep 15/2026	
O 416	Sep 15/2026	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

Subject/Page	Date	COC
27-31-32		
601	Jan 15/2026	
602	Sep 15/2025	
603	Sep 15/2025	
604	Sep 15/2025	
605	May 15/2024	
606	May 15/2024	
607	Jan 15/2026	
608	Jan 15/2025	
609	May 15/2024	
610	May 15/2024	
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614	May 15/2024	
615	May 15/2024	
616	May 15/2024	
617	May 15/2024	
618	May 15/2024	
27-31-34		
401	Jan 15/2025	
402	Jan 15/2025	
403	Sep 15/2025	
404	Sep 15/2025	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	Sep 15/2025	
414	Sep 15/2025	
R 415	Sep 15/2026	
416	May 15/2026	
417	May 15/2026	
418	May 15/2026	
419	May 15/2026	
420	May 15/2026	
421	May 15/2026	
422	Jan 15/2025	
R 423	Sep 15/2026	
O 424	Sep 15/2026	
425	Sep 15/2025	
426	Sep 15/2025	
427	Sep 15/2025	
428	Sep 15/2025	
429	Sep 15/2025	
430	Sep 15/2025	
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432	Sep 15/2025	
433	May 15/2024	
434	May 15/2024	
435	May 15/2024	
436	Jan 15/2025	
437	Jan 15/2025	
438	Jan 15/2025	
27-31-34		
R 601	Sep 15/2026	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
27-31-35		
R 401	Sep 15/2026	
O 402	Sep 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2025	
R 406	Sep 15/2026	
27-31-37		
201	May 15/2026	
202	May 15/2024	
203	May 15/2024	
204	May 15/2025	
R 205	Sep 15/2026	
O 206	Sep 15/2026	
O 207	Sep 15/2026	
O 208	Sep 15/2026	
27-31-37		
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R 402	Sep 15/2026	
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2026	
407	May 15/2024	
408	May 15/2024	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
O 413	Sep 15/2026	
O 414	Sep 15/2026	
R 415	Sep 15/2026	
O 416	Sep 15/2026	
27-31-37		
601	Sep 15/2025	
602	Sep 15/2025	
603	Sep 15/2025	
604	BLANK	
27-31-41		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-31-41 (cont)		
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
27-31-41		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
27-31-51		
401	May 15/2024	
402	May 15/2024	
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409	May 15/2024	
410	May 15/2024	
411	May 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
27-31-54		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
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406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	Jan 15/2025	
411	Jan 15/2025	
412	May 15/2024	
27-31-61		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
R 411	Sep 15/2026	
412	May 15/2026	
R 413	Sep 15/2026	
O 414	Sep 15/2026	
415	May 15/2026	
416	BLANK	
27-31-61		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	BLANK	
27-31-64		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
27-31-64 (cont)		
416	May 15/2024	
27-31-64		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
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27-31-67		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	Sep 15/2024	
R 406	Sep 15/2026	
407	May 15/2025	
408	May 15/2026	
409	May 15/2026	
410	May 15/2026	
27-31-67		
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502	May 15/2026	
503	May 15/2026	
504	May 15/2026	
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506	May 15/2026	
507	May 15/2026	
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510	May 15/2026	
511	May 15/2026	
512	BLANK	
27-31-71		
401	Jan 15/2025	
402	May 15/2026	
403	Sep 15/2025	
404	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-31-71 (cont)		
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	Jan 15/2025	
412	Sep 15/2025	
413	Sep 15/2025	
414	Jan 15/2025	
27-31-74		
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406	Sep 15/2025	
407	Sep 15/2024	
408	Sep 15/2024	
409	Sep 15/2025	
410	Sep 15/2024	
27-31-81		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
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405	May 15/2024	
406	May 15/2024	
407	Jan 15/2025	
408	Jan 15/2025	
409	May 15/2024	
410	BLANK	
27-31-91		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2024	

Subject/Page	Date	COC
27-31-91 (cont)		
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	Jan 15/2025	
412	May 15/2024	
413	Jan 15/2025	
414	May 15/2024	
27-31-91		
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602	May 15/2024	
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604	May 15/2024	
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606	BLANK	
27-31-94		
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402	May 15/2025	
403	May 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
R 408	Sep 15/2026	
409	May 15/2025	
410	Jan 15/2025	
411	May 15/2024	
412	BLANK	
27-32-00		
501	May 15/2025	
502	Jan 15/2025	
503	Jan 15/2025	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	

Subject/Page	Date	COC
27-32-00 (cont)		
R 507	Sep 15/2026	
R 508	Sep 15/2026	
R 509	Sep 15/2026	
R 510	Sep 15/2026	
R 511	Sep 15/2026	
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R 513	Sep 15/2026	
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27-32-21		
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404	Jan 15/2025	
27-32-31		
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402	May 15/2024	
403	Sep 15/2024	
404	Sep 15/2024	
405	Sep 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
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27-41-00		
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27-41-00 (cont)		
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O 419	Sep 15/2026	
O 420	Sep 15/2026	
O 421	Sep 15/2026	
O 422	Sep 15/2026	
O 423	Sep 15/2026	
O 424	Sep 15/2026	
O 425	Sep 15/2026	
426	May 15/2024	
27-41-21		
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403	May 15/2024	
404	Sep 15/2025	
27-41-31		
401	Jan 15/2026	
402	Jan 15/2026	
403	Jan 15/2026	
404	Jan 15/2026	
405	Jan 15/2026	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
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O 420	Sep 15/2026	
O 421	Sep 15/2026	
O 422	Sep 15/2026	
O 423	Sep 15/2026	
O 424	Sep 15/2026	
27-41-41		
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27-41-41		
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602	May 15/2024	
27-41-41 (cont)		
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27-41-51		
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27-41-61		
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27-41-71		
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R 402	Sep 15/2026	
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R 406	Sep 15/2026	
27-41-71		
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R 503	Sep 15/2026	
R 504	Sep 15/2026	
27-41-81		
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402	Jan 15/2026	
R 403	Sep 15/2026	
O 404	Sep 15/2026	
R 405	Sep 15/2026	
R 406	Sep 15/2026	
407	Sep 15/2025	
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27-41-81 (cont)		
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R 411	Sep 15/2026	
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R 413	Sep 15/2026	
R 414	Sep 15/2026	
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R 419	Sep 15/2026	
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R 424	Sep 15/2026	
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27-41-81		
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O 605	Sep 15/2026	
R 606	Sep 15/2026	
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R 609	Sep 15/2026	
R 610	Sep 15/2026	
O 611	Sep 15/2026	
O 612	Sep 15/2026	
O 613	Sep 15/2026	
O 614	Sep 15/2026	
R 615	Sep 15/2026	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-41-81 (cont)		
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A 617	Sep 15/2026	
A 618	Sep 15/2026	
A 619	Sep 15/2026	
A 620	Sep 15/2026	
27-41-82		
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27-41-91		
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27-41-91		
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27-41-91 (cont)		
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27-41-94		
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27-41-94		
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27-41-97		
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27-41-97		
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27-51-00		
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27-51-00		
501	Sep 15/2025	
502	Sep 15/2025	
503	Sep 15/2025	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
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R 510	Sep 15/2026	
O 511	Sep 15/2026	
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27-51-00 (cont)		
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575	May 15/2026	
27-51-00 (cont)		
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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

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27-51-04		
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27-51-06		
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27-51-08		
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27-51-08		
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27-51-11		
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R 404	Sep 15/2026	
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27-51-11 (cont)		
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R 416	Sep 15/2026	
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R 426	Sep 15/2026	
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R 428	Sep 15/2026	
R 429	Sep 15/2026	
R 430	Sep 15/2026	
27-51-11		
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517	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
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27-51-12		
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27-51-14		
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27-51-15		
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27-51-15		
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603	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
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507	May 15/2024	

Subject/Page	Date	COC
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27-51-21		
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410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
415	May 15/2026	
R 416	Sep 15/2026	
417	Jan 15/2026	
418	Jan 15/2026	
419	Jan 15/2026	
R 420	Sep 15/2026	
421	May 15/2024	
422	May 15/2024	
R 423	Sep 15/2026	
R 424	Sep 15/2026	
R 425	Sep 15/2026	
R 426	Sep 15/2026	
27-51-21		
501	May 15/2026	
R 502	Sep 15/2026	
O 503	Sep 15/2026	
R 504	Sep 15/2026	
R 505	Sep 15/2026	
R 506	Sep 15/2026	
O 507	Sep 15/2026	
O 508	Sep 15/2026	

Subject/Page	Date	COC
27-51-21 (cont)		
R 509	Sep 15/2026	
510	May 15/2026	
511	May 15/2026	
512	May 15/2026	
513	May 15/2026	
514	May 15/2026	
R 515	Sep 15/2026	
516	May 15/2026	
517	May 15/2026	
518	May 15/2026	
519	May 15/2026	
520	May 15/2026	
27-51-22		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
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405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
R 417	Sep 15/2026	
R 418	Sep 15/2026	
419	May 15/2026	
420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-51-22 (cont)		
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27-51-22		
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607	May 15/2026	
608	May 15/2026	
27-51-23		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	Sep 15/2025	
409	Jan 15/2026	
410	Sep 15/2025	
411	Sep 15/2025	
412	Sep 15/2025	
413	Sep 15/2025	
414	May 15/2024	
415	May 15/2024	
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417	May 15/2024	
418	Sep 15/2025	
419	Jan 15/2026	
420	Sep 15/2025	
421	Sep 15/2025	
422	BLANK	
27-51-23		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	

Subject/Page	Date	COC
27-51-23 (cont)		
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605	May 15/2024	
606	May 15/2024	
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610	May 15/2024	
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612	May 15/2024	
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614	May 15/2024	
615	May 15/2024	
616	May 15/2024	
27-51-24		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Sep 15/2025	
407	Sep 15/2025	
408	Sep 15/2025	
409	Sep 15/2025	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
27-51-25		
401	May 15/2024	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	Jan 15/2025	
409	Sep 15/2024	
410	Sep 15/2024	

Subject/Page	Date	COC
27-51-25 (cont)		
411	Jan 15/2025	
412	Sep 15/2024	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
416	May 15/2024	
417	Jan 15/2025	
418	Sep 15/2024	
419	Sep 15/2024	
420	BLANK	
27-51-25		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	
610	BLANK	
27-51-26		
401	May 15/2024	
402	Sep 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	Sep 15/2025	
413	Sep 15/2025	
414	Sep 15/2025	
415	Sep 15/2025	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-51-26 (cont)		
416	Sep 15/2025	
27-51-28		
401	May 15/2024	
402	Sep 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2025	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
O 413	Sep 15/2026	
O 414	Sep 15/2026	
O 415	Sep 15/2026	
O 416	Sep 15/2026	
O 417	Sep 15/2026	
O 418	Sep 15/2026	
O 419	Sep 15/2026	
O 420	Sep 15/2026	
O 421	Sep 15/2026	
O 422	Sep 15/2026	
R 423	Sep 15/2026	
R 424	Sep 15/2026	
R 425	Sep 15/2026	
R 426	Sep 15/2026	
R 427	Sep 15/2026	
O 428	Sep 15/2026	
O 429	Sep 15/2026	
O 430	Sep 15/2026	
A 431	Sep 15/2026	
A 432	BLANK	
27-51-28		
501	May 15/2024	
R 502	Sep 15/2026	

Subject/Page	Date	COC
27-51-28 (cont)		
R 503	Sep 15/2026	
R 504	Sep 15/2026	
505	May 15/2024	
506	May 15/2024	
R 507	Sep 15/2026	
508	May 15/2024	
509	May 15/2024	
510	May 15/2024	
R 511	Sep 15/2026	
R 512	Sep 15/2026	
R 513	Sep 15/2026	
514	May 15/2024	
R 515	Sep 15/2026	
516	May 15/2024	
517	May 15/2024	
518	BLANK	
27-51-31		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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413	May 15/2024	
414	May 15/2024	
415	Jan 15/2026	
416	Jan 15/2026	
417	Sep 15/2025	
418	Sep 15/2025	
27-51-32		
401	May 15/2024	

Subject/Page	Date	COC
27-51-32 (cont)		
402	May 15/2024	
403	Jan 15/2025	
404	May 15/2024	
405	Jan 15/2025	
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411	Jan 15/2025	
412	Jan 15/2025	
413	Jan 15/2025	
414	Jan 15/2025	
415	Sep 15/2025	
416	Sep 15/2025	
27-51-32		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	Sep 15/2024	
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614	May 15/2024	
615	May 15/2024	
616	May 15/2024	
617	May 15/2024	
618	May 15/2024	
619	May 15/2024	
620	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-51-32 (cont)		
621	Sep 15/2025	
622	Sep 15/2025	
623	May 15/2024	
624	May 15/2024	
625	May 15/2024	
626	Sep 15/2025	
627	Sep 15/2025	
628	May 15/2024	
629	May 15/2024	
630	May 15/2024	
27-51-41		
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419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
422	Sep 15/2025	
423	May 15/2024	
424	Jan 15/2026	
425	May 15/2024	

Subject/Page	Date	COC
27-51-41 (cont)		
426	May 15/2024	
27-51-42		
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403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	Jan 15/2025	
407	May 15/2024	
408	Jan 15/2025	
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412	May 15/2024	
413	May 15/2024	
414	Jan 15/2025	
415	May 15/2024	
416	May 15/2026	
417	May 15/2026	
418	May 15/2026	
27-51-42		
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610	Sep 15/2025	
611	Sep 15/2025	
612	May 15/2024	
613	May 15/2024	
614	May 15/2024	
615	Sep 15/2025	
616	Sep 15/2025	

Subject/Page	Date	COC
27-51-42 (cont)		
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619	May 15/2024	
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27-51-51		
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413	May 15/2024	
414	Jan 15/2026	
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417	Jan 15/2026	
418	May 15/2024	
419	May 15/2024	
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422	Jan 15/2026	
423	May 15/2024	
424	Jan 15/2026	
425	May 15/2024	
426	BLANK	
27-51-51		
501	May 15/2024	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-51-51 (cont)		
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27-51-52		
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402	May 15/2026	
403	May 15/2024	
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405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
408	BLANK	
27-51-53		
401	May 15/2026	
402	May 15/2026	
403	May 15/2024	
404	May 15/2026	
405	May 15/2026	
406	Jan 15/2026	
407	May 15/2024	
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27-51-55		
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402	May 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Jan 15/2026	
407	May 15/2024	
408	BLANK	
27-51-56		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Jan 15/2026	

Subject/Page	Date	COC
27-51-56 (cont)		
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27-51-57		
401	May 15/2026	
402	May 15/2026	
403	May 15/2024	
404	May 15/2026	
405	May 15/2026	
406	May 15/2026	
27-51-58		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2026	
424	BLANK	
27-51-58		
601	May 15/2024	
602	May 15/2024	

Subject/Page	Date	COC
27-51-58 (cont)		
R 603	Sep 15/2026	
O 604	Sep 15/2026	
605	May 15/2024	
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27-51-59		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
R 406	Sep 15/2026	
27-51-61		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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418	May 15/2024	
27-51-62		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-51-62 (cont)		
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409	May 15/2024	
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411	Jan 15/2026	
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27-51-63		
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403	May 15/2024	
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405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	BLANK	
27-51-63		
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502	May 15/2024	
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504	Sep 15/2025	
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506	Sep 15/2025	
507	Jan 15/2026	
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27-51-63 (cont)		
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27-51-65		
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402	May 15/2024	
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404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
O 410	Sep 15/2026	
R 411	Sep 15/2026	
412	May 15/2024	
27-51-65		
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503	Jan 15/2026	
504	May 15/2026	
505	May 15/2024	
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27-51-65		
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803	Jan 15/2026	
804	May 15/2026	
805	May 15/2024	
806	May 15/2024	
27-51-67		
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402	Jan 15/2025	
27-51-67 (cont)		
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404	Jan 15/2025	
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27-58-00		
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202	May 15/2024	
203	May 15/2024	
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27-58-01		
401	Sep 15/2025	
402	Sep 15/2025	
403	Jan 15/2026	
R 404	Sep 15/2026	
R 405	Sep 15/2026	
R 406	Sep 15/2026	
R 407	Sep 15/2026	
O 408	Sep 15/2026	
O 409	Sep 15/2026	
O 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
27-58-01		
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506	May 15/2024	
507	May 15/2024	
508	BLANK	
27-59-00		
501	May 15/2024	
502	May 15/2024	
503	Jan 15/2026	
504	May 15/2024	
505	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-59-00 (cont)		
506	May 15/2024	
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508	May 15/2024	
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27-59-01		
R 201	Sep 15/2026	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
R 205	Sep 15/2026	
206	Jan 15/2026	
R 207	Sep 15/2026	
208	May 15/2024	
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R 211	Sep 15/2026	
212	Jan 15/2026	
R 213	Sep 15/2026	
214	May 15/2024	
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216	May 15/2024	
R 217	Sep 15/2026	
218	Jan 15/2026	
R 219	Sep 15/2026	
O 220	Sep 15/2026	
221	May 15/2024	
222	May 15/2024	

Subject/Page	Date	COC
27-59-01 (cont)		
R 223	Sep 15/2026	
O 224	Sep 15/2026	
O 225	Sep 15/2026	
226	BLANK	
27-59-02		
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202	May 15/2024	
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R 205	Sep 15/2026	
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O 207	Sep 15/2026	
208	May 15/2024	
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211	May 15/2024	
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219	May 15/2026	
220	Jan 15/2026	
221	May 15/2024	
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223	May 15/2024	
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225	May 15/2024	
226	May 15/2026	
227	Jan 15/2026	
228	May 15/2024	
229	May 15/2024	
230	BLANK	
27-61-00		
201	Sep 15/2025	

Subject/Page	Date	COC
27-61-00 (cont)		
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203	May 15/2025	
204	May 15/2024	
205	May 15/2024	
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211	May 15/2024	
212	May 15/2024	
213	Sep 15/2025	
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27-61-00		
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512	Jan 15/2026	
513	May 15/2024	
514	Sep 15/2025	
515	Sep 15/2025	
516	Sep 15/2025	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-61-00 (cont)		
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518	Sep 15/2025	
519	Sep 15/2025	
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27-61-00		
601	May 15/2024	
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27-61-11		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	Jan 15/2025	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2025	
410	May 15/2025	
411	Jan 15/2025	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
415	Jan 15/2025	
416	BLANK	
27-61-12		
401	Sep 15/2024	
402	Sep 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Sep 15/2024	

Subject/Page	Date	COC
27-61-12 (cont)		
407	May 15/2024	
408	May 15/2024	
409	Sep 15/2024	
410	Sep 15/2024	
27-61-12		
501	May 15/2024	
502	May 15/2024	
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506	May 15/2024	
27-61-15		
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403	May 15/2024	
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407	Jan 15/2025	
408	Jan 15/2025	
27-61-44		
401	May 15/2024	
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405	May 15/2024	
406	May 15/2024	
27-61-45		
201	May 15/2024	
202	May 15/2024	
27-61-45		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	BLANK	

Subject/Page	Date	COC
27-61-45		
501	Jan 15/2026	
502	Jan 15/2026	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	
27-61-51		
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402	Sep 15/2025	
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404	Sep 15/2025	
405	Sep 15/2025	
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407	Sep 15/2025	
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414	Sep 15/2025	
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416	Sep 15/2025	
417	Sep 15/2025	
418	Sep 15/2025	
419	Sep 15/2025	
420	Sep 15/2025	
27-61-51		
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602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
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606	BLANK	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

Subject/Page	Date	COC
27-62-00		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	Sep 15/2025	
205	Sep 15/2025	
206	May 15/2024	
207	May 15/2024	
208	May 15/2025	
209	Sep 15/2025	
210	May 15/2024	
27-62-00		
501	May 15/2025	
502	May 15/2025	
R 503	Sep 15/2026	
504	May 15/2025	
505	May 15/2024	
506	May 15/2024	
507	May 15/2024	
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519	May 15/2024	
520	May 15/2024	
521	Jan 15/2026	
522	May 15/2024	
523	May 15/2024	
524	May 15/2024	
525	May 15/2024	

Subject/Page	Date	COC
27-62-00 (cont)		
526	May 15/2025	
527	May 15/2025	
528	May 15/2025	
529	May 15/2025	
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538	May 15/2025	
539	May 15/2025	
540	May 15/2025	
541	May 15/2025	
542	BLANK	
27-62-00		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
27-62-12		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	Jan 15/2025	
410	May 15/2024	

Subject/Page	Date	COC
27-62-12 (cont)		
411	May 15/2024	
412	Jan 15/2025	
413	Sep 15/2025	
414	Sep 15/2025	
27-62-15		
401	Jan 15/2025	
402	Jan 15/2025	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Jan 15/2025	
408	Jan 15/2025	
27-62-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
R 404	Sep 15/2026	
405	May 15/2024	
406	May 15/2024	
R 407	Sep 15/2026	
R 408	Sep 15/2026	
27-62-22		
R 401	Sep 15/2026	
O 402	Sep 15/2026	
R 403	Sep 15/2026	
O 404	Sep 15/2026	
R 405	Sep 15/2026	
O 406	Sep 15/2026	
O 407	Sep 15/2026	
408	May 15/2024	
409	May 15/2024	
R 410	Sep 15/2026	
411	May 15/2024	
412	May 15/2024	
R 413	Sep 15/2026	
414	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-62-22 (cont)		
415	May 15/2024	
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
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423	May 15/2024	
R 424	Sep 15/2026	
O 425	Sep 15/2026	
R 426	Sep 15/2026	
R 427	Sep 15/2026	
R 428	Sep 15/2026	
R 429	Sep 15/2026	
O 430	Sep 15/2026	
431	May 15/2026	
432	BLANK	
27-62-31		
401	Jan 15/2026	
402	May 15/2025	
403	Jan 15/2026	
404	Jan 15/2026	
405	Jan 15/2026	
406	Jan 15/2026	
407	Jan 15/2026	
408	Jan 15/2026	
R 409	Sep 15/2026	
410	Jan 15/2026	
411	Jan 15/2026	
412	Jan 15/2026	
27-62-32		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	

Subject/Page	Date	COC
27-62-32 (cont)		
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27-62-32		
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504	May 15/2024	
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27-62-37		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
R 408	Sep 15/2026	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
27-62-41		
401	Jan 15/2026	
402	Jan 15/2026	
403	May 15/2024	
404	Jan 15/2026	
405	May 15/2024	
406	Jan 15/2026	
407	Jan 15/2026	
408	Jan 15/2026	
27-62-71		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	

Subject/Page	Date	COC
27-62-71 (cont)		
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2025	
408	Sep 15/2025	
27-62-91		
401	Jan 15/2025	
402	Jan 15/2025	
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Jan 15/2025	
408	May 15/2024	
409	May 15/2024	
410	Jan 15/2025	
411	Sep 15/2025	
412	Sep 15/2025	
27-62-94		
401	May 15/2024	
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403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Sep 15/2025	
407	Jan 15/2026	
408	Sep 15/2025	
27-62-94		
501	Sep 15/2025	
502	Sep 15/2025	
27-62-94		
601	May 15/2026	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
27-81-00		
201	Sep 15/2025	
202	Sep 15/2025	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-81-00 (cont)		
203	Sep 15/2025	
204	Sep 15/2025	
205	Sep 15/2025	
206	Sep 15/2025	
207	Sep 15/2025	
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223	Sep 15/2025	
R 224	Sep 15/2026	
225	Sep 15/2025	
226	Sep 15/2025	
227	Sep 15/2025	
228	Sep 15/2025	
229	Sep 15/2025	
230	Sep 15/2025	
R 231	Sep 15/2026	
R 232	Sep 15/2026	
233	Sep 15/2025	
234	Jan 15/2026	
235	Sep 15/2025	
236	Jan 15/2026	
237	Sep 15/2025	
238	Sep 15/2025	
27-81-00 (cont)		
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27-81-00		
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502	May 15/2024	
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519	Jan 15/2025	
520	Sep 15/2025	
521	Sep 15/2025	
R 522	Sep 15/2026	
523	May 15/2024	
27-81-00 (cont)		
524	May 15/2024	
R 525	Sep 15/2026	
526	May 15/2026	
527	May 15/2026	
528	May 15/2026	
529	May 15/2026	
530	Jan 15/2025	
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553	Sep 15/2024	
554	Sep 15/2024	
555	May 15/2024	
556	May 15/2024	
557	May 15/2024	
558	May 15/2026	
559	May 15/2026	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-81-00 (cont)		
560	May 15/2026	
561	May 15/2026	
562	May 15/2024	
563	May 15/2024	
564	May 15/2024	
565	May 15/2024	
566	Jan 15/2026	
567	Jan 15/2026	
568	Jan 15/2026	
569	May 15/2024	
570	May 15/2024	
571	May 15/2024	
572	May 15/2024	
27-81-00		
601	May 15/2026	
602	Sep 15/2025	
603	Sep 15/2025	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2026	
609	May 15/2024	
610	May 15/2024	
611	May 15/2024	
612	May 15/2026	
27-81-11		
401	May 15/2024	
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404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	

Subject/Page	Date	COC
27-81-11 (cont)		
411	May 15/2024	
412	May 15/2024	
R 413	Sep 15/2026	
O 414	Sep 15/2026	
O 415	Sep 15/2026	
O 416	Sep 15/2026	
417	May 15/2025	
418	May 15/2025	
419	May 15/2026	
420	Sep 15/2025	
421	Sep 15/2025	
422	May 15/2024	
423	May 15/2024	
424	May 15/2024	
425	May 15/2024	
426	May 15/2024	
427	May 15/2024	
428	May 15/2024	
27-81-11		
601	Jan 15/2025	
602	Jan 15/2025	
603	Jan 15/2025	
604	Jan 15/2025	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
27-81-21		
401	May 15/2024	
402	Sep 15/2025	
403	Sep 15/2025	
404	Sep 15/2025	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	

Subject/Page	Date	COC
27-81-21 (cont)		
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2026	
415	May 15/2024	
416	Sep 15/2025	
417	Sep 15/2025	
418	Sep 15/2025	
419	Sep 15/2025	
420	Jan 15/2026	
421	Jan 15/2026	
422	Sep 15/2025	
423	May 15/2026	
424	May 15/2026	
425	May 15/2026	
426	May 15/2026	
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430	Jan 15/2026	
431	Sep 15/2025	
432	Sep 15/2025	
433	Sep 15/2025	
434	Sep 15/2025	
27-81-31		
401	Sep 15/2025	
402	May 15/2024	
403	Sep 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-81-31 (cont)		
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412	May 15/2024	
413	May 15/2024	
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27-81-31		
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606	May 15/2024	
27-81-41		
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414	Jan 15/2025	
415	Sep 15/2025	
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418	Jan 15/2025	
419	Jan 15/2025	
420	Sep 15/2025	
421	Sep 15/2025	
422	Sep 15/2025	
423	Sep 15/2025	
424	Sep 15/2025	

Subject/Page	Date	COC
27-81-41		
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602	May 15/2024	
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27-81-52		
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404	May 15/2024	
405	May 15/2026	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
27-81-52		
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602	May 15/2024	
603	May 15/2024	
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27-81-61		
401	May 15/2025	
402	May 15/2025	
403	May 15/2024	
404	May 15/2025	
405	May 15/2025	
406	May 15/2025	
407	Sep 15/2024	
408	Sep 15/2024	
27-81-71		
401	Sep 15/2025	
402	May 15/2024	
403	Sep 15/2025	
404	Sep 15/2025	
405	Sep 15/2025	
406	BLANK	

Subject/Page	Date	COC
27-81-81		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	Jan 15/2026	
407	Jan 15/2026	
408	May 15/2024	
27-81-81		
501	May 15/2024	
502	May 15/2024	
503	Jan 15/2026	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
27-81-91		
R 201	Sep 15/2026	
O 202	Sep 15/2026	
203	May 15/2024	
204	Sep 15/2024	
R 205	Sep 15/2026	
R 206	Sep 15/2026	
O 207	Sep 15/2026	
R 208	Sep 15/2026	
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27-83-00		
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502	May 15/2024	
503	May 15/2024	
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505	May 15/2026	
506	May 15/2024	
507	May 15/2024	
508	May 15/2024	

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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Subject/Page	Date	COC
27-83-00 (cont)		
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510	May 15/2024	
511	May 15/2024	
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27-83-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2026	
405	May 15/2024	
406	May 15/2024	
27-83-21		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
27-88-00		
501	May 15/2026	
502	May 15/2024	
503	May 15/2024	
504	May 15/2024	
505	May 15/2024	
506	May 15/2024	
27-88-11		
R 401	Sep 15/2026	
R 402	Sep 15/2026	
R 403	Sep 15/2026	
O 404	Sep 15/2026	
O 405	Sep 15/2026	
O 406	Sep 15/2026	
O 407	Sep 15/2026	
O 408	Sep 15/2026	
O 409	Sep 15/2026	
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O 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
27-88-11 (cont)		
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O 415	Sep 15/2026	
O 416	Sep 15/2026	
O 417	Sep 15/2026	
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O 419	Sep 15/2026	
O 420	Sep 15/2026	
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R 422	Sep 15/2026	
R 423	Sep 15/2026	
O 424	Sep 15/2026	
O 425	Sep 15/2026	
O 426	Sep 15/2026	
O 427	Sep 15/2026	
O 428	Sep 15/2026	
O 429	Sep 15/2026	
O 430	Sep 15/2026	
27-88-11		
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27-88-11 (cont)		
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532	May 15/2025	
533	May 15/2025	
534	May 15/2025	
27-88-21		
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402	May 15/2024	
403	May 15/2024	
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27-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FLIGHT CONTROLS - DDG MAINTENANCE PROCEDURES	27-00-00		901	PTF ALL
DDG 27-21-01 Rudder Trim Indicator Inoperative - Preparation TASK 27-00-00-040-812			901	PTF ALL
DDG 27-21-01 Rudder Trim Indicator Inoperative - Restoration TASK 27-00-00-440-812			902	PTF ALL
DDG 27-21-02 STBY RUD ON Light Inoperative - Preparation TASK 27-00-00-040-813			902	PTF ALL
DDG 27-21-02 STBY RUD ON Light Inoperative - Restoration TASK 27-00-00-440-813			904	PTF ALL
DDG 27-31-01 FEEL DIFF PRESS Light System Inoperative - Preparation TASK 27-00-00-040-814			905	PTF ALL
DDG 27-31-01 FEEL DIFF PRESS Light System Inoperative - Restoration TASK 27-00-00-440-814			921	PTF ALL
DDG 27-61-01 Flight Spoiler Systems Inoperative - Preparation TASK 27-00-00-040-811			921	PTF ALL
DDG 27-61-01 Flight Spoiler Systems Inoperative - Restoration TASK 27-00-00-440-811			924	PTF ALL
DDG 27-62-01 Auto Speed Brake System Inoperative - Preparation TASK 27-00-00-040-815			926	PTF ALL
DDG 27-62-01 Auto Speed Brake System Inoperative - Restoration TASK 27-00-00-440-815			927	PTF ALL
DDG 27-62-02 SPEEDBRAKES EXTENDED Light Inoperative - Preparation TASK 27-00-00-040-816			927	PTF ALL
DDG 27-62-02 SPEEDBRAKES EXTENDED Light Inoperative - Restoration TASK 27-00-00-440-816			929	PTF ALL

27-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
DDG 27-83-02 AUTO SLAT FAIL Light System Inoperative - Preparation TASK 27-00-00-040-817			929	PTF ALL
DDG 27-83-02 AUTO SLAT FAIL Light System Inoperative - Restoration TASK 27-00-00-440-817			930	PTF ALL
DDG 27-88-01 Leading Edge Flap/Slat Position Light System Inoperative - Preparation TASK 27-00-00-040-818			931	PTF ALL
DDG 27-88-01 Leading Edge Flap/Slat Position Light System Inoperative - Restoration TASK 27-00-00-440-818			935	PTF ALL
FLIGHT CONTROLS CABLES - MAINTENANCE PRACTICES	27-09-14		201	PTF ALL
Control Cables Identification and Assembly Specifications TASK 27-09-14-860-801			201	PTF ALL
Control Cables - Rigging TASK 27-09-14-820-801			232	PTF ALL
Cable Control - Tension Measurement TASK 27-09-14-860-802			233	PTF ALL
FLIGHT CONTROL CABLES AIR PRESSURE SEALS - MAINTENANCE PRACTICES	27-09-17		201	PTF ALL
Location of the Flight Control Cables Air Pressure Seals TASK 27-09-17-860-801			201	PTF ALL
FLIGHT CONTROL CABLE GUARD - REMOVAL/INSTALLATION	27-09-18		401	PTF ALL
Control Cable Guard Removal TASK 27-09-18-000-801			401	PTF ALL
Control Cable Guard Installation TASK 27-09-18-400-801			407	PTF ALL
FLIGHT CONTROLS HYDRAULIC MODULE - REMOVAL/INSTALLATION	27-09-21		401	PTF ALL
Flight Controls Hydraulic Module Removal TASK 27-09-21-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Flight Controls Hydraulic Module Installation TASK 27-09-21-400-801			407	PTF ALL
FLIGHT CONTROLS HYDRAULIC MODULE - ADJUSTMENT/TEST	27-09-21		501	PTF ALL
Flight Controls Hydraulic Module Test TASK 27-09-21-700-801			501	PTF ALL
FLIGHT CONTROLS SHUTOFF VALVE AND VALVE MOTOR - REMOVAL/INSTALLATION	27-09-31		401	PTF ALL
Flight Controls Shutoff Valve Removal TASK 27-09-31-000-801			401	PTF ALL
Flight Controls Shutoff Valve Installation TASK 27-09-31-400-801			407	PTF ALL
Flight Controls Shutoff Valve Motor Removal TASK 27-09-31-000-802			408	PTF ALL
Flight Controls Shutoff Valve Motor Installation TASK 27-09-31-400-802			410	PTF ALL
FLIGHT CONTROLS SHUTOFF VALVE - ADJUSTMENT/TEST	27-09-31		501	PTF ALL
Flight Controls Shutoff Valve Test TASK 27-09-31-710-801			501	PTF ALL
FLIGHT CONTROLS PANEL (P5-3) - REMOVAL/INSTALLATION	27-09-45		401	PTF ALL
Flight Controls Panel Removal TASK 27-09-45-000-801			401	PTF ALL
Flight Controls Panel Installation TASK 27-09-45-400-801			404	PTF ALL
LOW PRESSURE WARNING SWITCH - REMOVAL/INSTALLATION	27-09-61		401	PTF ALL
Low Pressure Warning Switch Removal TASK 27-09-61-000-801			401	PTF ALL
Low Pressure Warning Switch Installation TASK 27-09-61-400-801			406	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LOW PRESSURE WARNING SWITCH - ADJUSTMENT/TEST	27-09-61		501	PTF ALL
Low Pressure Warning Switch Test TASK 27-09-61-700-801			501	PTF ALL
FLIGHT CONTROLS TRIM CORRECTION - ADJUSTMENT/TEST	27-09-71		501	PTF ALL
Trailing Edge Flap Adjustment for Lateral Trim and Test TASK 27-09-71-820-801			501	PTF ALL
Aileron Balance Tab Adjustment to Equalize Lateral Manual Reversion Forces TASK 27-09-71-820-802			527	PTF ALL
Rudder Trim Adjustment for Power On Discrepancy Correction TASK 27-09-71-860-801			533	PTF ALL
Rudder Trim Adjustment for Power Off Discrepancy Correction TASK 27-09-71-860-802			534	PTF ALL
Flight Controls Adjustment Record TASK 27-09-71-970-801			534	PTF ALL
FLIGHT CONTROLS SURFACES - INSPECTION/CHECK	27-09-91		601	PTF ALL
Aileron - Inspection TASK 27-09-91-200-801			601	PTF ALL
Aileron Balance Tab Freeplay - Inspection TASK 27-09-91-200-802			604	PTF ALL
Rudder Inspection TASK 27-09-91-200-803			608	PTF ALL
Horizontal Stabilizer - Inspection TASK 27-09-91-200-804			611	PTF ALL
Spoiler Inspection TASK 27-09-91-200-805			629	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
AILERON AND AILERON TRIM CONTROL SYSTEM - MAINTENANCE PRACTICES	27-11-00		201	PTF ALL
Pressure from the Aileron Hydraulic Systems A and B - Deactivation TASK 27-11-00-860-801			201	PTF ALL
Pressure to the Aileron Hydraulic Systems A and B - Activation TASK 27-11-00-860-802			209	PTF ALL
AILERON - ADJUSTMENT/TEST	27-11-00		501	PTF ALL
Aileron Cables Adjustment TASK 27-11-00-820-801			501	PTF ALL
Pogo and Power Control Unit (PCU) Adjustment TASK 27-11-00-820-802			514	PTF ALL
Aileron Balance Tab Adjustment (Recommended Method) TASK 27-11-00-820-803			523	PTF ALL
Aileron Feel and Centering Unit Adjustment TASK 27-11-00-820-805			528	PTF ALL
Do A Check of the Aileron System Adjustment TASK 27-11-00-820-806			534	PTF ALL
Aileron Surface Friction and Motion Test TASK 27-11-00-700-801			539	PTF ALL
Manual Reversion Control Friction Test TASK 27-11-00-700-803			543	PTF ALL
Aileron Response (Operational) Test TASK 27-11-00-710-801			545	PTF ALL
Control Wheel Centering Check TASK 27-11-00-700-804			546	PTF ALL
Aileron Control Wheel Force Test TASK 27-11-00-700-805			547	PTF ALL
Aileron Trim Response Test TASK 27-11-00-730-801			549	PTF ALL
Transfer Mechanism Operational Test TASK 27-11-00-710-802			553	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Transfer Mechanism Functional Test TASK 27-11-00-700-806			555	PTF ALL
Control Wheel Travel Stop Test TASK 27-11-00-700-807			560	PTF ALL
Control Wheel Travel Interference Test TASK 27-11-00-700-808			561	PTF ALL
Aileron Balance Tab Position Test TASK 27-11-00-700-809			562	PTF ALL
Aileron Travel Test TASK 27-11-00-700-810			563	PTF ALL
Aileron Power Control Unit (PCU) Pogos - Functional Test TASK 27-11-00-720-801			565	PTF ALL
Flight Test to Find and Correct Asymmetrical Flight Control Forces TASK 27-11-00-750-801			569	PTF ALL
AILERON AND AILERON TRIM CONTROL SYSTEM - INSPECTION/CHECK	27-11-00		601	PTF ALL
Aileron Mechanical Components Inspection - Forward components TASK 27-11-00-210-801			601	PTF ALL
Aileron Mechanical Components Inspection - From the Aileron PCUs to the Aileron TASK 27-11-00-210-802			605	PTF ALL
AILERON - REMOVAL/INSTALLATION	27-11-11		401	PTF ALL
Aileron Removal TASK 27-11-11-000-801			401	PTF ALL
Aileron Installation TASK 27-11-11-400-801			419	PTF ALL
AILERON - INSPECTION/CHECK	27-11-11		601	PTF ALL
Aileron Inspection TASK 27-11-11-200-801			601	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
AILERON POWER CONTROL UNIT LINKAGE - REMOVAL/INSTALLATION	27-11-14		401	PTF ALL
Aileron Power Control Unit Linkage Removal TASK 27-11-14-000-801			401	PTF ALL
Aileron Power Control Unit Linkage Installation TASK 27-11-14-400-801			407	PTF ALL
AILERON BALANCE TAB - REMOVAL/INSTALLATION	27-11-21		401	PTF ALL
Aileron Balance Tab Removal TASK 27-11-21-000-801			401	PTF ALL
Aileron Balance Tab Installation TASK 27-11-21-400-801			407	PTF ALL
AILERON BALANCE TAB - INSPECTION/CHECK	27-11-21		601	PTF ALL
Aileron Balance Tab Inspection TASK 27-11-21-200-801			601	PTF ALL
AILERON WING QUADRANT - REMOVAL/INSTALLATION	27-11-24		401	PTF ALL
Aileron Wing Quadrant Removal TASK 27-11-24-000-801			401	PTF ALL
Aileron Wing Quadrant Installation TASK 27-11-24-400-801			409	PTF ALL
AILERON BALANCE PANELS - REMOVAL/INSTALLATION	27-11-31		401	PTF ALL
Aileron Balance Panels Removal TASK 27-11-31-000-801			401	PTF ALL
Aileron Balance Panels Installation TASK 27-11-31-400-801			406	PTF ALL
AILERON BALANCE PANEL - INSPECTION/CHECK	27-11-31		601	PTF ALL
Aileron Balance Panel - Inspection TASK 27-11-31-220-801			601	PTF ALL
Aileron Balance Panel Seals Detailed - Inspection TASK 27-11-31-210-801			605	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
AILERON TAB CONTROL RODS - REMOVAL/INSTALLATION	27-11-34		401	PTF ALL
Aileron Tab Control Rods Removal TASK 27-11-34-000-801			401	PTF ALL
Aileron Tab Control Rods Installation TASK 27-11-34-400-801			405	PTF ALL
AILERON TAB CONTROL RODS - INSPECTION/CHECK	27-11-34		601	PTF ALL
Aileron Tab Control Rods Inspection TASK 27-11-34-200-801			601	PTF ALL
AILERON CONTROL WHEEL - REMOVAL/INSTALLATION	27-11-41		401	PTF ALL
Aileron Control Wheel Removal TASK 27-11-41-000-801			401	PTF ALL
Aileron Control Wheel Installation TASK 27-11-41-400-801			407	PTF ALL
AILERON CONTROL WHEEL DRUM - REMOVAL/INSTALLATION	27-11-51		401	PTF ALL
Aileron Control Wheel Drum Removal TASK 27-11-51-000-801			401	PTF ALL
Aileron Control Wheel Drum Installation TASK 27-11-51-400-801			409	PTF ALL
AILERON TRIM ACTUATOR - REMOVAL/INSTALLATION	27-11-54		401	PTF ALL
Aileron Trim Actuator Removal TASK 27-11-54-000-801			401	PTF ALL
Aileron Trim Actuator Installation TASK 27-11-54-400-801			405	PTF ALL
AILERON TRANSFER MECHANISM - REMOVAL/INSTALLATION	27-11-61		401	PTF ALL
Aileron Transfer Mechanism Removal TASK 27-11-61-000-801			401	PTF ALL
Aileron Transfer Mechanism Installation TASK 27-11-61-400-801			408	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
AILERON TRANSFER MECHANISM - INSPECTION/CHECK	27-11-61		601	PTF ALL
Aileron Transfer Mechanism Inspection TASK 27-11-61-210-801			601	PTF ALL
AILERON POWER CONTROL UNIT - REMOVAL/INSTALLATION	27-11-71		401	PTF ALL
Aileron Power Control Unit Removal TASK 27-11-71-000-801			401	PTF ALL
Aileron Power Control Unit Installation TASK 27-11-71-400-801			407	PTF ALL
AILERON FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION	27-11-81		401	PTF ALL
Aileron Feel and Centering Unit - Removal TASK 27-11-81-000-801			401	PTF ALL
Aileron Feel and Centering Unit Installation TASK 27-11-81-400-801			410	PTF ALL
Cam Follower Bearing Removal TASK 27-11-81-000-802			413	PTF ALL
Cam Follower Bearing Installation TASK 27-11-81-400-802			418	PTF ALL
AILERON FEEL AND CENTERING UNIT - INSPECTION/CHECK	27-11-81		601	PTF ALL
Aileron Feel and Centering Unit Springs Detailed Inspection TASK 27-11-81-210-801			601	PTF ALL
AILERON/RUDDER TRIM CONTROL MODULE - REMOVAL/INSTALLATION	27-11-94		401	PTF ALL
Aileron and Rudder Trim Control Module Removal TASK 27-11-94-000-801			401	PTF ALL
Aileron and Rudder Trim Control Module Installation TASK 27-11-94-400-801			404	PTF ALL
Trim Control Switches Removal TASK 27-11-94-000-802			404	PTF ALL
Trim Control Switches Installation TASK 27-11-94-400-802			405	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Rudder Trim Indicator Removal TASK 27-11-94-000-803			407	PTF ALL
Rudder Trim Indicator Installation TASK 27-11-94-400-803			407	PTF ALL
RUDDER AND RUDDER TRIM CONTROL SYSTEM - MAINTENANCE PRACTICES	27-21-00		201	PTF ALL
Rudder Hydraulic System A, B, or Standby Pressurization TASK 27-21-00-800-801			201	PTF ALL
Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization TASK 27-21-00-840-801			205	PTF ALL
Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation TASK 27-21-00-800-802			205	PTF ALL
Pressure to the Rudder Systems A, B, and Standby - Activation TASK 27-21-00-840-802			213	PTF ALL
RUDDER AND RUDDER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST	27-21-00		501	PTF ALL
Rudder Forward Quadrant Adjustment TASK 27-21-00-820-801			501	PTF ALL
Rudder Pedals Adjustment TASK 27-21-00-820-802			510	PTF ALL
Rudder Control Cables RA and RB Adjustment TASK 27-21-00-820-803			512	PTF ALL
Control Rod for Main Rudder Power Control Unit Adjustment TASK 27-21-00-820-804			518	PTF ALL
Control Rod for Standby Rudder Power Control Unit Adjustment TASK 27-21-00-820-805			532	PTF ALL
Rudder Pedal Adjustment and Limit Travel Test TASK 27-21-00-700-801			536	PTF ALL
Rudder Hinge Bearing and PCU Bearing Freeplay TASK 27-21-00-700-802			543	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Rudder Centering Test TASK 27-21-00-700-803			547	PTF ALL
Rudder Pedal Forces Test TASK 27-21-00-700-804			549	PTF ALL
Rudder Travel Test TASK 27-21-00-700-805			555	PTF ALL
Standby Rudder Power Control Unit Deadband Test TASK 27-21-00-700-806			559	PTF ALL
Rudder Trim System Test TASK 27-21-00-700-807			560	PTF ALL
Wheel to Rudder Interconnect System Test TASK 27-21-00-700-808			564	PTF ALL
Rudder Hinge Friction Test TASK 27-21-00-700-809			569	PTF ALL
Main Rudder PCU Internal Hydraulic Leakage Test TASK 27-21-00-700-810			572	PTF ALL
Rudder Override Operational Test TASK 27-21-00-700-811			581	PTF ALL
Rudder Main PCU Force Fight Monitor Test TASK 27-21-00-700-812			584	PTF ALL
RUDDER - REMOVAL/INSTALLATION	27-21-11		401	PTF ALL
Rudder Removal TASK 27-21-11-000-801			401	PTF ALL
Rudder Installation TASK 27-21-11-400-801			416	PTF ALL
RUDDER INDEX PLATE - REMOVAL/INSTALLATION	27-21-17		401	PTF ALL
Rudder Index Plate Removal TASK 27-21-17-000-801			401	PTF ALL
Rudder Index Plate Installation TASK 27-21-17-400-801			403	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
STANDBY RUDDER POWER CONTROL UNIT - REMOVAL/INSTALLATION	27-21-24		401	PTF ALL
Standby Rudder Power Control Unit Removal TASK 27-21-24-000-801			401	PTF ALL
Standby Rudder Power Control Unit Installation TASK 27-21-24-400-801			405	PTF ALL
STANDBY RUDDER POWER CONTROL UNIT - INSPECTION/CHECK	27-21-24		601	PTF ALL
Standby Rudder Power Control Unit (with Hydraulic Power on) General Visual Inspection TASK 27-21-24-210-801			601	PTF ALL
Standby Rudder Power Control Unit Detail Visual Inspection TASK 27-21-24-210-802			602	PTF ALL
RUDDER PEDALS - REMOVAL/INSTALLATION	27-21-31		401	PTF ALL
Rudder Pedals Removal TASK 27-21-31-000-801			401	PTF ALL
Rudder Pedals Installation TASK 27-21-31-400-801			412	PTF ALL
RUDDER PEDAL ADJUSTMENT CRANK - REMOVAL/INSTALLATION	27-21-41		401	PTF ALL
Rudder Pedal Adjustment Crank Removal TASK 27-21-41-000-801			401	PTF ALL
Rudder Pedal Adjustment Crank Installation TASK 27-21-41-400-801			407	PTF ALL
RUDDER FORWARD QUADRANT - REMOVAL/INSTALLATION	27-21-51		401	PTF ALL
Rudder Forward Quadrant Removal TASK 27-21-51-000-801			401	PTF ALL
Rudder Forward Quadrant Installation TASK 27-21-51-400-801			418	PTF ALL
RUDDER FORWARD QUADRANT - INSPECTION/CHECK	27-21-51		601	PTF ALL
Rudder Mechanical Control Path Inspection TASK 27-21-51-210-801			601	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
RUDDER AFT QUADRANT - REMOVAL/INSTALLATION	27-21-61		401	PTF ALL
Rudder Aft Quadrant Removal TASK 27-21-61-000-801			401	PTF ALL
Rudder Aft Quadrant Installation TASK 27-21-61-400-801			409	PTF ALL
RUDDER AFT QUADRANT - INSPECTION/CHECK	27-21-61		601	PTF ALL
Rudder Aft Quadrant Detail Visual Inspection TASK 27-21-61-210-801			601	PTF ALL
Rudder Aft Quadrant General Visual Inspection TASK 27-21-61-210-802			606	PTF ALL
RUDDER CONTROL TORQUE TUBE - REMOVAL/INSTALLATION	27-21-71		401	PTF ALL
Rudder Control Torque Tube Removal TASK 27-21-71-000-801			401	PTF ALL
Rudder Control Torque Tube Installation TASK 27-21-71-400-801			406	PTF ALL
RUDDER FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION	27-21-82		401	PTF ALL
Rudder Feel and Centering Unit Removal TASK 27-21-82-000-801			401	PTF ALL
Rudder Feel and Centering Unit Installation TASK 27-21-82-400-801			409	PTF ALL
Rudder Feel and Centering Unit Spring Assembly Removal TASK 27-21-82-000-802			412	PTF ALL
Rudder Feel and Centering Unit Spring Assembly Installation TASK 27-21-82-400-802			416	PTF ALL
RUDDER FEEL AND CENTERING UNIT SPRING ASSEMBLY - INSPECTION/CHECK	27-21-82		601	PTF ALL
Rudder Feel and Centering Unit Inner Spring - Inspection TASK 27-21-82-211-801			601	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
RUDDER TRIM ACTUATOR - REMOVAL/INSTALLATION	27-21-87		401	PTF ALL
Rudder Trim Actuator Removal TASK 27-21-87-000-801			401	PTF ALL
Rudder Trim Actuator Installation TASK 27-21-87-400-801			405	PTF ALL
MAIN RUDDER POWER CONTROL UNIT - REMOVAL/INSTALLATION	27-21-91		401	PTF ALL
Main Rudder Power Control Unit Removal TASK 27-21-91-000-801			401	PTF ALL
Main Rudder Power Control Unit Installation TASK 27-21-91-400-801			410	PTF ALL
Main Rudder PCU Load Limiter Solenoid Valve Removal TASK 27-21-91-000-802			415	PTF ALL
Main Rudder PCU Load Limiter Solenoid Valve Installation TASK 27-21-91-400-802			419	PTF ALL
STANDBY RUDDER POWER CONTROL UNIT SHUTOFF VALVE - MAINTENANCE PRACTICES	27-21-94		201	PTF ALL
Rudder Valve Motor Removal TASK 27-21-94-000-801			201	PTF ALL
Rudder Valve Motor Installation TASK 27-21-94-400-801			205	PTF ALL
Standby Rudder PCU Shutoff Valve Removal TASK 27-21-94-000-802			206	PTF ALL
Standby Rudder PCU Shutoff Valve Installation TASK 27-21-94-400-802			207	PTF ALL
Standby Rudder PCU Shutoff Valve Installation Test TASK 27-21-94-710-801			209	PTF ALL
RUDDER HYDRAULIC FUSE - REMOVAL/INSTALLATION	27-21-95		401	PTF ALL
Rudder Hydraulic Fuse Removal TASK 27-21-95-000-801			401	PTF ALL

27-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Rudder Hydraulic Fuse Installation TASK 27-21-95-400-801			404	PTF ALL
ELEVATOR AND TAB CONTROL SYSTEM - MAINTENANCE PRACTICES	27-31-00		201	PTF ALL
Elevator Hydraulic System A and B - Pressurization TASK 27-31-00-800-801			201	PTF ALL
Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization TASK 27-31-00-840-801			205	PTF ALL
Pressure from the Elevator Hydraulic Systems A and B Removal TASK 27-31-00-800-802			205	PTF ALL
Putting the Elevator Systems A and B Back to the Condition Before the Pressure Removal TASK 27-31-00-840-802			210	PTF ALL
Null Procedure - Mach Trim Actuator TASK 27-31-00-700-801			211	PTF ALL
Elevator Control Pushrod - Detailed Visual Inspection TASK 27-31-00-200-801			214	PTF ALL
Elevator Feel and Centering Unit Detailed Inspection TASK 27-31-00-200-802			215	PTF ALL
Elevator Feel and Centering Unit Output Rod - Detailed Visual Inspection TASK 27-31-00-211-801			218	PTF ALL
Elevator Tab and Tab Mechanism - Detailed Visual Inspection TASK 27-31-00-220-801			221	PTF ALL
Elevator and Tab Control System - Deactivation TASK 27-31-00-000-801			234	PTF ALL
Elevator and Tab Control System - Activation TASK 27-31-00-400-801			239	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ELEVATOR AND ELEVATOR TRIM CONTROL SYSTEM - ADJUSTMENT/TEST	27-31-00		501	PTF ALL
Elevator and Elevator Trim Control System - Operational Test TASK 27-31-00-710-801			502	PTF ALL
Elevator Tab Control System - Operational Test TASK 27-31-00-700-802			503	PTF ALL
Elevator Control Column Override - Operational Test TASK 27-31-00-710-802			505	PTF ALL
Elevator Control Column Override - Functional Test TASK 27-31-00-720-801			506	PTF ALL
Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment TASK 27-31-00-820-801			511	PTF ALL
Pitch Force Transducer Operational Test TASK 27-31-00-710-805			521	PTF ALL
Elevator Control Pushrods - Adjustment TASK 27-31-00-820-802			524	PTF ALL
Input Rods of the Elevator Power Control Unit - Adjustment TASK 27-31-00-820-803			530	PTF ALL
Input Rods of the Elevator Neutral Shift Mechanism and Output Rod of the Feel and Centering Unit - Adjustment TASK 27-31-00-820-804			536	PTF ALL
Elevator Tab Pushrods - Adjustment TASK 27-31-00-820-805			543	PTF ALL
Input Rod of the Elevator Feel Computer - Adjustment TASK 27-31-00-820-806			553	PTF ALL
Elevator Surface Friction and Balance - Test TASK 27-31-00-700-803			557	PTF ALL
Elevator Balance Panels - Test TASK 27-31-00-700-804			560	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Elevator Travel Limit - Test TASK 27-31-00-700-805			561	PTF ALL
Elevator Output Torque Tube Buss Linkage Test TASK 27-31-00-700-806			564	PTF ALL
Elevator Tab Linkage - Test TASK 27-31-00-700-807			567	PTF ALL
Manual Mode - Test TASK 27-31-00-700-808			573	PTF ALL
Control Column Travel and Centering - Test TASK 27-31-00-700-809			576	PTF ALL
Control Column Power Force - Test TASK 27-31-00-700-810			581	PTF ALL
Neutral Shift - Test TASK 27-31-00-700-811			584	PTF ALL
Feel Differential Pressure Light - Test TASK 27-31-00-700-812			588	PTF ALL
Pitot System Leak - Test TASK 27-31-00-790-801			590	PTF ALL
Feel Force System - Test TASK 27-31-00-700-813			596	PTF ALL
Sensor Null - Test TASK 27-31-00-700-814			598.5	PTF ALL
Elevator Power Control Unit (PCU) Input Rod (Pogo) - Functional Test TASK 27-31-00-820-807			598.8	PTF ALL
Elevator Power-Off Flight Test TASK 27-31-00-710-803			598.13	PTF ALL
Elevator Tab Pushrod - Adjustment after the Elevator Power-Off Flight Test TASK 27-31-00-820-808			598.27	PTF ALL
Elevator Jam Landing Assist - Test TASK 27-31-00-710-804			598.33	PTF ALL
ELEVATOR - REMOVAL/INSTALLATION	27-31-11		401	PTF ALL
Elevator - Removal TASK 27-31-11-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Elevator - Installation TASK 27-31-11-400-801			426	PTF ALL
Elevator Control Pushrod - Removal TASK 27-31-11-000-802			436	PTF ALL
Elevator Control Pushrod - Installation TASK 27-31-11-400-802			443	PTF ALL
ELEVATOR POWER CONTROL UNIT - REMOVAL/INSTALLATION	27-31-14		401	PTF ALL
Elevator Power Control Unit - Removal TASK 27-31-14-000-801			401	PTF ALL
Elevator Power Control Unit Installation TASK 27-31-14-400-801			412	PTF ALL
ELEVATOR POWER CONTROL UNIT - INSPECTION/CHECK	27-31-14		601	PTF ALL
Elevator Power Control Unit Visual Inspection TASK 27-31-14-210-801			601	PTF ALL
Elevator Power Control Unit - Inspection TASK 27-31-14-200-801			605	PTF ALL
ELEVATOR PITOT-STATIC SYSTEM - MAINTENANCE PRACTICES	27-31-17		201	PTF ALL
Elevator Pitot-Static System Flushing TASK 27-31-17-170-801			201	PTF ALL
Elevator Pitot-Static System Drainage - Inspection TASK 27-31-17-200-801			209	PTF ALL
Elevator Pitot Probe - Inspection TASK 27-31-17-200-802			211	PTF ALL
Elevator Pitot System Leak - Test TASK 27-31-17-790-801			212	PTF ALL
Elevator Pitot System - Pressurization TASK 27-31-17-780-801			217	PTF ALL
ELEVATOR FEEL SHIFT MODULE - REMOVAL/INSTALLATION	27-31-20		401	PTF ALL
Elevator Feel Shift Module Removal TASK 27-31-20-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Elevator Feel Shift Module - Installation TASK 27-31-20-400-801			406	PTF ALL
ELEVATOR FEEL SHIFT MODULE - ADJUSTMENT/TEST	27-31-20		501	PTF ALL
Elevator Feel Shift Module - Test TASK 27-31-20-740-801			501	PTF ALL
ELEVATOR TAB - REMOVAL/INSTALLATION	27-31-31		401	PTF ALL
Elevator Tab Removal TASK 27-31-31-000-801			401	PTF ALL
Elevator Tab Installation TASK 27-31-31-400-801			411	PTF ALL
ELEVATOR AND ELEVATOR TAB FREEPLAY - INSPECTION/CHECK	27-31-32		601	PTF ALL
Elevator Output Torque Tube Buss Linkage Freeplay - Check TASK 27-31-32-200-801			601	PTF ALL
Elevator Tab Freeplay - Check TASK 27-31-32-200-802			607	PTF ALL
ELEVATOR TAB CONTROL MECHANISM - REMOVAL/INSTALLATION	27-31-34		401	PTF ALL
Elevator Tab Control Mechanism - Removal TASK 27-31-34-000-801			401	PTF ALL
Elevator Tab Control Mechanism - Installation TASK 27-31-34-400-801			415	PTF ALL
Elevator Tab Pushrod - Removal TASK 27-31-34-000-802			419	PTF ALL
Elevator Tab Pushrod - Installation TASK 27-31-34-400-802			423	PTF ALL
Elevator Tab Lock Actuator - Removal TASK 27-31-34-000-803			426	PTF ALL
Elevator Tab Lock Actuator - Installation TASK 27-31-34-400-803			429	PTF ALL
Elevator Tab Control Mechanism Springs - Removal TASK 27-31-34-000-804			430	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Elevator Tab Control Mechanism Springs - Installation TASK 27-31-34-400-804			436	PTF ALL
ELEVATOR TAB CONTROL MECHANISM - INSPECTION/CHECK	27-31-34		601	PTF ALL
Elevator Balance Weight Installation and Elevator Tab Control Mechanism - General Visual Inspection TASK 27-31-34-210-801			601	PTF ALL
ELEVATOR TAB CONTROL VALVE - REMOVAL/INSTALLATION	27-31-35		401	PTF ALL
Elevator Tab Control Valve - Removal TASK 27-31-35-000-801			401	PTF ALL
Elevator Tab Control Valve - Installation TASK 27-31-35-400-801			405	PTF ALL
ELEVATOR FEEL COMPUTER - MAINTENANCE PRACTICES	27-31-37		201	PTF ALL
Elevator Feel Computer Solenoid Valve - Removal TASK 27-31-37-000-802			201	PTF ALL
Elevator Feel Computer Solenoid Valve - Installation TASK 27-31-37-400-802			204	PTF ALL
ELEVATOR FEEL COMPUTER - REMOVAL/INSTALLATION	27-31-37		401	PTF ALL
Elevator Feel Computer Removal TASK 27-31-37-000-801			401	PTF ALL
Elevator Feel Computer Installation TASK 27-31-37-400-801			408	PTF ALL
ELEVATOR FEEL COMPUTER - INSPECTION/CHECK	27-31-37		601	PTF ALL
Elevator Feel Computer Leakage Check TASK 27-31-37-200-801			601	PTF ALL
Elevator Feel Computer Solenoid Valve Inspection TASK 27-31-37-200-802			602	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ELEVATOR BALANCE PANEL - REMOVAL/INSTALLATION	27-31-41		401	PTF ALL
Elevator Balance Panel - Removal TASK 27-31-41-000-801			401	PTF ALL
Elevator Balance Panel - Installation TASK 27-31-41-400-801			409	PTF ALL
ELEVATOR BALANCE PANEL - INSPECTION/CHECK	27-31-41		601	PTF ALL
Elevator Balance Panel - Inspection TASK 27-31-41-200-801			601	PTF ALL
CONTROL COLUMN - REMOVAL/INSTALLATION	27-31-51		401	PTF ALL
Control Column - Removal TASK 27-31-51-000-801			401	PTF ALL
Control Column - Installation TASK 27-31-51-400-801			411	PTF ALL
CONTROL COLUMN SUPPORT BEARING - REMOVAL/INSTALLATION	27-31-54		401	PTF ALL
Control Column Support Bearing Removal TASK 27-31-54-000-801			401	PTF ALL
Control Column Support Bearing Installation TASK 27-31-54-400-801			410	PTF ALL
ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY - REMOVAL/INSTALLATION	27-31-61		401	PTF ALL
Elevator Forward Control Quadrant Assembly Removal TASK 27-31-61-000-801			401	PTF ALL
Elevator Forward Control Quadrant Assembly Installation TASK 27-31-61-400-801			412	PTF ALL
ELEVATOR FORWARD CONTROL QUADRANT ASSEMBLY - INSPECTION/CHECK	27-31-61		601	PTF ALL
Elevator Mechanical Control Path - Inspection TASK 27-31-61-210-801			601	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ELEVATOR FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION	27-31-64		401	PTF ALL
Elevator Feel and Centering Unit Removal TASK 27-31-64-000-801			401	PTF ALL
Elevator Feel and Centering Unit Installation TASK 27-31-64-400-801			406	PTF ALL
Elevator Feel and Centering Unit Spring Removal TASK 27-31-64-000-802			408	PTF ALL
Elevator Feel and Centering Unit Spring Installation TASK 27-31-64-400-802			414	PTF ALL
ELEVATOR FEEL AND CENTERING UNIT - INSPECTION/CHECK	27-31-64		601	PTF ALL
Elevator Feel and Centering Unit Inspection TASK 27-31-64-200-801			601	PTF ALL
ELEVATOR FEEL ACTUATOR - REMOVAL/INSTALLATION	27-31-67		401	PTF ALL
Elevator Feel Actuator Removal TASK 27-31-67-000-801			401	PTF ALL
Elevator Feel Actuator Installation TASK 27-31-67-400-801			405	PTF ALL
ELEVATOR FEEL ACTUATOR - DIFFERENTIAL PRESSURE TEST	27-31-67		501	PTF ALL
Elevator Feel Actuator - Differential Pressure Test TASK 27-31-67-700-801			501	PTF ALL
ELEVATOR INPUT TORQUE TUBE - REMOVAL/INSTALLATION	27-31-71		401	PTF ALL
Elevator Input Torque Tube Removal TASK 27-31-71-000-801			401	PTF ALL
Elevator Input Torque Tube Installation TASK 27-31-71-400-801			411	PTF ALL
ELEVATOR OUTPUT TORQUE TUBE - REMOVAL/INSTALLATION	27-31-74		401	PTF ALL
Elevator Output Torque Tube Removal TASK 27-31-74-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Elevator Output Torque Tube Installation TASK 27-31-74-400-801			408	PTF ALL
ELEVATOR INDEX PLATE - REMOVAL/INSTALLATION	27-31-81		401	PTF ALL
Elevator Index Plate Removal TASK 27-31-81-000-801			401	PTF ALL
Elevator Index Plate Installation TASK 27-31-81-400-801			407	PTF ALL
ELEVATOR NEUTRAL SHIFT MECHANISM - REMOVAL/INSTALLATION	27-31-91		401	PTF ALL
Elevator Neutral Shift Mechanism Removal TASK 27-31-91-000-801			401	PTF ALL
Elevator Neutral Shift Mechanism Installation TASK 27-31-91-400-801			411	PTF ALL
ELEVATOR NEUTRAL SHIFT MECHANISM - INSPECTION/CHECK	27-31-91		601	PTF ALL
Elevator Neutral Shift Mechanism - Inspection TASK 27-31-91-200-801			601	PTF ALL
FEEL SYSTEM PITOT PROBE - REMOVAL/INSTALLATION	27-31-94		401	PTF ALL
Feel System Pitot Probe Removal TASK 27-31-94-000-801			401	PTF ALL
Feel System Pitot Probe Installation TASK 27-31-94-400-801			408	PTF ALL
STALL WARNING SYSTEM - ADJUSTMENT/TEST	27-32-00		501	PTF ALL
Stall Warning System - Operational Test TASK 27-32-00-710-801			501	PTF ALL
Stall Management Yaw Damper (SMYD) BITE Test - Existing Faults TASK 27-32-00-740-801			502	PTF ALL
Stall Management Yaw Damper (SMYD) BITE Test - Fault History TASK 27-32-00-740-802			506	PTF ALL
Stall Management Yaw Damper (SMYD) BITE Test TASK 27-32-00-740-803			507	PTF ALL

27-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
STALL WARNING TEST MODULE - REMOVAL/INSTALLATION	27-32-21		401	PTF ALL
Stall Warning Test Module Removal TASK 27-32-21-000-801			401	PTF ALL
Stall Warning Test Module Installation TASK 27-32-21-400-801			403	PTF ALL
CONTROL COLUMN SHAKER - REMOVAL/INSTALLATION	27-32-31		401	PTF ALL
Control Column Shaker Removal TASK 27-32-31-000-801			401	PTF ALL
Control Column Shaker Installation TASK 27-32-31-400-801			404	PTF ALL
STALL MANAGEMENT YAW DAMPER (SMYD) - REMOVAL/INSTALLATION	27-32-42		401	PTF ALL
Stall Management Yaw Damper (SMYD) Removal TASK 27-32-42-000-801			401	PTF ALL
Stall Management Yaw Damper (SMYD) Installation TASK 27-32-42-400-801			405	PTF ALL
HORIZONTAL STABILIZER TRIM CONTROL SYSTEM - MAINTENANCE PRACTICES	27-41-00		201	PTF ALL
Horizontal Stabilizer Trim Control System - Deactivation TASK 27-41-00-800-801			201	PTF ALL
Horizontal Stabilizer Trim Control System - Activation TASK 27-41-00-800-802			205	PTF ALL
HORIZONTAL STABILIZER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST	27-41-00		501	PTF ALL
Stabilizer Control Cable and Chain - Adjustment TASK 27-41-00-820-801			501	PTF ALL
Stabilizer Manual Trim and Trim Indicator - Test TASK 27-41-00-700-801			521	PTF ALL
Horizontal Stabilizer Trim Control System Friction Test TASK 27-41-00-700-802			526	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Stabilizer Electric Trim System Test TASK 27-41-00-700-803			529	PTF ALL
Stabilizer Trim Wheel Free Movement Test TASK 27-41-00-700-804			533	PTF ALL
Stabilizer Trim Limit Switches Test TASK 27-41-00-700-805			535	PTF ALL
Stabilizer Clutch Friction System Test TASK 27-41-00-700-806			542	PTF ALL
Column Actuated Stabilizer Trim Cutout Switch Override Test TASK 27-41-00-700-807			543	PTF ALL
Stabilizer Trim Secondary Brake System Test TASK 27-41-00-700-808			545	PTF ALL
Stabilizer Trim Cutout Test TASK 27-41-00-700-809			550	PTF ALL
Column Cutout Switches Test TASK 27-41-00-710-801			552	PTF ALL
HORIZONTAL STABILIZER - REMOVAL/INSTALLATION	27-41-11		401	PTF ALL
Horizontal Stabilizer Removal TASK 27-41-11-000-801			401	PTF ALL
Horizontal Stabilizer Installation TASK 27-41-11-400-801			418	PTF ALL
HORIZONTAL STABILIZER LEADING EDGE - REMOVAL/INSTALLATION	27-41-21		401	PTF ALL
Horizontal Stabilizer Leading Edge Removal TASK 27-41-21-000-801			401	PTF ALL
Horizontal Stabilizer Leading Edge Installation TASK 27-41-21-400-801			403	PTF ALL
HORIZONTAL STABILIZER CENTER SECTION HINGE FITTING - REMOVAL/INSTALLATION	27-41-31		401	PTF ALL
Horizontal Stabilizer Center Section Hinge Fitting Removal TASK 27-41-31-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Horizontal Stabilizer Center Section Hinge Fitting Installation TASK 27-41-31-400-801			419	PTF ALL
STABILIZER FORWARD CONTROL MECHANISM - REMOVAL/INSTALLATION	27-41-41		401	PTF ALL
Stabilizer Forward Control Mechanism Removal TASK 27-41-41-000-801			401	PTF ALL
Stabilizer Forward Control Mechanism - Installation TASK 27-41-41-400-801			410	PTF ALL
STABILIZER FORWARD CONTROL MECHANISM - INSPECTION/CHECK	27-41-41		601	PTF ALL
Stabilizer Control Mechanism Detail Visual Inspection TASK 27-41-41-210-801			601	PTF ALL
Upper and Lower Stabilizer Trim Forward Mechanism Retention Turnbuckles and Turnbuckle Attachment Points Detail Visual Inspection TASK 27-41-41-210-802			603	PTF ALL
Stabilizer Forward Trim Control Mechanism Inspection TASK 27-41-41-210-803			603	PTF ALL
COLUMN CUTOUT SWITCH - REMOVAL/INSTALLATION	27-41-51		401	PTF ALL
Column Cutout Switch Removal TASK 27-41-51-000-801			401	PTF ALL
Column Cutout Switch Installation TASK 27-41-51-400-801			406	PTF ALL
HORIZONTAL STABILIZER TRIM CONTROL WHEEL - REMOVAL/INSTALLATION	27-41-61		401	PTF ALL
Stabilizer Trim Control Wheel Removal TASK 27-41-61-010-801			401	PTF ALL
Stabilizer Trim Control Wheel Installation TASK 27-41-61-410-801			404	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
STABILIZER TRIM MOTOR - REMOVAL/INSTALLATION	27-41-71		401	PTF ALL
Stabilizer Trim Motor Removal TASK 27-41-71-000-801			401	PTF ALL
Stabilizer Trim Motor Installation TASK 27-41-71-400-801			405	PTF ALL
STABILIZER TRIM MOTOR - ADJUSTMENT/TEST	27-41-71		501	PTF ALL
Stabilizer Trim Motor Test TASK 27-41-71-700-801			501	PTF ALL
STABILIZER BALL NUT AND JACKSCREW GEARBOX - REMOVAL/INSTALLATION	27-41-81		401	PTF ALL
Stabilizer Ball Nut and Jackscrew Gearbox Removal TASK 27-41-81-000-801			401	PTF ALL
Stabilizer Ball Nut and Jackscrew Gearbox Installation TASK 27-41-81-400-801			419	PTF ALL
STABILIZER BALL NUT AND JACKSCREW GEARBOX - INSPECTION/CHECK	27-41-81		601	PTF ALL
Stabilizer Trim Ball Nut and Jackscrew Gearbox Detail Visual Inspection TASK 27-41-81-210-801			601	PTF ALL
Stabilizer Jackscrew Gearbox Backlash Inspection TASK 27-41-81-210-802			611	PTF ALL
Upper and Lower Gimbal Wear Limits Inspection TASK 27-41-81-200-801			615	PTF ALL
STABILIZER CONTROL CABLE - REMOVAL/INSTALLATION	27-41-82		401	PTF ALL
Stabilizer Control Cable Removal TASK 27-41-82-000-801			401	PTF ALL
Stabilizer Control Cable Installation TASK 27-41-82-400-801			415	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
STABILIZER TRIM CUTOUT SWITCHES - REMOVAL/INSTALLATION	27-41-91		401	PTF ALL
Stabilizer Trim Cutout Switch Removal TASK 27-41-91-000-801			401	PTF ALL
Stabilizer Trim Cutout Switch Installation TASK 27-41-91-400-801			403	PTF ALL
STABILIZER TRIM CUTOUT SWITCHES - ADJUSTMENT/TEST	27-41-91		501	PTF ALL
Primary Stabilizer Trim Cutout Switch Test TASK 27-41-91-700-801			501	PTF ALL
Back Up Stabilizer Trim Cutout Switch Test TASK 27-41-91-700-802			502	PTF ALL
STABILIZER TRIM LIMIT SWITCHES - REMOVAL/INSTALLATION	27-41-94		401	PTF ALL
Stabilizer Trim Limit Switch Removal TASK 27-41-94-000-801			401	PTF ALL
Stabilizer Trim Limit Switch Installation TASK 27-41-94-400-801			406	PTF ALL
STABILIZER TRIM LIMIT SWITCHES - ADJUSTMENT/TEST	27-41-94		501	PTF ALL
Stabilizer Trim Limit Switches Adjustment TASK 27-41-94-820-801			501	PTF ALL
Stabilizer Trim Limit Switches Test TASK 27-41-94-700-801			510	PTF ALL
STABILIZER TRIM CONTROL SWITCH - REMOVAL/INSTALLATION	27-41-97		401	PTF ALL
Stabilizer Trim Control Switch Removal TASK 27-41-97-000-801			401	PTF ALL
Stabilizer Trim Control Switch Installation TASK 27-41-97-400-801			405	PTF ALL
STABILIZER TRIM CONTROL SWITCH - ADJUSTMENT/TEST	27-41-97		501	PTF ALL
Stabilizer Trim Control Switch Test TASK 27-41-97-700-801			501	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
TRAILING EDGE FLAP SYSTEM - MAINTENANCE PRACTICES	27-51-00		201	PTF ALL
Trailing Edge Flap System Deactivation TASK 27-51-00-040-801			201	PTF ALL
Trailing Edge Flap System Reactivation TASK 27-51-00-440-801			208	PTF ALL
Trailing Edge Flap System Operation With Primary Control TASK 27-51-00-860-801			209	PTF ALL
Trailing Edge Flap System Operation With Alternate Control TASK 27-51-00-860-802			212	PTF ALL
Trailing Edge Flap System Manual Operation TASK 27-51-00-980-801			213	PTF ALL
Extend the Trailing Edge Flaps TASK 27-51-00-860-803			218	PTF ALL
Retract the Trailing Edge Flaps TASK 27-51-00-860-804			218	PTF ALL
Trailing Edge Flap System Operation With Leading Edge Flaps and Slats Disabled TASK 27-51-00-860-805			219	PTF ALL
Flap Control Handle Lockout Assembly Removal TASK 27-51-00-000-801			230	PTF ALL
Flap Control Handle Lockout Assembly Installation TASK 27-51-00-400-801			230	PTF ALL
TRAILING EDGE FLAP SYSTEM - ADJUSTMENT/TEST	27-51-00		501	PTF ALL
Trailing Edge Flap System Operational Test TASK 27-51-00-710-801			501	PTF ALL
Trailing Edge Flap Control System Adjustment and Test TASK 27-51-00-820-801			510	PTF ALL
Trailing Edge Flap Drive System Adjustment TASK 27-51-00-820-802			531	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Trailing Edge Flap System Test TASK 27-51-00-730-801			542	PTF ALL
FSEU Display and Existing Faults Test TASK 27-51-00-740-801			550	PTF ALL
Trailing Edge Flap Asymmetry Skew System BITE Test TASK 27-51-00-740-802			554	PTF ALL
Trailing Edge Flap Load Relief System BITE Test TASK 27-51-00-740-803			558	PTF ALL
Flap Priority Valve Test TASK 27-51-00-720-801			560	PTF ALL
Flap Control Lever Force Test TASK 27-51-00-720-802			561	PTF ALL
Flap Control Lever Feel Test TASK 27-51-00-720-805			563	PTF ALL
Alternate Flap Control System Functional Test TASK 27-51-00-720-803			564	PTF ALL
Transmission No-Back Brake Operational Test TASK 27-51-00-710-802			566	PTF ALL
Trailing Edge Flap Load Relief System Functional Test TASK 27-51-00-720-804			572	PTF ALL
TRAILING EDGE FLAP SYSTEM - INSPECTION/CHECK	27-51-00		601	PTF ALL
Trailing Edge Flap Drive System Inspection TASK 27-51-00-210-801			601	PTF ALL
Trailing Edge Flap Actuation System Inspection TASK 27-51-00-210-802			610	PTF ALL
FLAP/SLAT ELECTRONICS UNIT (FSEU) - REMOVAL/INSTALLATION	27-51-01		401	PTF ALL
Flap/Slat Electronics Unit Removal TASK 27-51-01-000-801			401	PTF ALL
Flap/Slat Electronics Unit Installation TASK 27-51-01-400-801			404	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
TRAILING EDGE FLAP BYPASS VALVE - REMOVAL/INSTALLATION	27-51-02		401	PTF ALL
Trailing Edge Flap Bypass Valve Assembly Removal TASK 27-51-02-000-801			401	PTF ALL
Trailing Edge Flap Bypass Valve Assembly Installation TASK 27-51-02-400-801			407	PTF ALL
Trailing Edge Flap Bypass Valve Motor Removal TASK 27-51-02-000-803			409	PTF ALL
Trailing Edge Flap Bypass Valve Motor Installation TASK 27-51-02-400-803			410	PTF ALL
TRAILING EDGE FLAP BYPASS VALVE - ADJUSTMENT/TEST	27-51-02		501	PTF ALL
Trailing Edge Flap Bypass Valve Functional Test TASK 27-51-02-720-801			501	PTF ALL
FLAP PRIORITY VALVE - REMOVAL/INSTALLATION	27-51-04		401	PTF ALL
Flap Priority Valve Removal TASK 27-51-04-000-801			401	PTF ALL
Flap Priority Valve Installation TASK 27-51-04-400-801			405	PTF ALL
FLAP CONTROL LEVER POSITION SENSOR - MAINTENANCE PRACTICES	27-51-06		201	PTF ALL
Flap Control Lever Position Sensor Removal TASK 27-51-06-000-801			201	PTF ALL
Flap Control Lever Position Sensor Installation TASK 27-51-06-400-801			206	PTF ALL
Flap Control Lever Position Sensor Adjustment and Test TASK 27-51-06-820-801			207	PTF ALL
Flap Control Lever Position Sensor System Test TASK 27-51-06-730-801			212	PTF ALL
Flap Control Lever Position Sensor and Mechanism Detailed Visual Inspection TASK 27-51-06-210-801			213	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FLAP LEVER POSITION SWITCHES - REMOVAL/INSTALLATION	27-51-08		401	PTF ALL
Flap Lever Position Switch Removal TASK 27-51-08-000-801			401	PTF ALL
Flap Lever Position Switch Installation TASK 27-51-08-400-801			408	PTF ALL
FLAP LEVER POSITION SWITCHES - ADJUSTMENT/TEST	27-51-08		501	PTF ALL
Flap Lever Position Switch Adjustment TASK 27-51-08-820-801			501	PTF ALL
INBOARD TRAILING EDGE FLAP - REMOVAL/INSTALLATION	27-51-11		401	PTF ALL
Inboard Trailing Edge Flap Removal TASK 27-51-11-000-801			401	PTF ALL
Inboard Trailing Edge Flap Installation TASK 27-51-11-400-801			413	PTF ALL
Inboard Trailing Edge Flap Bullnose Seal Removal TASK 27-51-11-000-802			423	PTF ALL
Inboard Trailing Edge Flap Bullnose Seal Installation TASK 27-51-11-400-802			423	PTF ALL
Inboard Trailing Edge Flap Outer Seal Removal TASK 27-51-11-000-803			426	PTF ALL
Inboard Trailing Edge Flap Outer Seal Installation TASK 27-51-11-400-803			429	PTF ALL
INBOARD TRAILING EDGE FLAP - ADJUSTMENT/TEST	27-51-11		501	PTF ALL
Inboard Trailing Edge Flap Adjustment TASK 27-51-11-820-801			501	PTF ALL
INBOARD AFT FLAP - REMOVAL/INSTALLATION	27-51-12		401	PTF ALL
Inboard Aft Flap Removal TASK 27-51-12-000-801			401	PTF ALL
Inboard Aft Flap Installation TASK 27-51-12-400-801			408	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
INBOARD AFT FLAP - INSPECTION/CHECK	27-51-12		601	PTF ALL
Inboard Aft Flap Inspection TASK 27-51-12-200-801			601	PTF ALL
INBOARD FLAP CARRIAGES - REMOVAL/INSTALLATION	27-51-13		401	PTF ALL
Inboard Flap Carriage Removal (Selection) TASK 27-51-13-000-801			401	PTF ALL
Inboard Flap Carriage Installation (Selection) TASK 27-51-13-400-801			401	PTF ALL
Inboard Flap Outboard Carriage Removal TASK 27-51-13-000-802			402	PTF ALL
Inboard Flap Outboard Carriage Installation TASK 27-51-13-400-802			405	PTF ALL
Inboard Flap Inboard Carriage Removal TASK 27-51-13-000-803			406	PTF ALL
Inboard Flap Inboard Carriage Installation TASK 27-51-13-400-803			411	PTF ALL
INBOARD FLAP CARRIAGES - INSPECTION/CHECK	27-51-13		601	PTF ALL
Inboard Flap Carriage Inspection (Selection) TASK 27-51-13-200-801			601	PTF ALL
Inboard Flap Outboard Carriage Inspection TASK 27-51-13-200-802			601	PTF ALL
Inboard Flap Inboard Carriage Inspection TASK 27-51-13-200-803			606	PTF ALL
INBOARD AFT FLAP ACTUATING MECHANISM - REMOVAL/INSTALLATION	27-51-14		401	PTF ALL
Inboard Aft Flap Actuating Mechanism Removal TASK 27-51-14-000-801			401	PTF ALL
Inboard Aft Flap Actuating Mechanism Installation TASK 27-51-14-400-801			409	PTF ALL
INBOARD FLAP TRACKS - REMOVAL/INSTALLATION	27-51-15		401	PTF ALL
Inboard Flap Track Removal (Selection) TASK 27-51-15-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Inboard Flap Track Installation (Selection) TASK 27-51-15-400-801			401	PTF ALL
Inboard Flap Outboard Track Removal TASK 27-51-15-000-802			401	PTF ALL
Inboard Flap Outboard Track Installation TASK 27-51-15-400-802			409	PTF ALL
Inboard Flap Inboard Track Removal TASK 27-51-15-000-803			413	PTF ALL
Inboard Flap Inboard Track Installation TASK 27-51-15-400-803			421	PTF ALL
INBOARD FLAP TRACKS - INSPECTION/CHECK	27-51-15		601	PTF ALL
Inboard Flap Track Inspection (Selection) TASK 27-51-15-200-801			601	PTF ALL
Inboard Flap Outboard Track Inspection TASK 27-51-15-200-802			601	PTF ALL
Inboard Flap Inboard Track Inspection TASK 27-51-15-200-803			605	PTF ALL
INBOARD AFT FLAP TRACKS - REMOVAL/INSTALLATION	27-51-16		401	PTF ALL
Inboard Aft Flap Track Removal TASK 27-51-16-000-801			401	PTF ALL
Inboard Aft Flap Track Installation TASK 27-51-16-400-801			404	PTF ALL
INBOARD FLAP SUPPORT FAIRING - REMOVAL/INSTALLATION	27-51-18		401	PTF ALL
Inboard Flap Support Fairing Removal (Selection) TASK 27-51-18-000-803			401	PTF ALL
Inboard Flap Support Fairing Installation (Selection) TASK 27-51-18-400-803			401	PTF ALL
Inboard Flap Support Forward Fairing Removal TASK 27-51-18-000-801			401	PTF ALL
Inboard Flap Support Forward Fairing Installation TASK 27-51-18-400-801			408	PTF ALL

27-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Inboard Flap Support Aft Fairing Removal TASK 27-51-18-000-802			409	PTF ALL
Inboard Flap Support Aft Fairing Installation TASK 27-51-18-400-802			411	PTF ALL
INBOARD FLAP SUPPORT FAIRING - ADJUSTMENT/TEST	27-51-18		501	PTF ALL
Inboard Flap Support Fairing Adjustment TASK 27-51-18-820-801			501	PTF ALL
OUTBOARD TRAILING EDGE FLAP - REMOVAL/INSTALLATION	27-51-21		401	PTF ALL
Outboard Trailing Edge Flap Removal TASK 27-51-21-000-801			401	PTF ALL
Outboard Trailing Edge Flap Installation TASK 27-51-21-400-801			413	PTF ALL
Outboard Trailing Edge Flap Outer Seal Removal TASK 27-51-21-000-802			423	PTF ALL
Outboard Trailing Edge Flap Outer Seal Installation TASK 27-51-21-400-802			425	PTF ALL
OUTBOARD TRAILING EDGE FLAP - ADJUSTMENT/TEST	27-51-21		501	PTF ALL
Outboard Trailing Edge Flap Adjustment TASK 27-51-21-820-801			501	PTF ALL
OUTBOARD AFT FLAP - REMOVAL/INSTALLATION	27-51-22		401	PTF ALL
Outboard Aft Flap Removal TASK 27-51-22-000-801			401	PTF ALL
Outboard Aft Flap Installation TASK 27-51-22-400-801			411	PTF ALL
OUTBOARD AFT FLAP - INSPECTION/CHECK	27-51-22		601	PTF ALL
Outboard Aft Flap Inspection TASK 27-51-22-200-801			601	PTF ALL
OUTBOARD FLAP CARRIAGES - REMOVAL/INSTALLATION	27-51-23		401	PTF ALL
Outboard Flap Carriage Removal (Selection) TASK 27-51-23-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Outboard Flap Carriage Installation (Selection) TASK 27-51-23-400-801			401	PTF ALL
Outboard Flap Outboard Carriage Removal TASK 27-51-23-000-802			401	PTF ALL
Outboard Flap Outboard Carriage Installation TASK 27-51-23-400-802			408	PTF ALL
Outboard Flap Inboard Carriage Removal TASK 27-51-23-000-803			411	PTF ALL
Outboard Flap Inboard Carriage Installation TASK 27-51-23-400-803			418	PTF ALL
OUTBOARD FLAP CARRIAGES - INSPECTION/CHECK	27-51-23		601	PTF ALL
Outboard Flap Carriage Inspection (Selection) TASK 27-51-23-200-801			601	PTF ALL
Outboard Flap Outboard Carriage Inspection TASK 27-51-23-200-802			601	PTF ALL
Outboard Flap Inboard Carriage Inspection TASK 27-51-23-200-803			609	PTF ALL
OUTBOARD AFT FLAP ACTUATING MECHANISM - REMOVAL/INSTALLATION	27-51-24		401	PTF ALL
Outboard Aft Flap Actuating Mechanism Removal TASK 27-51-24-000-801			401	PTF ALL
Outboard Aft Flap Actuating Mechanism Installation TASK 27-51-24-400-801			406	PTF ALL
OUTBOARD FLAP TRACKS - REMOVAL/INSTALLATION	27-51-25		401	PTF ALL
Outboard Flap Track Removal (Selection) TASK 27-51-25-000-801			401	PTF ALL
Outboard Flap Track Installation (Selection) TASK 27-51-25-400-801			401	PTF ALL
Outboard Flap Outboard Track Removal TASK 27-51-25-000-802			402	PTF ALL
Outboard Flap Outboard Track Installation TASK 27-51-25-400-802			408	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Outboard Flap Inboard Track Removal TASK 27-51-25-000-803			411	PTF ALL
Outboard Flap Inboard Track Installation TASK 27-51-25-400-803			417	PTF ALL
OUTBOARD FLAP TRACKS - INSPECTION/CHECK	27-51-25		601	PTF ALL
Outboard Flap Track Inspection (Selection) TASK 27-51-25-200-801			601	PTF ALL
Outboard Flap Outboard Track Inspection TASK 27-51-25-200-802			601	PTF ALL
Outboard Flap Inboard Track Inspection TASK 27-51-25-200-803			606	PTF ALL
OUTBOARD AFT FLAP TRACKS - REMOVAL/INSTALLATION	27-51-26		401	PTF ALL
Outboard Aft Flap Track Removal TASK 27-51-26-000-801			401	PTF ALL
Outboard Aft Flap Track Installation TASK 27-51-26-400-801			412	PTF ALL
OUTBOARD FLAP SUPPORT FAIRINGS - REMOVAL/INSTALLATION	27-51-28		401	PTF ALL
Outboard Flap Support Fairings Removal (Selection) TASK 27-51-28-000-801			401	PTF ALL
Outboard Flap Support Fairings Installation (Selection) TASK 27-51-28-400-801			401	PTF ALL
Outboard Flap Outboard Support Forward Fairing Removal TASK 27-51-28-000-802			402	PTF ALL
Outboard Flap Outboard Support Forward Fairing Installation TASK 27-51-28-400-802			408	PTF ALL
Outboard Flap Outboard Support Aft Fairing Removal TASK 27-51-28-000-803			410	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Outboard Flap Outboard Support Aft Fairing Installation TASK 27-51-28-400-803			412	PTF ALL
Outboard Flap Inboard Support Forward Fairing Removal TASK 27-51-28-000-804			416	PTF ALL
Outboard Flap Inboard Support Forward Fairing Installation TASK 27-51-28-400-804			423	PTF ALL
Outboard Flap Inboard Support Aft Fairing Removal TASK 27-51-28-000-805			425	PTF ALL
Outboard Flap Inboard Support Aft Fairing Installation TASK 27-51-28-400-805			428	PTF ALL
OUTBOARD FLAP SUPPORT FAIRINGS - ADJUSTMENT/TEST	27-51-28		501	PTF ALL
Outboard Flap Support Fairings Adjustment (Selection) TASK 27-51-28-800-801			501	PTF ALL
Outboard Flap Outboard Support Fairing Adjustment TASK 27-51-28-820-801			501	PTF ALL
Outboard Flap Inboard Support Fairing Adjustment TASK 27-51-28-820-802			510	PTF ALL
INBOARD FLAP TRANSMISSIONS - REMOVAL/INSTALLATION	27-51-31		401	PTF ALL
Inboard Flap Transmission Removal (Selection) TASK 27-51-31-000-801			401	PTF ALL
Inboard Flap Transmission Installation (Selection) TASK 27-51-31-400-801			401	PTF ALL
Inboard Flap Outboard Transmission Removal TASK 27-51-31-000-802			402	PTF ALL
Inboard Flap Outboard Transmission Installation TASK 27-51-31-400-802			407	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Inboard Flap Inboard Transmission Removal TASK 27-51-31-000-803			410	PTF ALL
Inboard Flap Inboard Transmission Installation TASK 27-51-31-400-803			415	PTF ALL
INBOARD FLAP BALLSCREWS AND GIMBALS - REMOVAL/INSTALLATION	27-51-32		401	PTF ALL
Inboard Flap Ballscrew and Gimbal Removal (Selection) TASK 27-51-32-000-801			401	PTF ALL
Inboard Flap Ballscrew and Gimbal Installation (Selection) TASK 27-51-32-400-801			401	PTF ALL
Inboard Flap Outboard Ballscrew and Gimbal Removal TASK 27-51-32-000-802			402	PTF ALL
Inboard Flap Outboard Ballscrew and Gimbal Installation TASK 27-51-32-400-802			406	PTF ALL
Inboard Flap Inboard Ballscrew and Gimbal Removal TASK 27-51-32-000-803			408	PTF ALL
Inboard Flap Inboard Ballscrew and Gimbal Installation TASK 27-51-32-400-803			414	PTF ALL
INBOARD FLAP BALLSCREWS AND GIMBALS - INSPECTION/CHECK	27-51-32		601	PTF ALL
Inboard Flap Outboard Ballscrew Inspection TASK 27-51-32-220-801			601	PTF ALL
Inboard Flap Inboard Ballscrew Inspection TASK 27-51-32-220-802			607	PTF ALL
Inboard Flap Outboard Carriage Bushing and Gimbal Bushing Inspection TASK 27-51-32-200-801			613	PTF ALL
Inboard Flap Inboard Carriage Bushing and Gimbal Bushing Inspection TASK 27-51-32-200-802			617	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Inboard Flap Inboard Ballscrew Detailed Lubrication Inspection TASK 27-51-32-200-803			621	PTF ALL
Inboard Flap Outboard Ballscrew Detailed Lubrication Inspection TASK 27-51-32-200-804			626	PTF ALL
OUTBOARD FLAP TRANSMISSIONS - REMOVAL/INSTALLATION	27-51-41		401	PTF ALL
Outboard Flap Transmission Removal (Selection) TASK 27-51-41-000-801			401	PTF ALL
Outboard Flap Transmission Installation (Selection) TASK 27-51-41-400-801			401	PTF ALL
Outboard Flap Outboard Transmission Removal TASK 27-51-41-000-802			402	PTF ALL
Outboard Flap Outboard Transmission Installation TASK 27-51-41-400-802			410	PTF ALL
Outboard Flap Inboard Transmission Removal TASK 27-51-41-000-803			414	PTF ALL
Outboard Flap Inboard Transmission Installation TASK 27-51-41-400-803			422	PTF ALL
OUTBOARD FLAP BALLSCREWS AND GIMBALS - REMOVAL/INSTALLATION	27-51-42		401	PTF ALL
Outboard Flap Ballscrew and Gimbal Removal (Selection) TASK 27-51-42-000-801			401	PTF ALL
Outboard Flap Ballscrew and Gimbal Installation (Selection) TASK 27-51-42-400-801			401	PTF ALL
Outboard Flap Outboard Ballscrew and Gimbal Removal TASK 27-51-42-000-802			402	PTF ALL
Outboard Flap Outboard Ballscrew and Gimbal Installation TASK 27-51-42-400-802			408	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Outboard Flap Inboard Ballscrew and Gimbal Removal TASK 27-51-42-000-803			410	PTF ALL
Outboard Flap Inboard Ballscrew and Gimbal Installation TASK 27-51-42-400-803			416	PTF ALL
OUTBOARD FLAP BALLSCREWS AND GIMBALS - INSPECTION/CHECK	27-51-42		601	PTF ALL
Outboard Flap Outboard Ballscrew Inspection TASK 27-51-42-220-801			601	PTF ALL
Outboard Flap Inboard Ballscrew Inspection TASK 27-51-42-220-802			607	PTF ALL
Outboard Flap Inboard Ballscrew Detailed Lubrication Inspection TASK 27-51-42-200-801			610	PTF ALL
Outboard Flap Outboard Ballscrew Detailed Lubrication Inspection TASK 27-51-42-200-802			615	PTF ALL
FLAP POWER DRIVE UNIT (PDU) - REMOVAL/INSTALLATION	27-51-51		401	PTF ALL
Flap Power Drive Unit (PDU) Removal TASK 27-51-51-000-801			401	PTF ALL
Flap Power Drive Unit (PDU) Installation TASK 27-51-51-400-801			409	PTF ALL
Flap Power Drive Unit Manual Drive Shearout Replacement TASK 27-51-51-960-801			413	PTF ALL
Flap Power Drive Unit (PDU) Seal Removal TASK 27-51-51-000-802			415	PTF ALL
Flap Power Drive Unit (PDU) Seal Installation TASK 27-51-51-400-802			421	PTF ALL
FLAP POWER DRIVE UNIT - ADJUSTMENT/TEST	27-51-51		501	PTF ALL
Flap Power Drive Unit Functional Test TASK 27-51-51-720-801			501	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FLAP HYDRAULIC MOTOR - REMOVAL/INSTALLATION	27-51-52		401	PTF ALL
Flap Hydraulic Motor Removal TASK 27-51-52-000-801			401	PTF ALL
Flap Hydraulic Motor Installation TASK 27-51-52-400-801			405	PTF ALL
FLAP ELECTRIC MOTOR - REMOVAL/INSTALLATION	27-51-53		401	PTF ALL
Flap Electric Motor Removal TASK 27-51-53-000-801			401	PTF ALL
Flap Electric Motor Installation TASK 27-51-53-400-801			405	PTF ALL
TEE ANGLE GEARBOX - REMOVAL/INSTALLATION	27-51-55		401	PTF ALL
Tee Angle Gearbox Removal TASK 27-51-55-000-801			401	PTF ALL
Tee Angle Gearbox Installation TASK 27-51-55-400-801			405	PTF ALL
MLG BEAM ANGLE GEARBOX - REMOVAL/INSTALLATION	27-51-56		401	PTF ALL
Main Landing Gear Beam Angle Gearbox Removal TASK 27-51-56-000-801			401	PTF ALL
Main Landing Gear Beam Angle Gearbox Installation TASK 27-51-56-400-801			405	PTF ALL
SEAL RIB ANGLE GEARBOX - REMOVAL/INSTALLATION	27-51-57		401	PTF ALL
Seal Rib Angle Gearbox Removal TASK 27-51-57-000-801			401	PTF ALL
Seal Rib Angle Gearbox Installation TASK 27-51-57-400-801			404	PTF ALL
FLAP TORQUE TUBES - REMOVAL/INSTALLATION	27-51-58		401	PTF ALL
Flap Torque Tube Removal (Selection) TASK 27-51-58-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Flap Torque Tube Installation (Selection) TASK 27-51-58-400-801			402	PTF ALL
Outboard Wing Flap Torque Tube Removal TASK 27-51-58-000-802			402	PTF ALL
Outboard Wing Flap Torque Tube Installation TASK 27-51-58-400-802			408	PTF ALL
Inboard Wing Flap Torque Tube Removal TASK 27-51-58-000-803			410	PTF ALL
Inboard Wing Flap Torque Tube Installation TASK 27-51-58-400-803			415	PTF ALL
Wheel Well Flap Torque Tube Removal TASK 27-51-58-000-804			417	PTF ALL
Wheel Well Flap Torque Tube Installation TASK 27-51-58-400-804			422	PTF ALL
FLAP TORQUE TUBES - INSPECTION/CHECK	27-51-58		601	PTF ALL
Flap Torque Tube - Inspection TASK 27-51-58-200-801			601	PTF ALL
FLAP TORQUE TUBE SUPPORT - REMOVAL/INSTALLATION	27-51-59		401	PTF ALL
Flap Torque Tube Support Removal TASK 27-51-59-000-801			401	PTF ALL
Flap Torque Tube Support Installation TASK 27-51-59-400-801			405	PTF ALL
FLAP CONTROL UNIT - REMOVAL/INSTALLATION	27-51-61		401	PTF ALL
Flap Control Unit Removal TASK 27-51-61-000-801			401	PTF ALL
Flap Control Unit Installation TASK 27-51-61-400-801			412	PTF ALL
FLAP CONTROL QUADRANT - REMOVAL/INSTALLATION	27-51-62		401	PTF ALL
Flap Control Quadrant Removal TASK 27-51-62-000-801			401	PTF ALL
Flap Control Quadrant Installation TASK 27-51-62-400-801			409	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FLAP LIMIT SWITCHES - REMOVAL/INSTALLATION	27-51-63		401	PTF ALL
Flap Limit Switch Removal TASK 27-51-63-000-801			401	PTF ALL
Flap Limit Switch Installation TASK 27-51-63-400-801			407	PTF ALL
FLAP LIMIT SWITCHES - ADJUSTMENT/TEST	27-51-63		501	PTF ALL
Flaps Limit Switch Adjustment TASK 27-51-63-820-801			501	PTF ALL
Flaps UP Limit Switch, S245 Functional Test TASK 27-51-63-720-801			512	PTF ALL
Flaps DOWN Limit Switch, S246 Functional Test TASK 27-51-63-720-802			518	PTF ALL
Flaps UP Limit Switch, S1051 Operational Test TASK 27-51-63-710-801			519	PTF ALL
TRAILING EDGE FLAP CONTROL VALVE - REMOVAL/INSTALLATION	27-51-65		401	PTF ALL
Trailing Edge Flap Control Valve Removal TASK 27-51-65-000-801			401	PTF ALL
Trailing Edge Flap Control Valve Installation TASK 27-51-65-400-801			407	PTF ALL
Flap Load Limiter Solenoid Valve Removal TASK 27-51-65-000-802			409	PTF ALL
Flap Load Limiter Solenoid Valve Installation TASK 27-51-65-400-802			411	PTF ALL
TRAILING EDGE FLAP CONTROL VALVE - ADJUSTMENT/TEST	27-51-65		501	PTF ALL
Trailing Edge Flap Control Valve Adjustment TASK 27-51-65-820-801			501	PTF ALL
TRAILING EDGE FLAP CONTROL VALVE - REPAIRS	27-51-65		801	PTF ALL
Trailing Edge Flap Control Valve Internal Leakage Repair TASK 27-51-65-360-801			801	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
TRAILING EDGE FLAP FLOW LIMITER - REMOVAL/INSTALLATION	27-51-67		401	PTF ALL
Trailing Edge Flap Flow Limiter Removal TASK 27-51-67-000-801			401	PTF ALL
Trailing Edge Flap Flow Limiter Installation TASK 27-51-67-400-801			404	PTF ALL
FLAP POSITION TRANSMITTERS AND INDICATORS - MAINTENANCE PRACTICES	27-58-00		201	PTF ALL
Flap Position Indicating System - Deactivation TASK 27-58-00-040-801			201	PTF ALL
Flap Position Indicating System - Activation TASK 27-58-00-440-801			202	PTF ALL
FLAP POSITION TRANSMITTER - REMOVAL/INSTALLATION	27-58-01		401	PTF ALL
Flap Position Transmitter Removal TASK 27-58-01-000-801			401	PTF ALL
Flap Position Transmitter Installation TASK 27-58-01-400-801			407	PTF ALL
FLAP POSITION TRANSMITTER - ADJUSTMENT/TEST	27-58-01		501	PTF ALL
Flap Position Transmitter Adjustment and Test TASK 27-58-01-820-801			501	PTF ALL
FLAP SKEW DETECTION SYSTEM - ADJUSTMENT/TEST	27-59-00		501	PTF ALL
Flap Skew Detection System Adjustment TASK 27-59-00-820-801			501	PTF ALL
Flap Skew Detection System Functional Test TASK 27-59-00-730-801			512	PTF ALL
FLAP SKEW SENSORS - MAINTENANCE PRACTICES	27-59-01		201	PTF ALL
Outboard Flap Outboard Skew Sensor Removal TASK 27-59-01-000-801			201	PTF ALL
Outboard Flap Outboard Skew Sensor Installation TASK 27-59-01-400-801			205	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Outboard Flap Inboard Skew Sensor Removal TASK 27-59-01-000-802			207	PTF ALL
Outboard Flap Inboard Skew Sensor Installation TASK 27-59-01-400-802			211	PTF ALL
Inboard Flap Outboard Skew Sensor Removal TASK 27-59-01-000-803			213	PTF ALL
Inboard Flap Outboard Skew Sensor Installation TASK 27-59-01-400-803			217	PTF ALL
Inboard Flap Inboard Skew Sensor Removal TASK 27-59-01-000-804			219	PTF ALL
Inboard Flap Inboard Skew Sensor Installation TASK 27-59-01-400-804			223	PTF ALL
FLAP SKEW MECHANISM - MAINTENANCE PRACTICES	27-59-02		201	PTF ALL
Outboard Flap Outboard Skew Mechanism Removal TASK 27-59-02-000-801			201	PTF ALL
Outboard Flap Outboard Skew Mechanism Installation TASK 27-59-02-400-801			205	PTF ALL
Outboard Flap Inboard Skew Mechanism Removal TASK 27-59-02-000-802			208	PTF ALL
Outboard Flap Inboard Skew Mechanism Installation TASK 27-59-02-400-802			212	PTF ALL
Inboard Flap Outboard Skew Mechanism Removal TASK 27-59-02-000-803			215	PTF ALL
Inboard Flap Outboard Skew Mechanism Installation TASK 27-59-02-400-803			219	PTF ALL
Inboard Flap Inboard Skew Mechanism Removal TASK 27-59-02-000-804			222	PTF ALL
Inboard Flap Inboard Skew Mechanism Installation TASK 27-59-02-400-804			226	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SPOILER CONTROL SYSTEM - MAINTENANCE PRACTICES	27-61-00		201	PTF ALL
Spoiler Control System - Deactivation TASK 27-61-00-040-801			201	PTF ALL
Spoiler Control System - Activation TASK 27-61-00-440-801			208	PTF ALL
Apply Pressure to the Spoiler Hydraulic Systems A and B TASK 27-61-00-800-801			210	PTF ALL
Shut Down the Spoiler Hydraulic Systems A and B TASK 27-61-00-840-801			213	PTF ALL
Depressurization of the Spoiler Hydraulic Systems A and B TASK 27-61-00-800-802			213	PTF ALL
Pressurization of the Spoiler Hydraulic Systems A and B to its Usual Condition TASK 27-61-00-840-802			219	PTF ALL
SPOILER CONTROL SYSTEM - ADJUSTMENT/TEST	27-61-00		501	PTF ALL
Spoiler Control System Operational Test TASK 27-61-00-710-801			501	PTF ALL
Flight Spoiler Adjustment TASK 27-61-00-820-802			502	PTF ALL
Flight Spoiler System Test TASK 27-61-00-730-801			512	PTF ALL
Spoiler Manual Operation Test TASK 27-61-00-000-801			516	PTF ALL
SPOILER CONTROL SYSTEM - INSPECTION/CHECK	27-61-00		601	PTF ALL
Spoiler PCU and Wiring Integrity Verification - Insulation Resistance Check TASK 27-61-00-700-801			601	PTF ALL
FLIGHT SPOILER - REMOVAL/INSTALLATION	27-61-11		401	PTF ALL
Flight Spoiler Removal TASK 27-61-11-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Flight Spoiler Installation TASK 27-61-11-400-801			411	PTF ALL
SPOILER SHUTOFF VALVE - REMOVAL/INSTALLATION	27-61-12		401	PTF ALL
Spoiler Shutoff Valve Removal TASK 27-61-12-000-801			401	PTF ALL
Spoiler Shutoff Valve Installation TASK 27-61-12-400-801			407	PTF ALL
Spoiler Shutoff Valve Motor Removal TASK 27-61-12-000-802			408	PTF ALL
Spoiler Shutoff Valve Motor Installation TASK 27-61-12-400-802			409	PTF ALL
SPOILER SHUTOFF VALVE - ADJUSTMENT/TEST	27-61-12		501	PTF ALL
Spoiler Shutoff Valve Test TASK 27-61-12-400-803			501	PTF ALL
FLIGHT SPOILER HINGE LINK - REMOVAL/INSTALLATION	27-61-15		401	PTF ALL
Flight Spoiler Hinge Link Removal TASK 27-61-15-000-801			401	PTF ALL
Flight Spoiler Hinge Link Installation TASK 27-61-15-400-801			407	PTF ALL
CONTROL WHEEL POSITION TRANSDUCER - REMOVAL/INSTALLATION	27-61-44		401	PTF ALL
Control Wheel Position Transducer Removal TASK 27-61-44-000-801			401	PTF ALL
Control Wheel Position Transducer Installation TASK 27-61-44-400-801			405	PTF ALL
SPOILER CONTROL ELECTRONICS - MAINTENANCE PRACTICES	27-61-45		201	PTF ALL
SCE Software Installation with Onboard Network System TASK 27-61-45-470-801			201	PTF ALL
SCE Configuration Check TASK 27-61-45-750-801			202	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SPOILER CONTROL ELECTRONICS - REMOVAL/INSTALLATION	27-61-45		401	PTF ALL
Spoiler Control Electronics Unit Removal TASK 27-61-45-000-801			401	PTF ALL
Spoiler Control Electronics Unit Installation TASK 27-61-45-400-801			404	PTF ALL
SPOILER CONTROL ELECTRONICS - ADJUSTMENT/TEST	27-61-45		501	PTF ALL
Spoiler Control Electronics Self Test TASK 27-61-45-730-801			501	PTF ALL
Speedbrake Lever Transducer Test TASK 27-61-45-730-802			502	PTF ALL
First Officer's Wheel Transducer Test TASK 27-61-45-730-803			504	PTF ALL
Ground Spoiler Control Module Replacement Test TASK 27-61-45-730-804			505	PTF ALL
Flight Spoiler Test TASK 27-61-45-730-805			507	PTF ALL
FLIGHT SPOILER ACTUATOR - REMOVAL/INSTALLATION	27-61-51		401	PTF ALL
Flight Spoiler Actuator - Removal TASK 27-61-51-000-801			401	PTF ALL
Flight Spoiler Actuator - Installation TASK 27-61-51-400-801			409	PTF ALL
Electrohydraulic Servo Valve (EHSV) - Removal TASK 27-61-51-000-802			413	PTF ALL
Electrohydraulic Servo Valve (EHSV) - Installation TASK 27-61-51-400-802			418	PTF ALL
FLIGHT SPOILER ACTUATOR - INSPECTION/CHECK	27-61-51		601	PTF ALL
Flight Spoiler Actuator Inspection TASK 27-61-51-200-801			601	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SPEED BRAKE CONTROL SYSTEM - MAINTENANCE PRACTICES	27-62-00		201	PTF ALL
Apply Pressure to the Speed Brake Hydraulic Systems A and B TASK 27-62-00-800-801			201	PTF ALL
Shut Down the Speed Brake Hydraulic Systems A and B TASK 27-62-00-840-801			204	PTF ALL
Depressurization of the Speed Brake Hydraulic Systems A and B TASK 27-62-00-800-802			204	PTF ALL
Pressurization of the Speed Brake Hydraulic Systems A and B to its Usual Condition TASK 27-62-00-840-802			209	PTF ALL
SPEED BRAKE CONTROL SYSTEM - ADJUSTMENT/TEST	27-62-00		501	PTF ALL
Speed Brake Control System Operational Test TASK 27-62-00-710-801			501	PTF ALL
Speed Brake Control Lever Adjustment TASK 27-62-00-820-801			502	PTF ALL
Ground Spoiler Adjustment TASK 27-62-00-820-802			512	PTF ALL
Speed Brake Lever Friction Test TASK 27-62-00-800-803			521	PTF ALL
Spoiler Operation Time Test TASK 27-62-00-820-803			523	PTF ALL
Speed Brake Throttle Test TASK 27-62-00-820-804			526	PTF ALL
Speed Brake Antiskid Inoperative Test TASK 27-62-00-820-805			533	PTF ALL
Speed Brake RTO Test TASK 27-62-00-820-807			536	PTF ALL
Speed Brake Extended Light Test TASK 27-62-00-820-808			539	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
SPEED BRAKE CONTROL SYSTEM - INSPECTION/CHECK	27-62-00		601	PTF ALL
Speedbrake System Components - Inspection TASK 27-62-00-210-801			601	PTF ALL
Spoiler Actuator - Inspection TASK 27-62-00-210-802			605	PTF ALL
INBOARD GROUND SPOILER - REMOVAL/INSTALLATION	27-62-12		401	PTF ALL
Inboard Ground Spoiler Removal TASK 27-62-12-000-801			401	PTF ALL
Inboard Ground Spoiler Installation TASK 27-62-12-400-801			409	PTF ALL
Inboard Ground Spoiler Functional Test TASK 27-62-12-720-801			413	PTF ALL
GROUND SPOILER HINGE LINK - REMOVAL/INSTALLATION	27-62-15		401	PTF ALL
Ground Spoiler Hinge Link Removal TASK 27-62-15-000-801			401	PTF ALL
Ground Spoiler Hinge Link Installation TASK 27-62-15-400-801			407	PTF ALL
SPEEDBRAKE MECHANISM - REMOVAL/INSTALLATION	27-62-21		401	PTF ALL
Speedbrake Mechanism Removal TASK 27-62-21-000-801			401	PTF ALL
Speedbrake Mechanism Installation TASK 27-62-21-400-801			407	PTF ALL
SPEEDBRAKE LEVER - REMOVAL/INSTALLATION	27-62-22		401	PTF ALL
Speedbrake Lever - Removal TASK 27-62-22-000-801			401	PTF ALL
Speedbrake Lever - Installation TASK 27-62-22-400-801			424	PTF ALL
AUTO SPEED BRAKE ACTUATOR - REMOVAL/INSTALLATION	27-62-31		401	PTF ALL
Auto Speed Brake Actuator Removal TASK 27-62-31-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Auto Speed Brake Actuator Installation TASK 27-62-31-400-801			409	PTF ALL
SPEED BRAKE REFUSED TAKEOFF (RTO) SWITCH - REMOVAL/INSTALLATION	27-62-32		401	PTF ALL
Speed Brake Refused Takeoff (RTO) Switch Removal TASK 27-62-32-000-801			401	PTF ALL
Speed Brake Refused Takeoff (RTO) Switch Installation TASK 27-62-32-400-801			405	PTF ALL
SPEED BRAKE REFUSED TAKEOFF (RTO) SWITCH - ADJUSTMENT/TEST	27-62-32		501	PTF ALL
Speed Brake Refused Takeoff (RTO) Switch Adjustment TASK 27-62-32-820-801			501	PTF ALL
SPEEDBRAKE LEVER TRANSDUCER - REMOVAL/INSTALLATION	27-62-37		401	PTF ALL
Speedbrake Lever Transducer Removal TASK 27-62-37-000-801			401	PTF ALL
Speedbrake Lever Transducer Installation TASK 27-62-37-400-801			405	PTF ALL
Speedbrake Lever Transducer Adjustment TASK 27-62-37-200-801			406	PTF ALL
GROUND SPOILER CONTROL MODULE-REMOVAL/INSTALLATION	27-62-41		401	PTF ALL
Ground Spoiler Control Module Removal TASK 27-62-41-000-801			401	PTF ALL
Ground Spoiler Control Module Installation TASK 27-62-41-400-801			405	PTF ALL
Ground Spoiler Control Module Functional Test TASK 27-62-41-720-801			407	PTF ALL
INBOARD GROUND SPOILER ACTUATOR - REMOVAL/INSTALLATION	27-62-71		401	PTF ALL
Inboard Ground Spoiler Actuator Removal TASK 27-62-71-000-801			401	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Inboard Ground Spoiler Actuator Installation TASK 27-62-71-400-801			405	PTF ALL
Inboard Ground Spoiler Actuator Functional Test TASK 27-62-71-700-801			407	PTF ALL
OUTBOARD GROUND SPOILER - REMOVAL/INSTALLATION	27-62-91		401	PTF ALL
Outboard Ground Spoiler Removal TASK 27-62-91-000-801			401	PTF ALL
Outboard Ground Spoiler Installation TASK 27-62-91-400-801			407	PTF ALL
Outboard Ground Spoiler Test TASK 27-62-91-000-802			411	PTF ALL
OUTBOARD GROUND SPOILER ACTUATOR - REMOVAL/INSTALLATION	27-62-94		401	PTF ALL
Outboard Ground Spoiler Actuator Removal TASK 27-62-94-000-801			401	PTF ALL
Outboard Ground Spoiler Actuator Installation TASK 27-62-94-400-801			406	PTF ALL
OUTBOARD GROUND SPOILER ACTUATOR - ADJUSTMENT/TEST	27-62-94		501	PTF ALL
Outboard Ground Spoiler Actuator Test TASK 27-62-94-720-801			501	PTF ALL
OUTBOARD GROUND SPOILER ACTUATOR - INSPECTION/CHECK	27-62-94		601	PTF ALL
Outboard Ground Spoiler Actuator Inspection TASK 27-62-94-200-801			601	PTF ALL
LEADING EDGE FLAP AND SLAT CONTROL SYSTEM - MAINTENANCE PRACTICES	27-81-00		201	PTF ALL
Leading Edge Flaps and Slats - Deactivation TASK 27-81-00-040-801			201	PTF ALL
Leading Edge Flaps and Slats - Activation TASK 27-81-00-440-801			224	PTF ALL
Leading Edge Flap and Slat System Operation With Primary Control TASK 27-81-00-860-801			229	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Leading Edge Flap and Slat System Operation With Alternate Control TASK 27-81-00-860-802			231	PTF ALL
Leading Edge Flaps and Slats Extension TASK 27-81-00-860-803			232	PTF ALL
Leading Edge Flaps and Slats Retraction TASK 27-81-00-860-804			233	PTF ALL
Leading Edge Flaps and Slats Manual Extension TASK 27-81-00-800-801			234	PTF ALL
Leading Edge Flaps and Slats Manual Retraction TASK 27-81-00-800-802			240	PTF ALL
Leading Edge Flap and Slat Locks Installation TASK 27-81-00-480-801			247	PTF ALL
Leading Edge Flap and Slat Locks Removal TASK 27-81-00-080-801			248	PTF ALL
LEADING EDGE FLAP AND SLAT - ADJUSTMENT/TEST	27-81-00		501	PTF ALL
Leading Edge Flaps Adjustment TASK 27-81-00-820-801			501	PTF ALL
Leading Edge Slat Actuator Adjustment TASK 27-81-00-820-802			511	PTF ALL
Leading Edge Slats Adjustment TASK 27-81-00-820-803			519	PTF ALL
Leading Edge Flap and Slat Operational Test in Primary Mode TASK 27-81-00-700-801			548	PTF ALL
Leading Edge Flap and Slat Timing Test in Primary Mode TASK 27-81-00-700-802			559	PTF ALL
Leading Edge Flap and Slat System Standby Actuation Operational Test TASK 27-81-00-710-801			561	PTF ALL
Leading Edge Flap and Slat Timing Test in the Standby Mode TASK 27-81-00-700-803			563	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Leading Edge Cruise Depressurization Test TASK 27-81-00-730-801			566	PTF ALL
Leading Edge Uncommanded Motion Protection Test TASK 27-81-00-700-804			568	PTF ALL
Standby Hydraulic System Leading Edge Uncommanded Motion Test TASK 27-81-00-710-802			570	PTF ALL
LEADING EDGE FLAP AND SLAT CONTROL SYSTEM - INSPECTION/CHECK	27-81-00		601	PTF ALL
Leading Edge Flap and Slat Actuators and Actuation Mechanism - Inspection TASK 27-81-00-210-801			601	PTF ALL
Leading Edge Flap and Slat System Internal Hydraulic Leakage Check TASK 27-81-00-790-801			608	PTF ALL
LEADING EDGE FLAPS - REMOVAL/INSTALLATION	27-81-11		401	PTF ALL
Leading Edge Flap Removal TASK 27-81-11-000-801			401	PTF ALL
Leading Edge Flap Installation TASK 27-81-11-400-801			413	PTF ALL
LEADING EDGE FLAP - INSPECTION/CHECK	27-81-11		601	PTF ALL
Leading Edge Flap Inspection TASK 27-81-11-200-801			601	PTF ALL
LEADING EDGE SLAT - REMOVAL/INSTALLATION	27-81-21		401	PTF ALL
Leading Edge Slat Removal TASK 27-81-21-000-801			401	PTF ALL
Leading Edge Slat Installation TASK 27-81-21-400-801			416	PTF ALL
Leading Edge Slat Main Track Removal TASK 27-81-21-000-802			425	PTF ALL
Leading Edge Slat Main Track Installation TASK 27-81-21-400-802			427	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LEADING EDGE FLAP ACTUATOR - REMOVAL/INSTALLATION	27-81-31		401	PTF ALL
Leading Edge Flap Actuator Removal TASK 27-81-31-000-801			401	PTF ALL
Leading Edge Flap Actuator Installation TASK 27-81-31-400-801			408	PTF ALL
LEADING EDGE FLAP ACTUATOR - INSPECTION/CHECK	27-81-31		601	PTF ALL
Leading Edge Flap Actuator Inspection TASK 27-81-31-200-801			601	PTF ALL
LEADING EDGE SLAT ACTUATOR - REMOVAL/INSTALLATION	27-81-41		401	PTF ALL
Leading Edge Slat Actuator Removal TASK 27-81-41-000-801			401	PTF ALL
Leading Edge Slat Actuator Installation TASK 27-81-41-400-801			415	PTF ALL
LEADING EDGE SLAT ACTUATOR - INSPECTION/CHECK	27-81-41		601	PTF ALL
Leading Edge Slat Actuator Lock Indicator Visual Check TASK 27-81-41-200-801			601	PTF ALL
LEADING EDGE UNCOMMANDED MOTION (UCM) VALVE AND LEADING EDGE CRUISE DEPRESSURIZATION VALVE - REMOVAL/INSTALLATION	27-81-52		401	PTF ALL
LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Removal TASK 27-81-52-000-801			401	PTF ALL
LE Uncommanded Motion (UCM) or LE Cruise Depressurization Valve Installation TASK 27-81-52-400-801			405	PTF ALL
LEADING EDGE CRUISE DEPRESSURIZATION VALVE - INSPECTION/CHECK	27-81-52		601	PTF ALL
Leading Edge Cruise Depressurization Valve Hydraulic Internal Leakage Check TASK 27-81-52-790-801			601	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LEADING EDGE FLAP AND SLAT HYDRAULIC FUSE - REMOVAL/INSTALLATION	27-81-61		401	PTF ALL
Leading Edge Flap and Slat Hydraulic Fuse Removal (LE System B or Standby System Fuses) TASK 27-81-61-000-801			401	PTF ALL
Leading Edge Flap and Slat Hydraulic Fuse Installation (LE System B or Standby System Fuses) TASK 27-81-61-400-801			405	PTF ALL
Flap and Slat Hydraulic Fuses (System B) Reset TASK 27-81-61-860-801			406	PTF ALL
Flap and Slat Hydraulic Fuse (Standby System) Reset TASK 27-81-61-860-802			407	PTF ALL
LEADING EDGE FLAP FLOW LIMITING VALVE - REMOVAL/INSTALLATION	27-81-71		401	PTF ALL
Leading Edge Flap Flow Limiting Valve Removal TASK 27-81-71-000-801			401	PTF ALL
Leading Edge Flap Flow Limiting Valve Installation TASK 27-81-71-400-801			404	PTF ALL
LEADING EDGE FLAP AND SLAT CONTROL VALVE - REMOVAL/INSTALLATION	27-81-81		401	PTF ALL
Leading Edge Flap and Slat Control Valve Removal TASK 27-81-81-000-801			401	PTF ALL
Leading Edge Flap and Slat Control Valve Installation TASK 27-81-81-400-801			406	PTF ALL
LEADING EDGE FLAP AND SLAT CONTROL VALVE - ADJUSTMENT/TEST	27-81-81		501	PTF ALL
Leading Edge Flap and Slat Control Valve Adjustment TASK 27-81-81-820-801			501	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LEADING EDGE STANDBY SHUTOFF VALVE - MAINTENANCE PRACTICES	27-81-91		201	PTF ALL
Leading Edge Standby Shutoff Valve Removal TASK 27-81-91-000-801			201	PTF ALL
Leading Edge Standby Shutoff Valve Installation TASK 27-81-91-400-801			205	PTF ALL
Standby Shutoff Valve Motor Removal TASK 27-81-91-000-802			206	PTF ALL
Standby Shutoff Valve Motor Installation TASK 27-81-91-400-802			208	PTF ALL
Leading Edge Standby Shutoff Valve Test TASK 27-81-91-720-801			209	PTF ALL
LEADING EDGE AUTOSLAT SYSTEM - ADJUSTMENT/TEST	27-83-00		501	PTF ALL
Autoslat System Indicator Light - Functional Test TASK 27-83-00-720-801			501	PTF ALL
Leading Edge Autoslat System - Functional Test TASK 27-83-00-710-801			505	PTF ALL
AUTOSLAT CONTROL VALVE - REMOVAL/INSTALLATION	27-83-21		401	PTF ALL
Autoslat Control Valve Removal TASK 27-83-21-000-801			401	PTF ALL
Autoslat Control Valve Installation TASK 27-83-21-400-801			404	PTF ALL
AUTOSLAT CONTROL VALVE - INSPECTION/CHECK	27-83-21		601	PTF ALL
Autoslat Valve Internal Hydraulic Leakage Check TASK 27-83-21-790-801			601	PTF ALL
LEADING EDGE FLAP AND SLAT POSITION INDICATING SYSTEM - ADJUSTMENT/TEST	27-88-00		501	PTF ALL
Leading Edge Flap and Slat Position Indicating System - Operational Test TASK 27-88-00-710-801			501	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 27
FLIGHT CONTROLS

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
LEADING EDGE FLAP AND SLAT PROXIMITY SENSOR - REMOVAL/INSTALLATION	27-88-11		401	PTF ALL
Leading Edge Slat Proximity Sensor Removal TASK 27-88-11-000-801			401	PTF ALL
Leading Edge Slat Proximity Sensor Installation TASK 27-88-11-400-801			412	PTF ALL
Leading Edge Flap Proximity Sensor Removal TASK 27-88-11-000-802			424	PTF ALL
Leading Edge Flap Proximity Sensor Installation TASK 27-88-11-400-802			428	PTF ALL
LEADING EDGE FLAP AND SLAT PROXIMITY SENSOR - ADJUSTMENT/TEST	27-88-11		501	PTF ALL
Leading Edge Flap and Slat Proximity Sensor Test TASK 27-88-11-700-801			501	PTF ALL
Leading Edge Flap and Slat Proximity Sensor Test Using a LCR Meter TASK 27-88-11-760-801			508	PTF ALL
Leading Edge Flap and Slat Proximity Sensor Test Using a Breakout Box and LCR Meter TASK 27-88-11-720-801			520	PTF ALL
P5-12 OVERHEAD LEADING EDGE DEVICE ANNUNCIATOR PANEL - REMOVAL/INSTALLATION	27-88-21		401	PTF ALL
P5-12 Overhead Leading Edge Device Annunciator Panel Removal TASK 27-88-21-000-801			401	PTF ALL
P5-12 Overhead Leading Edge Device Annunciator Panel Installation TASK 27-88-21-400-801			403	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks are used to prepare the airplane for flight with certain systems/components inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. These are the tasks for the components in the flight controls system:

(1) DDG 27-21-01 Rudder Trim Indicator Inoperative - Preparation

(2) DDG 27-21-01 Rudder Trim Indicator Inoperative - Restoration

(3) DDG 27-21-02 STBY RUD ON Light Inoperative - Preparation

(4) DDG 27-21-02 STBY RUD ON Light Inoperative - Restoration

(5) DDG 27-31-01 FEEL DIFF PRESS Light System Inoperative - Preparation

(6) DDG 27-31-01 FEEL DIFF PRESS Light System Inoperative - Restoration

(7) DDG 27-61-01 Flight Spoiler Systems Inoperative - Preparation

(8) DDG 27-61-01 Flight Spoiler Systems Inoperative - Restoration

(9) DDG 27-62-01 Auto Speedbrake System Inoperative - Preparation

(10) DDG 27-62-01 Auto Speedbrake System Inoperative - Restoration

(11) DDG 27-62-02 SPEEDBRAKES EXTENDED Light Inoperative - Preparation

(12) DDG 27-62-02 SPEEDBRAKES EXTENDED Light Inoperative - Restoration

(13) DDG 27-83-02 AUTO SLAT FAIL Light System Inoperative - Preparation

(14) DDG 27-83-02 AUTO SLAT FAIL Light System Inoperative - Restoration

(15) DDG 27-88-01 Leading Edge Flap/Slat Position Light Systems - Leading Edge Indications for Slats 1, 2, 7 and 8 Inoperative - Preparation

(16) DDG 27-88-01 Leading Edge Flap/Slat Position Light Systems - Leading Edge Indications for Slats 1, 2, 7 and 8 Inoperative - Restoration

TASK 27-00-00-040-812

2. DDG 27-21-01 Rudder Trim Indicator Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Rudder Trim Indicator inoperative.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Rudder Trim Indicator Deactivation

SUBTASK 27-00-00-860-063

(1) Do this task to pressurize the rudder hydraulic systems A and B: TASK 27-21-00-800-801.

SUBTASK 27-00-00-720-001

(2) Do these steps to make sure that the rudder trim actuator operates normally:

(a) Set the rudder trim knob to the NOSE LEFT position.

1) Make sure that the captain's left rudder pedal moves forward, and the right rudder pedal moves aft.

(b) Set the rudder trim knob to the NOSE RIGHT position.

1) Make sure that the captain's left rudder pedal moves aft, and the right rudder pedal moves forward.

SUBTASK 27-00-00-860-064

(3) Do this task to return the aircraft to its condition before rudder hydraulic system pressurization: TASK 27-21-00-840-801.

SUBTASK 27-00-00-410-004

(4) Install an INOP placard next to the deactivated Rudder Trim indicator.

——— END OF TASK ———

TASK 27-00-00-440-812

3. DDG 27-21-01 Rudder Trim Indicator Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Rudder Trim Indicator inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Rudder Trim Indicator Restoration

SUBTASK 27-00-00-010-005

(1) Remove the INOP placard next to the Rudder Trim Indicator.

——— END OF TASK ———

TASK 27-00-00-040-813

4. DDG 27-21-02 STBY RUD ON Light Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the STBY RUD ON indicator light inoperative.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

D. STBY RUD ON Light Deactivation

SUBTASK 27-00-00-860-001

WARNING: MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task to pressurize the rudder hydraulic systems A and B: TASK 27-21-00-800-801.

SUBTASK 27-00-00-810-001

(2) Do these steps to make sure that the rudder operates normally and independently on hydraulic system A and B:

(a) Set the FLT CONTROL A switch to ON and the FLT CONTROL B switch to OFF.

(b) While holding the tiller to prevent tire scrub, make sure that the rudder pedal forces are normal by pushing the captain’s left rudder pedal forward until it touches the stop and hold.

1) Have a ground observer make sure that the rudder moves left.

(c) While continuing to hold the tiller, release the left rudder pedal and confirm normal rudder pedal forces by pushing the captain’s right rudder pedal forward until it touches the stop and hold.

1) Have a ground observer make sure that the rudder moves right.

(d) While continuing to hold the tiller, release the captain’s right rudder pedal.

1) Have a ground observer make sure that the rudder moves back to neutral.

(e) Set the FLT CONTROL B switch to ON and the FLT CONTROL A switch to OFF.

(f) While holding the tiller to prevent tire scrub, make sure that the rudder pedal forces are normal by pushing the captain’s left rudder pedal forward until it touches the stop and hold.

1) Have a ground observer make sure that the rudder moves left.

(g) While continuing to hold the tiller, release the left rudder pedal and confirm normal rudder pedal forces by pushing the captain’s right rudder pedal forward until it touches the stop and hold.

1) Have a ground observer make sure that the rudder moves right.

(h) While continuing to hold the tiller, release the captain’s right rudder pedal.

1) Have a ground observer make sure that the rudder moves back to neutral.

SUBTASK 27-00-00-810-002

(3) Do these steps to make sure that the standby hydraulic pump is operating normally:

(a) Set the FLT CONTROL A switch to STBY RUD and the FLT CONTROL B switch to OFF.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) While holding the tiller to prevent tire scrub, make sure that the rudder pedal forces are normal by pushing the captain's left rudder pedal forward until it touches the stop and hold.

1) Have a ground observer make sure that the rudder moves left.

(c) While continuing to hold the tiller, release the left rudder pedal and confirm normal rudder pedal forces by pushing the captain's right rudder pedal forward until it touches the stop and hold.

1) Have a ground observer make sure that the rudder moves right.

(d) While continuing to hold the tiller, release the captain's right rudder pedal.

1) Have a ground observer make sure that the rudder moves back to neutral.

SUBTASK 27-00-00-860-002

(4) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

(a) Make sure that the STBY RUD ON light is off.

SUBTASK 27-00-00-860-066

(5) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-00-00-930-001

(6) Attach an INDICATOR INOP placard to the STBY RUD ON indicator light on the P6-2 panel.

——— END OF TASK ———

TASK 27-00-00-440-813

5. DDG 27-21-02 STBY RUD ON Light Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the STBY RUD ON indicator light inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

C. STBY RUD ON Light Restoration

SUBTASK 27-00-00-860-003

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

SUBTASK 27-00-00-930-002

(2) Remove the INDICATOR INOP placard from the STBY RUD ON indicator on the P6-2 panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-00-00-040-814

6. DDG 27-31-01 FEEL DIFF PRESS Light System Inoperative - Preparation

(Figure 901, Figure 902, Figure 903)

A. General

(1) This task gives the maintenance steps that prepare the airplane for flight with the FEEL DIFF PRESS light system inoperative.

B. References

Reference	Title
27-31-00-700-801	Null Procedure - Mach Trim Actuator (P/B 201)
27-31-00-800-801	Elevator Hydraulic System A and B - Pressurization (P/B 201)
27-31-00-840-801	Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1914	Test Set - Air Data Model FLMTS (Flight Line Maintenance) Part #: ADSE 650 Supplier: 3BSK6 Part #: ADTS2000 Supplier: 52892 Part #: ADTS3350ER Supplier: 52892 Part #: ADTS405MKII Supplier: U0427 Part #: ADTS530 Supplier: U0427 Part #: ADTS552F Supplier: U0427 Part #: ADTS553F Supplier: U0427 Part #: ADTS554F Supplier: U0427 Part #: D60340MK Supplier: K1474 Part #: DPS1000 Supplier: 21844 Part #: MODEL 6300 Supplier: 0RDZ5 Part #: MPS34C Supplier: 48RQ2 Part #: MPS43B Supplier: A0197 Part #: MPS45 Supplier: 48RQ2 Part #: MPS49 Supplier: 48RQ2 Part #: TES9463 Supplier: 88277 Opt Part #: 01-0987-00 Supplier: 41364 Opt Part #: 18910480000 Supplier: 89944 Opt Part #: 18910920000 Supplier: 89944 Opt Part #: ADTS405F Supplier: U0427 Opt Part #: ADTS505 Supplier: U0427 Opt Part #: D60302 Supplier: K1474 Opt Part #: D60340 Supplier: K1474 Opt Part #: D60383 Supplier: K1474 Opt Part #: DPS350 Supplier: 21844 Opt Part #: DPS450 Supplier: 21844 Opt Part #: DPS500 Supplier: 21844 Opt Part #: MPS31C Supplier: 48RQ2 Opt Part #: MPS43 Supplier: A0197

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-1916	Adapter - Pitot Test (Typically included in Air Data Accessory Kit) Part #: CSA75700HT-3 Supplier: 3BSK6 Part #: P75701-4 Supplier: 38002 Part #: P75701M2-3 Supplier: 38002
COM-16577	Plug - Pitot Part #: BACP20AU6D Supplier: 81205
COM-25177	Accessory Kit - Air Data, 737 (Pitot/Static adapters, hoses, etc.) Part #: ADA737-678 Supplier: 38002 Part #: AK737-900 Supplier: 3BSK6 Part #: PA-0120-A Supplier: 3BSK6
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1677	Trammel Bar Assembly - Stabilizer Trim Actuator Part #: F80055-36 Supplier: 81205 Opt Part #: F80055-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205
SPL-1742	Regulator - Air Pressure, Elevator Feel Computer Part #: F72928-109 Supplier: 81205 Part #: F72928-110 Supplier: 81205 Part #: F72928-111 Supplier: 81205 Opt Part #: F72928-58 Supplier: 81205 Opt Part #: F72928-62 Supplier: 81205 Opt Part #: F72928-63 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-1100	Gauge - Push/Pull, Dial, 0-50 Lb Capacity

D. Consumable Materials

Reference	Description	Specification
G00252	Film - Polyethylene Film And Sheeting	ASTM D2103 (Supersedes L-P-512)
G00270	Tape - Scotch Flatback Masking 250	ASTM D6123 (Supersedes A-A-883)
G02219	Tape - Yellow Vinyl Adhesive, Scotch Brand No.471, 1.5 Inches (38.1 mm) Wide	
G02443	Tape - Barricade, Non-Adhesive, Orange, 3 (76 mm) Inches Wide, 4 mils (0.102 mm) Thick, "REMOVE BEFORE FLIGHT"	
G02447	Tag - Red Paper, "PITOT PROBES COVERED" - 3 inches (76.2 mm) Wide, 6 inches (152.4 mm) Long	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
323	Vertical Fin - Front Spar To Rear Spar

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

G. Elevator Feel System Test

SUBTASK 27-00-00-860-005

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task to pressurize the elevator hydraulic systems A and B: Elevator Hydraulic System A and B - Pressurization, TASK 27-31-00-800-801.

SUBTASK 27-00-00-860-006

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

SUBTASK 27-00-00-730-004

(3) Set the Proximity Sensor Electronics Unit (PSEU) to the ground mode.

(a) Set the GND TEST switch, on the P61-4 panel, to the ENABLE position.

(b) Press SYS button on the Multi Function Display (MFD) panel.

(c) Select MAINT -> ONBD MAINT -> OTHER FUNCTIONS -> SPECIAL FUNCTIONS -> ATA System 32: PSEU -> PSEU Air/Ground Override -> CONTINUE -> OPERATE FUNCTION -> CONTINUE -> START FUNCTION.

(d) Select SET SYS 1 & 2 ON GND.

(e) SELECT CONTINUE -> STOP FUNCTION.

SUBTASK 27-00-00-860-007

(4) Use the STAB TRIM switches to set the stabilizer to 4 units of trim.

SUBTASK 27-00-00-860-008

(5) Put the FLT CONTROL A and B switches to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-00-00-860-009

CAUTION

MAKE SURE TO GIVE PROTECTION TO THE WIRE BUNDLES WHEN YOU USE THE TRAMMEL BAR, SPL-1677. THE TRAMMEL BAR HAS SHARP EDGES THAT CAN CAUSE DAMAGE TO THE WIRE BUNDLE INSULATION.

(6) Use the trammel bar, SPL-1677, to make sure that the horizontal stabilizer "B" dimension is 39.17 ±0.01 in. (994.92 ±0.26 mm) (Figure 903).

SUBTASK 27-00-00-860-010

(7) Do this task: Null Procedure - Mach Trim Actuator, TASK 27-31-00-700-801.

SUBTASK 27-00-00-010-001

(8) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-00-00-950-002

(9) Apply a seal to the drain hole in each pitot probe on the vertical fin.

(a) Use vinyl adhesive Scotch Brand No.471 tape, G02219, and polyethylene film, G00252, to cover and seal the drain holes.

SUBTASK 27-00-00-930-018

CAUTION

WHENEVER AN OPENING IS COVERED, MAKE SURE THAT CONDITION IS VISIBLE FROM GROUND. ENGINES SHOULD NOT BE OPERATED WITH COVERS IN PLACE BECAUSE COVERS CAN COME OFF AND DAMAGE ENGINES.

(10) Attach a "PITOT PROBES COVERED" tag, G02447, that has "PITOT PROBES COVERED" printed on it in black letters, to the top of the left control wheel in the flight deck with wire.

WARNING

WHEN THE DRAIN HOLES ARE COVERED, MAKE SURE THAT CONDITION IS VISIBLE FROM THE GROUND. IN ADDITION, ATTACH A TAG TO THE LEFT CONTROL WHEEL IN THE FLIGHT DECK AS A REMINDER THAT THE STATIC PORTS ARE COVERED. FAILURE TO OBSERVE AND REMOVE COVERINGS OVER STATIC PORTS BEFORE FLIGHT CAN CAUSE LARGE ERRORS IN AIRSPEED-SENSING AND ALTITUDE-SENSING SIGNALS. THIS CAN LEAD TO LOSS OF SAFE FLIGHT.

(a) Put one end of a 4 ft (1.22 m) or longer piece of the orange barricade tape, G02443, over the black polyethylene sheet covering the drain holes of the pitot probes and secure the upper edge with a 5 in. (12.70 cm) strip of the adhesive Scotch Brand No.471 tape, G02219.

1) Smooth Scotch Brand No.471 tape, G02219, on the airplane surface to make sure that the bond is satisfactory.

SUBTASK 27-00-00-010-004

(11) Remove the restrictor valves from the systems A and B pitot lines.

NOTE: The systems A and B pitot lines are below the elevator feel computer.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-00-00-860-073

(12) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-00-00-480-002

(13) Install the regulator, SPL-1742, or equivalent air data model test set, COM-1914, with the pitot test adapter, COM-1916.

(a) Do this step for the leak test with regulator, SPL-1742:

NOTE: The DPS 400 Test Set can be used as an alternate tool to the DPS 500.

1) Connect the pressure gage lines from the regulator, SPL-1742, to the drain holes of the pitot lines.

(b) Do these steps for the leak test with the air data model test set, COM-1914:

1) Install the pitot plugs, COM-16577, over the ends of the pitot lines.

2) Install a highly visible streamer on the opening of the aft unpressurized compartment.

a) Make sure that you can easily see the streamer from the ground.

NOTE: This is to remind you to remove the plugs and install the restrictor valves again when the test is completed.

SUBTASK 27-00-00-790-002

(14) Do a check for pressure leakage in the feel pitot system.

CAUTION

MAKE SURE THAT THE PITOT PROBE HAS NO ADDED WEIGHT ON IT FROM THE TEST HOSE. THE WEIGHT OF THE TEST HOSE CAN BEND OR TWIST THE PITOT PROBE OUT OF TOLERANCE.

(a) Connect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the two vertical fin pitot probes.

1) Do this step for the regulator, SPL-1742:

a) Connect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, to the two vertical fin pitot probes.

2) Do this step for the air data model test set, COM-1914:

a) Use the air data accessory kit, COM-25177, to connect the air data model test set, COM-1914, to the pitot test adapter, COM-1916.

NOTE: The air data accessory kit, COM-25177, contains the hoses to connect to the air data model test set, COM-1914, and the pitot test adapter, COM-1916.

CAUTION

DO NOT INCREASE THE PRESSURE TO MORE THAN 5 PSIG (437 KCAS) DURING THE TEST. IF THE PRESSURE IS MORE THAN 5 PSIG, IT CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(b) Operate the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to apply pressure 0.9 psig (6.21 kPa) - 0.97 psig (6.69 kPa) to simulate 194 to 201 kcas.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Set the shutoff valve in the air data model test set, COM-1914, or regulator, SPL-1742 to the OFF position.

(d) Make sure that the pressure does not decrease more than 0.2 psig (1.4 kPa) (8 kcas) during a 2-minute time.

NOTE: The low pressure pressurization steps above make sure that there is no system leak before you apply full pressure in the following steps.

1) If it is necessary, repair the pitot system or pressurization equipment leak.

(e) Operate the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to pressurize the pitot system to 4.7 psig (32.4 kPa) - 4.9 psig (33.8 kPa).

(f) Set the shutoff valve in the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to the OFF position.

(g) Make sure that the pressure does not decrease more than 0.3 psig (2.1 kPa) during a 2-minute time.

(h) Operate the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to put the system back to ambient pressure (0 psig (0.0 kPa)).

SUBTASK 27-00-00-720-002

(15) Test the control column force, use one of the two methods.

SUBTASK 27-00-00-740-007

(16) Method 1 (recommended): test the control column force, use the MFD FLIGHT CONTROL display.

(a) Set up the MAX Display System (MDS).

(b) Select MAINT -> MAINT DATA PGS -> 27 FLIGHT CONTROL DISPLAY (Figure 902).

(c) Make sure that the FLT CONTROL A and B switches are in the ON position.

CAUTION

DO NOT INCREASE THE PRESSURE TO MORE THAN 5 PSIG (437 KCAS) DURING THE TEST. IF THE PRESSURE IS MORE THAN 5 PSIG, IT CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(d) Use the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to pressurize the pitot system to 2.47 ±0.05 psi (17.03 ±0.34 kPa).

(e) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

1) Attach DO NOT OPERATE tags, STD-858, on the control columns.

(f) Record right elevator position that is displayed on the screen.

NOTE: This value shows the neutral position of the right elevator.

1) Make sure that the right elevator position is in the range of 4.0 ± 0.3 degree.

(g) Calculate the trailing edge up test target position value, as follows:

1) Take away 2.0 degrees from the value of the right elevator neutral position.

NOTE: This value shows the trailing edge up test target position.

(h) Put FLT CONTROL B switch to the OFF position and leave FLT CONTROL A switch in the ON position.

(i) Remove the DO NOT OPERATE tags, from the control columns.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Move the captain's control column slowly aft with a push/pull gauge (0-50 lbs), STD-1100, until the MFD display right elevator value will be equal to the trailing edge up test target position value.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.

a) If a column overshoot occurs, return the column to neutral and repeat the step.

2) Make sure that the column is not held in place when a force is read.

a) Read the column force at the moment when the column reaches the correct position.

3) Record the control column force.

4) Make sure that the force to hold the control column is at 27.5 lbf (122.3 N) - 47.5 lbf (211.3 N).

(k) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

(l) Calculate the trailing edge down test target position value, as follows:

1) Add 2.0 degrees to the value of the right elevator neutral position.

NOTE: This value shows the trailing edge down test target position.

(m) Move the captain's control column slowly forward with a push/pull gauge (0-50 lbs), STD-1100, until the MFD display right elevator value will be equal to the trailing edge down test target position value.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.

a) If a column overshoot occurs, return the column to neutral and repeat the step.

2) Make sure that the column is not held in place when a force is read.

a) Read the column force at the moment when the column reaches the correct position.

3) Record the control column force.

4) Make sure that the force to hold the control column at 29 lbf (129 N) - 52 lbf (231 N).

(n) Put FLT CONTROL A switch to the OFF position and FLT CONTROL B switch to the ON position.

(o) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

(p) Move the captain's control column slowly aft with a push/pull gauge (0-50 lbs), STD-1100, until the MFD display right elevator value will be equal to the trailing edge up test target position value.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.

a) If a column overshoot occurs, return the column to neutral and repeat the step.

2) Make sure that the column is not held in place when a force is read.

a) Read the column force at the moment when the column reaches the correct position.

3) Record the control column force.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Make sure that the force to hold the control column is at 27.5 lbf (122.3 N) - 47.5 lbf (211.3 N).

(q) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

(r) Move the captain's control column slowly forward with a push/pull gauge (0-50 lbs), STD-1100, until the MFD display right elevator value will be equal to the trailing edge down test target position value.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.

a) If a column overshoot occurs, return the column to neutral and repeat the step.

2) Make sure that the column is not held in place when a force is read.

a) Read the column force at the moment when the column reaches the correct position.

3) Record the control column force.

4) Make sure that the force to hold the control column at 29 lbf (129 N) - 52 lbf (231 N).

SUBTASK 27-00-00-730-001

(17) Method 2 (optional): test the control column force, use masking tape on the tail cone.

(a) Make sure that the FLT CONTROL A and B switches are in the ON position.

CAUTION

DO NOT INCREASE THE PRESSURE TO MORE THAN 5 PSIG (437 KCAS) DURING THE TEST. IF THE PRESSURE IS MORE THAN 5 PSIG, IT CAN CAUSE DAMAGE TO THE ELEVATOR FEEL COMPUTER.

(b) Operate the regulator, SPL-1742, or equivalent air data model test set, COM-1914, to pressurize the pitot systems to 2.47 ±0.05 psi (17.03 ±0.34 kPa).

(c) Install a piece of the Scotch Flatback Masking Tape 250, G00270, on the tail cone along the direction of the right elevator trailing edge travel.

NOTE: Elevator travel measuring tool can be used instead of tape method.

(d) Attach the adapter, SPL-1674, to the captain's control column.

(e) Move the captain's control column lightly to put the elevator system in the neutral position.

1) Attach DO NOT OPERATE tags, STD-858, on the control columns.

(f) Make a mark of the neutral position of the elevator trailing edge on Scotch Flatback Masking Tape 250, G00270 (Figure 901).

1) Use this mark of the neutral position as a reference.

(g) Remove the DO NOT OPERATE tags, from the control columns.

(h) Measure and mark 1.20 in. (30.48 mm) above the neutral mark.

NOTE: This mark will be the trailing edge up test target position.

(i) Measure and mark 1.20 in. (30.48 mm) below the neutral mark.

NOTE: This mark will be used as the trailing edge down test target position.

(j) Put FLT CONTROL B switch to the OFF position and leave FLT CONTROL A switch in the ON position.

(k) Move the captain's control column aft with a push/pull gauge (0-50 lbs), STD-1100, to set the elevator trailing edge to 1.20 ±0.05 in. (30.48 ±1.27 mm) up from the neutral position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.
   a) If a column overshoot occurs, return the column to neutral and repeat the step.
2) Make sure that the column is not held in place when a force is read.
   a) Read the column force at the moment when the column reaches the correct position.
3) Make sure that the force to hold the control column at this position is 27.5 lbf (122.3 N) - 47.5 lbf (211.3 N).

(l) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

(m) Move the captain's control column forward with a push/pull gauge (0-50 lbs), STD-1100, to set the elevator trailing edge to 1.20 ±0.05 in. (30.48 ±1.27 mm) down from the neutral position.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.
   a) If a column overshoot occurs, return the column to neutral and repeat the step.
2) Make sure that the column is not held in place when a force is read.
   a) Read the column force at the moment when the column reaches the correct position.
3) Make sure that the force to hold the control column at this position is 29 lbf (129 N) - 52 lbf (231 N).

(n) Put FLT CONTROL A switch to the OFF position and FLT CONTROL B switch to the ON position.

(o) Move the captain's control column aft with a push/pull gauge (0-50 lbs), STD-1100, to set the elevator trailing edge to 1.20 ±0.05 in. (30.48 ±1.27 mm) up from the neutral position.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.
   a) If a column overshoot occurs, return the column to neutral and repeat the step.
2) Make sure that the column is not held in place when a force is read.
   a) Read the column force at the moment when the column reaches the correct position.
3) Make sure that the force to hold the control column at this position is 27.5 lbf (122.3 N) - 47.5 lbf (211.3 N).

(p) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

(q) Move the captain's control column forward with a push/pull gauge (0-50 lbs), STD-1100, to set the elevator trailing edge to 1.20 ±0.05 in. (30.48 ±1.27 mm) down from the neutral position.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.
   a) If a column overshoot occurs, return the column to neutral and repeat the step.
2) Make sure that the column is not held in place when a force is read.
   a) Read the column force at the moment when the column reaches the correct position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Make sure that the force to hold the control column at this position is 29 lbf (129 N) - 52 lbf (231 N).

SUBTASK 27-00-00-730-003

(18) Methods 1 and 2: do the feel limit functionality test, as follows:

(a) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

(b) Make sure that the pitot pressure is in range 2.42 psig (16.69 kPa) - 2.52 psig (17.37 kPa).

(c) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

1) Wait at least 15 seconds.

(d) Move the captain’s control column slowly aft to the trailing edge up test target position.

NOTE: The value of the trailing edge up test target position depends on the method you use to test control column force.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.

a) If a column overshoot occurs, return the column to neutral and repeat the step.

2) Make sure that the column is not held in place when a force is read.

a) Read the column force at the moment when the column reaches the correct position.

3) Record the control column force.

(e) Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

(f) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

1) Wait at least 20 seconds.

2) Ignore the FEEL DIFF PRESS light, on the P5-3 panel.

(g) Move the captain’s control column slowly aft to the trailing edge up test target position.

NOTE: The value of the trailing edge up test target position depends on the method you used to test control column force.

1) Do not allow the column to move beyond the target elevator positions and back in the direction of neutral before the force is recorded.

a) If a column overshoot occurs, return the column to neutral and repeat the step.

2) Make sure that the column is not held in place when a force is read.

a) Read the column force at the moment when the column reaches the correct position.

3) Record the control column force.

(h) Calculate the difference of forces when the circuit breaker is opened and closed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the difference in forces at least 20 lbf (89 N).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-00-00-840-002

(1) Move the captain's control column back to neutral position.

SUBTASK 27-00-00-840-001

(2) Decrease the pitot pressure to pitot tubes to 0 psig (0 kPa).

SUBTASK 27-00-00-080-001

(3) Remove the adapter, SPL-1674, protractor, SPL-1680, and any other tools from the captain's control column.

SUBTASK 27-00-00-080-002

(4) Remove any Scotch Flatback Masking Tape 250, G00270, from the aircraft.

SUBTASK 27-00-00-080-003

(5) Remove vinyl adhesive Scotch Brand No.471 tape, G02219, and black polyethylene sheet polyethylene film, G00252, used to cover and seal the drain holes, and the 4 ft (1.22 m) or longer piece of the orange barricade tape, G02443, from the drain hole in each vertical fin pitot probe.

SUBTASK 27-00-00-080-004

(6) Disconnect the regulator, SPL-1742, or equivalent air data model test set, COM-1914, from the two vertical fin pitot probes.

(a) Do this step for the regulator, SPL-1742:

1) Disconnect the pressure lines from the regulator, SPL-1742, with the pitot test adapter, COM-1916, from the two vertical fin pitot probes.

(b) Do this step for the air data model test set, COM-1914:

1) Disconnect the air data model test set, COM-1914, with the pitot test adapter, COM-1916 from the two vertical fin pitot probes.

SUBTASK 27-00-00-080-005

(7) Do these steps:

(a) For the regulator, SPL-1742, do this step:

1) Disconnect the pressure gage lines of the regulator, SPL-1742, from the drain holes of the pitot lines.

(b) For the air data model test set, COM-1914, do these steps:

1) Remove the pitot plugs, COM-16577, that were installed over the ends of the pitot lines.

2) Remove the highly visible streamer on the opening of the aft unpressurized compartment.

SUBTASK 27-00-00-860-074

(8) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	4	C00236	HEATERS ELEV PITOT LEFT
D	4	C00237	HEATERS ELEV PITOT RIGHT

SUBTASK 27-00-00-420-001

(9) Install the restrictor valves to the systems A and B drain lines.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-00-00-860-011

(10) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	10	C01676	ELEVATOR FEEL LIMIT

SUBTASK 27-00-00-410-001

(11) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-00-00-860-012

(12) Do this task: Put the Elevator Hydraulic Systems A and B Back to the Condition Before the Pressurization, TASK 27-31-00-840-801.

SUBTASK 27-00-00-730-005

(13) Set the Proximity Switch Electronic Unit (PSEU) to the normal mode (TASK 32-09-00-860-802).

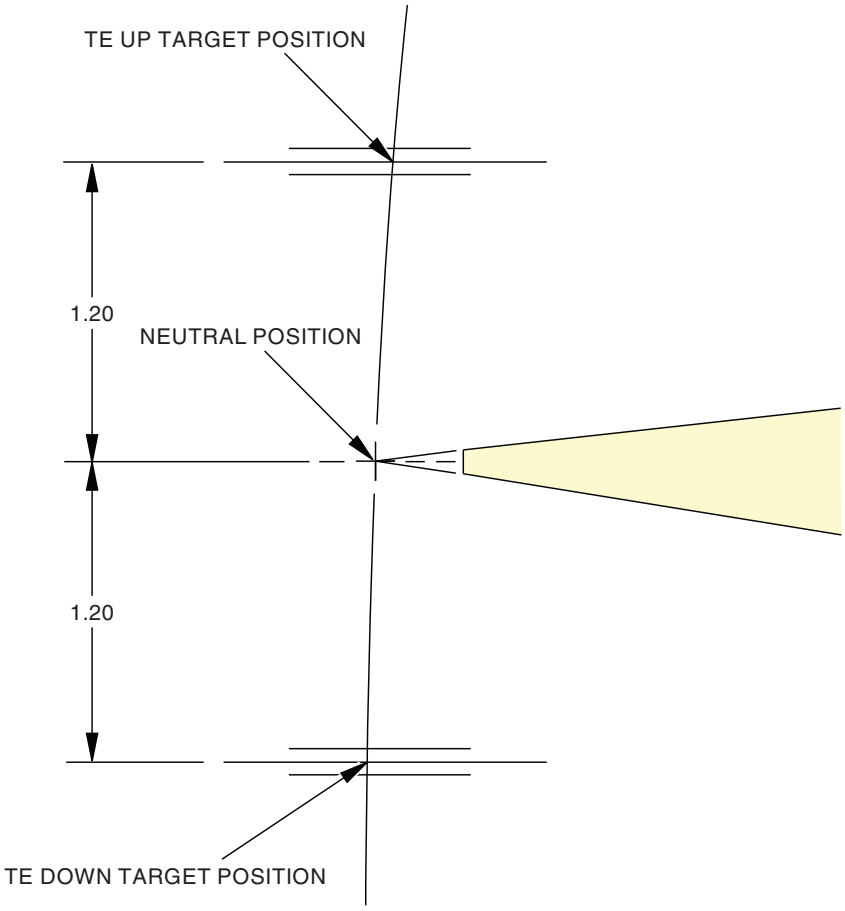
I. FEEL DIFF PRESS Light Deactivation

SUBTASK 27-00-00-930-003

(1) Install a LIGHT INOP placard to the FEEL DIFF PRESS light.

——— END OF TASK ———

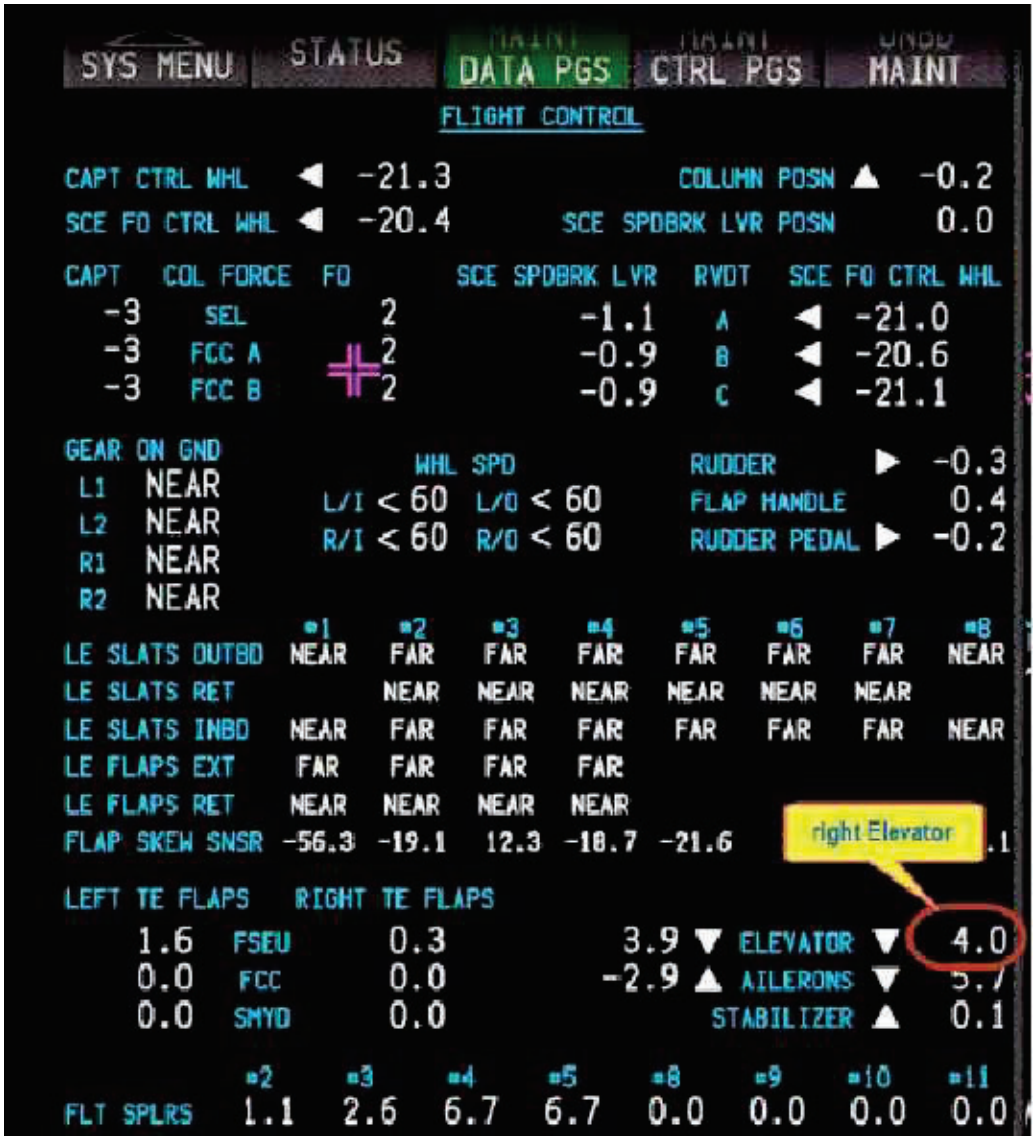
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2949741 S0000726386_V1

Control Column Force Test (Tape method)
Figure 901/27-00-00-990-803

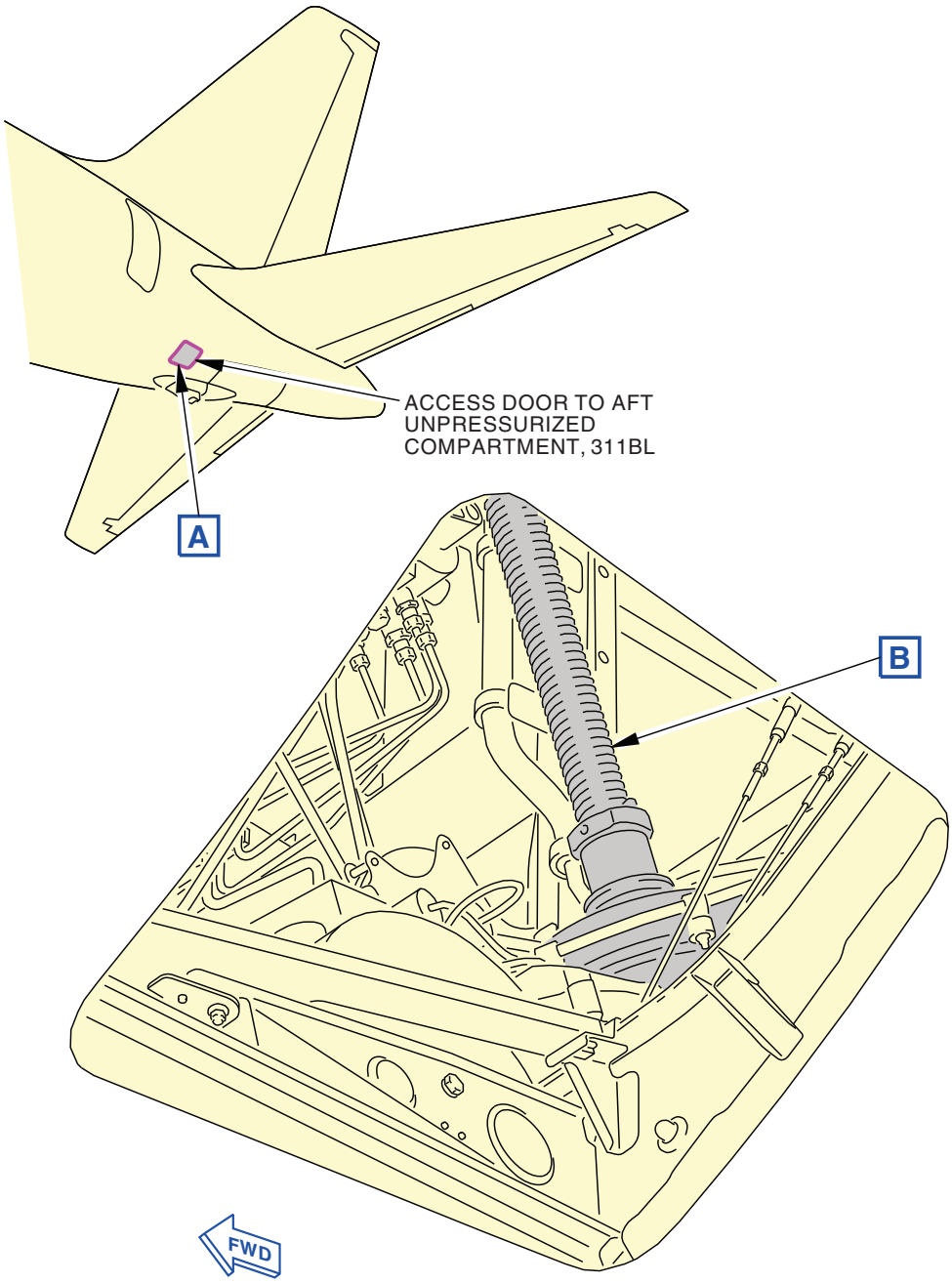
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2949746 S0000726387_V1

Control Column Force Test (MFD method)
Figure 902/27-00-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details



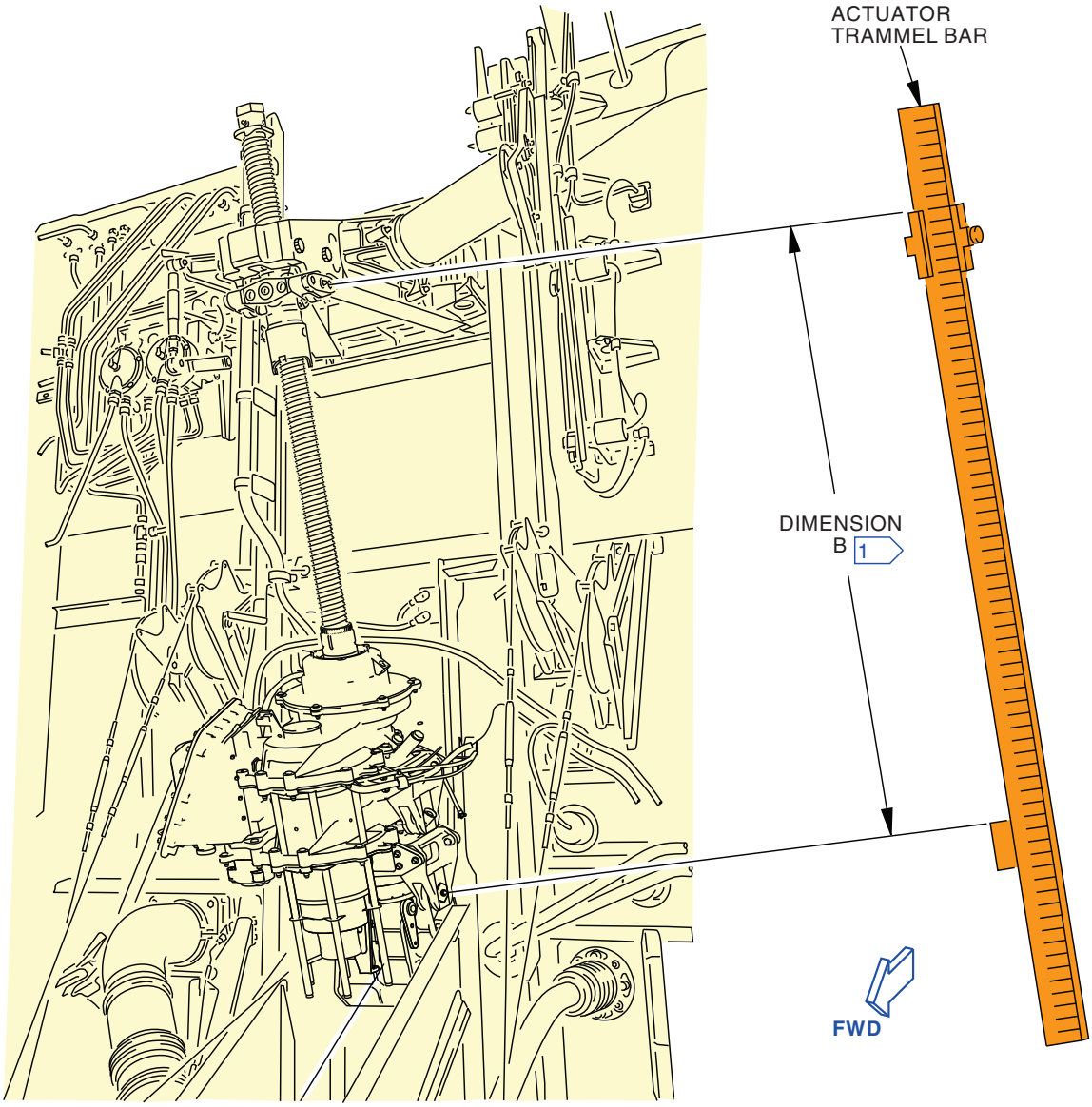
**VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL**

A

2412457 S00061530716_V1

**Stabilizer Trim Jackscrew Setting
Figure 903/27-00-00-990-805 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STABILIZER TRIM ACTUATOR

B

NOTE:

THE STABILIZER TRIM JACKSCREW IS SHOWN WITH THE STABILIZER LEADING EDGE AT ZERO DEGREES.

1 THE DIMENSION B IS MEASURED BETWEEN THE CENTER OF THE UPPER AND LOWER GIMBAL PINS, (CENTER OF GREASE FITTINGS).

2412458 S00061530717_V2

Stabilizer Trim Jackscrew Setting
Figure 903/27-00-00-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-00-00-440-814

7. DDG 27-31-01 FEEL DIFF PRESS Light System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the FEEL DIFF PRESS light system inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. FEEL DIFF PRESS Light Restoration

SUBTASK 27-00-00-930-004

(1) Remove the LIGHT INOP placard from the FEEL DIFF PRESS light.

——— END OF TASK ———

TASK 27-00-00-040-811

8. DDG 27-61-01 Flight Spoiler Systems Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with a single flight spoiler pair inoperative.

NOTE: The flight spoilers include spoiler pairs 2/11, 3/10, 4/9 and 5/8. Flight spoiler pair 5/8 must stay serviceable.

B. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-45-730-805	Flight Spoiler Test (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the procedure

SUBTASK 27-00-00-860-051

(1) Extend trailing edge flaps to the full extend position to gain access to the PCUs (TASK 27-51-00-860-803).

SUBTASK 27-00-00-860-052

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-00-00-860-053

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-00-00-860-067

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

E. Procedure

SUBTASK 27-00-00-820-004

(1) Operate the manual release cam on the bottom of the PCU and manually move the spoiler panel up as needed to gain access to the PCU connector.

SUBTASK 27-00-00-020-002

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. SUDDEN MOVEMENT CAN OCCUR WHEN YOU DISCONNECT THE OPPOSITE SPOILERS PCU-ELECTRICAL-CONNECTOR. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Disconnect, cap and stow the electrical connector from the PCU.

SUBTASK 27-00-00-820-003

(3) Manually move the spoiler panel to the stowed position.

SUBTASK 27-00-00-020-003

(4) Do the steps again for the opposite spoiler.

NOTE: Spoilers pairs are 2/11, 3/10 and 4/9.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-00-00-860-054

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-00-00-860-068

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-00-00-860-055

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-00-00-860-056

(4) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

G. Spoiler Operational Check

SUBTASK 27-00-00-710-007

(1) Make sure the deactivated spoiler panels remain in the stowed position.

(a) Turn the control wheel full counterclockwise. Make sure that the serviceable left wing flight spoilers extend.

(b) Make sure that the inoperative left wing spoiler panel stays down.

(c) Center the control wheel. Make sure that the serviceable left wing flight spoilers retract.

(d) Turn the control wheel full clockwise. Make sure that the serviceable right wing flight spoilers extend.

(e) Make sure that the inoperative right wing spoiler panel stays down.

(f) Center the control wheel. Make sure that the serviceable right wing spoilers retract.

SUBTASK 27-00-00-700-004

(2) Use the Onboard Maintenance Function (OMF) to perform the ground test for the flight spoilers. Do this task: Flight Spoiler Test, TASK 27-61-45-730-805.

NOTE: This will extinguish the SPOILERS light on the P5-3 panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-00-00-810-010

(3) If necessary, extinguish the MAINT light on the P5-1 panel. Do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

TASK 27-00-00-440-811

9. DDG 27-61-01 Flight Spoiler Systems Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with a single flight spoiler pair inoperative.

NOTE: The flight spoilers include spoiler pairs 2/11, 3/10, 4/9 and 5/8.

B. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-61-45-730-805	Flight Spoiler Test (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)

C. Location Zones

Zone	Area
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11

D. Prepare for the procedure

SUBTASK 27-00-00-860-057

(1) Extend trailing edge flaps to the full extend position to gain access to the PCUs (TASK 27-51-00-860-803).

SUBTASK 27-00-00-860-058

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-00-00-860-059

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-00-00-860-069

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

E. Procedure

SUBTASK 27-00-00-820-005

(1) Operate the manual release cam on the bottom of the PCU and manually move the spoiler panel up as needed to gain access to the PCU connector.

SUBTASK 27-00-00-910-001

(2) Before connecting the electrical connector, do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

SUBTASK 27-00-00-020-004

(3) Connect the electrical connector for the PCU.

SUBTASK 27-00-00-820-006

(4) Manually move the spoiler panel to the stowed position.

SUBTASK 27-00-00-020-005

(5) Do the steps again for the opposite spoiler.

NOTE: Spoilers pairs are 2/11, 3/10 and 4/9.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-00-00-860-060

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-00-00-860-070

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	11	C04040	GROUND SPOILER CONTROL
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER

SUBTASK 27-00-00-860-061

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-00-00-860-062

(4) Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

G. Spoiler Operational Check

SUBTASK 27-00-00-700-003

(1) Use the Onboard Maintenance Function (OMF) to perform the ground test for the flight spoilers. Do this task: Flight Spoiler Test, TASK 27-61-45-730-805.

NOTE: The deactivated spoiler pair will be restored after this test is performed.

H. MAINT Light Status Message

SUBTASK 27-00-00-750-001

(1) Uncheck the applicable MAINT light Status Message(s) on the MAINT LIGHT menu (TASK 31-65-00-750-801).

END OF TASK

TASK 27-00-00-040-815

10. DDG 27-62-01 Auto Speed Brake System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Auto Speed Brake System inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Auto Speed Brake System Deactivation

SUBTASK 27-00-00-710-005

(1) Make sure that the manual speed brake operates correctly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-00-00-010-002

(2) Open this circuit breaker and install safety lock:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 27-00-00-930-009

(3) Attach an AUTO SPOILER INOP placard to the speed brake control lever.

——— END OF TASK ———

TASK 27-00-00-440-815

11. DDG 27-62-01 Auto Speed Brake System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the Auto Speed Brake System inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Auto Speed Brake System Restoration

SUBTASK 27-00-00-860-021

(1) Remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE

SUBTASK 27-00-00-930-011

(2) Remove the AUTO SPOILER INOP placard from the speed brake control lever.

——— END OF TASK ———

TASK 27-00-00-040-816

12. DDG 27-62-02 SPEEDBRAKES EXTENDED Light Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the SPEEDBRAKES EXTENDED indicator light inoperative.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
27-62-00-800-801	Apply Pressure to the Speed Brake Hydraulic Systems A and B (P/B 201)
27-62-00-840-801	Shut Down the Speed Brake Hydraulic Systems A and B (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. SPEEDBRAKES EXTENDED Indicator Light Deactivation

SUBTASK 27-00-00-860-022

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task to pressurize the speed brake hydraulic systems A and B: TASK 27-62-00-800-801.

SUBTASK 27-00-00-040-002

(2) Do these steps to test the speedbrake actuator:

(a) Make sure that the wheel chocks are installed (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

(b) Put the speedbrake lever in the DOWN position.

NOTE: The spoilers will move down during this step.

(c) Make sure that the flight and ground spoilers are all down.

(d) Put the speedbrake lever in the FLIGHT DETENT.

NOTE: The spoilers will move up during this step.

(e) Make sure that the flight and ground spoilers deploy.

(f) Put the speedbrake lever in the DOWN position.

NOTE: The spoilers will move down during this step.

(g) Make sure that the flight and ground spoilers are all down.

(h) Make sure that the stabilizer trim is in the green band.

(i) Put the flap lever to a takeoff position.

(j) Release the parking brake.

WARNING MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE SERIOUS INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(k) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

(l) With engines not running, move both thrust levers beyond 60 degrees.

(m) Make sure that the Takeoff Configuration Warning Horn does not sound.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(n) Move the thrust levers to idle.

(o) Set the parking brake.

(p) Put the flaps to the retract position.

(q) If the SPEEDBRAKES EXTENDED light remains illuminated while on the ground, remove the light bulb.

(r) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

SUBTASK 27-00-00-930-012

(3) Attach an INDICATOR INOP placard to the SPEEDBRAKES EXTENDED indicator light.

SUBTASK 27-00-00-860-023

(4) Do this task: Shut Down the Speed Brake Hydraulic Systems A and B, TASK 27-62-00-840-801.

——— END OF TASK ———

TASK 27-00-00-440-816

13. DDG 27-62-02 SPEEDBRAKES EXTENDED Light Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the SPEEDBRAKES EXTENDED indicator light.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

C. SPEEDBRAKES EXTENDED Indicator Light Restoration

SUBTASK 27-00-00-410-003

(1) If necessary, install the SPEEDBRAKES EXTENDED indicator light bulb.

SUBTASK 27-00-00-930-013

(2) Remove the INDICATOR INOP placard from the SPEEDBRAKES EXTENDED indicator.

——— END OF TASK ———

TASK 27-00-00-040-817

14. DDG 27-83-02 AUTO SLAT FAIL Light System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with an AUTO SLAT FAIL light on the forward annunciator panel inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. AUTO SLAT FAIL Light Deactivation

SUBTASK 27-00-00-010-003

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-00-00-740-002

(2) Do the EXISTING FAULTS test on the SMYD BITE:

(a) Push the ON/OFF switch.

NOTE: The display will show “EXISTING FAULTS?”.

(b) Push the YES switch.

1) Make sure that the display shows NO FAULTS.

(c) Push the MENU switch to return to the “EXISTING FAULTS?” display.

(d) Push the ON/OFF switch to turn the SMYD display off.

SUBTASK 27-00-00-410-002

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-00-00-930-014

(4) Attach a LIGHT INOP placard to the AUTO SLAT FAIL light.

——— END OF TASK ———

TASK 27-00-00-440-817

15. DDG 27-83-02 AUTO SLAT FAIL Light System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with an AUTO SLAT FAIL light on the forward annunciator panel inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. AUTO SLAT FAIL Light Restoration

SUBTASK 27-00-00-930-015

(1) Remove the LIGHT INOP placard from the AUTO SLAT FAIL light.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-00-00-040-818

16. DDG 27-88-01 Leading Edge Flap/Slat Position Light System Inoperative - Preparation

(Figure 904)

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Leading Edge (LE) slat indications inoperative.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Leading Edge Flap/Slat Position Light System Deactivation

SUBTASK 27-00-00-860-018

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize the hydraulic system B to 3000 ±100 psi (211 ±7 kg/cm^2) (TASK 29-11-00-860-801).

SUBTASK 27-00-00-710-002

(2) Cycle the flaps.

SUBTASK 27-00-00-210-001

(3) Make sure that the remaining LE devices annunciator panel, P5-12 lights operate normally during flap extension and retraction.

SUBTASK 27-00-00-740-001

(4) Check to see which Trailing Edge (TE) flap lever positions result in an inoperative LE indication.

SUBTASK 27-00-00-710-003

(5) Set the flap lever to the position which makes the faulty indication.

SUBTASK 27-00-00-930-008

(6) Make sure that the LE and TE devices are in the correct position.

SUBTASK 27-00-00-710-004

(7) Do these steps to make sure that both stall warning systems operate normally:

(a) Make sure that hydraulic system B pressure is between 3000 ±100 psi (211 ±7 kg/cm^2).

(b) Push and hold the STALL WARNING TEST NO. 1 and NO. 2 switches on the stall warning test panel, P5 Overhead Panel.

(c) Make sure that the control column stick shakers operate normally.

I (8) If the inoperative slat indication occurs at more than one flap position, repeat the above procedure to make sure both Stall Warning Systems at Flaps 0, Flaps1, and Flaps 10 operate normally.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-00-00-860-065

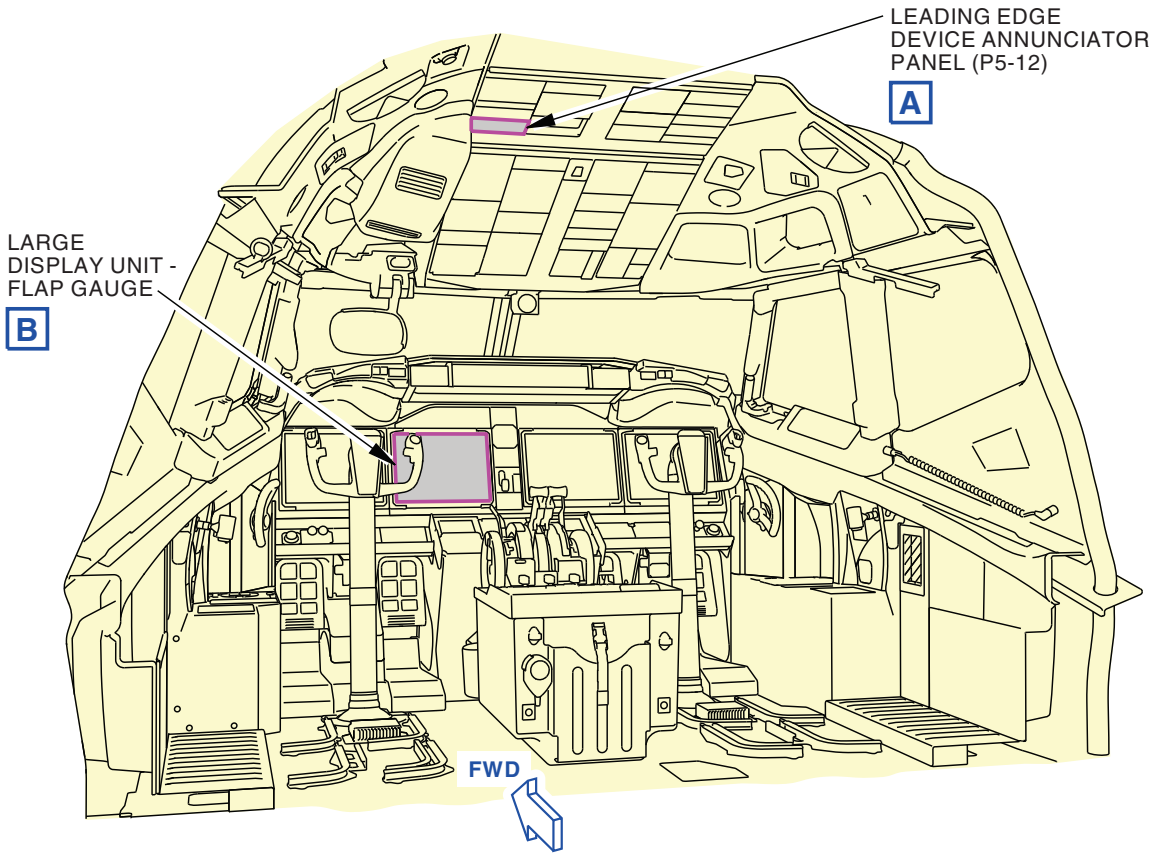
(9) Remove power from hydraulic system B (TASK 29-11-00-860-805).

SUBTASK 27-00-00-410-005

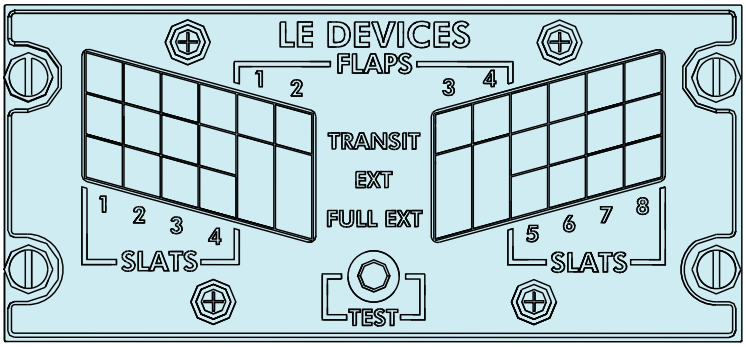
(10) Install a LIGHT INOP placard next to the applicable deactivated LE slat indications.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



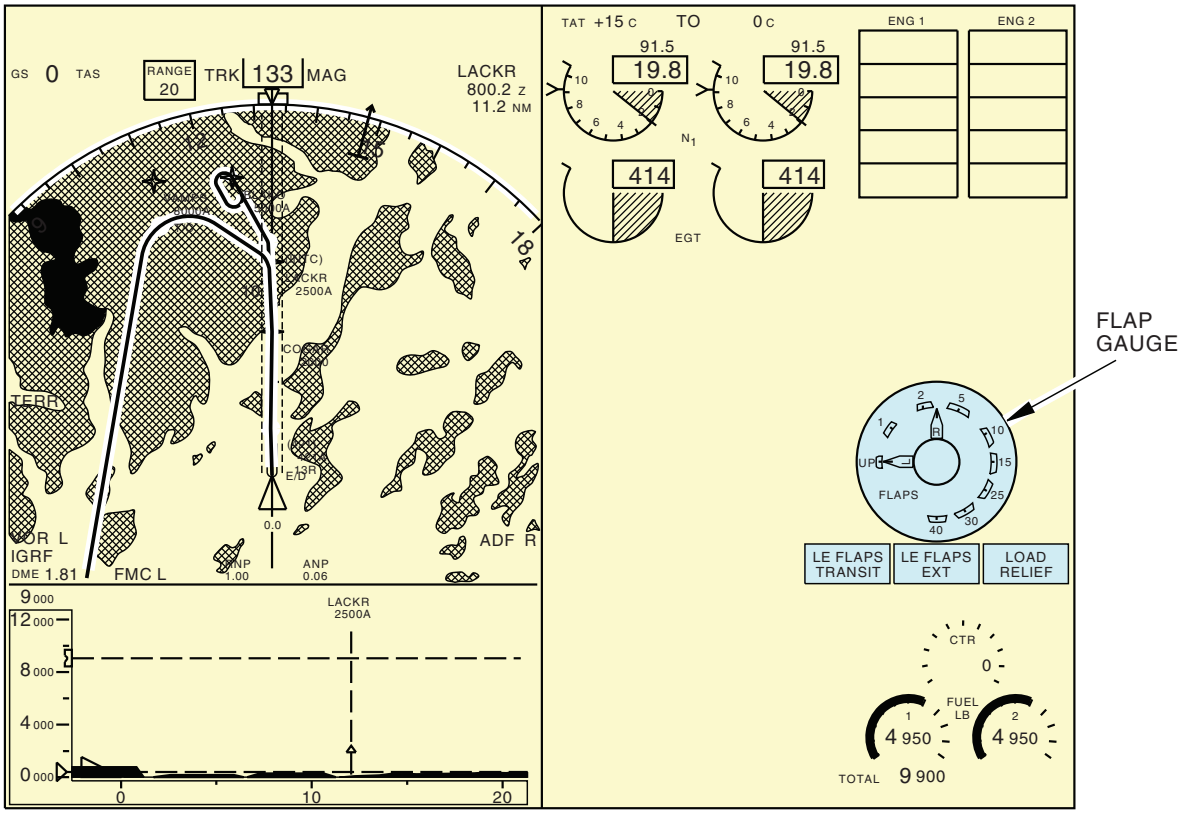
**LEADING EDGE DEVICE
ANNUNCIATOR PANEL (P5-12)**

A

2412194 S00061530172_V1

**Leading Edge Flap and Slat Position Light Location
Figure 904/27-00-00-990-802 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LARGE DISPLAY UNIT - FLAP GAUGE

B

2503148 S0000587533_V1

Leading Edge Flap and Slat Position Light Location
Figure 904/27-00-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-00-00-440-818

17. DDG 27-88-01 Leading Edge Flap/Slat Position Light System Inoperative - Restoration

(Figure 904)

A. General

(1) This task puts the airplane back to its usual condition after operation with the Leading Edge (LE) slat indications inoperative.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Leading Edge Flap/Slat Position Light System Restoration

SUBTASK 27-00-00-930-010

(1) Remove the LIGHT INOP placard from the applicable LE slat indications.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS CABLES - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) The identification and assembly specifications of the control cables.

(2) The rigging of the control cables

(3) The tension measurement of the control cables.

TASK 27-09-14-860-801

2. Control Cables Identification and Assembly Specifications

(Figure 201, Figure 202, Figure 203, Figure 204, Figure 205, Figure 206)

A. References

Reference	Title
20-10-91-000-801	Control Cables Removal (P/B 401)
20-10-91-400-801	Control Cables Installation (P/B 401)
27-11-00-700-804	Control Wheel Centering Check (P/B 501)
27-11-00-700-805	Aileron Control Wheel Force Test (P/B 501)
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-730-801	Aileron Trim Response Test (P/B 501)
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-21-00-700-803	Rudder Centering Test (P/B 501)
27-21-00-700-804	Rudder Pedal Forces Test (P/B 501)
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-820-803	Rudder Control Cables RA and RB Adjustment (P/B 501)
27-31-00-700-805	Elevator Travel Limit - Test (P/B 501)
27-31-00-700-809	Control Column Travel and Centering - Test (P/B 501)
27-31-00-700-810	Control Column Power Force - Test (P/B 501)
27-31-00-710-801	Elevator and Elevator Trim Control System - Operational Test (P/B 501)
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-41-00-700-801	Stabilizer Manual Trim and Trim Indicator - Test (P/B 501)
27-41-00-700-802	Horizontal Stabilizer Trim Control System Friction Test (P/B 501)
27-41-00-700-803	Stabilizer Electric Trim System Test (P/B 501)
27-41-00-700-804	Stabilizer Trim Wheel Free Movement Test (P/B 501)
27-41-00-700-806	Stabilizer Clutch Friction System Test (P/B 501)
27-41-00-820-801	Stabilizer Control Cable and Chain - Adjustment (P/B 501)
27-51-00-720-803	Alternate Flap Control System Functional Test (P/B 501)
27-51-00-730-801	Trailing Edge Flap System Test (P/B 501)
27-51-00-820-801	Trailing Edge Flap Control System Adjustment and Test (P/B 501)
27-51-14-400-801	Inboard Aft Flap Actuating Mechanism Installation (P/B 401)
SOPM 20-50-16	Assembly of Swaged Cables

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Control Cables Identification and Assembly Specifications

SUBTASK 27-09-14-960-001

(1) If it is necessary to replace the cables, do these tasks:

- Control Cables Removal, TASK 20-10-91-000-801
- Control Cables Installation, TASK 20-10-91-400-801.

SUBTASK 27-09-14-700-001

(2) If the aileron control cables were replaced, do these tasks:

- Aileron Cables Adjustment, TASK 27-11-00-820-801
- Aileron Trim Response Test, TASK 27-11-00-730-801
- Aileron Travel Test, TASK 27-11-00-700-810
- Control Wheel Centering Check, TASK 27-11-00-700-804
- Aileron Control Wheel Force Test, TASK 27-11-00-700-805.

SUBTASK 27-09-14-700-002

(3) If the rudder control cables were replaced, do these tasks:

- Rudder Control Cables RA and RB Adjustment, TASK 27-21-00-820-803
- Rudder Travel Test, TASK 27-21-00-700-805
- Rudder Centering Test, TASK 27-21-00-700-803
- Rudder Pedal Forces Test, TASK 27-21-00-700-804.

SUBTASK 27-09-14-700-003

(4) If the elevator control cables were replaced, do these tasks:

- Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment, TASK 27-31-00-820-801
- Elevator and Elevator Trim Control System - Operational Test, TASK 27-31-00-710-801
- Elevator Travel Limit - Test, TASK 27-31-00-700-805
- Control Column Travel and Centering - Test, TASK 27-31-00-700-809
- Control Column Power Force - Test, TASK 27-31-00-700-810.

SUBTASK 27-09-14-700-004

(5) If the trailing edge flap control cables were replaced, do these tasks:

- Trailing Edge Flap Control System Adjustment and Test, TASK 27-51-00-820-801
- Trailing Edge Flap System Test, TASK 27-51-00-730-801
- Alternate Flap Control System Functional Test, TASK 27-51-00-720-803.

SUBTASK 27-09-14-700-005

(6) If the inboard trailing edge flap/aft flap drive cables were replaced, do the cable adjustment steps in this task:

- Inboard Aft Flap Actuating Mechanism Installation, TASK 27-51-14-400-801.

SUBTASK 27-09-14-700-006

(7) If the horizontal stabilizer control cables were replaced, do these tasks:

- Stabilizer Control Cable and Chain - Adjustment, TASK 27-41-00-820-801
- Stabilizer Manual Trim and Trim Indicator - Test, TASK 27-41-00-700-801
- Horizontal Stabilizer Trim Control System Friction Test, TASK 27-41-00-700-802
- Stabilizer Electric Trim System Test, TASK 27-41-00-700-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Stabilizer Trim Wheel Free Movement Test, TASK 27-41-00-700-804
- Stabilizer Clutch Friction System Test, TASK 27-41-00-700-806.

SUBTASK 27-09-14-860-001

(8) Identify and make control cable assemblies for these flight control systems, as necessary (SOPM 20-50-16):

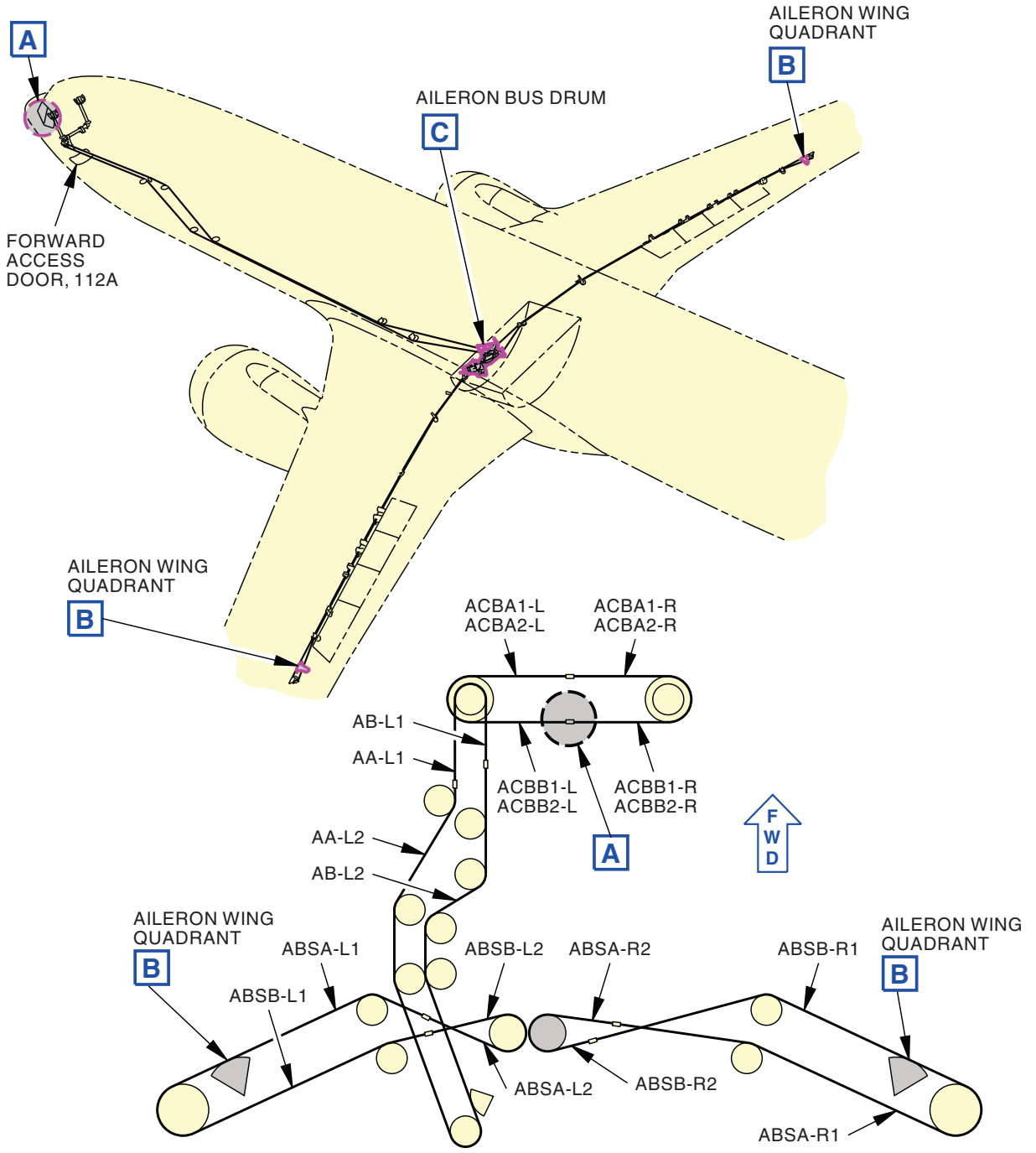
- Figure 201
- Figure 202
- Figure 203
- Figure 204
- Figure 205
- Figure 206.

SUBTASK 27-09-14-860-002

(9) Swage control cables with dies for the next larger size AN standard terminals for all BACT14A terminals.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

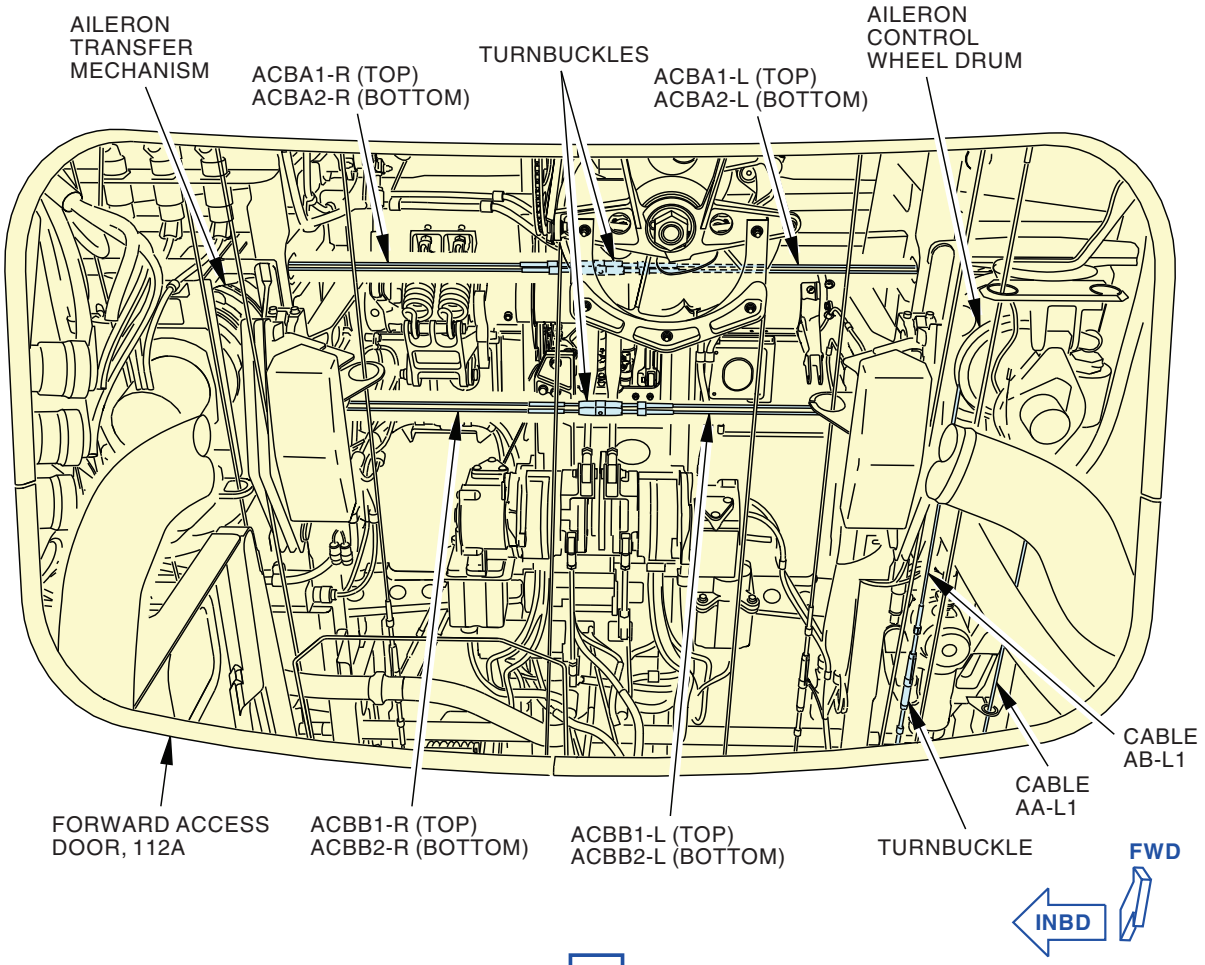


AILERON CONTROL CABLE SCHEMATIC

2410984 S00061526723_V3

Aileron Control Cables
Figure 201/27-09-14-990-801 (Sheet 1 of 6)

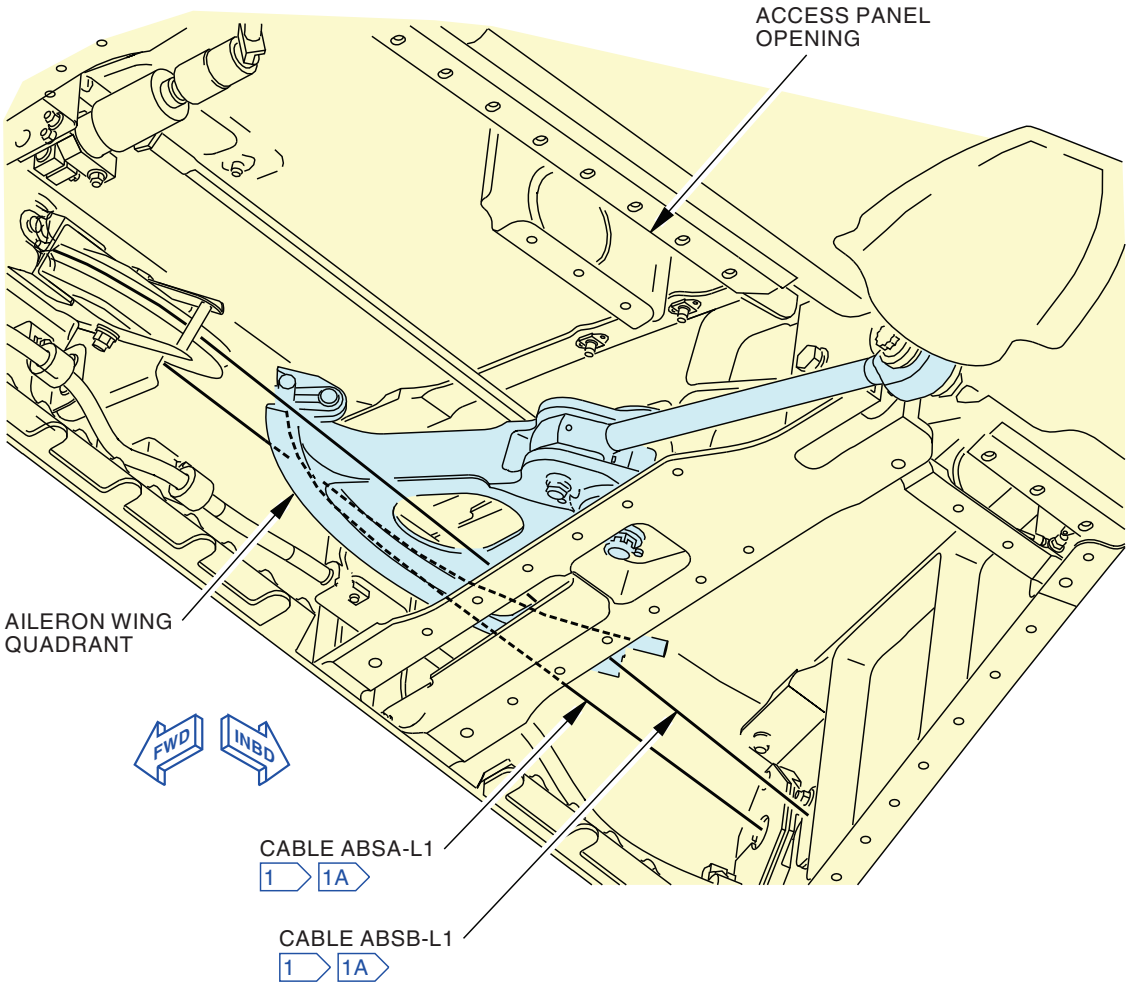
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410985 S00061526724_V2

Aileron Control Cables
Figure 201/27-09-14-990-801 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT AILERON WING QUADRANT
(RIGHT AILERON WING QUADRANT IS EQUIVALENT)

B

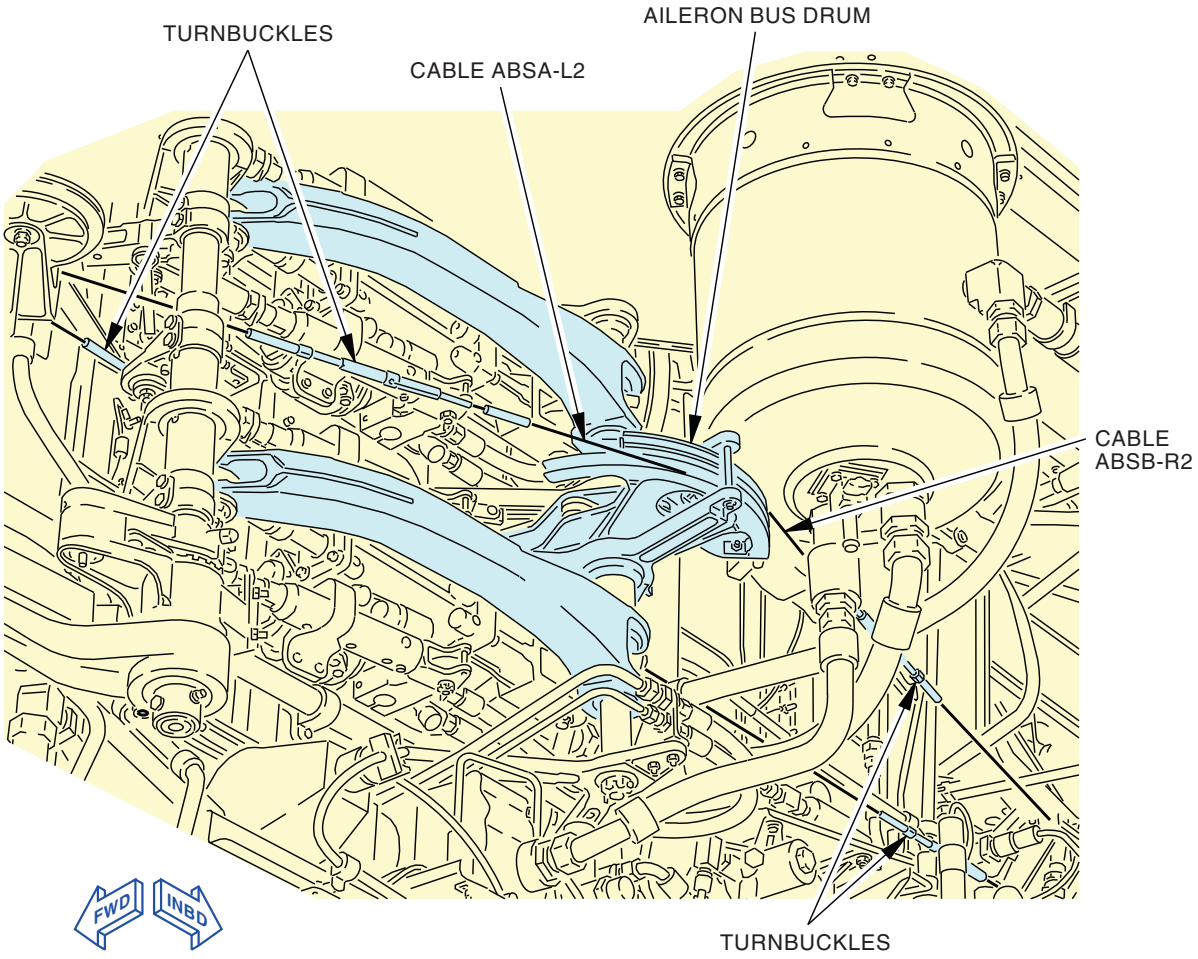
1 FOR THE RIGHT AILERON WING QUADRANT, INSTALL CABLE ABSB-R1 BELOW CABLE ABSA-R1.

1A FIT ABSA-L1 TO THE LOWER GROOVE OF THE QUADRANT
FIT ABSB-L1 TO THE UPPER GROOVE OF THE QUADRANT.

2410986 S00061526725_V2

Aileron Control Cables
Figure 201/27-09-14-990-801 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



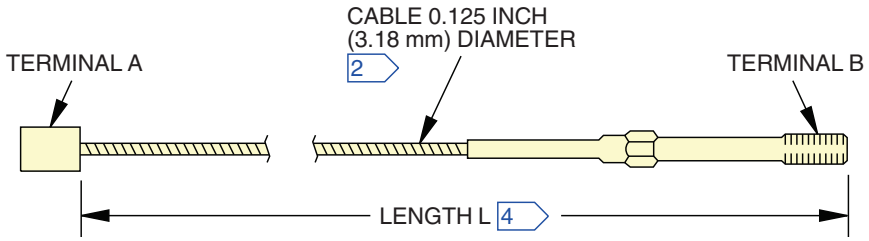
AILERON BUS DRUM
(MAIN LANDING GEAR WHEEL WELL)
(EXAMPLE)

C

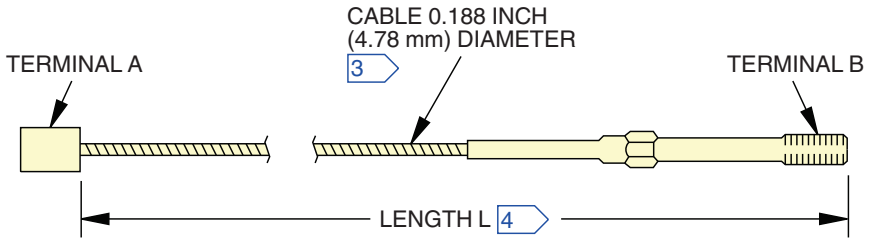
2410987 S00061526726_V4

Aileron Control Cables
Figure 201/27-09-14-990-801 (Sheet 4 of 6)

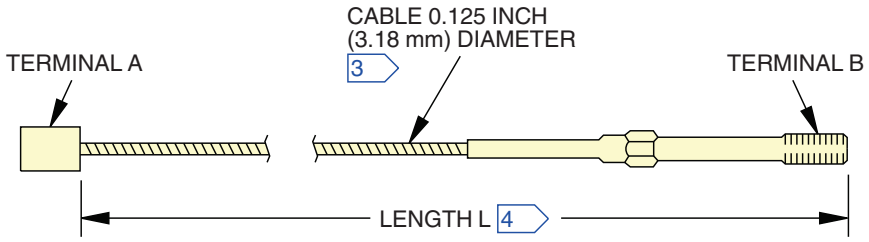
ECCN 9E991 BOEING PROPRIETARY - See title page for details



AA-L1, AA-L2, AB-L1, AB-L2



**ABSA-L1, ABSA-L2, ABSA-R1, ABSA-R2,
ABSB-L1, ABSB-L2, ABSB-R1, ABSB-R2**



**ACBA1-L, ACBA1-R, ACBA2-L, ACBA2-R,
ACBB1-L, ACBB1-R, ACBB2-L, ACBB2-R**

2 CABLE CONSTRUCTION IS CARBON STEEL:
BMS 7-265, TYPE 1, COMPOSITION A (TIN OVER ZINC), 7 X 19.

3 CABLE CONSTRUCTION IS CARBON STEEL: BMS 7-265, COMPOSITION A (ZINC), 7 X 19.

4 MEASURE CABLE WITH A LOAD OF 40 ±3 POUNDS (178 ±13 NEWTONS).

2410988 S00061526727_V3

**Aileron Control Cables
Figure 201/27-09-14-990-801 (Sheet 5 of 6)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABLE NAME	LENGTH L [4] INCHES (MILLIMETERS)	TERMINAL A	TERMINAL B
AA-L1 AA-L2	38.5 ±0.12 (978 ±3) 671.8 ±0.25 (17,064 ±6)	BACT14A4 BACT14A4	MS21260L4RH MS21260L4LH
AB-L1 AB-L2	26.2 ±0.12 (665 ±3) 652.3 ±0.25 (16,568 ±6)	BACT14A4 BACT14A4	MS21260L4LH MS21260L4RH
ABSA-L1 ABSA-L2	439.8 ±0.20 (11,171 ±5) 18.4 ±0.12 (467 ±3)	BACT14A6 BACT14A6	MS21260L6LH MS21260L6RH
ABSA-R1 ABSA-R2	462.9 ±0.20 (11,758 ±5) 35.4 ±0.12 (899 ±3)	BACT14A6 BACT14A6	MS21260L6LH MS21260L6RH
ABSB-L1 ABSB-L2	466.9 ±0.20 (11,859 ±5) 21.7 ±0.12 (551 ±3)	BACT14A6 BACT14A6	MS21260L6RH MS21260L6LH
ABSB-R1 ABSB-R2	449.9 ±0.20 (11,427 ±5) 17.4 ±0.12 (442 ±3)	BACT14A6 BACT14A6	MS21260L6RH MS21260L6LH
ACBA1-L	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4LH
ACBA1-R	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4RH
ACBA2-L	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4LH
ACBA2-R	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4RH
ACBB1-L	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4RH
ACBB1-R	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4LH
ACBB2-L	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4RH
ACBB2-R	32.1 ±0.12 (815 ±3)	BACT14A4	MS21260S4LH

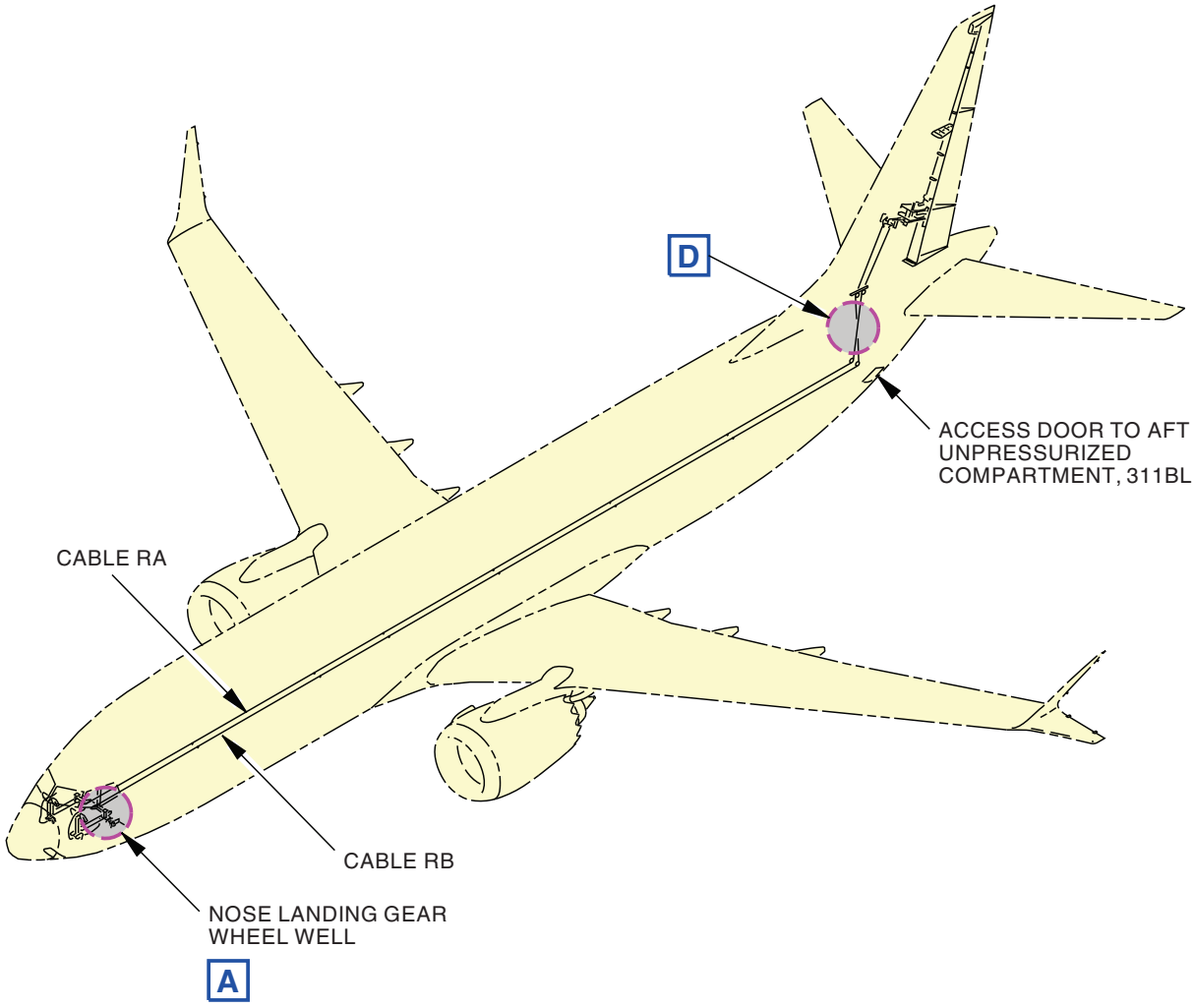
TABLE A

[4] MEASURE CABLE WITH A LOAD OF 40 ±3 POUNDS (178 ±13 NEWTONS).

2410989 S00061526728_V2

Aileron Control Cables
Figure 201/27-09-14-990-801 (Sheet 6 of 6)

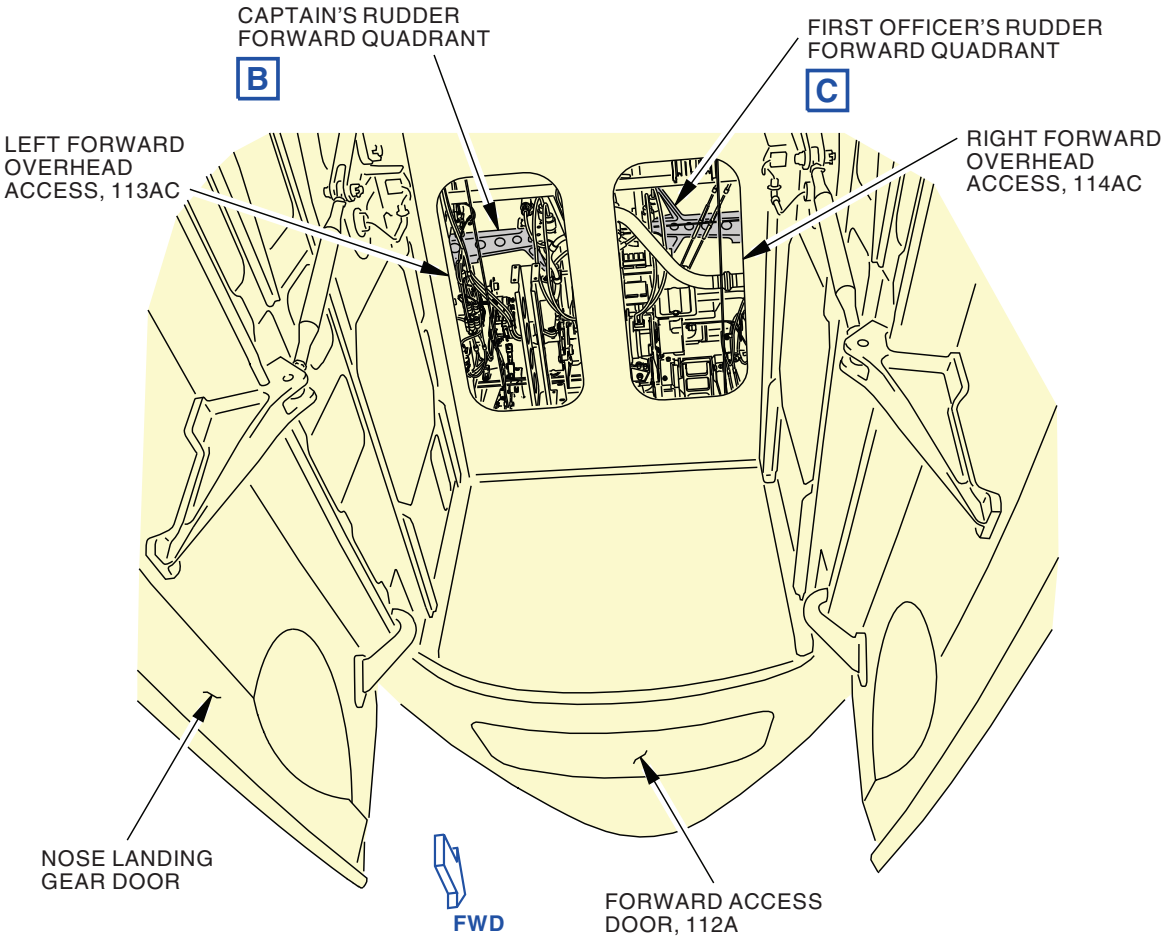
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410990 S00061526729_V2

Rudder Control Cables
Figure 202/27-09-14-990-802 (Sheet 1 of 5)

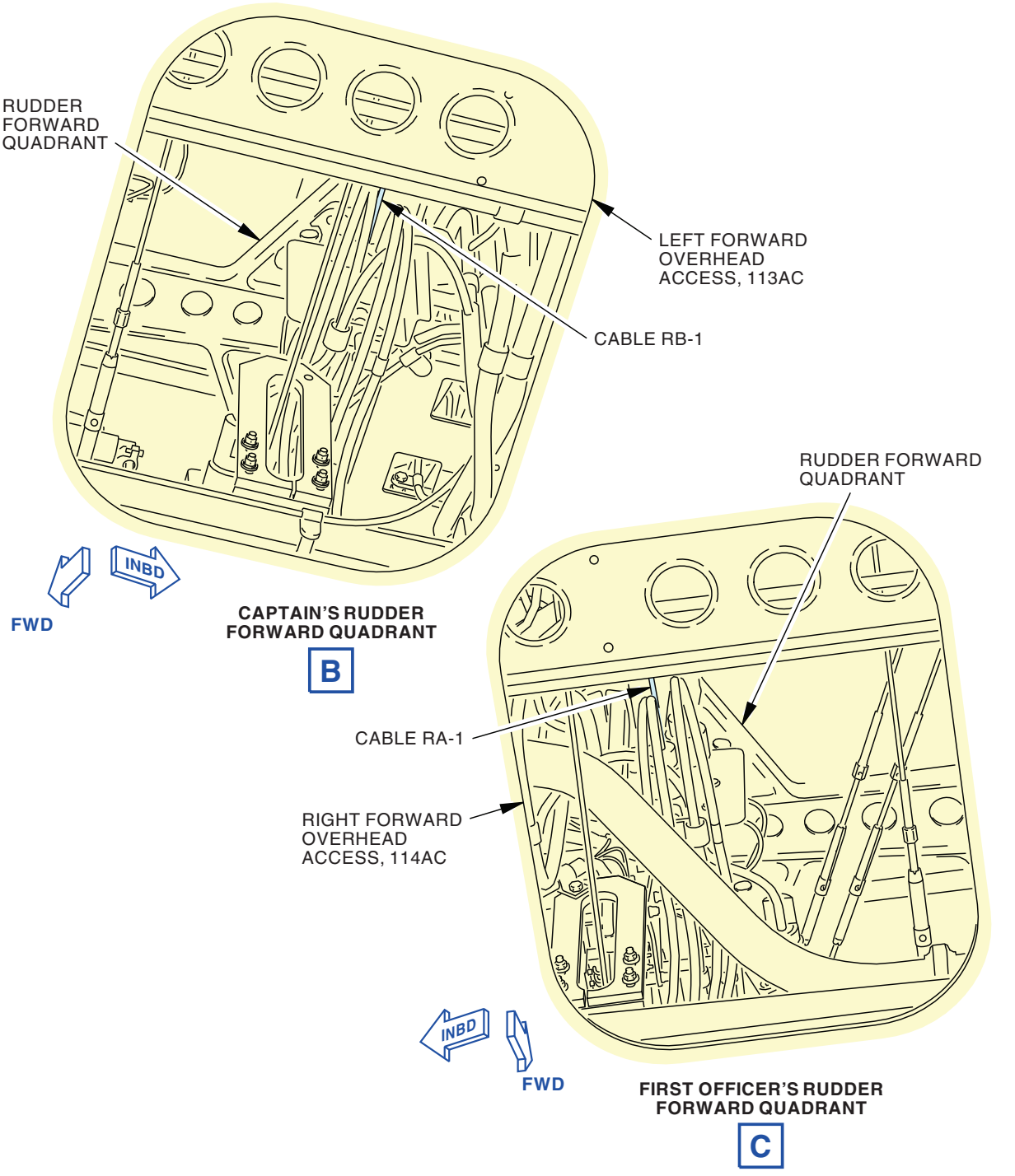
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410991 S00061526730_V2

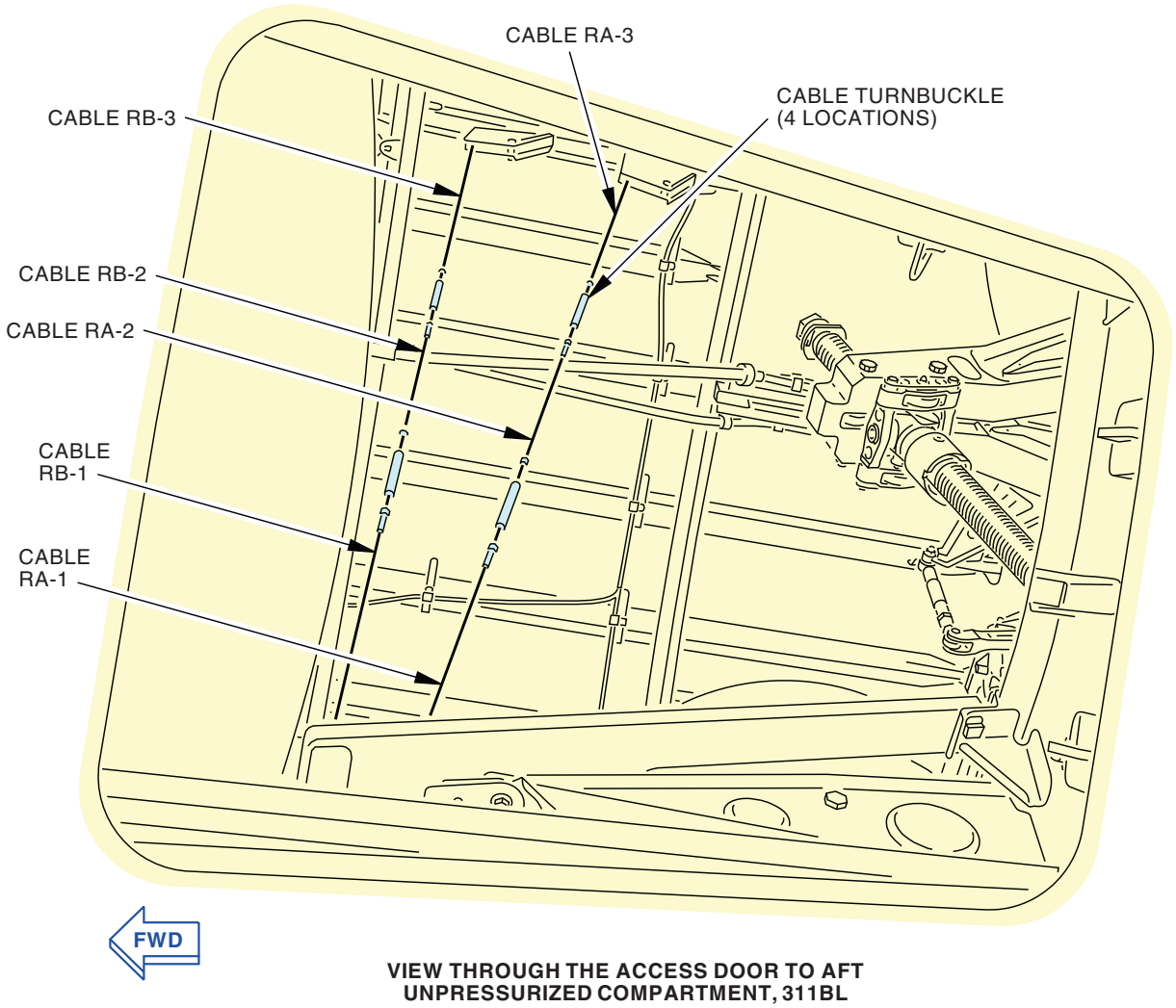
Rudder Control Cables
Figure 202/27-09-14-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Control Cables
Figure 202/27-09-14-990-802 (Sheet 3 of 5)

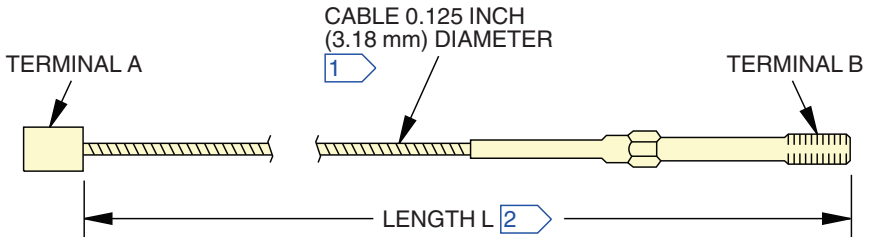
ECCN 9E991 BOEING PROPRIETARY - See title page for details



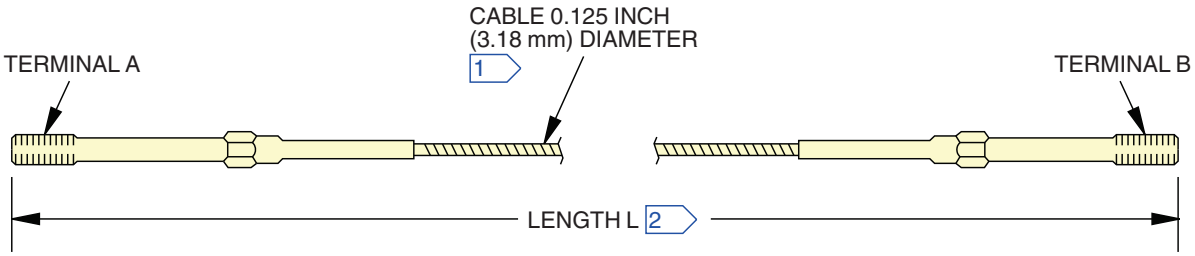
2410993 S00061526732_V1

Rudder Control Cables
Figure 202/27-09-14-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RA-1, RA-3, RB-1, RB-3



RA-2, RB-2

CABLE NAME		LENGTH L 2 INCHES (MILLIMETERS)	TERMINAL A	TERMINAL B
RIGHT SIDE	RA-1	1272.5 ±0.50 (32,321 ±13)	BACT14A4	MS21260L4RH
	RA-2	18.8 ±0.12 (478 ±3)	MS21260L4RH	MS21260L4LH
	RA-3	111.0 ±0.12 (2,819 ±3)	BACT14A4	MS21260L4LH
LEFT SIDE	RB-1	1265.2 ±0.50 (32,136 ±13)	BACT14A4	MS21260L4LH
	RB-2	18.8 ±0.12 (478 ±3)	MS21260L4RH	MS21260L4LH
	RB-3	123.7 ±0.12 (3,142 ±3)	BACT14A4	MS21260L4RH

TABLE A

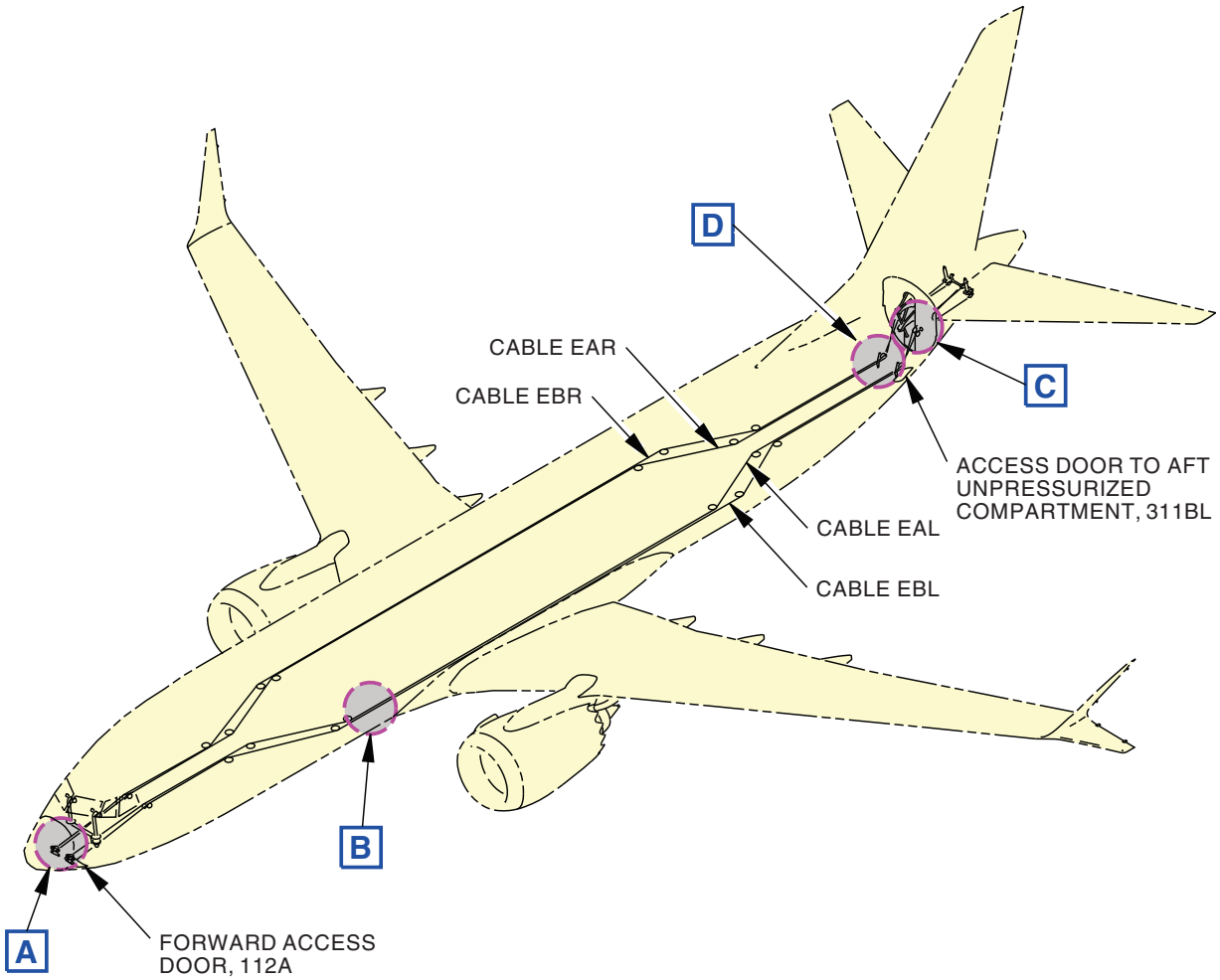
1 CABLE CONSTRUCTION IS CARBON STEEL: BMS 7-265, TYPE 1, COMPOSITION A (TIN OVER ZINC), 7 X 19

2 MEASURE CABLE WITH A LOAD OF 40 ±3 POUNDS (178 ±13 NEWTONS).

2410994 S00061526733_V2

Rudder Control Cables
Figure 202/27-09-14-990-802 (Sheet 5 of 5)

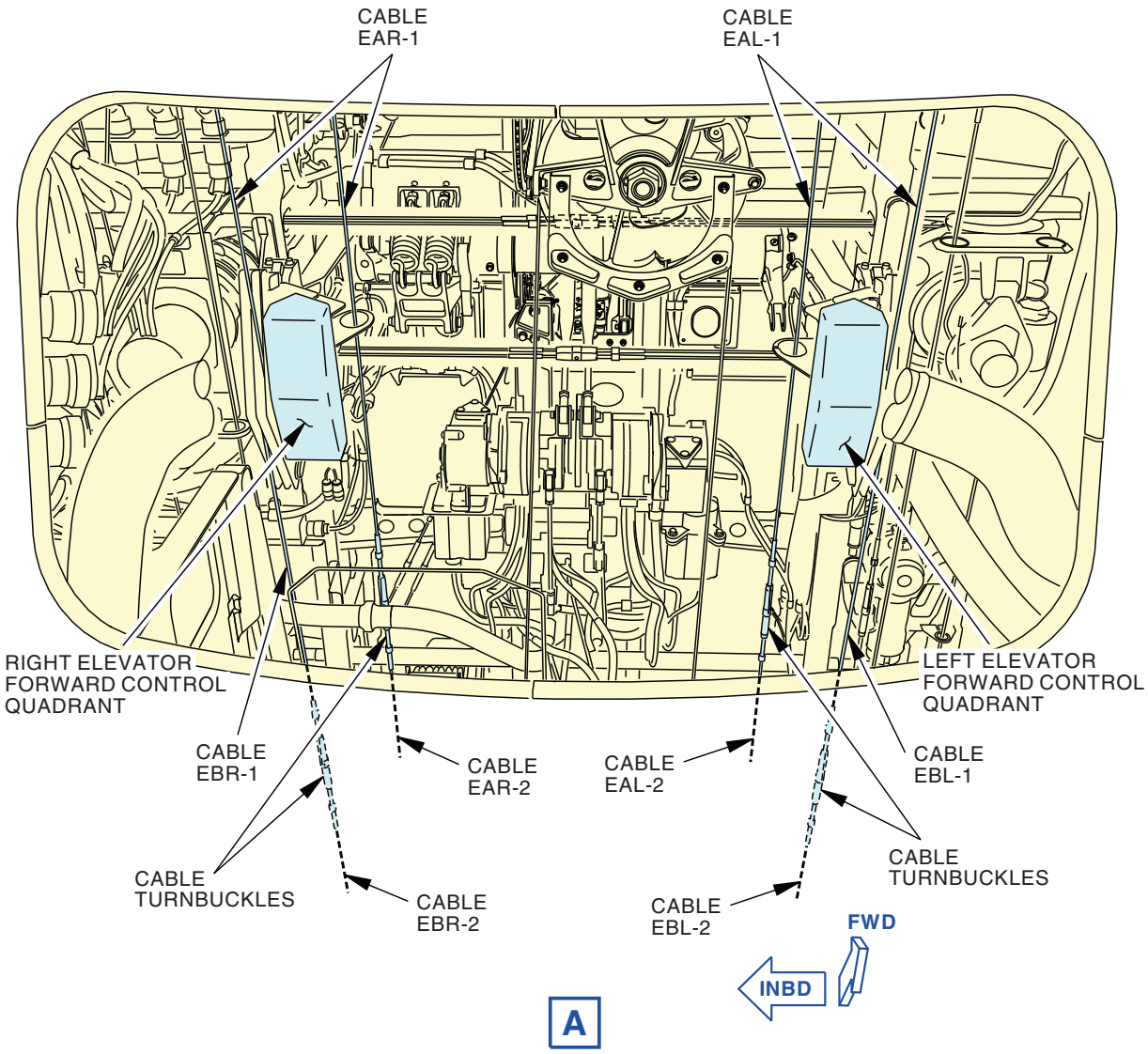
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410995 S00061526734_V1

Elevator Control Cables
Figure 203/27-09-14-990-803 (Sheet 1 of 5)

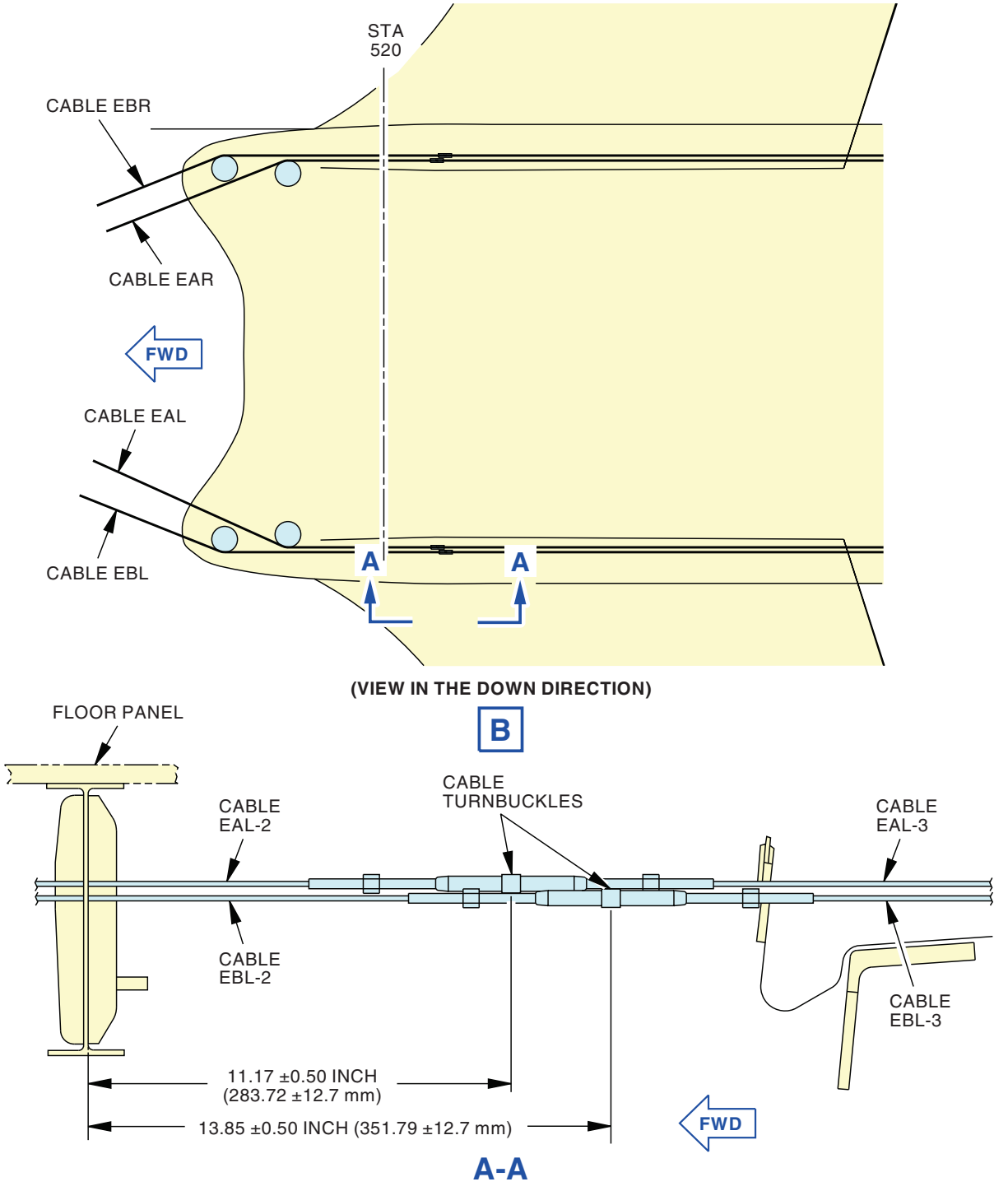
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2410996 S00061526735_V3

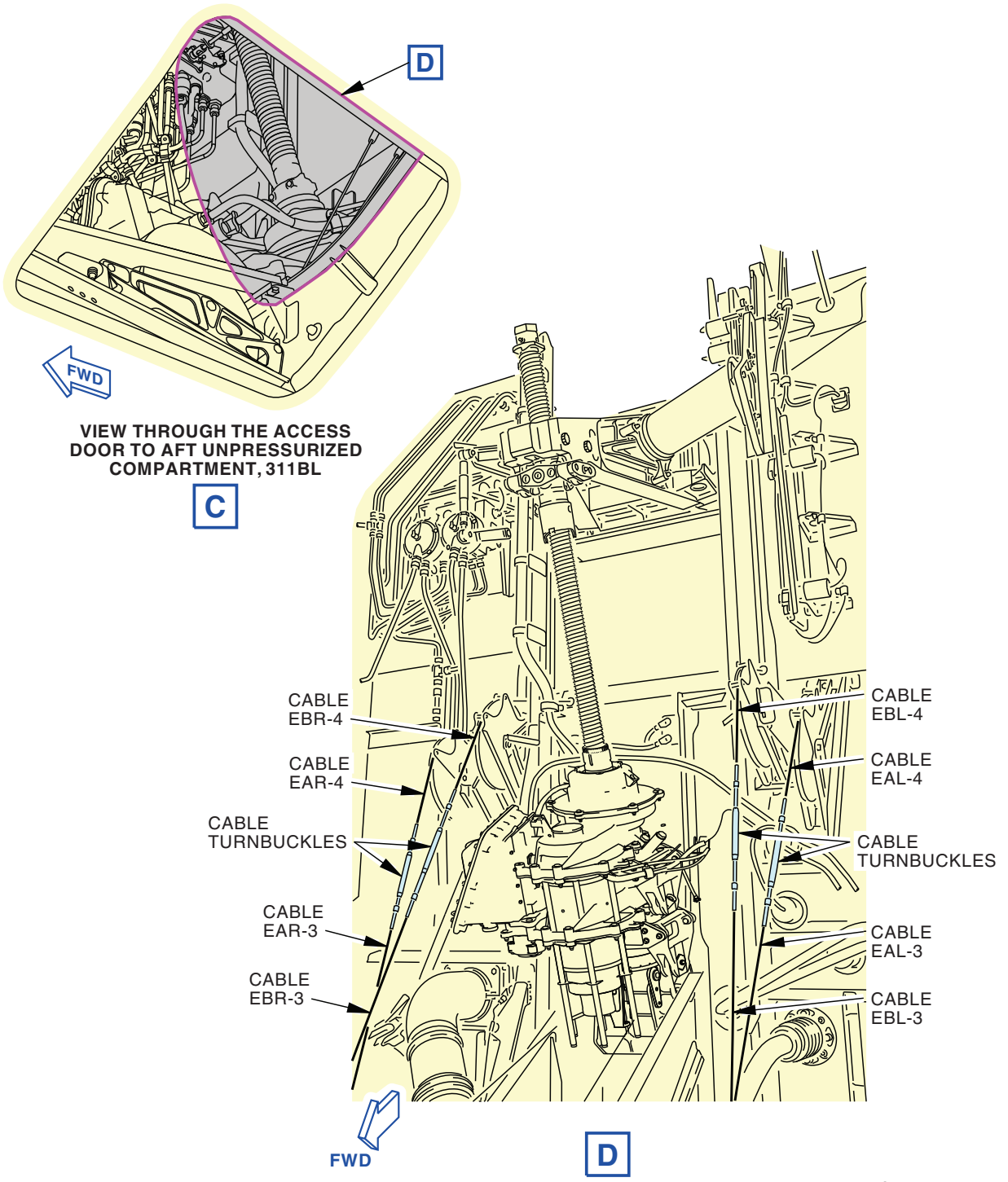
Elevator Control Cables
Figure 203/27-09-14-990-803 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



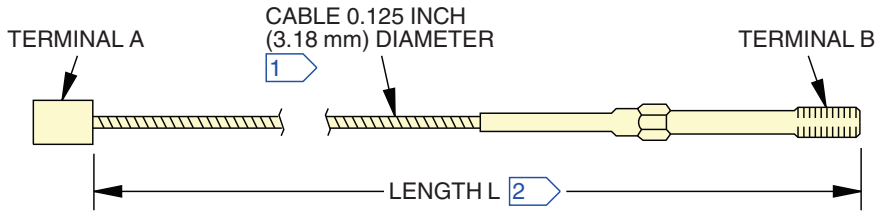
Elevator Control Cables
Figure 203/27-09-14-990-803 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

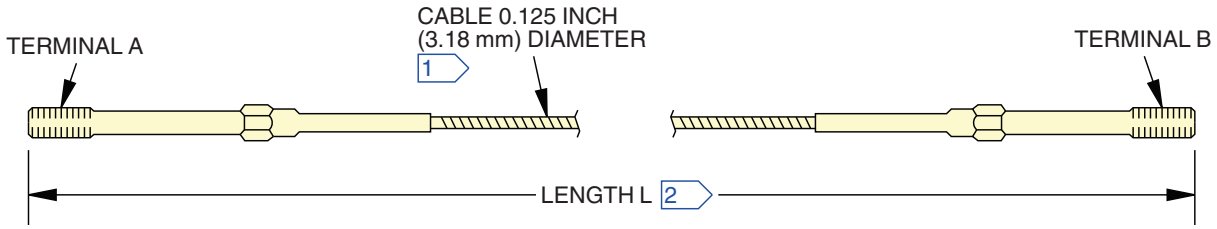


Elevator Control Cables
Figure 203/27-09-14-990-803 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EAL-1, EAL-4, EAR-1, EAR-4, EBL-1, EBL-4, EBR-1, EBR-4



EAL-2, EAL-3, EAR-2, EAR-3, EBL-2, EBL-3, EBR-2, EBR-3

CABLE NAME		LENGTH L 2 INCHES (MILLIMETERS)	TERMINAL A	TERMINAL B
LEFT SIDE	EAL-1	97.6 ±0.12 (2,479 ±3)	BACT14A4	MS21260L4LH
	EAL-2	502.2 ±0.25 (12,756 ±6)	MS21260L4RH	MS21260L4LH
	EAL-3	780.0 ±0.38 (19,812 ±9)	MS21260L4RH	MS21260L4RH
	EAL-4	102.8 ±0.12 (2,611 ±3)	BACT14A4	MS21260L4LH
RIGHT SIDE	EAR-1	97.6 ±0.12 (2,479 ±3)	BACT14A4	MS21260L4LH
	EAR-2	502.2 ±0.25 (12,756 ±6)	MS21260L4RH	MS21260L4LH
	EAR-3	782.1 ±0.38 (19,865 ±9)	MS21260L4RH	MS21260L4RH
	EAR-4	102.8 ±0.12 (2,611 ±3)	BACT14A4	MS21260L4LH
LEFT SIDE	EBL-1	31.5 ±0.12 (800 ±3)	BACT14A4	MS21260L4RH
	EBL-2	495.8 ±0.25 (12,593 ±6)	MS21260L4RH	MS21260L4LH
	EBL-3	778.4 ±0.38 (19,771 ±9)	MS21260L4LH	MS21260L4LH
	EBL-4	102.1 ±0.12 (2,593 ±3)	BACT14A4	MS21260L4RH
RIGHT SIDE	EBR-1	31.5 ±0.12 (800 ±3)	BACT14A4	MS21260L4RH
	EBR-2	495.8 ±0.25 (12,593 ±6)	MS21260L4RH	MS21260L4LH
	EBR-3	781.8 ±0.38 (19,858 ±9)	MS21260L4LH	MS21260L4LH
	EBR-4	102.1 ±0.12 (2,593 ±3)	BACT14A4	MS21260L4RH

TABLE A

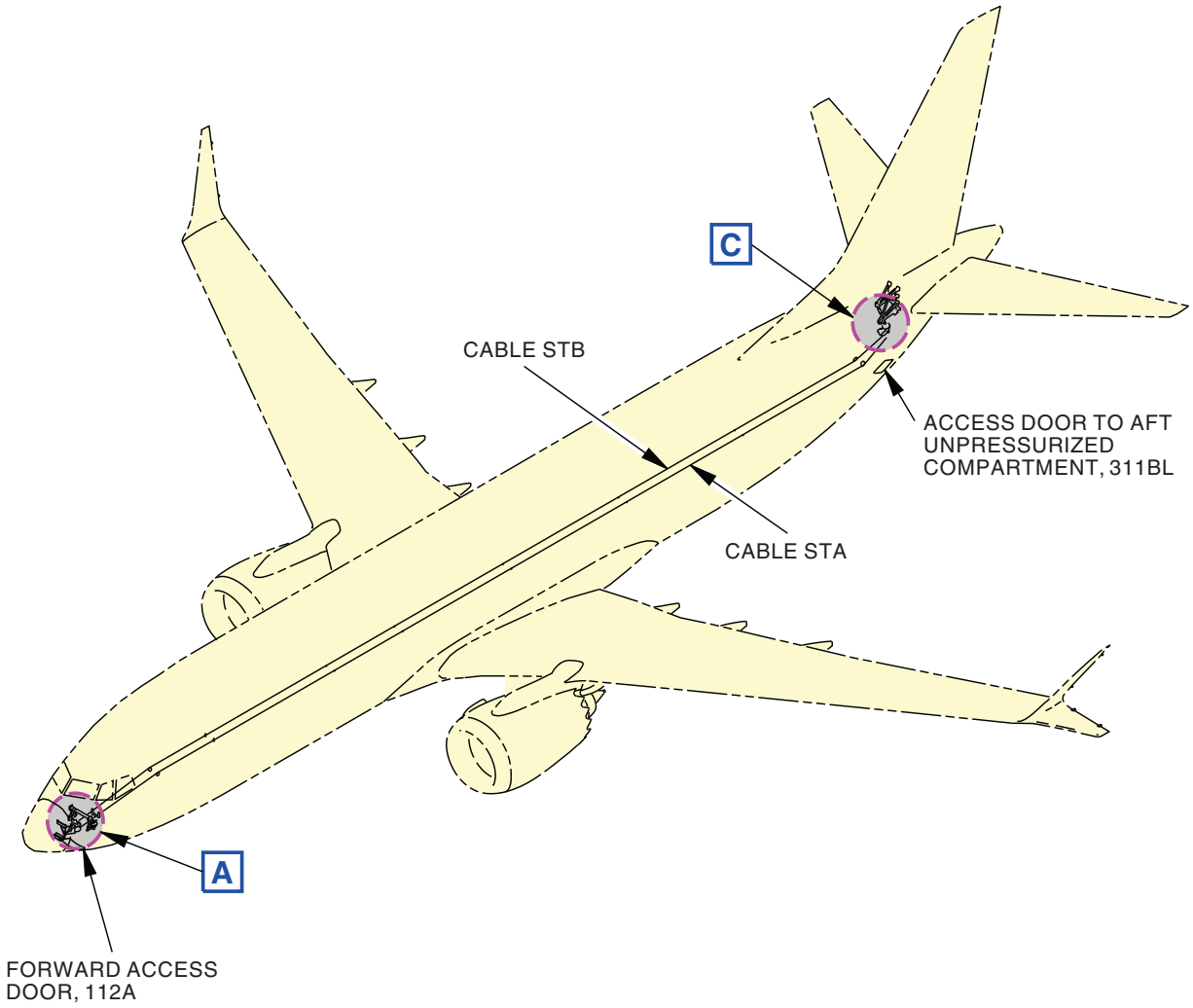
1 CABLE CONSTRUCTION IS CARBON STEEL: BMS 7-265, TYPE 1, COMPOSITION A (TIN OVER ZINC), 7 X 19.

2 MEASURE CABLE WITH A LOAD OF 40 ±3 POUNDS (178 ±13 NEWTONS).

2410999 S00061526738_V3

Elevator Control Cables
Figure 203/27-09-14-990-803 (Sheet 5 of 5)

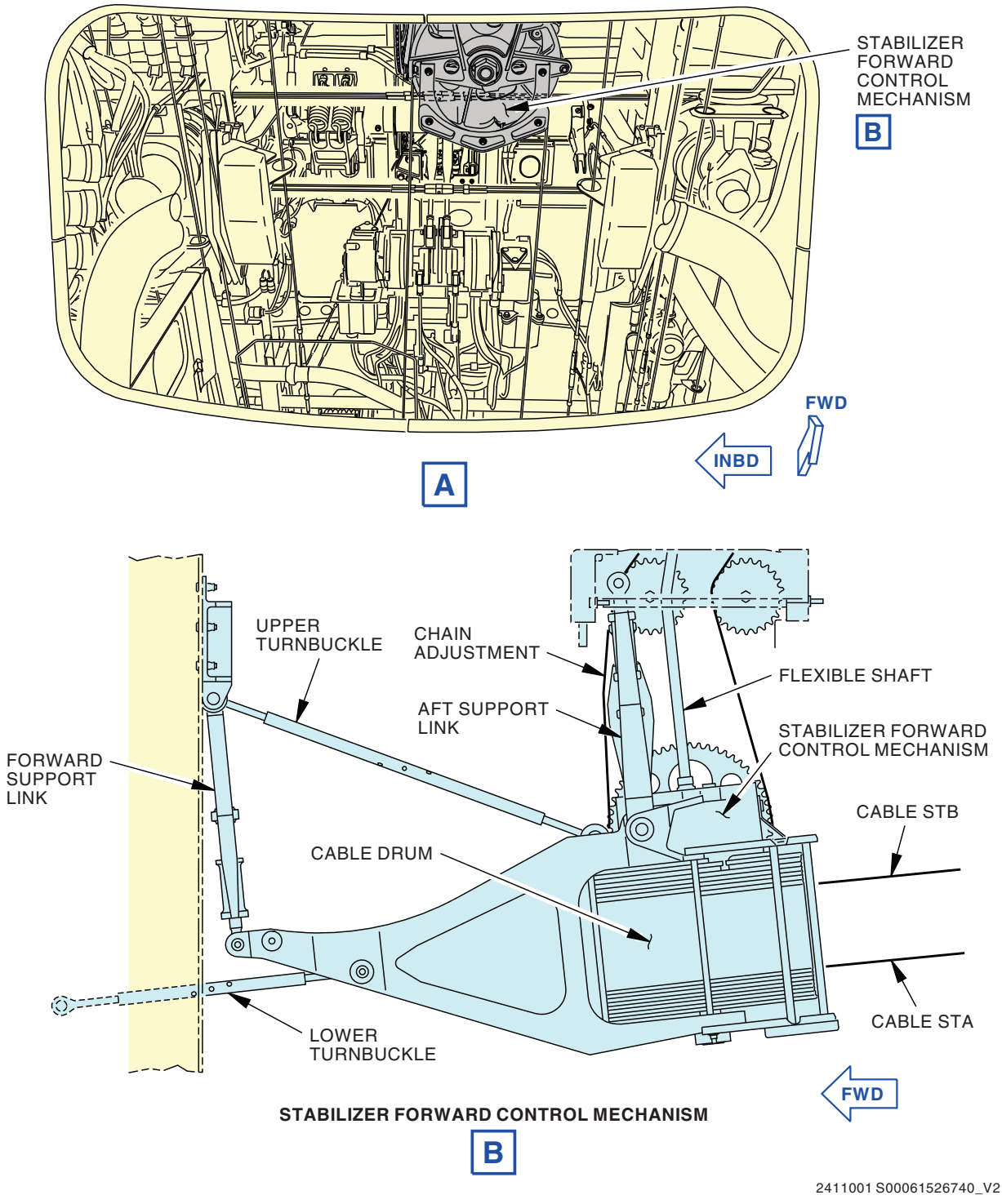
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411000 S00061526739_V1

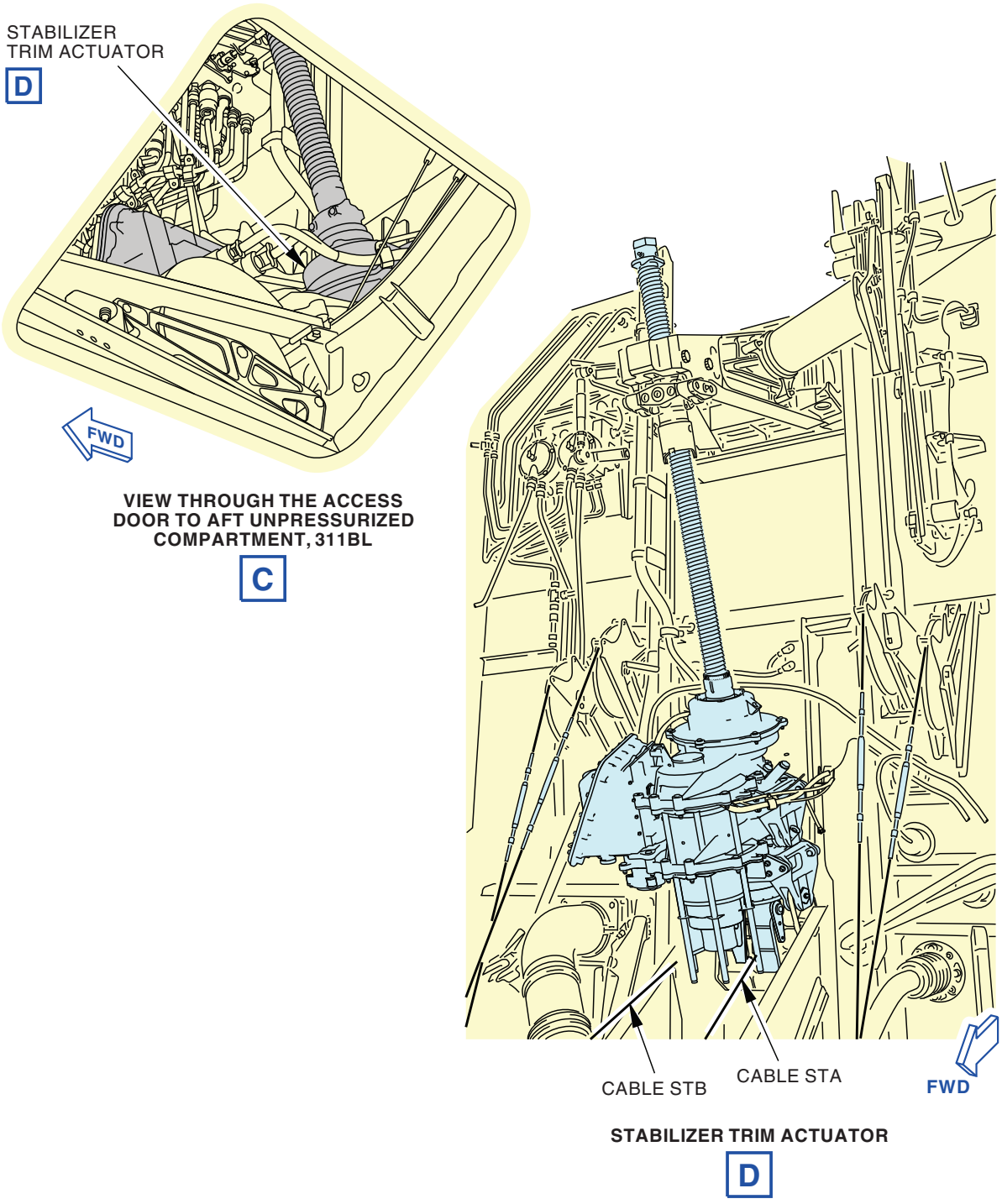
Stabilizer Control Cables
Figure 204/27-09-14-990-804 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



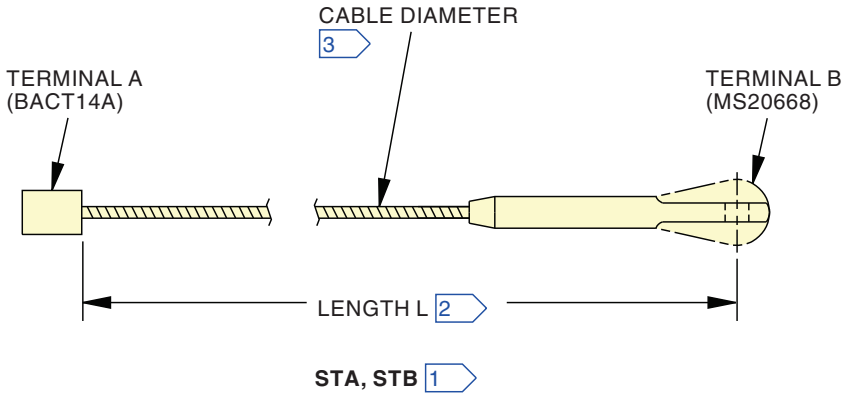
Stabilizer Control Cables
Figure 204/27-09-14-990-804 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Stabilizer Control Cables
Figure 204/27-09-14-990-804 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 CABLE CONSTRUCTION IS CARBON STEEL: BMS 7-265, TYPE 1, (TIN OVER ZINC), 7 X 19.

2 MEASURE CABLE LENGTH WITH A LOAD OF 140 ±5 POUNDS (623 ±22 NEWTONS) OF FORCE. SEE TABLE A.

3 MEASURE THE NEW CABLE DIAMETER AT 3 PLACES AND AVERAGE. THIS CABLE DIAMETER MEASUREMENT SHOULD NOT BE TAKEN WHILE THE CABLE IS UNDER LOAD.

2411003 S00061526742_V1

Stabilizer Control Cables
Figure 204/27-09-14-990-804 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

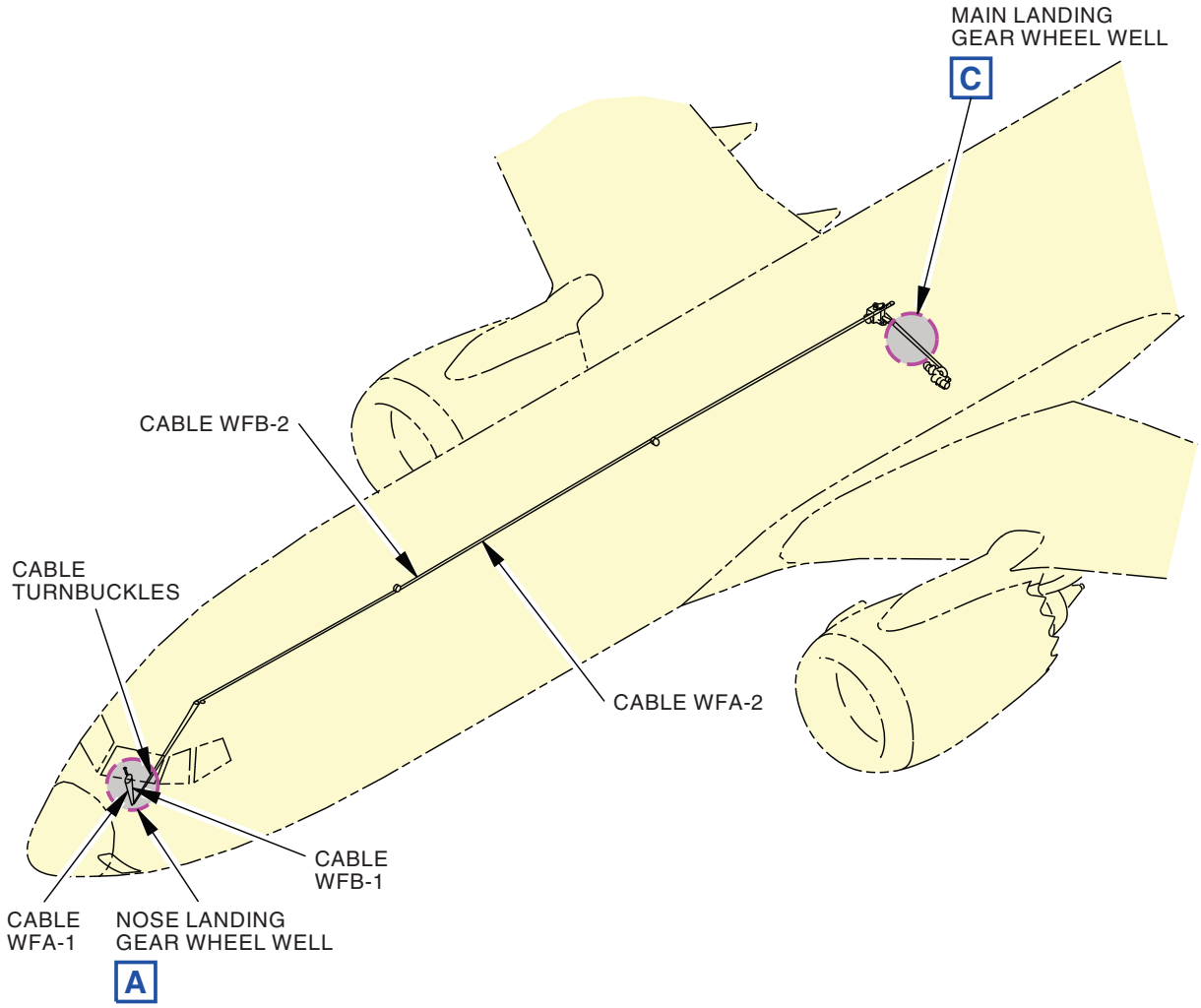
CABLE DIAMETER 3	CABLE LENGTH L INCHES (METERS) 2	
	STA (LEFT)	STB (RIGHT)
0.125	2199.55 ±0.50 (55.869 ±0.013)	2186.02 ±0.50 (55.525 ±0.013)
0.126	2199.69 ±0.50 (55.872 ±0.013)	2186.16 ±0.50 (55.528 ±0.013)
0.127	2199.83 ±0.50 (55.876 ±0.013)	2186.30 ±0.50 (55.532 ±0.013)
0.128	2199.97 ±0.50 (55.879 ±0.013)	2186.44 ±0.50 (55.536 ±0.013)
0.129	2200.11 ±0.50 (55.883 ±0.013)	2186.58 ±0.50 (55.539 ±0.013)
0.130	2200.25 ±0.50 (55.886 ±0.013)	2186.72 ±0.50 (55.543 ±0.013)
0.131	2200.39 ±0.50 (55.890 ±0.013)	2186.86 ±0.50 (55.546 ±0.013)
0.132	2200.53 ±0.50 (55.893 ±0.013)	2187.00 ±0.50 (55.550 ±0.013)
0.133	2200.67 ±0.50 (55.897 ±0.013)	2187.14 ±0.50 (55.553 ±0.013)
0.134	2200.81 ±0.50 (55.901 ±0.013)	2187.28 ±0.50 (55.557 ±0.013)
0.135	2200.95 ±0.50 (55.904 ±0.013)	2187.42 ±0.50 (55.560 ±0.013)
0.136	2201.09 ±0.50 (55.907 ±0.013)	2187.56 ±0.50 (55.564 ±0.013)
0.137	2201.23 ±0.50 (55.911 ±0.013)	2187.70 ±0.50 (55.568 ±0.013)
0.138	2201.37 ±0.50 (55.915 ±0.013)	2187.84 ±0.50 (55.571 ±0.013)
0.139	2201.51 ±0.50 (55.918 ±0.013)	2187.98 ±0.50 (55.575 ±0.013)

TABLE A

2411004 S00061526743_V1

Stabilizer Control Cables
Figure 204/27-09-14-990-804 (Sheet 5 of 5)

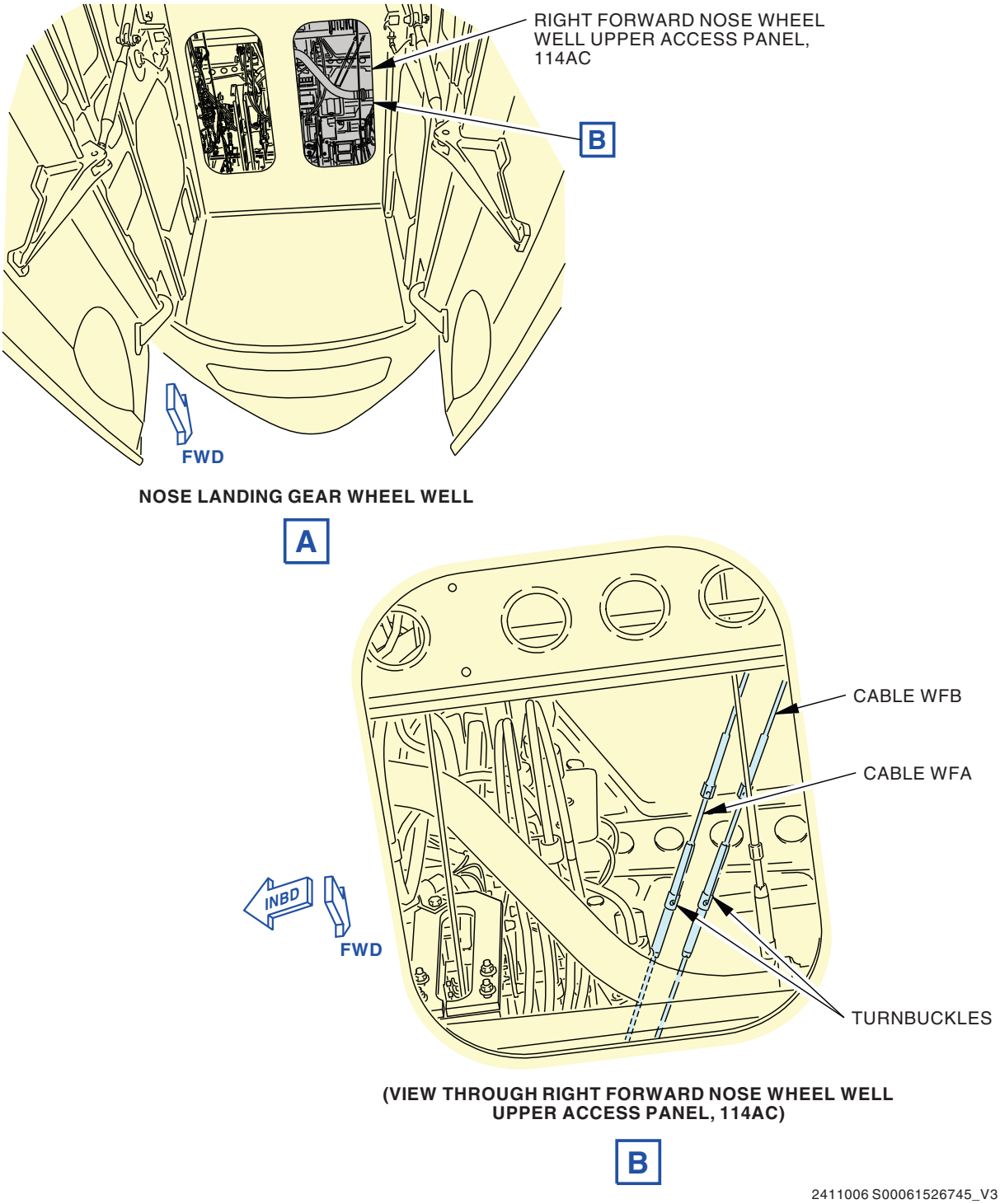
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411005 S00061526744_V2

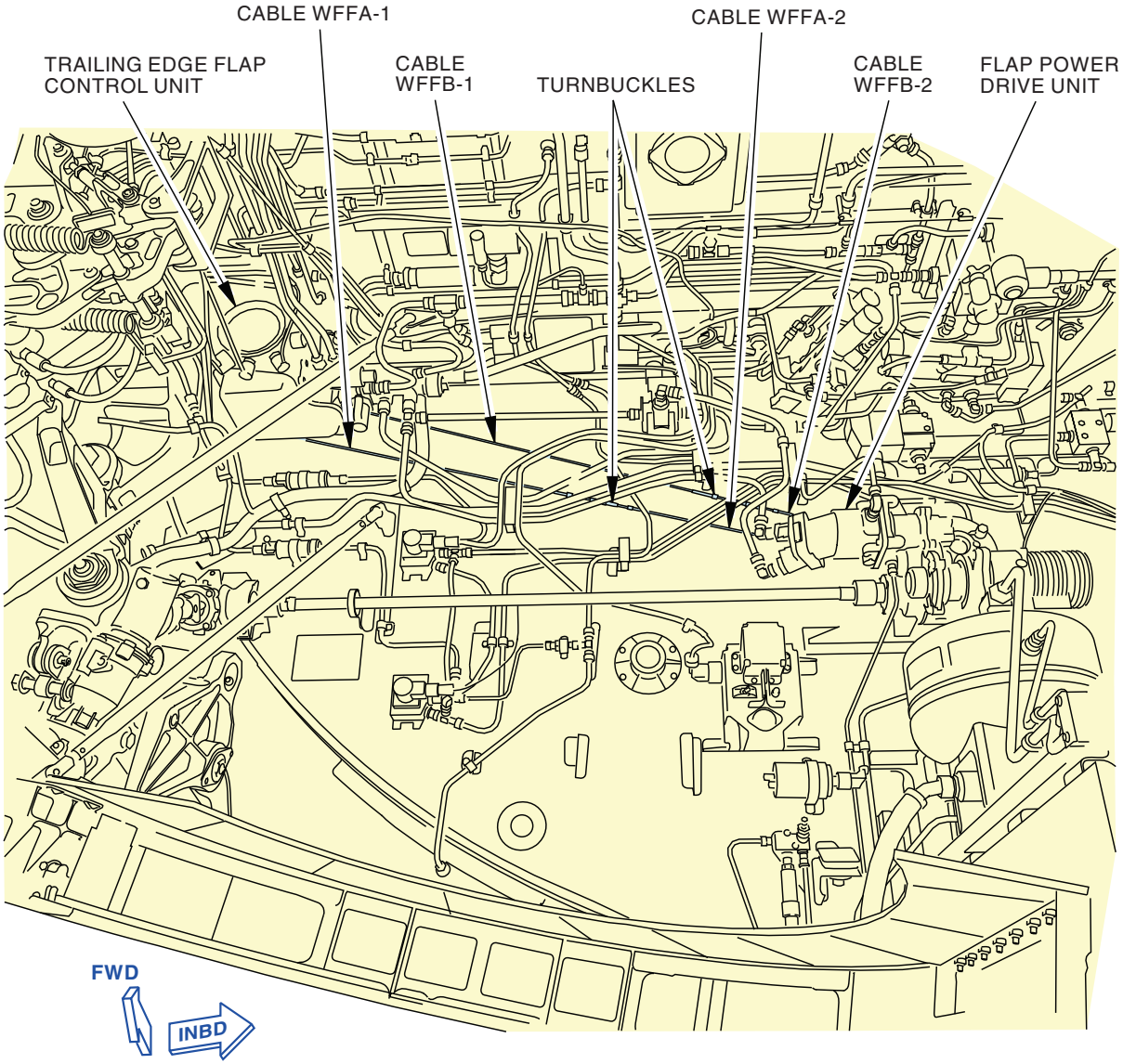
Trailing Edge Flap Control Cables
Figure 205/27-09-14-990-805 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Trailing Edge Flap Control Cables
Figure 205/27-09-14-990-805 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



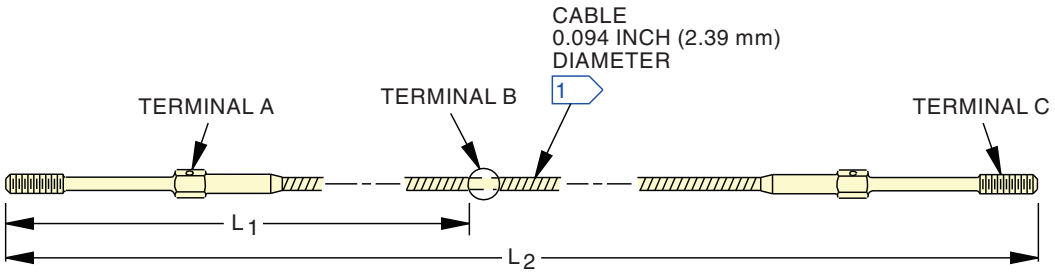
MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)
(EXAMPLE)

C

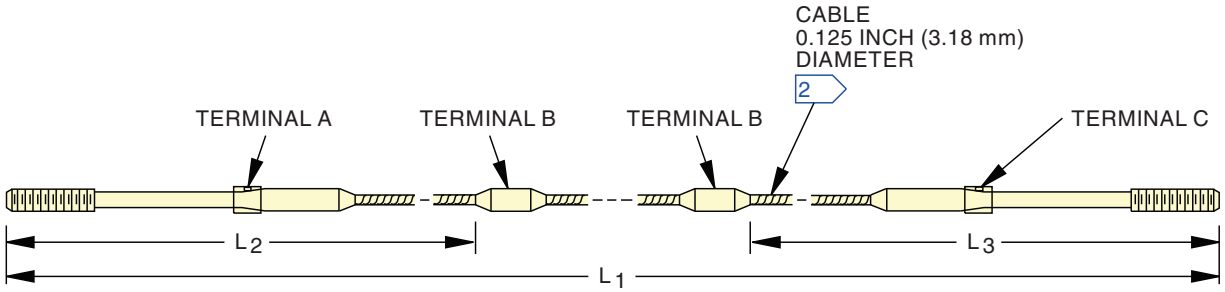
2411007 S00061526746_V2

Trailing Edge Flap Control Cables
Figure 205/27-09-14-990-805 (Sheet 3 of 5)

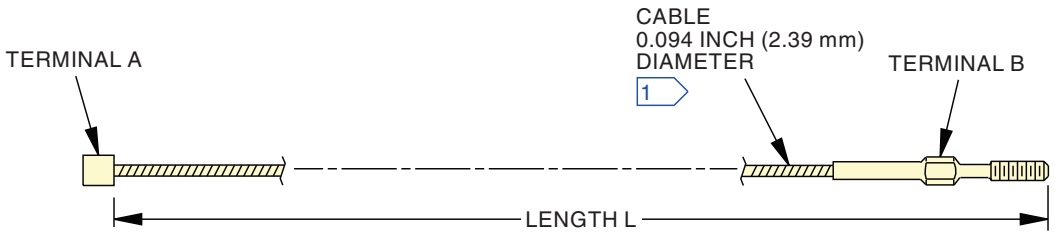
ECCN 9E991 BOEING PROPRIETARY - See title page for details



WFA-1, WFB-1



WFA-2, WFB-2



**WFFA-1, WFFA-2,
WFFB-1, WFFB-2**

1 CABLE CONSTRUCTION IS CARBON STEEL:
BMS 7-265, TYPE 1, COMPOSITION A (TIN OVER ZINC), 7 X 7

2 CABLE CONSTRUCTION IS CARBON STEEL:
BMS 7-265, TYPE 1, COMPOSITION A (TIN OVER ZINC), 7 X 19

2411008 S00061526747_V2

**Trailing Edge Flap Control Cables
Figure 205/27-09-14-990-805 (Sheet 4 of 5)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CABLE NAME	LENGTH L INCHES (MILLIMETERS)	TERMINAL A	TERMINAL B	TERMINAL C
WFA-1 WFB-1	L_1 - 41.5 ±0.12 (1,054 ±3) 1 4 L_2 - 84.8 ±0.25 (2,154 ±6) 1 4	MS21260L3LH	BACT14B3	MS21260L3RH
WFA-2 5 WFB-2 5	L_1 - 1418.8 (35,758) 2 3 L_2 - 712.4 (18,095) 2 3 L_3 - 695.4 (17,663) 2 3	69-38195-2	BACT14A4	69-38195-1
WFFA-1	30.4 ±0.12 (772 ±3) 1 4	BACT14A3	MS21260S3RH	
WFFB-1	32.9 ±0.12 (836 ±3) 1 4	BACT14A3	MS21260S3RH	
WFFA-2 WFFB-2	46.8 ±0.12 (1,189 ±3) 1 4	BACT14A3	MS21260S3LH	

TABLE A

1 CABLE CONSTRUCTION IS CARBON STEEL: BMS 7-265, TYPE 1, COMPOSITION A (TIN OVER ZINC), 7 X 7

2 CABLE CONSTRUCTION IS CARBON STEEL: BMS 7-265, TYPE 1, COMPOSITION A (TIN OVER ZINC), 7 X 19

3 MEASURE CABLE WITH A LOAD OF 40 ±3 POUNDS (178 ±13 NEWTONS)

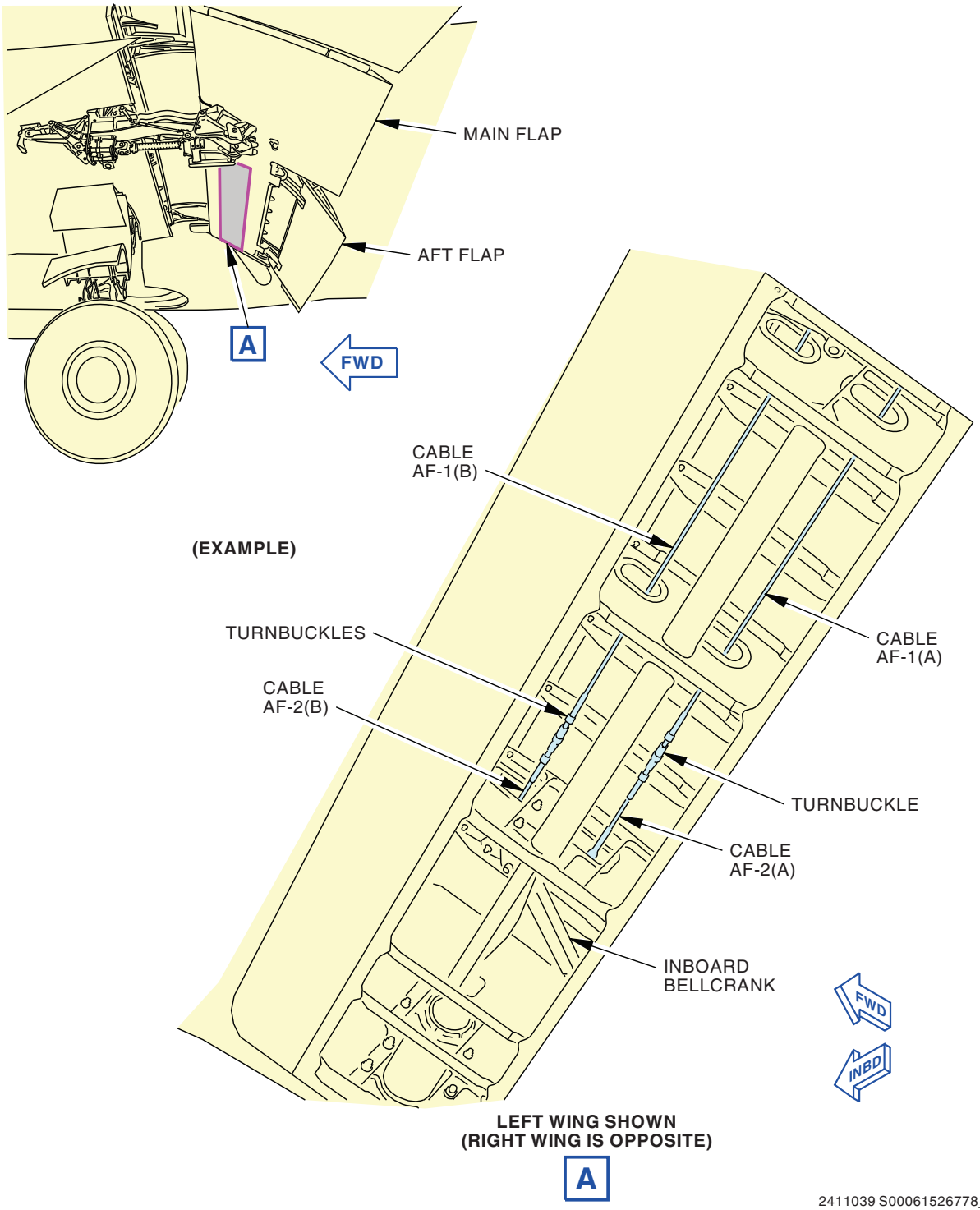
4 MEASURE CABLE WITH A LOAD OF 20 ±3 POUNDS (89 ±13 NEWTONS)

5 PROOF LOAD THESE CABLES TO 552 -0/+25 POUNDS (250 -0/+11 KG)

2411009 S00061526748_V3

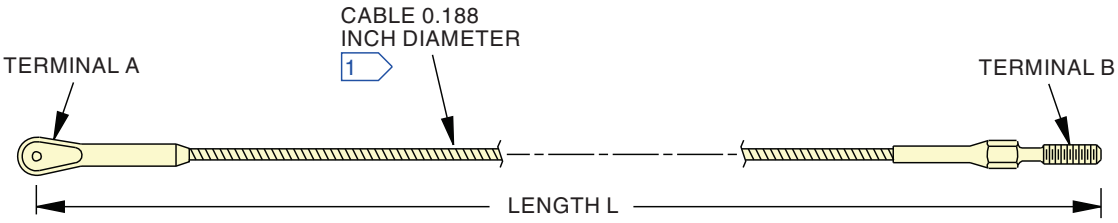
Trailing Edge Flap Control Cables
Figure 205/27-09-14-990-805 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Flap Drive Cables
Figure 206/27-09-14-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**AF-1(A), AF-1(B),
AF-2(A), AF-2(B)**

CABLE NAME	LENGTH L (INCHES) 1	TERMINAL A	TERMINAL B
AF-1(A) AF-1(B)	47.3384	113A2641-1	MS21260S6RH
AF-2(A) AF-2(B)	10.6360	113A2641-1	MS21260S6LH

TABLE A

1 CABLE CONSTRUCTION IS CORROSION RESISTANT STEEL: MIL-W-83420, COMPOSITION B, 7 X 19

2 MEASURE CABLE WITH A LOAD OF 40 ±3 POUNDS.

2411040 S00061526779_V1

**Aft Flap Drive Cables
Figure 206/27-09-14-990-806 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-14-820-801

3. Control Cables - Rigging

A. General

(1) Before you do cable adjustment, permit a minimum of one hour at a constant ambient temperature of ± 5 degrees F (± 2.8 degrees C) for the airframe temperature to become stable.

(a) It is recommended to do the cable adjustments in a controlled environment such as a hangar.

NOTE: Direct sunlight on the airframe will cause the cable readings to be inaccurate.

(b) All temperature readings will be taken as ambient outside but adjacent to the applicable area of the airframe.

(c) The cable tension values will not be correct when there are temperature differences along the cable run.

(2) Before you do cable adjustment, allow the tensiometer to stabilize to the same ambient temperature as the airplane.

(3) For cables retensioning, it is recommended to adjust the cables to the upper limit values, to account for tension loss in service.

B. References

Reference	Title
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-21-00-820-803	Rudder Control Cables RA and RB Adjustment (P/B 501)
27-31-00-820-801	Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment (P/B 501)
27-41-00-820-801	Stabilizer Control Cable and Chain - Adjustment (P/B 501)
27-51-00-820-801	Trailing Edge Flap Control System Adjustment and Test (P/B 501)
27-51-14-400-801	Inboard Aft Flap Actuating Mechanism Installation (P/B 401)

C. Adjust the Control Cables

SUBTASK 27-09-14-820-001

(1) To adjust the aileron cable system, do this task: Aileron Cables Adjustment, TASK 27-11-00-820-801.

SUBTASK 27-09-14-820-002

(2) To adjust the rudder cable system, do this task: Rudder Control Cables RA and RB Adjustment, TASK 27-21-00-820-803.

SUBTASK 27-09-14-820-003

(3) To adjust the elevator cable system, do this task: Elevator Control Cables EA and EB and Pitch Force Transducers - Adjustment, TASK 27-31-00-820-801.

SUBTASK 27-09-14-820-004

(4) To adjust the stabilizer cable system, do this task: Stabilizer Control Cable and Chain - Adjustment, TASK 27-41-00-820-801.

SUBTASK 27-09-14-820-005

(5) To adjust the trailing edge flaps cable system, do this task: Trailing Edge Flap Control System Adjustment and Test, TASK 27-51-00-820-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-14-820-006

(6) To adjust the trailing edge flap boom cables, do this task: Inboard Aft Flap Actuating Mechanism Installation, TASK 27-51-14-400-801.

——— END OF TASK ———

TASK 27-09-14-860-802

4. Cable Control - Tension Measurement

A. General

(1) This task gives the instructions to measure the tension of the control cables.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
COM-1554	Tensiometer - Cable, High Tension (200 lbs and above) Part #: 102-03108 Supplier: 21844 Part #: 102-03120 Supplier: 21844 Part #: ACM-300 Supplier: 13331 Part #: ACM-600 Supplier: 13331 Part #: ACX-500-1 Supplier: 13331 Part #: T5-8008-106-00 Supplier: 0N8U4 Part #: T60-1001-C9-1A Supplier: 0N8U4

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
324	Vertical Fin - Rear Spar To Trailing Edge

D. Prepare for the Measurement

SUBTASK 27-09-14-860-006

(1) Before you do cable tension measurement, permit a minimum of one hour at a constant ambient temperature of ±5°F (±2.8°C) for the airframe temperature to become stable.

(a) It is recommended to do the cable tension measurement in a controlled environment such as a hangar.

NOTE: Direct sunlight on the airframe will cause the cable readings to be inaccurate.

(b) All temperature readings will be taken as ambient outside but adjacent to the applicable area of the airframe.

(c) The cable tension values will not be correct when there are temperature differences along the cable run.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) Before you do cable tension measurement, allow the tensiometer to stabilize to the same ambient temperature as the airplane.

E. Control Cable Tension Measurement

SUBTASK 27-09-14-860-003

(1) For the aileron control cables use a high tension cable tensiometer, COM-1554, or low tension cable tensiometer, COM-1553, to measure the cable tension (Table 201).

Table 201/27-09-14-993-801 Aileron Cable Rigging Load

Temperature	AA & AB*[1]	ABSA & ABSB*[1]	ACBA1 & ACBA2*[1] ACBB1 & ACBB2*[1]
108°F (42.2°C)	183 +10 / -30 lb (83 +5 / -14 kg)	271 +10 / -30 lb (123 +5 / -14 kg)	76 +10 / -30 lb (34 +5 / -14 kg)
106°F (41.1°C)	181 +10 / -30 lb (82 +5 / -14 kg)	269 +10 / -30 lb (122 +5 / -14 kg)	75 +10 / -30 lb (34 +5 / -14 kg)
104°F (40.0°C)	180 +10 / -30 lb (82 +5 / -14 kg)	267 +10 / -30 lb (121 +5 / -14 kg)	74 +10 / -30 lb (34 +5 / -14 kg)
102°F (38.9°C)	178 +10 / -30 lb (81 +5 / -14 kg)	264 +10 / -30 lb (120 +5 / -14 kg)	74 +10 / -30 lb (34 +5 / -14 kg)
100°F (37.8°C)	177 +10 / -30 lb (80 +5 / -14 kg)	262 +10 / -30 lb (119 +5 / -14 kg)	73 +10 / -30 lb (33 +5 / -14 kg)
98°F (36.7°C)	175 +10 / -30 lb (79 +5 / -14 kg)	260 +10 / -30 lb (118 +5 / -14 kg)	73 +10 / -30 lb (33 +5 / -14 kg)
96°F (35.6°C)	174 +10 / -30 lb (79 +5 / -14 kg)	258 +10 / -30 lb (117 +5 / -14 kg)	72 +10 / -30 lb (33 +5 / -14 kg)
94°F (34.4°C)	172 +10 / -30 lb (78 +5 / -14 kg)	256 +10 / -30 lb (116 +5 / -14 kg)	72 +10 / -30 lb (33 +5 / -14 kg)
92°F (33.3°C)	171 +10 / -30 lb (78 +5 / -14 kg)	254 +10 / -30 lb (115 +5 / -14 kg)	71 +10 / -30 lb (32 +5 / -14 kg)
90°F (32.2°C)	169 +10 / -50 lb (77 +5 / -23 kg)	251 +10 / -50 lb (114 +5 / -23 kg)	71 +10 / -50 lb (32 +5 / -23 kg)
88°F (31.1°C)	168 +10 / -50 lb (76 +5 / -23 kg)	249 +10 / -50 lb (113 +5 / -23 kg)	70 +10 / -50 lb (32 +5 / -23 kg)
86°F (30.0°C)	166 +10 / -50 lb (75 +5 / -23 kg)	247 +10 / -50 lb (112 +5 / -23 kg)	70 +10 / -50 lb (32 +5 / -23 kg)
84°F (28.9°C)	165 +10 / -50 lb (75 +5 / -23 kg)	245 +10 / -50 lb (111 +5 / -23 kg)	69 +10 / -50 lb (31 +5 / -23 kg)
82°F (27.8°C)	163 +10 / -50 lb (74 +5 / -23 kg)	243 +10 / -50 lb (110 +5 / -23 kg)	69 +10 / -50 lb (31 +5 / -23 kg)
80°F (26.7°C)	162 +10 / -50 lb (73 +5 / -23 kg)	241 +10 / -50 lb (109 +5 / -23 kg)	68 +10 / -50 lb (31 +5 / -23 kg)
78°F (25.6°C)	160 +10 / -50 lb (73 +5 / -23 kg)	238 +10 / -50 lb (108 +5 / -23 kg)	68 +10 / -50 lb (31 +5 / -23 kg)
76°F (24.4°C)	159 +10 / -50 lb (72 +5 / -23 kg)	236 +10 / -50 lb (107 +5 / -23 kg)	67 +10 / -50 lb (30 +5 / -23 kg)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 201/27-09-14-993-801 Aileron Cable Rigging Load (Continued)

Temperature	AA & AB*[1]	ABSA & ABSB*[1]	ACBA1 & ACBA2*[1] ACBB1 & ACBB2*[1]
74°F (23.3°C)	157 +10 / -50 lb (71 +5 / -23 kg)	234 +10 / -50 lb (106 +5 / -23 kg)	67 +10 / -50 lb (30 +5 / -23 kg)
72°F (22.2°C)	156 +10 / -50 lb (71 +5 / -23 kg)	232 +10 / -50 lb (105 +5 / -23 kg)	66 +10 / -50 lb (30 +5 / -23 kg)
70°F (21.1°C)	155 +10 / -50 lb (70 +5 / -23 kg)	230 +10 / -50 lb (104 +5 / -23 kg)	66 +10 / -50 lb (30 +5 / -23 kg)
68°F (20.0°C)	153 +10 / -50 lb (69 +5 / -23 kg)	228 +10 / -50 lb (103 +5 / -23 kg)	65 +10 / -50 lb (29 +5 / -23 kg)
66°F (18.9°C)	152 +10 / -50 lb (69 +5 / -23 kg)	225 +10 / -50 lb (102 +5 / -23 kg)	65 +10 / -50 lb (29 +5 / -23 kg)
64°F (17.8°C)	150 +10 / -50 lb (68 +5 / -23 kg)	223 +10 / -50 lb (101 +5 / -23 kg)	64 +10 / -50 lb (29 +5 / -23 kg)
62°F (16.7°C)	149 +10 / -50 lb (68 +5 / -23 kg)	221 +10 / -50 lb (100 +5 / -23 kg)	64 +10 / -50 lb (29 +5 / -23 kg)
60°F (15.6°C)	148 +10 / -50 lb (67 +5 / -23 kg)	219 +10 / -50 lb (99 +5 / -23 kg)	63 +10 / -50 lb (29 +5 / -23 kg)
58°F (14.4°C)	146 +10 / -50 lb (66 +5 / -23 kg)	217 +10 / -50 lb (98 +5 / -23 kg)	63 +10 / -50 lb (29 +5 / -23 kg)
56°F (13.3°C)	145 +10 / -50 lb (66 +5 / -23 kg)	215 +10 / -50 lb (98 +5 / -23 kg)	62 +10 / -50 lb (28 +5 / -23 kg)
54°F (12.2°C)	144 +10 / -50 lb (65 +5 / -23 kg)	213 +10 / -50 lb (97 +5 / -23 kg)	62 +10 / -50 lb (28 +5 / -23 kg)
52°F (11.1°C)	142 +10 / -50 lb (64 +5 / -23 kg)	210 +10 / -50 lb (95 +5 / -23 kg)	61 +10 / -50 lb (28 +5 / -23 kg)
50°F (10.0°C)	141 +10 / -50 lb (64 +5 / -23 kg)	208 +10 / -50 lb (94 +5 / -23 kg)	61 +10 / -50 lb (28 +5 / -23 kg)
48°F (8.9°C)	140 +10 / -50 lb (64 +5 / -23 kg)	206 +10 / -50 lb (93 +5 / -23 kg)	60 +10 / -50 lb (27 +5 / -23 kg)
46°F (7.8°C)	138 +10 / -50 lb (63 +5 / -23 kg)	204 +10 / -50 lb (93 +5 / -23 kg)	60 +10 / -50 lb (27 +5 / -23 kg)
44°F (6.7°C)	137 +10 / -50 lb (62 +5 / -23 kg)	202 +10 / -50 lb (92 +5 / -23 kg)	59 +10 / -50 lb (27 +5 / -23 kg)
42°F (5.6°C)	136 +10 / -50 lb (62 +5 / -23 kg)	200 +10 / -50 lb (91 +5 / -23 kg)	59 +10 / -50 lb (27 +5 / -23 kg)
40°F (4.4°C)	134 +10 / -50 lb (61 +5 / -23 kg)	197 +10 / -50 lb (89 +5 / -23 kg)	58 +10 / -50 lb (26 +5 / -23 kg)
38°F (3.3°C)	133 +10 / -30 lb (60 +5 / -14 kg)	195 +10 / -30 lb (88 +5 / -14 kg)	58 +10 / -30 lb (26 +5 / -14 kg)
36°F (2.2°C)	132 +10 / -30 lb (60 +5 / -14 kg)	193 +10 / -30 lb (88 +5 / -14 kg)	57 +10 / -30 lb (26 +5 / -14 kg)
34°F (1.1°C)	131 +10 / -30 lb (59 +5 / -14 kg)	191 +10 / -30 lb (87 +5 / -14 kg)	57 +10 / -30 lb (26 +5 / -14 kg)
32°F (0.0°C)	129 +10 / -30 lb (59 +5 / -14 kg)	189 +10 / -30 lb (86 +5 / -14 kg)	56 +10 / -30 lb (25 +5 / -14 kg)
30°F (-1.1°C)	128 +10 / -30 lb (58 +5 / -14 kg)	187 +10 / -30 lb (85 +5 / -14 kg)	56 +10 / -30 lb (25 +5 / -14 kg)
28°F (-2.2°C)	127 +10 / -30 lb (58 +5 / -14 kg)	184 +10 / -30 lb (83 +5 / -14 kg)	55 +10 / -30 lb (25 +5 / -14 kg)
26°F (-3.3°C)	126 +10 / -30 lb (57 +5 / -14 kg)	182 +10 / -30 lb (83 +5 / -14 kg)	55 +10 / -30 lb (25 +5 / -14 kg)
24°F (-4.4°C)	125 +10 / -30 lb (57 +5 / -14 kg)	180 +10 / -30 lb (82 +5 / -14 kg)	54 +10 / -30 lb (24 +5 / -14 kg)
22°F (-5.6°C)	123 +10 / -30 lb (56 +5 / -14 kg)	178 +10 / -30 lb (81 +5 / -14 kg)	53 +10 / -30 lb (24 +5 / -14 kg)
20°F (-6.7°C)	122 +10 / -30 lb (55 +5 / -14 kg)	176 +10 / -30 lb (80 +5 / -14 kg)	53 +10 / -30 lb (24 +5 / -14 kg)
18°F (-7.8°C)	121 +10 / -30 oz (3 +0 / -1 kg)	174 +10 / -30 lb (79 +5 / -14 kg)	52 +10 / -30 lb (24 +5 / -14 kg)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 201/27-09-14-993-801 Aileron Cable Rigging Load (Continued)

Temperature	AA & AB*[1]	ABSA & ABSB*[1]	ACBA1 & ACBA2*[1] ACBB1 & ACBB2*[1]
16°F (-8.9°C)	120 +10 / -30 lb (54 +5 / -14 kg)	171 +10 / -30 lb (78 +5 / -14 kg)	52 +10 / -30 lb (24 +5 / -14 kg)
14°F (-10.0°C)	119 +10 / -30 lb (54 +5 / -14 kg)	169 +10 / -30 lb (77 +5 / -14 kg)	51 +10 / -30 lb (23 +5 / -14 kg)
12°F (-11.1°C)	118 +10 / -30 lb (54 +5 / -14 kg)	167 +10 / -30 lb (76 +5 / -14 kg)	51 +10 / -30 lb (23 +5 / -14 kg)
10°F (-12.2°C)	117 +10 / -30 lb (53 +5 / -14 kg)	165 +10 / -30 lb (75 +5 / -14 kg)	50 +10 / -30 lb (23 +5 / -14 kg)
8°F (-13.3°C)	115 +10 / -30 lb (52 +5 / -14 kg)	163 +10 / -30 lb (74 +5 / -14 kg)	50 +10 / -30 lb (23 +5 / -14 kg)
6°F (-14.4°C)	114 +10 / -30 lb (52 +5 / -14 kg)	161 +10 / -30 lb (73 +5 / -14 kg)	49 +10 / -30 lb (22 +5 / -14 kg)
4°F (-15.6°C)	113 +10 / -30 lb (51 +5 / -14 kg)	159 +10 / -30 lb (72 +5 / -14 kg)	49 +10 / -30 lb (22 +5 / -14 kg)
2°F (-16.7°C)	112 +10 / -30 lb (51 +5 / -14 kg)	156 +10 / -30 lb (71 +5 / -14 kg)	48 +10 / -30 lb (22 +5 / -14 kg)
0°F (-17.8°C)	111 +10 / -30 lb (50 +5 / -14 kg)	154 +10 / -30 lb (70 +5 / -14 kg)	48 +10 / -30 lb (22 +5 / -14 kg)
-2°F (-18.9°C)	110 +10 / -30 lb (50 +5 / -14 kg)	152 +10 / -30 lb (69 +5 / -14 kg)	47 +10 / -30 lb (21 +5 / -14 kg)
-4°F (-20.0°C)	109 +10 / -30 lb (49 +5 / -14 kg)	150 +10 / -30 lb (68 +5 / -14 kg)	47 +10 / -30 lb (21 +5 / -14 kg)
-6°F (-21.1°C)	108 +10 / -30 lb (49 +5 / -14 kg)	148 +10 / -30 lb (67 +5 / -14 kg)	46 +10 / -30 lb (21 +5 / -14 kg)
-8°F (-22.2°C)	107 +10 / -30 lb (49 +5 / -14 kg)	146 +10 / -30 lb (66 +5 / -14 kg)	46 +10 / -30 lb (21 +5 / -14 kg)
-10°F (-23.3°C)	106 +10 / -30 lb (48 +5 / -14 kg)	143 +10 / -30 lb (65 +5 / -14 kg)	46 +10 / -30 lb (21 +5 / -14 kg)
-12°F (-24.4°C)	105 +10 / -30 lb (48 +5 / -14 kg)	141 +10 / -30 lb (64 +5 / -14 kg)	45 +10 / -30 lb (20 +5 / -14 kg)
-14°F (-25.6°C)	104 +10 / -30 lb (47 +5 / -14 kg)	139 +10 / -30 lb (63 +5 / -14 kg)	45 +10 / -30 lb (20 +5 / -14 kg)
-16°F (-26.7°C)	103 +10 / -30 lb (47 +5 / -14 kg)	137 +10 / -30 lb (62 +5 / -14 kg)	44 +10 / -30 lb (20 +5 / -14 kg)
-18°F (-27.8°C)	102 +10 / -30 lb (46 +5 / -14 kg)	135 +10 / -30 lb (61 +5 / -14 kg)	44 +10 / -30 lb (20 +5 / -14 kg)
-20°F (-28.9°C)	101 +10 / -30 lb (46 +5 / -14 kg)	133 +10 / -30 lb (60 +5 / -14 kg)	43 +10 / -30 lb (20 +5 / -14 kg)
-22°F (-30.0°C)	100 +10 / -30 lb (45 +5 / -14 kg)	130 +10 / -30 lb (59 +5 / -14 kg)	43 +10 / -30 lb (20 +5 / -14 kg)
-24°F (-31.1°C)	99 +10 / -30 lb (45 +5 / -14 kg)	128 +10 / -30 lb (58 +5 / -14 kg)	42 +10 / -30 lb (19 +5 / -14 kg)
-26°F (-32.2°C)	98 +10 / -30 lb (44 +5 / -14 kg)	126 +10 / -30 lb (57 +5 / -14 kg)	42 +10 / -30 lb (19 +5 / -14 kg)
-28°F (-33.3°C)	97 +10 / -30 lb (44 +5 / -14 kg)	124 +10 / -30 lb (56 +5 / -14 kg)	42 +10 / -30 lb (19 +5 / -14 kg)
-30°F (-34.4°C)	96 +10 / -30 lb (44 +5 / -14 kg)	122 +10 / -30 lb (55 +5 / -14 kg)	41 +10 / -30 lb (19 +5 / -14 kg)
-32°F (-35.6°C)	95 +10 / -30 lb (43 +5 / -14 kg)	120 +10 / -30 lb (54 +5 / -14 kg)	41 +10 / -30 lb (19 +5 / -14 kg)
-34°F (-36.7°C)	94 +10 / -30 lb (43 +5 / -14 kg)	117 +10 / -30 lb (53 +5 / -14 kg)	40 +10 / -30 lb (18 +5 / -14 kg)
-36°F (-37.8°C)	93 +10 / -30 lb (42 +5 / -14 kg)	115 +10 / -30 lb (52 +5 / -14 kg)	40 +10 / -30 lb (18 +5 / -14 kg)
-38°F (-38.9°C)	92 +10 / -30 lb (42 +5 / -14 kg)	113 +10 / -30 lb (51 +5 / -14 kg)	39 +10 / -30 lb (18 +5 / -14 kg)
-40°F (-40.0°C)	91 +10 / -30 lb (41 +5 / -14 kg)	111 +10 / -30 lb (50 +5 / -14 kg)	39 +10 / -30 lb (18 +5 / -14 kg)

*[1] The minimum cable tension applies only during maintenance procedure. This value must not be used as an in-service operational limit for cable tension.

SUBTASK 27-09-14-860-004

(2) For the elevator control cables use a high tension cable tensiometer, COM-1554, or low tension cable tensiometer, COM-1553, to measure the cable tension (Table 202).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 202/27-09-14-993-806 Elevator Control Cable Rigging Tension

Temperature Deg F (C)	Maximum Lbs (Kg)	Minimum Lbs (Kg)*[1]
-40°F (-40°C)	85 lb (39 kg)	50 lb (23 kg)
-38°F (-39°C)	85 lb (39 kg)	50 lb (23 kg)
-36°F (-38°C)	86 lb (39 kg)	51 lb (23 kg)
-34°F (-37°C)	87 lb (39 kg)	52 lb (24 kg)
-32°F (-36°C)	88 lb (40 kg)	53 lb (24 kg)
-30°F (-34°C)	90 lb (41 kg)	55 lb (25 kg)
-28°F (-33°C)	91 lb (41 kg)	56 lb (25 kg)
-26°F (-32°C)	92 lb (42 kg)	57 lb (26 kg)
-24°F (-31°C)	93 lb (42 kg)	58 lb (26 kg)
-22°F (-30°C)	94 lb (43 kg)	59 lb (27 kg)
-20°F (-29°C)	95 lb (43 kg)	60 lb (27 kg)
-18°F (-28°C)	96 lb (44 kg)	61 lb (28 kg)
-16°F (-27°C)	97 lb (44 kg)	62 lb (28 kg)
-14°F (-26°C)	98 lb (44 kg)	63 lb (29 kg)
-12°F (-24°C)	99 lb (45 kg)	64 lb (29 kg)
-10°F (-23°C)	100 lb (45 kg)	65 lb (29 kg)
-8°F (-22°C)	101 lb (46 kg)	66 lb (30 kg)
-6°F (-21°C)	103 lb (47 kg)	68 lb (31 kg)
-4°F (-20°C)	104 lb (47 kg)	69 lb (31 kg)
-2°F (-19°C)	105 lb (48 kg)	70 lb (32 kg)
0°F (-18°C)	106 lb (48 kg)	71 lb (32 kg)
2°F (-17°C)	107 lb (49 kg)	72 lb (33 kg)
4°F (-16°C)	108 lb (49 kg)	73 lb (33 kg)
6°F (-14°C)	110 lb (50 kg)	75 lb (34 kg)
8°F (-13°C)	111 lb (50 kg)	76 lb (34 kg)
10°F (-12°C)	112 lb (51 kg)	77 lb (35 kg)
12°F (-11°C)	113 lb (51 kg)	78 lb (35 kg)
14°F (-10°C)	115 lb (52 kg)	80 lb (36 kg)
16°F (-9°C)	116 lb (53 kg)	81 lb (37 kg)
18°F (-8°C)	117 lb (53 kg)	82 lb (37 kg)
20°F (-7°C)	118 lb (54 kg)	83 lb (38 kg)
22°F (-6°C)	120 lb (54 kg)	85 lb (39 kg)
24°F (-4°C)	121 lb (55 kg)	86 lb (39 kg)
26°F (-3°C)	122 lb (55 kg)	87 lb (39 kg)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 202/27-09-14-993-806 Elevator Control Cable Rigging Tension (Continued)

Temperature Deg F (C)	Maximum Lbs (Kg)	Minimum Lbs (Kg)*[1]
28°F (-2°C)	124 lb (56 kg)	89 lb (40 kg)
30°F (-1°C)	125 lb (57 kg)	90 lb (41 kg)
32°F (0°C)	126 lb (57 kg)	91 lb (41 kg)
34°F (1°C)	128 lb (58 kg)	93 lb (42 kg)
36°F (2°C)	129 lb (59 kg)	94 lb (43 kg)
38°F (3°C)	130 lb (59 kg)	95 lb (43 kg)
40°F (4°C)	132 lb (60 kg)	82 lb (37 kg)
42°F (6°C)	133 lb (60 kg)	83 lb (38 kg)
44°F (7°C)	134 lb (61 kg)	84 lb (38 kg)
46°F (8°C)	136 lb (62 kg)	86 lb (39 kg)
48°F (9°C)	137 lb (62 kg)	87 lb (39 kg)
50°F (10°C)	139 lb (63 kg)	89 lb (40 kg)
52°F (11°C)	140 lb (64 kg)	90 lb (41 kg)
54°F (12°C)	142 lb (64 kg)	92 lb (42 kg)
56°F (13°C)	143 lb (65 kg)	93 lb (42 kg)
58°F (14°C)	145 lb (66 kg)	95 lb (43 kg)
60°F (16°C)	146 lb (66 kg)	96 lb (44 kg)
62°F (17°C)	148 lb (67 kg)	98 lb (44 kg)
64°F (18°C)	149 lb (68 kg)	99 lb (45 kg)
66°F (19°C)	151 lb (68 kg)	101 lb (46 kg)
68°F (20°C)	152 lb (69 kg)	102 lb (46 kg)
70°F (21°C)	154 lb (70 kg)	104 lb (47 kg)
72°F (22°C)	155 lb (70 kg)	105 lb (48 kg)
74°F (23°C)	157 lb (71 kg)	107 lb (49 kg)
76°F (24°C)	158 lb (72 kg)	108 lb (49 kg)
78°F (26°C)	160 lb (73 kg)	110 lb (50 kg)
80°F (27°C)	162 lb (73 kg)	112 lb (51 kg)
82°F (28°C)	163 lb (74 kg)	113 lb (51 kg)
84°F (29°C)	165 lb (75 kg)	115 lb (52 kg)
86°F (30°C)	166 lb (75 kg)	116 lb (53 kg)
88°F (31°C)	168 lb (76 kg)	118 lb (54 kg)
90°F (32°C)	170 lb (77 kg)	120 lb (54 kg)
92°F (33°C)	171 lb (78 kg)	136 lb (62 kg)
94°F (34°C)	173 lb (78 kg)	138 lb (63 kg)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 202/27-09-14-993-806 Elevator Control Cable Rigging Tension (Continued)

Temperature Deg F (C)	Maximum Lbs (Kg)	Minimum Lbs (Kg)*[1]
96°F (36°C)	175 lb (79 kg)	140 lb (64 kg)
98°F (37°C)	176 lb (80 kg)	141 lb (64 kg)
100°F (38°C)	178 lb (81 kg)	143 lb (65 kg)
102°F (39°C)	180 lb (82 kg)	145 lb (66 kg)
104°F (40°C)	181 lb (82 kg)	146 lb (66 kg)
106°F (41°C)	183 lb (83 kg)	148 lb (67 kg)
108°F (42°C)	185 lb (84 kg)	150 lb (68 kg)
110°F (43°C)	187 lb (85 kg)	152 lb (69 kg)

*[1] The minimum cable tension applies only during maintenance procedure. This value must not be used as an in-service operational limit for cable tension.

SUBTASK 27-09-14-860-005

(3) For the rudder control cables use a high tension cable tensiometer, COM-1554, or low tension cable tensiometer, COM-1553, to measure the cable tension (Table 203).

Table 203/27-09-14-993-807 Rudder Cable Rigging Load

Temp Deg F (C)	Nominal Lbs (Kg)	Minimum Lbs (Kg)*[1]	Maximum Lbs (Kg)
110°F (43.3°C)	229 lb (103.9 kg)	199 lb (90.3 kg)	234 lb (106.1 kg)
108°F (42.2°C)	228 lb (103.4 kg)	198 lb (89.8 kg)	233 lb (105.7 kg)
106°F (41.1°C)	227 lb (103.0 kg)	197 lb (89.4 kg)	232 lb (105.2 kg)
104°F (40.0°C)	226 lb (102.5 kg)	196 lb (88.9 kg)	231 lb (104.8 kg)
102°F (38.9°C)	224 lb (101.6 kg)	194 lb (88.0 kg)	229 lb (103.9 kg)
100°F (37.8°C)	223 lb (101.2 kg)	193 lb (87.5 kg)	228 lb (103.4 kg)
98°F (36.7°C)	222 lb (100.7 kg)	192 lb (87.1 kg)	227 lb (103.0 kg)
96°F (35.6°C)	220 lb (99.8 kg)	190 lb (86.2 kg)	225 lb (102.1 kg)
94°F (34.4°C)	219 lb (99.3 kg)	189 lb (85.7 kg)	224 lb (101.6 kg)
92°F (33.3°C)	218 lb (98.9 kg)	188 lb (85.3 kg)	223 lb (101.2 kg)
90°F (32.2°C)	217 lb (98.4 kg)	137 lb (62.1 kg)	222 lb (100.7 kg)
88°F (31.1°C)	215 lb (97.5 kg)	135 lb (61.2 kg)	220 lb (99.8 kg)
86°F (30.0°C)	214 lb (97.1 kg)	134 lb (60.8 kg)	219 lb (99.3 kg)
84°F (28.9°C)	213 lb (96.6 kg)	133 lb (60.3 kg)	218 lb (98.9 kg)
82°F (27.8°C)	211 lb (95.7 kg)	131 lb (59.4 kg)	216 lb (98.0 kg)
80°F (26.7°C)	210 lb (95.3 kg)	130 lb (59.0 kg)	215 lb (97.5 kg)
78°F (25.6°C)	209 lb (94.8 kg)	129 lb (58.5 kg)	214 lb (97.1 kg)
76°F (24.4°C)	208 lb (94.3 kg)	128 lb (58.1 kg)	213 lb (96.6 kg)
74°F (23.3°C)	206 lb (93.4 kg)	126 lb (57.2 kg)	211 lb (95.7 kg)
72°F (22.2°C)	205 lb (93.0 kg)	125 lb (56.7 kg)	210 lb (95.3 kg)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 203/27-09-14-993-807 Rudder Cable Rigging Load (Continued)

Temp Deg F (C)	Nominal Lbs (Kg)	Minimum Lbs (Kg)*[1]	Maximum Lbs (Kg)
70°F (21.1°C)	204 lb (92.5 kg)	124 lb (56.2 kg)	209 lb (94.8 kg)
68°F (20.0°C)	203 lb (92.1 kg)	123 lb (55.8 kg)	208 lb (94.3 kg)
66°F (18.9°C)	201 lb (91.2 kg)	121 lb (54.9 kg)	206 lb (93.4 kg)
64°F (17.8°C)	200 lb (90.7 kg)	120 lb (54.4 kg)	205 lb (93.0 kg)
62°F (16.7°C)	199 lb (90.3 kg)	119 lb (54.0 kg)	204 lb (92.5 kg)
60°F (15.6°C)	197 lb (89.4 kg)	117 lb (53.1 kg)	202 lb (91.6 kg)
58°F (14.4°C)	196 lb (88.9 kg)	116 lb (52.6 kg)	201 lb (91.2 kg)
56°F (13.3°C)	195 lb (88.5 kg)	115 lb (52.2 kg)	200 lb (90.7 kg)
54°F (12.2°C)	194 lb (88.0 kg)	114 lb (51.7 kg)	199 lb (90.3 kg)
52°F (11.1°C)	192 lb (87.1 kg)	112 lb (50.8 kg)	197 lb (89.4 kg)
50°F (10.0°C)	191 lb (86.6 kg)	111 lb (50.3 kg)	196 lb (88.9 kg)
48°F (8.9°C)	190 lb (86.2 kg)	110 lb (49.9 kg)	195 lb (88.5 kg)
46°F (7.8°C)	188 lb (85.3 kg)	108 lb (49.0 kg)	193 lb (87.5 kg)
44°F (6.7°C)	187 lb (84.8 kg)	107 lb (48.5 kg)	192 lb (87.1 kg)
42°F (5.6°C)	186 lb (84.4 kg)	106 lb (48.1 kg)	191 lb (86.6 kg)
40°F (4.4°C)	185 lb (83.9 kg)	105 lb (47.6 kg)	190 lb (86.2 kg)
38°F (3.3°C)	183 lb (83.0 kg)	153 lb (69.4 kg)	188 lb (85.3 kg)
36°F (2.2°C)	182 lb (82.6 kg)	152 lb (68.9 kg)	187 lb (84.8 kg)
34°F (1.1°C)	181 lb (82.1 kg)	151 lb (68.5 kg)	186 lb (84.4 kg)
32°F (0.0°C)	179 lb (81.2 kg)	149 lb (67.6 kg)	184 lb (83.5 kg)
30°F (-1.1°C)	178 lb (80.7 kg)	148 lb (67.1 kg)	183 lb (83.0 kg)
28°F (-2.2°C)	177 lb (80.3 kg)	147 lb (66.7 kg)	182 lb (82.6 kg)
26°F (-3.3°C)	176 lb (79.8 kg)	146 lb (66.2 kg)	181 lb (82.1 kg)
24°F (-4.4°C)	174 lb (78.9 kg)	144 lb (65.3 kg)	179 lb (81.2 kg)
22°F (-5.6°C)	173 lb (78.5 kg)	143 lb (64.9 kg)	178 lb (80.7 kg)
20°F (-6.7°C)	172 lb (78.0 kg)	142 lb (64.4 kg)	177 lb (80.3 kg)
18°F (-7.8°C)	171 lb (77.6 kg)	141 lb (64.0 kg)	176 lb (79.8 kg)
16°F (-8.9°C)	169 lb (76.7 kg)	139 lb (63.0 kg)	174 lb (78.9 kg)
14°F (-10.0°C)	168 lb (76.2 kg)	138 lb (62.6 kg)	173 lb (78.5 kg)
12°F (-11.1°C)	167 lb (75.7 kg)	137 lb (62.1 kg)	172 lb (78.0 kg)
10°F (-12.2°C)	165 lb (74.8 kg)	135 lb (61.2 kg)	170 lb (77.1 kg)
8°F (-13.3°C)	164 lb (74.4 kg)	134 lb (60.8 kg)	169 lb (76.7 kg)
6°F (-14.4°C)	163 lb (73.9 kg)	133 lb (60.3 kg)	168 lb (76.2 kg)
4°F (-15.6°C)	162 lb (73.5 kg)	132 lb (59.9 kg)	167 lb (75.7 kg)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 203/27-09-14-993-807 Rudder Cable Rigging Load (Continued)

Temp Deg F (C)	Nominal Lbs (Kg)	Minimum Lbs (Kg)*[1]	Maximum Lbs (Kg)
2°F (-16.7°C)	160 lb (72.6 kg)	130 lb (59.0 kg)	165 lb (74.8 kg)
0°F (-17.8°C)	159 lb (72.1 kg)	129 lb (58.5 kg)	164 lb (74.4 kg)
-2°F (-18.9°C)	158 lb (71.7 kg)	128 lb (58.1 kg)	163 lb (73.9 kg)
-4°F (-20.0°C)	156 lb (70.8 kg)	126 lb (57.2 kg)	161 lb (73.0 kg)
-6°F (-21.1°C)	155 lb (70.3 kg)	125 lb (56.7 kg)	160 lb (72.6 kg)
-8°F (-22.2°C)	154 lb (69.9 kg)	124 lb (56.2 kg)	159 lb (72.1 kg)
-10°F (-23.3°C)	153 lb (69.4 kg)	123 lb (55.8 kg)	158 lb (71.7 kg)
-12°F (-24.4°C)	151 lb (68.5 kg)	121 lb (54.9 kg)	156 lb (70.8 kg)
-14°F (-25.6°C)	150 lb (68.0 kg)	120 lb (54.4 kg)	155 lb (70.3 kg)
-16°F (-26.7°C)	149 lb (67.6 kg)	119 lb (54.0 kg)	154 lb (69.9 kg)
-18°F (-27.8°C)	147 lb (66.7 kg)	117 lb (53.1 kg)	152 lb (68.9 kg)
-20°F (-28.9°C)	146 lb (66.2 kg)	116 lb (52.6 kg)	151 lb (68.5 kg)
-22°F (-30.0°C)	145 lb (65.8 kg)	115 lb (52.2 kg)	150 lb (68.0 kg)
-24°F (-31.1°C)	144 lb (65.3 kg)	114 lb (51.7 kg)	149 lb (67.6 kg)
-26°F (-32.2°C)	142 lb (64.4 kg)	112 lb (50.8 kg)	147 lb (66.7 kg)
-28°F (-33.3°C)	141 lb (64.0 kg)	111 lb (50.3 kg)	146 lb (66.2 kg)
-30°F (-34.4°C)	140 lb (63.5 kg)	110 lb (49.9 kg)	145 lb (65.8 kg)
-32°F (-35.6°C)	139 lb (63.0 kg)	109 lb (49.4 kg)	144 lb (65.3 kg)
-34°F (-36.7°C)	137 lb (62.1 kg)	107 lb (48.5 kg)	142 lb (64.4 kg)
-36°F (-37.8°C)	136 lb (61.7 kg)	106 lb (48.1 kg)	141 lb (64.0 kg)
-38°F (-38.9°C)	135 lb (61.2 kg)	105 lb (47.6 kg)	140 lb (63.5 kg)
-40°F (-40.0°C)	133 lb (60.3 kg)	103 lb (46.7 kg)	138 lb (62.6 kg)

*[1] The minimum cable tension applies only during maintenance procedure. This value must not be used as an in-service operational limit for cable tension.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROL CABLES AIR PRESSURE SEALS - MAINTENANCE PRACTICES

1. General

A. This procedure gives the location of the air pressure seals for the control cables of the flight control systems.

TASK 27-09-17-860-801

2. Location of the Flight Control Cables Air Pressure Seals

(Figure 201)

A. References

Reference	Title
20-10-04-000-802	Control Cable Half Ball Pressure Seal Removal (P/B 401)
20-10-04-400-802	Control Cable Half Ball Pressure Seal Installation (P/B 401)

B. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

C. Location of the FLight Control Cables Air Pressure Seals

SUBTASK 27-09-17-860-001

(1) Find the air pressure seals (Table 201).

Table 201/27-09-17-993-801 Air Pressure Seals

Cable System	Number of Seals	Location	Access Panel Number
Left Elevator Cables	2	Aft Pressure Bulkhead	311BL
Right Elevator Cables	2	Aft Pressure Bulkhead	311BL
Rudder Cables	2	Aft Pressure Bulkhead	311BL
Stabilizer Trim Cables	2	Aft Pressure Bulkhead	311BL

SUBTASK 27-09-17-960-001

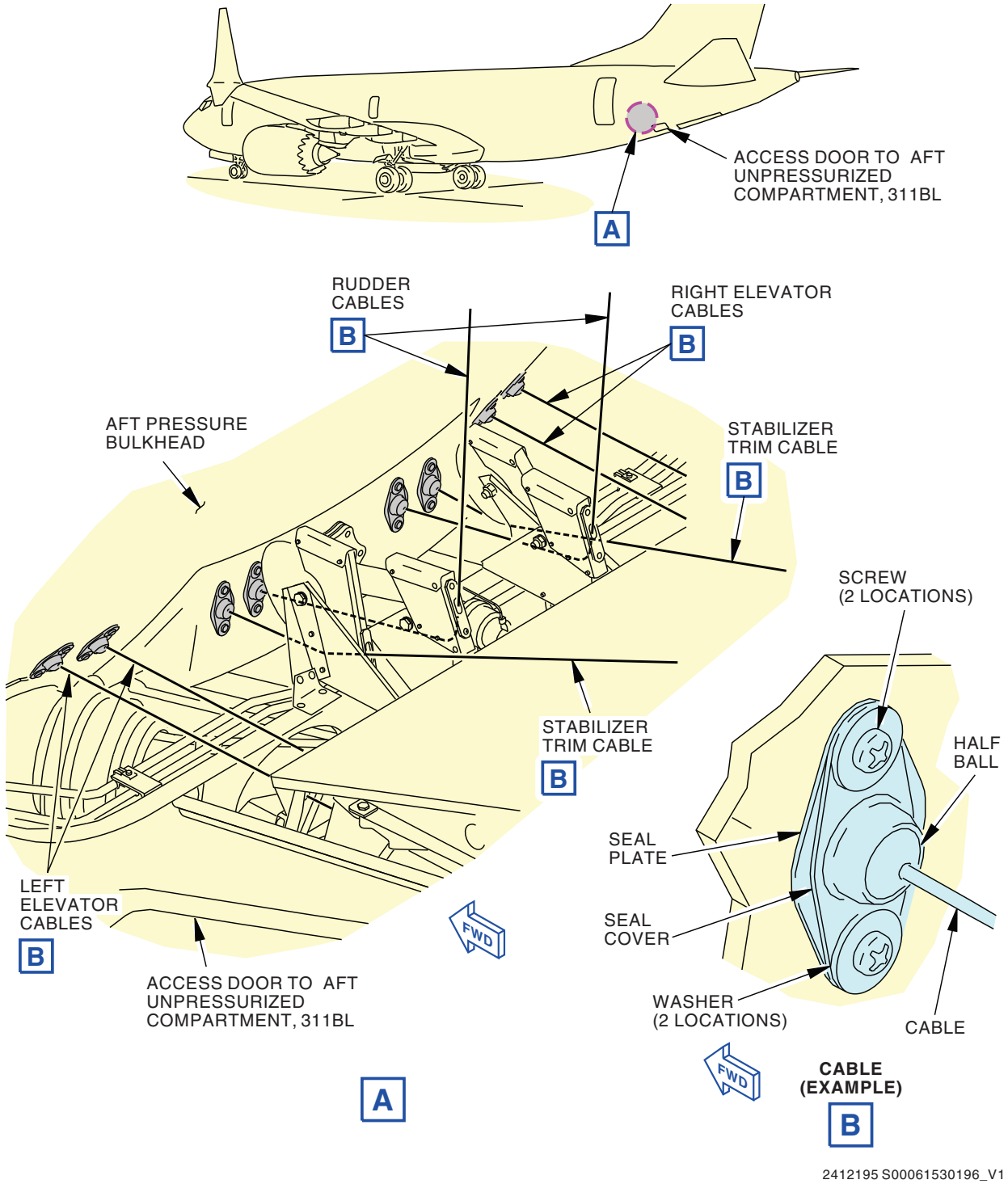
(2) If it is necessary to remove the seals, do this task: Control Cable Half Ball Pressure Seal Removal, TASK 20-10-04-000-802.

SUBTASK 27-09-17-960-002

(3) If it is necessary to install the seals, do this task: Control Cable Half Ball Pressure Seal Installation, TASK 20-10-04-400-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Control Cable Air Pressure Seals
Figure 201/27-09-17-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROL CABLE GUARD - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) The removal of a cable guard.

(2) The installation of a cable guard.

TASK 27-09-18-000-801

2. Control Cable Guard Removal

(Figure 401)

A. References

Reference	Title
25-22-00-000-801	Passenger Seat Removal (P/B 401)
25-27-15-000-801	Carpet Removal (P/B 401)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
38-42-00-800-801	Potable Water System - Deactivation (P/B 201)
50-10-03-000-801	Cargo Compartment Ceiling Liner Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

C. Access Panels

Number	Name/Location
822	Aft Cargo Door
822AR	Access Panel on Aft Cargo Door - External
822BR	Access Panel on Aft Cargo Door

D. Procedure

SUBTASK 27-09-18-010-001

(1) Open these access panels:

Number	Name/Location
822	Aft Cargo Door
822AR	Access Panel on Aft Cargo Door - External
822BR	Access Panel on Aft Cargo Door

SUBTASK 27-09-18-860-001

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-09-18-860-002

(3) Do this task: Potable Water System - Deactivation, TASK 38-42-00-800-801.

SUBTASK 27-09-18-940-001

(4) Attach DO-NOT-OPERATE tags to the control columns.

SUBTASK 27-09-18-010-002

(5) Get access to the aft cable guards.

(a) Remove the aft bulkhead lining in the aft cargo area.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-18-020-001

(6) Remove the aft cable guards (Figure 401).

(a) Loosen the clamps and remove the cable guards.

SUBTASK 27-09-18-010-003

CAUTION

DO NOT REMOVE THE TAPE FROM THE SCREWS ON THE CEILING LINING. IF YOU REMOVE THE TAPE, YOU CAN CAUSE DAMAGE TO THE CEILING LINING.

(7) Get access to the cable guards, hydraulic tubes and the water tubes in the cargo ceiling.

NOTE: Go through the tape with a screwdriver to loosen the screws.

(a) If it is necessary, remove the ceiling lining in the aft cargo area (TASK 50-10-03-000-801).

SUBTASK 27-09-18-010-004

(8) Get access to the hydraulic tube.

(a) If it is necessary, remove the seats on the left side of the passenger area (TASK 25-22-00-000-801).

(b) If it is necessary, remove the carpets on the left side of the passenger area (TASK 25-27-15-000-801).

(c) If it is necessary, remove the floor panels on the left side of the passenger area (TASK 53-21-00-000-801).

SUBTASK 27-09-18-020-002

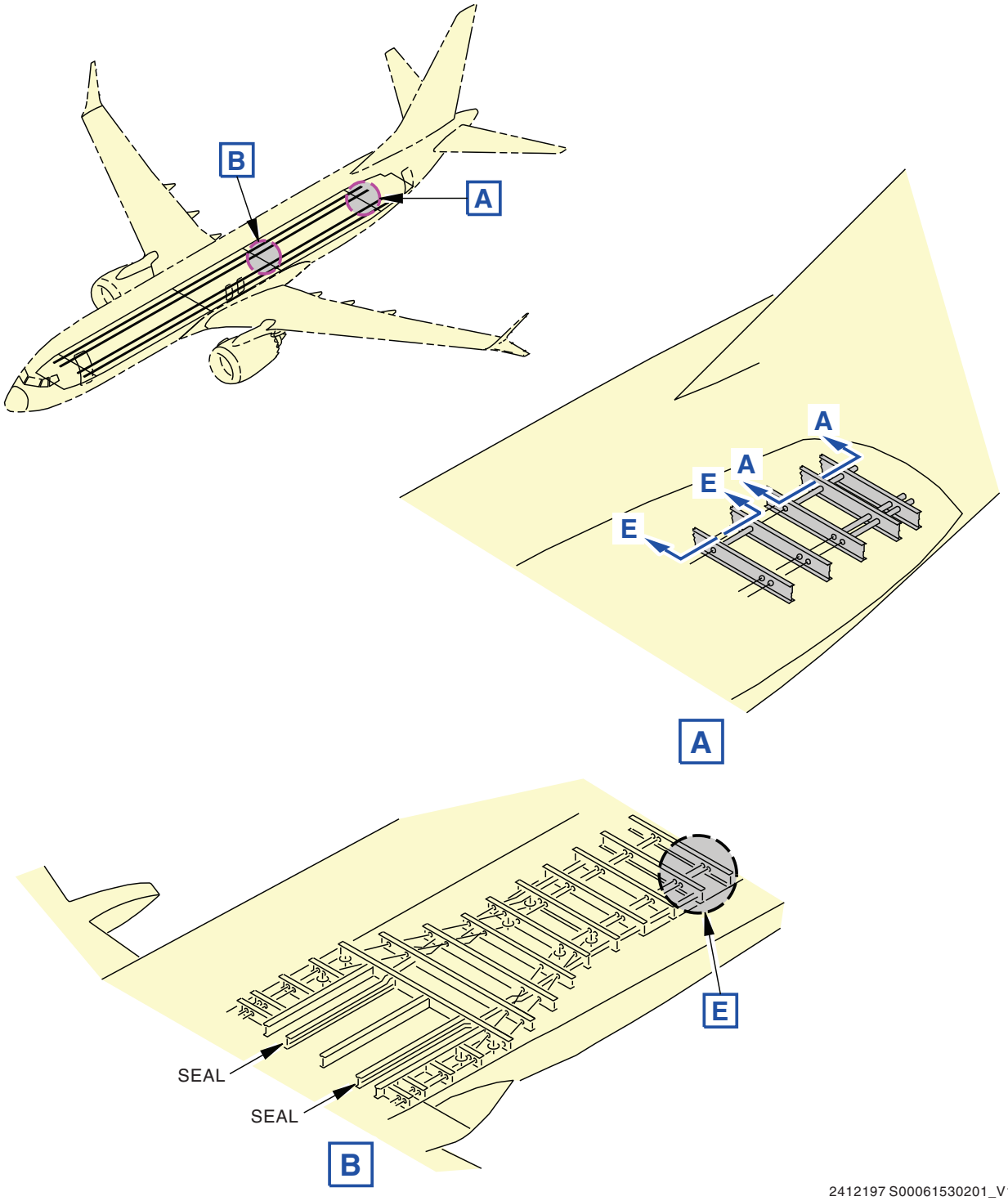
(9) Remove the water pressure tube in the aft cargo area (Figure 401).

SUBTASK 27-09-18-020-003

(10) Remove the hydraulic tube and the cable guards in the aft cargo area (Figure 401).

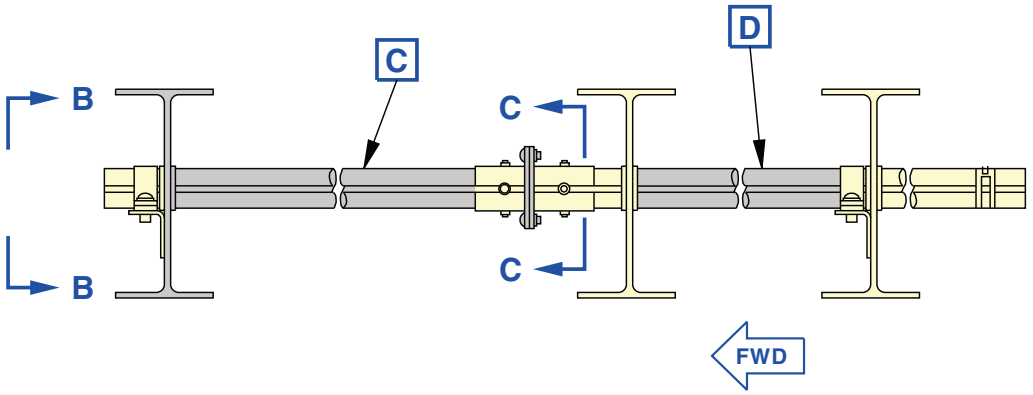
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

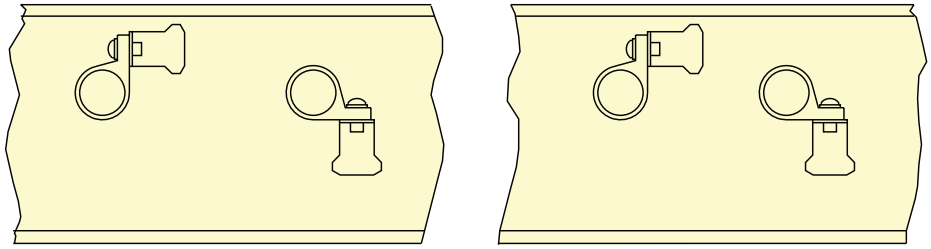


Control Cable Guard Installation
Figure 401/27-09-18-990-801 (Sheet 1 of 4)

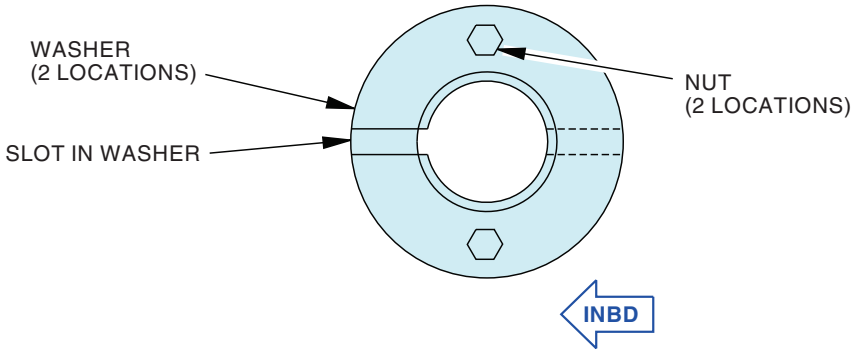
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

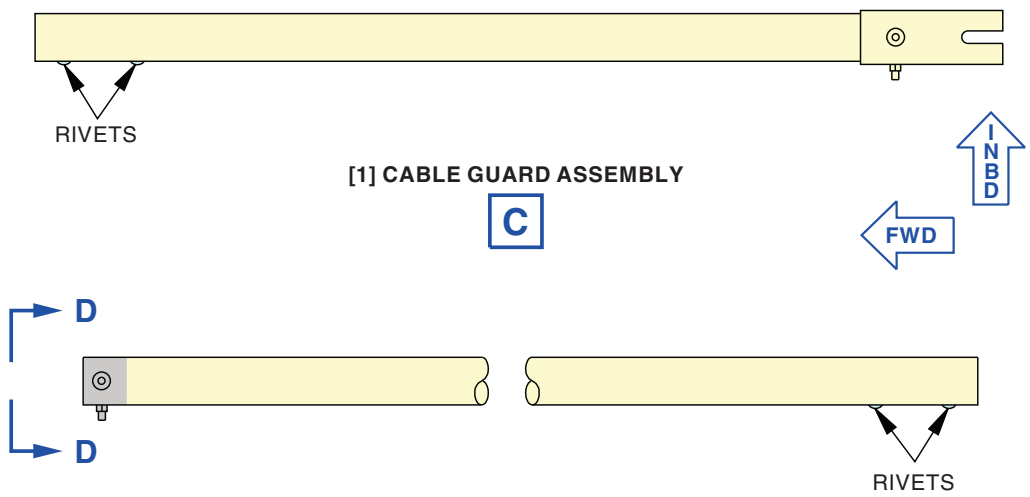


C-C

2412198 S00061530202_V2

Control Cable Guard Installation
Figure 401/27-09-18-990-801 (Sheet 2 of 4)

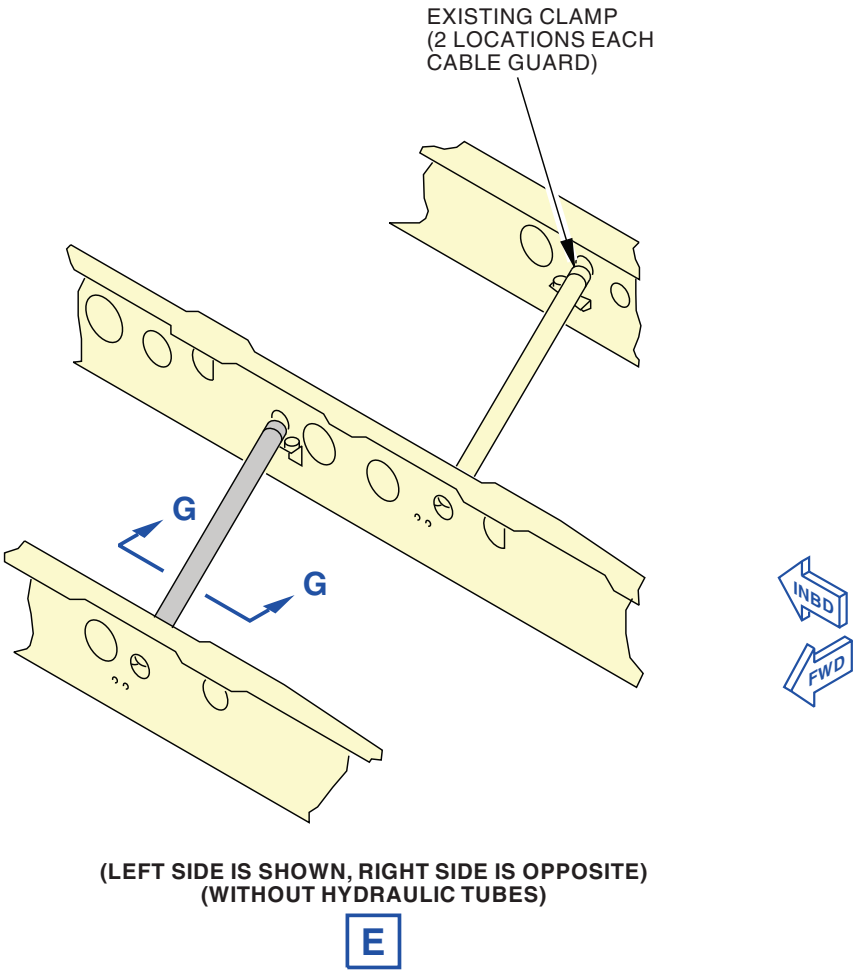
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412199 S00061530203_V2

Control Cable Guard Installation
Figure 401/27-09-18-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)
(WITHOUT HYDRAULIC TUBES)

E

2412200 S00061530204_V1

Control Cable Guard Installation
Figure 401/27-09-18-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-18-400-801

3. Control Cable Guard Installation

A. References

Reference	Title
12-26-00-600-801	Control Cable Lubrication (P/B 301)
25-22-00-400-801	Passenger Seat Installation (P/B 401)
25-27-15-400-801	Carpet - Installation (P/B 401)
27-31-00 P/B 501	ELEVATOR AND ELEVATOR TRIM CONTROL SYSTEM - ADJUSTMENT/TEST
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
38-42-00-800-802	Potable Water System - Activation (P/B 201)
50-10-03-400-801	Cargo Compartment Ceiling Liner Installation (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
G50344	Tape - Nitto P-212HD WHT (Formerly Permacel) Glass Cloth	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cable guard assembly	27-09-18-02-045	PTF ALL
		27-09-18-02-070	PTF ALL
		27-09-18-08-015	PTF ALL
2	Cable guard assembly	27-09-18-02-010	PTF ALL
		27-09-18-02-015	PTF ALL
		27-09-18-08-020	PTF ALL
3	Cable guard assembly	27-09-18-01-025	PTF ALL

D. Location Zones

Zone	Area
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

E. Access Panels

Number	Name/Location
822	Aft Cargo Door
822AR	Access Panel on Aft Cargo Door - External
822BR	Access Panel on Aft Cargo Door

F. Procedure

SUBTASK 27-09-18-640-001

(1) Lubricate the cables in the cable guard area. Do this task: Control Cable Lubrication, TASK 12-26-00-600-801.

SUBTASK 27-09-18-420-001

(2) Install the hydraulic tubes (Figure 401).

SUBTASK 27-09-18-420-002

(3) Install the aft cable guards (Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Put the cable guard assembly [1] in its position.

(b) Install the washers on the sleeve of the cable guard assembly [1].

1) Connect each set of washers with a screw and a nut.

(c) Put the cable guard assembly [2] into the sleeve of the cable guard assembly [1].

1) Install the three bolts on the cable guard assembly [2] into the slots on the sleeve of the cable guard assembly [1].

2) Tighten the nut until the bolt does not turn easily.

(d) Put the clamp on the end of the cable guard assembly [2] between the two rivets.

1) Tighten the clamp.

(e) Install the cable guard assembly [3] with the slot on the assembly inboard.

1) Tighten the clamps to hold the guard assemblies.

2) Make sure a minimum clearance of 0.10 in. (2.5 mm) (adjust tubing as necessary) is kept between the cable guard and the hydraulic or the water tubings.

SUBTASK 27-09-18-940-002

(4) Remove the DO-NOT-OPERATE tags from the control columns.

SUBTASK 27-09-18-700-001

(5) Do a test of the cables to make sure there is free movement (PAGEBLOCK 27-31-00/501).

SUBTASK 27-09-18-860-003

(6) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-09-18-790-001

(7) Examine the hydraulic tubes for leaks.

SUBTASK 27-09-18-860-004

(8) Do this task: Potable Water System - Activation, TASK 38-42-00-800-802.

SUBTASK 27-09-18-860-005

(9) Examine the water pressure tube for leaks.

SUBTASK 27-09-18-410-004

(10) Install all floor panels that were removed (TASK 53-21-00-400-801).

SUBTASK 27-09-18-410-001

(11) Install all carpets that were removed (TASK 25-27-15-400-801).

SUBTASK 27-09-18-410-002

(12) Install all seats that were removed (TASK 25-22-00-400-801).

SUBTASK 27-09-18-860-006

(13) Install all cargo compartment lining that was removed (TASK 50-10-03-400-801).

SUBTASK 27-09-18-860-007

(14) Apply new Nitto P-212HD WHT tape, G50344 tape on the screws.

SUBTASK 27-09-18-410-003

(15) Close these access panels:

Number	Name/Location
822	Aft Cargo Door
822AR	Access Panel on Aft Cargo Door - External
822BR	Access Panel on Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS HYDRAULIC MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flight controls hydraulic module.

(2) An installation of the flight controls hydraulic module.

B. The flight controls hydraulic module is referred to as the "hydraulic module" in this procedure.

C. There are two hydraulic modules installed on the airplane. The hydraulic module for the hydraulic system A is found in the left main wheel well. The hydraulic module for the hydraulic system B is found in the right main wheel well.

D. Each hydraulic module contains these components:

(1) A check valve

(2) A low pressure warning switch

(3) A compensator cartridge

(4) A spoiler shutoff valve

(5) A flight controls shutoff valve.

TASK 27-09-21-000-801

2. Flight Controls Hydraulic Module Removal

(Figure 401)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 27-09-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-21-860-001

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-09-21-860-002

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 27-09-21-860-003

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

E. Flight Controls Hydraulic Module Removal

SUBTASK 27-09-21-020-001

(1) Remove the hydraulic module [1]:

(a) Disconnect the electrical connector [13] from the low pressure warning switch [15].

(b) Disconnect the electrical connector [14] from the flight controls shutoff valve [17].

(c) Disconnect the electrical connector [18] from the spoiler shutoff valve [16].

(d) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(e) If there was corrosion, refer to SWPM 20-60-03.

(f) Disconnect the hydraulic tubing from the hydraulic module [1].

1) Use a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110 to catch hydraulic fluid that drains from the hydraulic module [1].

(g) Remove the bolts [6] and washers [7] to disconnect the hydraulic module [1] from the top mounting bracket.

(h) Remove the bolts [8] and washers [9] to disconnect the hydraulic module [1] from the bottom mounting bracket.

(i) Remove the hydraulic module [1] from the airplane.

SUBTASK 27-09-21-420-001

(2) Install protective caps on the ends of the hydraulic tubing.

SUBTASK 27-09-21-020-002

(3) If you will install a new hydraulic module [1], do these steps:

(a) Remove the unions [3] and packings [2] in ports 1, 2 and 5 from the hydraulic module [1].

1) Discard the packing [2].

(b) Remove the unions [5] and packings [4] in ports 4 and 7 from the hydraulic module [1].

1) Discard the packings [4].

(c) Remove the check valve [10] and packing [4] in port 3 from the hydraulic module [1].

1) Discard the packing [4].

(d) Remove the reducer [11] and packing [12] in port 6 from the hydraulic module [1].

1) Discard the packing [12].

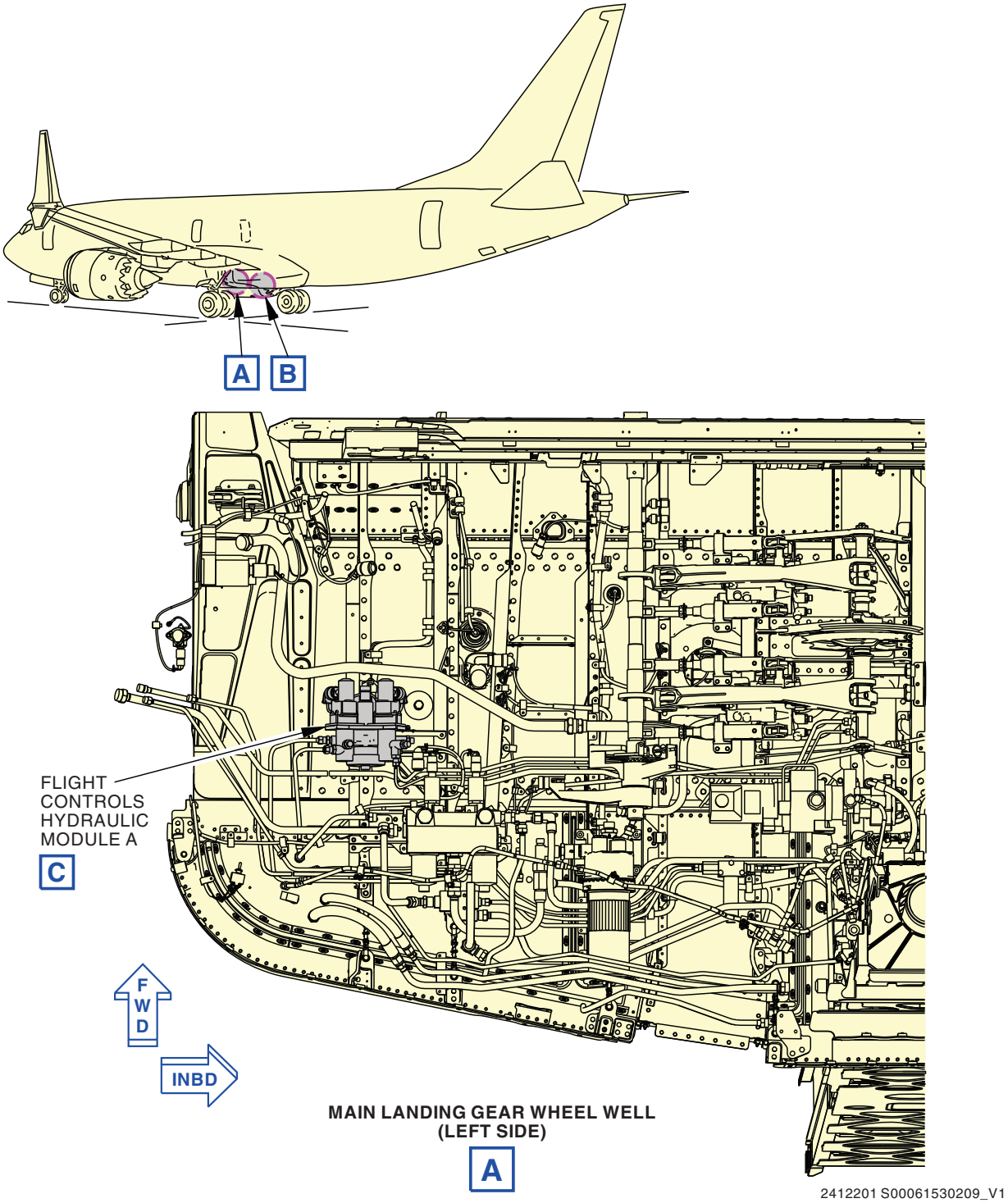
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-21-420-002

(4) Install protective caps in the open ports of the hydraulic module [1].

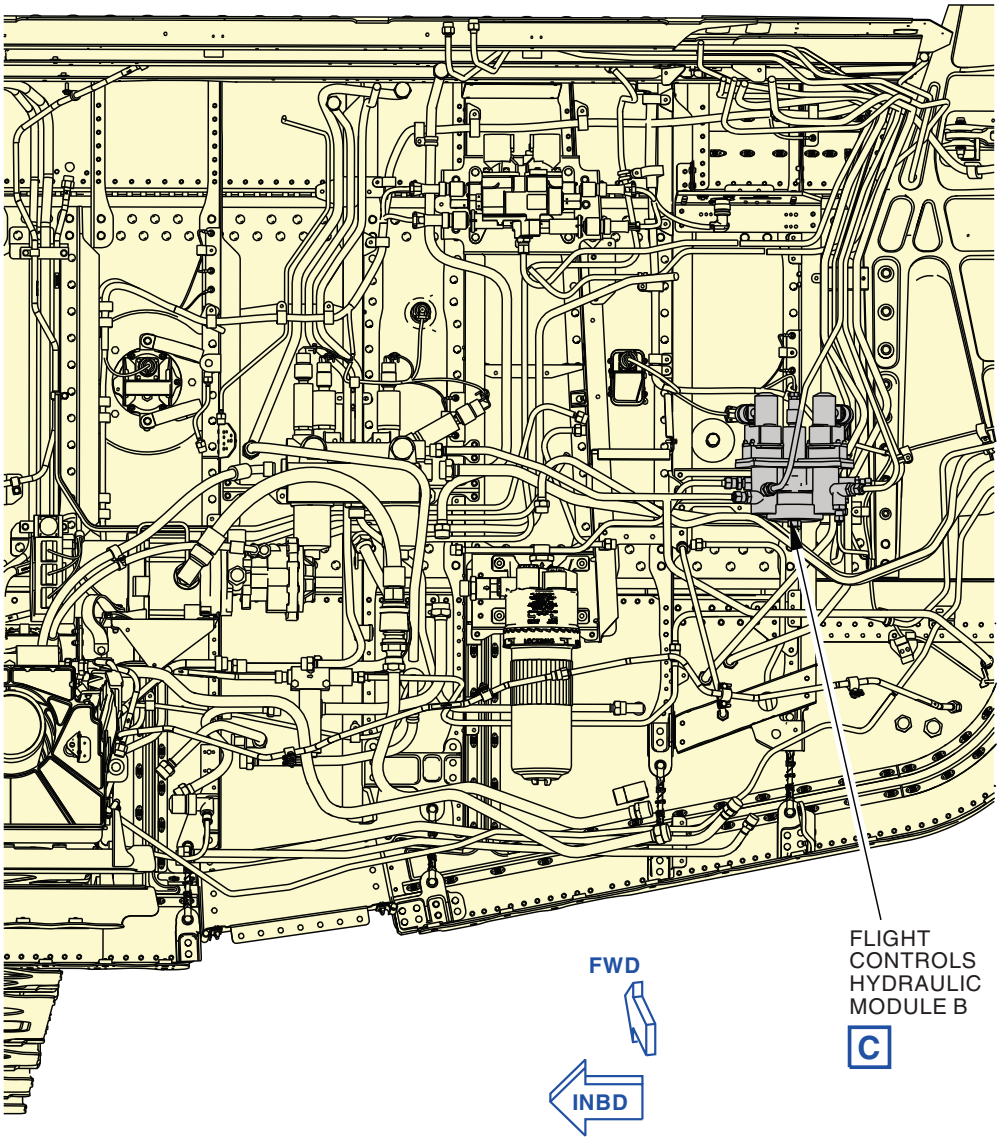
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



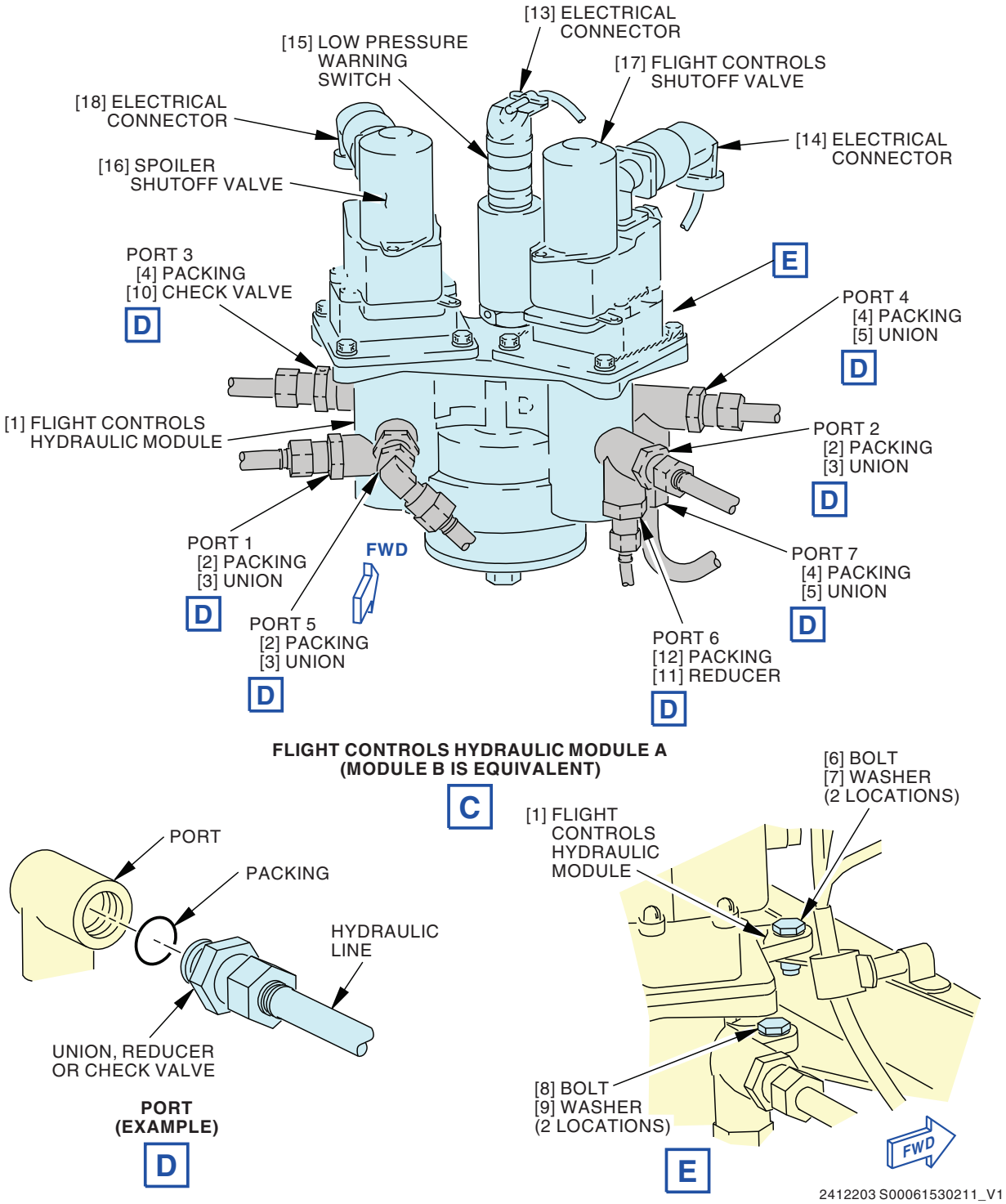
Flight Controls Hydraulic Module Installation
Figure 401/27-09-21-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Hydraulic Module Installation
Figure 401/27-09-21-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Hydraulic Module Installation
Figure 401/27-09-21-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-21-400-801

3. Flight Controls Hydraulic Module Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-09-21-700-801	Flight Controls Hydraulic Module Test (P/B 501)
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Hydraulic module	27-09-21-01-020	PTF ALL
		27-09-21-02-005	PTF ALL
2	Packing	27-09-21-01-045	PTF ALL
4	Packing	27-09-21-01-045	PTF ALL
12	Packing	27-09-21-01-045	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Flight Controls Hydraulic Module Installation

SUBTASK 27-09-21-020-003

(1) Remove protective caps from the open ports of the hydraulic module [1].

SUBTASK 27-09-21-420-003

(2) If they are not installed, install the unions [3], unions [5], check valve [10], reducer [11], packings [2], packings [4], and packing [12] on the hydraulic module [1]:

(a) Apply MCS 352B fluid, D00054 (recommended) or hydraulic fluid, D00153 (alternative) to the unions [3], unions [5], reducer [11] and new packings [2], new packings [4] and new packing [12].

(b) Install the new packings [2] and unions [3] in ports 1, 2 and 5 on the hydraulic module [1].

(c) Install the new packings [4] and unions [5] in ports 4 and 7 on the hydraulic module [1].

(d) Install the new packing [4] and check valve [10] in port 3 on the hydraulic module [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Install the check valve [10] with the free flow arrow pointed away from the hydraulic module [1].

(e) Install the new packing [12] and reducer [11] in port 6 on the hydraulic module [1].

SUBTASK 27-09-21-020-004

(3) Remove the protective caps from the hydraulic tubing.

SUBTASK 27-09-21-420-004

(4) Install the hydraulic module [1]:

(a) Put the hydraulic module [1] on its mounting bracket.

(b) Connect the hydraulic module [1] to the top mounting bracket with the bolts [6] and washers [7].

(c) Connect the hydraulic module [1] to the bottom mounting bracket with the bolts [8] and washers [9].

(d) Connect the hydraulic tubing to the hydraulic module [1].

1) Tighten the B-nuts on the hydraulic lines for ports 3 and 4 to 170 ±9 in-lb (19 ±1 N·m).

2) Tighten the B-nuts on the hydraulic lines for ports 1, 5, 2, and 7 to 270 ±14 in-lb (31 ±2 N·m).

3) Tighten the B-nut on the hydraulic line for port 6 to 140 ±7 in-lb (16 ±1 N·m).

(e) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

1) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

2) If there was corrosion, refer to SWPM 20-60-03, to correct the problem.

(f) Apply the D5026NS or ZC-026 compound, G50171 to the connectors (SWPM 20-60-03).

(g) Connect the electrical connector [18] to the spoiler shutoff valve [16].

(h) Connect the electrical connector [14] to the flight controls shutoff valve [17].

(i) Connect the electrical connector [13] to the low pressure warning switch [15].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-21-860-004

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-21-860-009

(2) Do this task: Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

SUBTASK 27-09-21-860-010

(3) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-21-610-001

(4) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-09-21-730-001

(5) Do this task: Flight Controls Hydraulic Module Test, TASK 27-09-21-700-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS HYDRAULIC MODULE - ADJUSTMENT/TEST

1. General

A. There is one task in this procedure which is the test of the flight controls hydraulic module.

B. The flight controls hydraulic module is referred to as the "hydraulic module" in this procedure.

C. The hydraulic module for hydraulic system A is found in the left main wheel well. The hydraulic module for hydraulic system B is found in the right main wheel well.

TASK 27-09-21-700-801

2. Flight Controls Hydraulic Module Test

(Figure 501, Figure 502)

A. General

(1) In this procedure, the two hydraulic modules [3] are tested at the same time.

B. References

Reference	Title
29-11-00 P/B 201	HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-09-21-860-006

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

SUBTASK 27-09-21-860-007

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize hydraulic systems A and B (PAGEBLOCK 29-11-00/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Flight Controls Hydraulic Module Test

SUBTASK 27-09-21-720-001

(1) Do the test of the hydraulic module [3]:

(a) Make sure the two FLT CONTROL switches, on the P5-3 flight controls panel, are in the ON position.

(b) Make sure the two SPOILER switches, on the P5-3 flight controls panel, are in the ON position.

(c) Make sure the manual override levers [4] of the spoiler shutoff valve [1] and flight controls shutoff valve [2] on the two hydraulic modules [3] are in position 1.

(d) Operate the aileron control wheel, control column, and rudder pedal sets to bleed the hydraulic system.

(e) Make sure there is no leakage of hydraulic fluid at the hydraulic modules [3].

(f) Set the two SPOILER switches to the OFF position.

(g) Make sure the manual override lever [4] of the spoiler shutoff valve [1] on both hydraulic modules [3] is in position 2.

(h) Turn the aileron control wheel.

(i) Make sure the spoilers do not operate.

(j) Set the two FLIGHT CONTROL switches to the OFF position.

(k) Make sure the manual override lever [4] of the flight controls shutoff valve [2] on both hydraulic modules [3] is in position 2.

(l) Make sure the two LOW PRESSURE warning lights, on the P5-3 flight controls panel, come on.

(m) Set the two FLIGHT CONTROL switches to the ON position.

(n) Make sure the two LOW PRESSURE warning lights go off.

(o) Set the two SPOILER switches to the ON position.

SUBTASK 27-09-21-860-008

(2) Remove pressure from hydraulic systems A and B (PAGEBLOCK 29-11-00/201).

SUBTASK 27-09-21-210-001

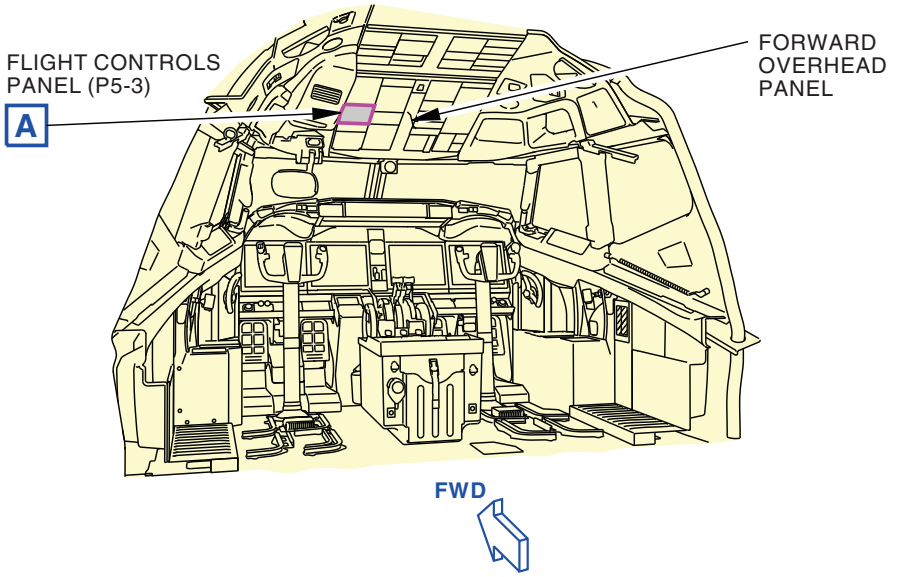
(3) Make sure the LOW PRESSURE lights come on.

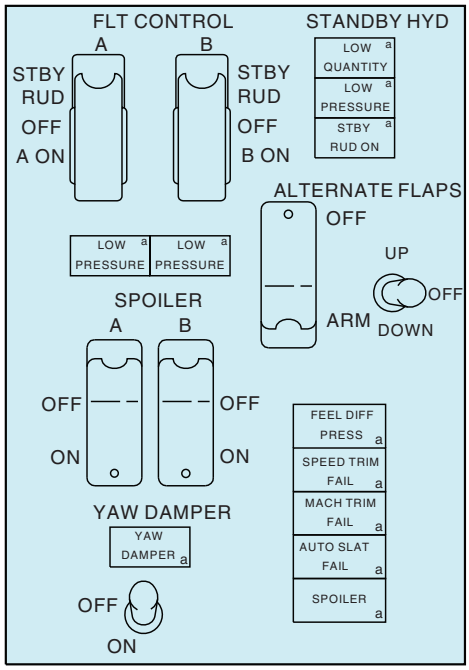
SUBTASK 27-09-21-210-002

(4) Make sure the master warning and caution lights and the flight controls warning light come on.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





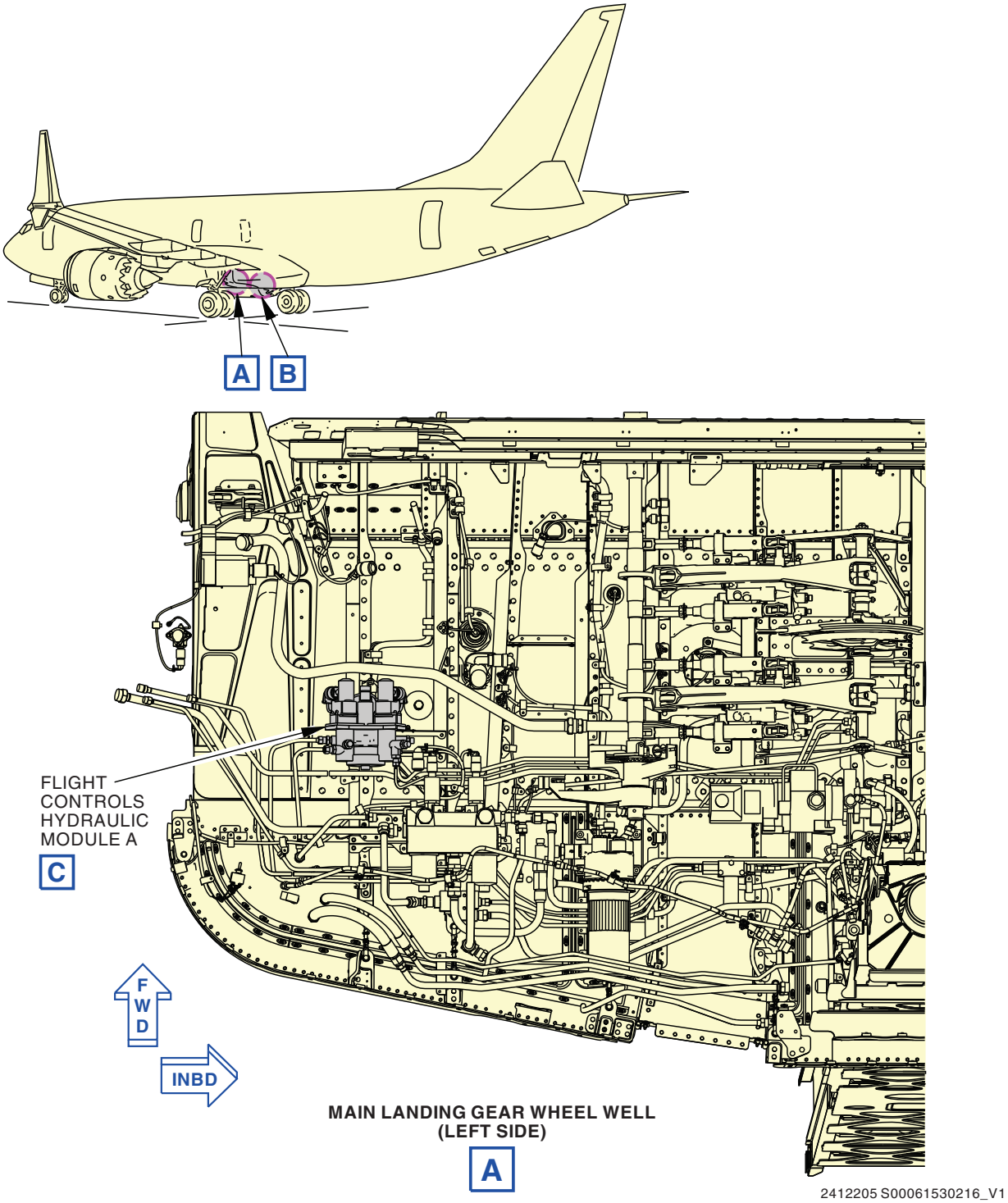
FLIGHT CONTROLS PANEL (P5-3)

A

2412204 S00061530215_V1

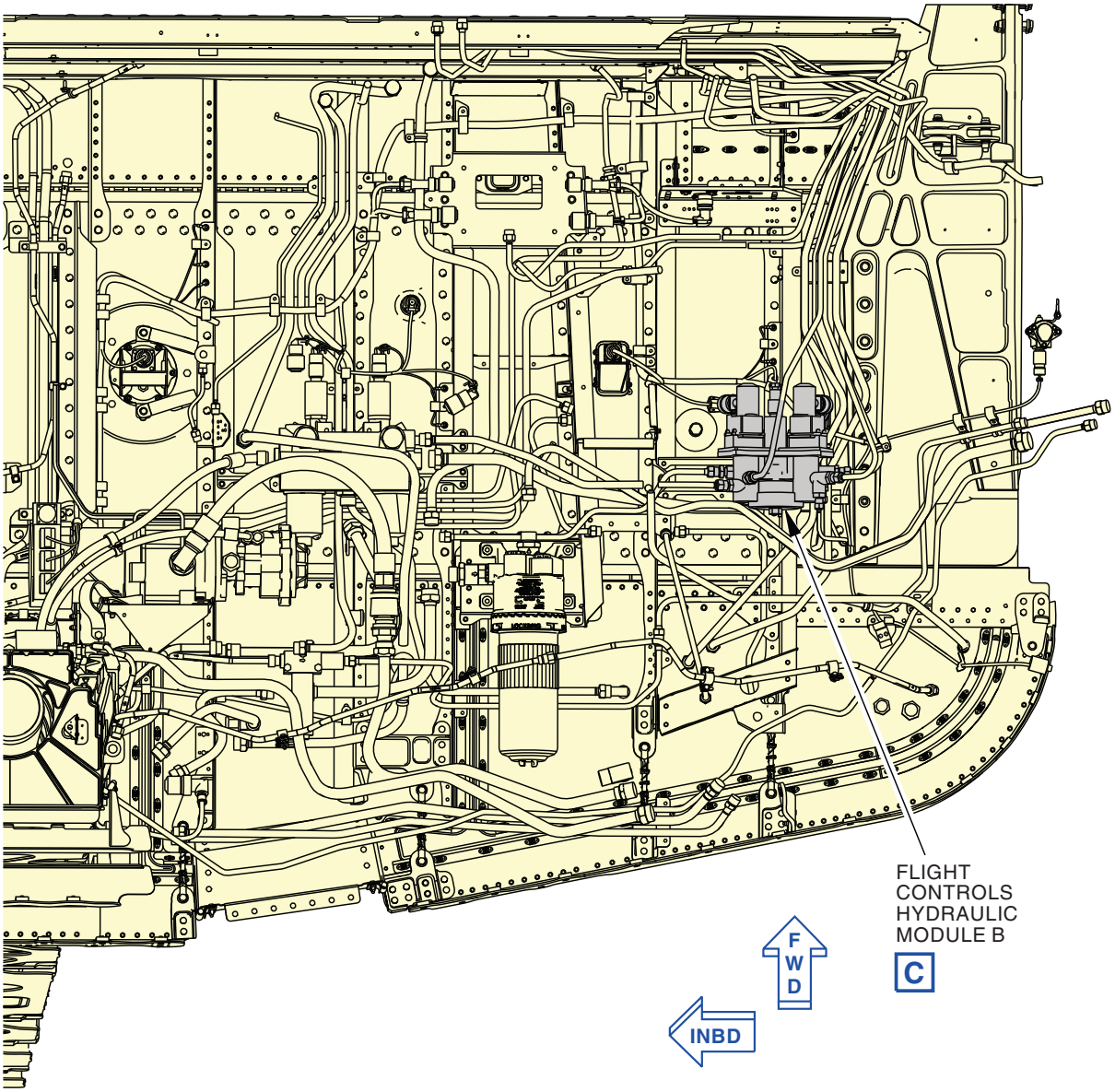
Flight Controls Panel
Figure 501/27-09-21-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



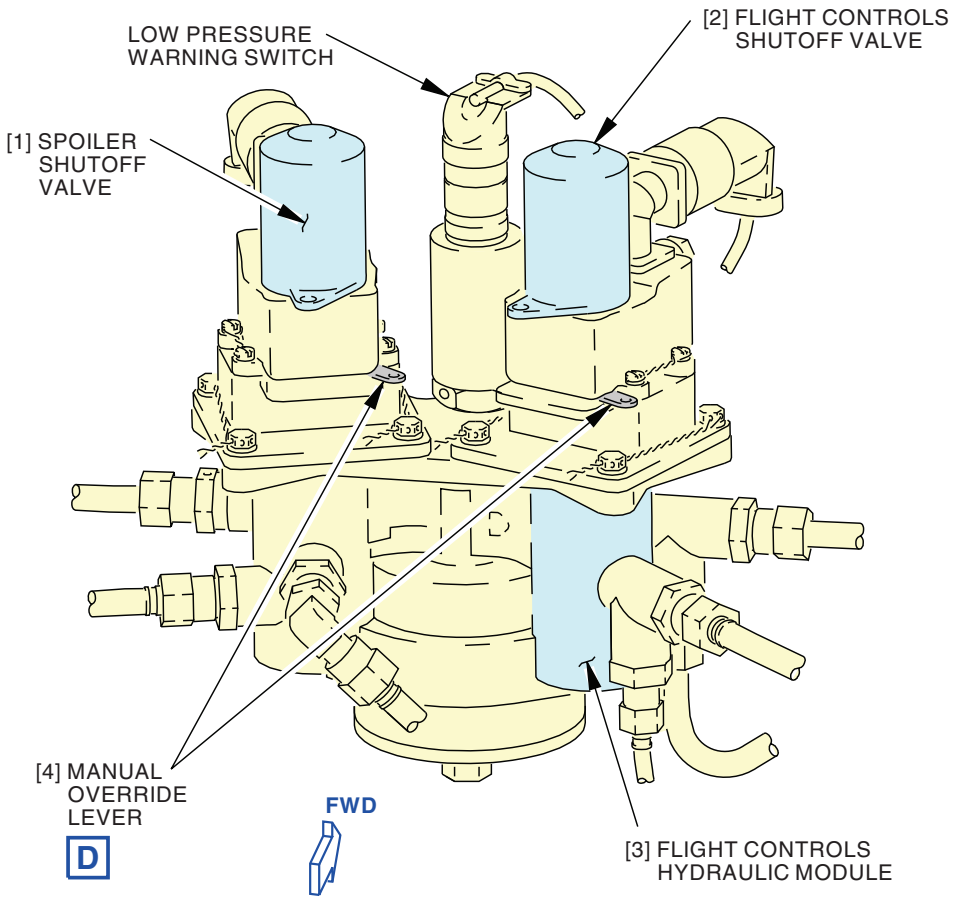
Flight Controls Shutoff Valve Test
Figure 502/27-09-21-990-803 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Shutoff Valve Test
Figure 502/27-09-21-990-803 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



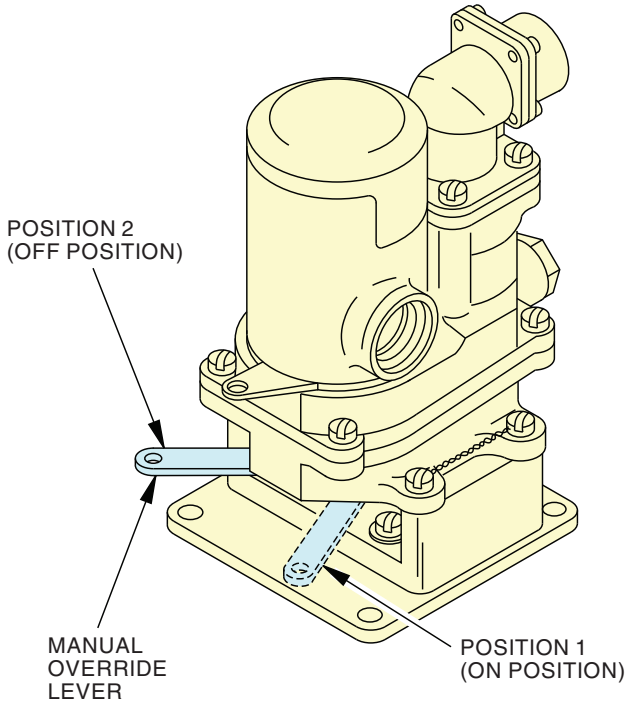
FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2412207 S00061530218_V2

Flight Controls Shutoff Valve Test
Figure 502/27-09-21-990-803 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**MANUAL OVERRIDE LEVER POSITION
(EXAMPLE)**

D

2844536 S0000662328_V3

**Flight Controls Shutoff Valve Test
Figure 502/27-09-21-990-803 (Sheet 4 of 4)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS SHUTOFF VALVE AND VALVE MOTOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the flight controls shutoff valve assembly (the shutoff valve and the shutoff valve motor).

(2) An installation of the flight controls shutoff valve assembly (the shutoff valve and the shutoff valve motor).

(3) A removal of the flight controls shutoff valve motor.

(4) An installation of the flight controls shutoff valve motor.

B. The flight controls shutoff valve is part of the flight controls hydraulic module. The hydraulic module for the hydraulic system A is found in the left main wheel well. The hydraulic module for the hydraulic system B is found in the right main wheel well.

C. The flight controls shutoff valve is referred to as the shutoff valve and the flight controls shutoff valve motor is referred to as the shutoff valve motor in this procedure.

D. You can replace the shutoff valve motor with the shutoff valve installed in the flight controls hydraulic module.

TASK 27-09-31-000-801

2. Flight Controls Shutoff Valve Removal

(Figure 401)

A. General

(1) This task is the removal of the flight controls shutoff valve assembly, that contains the shutoff valve motor and shutoff valve.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1154	Container - 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 27-09-31-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-09-31-860-001

(2) If there is power on hydraulic system A or B, remove hydraulic power for applicable hydraulic system (TASK 29-11-00-860-805).

NOTE: Remove power from hydraulic system A for the left shutoff valve. Remove power from hydraulic system B for the right shutoff valve.

SUBTASK 27-09-31-860-002

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 27-09-31-860-003

(4) If you remove the left shutoff valve, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-31-860-004

(5) If you remove the right shutoff valve, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

F. Flight Controls Shutoff Valve Removal

SUBTASK 27-09-31-010-001

(1) Remove the shutoff valve motor [6], do this task: Flight Controls Shutoff Valve Motor Removal, TASK 27-09-31-000-802.

SUBTASK 27-09-31-020-001

(2) Remove the shutoff valve [8].

(a) Remove the four bolts [2] and washers [3] that attach the shutoff valve [8] to the hydraulic module housing [5].

(b) Put a 5 gallon (19 liter) container, STD-1154, below the shutoff valve [8] to catch hydraulic fluid.

(c) Carefully lift the shutoff valve [8] straight up to remove it from the hydraulic module housing [5].

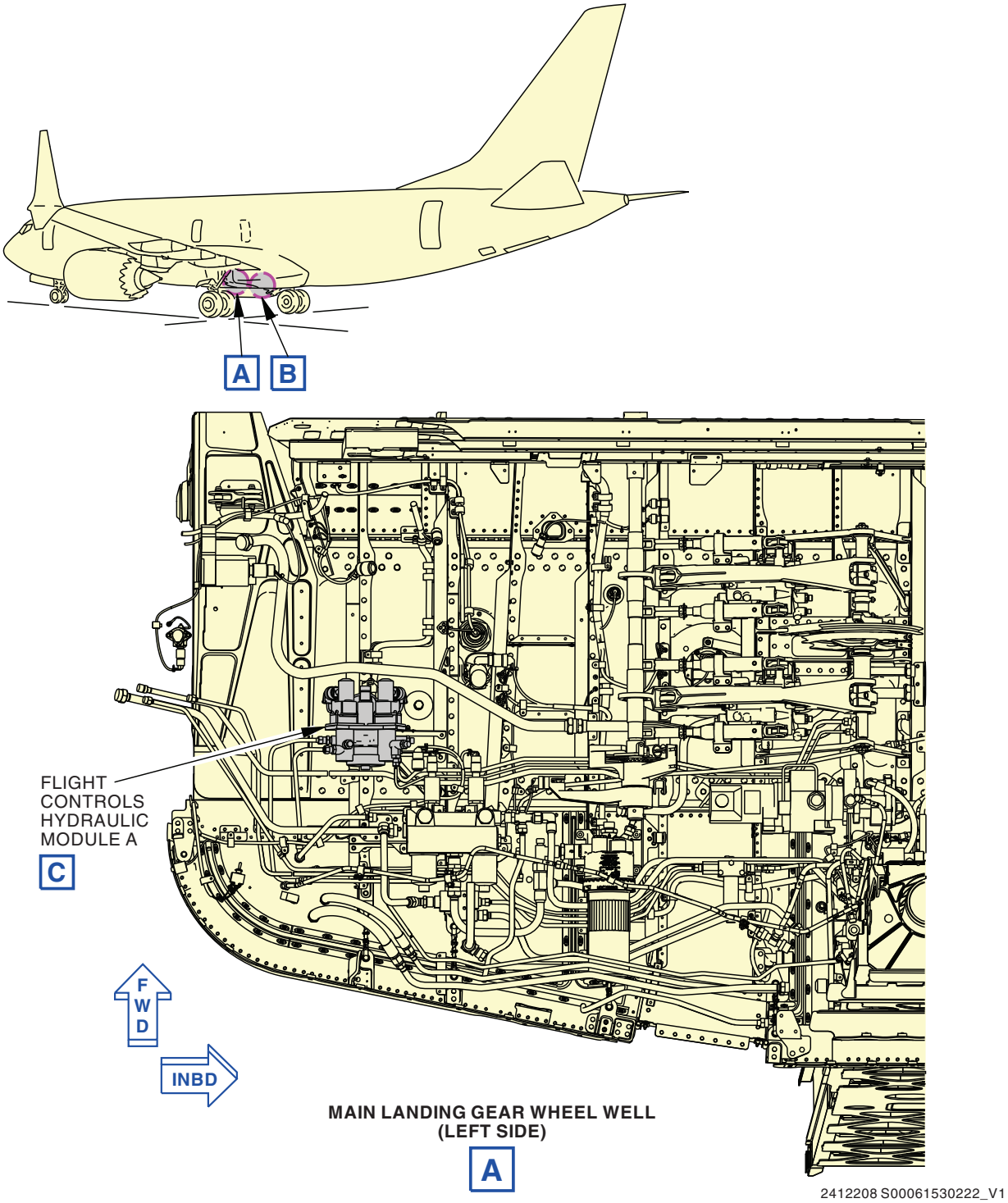
1) Discard the packings [10] and packing [12].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Put a cover on the hydraulic module housing [5] to keep contamination out of the open valve hole.

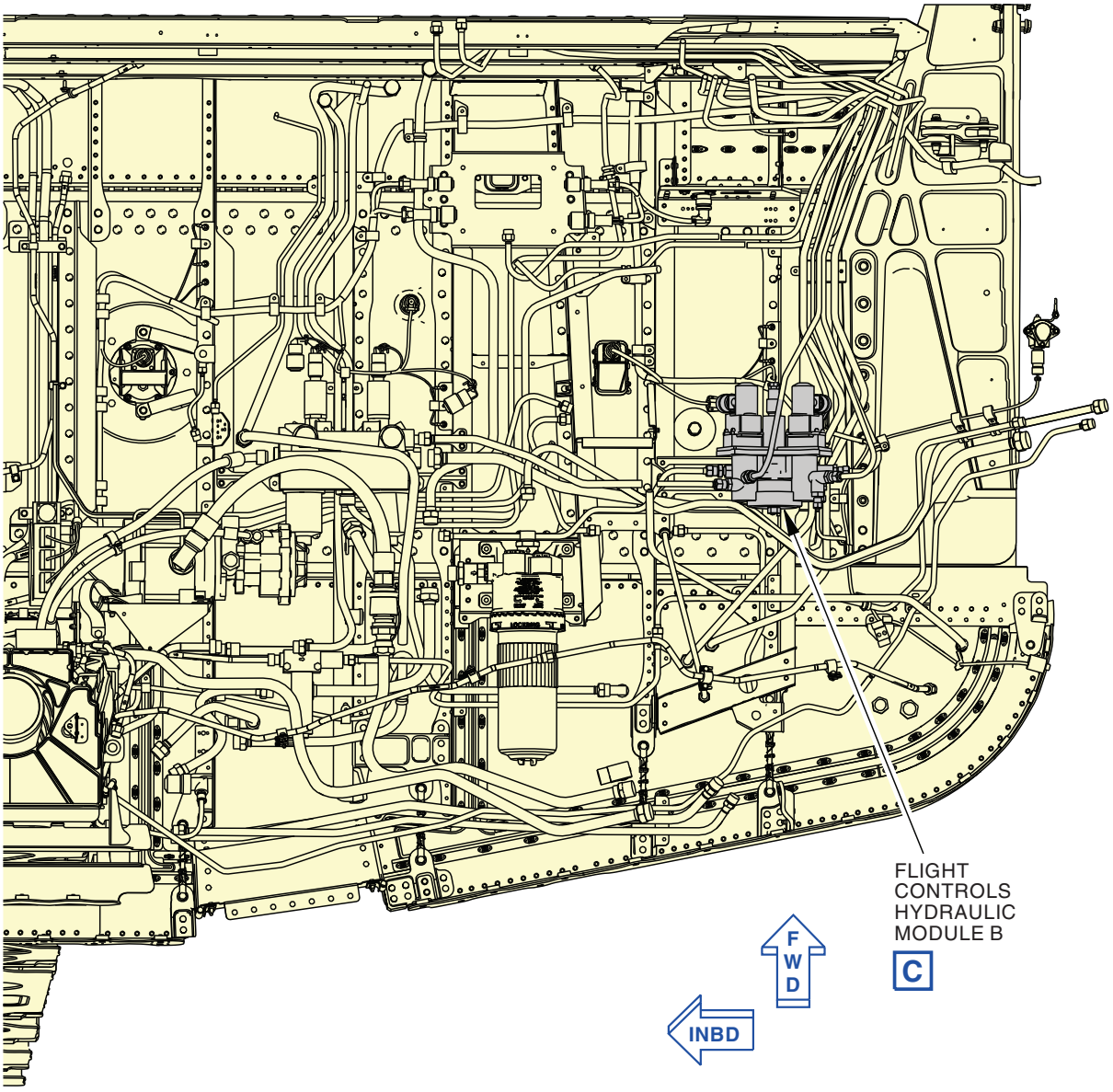
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



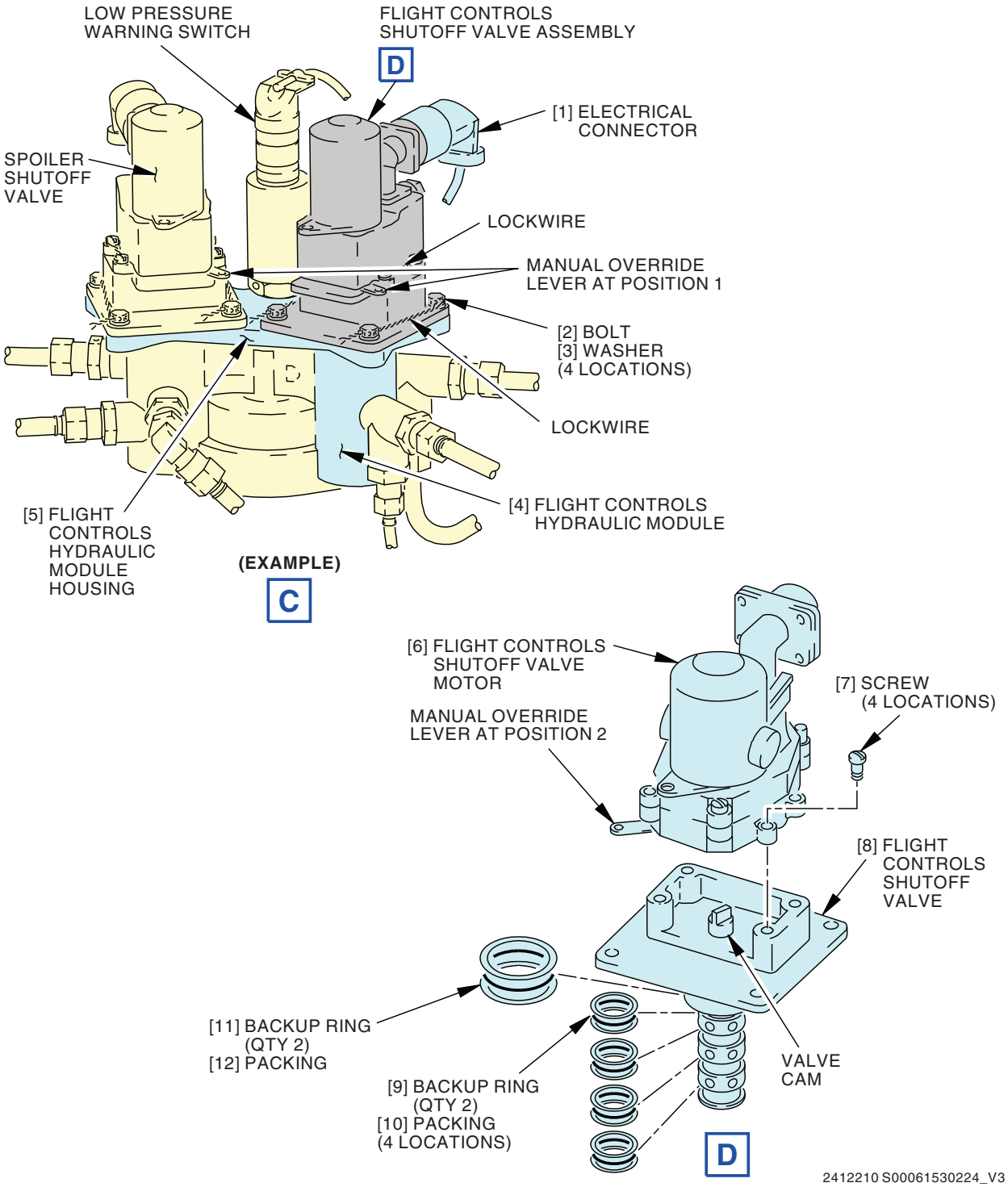
Flight Controls Shutoff Valve Installation
Figure 401/27-09-31-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Shutoff Valve Installation
Figure 401/27-09-31-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Shutoff Valve Installation
Figure 401/27-09-31-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-31-400-801

3. Flight Controls Shutoff Valve Installation

(Figure 401)

A. General

(1) This task is the installation of the flight controls shutoff valve assembly, that contains the shutoff valve motor and shutoff valve.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-09-31-710-801	Flight Controls Shutoff Valve Test (P/B 501)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Shutoff valve	27-09-31-01-065	PTF ALL
9	Backup ring	27-09-31-01-045	PTF ALL
10	Packing	27-09-31-01-060	PTF ALL
11	Backup ring	27-09-31-01-050	PTF ALL
12	Packing	27-09-31-01-055	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Flight Controls Shutoff Valve Installation

SUBTASK 27-09-31-420-001

(1) If they are not installed, install the packings [12], backup rings [11], packings [10], and backup rings [9].

(a) Apply MCS 352B fluid, D00054 (recommended), or hydraulic fluid, D00153 (alternative), to the packings [12], backup rings [11], packings [10], and backup rings [9].

(b) Install the packings [12], backup rings [11], packings [10], and backup rings [9] on the shutoff valve [8].

SUBTASK 27-09-31-420-002

(2) Install the shutoff valve [8].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Carefully install the shutoff valve [8] in the hydraulic module housing [5].

(b) Install the bolts [2] and washers [3] to attach the shutoff valve [8] to the hydraulic module housing [5].

(c) Tighten the bolts [2] to 35 ±5 in-lb (4.0 ±0.6 N·m).

(d) Install the MS20995NC32 lockwire, G01912, on the four bolts [2] (TASK 20-10-44-400-801).

SUBTASK 27-09-31-410-001

(3) Install the shutoff valve motor [6], do this task: Flight Controls Shutoff Valve Motor Installation, TASK 27-09-31-400-802.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-31-860-005

(1) If the left shutoff valve was removed, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-31-860-006

(2) If the right shutoff valve was removed, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-31-610-001

(3) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-09-31-860-016

(4) For the applicable hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-09-31-730-001

(5) Do this task: Flight Controls Shutoff Valve Test, TASK 27-09-31-710-801.

——— END OF TASK ———

TASK 27-09-31-000-802

4. Flight Controls Shutoff Valve Motor Removal

(Figure 401)

A. General

(1) The shutoff valve motor can be removed with the shutoff valve installed in the flight controls hydraulic module.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 27-09-31-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-09-31-860-007

(2) If the left motor will be removed, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-31-860-008

(3) If the right motor will be removed, do this step:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

E. Flight Controls Shutoff Valve Motor Removal

SUBTASK 27-09-31-020-002

(1) Remove the shutoff valve motor [6].

(a) Disconnect the electrical connector [1] from the shutoff valve motor [6].

(b) After the electrical connector [1] is disconnected, examine the electrical connector [1] for corrosion.

NOTE: The inspection of the electrical connector can be done after the removal step or before the installation step of the electrical connector. This is an operator option.

(c) If there was corrosion, correct the problem (SWPM 20-60-03).

(d) Set the manual override lever on the shutoff valve motor [6] to the position 2.

(e) Remove the lockwire from the screws [7].

(f) Remove the screws [7] that attach the shutoff valve motor [6] to the shutoff valve [8].

(g) Remove the shutoff valve motor [6] from the shutoff valve [8].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-31-400-802

5. Flight Controls Shutoff Valve Motor Installation

(Figure 401)

A. General

(1) The shutoff valve motor can be installed with the shutoff valve installed in the flight controls hydraulic module.

B. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-09-31-710-801	Flight Controls Shutoff Valve Test (P/B 501)

C. Consumable Materials

Reference	Description	Specification
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Shutoff valve	27-09-31-01-065	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Flight Controls Shutoff Valve Motor Installation

SUBTASK 27-09-31-420-005

(1) Install the shutoff valve motor [6].

(a) Make sure that the manual override lever on the shutoff valve motor [6] is set to the position 2.

(b) Set the shutoff valve motor [6] on the shutoff valve [8].

(c) Engage the shutoff valve motor [6] with the valve cam.

(d) Install the screws [7].

(e) Install MS20995NC20 lockwire, G02166, on the screws [7] (TASK 20-10-44-400-801).

(f) If the electrical connector [1] was not examined at the removal step, do these steps:

NOTE: The inspection of the electrical connector can be done after the removal step or before the installation step of the electrical connector. This is an operator option.

1) Before the electrical connector [1] is connected, examine the electrical connector [1] for corrosion.

2) If there was corrosion, correct the problem (SWPM 20-60-03).

(g) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [1] (SWPM 20-60-03).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) Connect the electrical connector [1] to the shutoff valve motor [6].

G. Flight Controls Shutoff Valve Motor Installation Test

SUBTASK 27-09-31-860-009

(1) Set the manual override lever on the shutoff valve motor [6] to the position 1.

SUBTASK 27-09-31-860-010

(2) If the left motor was removed, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-31-860-011

(3) If the right motor was removed, do this step:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-31-730-002

(4) Do this task: Flight Controls Shutoff Valve Test, TASK 27-09-31-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS SHUTOFF VALVE - ADJUSTMENT/TEST

1. General

A. This procedure has the task to do a test of the flight controls shutoff valve.

TASK 27-09-31-710-801

2. Flight Controls Shutoff Valve Test

A. General

(1) The flight controls shutoff valve is part of the flight controls hydraulic module. The hydraulic module for the hydraulic system A is found in the left main wheel well. The hydraulic module for the hydraulic system B is found in the right main wheel well.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-09-31-860-013

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-09-31-860-014

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

E. Flight Controls Shutoff Valve Test

SUBTASK 27-09-31-710-001

(1) Do a test of flight controls shutoff valve.

(a) Make sure that the FLT CONTROL switches, on the P5 flight controls panel, are in the ON position.

(b) Make sure that the manual override lever on the flight controls shutoff valve motor is in position 1 for hydraulic systems A and B.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Operate the aileron control wheel to bleed air from the hydraulic system lines and components.

1) Make sure that there is no leakage of hydraulic fluid at the flight control hydraulic modules.

(d) Operate the aileron control wheel, control column, and rudder pedals.

1) Make sure that the ailerons, elevators, and rudder operate satisfactory.

(e) Set the FLT CONTROL switches to the OFF position.

1) Make sure that the manual override lever on the flight controls shutoff valve motor is in position 2 for hydraulic systems A and B.

(f) Operate the rudder pedals.

NOTE: The rudder must not operate hydraulically, but the rudder pedals will operate manually. Some rudder surface movement can occur if the pedal input is large enough to mechanically displace the PCU actuating piston. Full rudder authority may be available.

1) Make sure that the force to operate the rudder pedals is higher now than when you set the FLT CONTROLS switches to the ON position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-31-860-015

(1) Set the FLT CONTROL switches to the ON position.

(a) Make sure that the manual override lever on the flight controls shutoff valve motor is in position 1 for hydraulic systems A and B.

SUBTASK 27-09-31-040-001

(2) Remove pressure from hydraulic systems A and B, do this task Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS PANEL (P5-3) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the flight controls panel, referred to as the flight controls panel

(2) An installation of the flight controls panel.

B. The flight controls panel (P5-3) is on the captain's overhead panel. The flight controls panel has the lights and switches used by the primary and auxiliary flight control system hydraulic supply valves.

TASK 27-09-45-000-801

2. Flight Controls Panel Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-09-45-860-002

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	2	C01045	AFCS SYS A FCC DC
D	6	C01354	YAW DAMPER 2 DC

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF
B	3	C01046	AFCS SYS B FCC DC
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
E	14	C00316	INDICATOR MASTER DIM SECT 4

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	10	C01283	GENERATOR CONT UNIT 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	11	C01284	GENERATOR CONT UNIT 2

C. Flight Controls Panel Removal

SUBTASK 27-09-45-020-001

(1) Remove the flight controls module assembly [1]:

(a) Loosen the quarter-turn fasteners [2] that hold the flight controls module assembly [1].

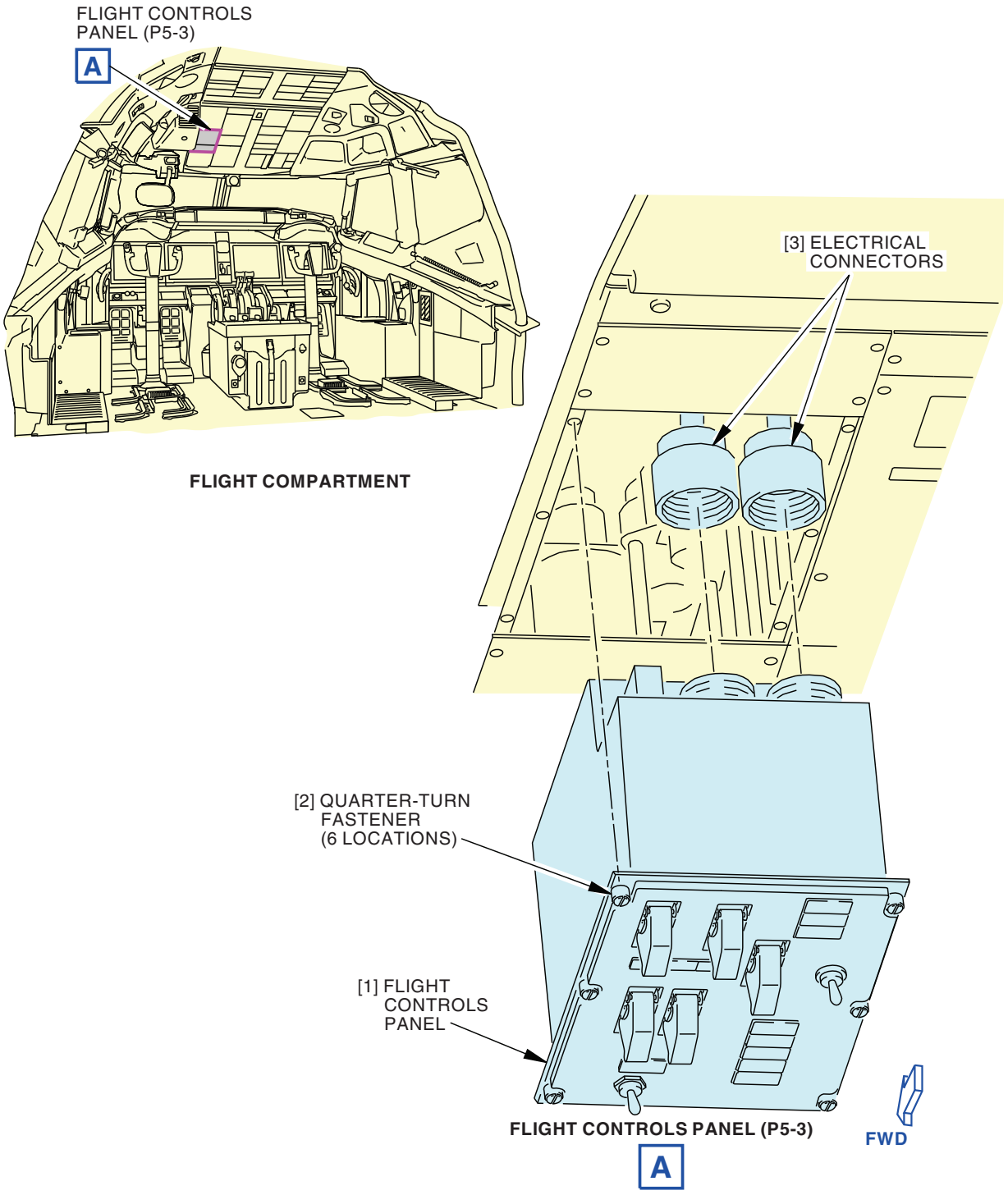
(b) Lower the flight controls module assembly [1].

(c) Disconnect the electrical connectors [3] from the flight controls module assembly [1].

(d) Remove the flight controls module assembly [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Panel (P5-3) Installation
Figure 401/27-09-45-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-45-400-801

3. Flight Controls Panel Installation

(Figure 401)

A. References

Reference	Title
29-11-00 P/B 201	HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Flight controls module assembly	27-09-45-01-050	PTF ALL

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Flight Controls Panel Installation

SUBTASK 27-09-45-420-001

(1) Install the flight controls module assembly [1]:

(a) Connect the electrical connectors [3] to the flight controls module assembly [1].

NOTE: Make sure that the electrical connectors [3] match the mates on the flight controls module assembly [1].

(b) Push the flight controls module assembly [1] into the captain's overhead panel.

(c) Tighten the quarter-turn fastener [2] that hold the flight controls module assembly [1].

SUBTASK 27-09-45-860-003

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
D	2	C01045	AFCS SYS A FCC DC
D	6	C01354	YAW DAMPER 2 DC

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	6	C00566	FLIGHT CONTROL FLAP LOAD RELIEF

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	3	C01046	AFCS SYS B FCC DC
B	12	C00204	FLIGHT CONTROL SHUTOFF VALVES SPOILER
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	12	C00132	MASTER CAUTION ANNUNCIATOR BUS 1
B	13	C00131	MASTER CAUTION ANNUNCIATOR BAT
D	12	C00310	INDICATOR MASTER DIM BAT
D	13	C00311	INDICATOR MASTER DIM BUS 1
D	14	C00312	INDICATOR MASTER DIM BUS 2
E	14	C00316	INDICATOR MASTER DIM SECT 4

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
F	10	C01283	GENERATOR CONT UNIT 1
F	11	C01284	GENERATOR CONT UNIT 2

SUBTASK 27-09-45-710-001

(3) Do this test of the hydraulic system valve control:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	15	C01401	LANDING GEAR AIR/GND RELAY

(b) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(c) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

NOTE: With these circuit breakers open the standby pump and the alternate flap drive motor will not operate.

(d) Make sure that these switches on the flight controls module assembly [1] are in the OFF position:

1) FLT CONTROL A, and FLT CONTROL B
2) SPOILER A, and SPOILER B
3) ALTERNATE FLAPS arm switch
4) ALTERNATE FLAPS control switch.

(e) Do this test of the control of the system A and system B flight control hydraulic valves from the flight controls module assembly [1]:

NOTE: The flight control hydraulic valves are installed on the left and right sides of the forward bulkhead of the landing gear wheel well.

1) Move the FLT CONTROL A switch to the ON position.
   a) Make sure that the system A flight control valve moves to position 1.
   b) Make sure that the system B flight control valve does not move (stays in position 2).
2) Move the FLT CONTROL B switch to the ON position.
   a) Make sure that the system B flight control valve moves to position 1.
3) Move the FLT CONTROL A switch to the OFF position.
   a) Make sure that the system A flight control valve moves to position 2.
   b) Make sure that the system B flight control valve does not move (stays in position 1).
4) Move the FLT CONTROL B switch to the OFF position.
   a) Make sure that the system B flight control valve moves to position 2.

(f) Do this test of the control of the system A and system B spoiler hydraulic valves from the flight controls module assembly [1]:

NOTE: The spoiler hydraulic valves are installed on the left and right sides of the forward bulkhead of the landing gear wheel well.

1) Move the SPOILER A switch to the ON position.
   a) Make sure that the system A spoiler valve moves to position 1.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Make sure that the system B spoiler valve does not move (stays in position 2).

2) Move the SPOILER B switch to the ON position.

a) Make sure that the system B spoiler valve moves to position 1.

3) Move the SPOILER A switch to the OFF position.

a) Make sure that the system A spoiler valve moves to position 2.

b) Make sure that the system B spoiler valve does not move (stays in position 1).

4) Move the SPOILER B switch to the OFF position.

a) Make sure that the system B spoiler valve moves to position 2.

(g) Set the SPOILER A and B switches to the ON position.

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(h) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

NOTE: If you will perform the "Test of the hydraulic system indications and lights" next, do not close the STANDBY HYDRAULIC PUMP circuit breaker on the P92 panel.

(i) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	13	C00841	FLIGHT CONTROL ALTN T.E. FLAP DRIVE

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	15	C01401	LANDING GEAR AIR/GND RELAY

SUBTASK 27-09-45-710-002

(4) Do this test of the hydraulic system indications and lights:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(a) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

(b) Move the FLT CONTROL A and B switches to the ON position.

1) Make sure that the FLT CONTROL - LOW PRESSURE A and B lights are ON.

(c) Move the ALTERNATE FLAPS arm switch to the ARM position.

1) Make sure that the STANDBY HYD - LOW PRESSURE light is ON.

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(d) Supply hydraulic power to systems A and B (PAGEBLOCK 29-11-00/201).

1) Make sure that the FLT CONTROL - LOW PRESSURE A and B lights are OFF.

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(e) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

1) Make sure that the STANDBY HYD - LOW PRESSURE light is OFF.

(f) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

(g) Remove the power from hydraulic systems A and B. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(h) Move the ALTERNATE FLAPS arm switch to the OFF position.

(i) Do this test of the STANDBY HYD - LOW QUANTITY light:

1) Open this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

2) Disconnect the electrical connector D678 from the standby system reservoir switch S49.

NOTE: The standby system reservoir is on the keel beam in the main wheel well.

3) Connect a jumper wire between pins 1 and 2 on electrical connector D678.

4) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

5) Make sure that the STANDBY HYD - LOW QUANTITY light is ON.

6) Open this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

7) Remove the jumper wire from electrical connector D678.

8) Connect the electrical connector D678 on the standby system reservoir switch S49.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

9) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

10) Make sure that the STANDBY HYD - LOW QUANTITY light is OFF.

SUBTASK 27-09-45-710-003

(5) Do a test of the lights on the flight controls module assembly [1]:

NOTE: Some lights might be ON.

(a) Move and hold the LIGHTS switch on the P1 Captain's Panel to the TEST position.

1) Make sure that these lights are ON:

a) STANDBY HYD - LOW QUANTITY
b) STANDBY HYD - LOW PRESSURE
c) STANDBY HYD - STBY RUD ON
d) FLT CONTROL - LOW PRESSURE A
e) FLT CONTROL - LOW PRESSURE B
f) FEEL DIFFERENTIAL PRESS
g) SPEED TRIM FAIL
h) MACH TRIM FAIL
i) AUTO SLAT FAIL
j) SPOILERS
k) YAW DAMPER.

(b) Move the LIGHTS switch back to BRT.

SUBTASK 27-09-45-710-004

(6) Do a test of the YAW DAMPER light for the yaw damper system:

(a) Align the ADIRS. Do this task: Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803.

(b) Make sure that the FLT CONTROL B switch is ON.

(c) Turn the YAW DAMPER switch to ON.

1) Make sure that the light for the YAW DAMPER goes off.

(d) Turn the YAW DAMPER switch to OFF.

1) Make sure that the light for the YAW DAMPER comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-45-710-005

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(7) Do this test of the alternate control system for the leading edge flaps and slats and the trailing edge flaps:

(a) Make sure that the flap control lever is in the FLAP UP detent.

NOTE: If the flap control lever is not in the FLAP UP detent, the flaps and slats might extend when hydraulic pressure is applied.

(b) Set the FLT CONTROL B switch to STBY RUD.

1) Make sure that the STANDBY HYD - LOW PRESSURE light comes on momentarily, and then goes off.

2) Make sure that the STBY RUD ON light on the P5-3 panel is illuminated.

(c) Set the ALTERNATE FLAPS arm switch to ARM.

1) Make sure that the leading edge flaps and slats and the trailing edge flaps do not extend.

WARNING

MAKE SURE THAT PERSONNEL AND EQUIPMENT STAY AWAY FROM THE LEADING EDGE SLATS, TRAILING EDGE FLAPS, AND DRIVE MECHANISMS. THE FLAPS, SLATS, AND DRIVE MECHANISMS MOVE QUICKLY. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT OPERATE THE TRAILING EDGE FLAPS WITH ELECTRICAL POWER MORE THAN ONE FULL EXTEND/RETRACT CYCLE IN 25 MINUTES. YOU CAN CAUSE DAMAGE TO THE FLAP ELECTRIC MOTORS IF YOU OPERATE THEM TOO FREQUENTLY.

(d) Move the ALTERNATE FLAPS control switch to the DOWN position.

1) Make sure that the leading edge flaps and slats and the trailing edge flaps extend.

NOTE: It is not necessary to fully extend the trailing edge flaps.

(e) Move the ALTERNATE FLAPS control switch to the UP position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the trailing edge flaps retract.

NOTE: It is not necessary to fully retract the trailing edge flaps with alternate control.

(f) Move the ALTERNATE FLAPS ARM switch to the OFF position.

(g) Supply hydraulic power to system B (PAGEBLOCK 29-11-00/201).

1) Make sure that the trailing edge flaps fully retract.

2) Make sure that the leading edge flaps and slats fully retract.

(h) Move the FLT CONTROL B switch to the OFF position.

(i) Move the ALTERNATE FLAPS ARM switch to the ARM position.

1) Make sure that the STANDBY HYD - LOW PRESSURE light is off.

(j) Move the ALTERNATE FLAPS ARM switch to the OFF position.

(k) Move the FLT CONTROL B switch to the ON position.

(l) Move the ALTERNATE FLAPS control switch to the OFF position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE WARNING SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the low pressure warning switch.

(2) An installation of the low pressure warning switch.

B. The low pressure warning switch is referred to as the "pressure switch" in this procedure.

C. The pressure switch is part of the flight controls hydraulic module. One flight controls hydraulic module each is installed in the hydraulic systems A and B. The hydraulic module for hydraulic system A is found in the left main wheel well. The hydraulic module for hydraulic system B is found in the right main wheel well.

TASK 27-09-61-000-801

2. Low Pressure Warning Switch Removal

(Figure 401)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-1110	Container - Hydraulic Fluid Resistant, 5 Gallon (19 Liter)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Low Pressure Warning Switch Removal

SUBTASK 27-09-61-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-09-61-860-001

(2) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-09-61-860-002

(3) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-61-860-003

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

SUBTASK 27-09-61-020-001

(5) Disconnect the electrical connector [4] from the low pressure warning switch [3].

(a) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(b) If there was corrosion, refer to SWPM 20-60-03, to correct the problem.

SUBTASK 27-09-61-020-002

(6) Remove the low pressure warning switch [3]:

(a) Use a 5 gallon (19 liter) hydraulic fluid resistant container, STD-1110, to catch hydraulic fluid that drains from the low pressure warning switch [3].

(b) Remove the lockwire from the low pressure warning switch [3].

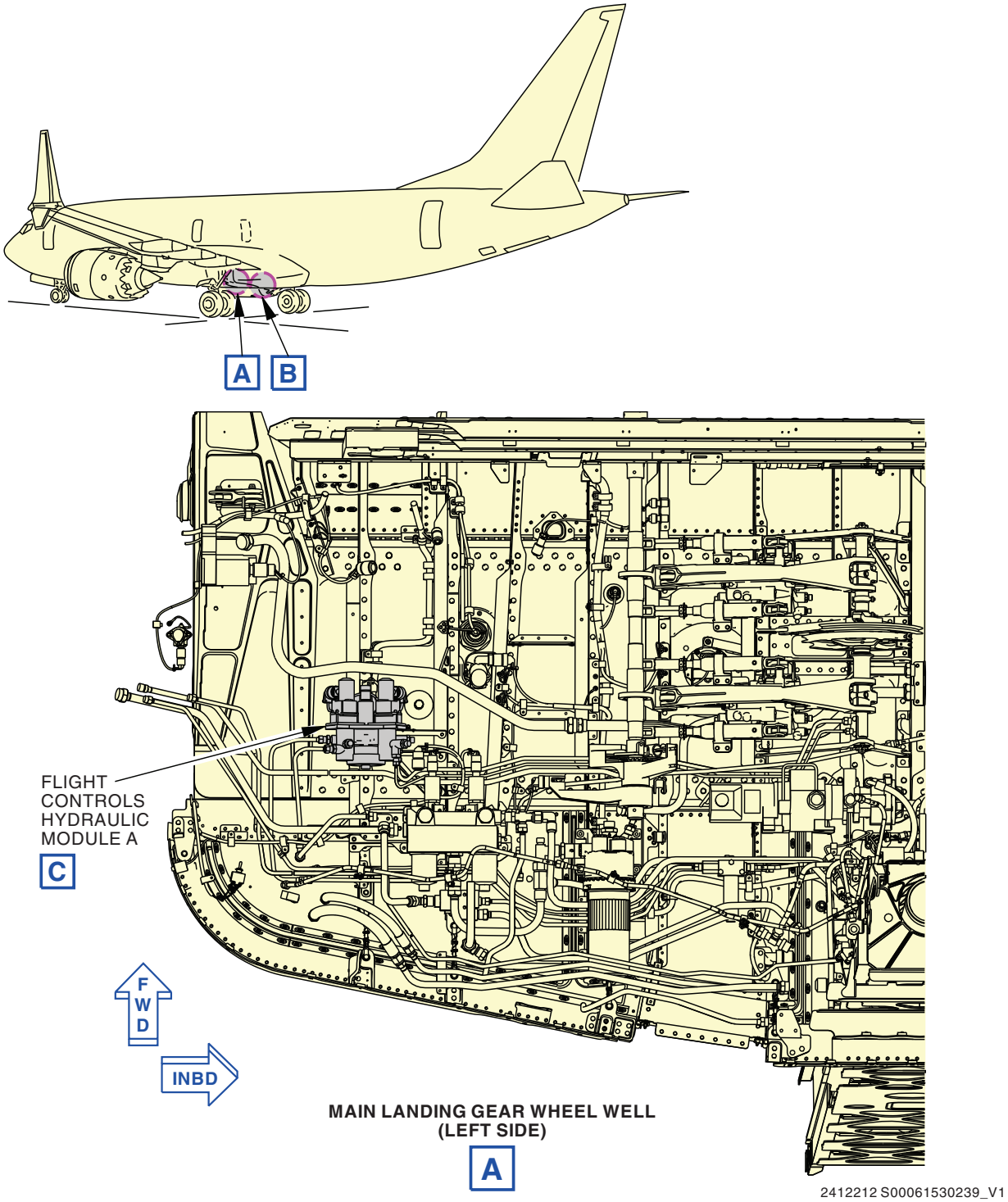
(c) Remove the low pressure warning switch [3] from the flight controls hydraulic module [5].

(d) Remove and discard the packing [2].

(e) Put a cover on the hydraulic module housing to keep unwanted material out of the open port.

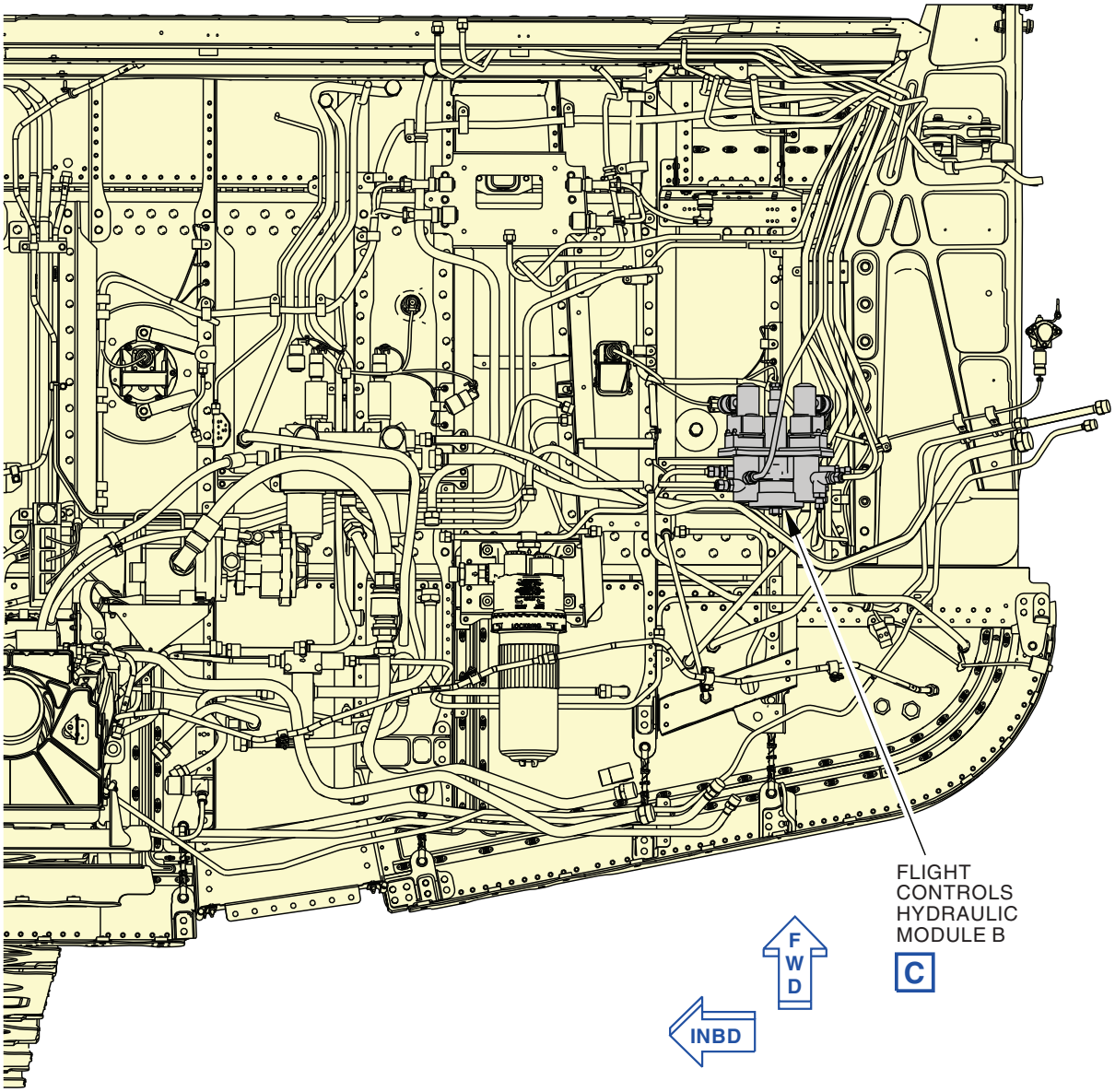
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



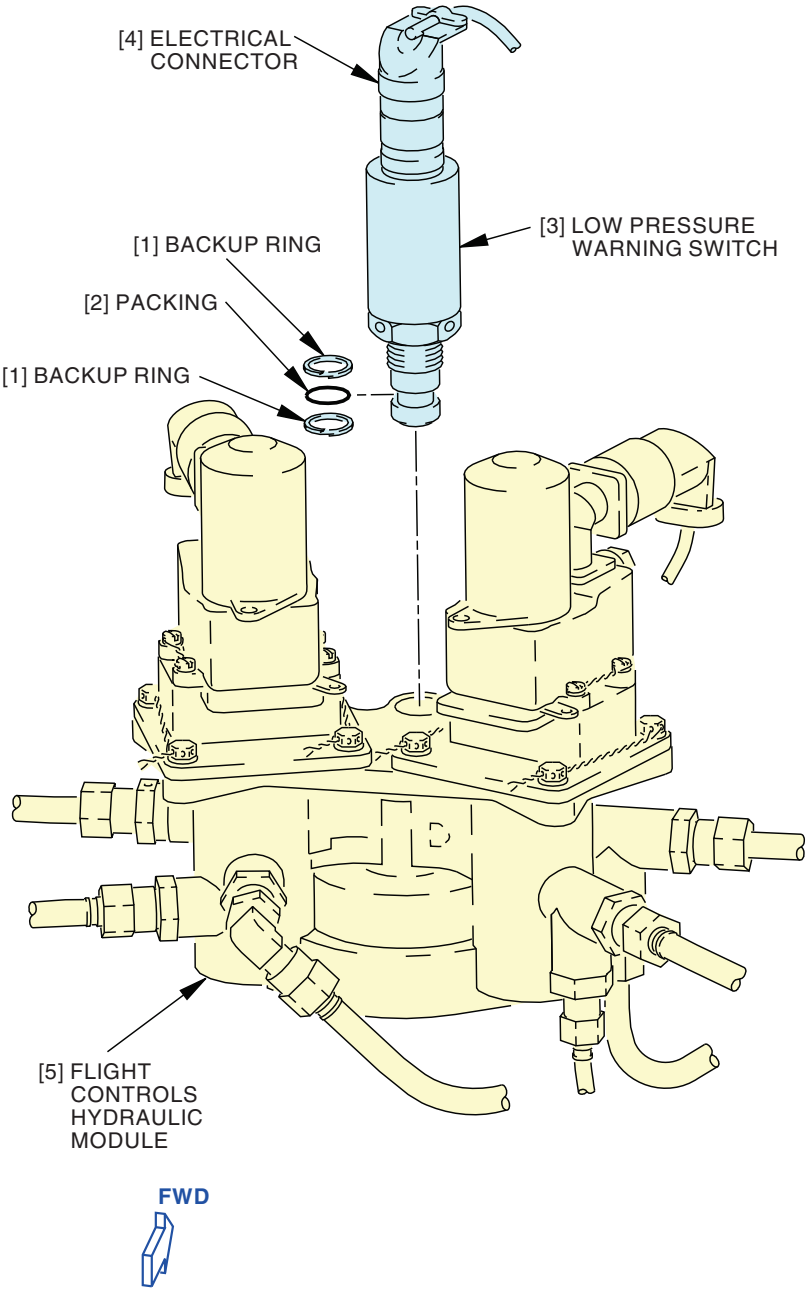
Low Pressure Warning Switch Installation
Figure 401/27-09-61-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Warning Switch Installation
Figure 401/27-09-61-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2412214 S00061530241_V1

Low Pressure Warning Switch Installation
Figure 401/27-09-61-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-61-400-801

3. Low Pressure Warning Switch Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
27-09-61-700-801	Low Pressure Warning Switch Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Backup ring	27-09-61-01-025	PTF ALL
2	Packing	27-09-61-01-030	PTF ALL
3	Low pressure warning switch	27-09-61-01-020	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Low Pressure Warning Switch Installation

SUBTASK 27-09-61-420-001

(1) If they are not installed, install the backup rings [1] and packing [2] on the low pressure warning switch [3].

(a) Apply MCS 352B fluid, D00054 (recommended), or hydraulic fluid, D00153 (alternative), to the backup ring [1] and new packing [2].

(b) Put the backup rings [1] and packing [2] in their position on the low pressure warning switch [3].

SUBTASK 27-09-61-420-002

(2) Install the low pressure warning switch [3].

(a) Install the low pressure warning switch [3] on the flight controls hydraulic module [5].

(b) Tighten the low pressure warning switch [3] to 362.5 ±12.5 in-lb (41.0 ±1.4 N·m).

(c) Install a MS20995NC32 lockwire, G01912, on the low pressure warning switch [3] and nearest bolt (TASK 20-10-44-400-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-61-420-003

(3) Connect the electrical connector [4].

(a) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

1) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

2) If there was corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the connector (SWPM 20-60-03).

(c) Connect the electrical connector [4] to the low pressure warning switch [3].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-61-860-004

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	14	C00316	INDICATOR MASTER DIM SECT 4

SUBTASK 27-09-61-610-001

(2) If servicing is necessary for hydraulic system A and B reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-09-61-710-001

(3) Do this task: Low Pressure Warning Switch Test, TASK 27-09-61-700-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE WARNING SWITCH - ADJUSTMENT/TEST

1. General

A. This procedure contains the test of the low pressure warning switch.

B. The low pressure warning switch is part of the flight controls hydraulic module. The flight controls hydraulic module for hydraulic system A is found in the left main wheel well. The flight controls hydraulic module for hydraulic system B is found in the right main wheel well.

TASK 27-09-61-700-801

2. Low Pressure Warning Switch Test

A. References

Reference	Title
29-11-00 P/B 201	HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-09-61-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

D. Do a Test of the Low Pressure Warning Switch

SUBTASK 27-09-61-710-002

(1) Do a test of the low pressure warning switch:

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Supply hydraulic power to systems A and B (PAGEBLOCK 29-11-00/201).

(b) Operate the aileron control wheel, the control column, and the rudder pedals to bleed the hydraulic system.

(c) Make sure that there is no leakage of hydraulic fluid at the flight controls hydraulic module.

(d) Make sure the LOW PRESSURE light on the P5 forward overhead panel is not on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Remove hydraulic power from systems A and B (PAGEBLOCK 29-11-00/201).

(f) Make sure the LOW PRESSURE light comes on.

(g) Make sure the MASTER CAUTION and the FLT CONT lights on the P7 lightshield panel are on.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS TRIM CORRECTION - ADJUSTMENT/TEST

1. General

A. This procedure contains these tasks:

(1) The adjustment of the trailing edge flaps for lateral trim

(2) The adjustment of the aileron balance tab to equalize lateral manual reversion forces

(3) The rudder trim adjustment for power on discrepancy correction

(4) The rudder trim adjustment for power off discrepancy correction

(5) The flight controls adjustment record

B. Make sure the control system components are correctly adjusted when you measure the trim necessary for level flight.

C. To measure the lateral trim error you must do a test flight with the flight profile detailed in the task.

D. It is not necessary to adjust the lateral trim if the aileron and rudder trim necessary to keep level flight are within the limits shown in the table found in the task below:

TASK 27-09-71-820-801

2. Trailing Edge Flap Adjustment for Lateral Trim and Test

(Figure 501, Figure 502, Figure 503, Figure 504, Figure 505, Figure 506, Figure 507)

A. General

(1) This task supplies the data necessary to adjust the trailing edge flaps to decrease the aileron trim necessary for straight and level flight during cruise, and with the trailing edge flaps extended.

(2) It is necessary to adjust the lateral trim if the aileron and rudder trim necessary to keep level flight are not in the limits shown in (Table 501).

NOTE: There are no specific limits or requirements for lateral trim outside of the flight test profile below. It is acceptable at other flight conditions to observe different trim settings from those specified.

Table 501/27-09-71-993-801 Lateral Trim Limits

Flight Condition	Aileron Trim	Rudder Trim	Airspeed	Altitude (feet)
Cruise (Flaps UP/Gear Up)	0.75 unit	0.5 unit left / 1.0 unit right	.78 mach	37,000 or higher
Low Speed (Flaps UP/Gear Up)	0.75 unit	0.75 unit	250 kias	10,000 to 17,000
Flaps Extended to 1 unit (Gear Up)	1.0 unit *[1]	1.0 unit	Flaps 1 Maneuver Speed	10,000 to 17,000
Flaps Extended to 15 units, Landing Gear Extended	1.0 unit *[1] *[2]	1.0 unit	Flaps 15 Maneuver Speed	10,000 to 17,000
Flaps Extended to 40 units, Landing Gear Extended	1.3 unit *[2]	1.0 unit	Vref+5	10,000 to 17,000

*[1] The aileron trim change between flap positions 1 and 15 must be less than 1.0 unit.

*[2] The aileron trim change between flap positions 15 and 40 must be less than 1.4 units. An aileron trim requirement shift from side to side is acceptable, provided it is within the limit of 1.4 units.

(3) There are three methods available to adjust the lateral trim.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) You can change the shims at the aft end of the carriages on the outboard flap to correct lateral trim during cruise.

NOTE: This method will have an effect on the lateral trim with the flaps extended.

(b) You can change length of the bellcranks on the aft flap drive for the outboard flap to correct lateral trim with the flaps extended.

NOTE: This method has the most effect with the flaps extended to the 40 unit position. There is no effect on the lateral trim with the flaps retracted, and a small effect with the flaps extended to 1 and 15 unit positions.

(c) You can change the compression of the bullnose seal on the inboard main flap. If the bullnose seals are not symmetric between the left and right wings, the airflow past the seals can cause lateral trim errors with the flaps extended to 40-units. This airflow can cause the aft flap bellcrank adjustment to be not effective.

NOTE: This method will have an effect on the lateral trim with the flaps fully extended and with the change in the lateral trim between the 15-unit and 40-unit positions.

(4) You can do more than one of the adjustments to correct for lateral trim problems. The charts shown in this task are for single adjustments. The results shown in the charts can be added together if you do multiple adjustments.

B. References

Reference	Title
27-11-00-820-806	Do A Check of the Aileron System Adjustment (P/B 501)
27-21-00-700-803	Rudder Centering Test (P/B 501)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
27-51-11-820-801	Inboard Trailing Edge Flap Adjustment (P/B 501)
27-51-21-820-801	Outboard Trailing Edge Flap Adjustment (P/B 501)
27-51-22-400-801	Outboard Aft Flap Installation (P/B 401)
27-51-28-820-801	Outboard Flap Outboard Support Fairing Adjustment (P/B 501)
27-51-28-820-802	Outboard Flap Inboard Support Fairing Adjustment (P/B 501)
27-61-00-820-802	Flight Spoiler Adjustment (P/B 501)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
543	Left Wing - Fairing Flap Support No. 2
544	Left Wing - Fairing Flap Support No. 1
567	Left Wing - Outboard Flap

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
643	Right Wing - Fairing Flap Support No. 7
644	Right Wing - Fairing Flap Support No. 8
667	Right Wing - Outboard Flap

E. Find the Necessary Flap Adjustment

SUBTASK 27-09-71-820-001

(1) Do a check of these other systems to make sure that they are not the cause of the lateral trim error.

(a) Make sure that the ailerons are adjusted correctly.

1) To check the ailerons, do this task: Do A Check of the Aileron System Adjustment, TASK 27-11-00-820-806.

2) Make sure that the aileron downrig is symmetric between the left and right aileron in the limit of 0.10 in. (2.54 mm).

(b) Make sure that the rudder is adjusted correctly.

1) To check the rudder, do this task: Rudder Centering Test, TASK 27-21-00-700-803.

(c) Make sure that the outboard trailing edge flaps and aft flaps are adjusted correctly.

1) For the outboard trailing edge flaps, do this task: Outboard Trailing Edge Flap Adjustment, TASK 27-51-21-820-801.

2) For the outboard aft flap, do this task: Outboard Aft Flap Installation, TASK 27-51-22-400-801.

a) Make sure that the main flap to aft flap mismatch trailing edge dimensions are in the limits.

(d) Make sure that the inboard trailing edge flaps are adjusted correctly, do this task: Inboard Trailing Edge Flap Adjustment, TASK 27-51-11-820-801.

1) Make sure that the torque tube door is sealed correctly.

NOTE: If the torque tube door is not sealed near the flap leading edge when the flaps are extended to flaps 40 units, lateral trim problems can occur.

(e) Make sure that the spoilers are adjusted correctly.

1) To check the spoilers, do this task: Flight Spoiler Adjustment, TASK 27-61-00-820-802.

2) Make sure that the spoilers on the left and right wings are adjusted symmetrically.

SUBTASK 27-09-71-750-001

(2) Do a test flight to find the lateral trim necessary for straight and level flight for all flight conditions.

NOTE: One unit of aileron trim is equal to six degrees of control wheel rotation.

NOTE: There are no specific limits or requirements for lateral trim outside of this flight test profile. It is acceptable at other flight conditions to observe different trim settings from those specified.

(a) To find the aileron trim necessary during cruise, fly with these conditions.

1) Flight speed: Approximately 0.78 mach.

2) Altitude: 37,000 feet or higher.

3) Fuel load: Balanced in the limit of 150 lb (68 kg).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Throttles: Balanced thrust in the limit of 1 percent N1 difference, with autothrottles OFF.

5) Autopilot engaged with in HDG SEL, with the wings level.

(b) Apply rudder trim to get zero heading change.

(c) Disengage the autopilot.

(d) Apply aileron trim to keep the wings level and maintain zero heading change.

(e) Write down the amount and direction of aileron trim necessary to maintain straight and level flight.

NOTE: This is the trim for the Cruise condition.

(f) To find the aileron trim necessary for low speed cruise, and with the flaps extended, fly with these conditions.

1) Altitude: Between 10,000 to 17,000 feet.

2) Fuel load: Balanced in the limit of 150 lb (68 kg).

3) Throttles: Balanced thrust in the limit of 1 percent N1 difference, with autothrottles OFF.

4) Autopilot OFF.

(g) Fly at 250 kias, with the flaps retracted.

(h) Apply rudder trim to get zero heading change.

(i) Apply aileron trim to keep the wings level and maintain zero heading change.

(j) Write down the amount and direction of aileron trim necessary to maintain straight and level flight.

NOTE: This is the trim for the Low Speed Flaps UP.

(k) Change the airspeed to flaps 1 maneuvering speed, and extend the flaps to the 1-unit position.

(l) Apply rudder trim to get zero heading change.

(m) Apply aileron trim to keep the wings level and maintain zero heading change.

(n) Write down the amount and direction of aileron trim necessary to maintain straight and level flight.

NOTE: This is the trim for the flaps 1 unit.

(o) Change the airspeed to flaps 15 maneuvering speed, and extend the flaps to the 15-unit position, and extend the landing gear.

(p) Apply rudder trim to get zero heading change.

(q) Apply aileron trim to keep the wings level and maintain zero heading change.

(r) Write down the amount and direction of aileron trim necessary to maintain straight and level flight.

NOTE: This is the trim for the flaps 15 units.

(s) Change the airspeed to Vref +5 knots, and extend the flaps to the 40-unit position.

(t) Apply rudder trim to get zero heading change.

(u) Apply aileron trim to keep the wings level and maintain zero heading change.

(v) Write down the amount and direction of aileron trim necessary to maintain straight and level flight.

NOTE: This is the trim for the flaps 40 units.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-71-800-001

(3) If the aileron trim necessary to maintain level flight is not in the limits shown in (Table 501) primarily for the cruise condition, then do the steps in Lateral Trim Correction for the Cruise Condition below.

SUBTASK 27-09-71-800-002

(4) If the aileron trim necessary to maintain level flight is not in the limits shown in (Table 501) primarily for the flaps extended to 40-units condition, then do the steps in Lateral Trim Correction for the Flaps Extended Condition below.

SUBTASK 27-09-71-800-003

(5) If the aileron trim necessary to maintain level flight is more than 1.0 units for most of the flight conditions shown in (Table 501), and the trim values are either left or right, then do the steps in Lateral Trim Correction for the Multiple Flight Conditions below.

SUBTASK 27-09-71-800-004

(6) If the aileron trim necessary to maintain level flight is not in the limits shown in (Table 501) for the flaps extended to 40-units condition, or the change in the trim between the 15-unit and 40-unit conditions is not in the limit, then make sure that the inboard main flap bullnose seals are in the correct position.

(a) Do the steps in the Inboard Flap Bullnose Seal Position Adjustment below.

SUBTASK 27-09-71-800-007

(7) You can combine the adjustments to adjust the lateral trim for low speed, flaps 1 unit, and flaps 15 unit conditions.

F. Lateral Trim Correction for the Cruise Condition

NOTE: It is necessary to use these steps if the aileron trim necessary to keep level flight is not in the limits for the cruise condition. This change will also have some effect on the other flight conditions. This procedure changes the thickness of the shims on the carriages for the outboard flap.

SUBTASK 27-09-71-970-001

(1) Use the graphs to find the total shim thickness change necessary to correct the lateral trim error (Figure 501).

SUBTASK 27-09-71-970-002

(2) Find the thickness and the location of the shims that you will change:

NOTE: The thickness of the shims added on one wing added to the thickness of the shims removed from the other wing must be equal to the total shim thickness change. If it is possible, the total shim thickness change should be divided equally between the left and right wings. If the total shim change is 0.064 in. (1.626 mm) or less, you can change the shims in only one wing. The maximum permitted thickness of the shims on a carriage is 0.50 in. (12.70 mm). The minimum permitted thickness of the shims on a carriage is 0.00 inch.

(a) Flight test condition; Right Trim.

NOTE: Right aileron trim indicates a left wing heavy condition.

1) Add approximately 1/2 of the total shim thickness change to the carriages on the right outboard flap.

2) Remove approximately 1/2 of the total shim thickness change from the carriages on the left outboard flap.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Flight test condition; Left Trim.

NOTE: Left aileron trim indicates a right wing heavy condition.

1) Add approximately 1/2 of the total shim thickness change to the carriages on the left outboard flap.

2) Remove approximately 1/2 of the total shim thickness change from the carriages on the right outboard flap.

SUBTASK 27-09-71-040-001

(3) Prepare to change the thickness of the shims on the carriages for the outboard flaps.

(a) Extend the trailing edge flaps to the 40-unit position, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

(b) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-09-71-820-002

(4) Change the thickness of the shims [5] at the aft link fitting on the inboard and outboard carriages (Figure 502).

NOTE: The change in the thickness of the shims must be the same for the inboard carriage and the outboard carriage on a flap.

(a) At the outboard carriage [6], change the thickness of the shims [5].

1) Remove the nuts [4], washers [3], and bolts [2] that attach the aft link fitting [1] to the outboard carriage [6].

2) Add or remove shims [5] between the aft link fitting [1] and outboard carriage [6] to get the correct shim thickness.

3) Apply sealant, A00247, to the mating surfaces of the shims [5].

4) Put the shims [5] in their location between the aft link fitting [1] and outboard carriage [6].

NOTE: The maximum total thickness of the shims is 0.50 in. (12.70 mm). The minimum total thickness of the shims is 0.00 inch.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

5) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolts [2].

a) Make sure that the bolts [2] are the correct length for installation with the new shim thickness.

6) Install the bolts [2], washers [3], and nuts [4] that attach the aft link fitting [1] to the outboard carriage [6].

(b) At the inboard carriage [11], change the thickness of the shims [5].

1) Remove the nuts [10], washers [9], and bolts [8] that attach the aft link fitting [7] to the inboard carriage [11].

2) Add or remove shims [5] between the aft link fitting [7] and inboard carriage [11] to get the correct shim thickness.

3) Apply sealant, A00247, to the mating surfaces of the shims [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Put the shims [5] in their location between the aft link fitting [7] and inboard carriage [11].

NOTE: The maximum thickness of the shims is 0.50 in. (12.70 mm). The minimum thickness of the shims is 0.00 inch.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

5) Apply Cor-Ban 27L Compound, G50237, to the shank and threads of the bolts [8].

6) Install the bolts [8], washers [9] and nuts [10] that attach the aft link fitting [7] to the inboard carriage [11].

SUBTASK 27-09-71-940-001

(5) Put the airplane back to its usual condition.

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Retract the trailing edge flaps to the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

(c) Do a check of the fair and alignment of the aft flap [51] with the main flap, do this task: Outboard Aft Flap Installation, TASK 27-51-22-400-801.

NOTE: It is necessary to use a tolerance for the mismatch between the main flap and the aft flap of 0.000 ±0.030 in. (0.000 ±0.762 mm). This tolerance is smaller than the normal adjustment tolerance. This increases the effect of the lateral trim adjustment.

(d) Do a check of the clearances between the aft fairings and flap.

1) To check the outboard fairing on the outboard flap, do this task: Outboard Flap Outboard Support Fairing Adjustment, TASK 27-51-28-820-801.

2) To check the inboard fairing on the outboard flap, do this task: Outboard Flap Inboard Support Fairing Adjustment, TASK 27-51-28-820-802.

(e) Do a check of the clearances between the spoilers and the flap, do this task: Flight Spoiler Adjustment, TASK 27-61-00-820-802.

G. Lateral Trim Correction for the Flaps Extended Conditions

NOTE: It is necessary to use these steps if the aileron trim necessary to keep level flight is not in the limits with the flaps extended. This change will primarily effect the trim at the flaps 40 position, with less effect at the other flap positions. This change will not have an affect on the lateral trim with the flaps retracted. This procedure uses the adjustment fitting to change the length of the bellcranks on the carriages for the outboard flap.

SUBTASK 27-09-71-970-003

(1) Use the graphs to find the total length change of the bellcrank [34] necessary to correct the lateral trim error with the flaps extended (Figure 503).

NOTE: It is necessary to use the nuts on the adjustment fitting at the end of the bellcrank to change the length of the bellcrank.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-71-970-004

(2) Find the dimension and direction to adjust the nuts [32] for the adjustment fittings [31] on the bellcranks [34] (Figure 504).

NOTE: The number of turns of the nuts for the adjustment fitting on one wing should be the same, but in the opposite direction of the nuts on the other wing.

(a) Flight test condition; Right Trim.

NOTE: Right aileron trim indicates a left wing heavy condition.

1) Decrease the length of the bellcranks [34] on the right wing by the 1/2 the dimension shown in the graph.

2) Increase the length of the bellcranks [34] on the left wing by the 1/2 the dimension shown in the graph.

(b) Flight test condition; Left Trim.

NOTE: Left aileron trim indicates a right wing heavy condition.

1) Decrease the length of the bellcranks [34] on the left wing by the 1/2 the dimension shown in the graph.

2) Increase the length of the bellcranks [34] on the right wing by the 1/2 the dimension shown in the graph.

SUBTASK 27-09-71-860-001

(3) Extend the trailing edge flaps to the 40-unit position, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

SUBTASK 27-09-71-040-002

(4) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

SUBTASK 27-09-71-820-003

(5) Change the length of the bellcranks [34] on the inboard and outboard carriages (Figure 504).

NOTE: The change in the length of the bellcranks must be the same for the inboard carriage and the outboard carriage on a flap in the limit of 0.05 in. (1.27 mm). It is important to adjust the aft pushrods after you adjust the length of the bellcranks.

(a) Turn the applicable nut [32] on the bellcrank [34] to change the position of the adjustment fitting [31].

NOTE: To increase the length of the bellcrank, loosen the lower nut the distance shown in the graph. To decrease the length of the bellcrank, loosen the upper nut the distance shown. One full turn of a nut will change the length of the bellcrank by 0.05 in. (1.27 mm). You can use the distance from the top of the bellcrank to the bottom flange of the adjustment fitting as a reference dimension.

(b) Move the adjustment fitting [31] in the bellcrank [34].

(c) Engage the locking washer [33] with the bellcrank [34].

(d) Tighten the other nut [32] to hold the adjustment fitting [31].

(e) Install lockwire on the nuts [32].

SUBTASK 27-09-71-820-004

(6) Adjust the aft pushrods [53] for the aft flap [51] (Figure 505).

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Retract the trailing edge flaps to the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(d) Measure the mismatch between the aft flap [51] and main flap at the inboard and outboard ends of the aft flap [51].

(e) If the mismatch at the inboard and outboard ends of the aft flap [51] is not in the limit of 0.000 ±0.030 in. (0.000 ±0.762 mm), then adjust the length of the applicable aft pushrod [53].

NOTE: The tolerance on the mismatch is smaller than the normal adjustment tolerance. This increases the effect of the lateral trim adjustment.

1) Loosen the nuts [52] on the aft pushrod [53].

2) Turn the applicable aft pushrod [53] until the mismatch is in the limit at the inboard and outboard ends of the aft flap [51].

NOTE: For each 1/4 turn of the aft pushrods, the trailing edge of the aft flap will move approximately 0.04 in. (1.02 mm).

3) Tighten the nut [52] on the aft pushrod [53].

4) Install lockwire on the nuts [52].

SUBTASK 27-09-71-940-002

(7) Put the airplane back to its usual condition.

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Retract the trailing edge flaps to the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

H. Inboard Flap Bullnose Seal Compression Adjustment

NOTE: Use these steps if the flap position 40 trim inputs are out of limits or if the aileron trim change between flap positions 15 and 40 is not in the limit of 1.4 units. This adjustment will typically affect only flap position 40 trim.

NOTE: The front of the bullnose seals are not installed on adjustable retainers, therefore it is necessary to add or remove shims to adjust the forward portion of the bullnose seal compression. After the shim thickness has been adjusted, the aft retainer must be adjusted to achieve consistent seal compression.

SUBTASK 27-09-71-970-015

(1) Find the side from which to decrease the bullnose seal shim [110].

(a) Flight test condition: Right Trim at the 40-unit flap condition.

NOTE: Right aileron trim shows a left wing heavy condition.

1) Decrease shim [110] thickness below the bullnose seal [107] on the right main flap.

(b) Flight test condition: Left Trim at the 40-unit flap condition.

NOTE: Left aileron trim shows a right wing heavy condition.

1) Decrease shim [110] thickness below the bullnose seal [107] on the left main flap.

SUBTASK 27-09-71-970-016

(2) Find the correct thickness of the bullnose shim [110] to replace the existing shim [110].

(a) Extend the trailing edge flaps to the 15-unit or 30-unit position, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

(b) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(c) If the nose and main landing gear downlock pins are not installed, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-09-71-020-004

(3) Remove the bullnose seal [107] (Figure 507).

(a) If necessary, move the torque tube door [94] up and inboard to get access to the bullnose seal [107].

(b) Remove the bolts [108] and washers [109] that attach the bullnose seal [107] to the inboard flap [71].

(c) Remove the bullnose seal [107] from the inboard flap [71].

SUBTASK 27-09-71-800-010

(4) Measure the thickness of the bullnose seal [107].

NOTE: The bullnose shim can be made of more than one part.

NOTE: The bullnose shim can be one part that has different thicknesses.

NOTE: If the initial shim is less than 0.10 in. (2.54 mm), then you can install the shorter bullnose seal alternative. The short bullnose seal is 0.24 in. (6.10 mm) shorter than the tall bullnose seal.

(a) If the bullnose shim [110] has different thickness measure the thickness at locations A, B and C.

1) Determine the maximum thickness.

(b) Record the maximum thickness of the bullnose shim [110].

SUBTASK 27-09-71-800-011

(5) Make a new shim [110] for the bullnose seal [107].

(a) Make the new shim 0.10 in. (2.54 mm) less thick than the maximum thickness of the used shim.

NOTE: The new shim can be constant thickness even if the original shim was variable thickness.

(b) Make the outline of the bullnose seal [107] on the shim material.

NOTE: The material for the shim is optional. Non-metal materials can be used as the shim material. Nylon is one such material.

(c) Cut out the shim [110] of the material.

1) Make sure that the shim aligns with the shape of the bullnose seal [107] in the limit of 0.0 ±0.1 in. (0.0 ±2.5 mm).

(d) Drill the holes in the shim [110] to align with the holes in the bullnose seal [107].

SUBTASK 27-09-71-420-004

(6) Install the bullnose seal [107] and the new shim [110].

(a) If the shim material is metal, apply a layer of sealant, A00247, between the shim and the inboard flap [71].

(b) Put the bullnose seal [107] and the shim in their position on the inboard flap [71].

(c) Install the bullnose seal [107] and the washers [109] to attach the bullnose seal [107].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the grip length of the bolts [108] is correct with the added thickness with the shims.

(d) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(e) Extend the trailing edge flaps to the 40-unit position, do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

(f) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

(g) Adjust the seal retainer at the aft end of the bullnose seal [107] until the seal compression on the upper aft end of the seal is the same as the compression at point B.

(h) Release the torque tube door [94] and let it close.

SUBTASK 27-09-71-940-007

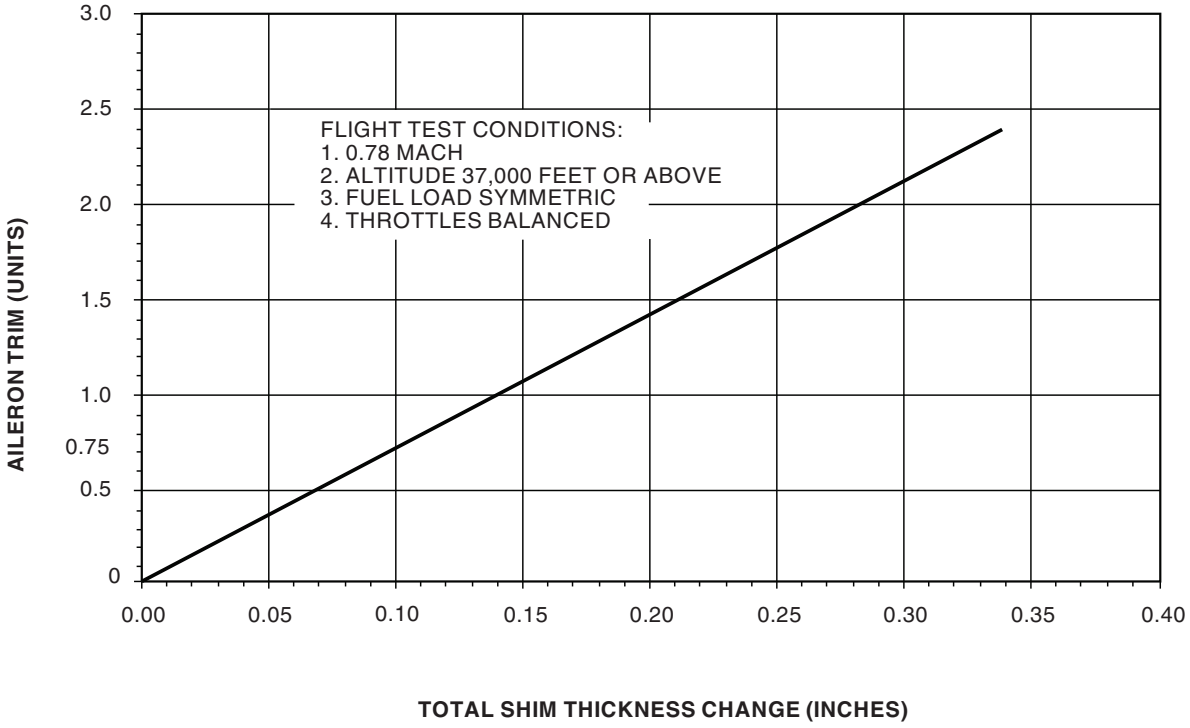
(7) Put the airplane back to its usual condition.

(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Retract the flaps to the retracted position, do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON TRIM (UNITS) _____ RIGHT TRIM / LEFT TRIM

TOTAL SHIM THICKNESS CHANGE (INCHES) _____

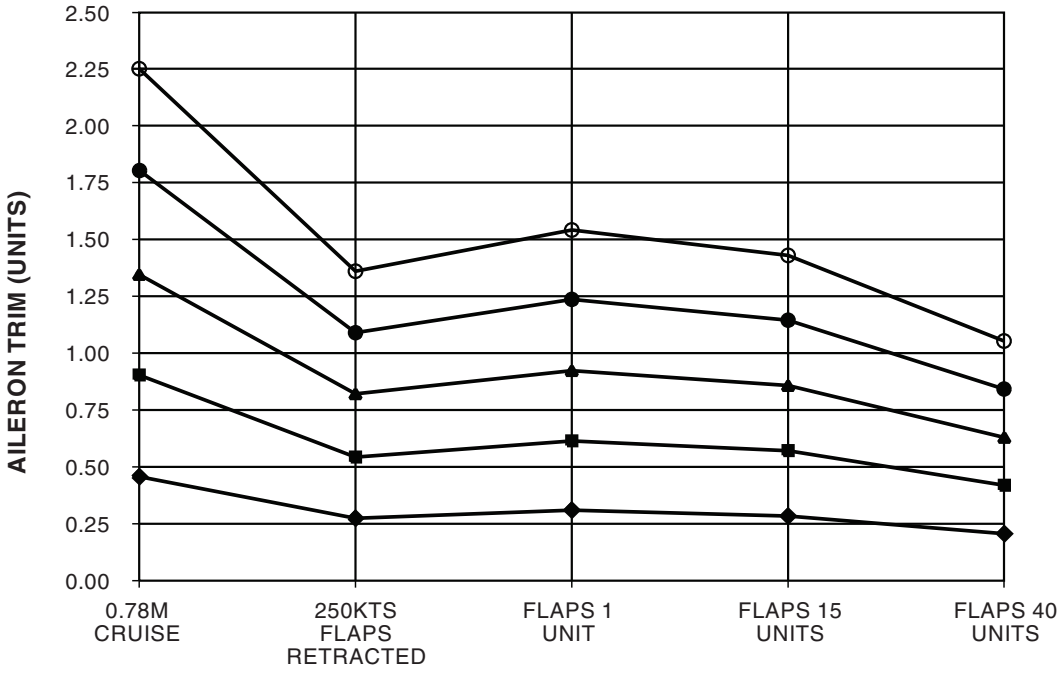
SHIM CHANGE (INCHES) LEFT WING _____

RIGHT WING _____

2412215 S00061530247_V1

Lateral Trim Correction for Cruise Condition
Figure 501/27-09-71-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON TRIM (UNITS) _____ RIGHT TRIM / LEFT TRIM

TOTAL SHIM THICKNESS CHANGE (INCHES) _____

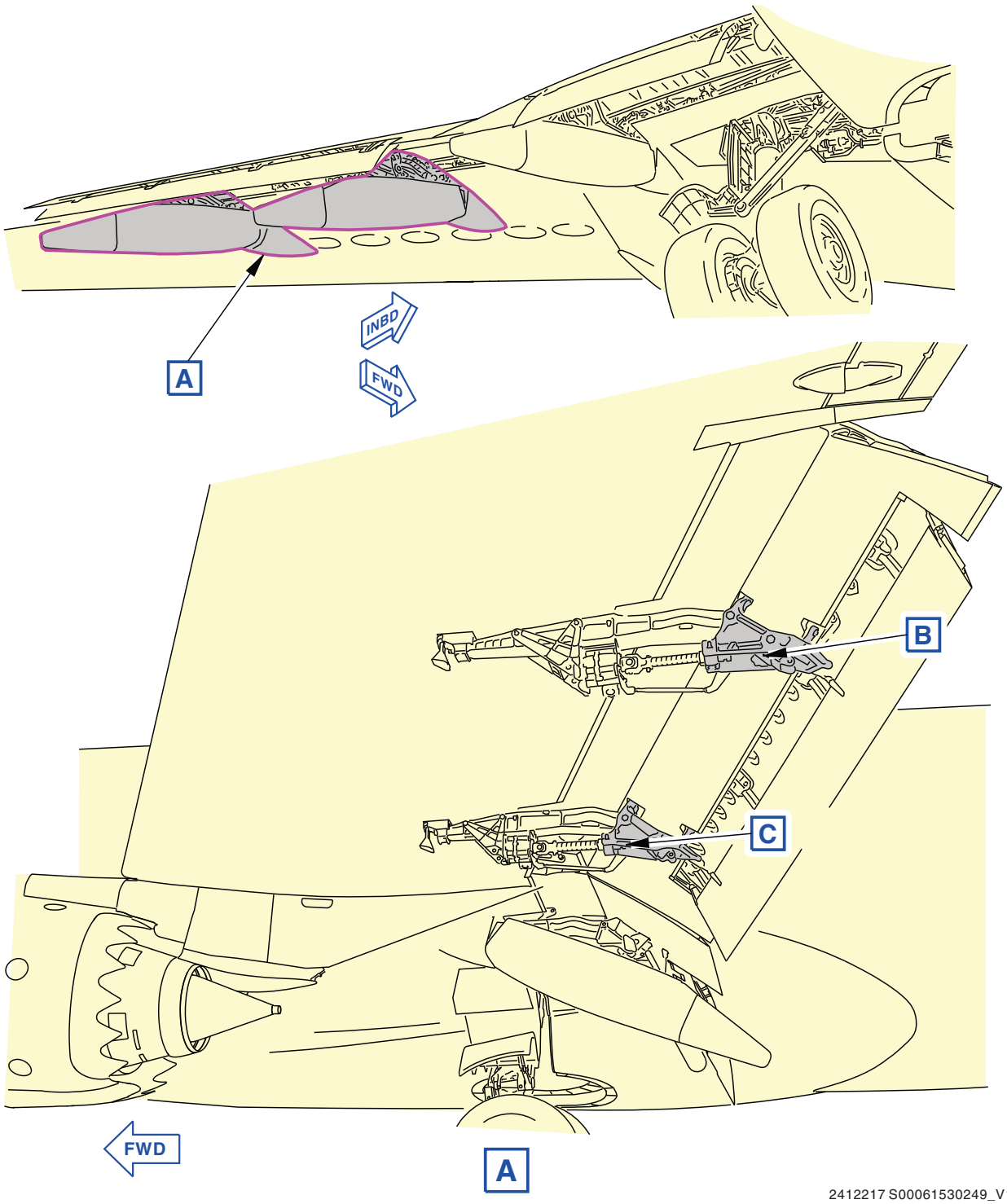
SHIM CHANGE (INCHES) LEFT WING _____

RIGHT WING _____

2412216 S00061530248_V1

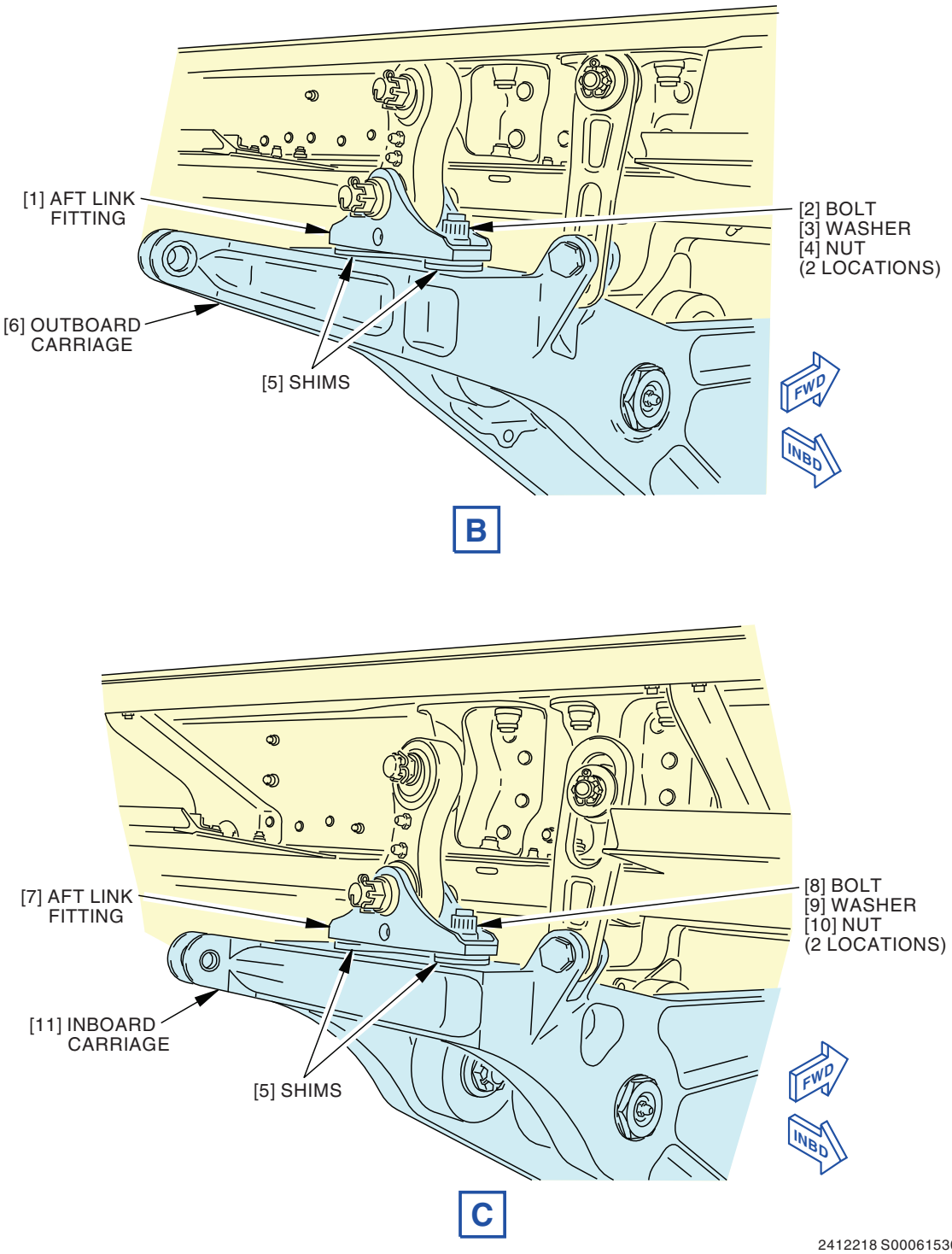
Lateral Trim Correction for Cruise Condition
Figure 501/27-09-71-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



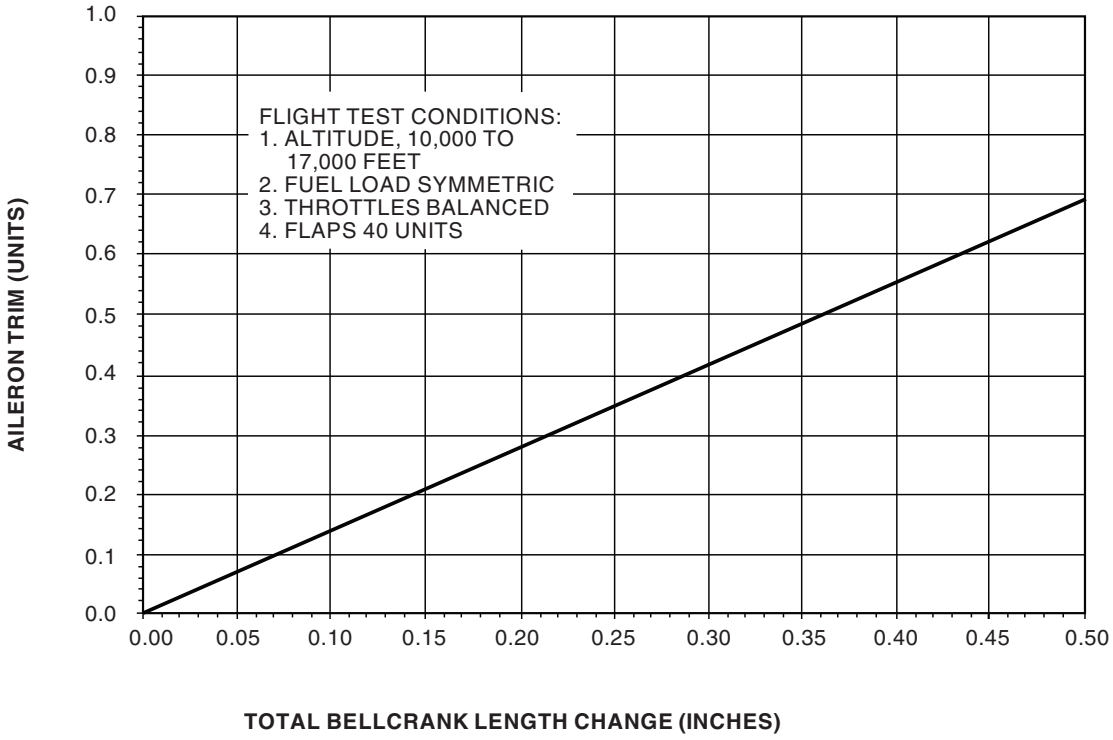
Outboard Flap Shim Thickness Change
Figure 502/27-09-71-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Flap Shim Thickness Change
Figure 502/27-09-71-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON TRIM (UNITS) REQUIRED WITH FLAPS AT 40 UNITS _____ RIGHT TRIM / LEFT TRIM

TOTAL BELLCRANK LENGTH CHANGE _____

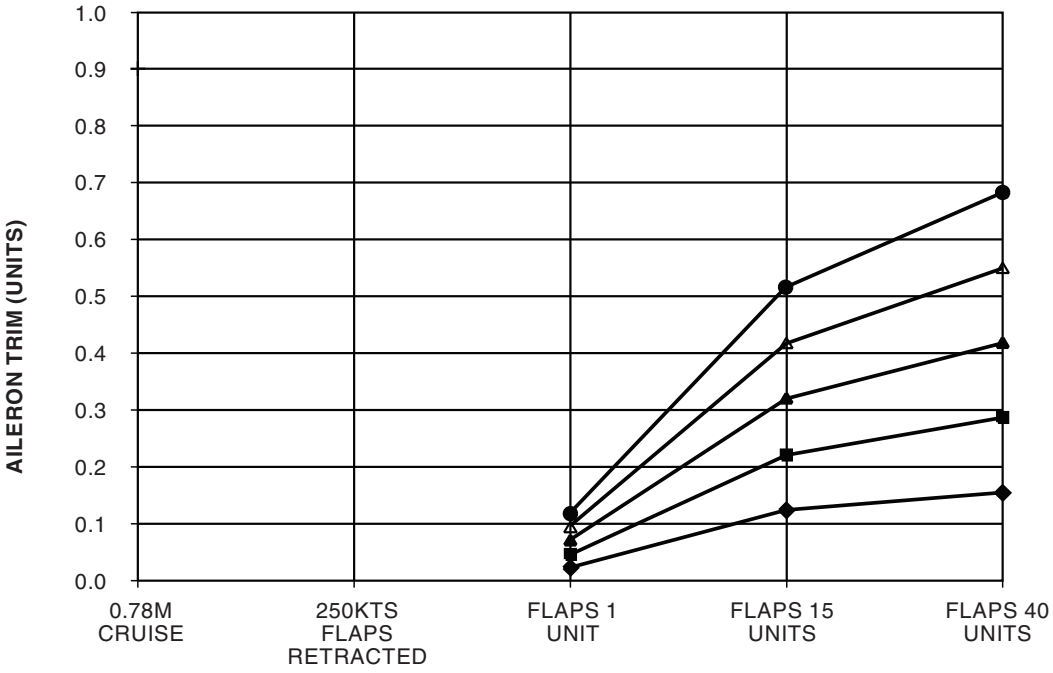
LEFT WING BELLCRANK LENGTH CHANGE _____

RIGHT WING BELLCRANK LENGTH CHANGE _____

2412219 S00061530251_V1

Lateral Trim Correction for Flaps 40 Unit Conditions
Figure 503/27-09-71-990-803 (Sheet 1 of 2)

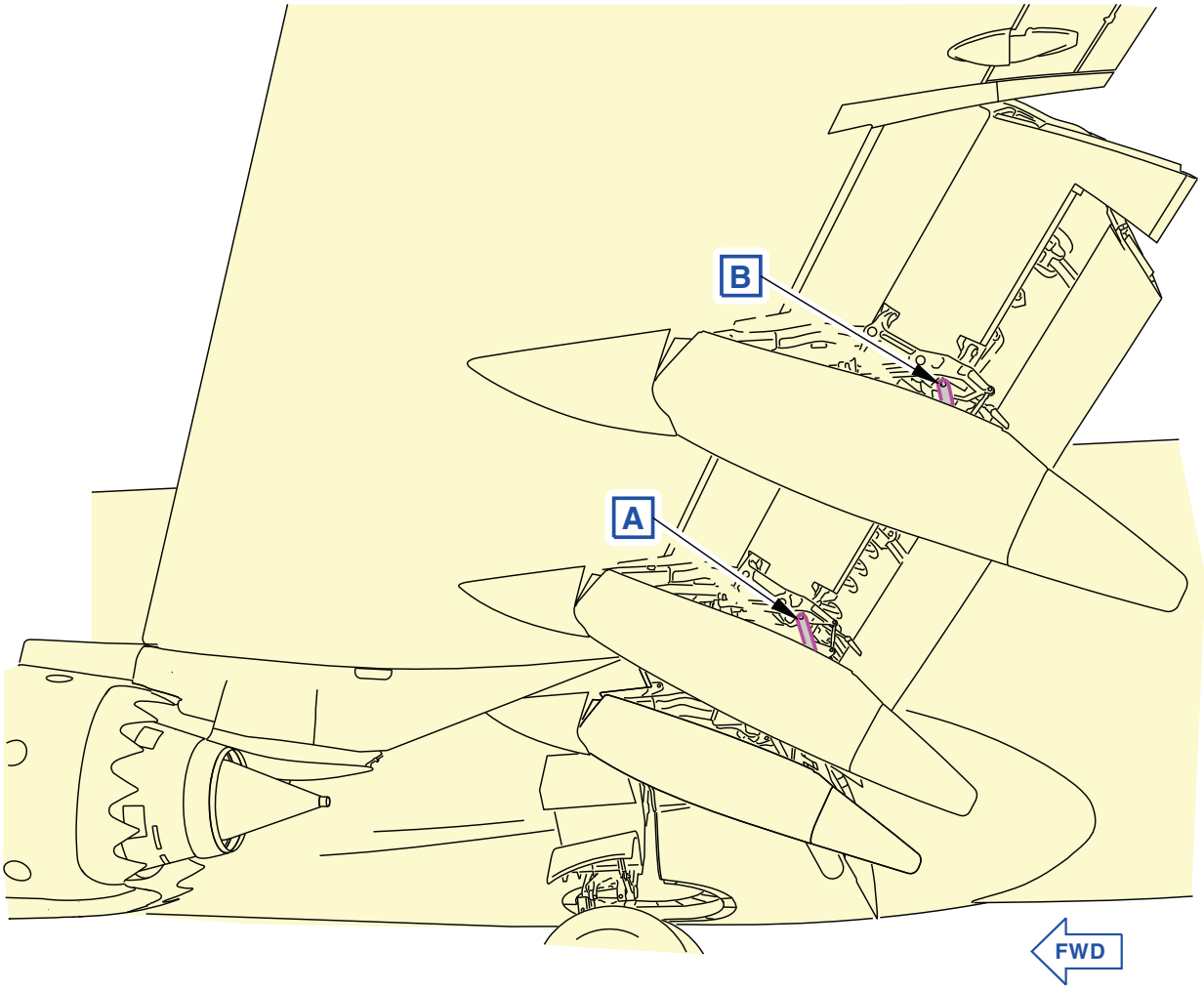
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412220 S00061530252_V1

Lateral Trim Correction for Flaps 40 Unit Conditions
Figure 503/27-09-71-990-803 (Sheet 2 of 2)

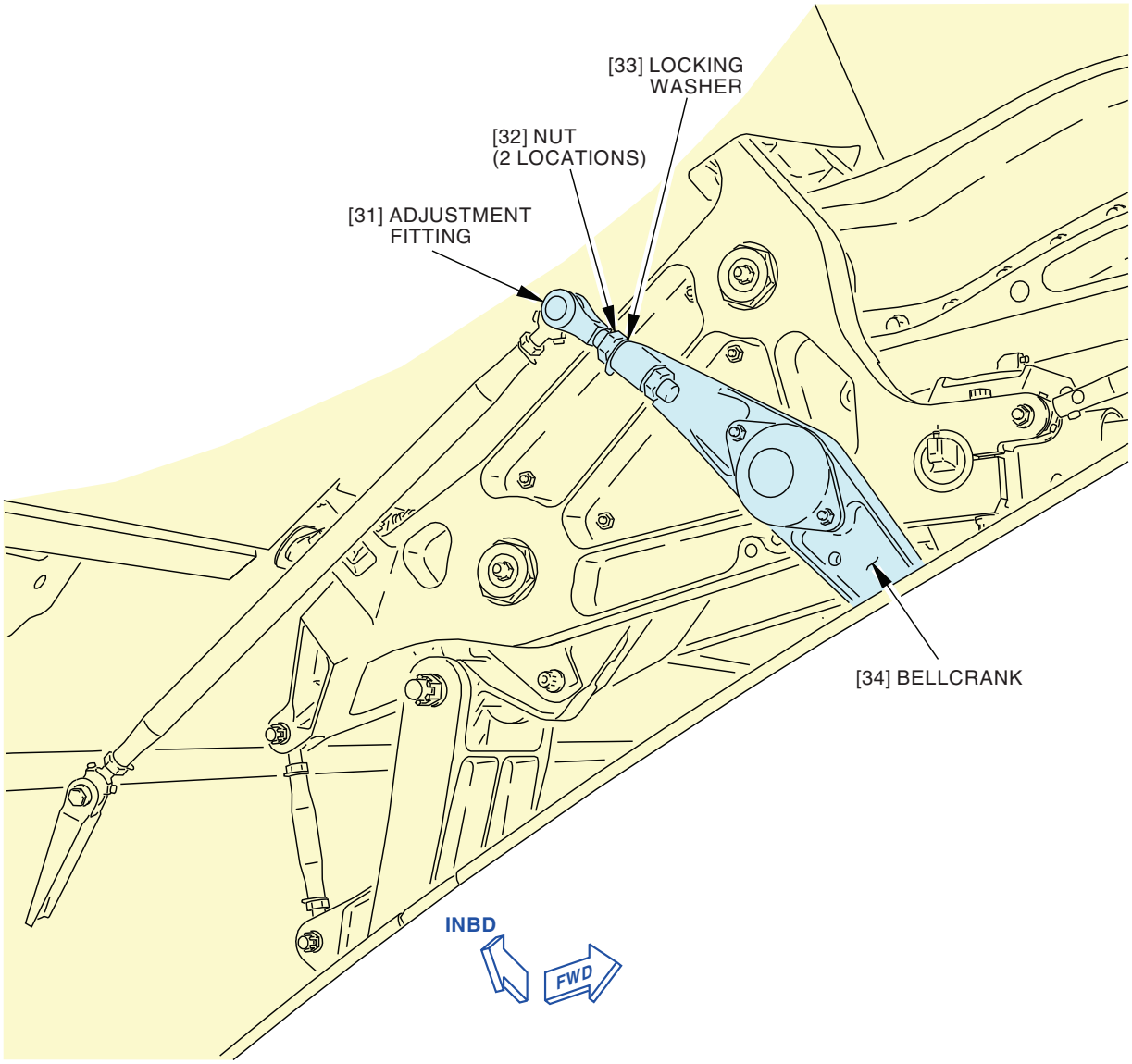
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412221 S00061530253_V1

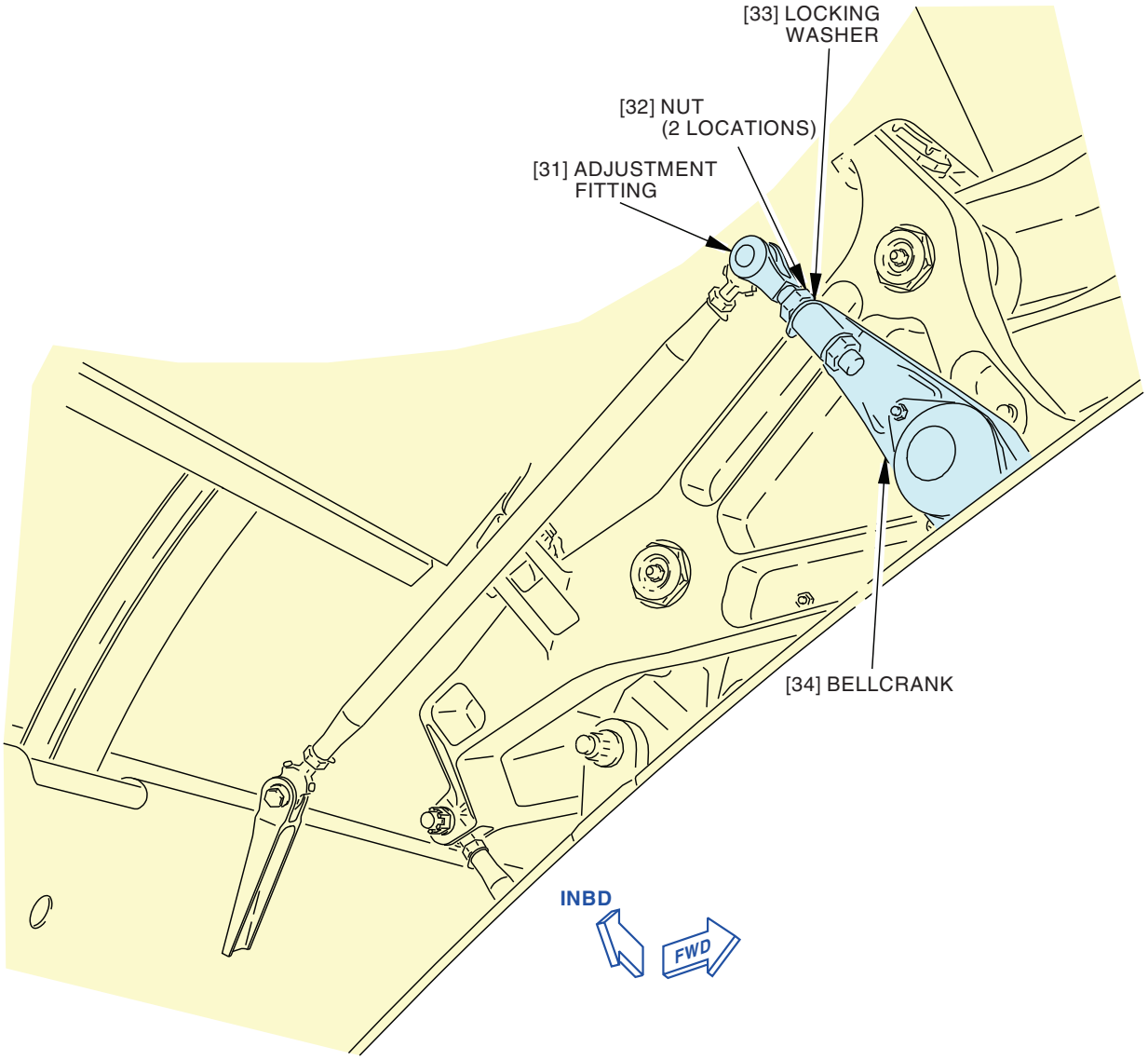
Outboard Aft Flap Bellcrank Adjustment
Figure 504/27-09-71-990-804 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outboard Aft Flap Bellcrank Adjustment
Figure 504/27-09-71-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



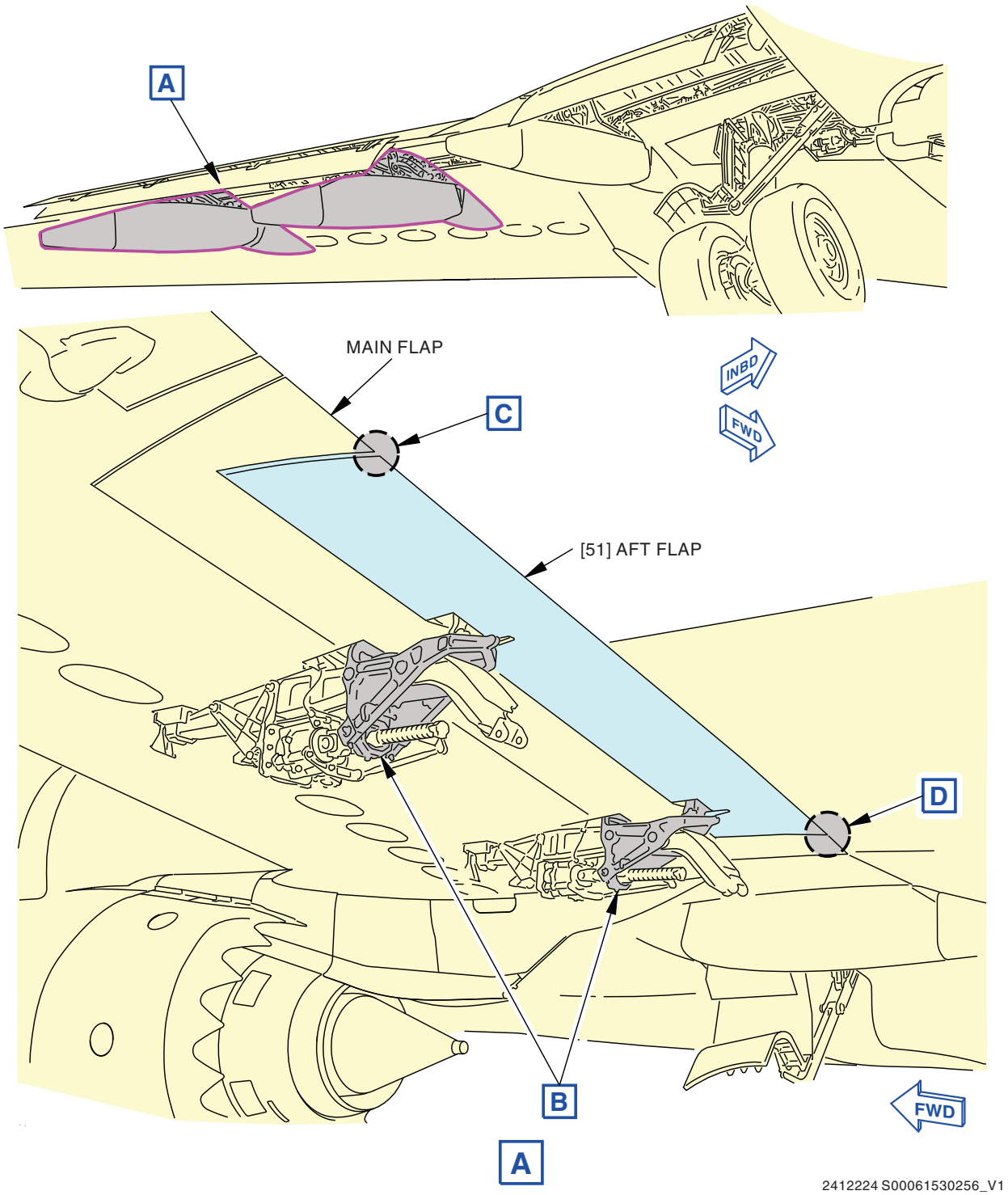
OUTBOARD AFT FLAP DRIVE

B

2412223 S00061530255_V1

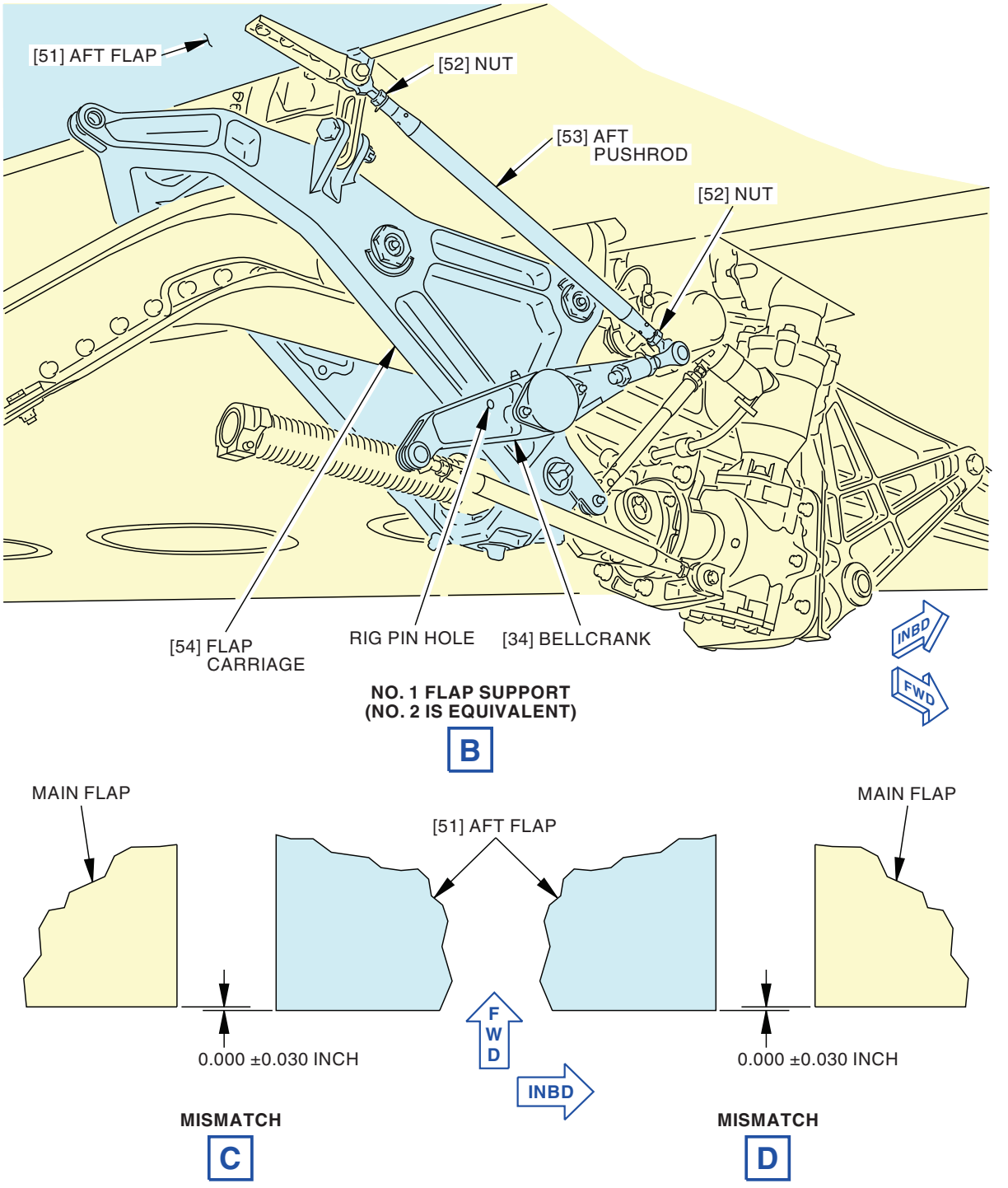
Outboard Aft Flap Bellcrank Adjustment
Figure 504/27-09-71-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



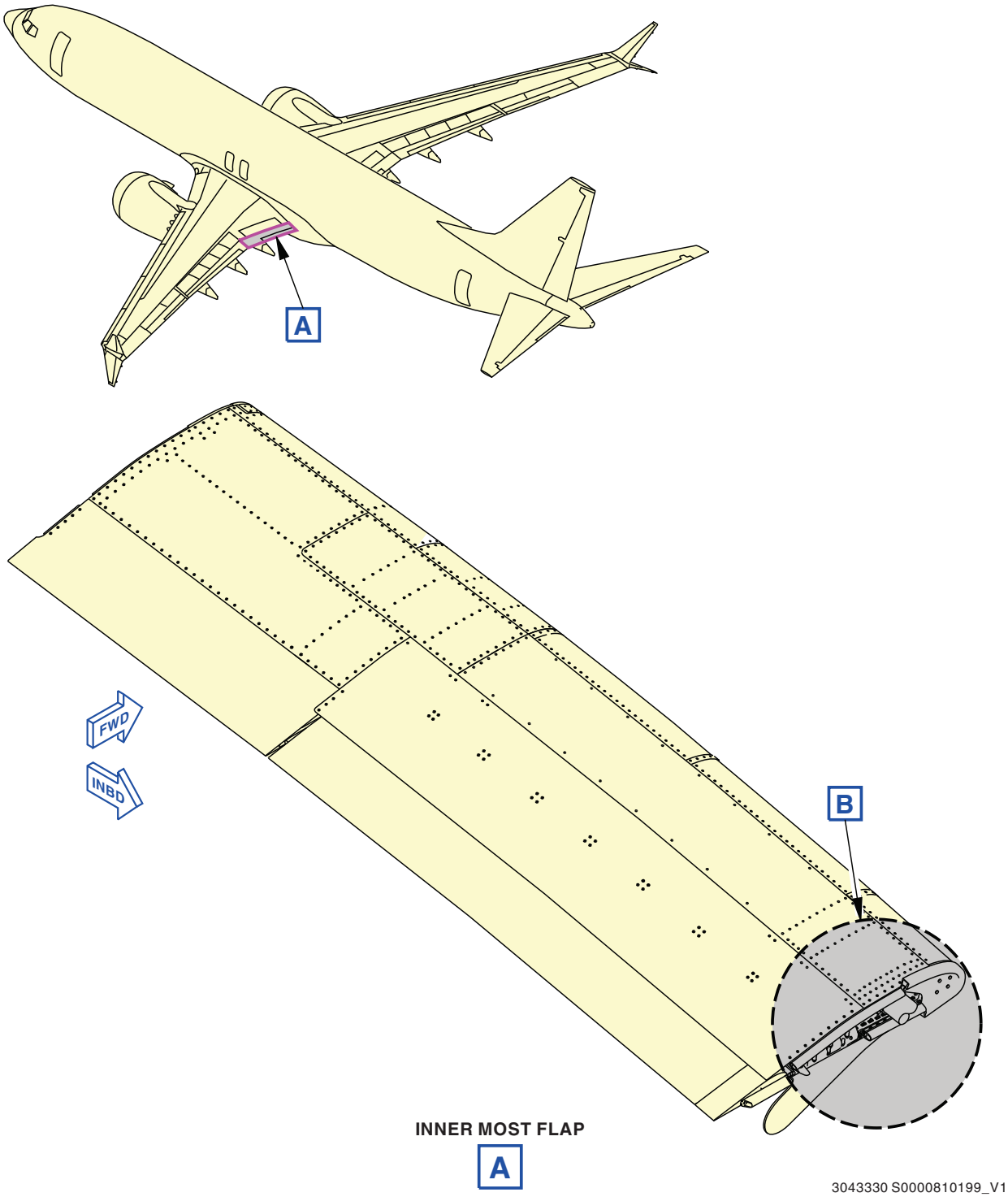
Aft Flap Pushrod Adjustment
Figure 505/27-09-71-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



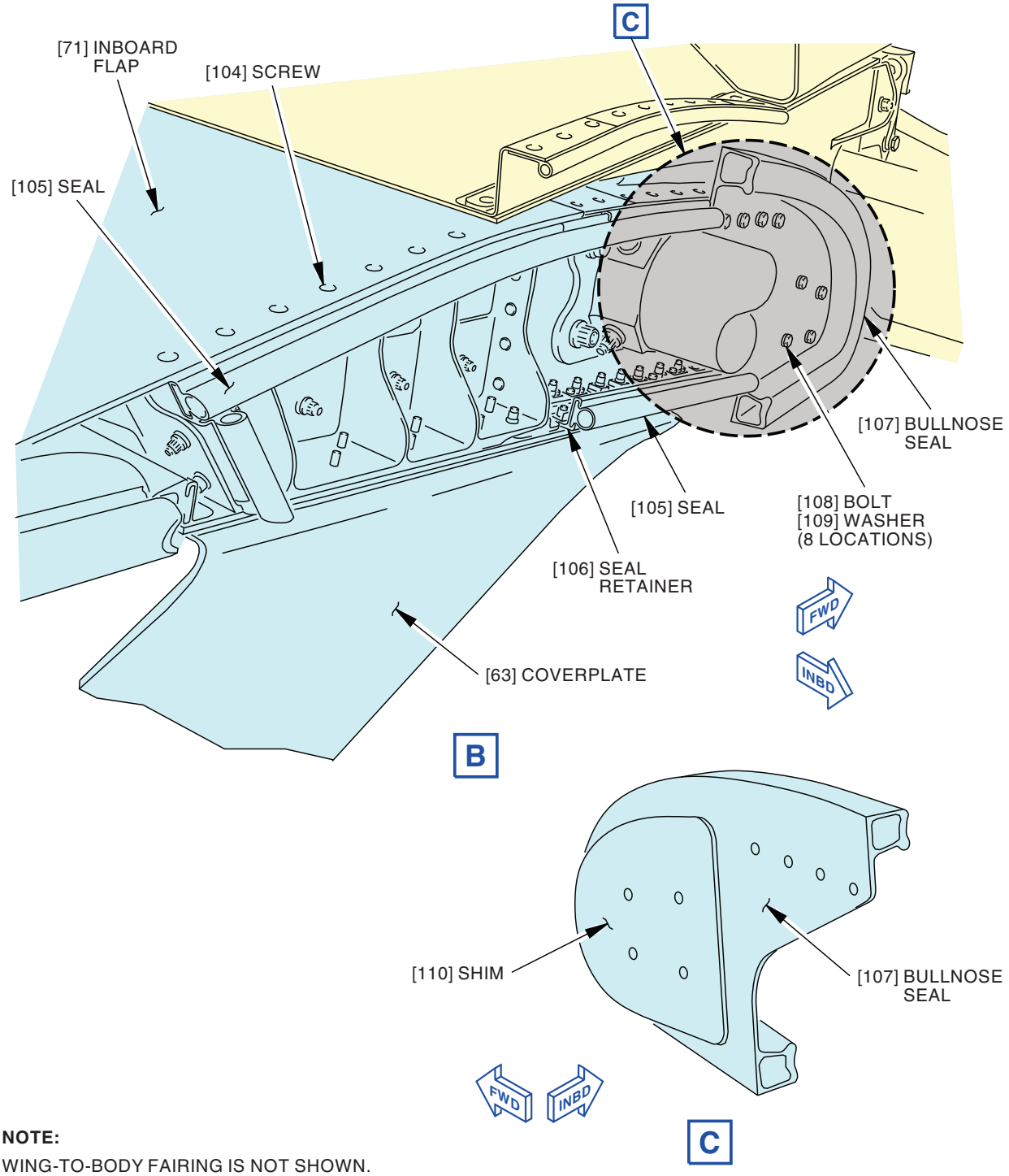
Aft Flap Pushrod Adjustment
Figure 505/27-09-71-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Inboard Trailing Edge Flap Adjustment
Figure 506/27-09-71-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



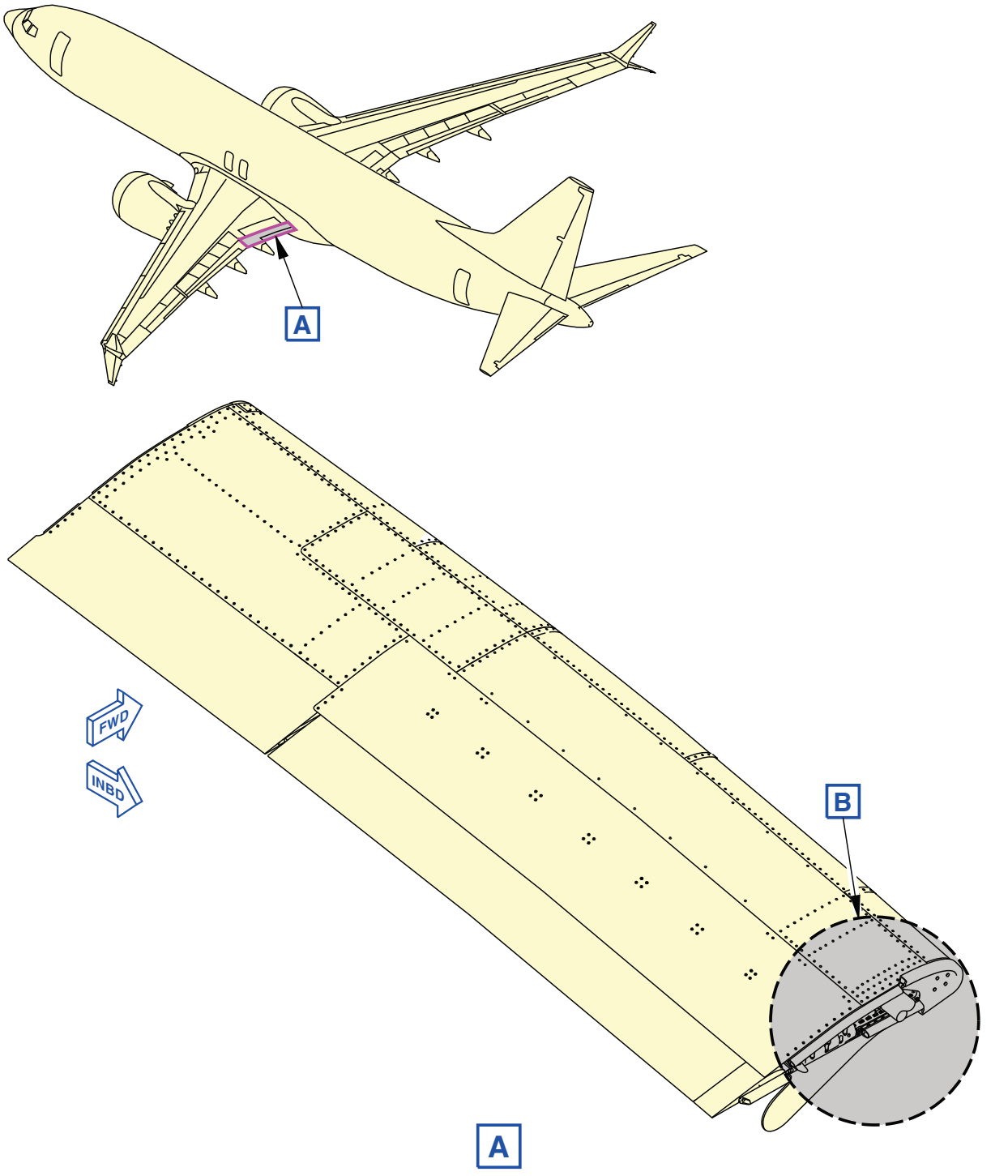
NOTE:

WING-TO-BODY FAIRING IS NOT SHOWN.

3043334 S0000802299_V1

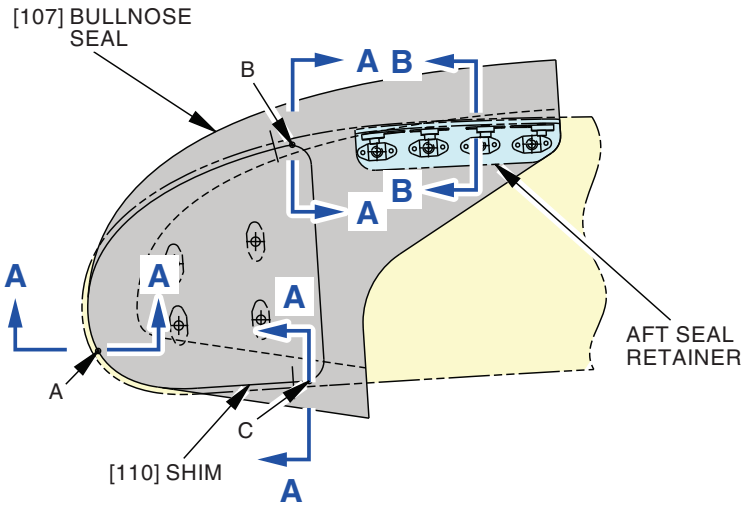
Inboard Trailing Edge Flap Adjustment
Figure 506/27-09-71-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



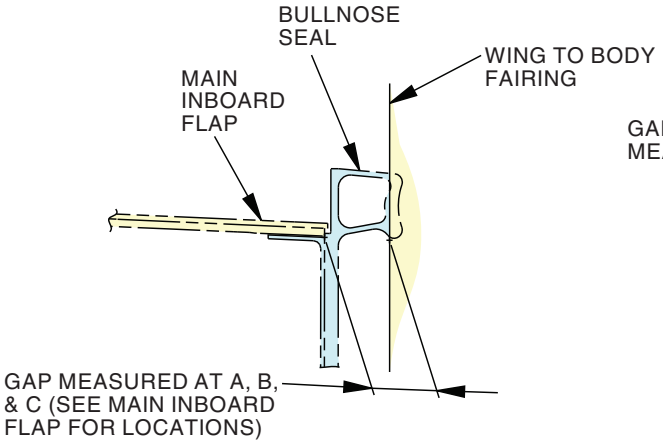
Bullnose Seal Measurement
Figure 507/27-09-71-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



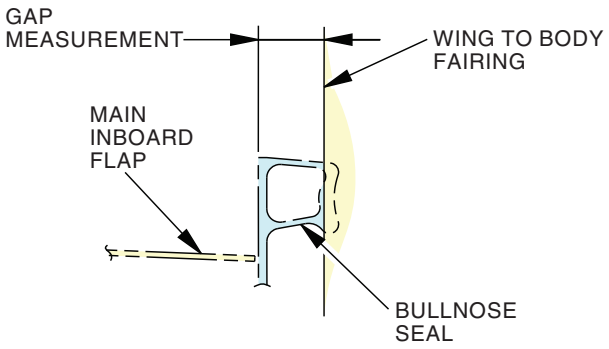
MAIN INBOARD FLAP

B



GAP MEASUREMENT - FORWARD END

A-A



GAP MEASUREMENT - AFT END

B-B

2412236 S00061530268_V2

Bullnose Seal Measurement
Figure 507/27-09-71-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-71-820-802

3. Aileron Balance Tab Adjustment to Equalize Lateral Manual Reversion Forces

A. General

(1) Use this procedure to adjust the aileron balance tab if the difference in the lateral manual reversion forces is more than the permitted 2 to 1 ratio.

(2) In this task the aileron tab control rods are referred to as the "control rods".

B. References

Reference	Title
27-11-00-700-803	Manual Reversion Control Friction Test (P/B 501)
27-11-00-820-802	Pogo and Power Control Unit (PCU) Adjustment (P/B 501)
27-11-00-820-803	Aileron Balance Tab Adjustment (Recommended Method) (P/B 501)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

D. Prepare for the Adjustment

SUBTASK 27-09-71-860-007

(1) Make sure the aileron balance tab is correctly adjusted.

(a) To do a check of the adjustment of the aileron balance tab, do this task: Aileron Balance Tab Adjustment (Recommended Method), TASK 27-11-00-820-803.

SUBTASK 27-09-71-860-008

(2) Make sure the aileron control wheel forces, with power off, are not more than the specified limits.

(a) To do a check of the aileron control wheel forces, do this task: Manual Reversion Control Friction Test, TASK 27-11-00-700-803.

SUBTASK 27-09-71-860-009

(3) Make sure the stop clearances at the quadrants of the power control unit are not out of tolerance.

(a) To do a check of the stop clearances, do this task: Pogo and Power Control Unit (PCU) Adjustment, TASK 27-11-00-820-802.

SUBTASK 27-09-71-860-010

(4) If the Flaps Up trim is more than 0.75 unit, make the trim correction first before you do the manual reversion correction.

(a) To adjust the trailing edge flap for lateral trim, do this task: Trailing Edge Flap Adjustment for Lateral Trim and Test, TASK 27-09-71-820-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-71-860-011

(5) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

E. Aileron Balance Tab Adjustment To Equalize Lateral Manual Reversion Forces

SUBTASK 27-09-71-970-010

(1) Make a record of the force necessary to turn the control wheel approximately 30 degrees in the clockwise direction.

SUBTASK 27-09-71-970-011

(2) Make a record of the force necessary to turn the control wheel approximately 30 degrees in the counterclockwise direction.

SUBTASK 27-09-71-970-012

(3) Divide the counterclockwise force by the clockwise force to get the force ratio, or divide the clockwise force by the counterclockwise force to get a force ratio value greater than 1.0.

SUBTASK 27-09-71-970-013

(4) Use the force ratio to find the correct length change for the control rod (Figure 508).

NOTE: For the left or right aileron balance tab, you must change the length of the control rods in sets. The adjustment of the two control rods must be the same amount. It is recommended that you split the adjustment equally between the left and right control rod sets.

(a) To correct high left forces (control wheel counterclockwise), do one of these steps:

1) Make the left control rod set shorter.
2) Make the right control rod set longer.
3) Make the left control rod set shorter and make the right control rod set longer.

(b) To correct high right forces (control wheel clockwise), do one of these steps:

1) Make the right control rod set shorter.
2) Make the left control rod set longer.
3) Make the right control rod set shorter and make the left control rod set longer.

SUBTASK 27-09-71-820-008

(5) Adjust the aileron balance tab [111] (Figure 509):

(a) Do these steps to get access to the aileron tab control rods [113]:

1) Move the aileron trailing edge to the down position.
2) Remove the bolts [122] that attach the forward removable fairing [123] to the aileron [112].
3) Remove the bolts [120] that attach the aft removable fairing [121] to the aileron balance tab [111].

(b) Loosen the jamnuts [124] and jamnuts [126] on the aileron tab control rods [113].

(c) Do these steps if coarse adjustment is necessary;

1) Remove the pins [119], nuts [118], washers [117], bushings [116], washers [115] and bolts [114] to disconnect the aileron tab control rods [113] from the aileron balance tab [111].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Turn the rod ends on the aileron tab control rods [113] for adjustment.

NOTE: One-half turn of the rod end lengthens the control rod 0.021 in. (0.53 mm). One-half turn of the rod end and the barrel [125] together changes the position of the tab trailing edge 0.025 in. (0.64 mm).

a) To raise the trailing edge of the aileron balance tab [111], lengthen the aileron tab control rods [113].

b) To lower the trailing edge of the aileron balance tab [111], shorten the aileron tab control rods [113].

3) Connect the aileron tab control rods [113] to the aileron balance tab [111];

a) Install the bushings [116], washers [115], washers [117], bolts [114] and nuts [118].

b) If there is binding between the aileron tab control rod [113] and the bolt [114] or deflection of the aileron tab control rods [113], adjust the control rod length.

c) Make sure there is a minimum of 0.015 in. (0.38 mm) gap between the bushing [116] and the washer [115].

d) Tighten the nuts [118] to 50 in-lb (5.6 N·m) - 80 in-lb (9.0 N·m).

e) Install the pins [119].

(d) Do these steps if fine adjustment is necessary:

1) Turn the barrel [125] on the aileron tab control rods [113].

NOTE: One turn of the barrel lengthens the control rod 0.008 in. (0.20 mm).

a) To raise the trailing edge of the aileron balance tab [111], lengthen the aileron tab control rods [113].

b) To lower the trailing edge of the aileron balance tab [111], shorten the aileron tab control rods [113].

2) Make sure the threads cover the inspection holes in the barrel [125].

(e) Tighten the jamnuts [124] to 100 in-lb (11.3 N·m) - 140 in-lb (15.8 N·m).

(f) Tighten the jamnuts [126] to 80 in-lb (9.0 N·m) - 100 in-lb (11.3 N·m).

(g) Install the bolts [120] that attach the aft removable fairing [121] to the aileron balance tab [111].

(h) Install the bolts [122] that attach the forward removable fairing [123] to the aileron [112].

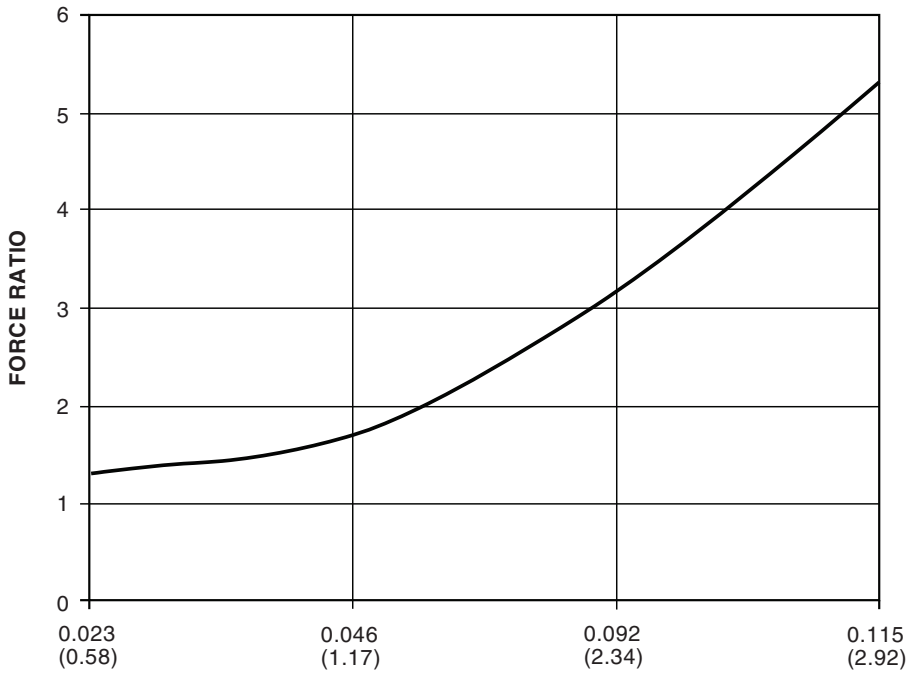
F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-71-860-012

(1) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

——— END OF TASK ———

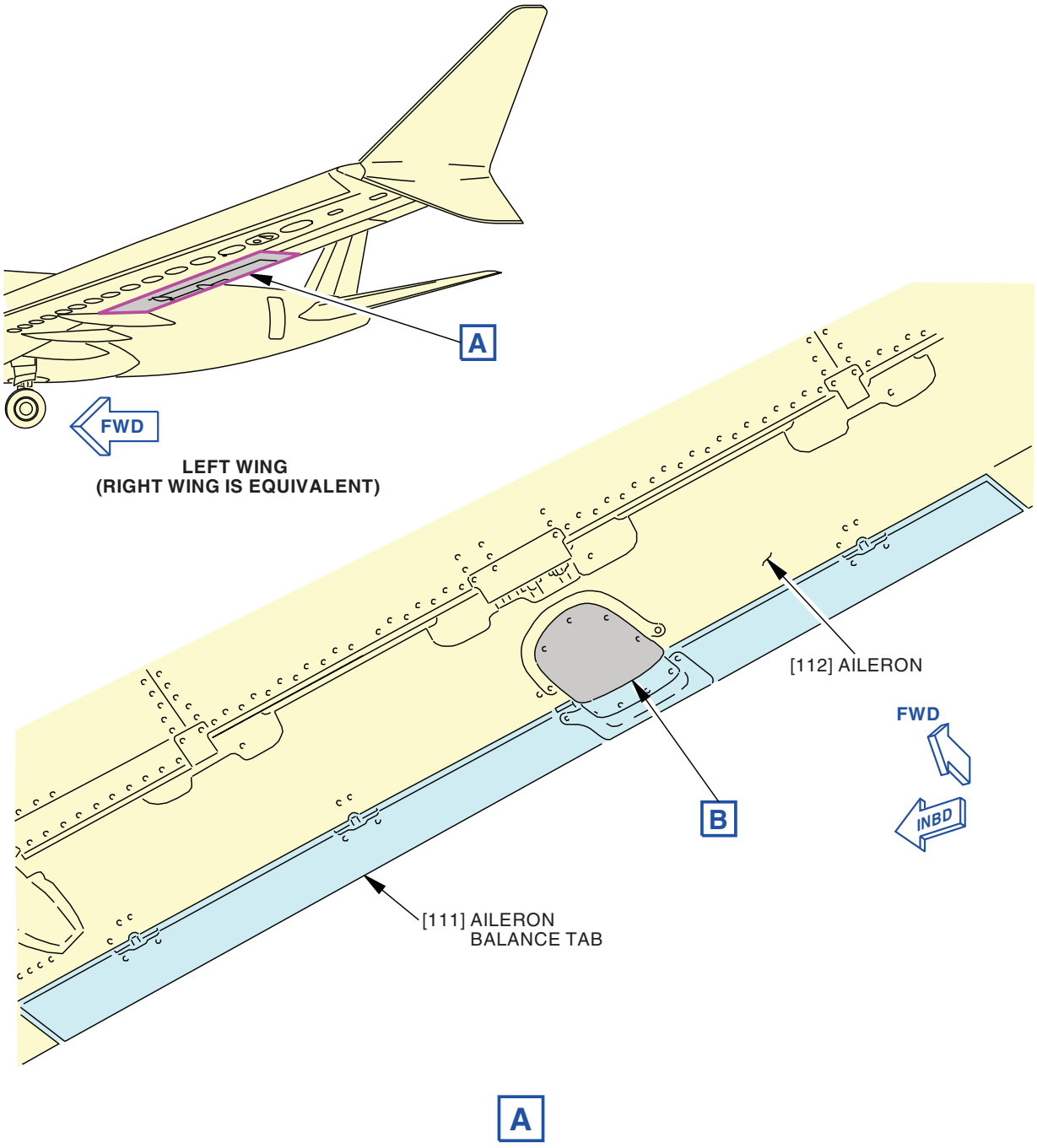
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412237 S00061530270_V1

Manual Reversion Force Correction
Figure 508/27-09-71-990-809

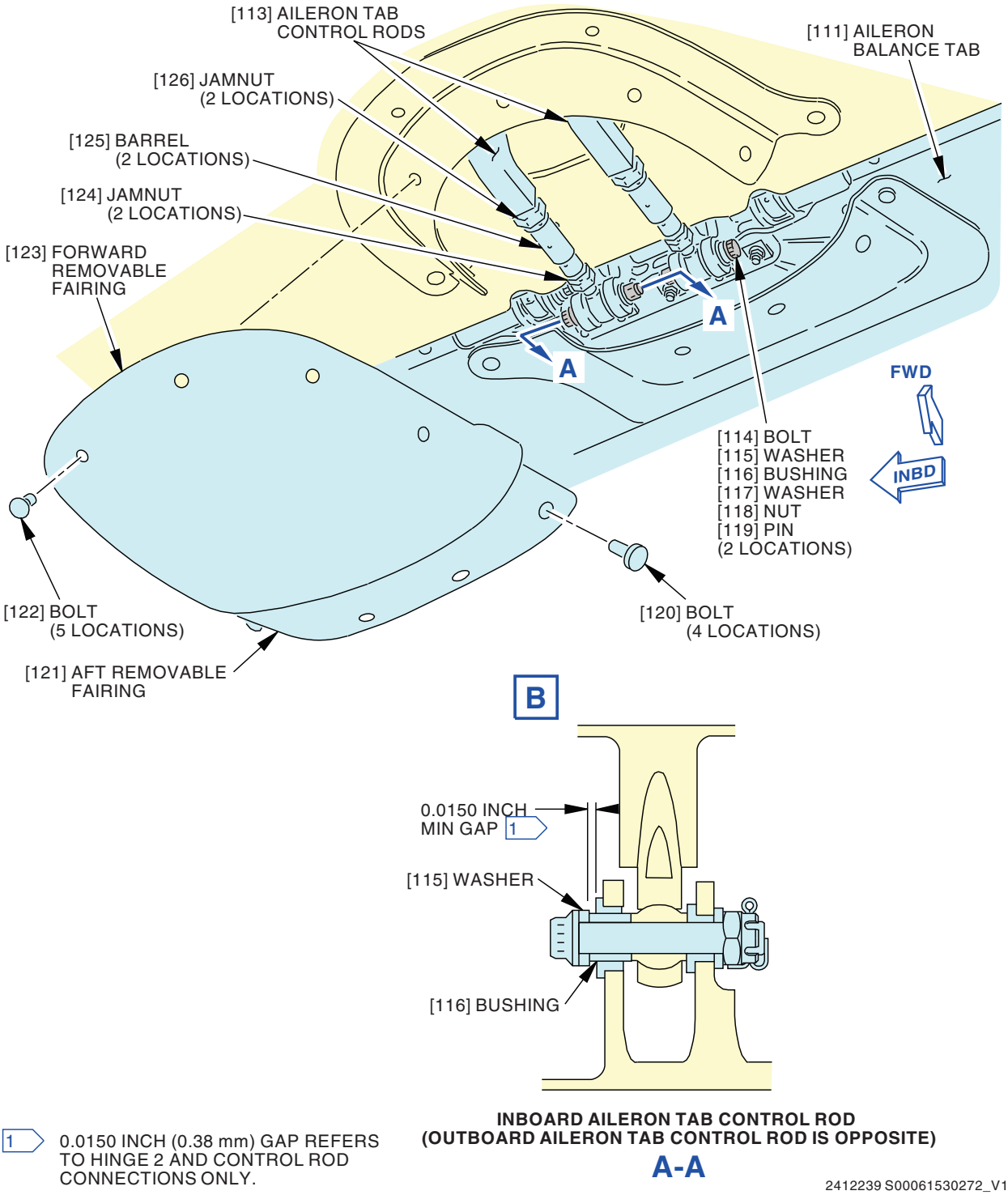
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412238 S00061530271_V1

Aileron Balance Tab Adjustment
Figure 509/27-09-71-990-810 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 0.0150 INCH (0.38 mm) GAP REFERS TO HINGE 2 AND CONTROL ROD CONNECTIONS ONLY.

INBOARD AILERON TAB CONTROL ROD
(OUTBOARD AILERON TAB CONTROL ROD IS OPPOSITE)

A-A

2412239 S00061530272_V1

Aileron Balance Tab Adjustment
Figure 509/27-09-71-990-810 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-71-860-801

4. Rudder Trim Adjustment for Power On Discrepancy Correction

A. General

(1) It is necessary to do the rudder power on discrepancy correction if the airplane does not meet the rudder trim requirements shown in the table found in this task (TASK 27-09-71-820-801).

B. References

Reference	Title
27-21-00-700-805	Rudder Travel Test (P/B 501)
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-820-804	Control Rod for Main Rudder Power Control Unit Adjustment (P/B 501)
27-21-00-820-805	Control Rod for Standby Rudder Power Control Unit Adjustment (P/B 501)

C. Location Zones

Zone	Area
325	Vertical Fin - Rudder

D. Procedure

SUBTASK 27-09-71-860-014

(1) Do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-09-71-820-010

(2) Adjust the rudder trim:

(a) Trim until the rudder trim indicator is ±0.25 units of the position noted on the flight squawk.

(b) Loosen the two index plate screws on the tailcone.

(c) Put the index plate at a position where the two sides of the rudder trailing edge are within the width of the tab on the index plate.

(d) Tighten the index plate screws.

SUBTASK 27-09-71-820-011

(3) Use the rudder trim knob to set the trim to indicate zero.

(a) Make sure you can install the rig pin, R-5.

SUBTASK 27-09-71-860-015

(4) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-09-71-820-012

(5) Adjust the input of the main power control unit (PCU) to align the rudder with the new index position. To do this, do this task: Control Rod for Main Rudder Power Control Unit Adjustment, TASK 27-21-00-820-804.

SUBTASK 27-09-71-820-013

(6) Adjust the input of the standby PCU to align with the main PCU. To do this, do this task: Control Rod for Standby Rudder Power Control Unit Adjustment, TASK 27-21-00-820-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-09-71-820-014

(7) Adjust the input of the rudder position sensor to align the rudder with the new index plate position.

SUBTASK 27-09-71-720-001

(8) Do this task: Rudder Travel Test, TASK 27-21-00-700-805.

——— END OF TASK ———

TASK 27-09-71-860-802

5. Rudder Trim Adjustment for Power Off Discrepancy Correction

A. References

Reference	Title
27-21-00-820-805	Control Rod for Standby Rudder Power Control Unit Adjustment (P/B 501)

B. Location Zones

Zone	Area
320	Subzone - Vertical Fin and Rudder
325	Vertical Fin - Rudder

C. Procedure

SUBTASK 27-09-71-820-015

(1) Do a check of the standby rudder standby power control unit adjustment. To do this, do this task: Control Rod for Standby Rudder Power Control Unit Adjustment, TASK 27-21-00-820-805.

——— END OF TASK ———

TASK 27-09-71-970-801

6. Flight Controls Adjustment Record

(Figure 510)

A. Location Zones

Zone	Area
190	Subzone - Wing-to-Body Fairing
300	Empennage
500	Left Wing
600	Right Wing

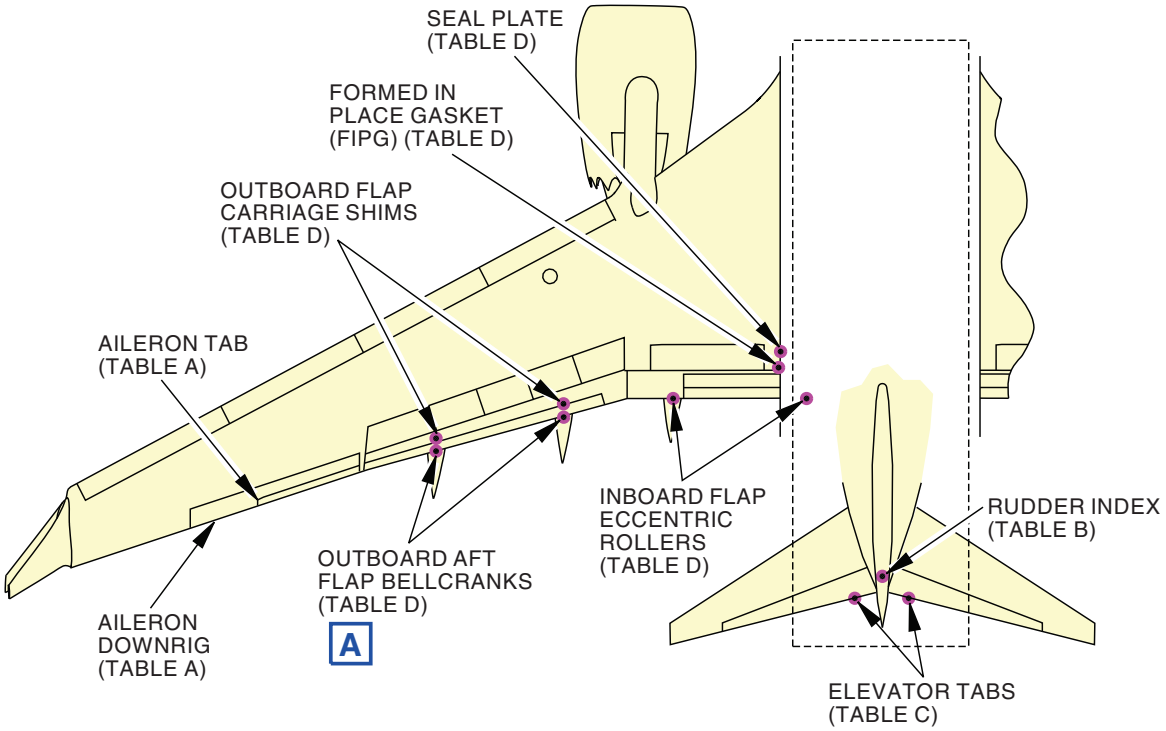
B. Procedure

SUBTASK 27-09-71-970-014

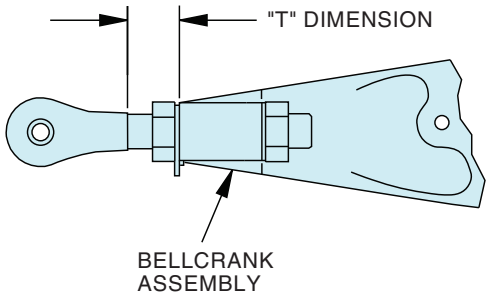
(1) You can use this figure to record the initial rigging condition of the flight control surfaces.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT WING
(RIGHT WING IS OPPOSITE)



OUTBOARD AFT FLAP BELLCRANK

A

2412240 S00061530277_V1

Flight Controls Surface Mismatch Record
Figure 510/27-09-71-990-811

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERONS		
	AILERON DOWNRIG:	LHS_____ RHS_____
3	AILERON TAB:	LHS_____ RHS_____

TABLE A

RUDDER	
	INDEX LOCATION TO APL ¢_____
	T.E. WITHIN INDEX YES_____ NO_____
	RUDDER TRIM_____

TABLE B

ELEVATORS (TAB RIGGING LIMITS)
1 SET STAB = 4 UNITS (B = 39.17 ±0.01)_____
SET ELEV TO ALIGN WITH INDEX = ±0.04_____ (HYDRAULIC POWER OFF)
1 FINAL TAB SETTING PRIOR TO DELIVERY (+0.20/+0.50) LEFT_____ RIGHT_____

TABLE C

LATERAL TRIM CONFIGURATION	
	SEAL PLATE: LHS: 149A7330-_____ THICKNESS_____ NOT USED_____ RHS: 149A7330-_____ THICKNESS_____ NOT USED_____
	FORMED IN PLACE GASKET (FIPG): LHS: USED_____ NOT USED_____ RHS: USED_____ NOT USED_____
4	INBD FLAP ECCENTRIC ROLLER: LHS OUTBD_____ LHS INBD_____ RHS OUTBD_____ RHS INBD_____
	OUTBD FLAP CARRIAGE SHIM: LHS OUTBD_____ LHS INBD_____ RHS OUTBD_____ RHS INBD_____
2	OUTBD AFT FLAP BELLCRANK "T": LHS OUTBD_____ LHS INBD_____ RHS OUTBD_____ RHS INBD_____--

TABLE D

1 (+) INDICATES CONTROL SURFACE ABOVE TRAILING EDGE.
(-) INDICATES CONTROL SURFACE BELOW TRAILING EDGE.

2 MUST AGREE WITHIN ±0.050.

3 MEASURE AILERON TAB (ALL MODELS) AT OUTBOARD END OF TAB, WITH THE AILERON IN THE DOWNRIG POSITION.

4 ALL FOUR ECCENTRIC ROLLERS ON THE SAME FLAP MUST BE AT THE SAME POSITION.

2412241 S00061530278_V2

Flight Controls Surface Mismatch Record
Figure 511/27-09-71-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLIGHT CONTROLS SURFACES - INSPECTION/CHECK

1. General

A. This procedure contains these tasks:

(1) The inspection of the aileron.

(2) The inspection of the aileron balance tab freeplay.

(3) The inspection of the rudder.

(4) The inspection of the horizontal stabilizer.

(5) The inspection of the spoilers.

TASK 27-09-91-200-801

2. Aileron - Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-11-11-200-801	Aileron Inspection (P/B 601)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy
STD-1238	Indicator - Dial
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator

C. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

D. Prepare for the Check

SUBTASK 27-09-91-860-001

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply power to the hydraulic systems A and B (TASK 29-11-00-860-801).

E. Do a Check of the Aileron Freeplay

SUBTASK 27-09-91-220-001

(1) Do a freeplay check of one aileron with no movement of the opposite aileron or the control wheel:

(a) Make sure the control wheel is at the center.

(b) Attach a mount, STD-1279, to the top wing structure (outboard of the aileron).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Attach a dial indicator, STD-1238 to take measurements from the outboard corner of the aileron trailing edge.

(d) Put the plunger of the dial indicator, STD-1238 within 0.1 in. (2.54 mm) forward of the trailing edge of the aileron.

1) Adjust the dial indicator, STD-1238 to zero.

CAUTION PUT SUFFICIENT PADDING BETWEEN THE FORCE SCALE, AND THE CONTROL SURFACE TO PREVENT DAMAGE TO THE CONTROL SURFACE. THE FORCE SCALE CAN CAUSE DAMAGE TO THE CONTROL SURFACE.

(e) With the push/pull scale 0-25 pound (0-11 kilogram), STD-753, push up with a force of approximately 5 lbf (22.24 N).

NOTE: Apply the load to the aileron balance tab cutout rib (adjacent to the outboard edge of the aileron tab) and centered within 0.5 in. (12.70 mm) of the trailing edge.

1) Make a record of the travel shown on the dial indicator.

2) Slowly remove the force.

(f) Do not adjust the dial indicator, STD-1238.

CAUTION PUT SUFFICIENT PADDING BETWEEN THE FORCE SCALE, AND THE CONTROL SURFACE TO PREVENT DAMAGE TO THE CONTROL SURFACE. THE FORCE SCALE CAN CAUSE DAMAGE TO THE CONTROL SURFACE.

(g) With the push/pull scale 0-25 pound (0-11 kilogram), STD-753, push down with a force of approximately 5 lbf (22.24 N).

NOTE: Apply the load to the aileron balance tab cutout rib (adjacent to the outboard edge of the aileron tab) and centered within 0.5 in. (12.70 mm) of the trailing edge.

1) Make a record of the travel shown on the dial indicator.

2) Slowly remove the force.

(h) Make sure the total up and down freeplay of the aileron is not more than 0.1 in. (2.54 mm).

(i) Remove the dial indicator, STD-1238.

(j) Remove the mount, STD-1279.

SUBTASK 27-09-91-200-002

(2) If the freeplay is not in the acceptable limits, do this task: Aileron Inspection, TASK 27-11-11-200-801.

SUBTASK 27-09-91-220-002

(3) Do the same freeplay check on the opposite aileron.

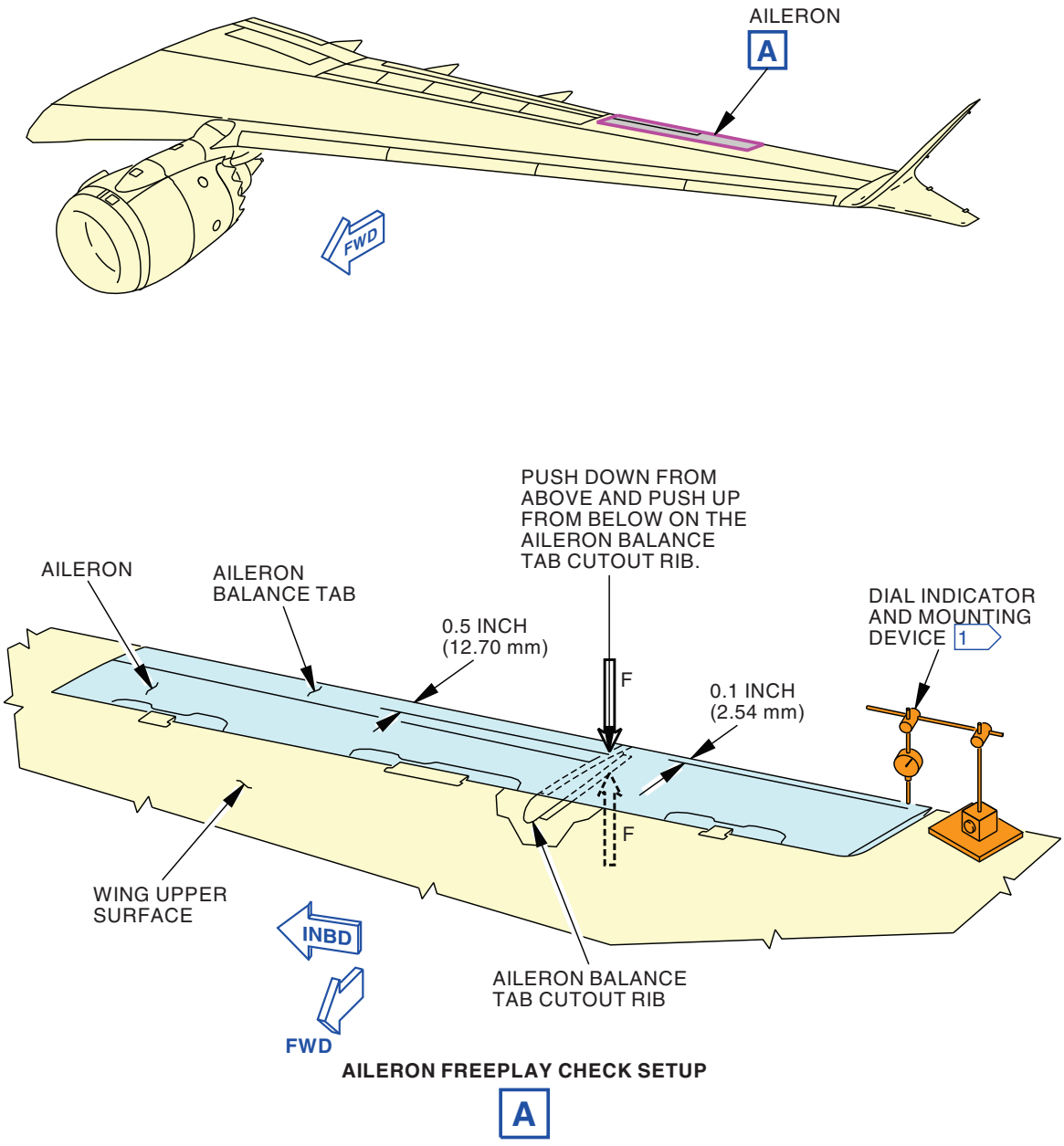
F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-91-000-001

(1) Remove power from the hydraulic systems A and B (TASK 29-11-00-860-805).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON FREEPLAY CHECK SETUP

A

1 THE PLUNGER SHOULD BE WITHIN 0.1 INCH (2.54 mm) FORWARD OF THE TRAILING EDGE OF THE AILERON.

2412242 S00061530282_V3

Aileron Freeplay Check
Figure 601/27-09-91-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

27-CMR-04

TASK 27-09-91-200-802

3. Aileron Balance Tab Freeplay - Inspection

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-11-21-200-801	Aileron Balance Tab Inspection (P/B 601)
27-11-34-200-801	Aileron Tab Control Rods Inspection (P/B 601)
29-11-00 P/B 201	HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy
STD-1107	Gauge - Feeler, 0.0 - 0.5 Inch, Readable to 1/1000th
STD-1238	Indicator - Dial
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator
STD-1303	Caliper - Dial

C. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

D. Prepare for the Check

SUBTASK 27-09-91-863-001

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply power to the hydraulic systems A and B (PAGEBLOCK 29-11-00/201).

E. Aileron Balance Tab Freeplay Check

SUBTASK 27-09-91-220-003

(1) Do a freeplay check of one aileron balance tab with no movement of the aileron:

(a) Make sure that the aileron does not move while you perform this freeplay check of the balance tab.

NOTE: If the aileron moves, the reading on the indicator will not be accurate.

(b) Make sure that the control wheel is at the center.

(c) Attach the mount, STD-1279, and dial indicator, STD-1238, to the aileron.

(d) Put the plunger of the dial indicator, STD-1238, within 0.1 in. (2.54 mm) forward of the aileron balance tab trailing edge, and opposite of the tab control rods.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Adjust the dial indicator, STD-1238, to zero.

CAUTION PUT SUFFICIENT PADDING BETWEEN THE FORCE SCALE, AND THE CONTROL SURFACE TO PREVENT DAMAGE TO THE CONTROL SURFACE. THE FORCE SCALE CAN CAUSE DAMAGE TO THE CONTROL SURFACE.

(e) With the push/pull scale 0-25 pound (0-11 kilogram), STD-753, push up with a force of approximately 5 lbf (22.24 N).

1) Apply the force 0.25 in. (6.35 mm) from the trailing edge of the aileron balance tab, directly under the plunger of the dial indicator, STD-1238.

2) Make a record of the travel shown on the dial indicator, STD-1238.

3) Slowly remove the force.

(f) Do not adjust the dial indicator, STD-1238.

CAUTION PUT SUFFICIENT PADDING BETWEEN THE FORCE SCALE, AND THE CONTROL SURFACE TO PREVENT DAMAGE TO THE CONTROL SURFACE. THE FORCE SCALE CAN CAUSE DAMAGE TO THE CONTROL SURFACE.

(g) With the push/pull scale 0-25 pound (0-11 kilogram), STD-753, push down with a force of approximately 5 lbf (22.24 N).

1) Apply the force 0.25 in. (6.35 mm) from the trailing edge of the aileron balance tab, 0.5 in. (12.70 mm) or less from the plunger of the dial indicator, STD-1238.

2) Make a record of the travel shown on the dial indicator, STD-1238.

3) Slowly remove the force.

(h) Add the up and down freeplay of the aileron balance tab to get the total up and down freeplay of the aileron balance tab.

1) Make sure that the total up and down freeplay of the aileron balance tab is not more than 0.06 in. (1.52 mm).

(i) Remove the dial indicator, STD-1238.

(j) Remove the mount, STD-1279.

SUBTASK 27-09-91-220-012

(2) Do these steps to make sure that the axial play (inboard-outboard direction) of the aileron balance tab is not more than 0.04 in. (1.02 mm).

(a) Hold the aileron balance tab around the trailing edge and with hand pressure, move it outboard.

(b) Use a 0.0 - 0.5 Inch feeler gauge, STD-1107, or a dial caliper, STD-1303, to measure the clearance between the aileron and the outboard end of the aileron balance tab.

1) Make a record of this clearance.

(c) Hold the aileron balance tab around the trailing edge and with hand pressure, move it inboard.

(d) Measure the clearance between the aileron and the outboard end of the aileron balance tab.

1) Make a record of the clearance dimension.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Make sure that the difference between the two measured clearances, or the axial play, is not more than 0.04 in. (1.02 mm).

SUBTASK 27-09-91-200-001

(3) If the freeplay of the aileron balance tab is not in the acceptable limits, do these tasks:

- Aileron Balance Tab Inspection, TASK 27-11-21-200-801
- Aileron Tab Control Rods Inspection, TASK 27-11-34-200-801.

SUBTASK 27-09-91-220-004

(4) Do the same freeplay check on the aileron balance tab of the opposite aileron.

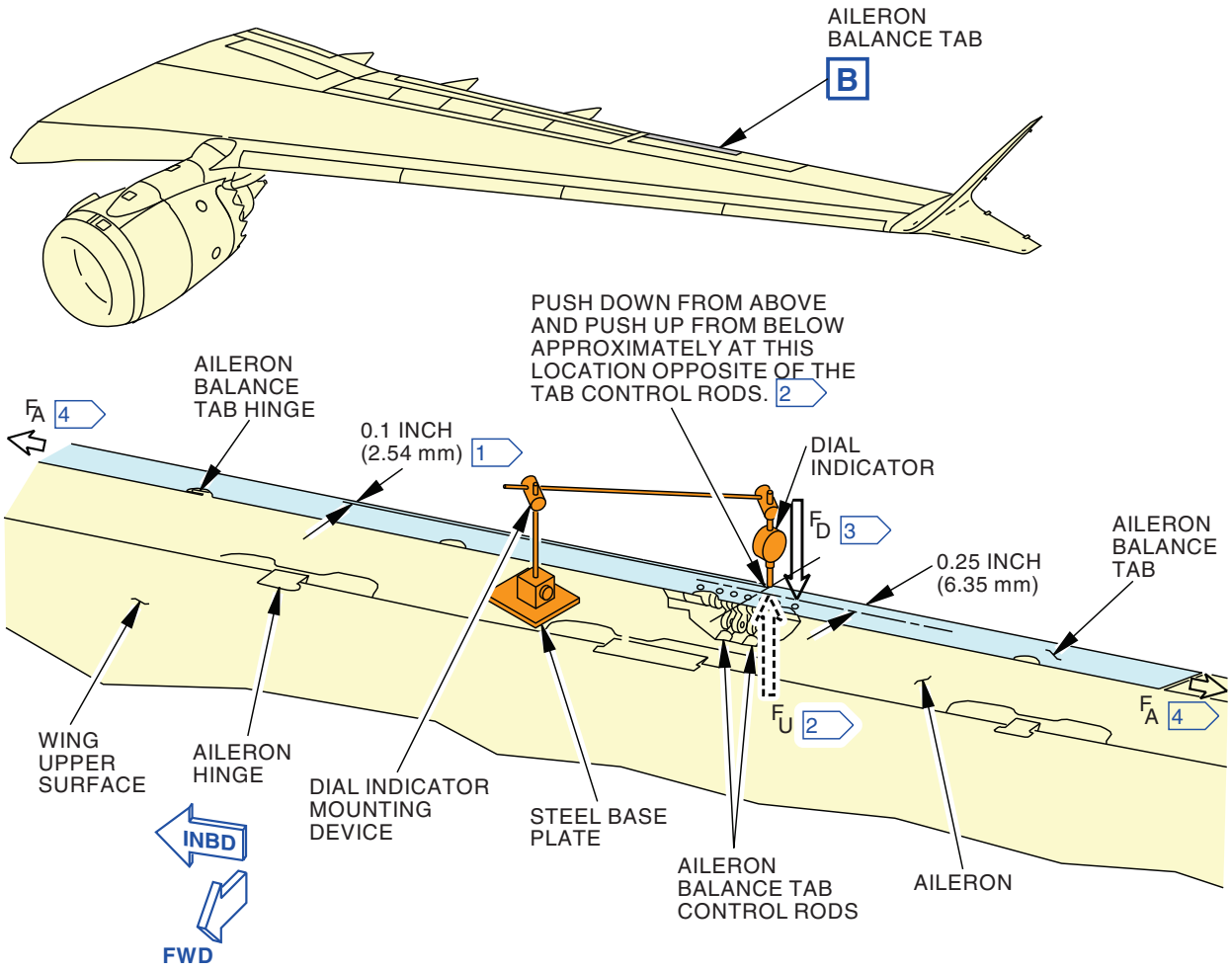
F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-91-860-007

(1) Remove power from hydraulic systems A and B (TASK 29-11-00-860-805).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT AILERON BALANCE TAB FREEPLAY CHECK
(RIGHT AILERON BALANCE TAB IS OPPOSITE)

B

1 THE PLUNGER SHOULD BE WITHIN 0.1 INCH (2.54 mm) FORWARD OF THE TRAILING EDGE OF THE AILERON BALANCE TAB, AND OPPOSITE OF THE TAB CONTROL RODS.

2 APPLY THE UPWARD FORCE, FU, 0.25 INCH (6.35 mm) FORWARD OF THE TRAILING EDGE OF THE AILERON BALANCE TAB, APPROXIMATELY UNDER THE PLUNGER OF THE DIAL INDICATOR.

3 APPLY THE DOWNWARD FORCE, FD, 0.25 INCH (6.35 mm) FORWARD OF THE TRAILING EDGE OF THE AILERON BALANCE TAB, 0.5 INCH (12.70 mm), OR LESS, FROM THE PLUNGER OF THE DIAL INDICATOR.

4 USE HAND PRESSURE TO MOVE THE AILERON OUTBOARD, THEN INBOARD TO DETERMINE THE AILERON BALANCE TAB AXIAL FREEPLAY.

2412243 S00061530284_V3

Aileron Balance Tab Freeplay Check
Figure 602/27-09-91-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-91-200-803

4. Rudder Inspection

(Figure 603)

A. General

(1) Use this task to do a check of the rudder control system freeplay.

B. References

Reference	Title
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
29-11-00 P/B 201	HYDRAULIC SYSTEMS A AND B - MAINTENANCE PRACTICES

C. Tools/Equipment

Reference	Description
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy
STD-1238	Indicator - Dial
STD-1278	Block - Loading, 1/8 Inch Thick, 3 Inch by 3 Inch Wood or Fiberglass Block with Pad to Prevent Damage to the Skin.
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

E. Procedure

SUBTASK 27-09-91-860-003

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Supply power to the hydraulic systems A and B (PAGEBLOCK 29-11-00/201).

SUBTASK 27-09-91-220-005

(2) Do the rudder hinge bearing and Power Control Unit (PCU) bearing freeplay check:

(a) Make sure the rudder and rudder pedals are in the neutral position.

(b) Make sure the rudder trim indicator is at the zero unit of trim.

(c) Attach the dial indicator, STD-1238 with mount, STD-1279 at the lower end of the rudder trailing edge.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION USE A LOADING BLOCK AT THE SPECIFIED LOCATION TO MOVE THE RUDDER SURFACE. TOO MUCH FORCE OR THE INCORRECT LOCATION WILL CAUSE DAMAGE TO THE RUDDER.

(d) Use a block, STD-1278 and push/pull scale 0-25 pound (0-11 kilogram), STD-753 to move the rudder to the left side with a force of 12 ±4 lbf (53.38 ±17.79 N) and release slowly.

NOTE: Apply the force approximately6 in. (152.40 mm) -8 in. (203.20 mm) from the bottom of the trailing edge and 3 in. (76.20 mm) forward of the trailing edge.

1) Make a record of the rudder position on the dial indicator.

CAUTION USE A LOADING BLOCK AT THE SPECIFIED LOCATION TO MOVE THE RUDDER SURFACE. TOO MUCH FORCE OR THE INCORRECT LOCATION WILL CAUSE DAMAGE TO THE RUDDER.

(e) Use a block, STD-1278 and push/pull scale 0-25 pound (0-11 kilogram), STD-753 to move the rudder to the right side with a force of 12 ±4 lbf (53.38 ±17.79 N) and release slowly.

NOTE: Apply the force approximately 6 in. (152.40 mm) - 8 in. (203.20 mm) from the bottom of the trailing edge and 3 in. (76.20 mm) forward of the trailing edge.

1) Make a record of the rudder position on the dial indicator, STD-1238.

(f) Make sure the difference between the left and right positions is not more than 0.05 in. (1.27 mm).

1) If the difference is more than 0.05 in. (1.27 mm), then replace the bearings on the rudder hinge or power control units.

(g) Use a block, STD-1278 and push/pull scale 0-25 pound (0-11 kilogram), STD-753 to move the rudder to the left side with a force of 12 ±4 lbf (53.38 ±17.79 N) and release slowly.

NOTE: Apply the force approximately 6 in. (152.40 mm) - 8 in. (203.20 mm) from the bottom of the trailing edge and 3 in. (76.20 mm) forward of the trailing edge.

1) Make a record of the rudder position on the dial indicator, STD-1238.

(h) Make sure the difference between the two left positions is not more than 0.03 in. (0.76 mm).

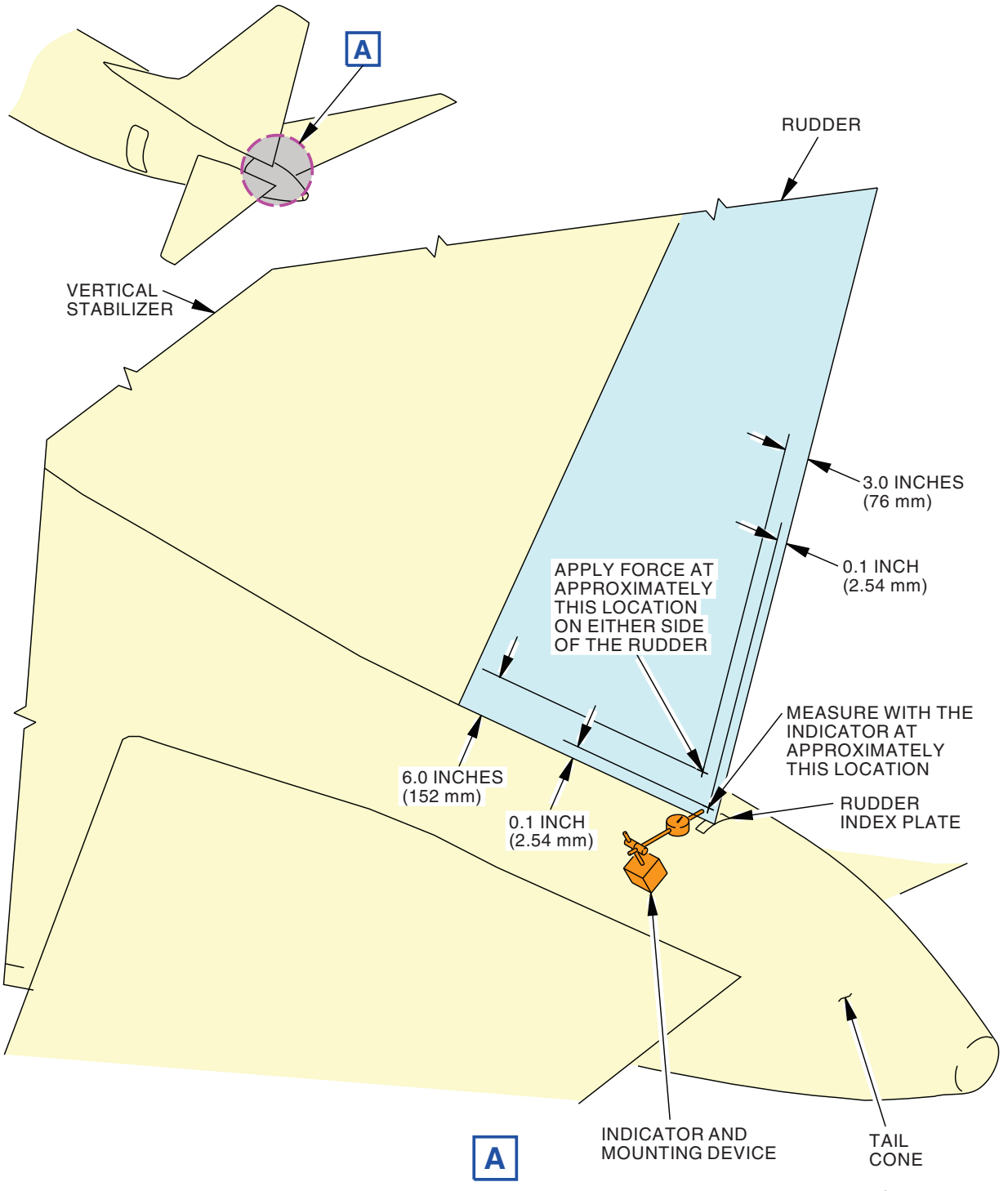
(i) Remove the dial indicator, STD-1238 from the rudder.

SUBTASK 27-09-91-860-004

(3) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Hinge and PCU Bearing Freeplay Check
Figure 603/27-09-91-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

27-CMR-10

TASK 27-09-91-200-804

5. Horizontal Stabilizer - Inspection

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task is to calculate the freeplay of the horizontal stabilizer (Figure 604). The freeplay is measured individually at three support locations that support the weight of the horizontal stabilizer: the jackscrew, the Left Hand Side (LHS) pivot hinge, and the Right Hand Side (RHS) pivot hinge. A total freeplay value will then be calculated using the freeplay values of the three locations.

(2) All three supports are under load due to the horizontal stabilizer static weight. To measure the freeplay at a support, an adequate amount of reverse load is needed. Weights are placed on top of the front spar area to create the required reverse load at the support where the freeplay is to be measured. Weights such as shotbags and persons are used as examples in this procedure.

NOTE: The user has the option to use other types of weights in the procedure.

(3) The freeplay is measured using a displacement gauge, such as a dial indicator, STD-1238, to check the movement (or slack) at the support mechanism when the load at the support is reversed.

NOTE: The user has the option to use other types of displacement gauges to replace the following tools in the procedure:

- dial indicator, STD-1238
- mount, STD-1279

(4) For the LHS pivot hinge or the RHS pivot hinge freeplay measurement, a table is provided to calculate each individual weight's contribution to the total reverse load. A total of 2500 ± 200 lbs of reverse load is required. This table is intended to facilitate the user in the placement of weights to meet the required reverse load at the pivot hinge support where the freeplay is to be measured (Table 601).

(5) For the jackscrew freeplay measurement, two sets of weights of a total of 1000 ± 40 lbs (500 ± 20 lbs per set) are required. The weights are applied in the area of the front spar of the horizontal stabilizer.

(6) For this task, only these dimensions will be used:

(a) Linear dimensions are in inches (in.)

(b) Weights are in pounds (lbs.)

NOTE: The tolerances of the pivot hinge total reverse load and the total jackscrew weights are provided. The variation of individual weights and distance measurements must meet those tolerances given.

B. References

Reference	Title
27-41-31-000-801	Horizontal Stabilizer Center Section Hinge Fitting Removal (P/B 401)
27-41-31-400-801	Horizontal Stabilizer Center Section Hinge Fitting Installation (P/B 401)
27-41-81-000-801	Stabilizer Ball Nut and Jackscrew Gearbox Removal (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-41-81-400-801	Stabilizer Ball Nut and Jackscrew Gearbox Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1238	Indicator - Dial
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
330	Subzone - Left Horizontal Stabilizer and Elevator
340	Subzone - Right Horizontal Stabilizer and Elevator

E. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

F. Prepare for the Check

SUBTASK 27-09-91-010-001

(1) On the stabilizer trim actuator, do this step:

(a) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-09-91-480-001

(2) Using either the STAB TRIM switches on the control wheels or using the stabilizer manual trim wheel on the control stand, set the horizontal stabilizer at 4 units of trim as indicated by the horizontal stabilizer position indicator on the control stand.

SUBTASK 27-09-91-860-008

(3) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand, to the CUTOUT position.

SUBTASK 27-09-91-710-001

(4) Attach a DO NOT OPERATE tag, STD-858, to the PRI switch and stabilizer trim wheel.

G. Horizontal Stabilizer Freeplay Inspection

SUBTASK 27-09-91-220-014

(1) Do a check of the freeplay for the pivot hinges (LHS and RHS) of the stabilizer center section (Figure 605).

(a) Measure the freeplay of the LHS pivot hinge (D1):

1) Attach a dial indicator, STD-1238, with a mount, STD-1279, to the LHS of the fuselage at the Body Station 1156 pivot bulkhead (Figure 605 (Sheet 1), View A).

a) Make sure that the plunger of the dial indicator, STD-1238, is approximately perpendicular to the stabilizer lower surface where it contacts the LHS retainer.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Position the dial indicator, STD-1238, to allow for subsequent plunger extension and dial travel of 0.13 inches minimum.

2) Set the dial indicator, STD-1238, to zero.

NOTE: Movement of equipment or personnel inside the fuselage can cause inaccuracy with the dial indicator readings. The dial indicator, STD-1238, should be stable at zero before weight application.

3) Gently place weights on top of the RHS stabilizer front spar in the area shown (Figure 605 (Sheet 3)).

a) Protect the upper surface of the horizontal stabilizer as necessary to make sure that the weights do not cause damage.

b) Apply weights per the "Weight Application Requirements and Calculation of Total Reverse Load (For Pivot Hinge Freeplay Measurements Only)" section of this task.

4) Record the measurement of the extension of the plunger of the dial indicator, STD-1238 as "D1".

NOTE: Movement of equipment or personnel inside the fuselage can cause inaccuracy with the dial indicator readings. The dial indicator, STD-1238, should show a stable reading when recording the freeplay measurement.

5) Gently remove the weights from the RHS stabilizer.

6) Make sure that the dial indicator value returns to 0 ± 0.003 inch.

a) If this is not met, check the plunger contact, mount, and function of the dial indicator, STD-1238, to troubleshoot.

b) If this is not met, redo the freeplay check of the LHS pivot hinge.

7) Remove the dial indicator, STD-1238, and mount, STD-1279, from the LHS of the fuselage.

(b) Measure the freeplay of the RHS pivot hinge (D2):

1) Attach a dial indicator, STD-1238, with a mount, STD-1279, to the RHS of the fuselage at the Body Station 1156 pivot bulkhead (Figure 605 (Sheet 1), View A).

a) Make sure that the plunger of the dial indicator, STD-1238, is approximately perpendicular to the stabilizer lower surface where it contacts the RHS retainer.

b) Position the dial indicator, STD-1238, to allow for subsequent plunger extension and dial travel of 0.13 inches minimum.

2) Set the dial indicator, STD-1238, to zero.

NOTE: Movement of equipment or personnel inside the fuselage can cause inaccuracy with the dial indicator readings. The dial indicator, STD-1238, should be stable at zero before weight application.

3) Gently place weights on top of the LHS stabilizer front spar in the area shown (Figure 605 (Sheet 3)).

a) Protect the upper surface of the horizontal stabilizer as necessary to make sure that the weights do not cause damage.

b) Apply weights per the "Weight Application Requirements and Calculation of Total Reverse Load (For Pivot Hinge Freeplay Measurements Only)" section of this task.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Record the measurement of the extension of the plunger of the dial indicator, STD-1238 as "D2".

NOTE: Movement of equipment or personnel inside the fuselage can cause inaccuracy with the dial indicator readings. The dial indicator, STD-1238, should show a stable reading when recording the freeplay measurement.

5) Gently remove the weights from the LHS stabilizer.

6) Make sure that the dial indicator value returns to 0 ± 0.003 inch.

a) If this is not met, check the plunger contact, mount, and function of the dial indicator, STD-1238, to troubleshoot.

b) If this is not met, redo the freeplay check of the RHS pivot hinge.

7) Remove the dial indicator, STD-1238, and mount, STD-1279, from the RHS of the fuselage.

SUBTASK 27-09-91-220-015

(2) Do a check of the freeplay for the stabilizer jackscrew (Figure 607).

(a) Measure the freeplay of the stabilizer jackscrew (D3):

1) Make sure that the stabilizer is at 4 units of trim.

2) Attach a dial indicator, STD-1238, with a mount, STD-1279, to the Body Station 1088 bulkhead structure (Figure 607, View B).

a) Make sure that the plunger of the dial indicator, STD-1238, is approximately perpendicular to the surface where it contacts the stabilizer center section (Texas Star structure).

b) Position the dial indicator, STD-1238, to allow for subsequent plunger contraction and dial travel of 0.13 inches minimum.

3) Set the dial indicator, STD-1238, to zero.

NOTE: Movement of equipment or personnel inside the fuselage can cause inaccuracy with the dial indicator readings. The dial indicator, STD-1238, should be stable at zero before weight application.

4) Gently place weights on top of the LHS and RHS stabilizer front spars (Figure 607, View C).

a) Protect the upper surface of the horizontal stabilizer as necessary to make sure that the weights do not cause damage.

b) Place weights within the weight application area shown. Distribute the weights to avoid localized high pressure at contact.

5) Record the measurement of the contraction of the plunger of the dial indicator, STD-1238, as "D3".

NOTE: Movement of equipment or personnel inside the fuselage can cause inaccuracy with the dial indicator readings. The dial indicator, STD-1238, should show a stable reading when recording the freeplay measurement.

6) Gently remove the weights from the LHS and RHS stabilizers.

7) Make sure that the dial indicator value returns to 0 ± 0.003 inch.

a) If this is not met, check the plunger contact, mount, and function of the dial indicator, STD-1238, to troubleshoot.

b) If this is not met, redo the freeplay check of the stabilizer jackscrew.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

8) Remove the dial indicator, STD-1238, and mount, STD-1279, from the Body Station 1088 bulkhead structure.

SUBTASK 27-09-91-220-016

(3) Calculate the total freeplay, referred as “H”, of the horizontal stabilizer:

(a) Use the values D1, D2, and D3 to calculate H:

H = 1.25 × D3 + [(D1 + D2)/2]

SUBTASK 27-09-91-220-017

(4) Do a check of the D1, D2, D3, and H values:

(a) If the "D1" value is more than 0.066 inch, then replace the LHS bearing.

1) Remove the hinge fitting. Do this task: Horizontal Stabilizer Center Section Hinge Fitting Removal, TASK 27-41-31-000-801.

2) Make sure that the bearing is replaced and that other related parts are serviceable.

3) Install a new or serviceable hinge fitting. Do this task: Horizontal Stabilizer Center Section Hinge Fitting Installation, TASK 27-41-31-400-801.

(b) If the "D2" value is more than 0.066 inch, then replace the RHS bearing.

1) Remove the hinge fitting. Do this task: Horizontal Stabilizer Center Section Hinge Fitting Removal, TASK 27-41-31-000-801.

2) Make sure that the bearing is replaced and that other related parts are serviceable.

3) Install a new or serviceable hinge fitting. Do this task: Horizontal Stabilizer Center Section Hinge Fitting Installation, TASK 27-41-31-400-801.

(c) If the "D3" value is more than 0.056 inch, then replace the stabilizer ball nut, ballscrew, and jackscrew gearbox.

1) Remove the stabilizer ball nut, ballscrew, and jackscrew gearbox. Do this task: Stabilizer Ball Nut and Jackscrew Gearbox Removal, TASK 27-41-81-000-801.

2) Install the new or serviceable stabilizer ball nut, ballscrew, and jackscrew gearbox. Do this task: Stabilizer Ball Nut and Jackscrew Gearbox Installation, TASK 27-41-81-400-801.

(d) If the "H" value is more than 0.078 inch, then do one or both of the following replacement options as necessary:

1) Replace the bearings at both the LHS and RHS pivot hinge support (Horizontal Stabilizer Center Section Hinge Fitting Removal, TASK 27-41-31-000-801 and Horizontal Stabilizer Center Section Hinge Fitting Installation, TASK 27-41-31-400-801) and other related parts that are not serviceable.

2) Replace the stabilizer ball nut, ballscrew, and jackscrew gearbox (Stabilizer Ball Nut and Jackscrew Gearbox Removal, TASK 27-41-81-000-801 and Stabilizer Ball Nut and Jackscrew Gearbox Installation, TASK 27-41-81-400-801).

NOTE: Fleet history shows that the hinge bearings wear significantly more than the jackscrew does, and this should be considered when making repairs to restore “H” to within limits.

(5) If any replacement(s) are done to restore D1, D2, D3, and H values to be within their respective limits, then perform the horizontal stabilizer freeplay inspection procedure again for verification.

(6) Record the inspection date and the following data of the freeplay check in a history log:

(a) D1, D2, D3, and H values.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Any replaced parts when restoring D1, D2, D3, and H values to be within their respective limits.

H. Weight Application Requirements and Calculation of Total Reverse Load (For Pivot Hinge Freeplay Measurements Only)

(Figure 606)

NOTE: This section is for providing additional information on how to apply weights and calculate the total reverse load.

SUBTASK 27-09-91-220-018

WARNING MAKE SURE THAT THE WEIGHTS ARE STABLE AND SAFELY SET. DO NOT LET PERSONNEL IN THE AREA BELOW THE STABILIZER. IF YOU DO NOT OBEY, THE WEIGHTS CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO THE AIRCRAFT.

(1) Apply enough weights during Pivot Hinge Freeplay Inspections so that a Total Reverse Load of 2500 ± 200 lbs is achieved (Figure 605 (Sheet 3)).

NOTE: Make sure that proper fall protection is in place when personnel are used as weights. The weight per person should not exceed 220 lbs.

(a) Make sure that any weights used have a base area to contact the stabilizer with in order to avoid localized loading point. Soft padding, if needed, can be used to distribute the load and protect the surface of the horizontal stabilizer.

(b) For any type of weights applied, make sure that the pressure at the contact area is less than 4 psi average.

SUBTASK 27-09-91-940-001

(2) Locate Point 0 on the horizontal stabilizer upper surface.

(a) Point 0 is the upper aft and inboard corner of the leading edge (LE) skin common to LE Station 198.23.

SUBTASK 27-09-91-970-001

(3) For each weight (or stack of weights) applied, do the following:

(a) Record the weight (or total of the stacked weights) in pounds.

(b) Record the distance in inches.

NOTE: This is the distance between Point 0 and the Center of Gravity (CG) of the applied weight (or stack of weights), along the direction of the front spar centerline.

(c) If the CG of the individual weight (or individual stack of weights) cannot be placed along the front spar, it is acceptable to shift the weight(s) in the AFT direction, so long as the CG of the weight(s) is approximately positioned atop an inspar rib.

SUBTASK 27-09-91-970-002

(4) Calculate the Total Reverse Load using the table below:

NOTE: In the Total Reverse Load table, weight and distance must be recorded in pounds and inches. Using metric values will cause the calculated Total Reverse Load to be in error.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-09-91-993-801 Total Reverse Load Table

Weight locations	(A) Enter the weight in lbs.	(B) Enter the distance in inches.	(C)	(D) Multiply (B) by (C)	(E)	(F) Add (E) to (D)	Reverse Load in lbs. Multiply (A) by (F)
1			x 0.016		+ 3.068		
2			x 0.016		+ 3.068		
3			x 0.016		+ 3.068		
4			x 0.016		+ 3.068		
5			x 0.016		+ 3.068		
6			x 0.016		+ 3.068		
7			x 0.016		+ 3.068		
8			x 0.016		+ 3.068		
9			x 0.016		+ 3.068		
10			x 0.016		+ 3.068		
11			x 0.016		+ 3.068		
12			x 0.016		+ 3.068		
	REFERENCE TOTAL WEIGHT (lbs):						TOTAL REVERSE LOAD (lbs):

(a) For example, refer to Figure 608.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-91-080-001

(1) Remove the dial indicator, STD-1238.

SUBTASK 27-09-91-080-002

(2) Remove the mount, STD-1279.

SUBTASK 27-09-91-410-001

(3) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 27-09-91-710-002

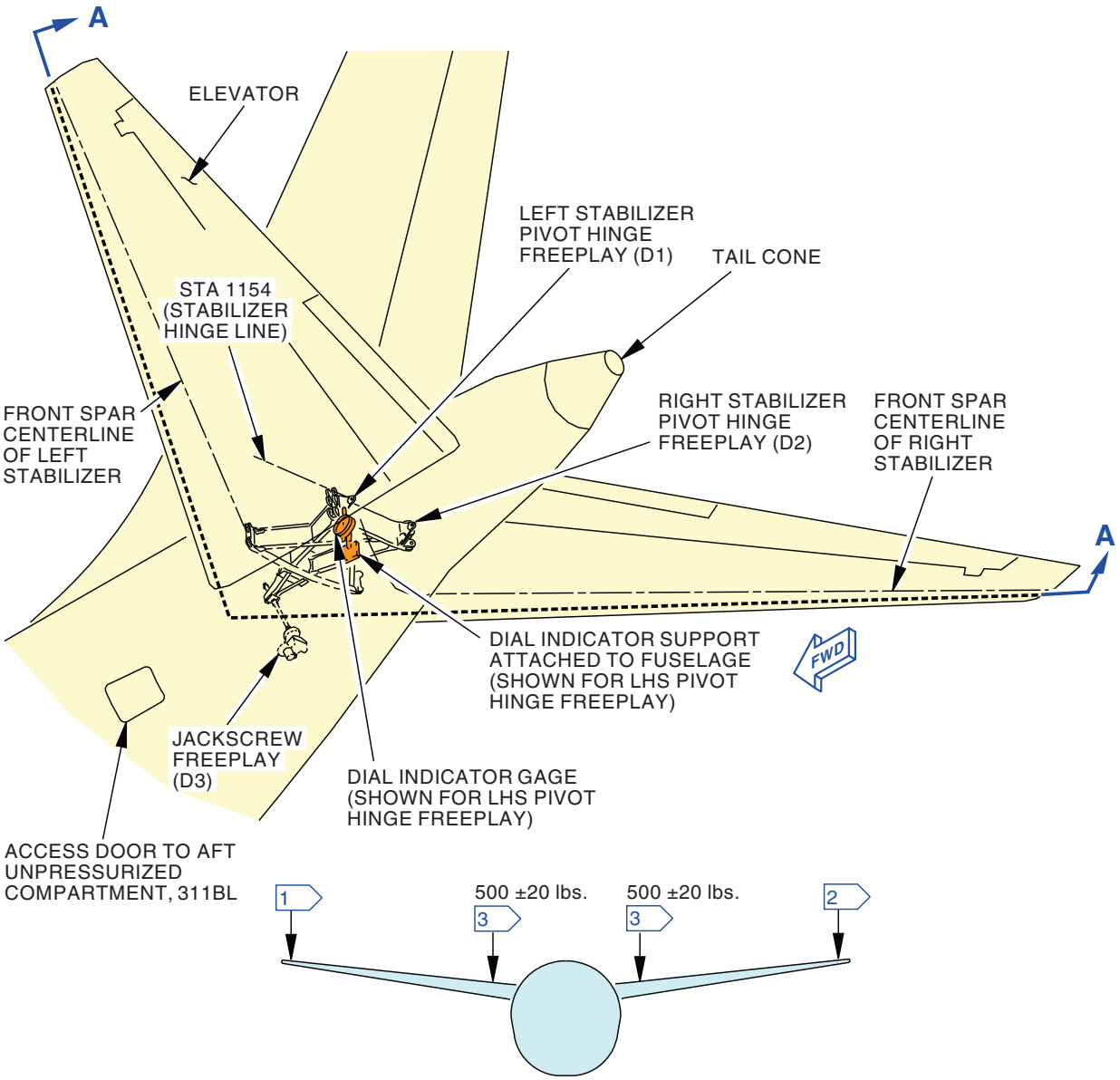
(4) Remove the DO NOT OPERATE tag, STD-858, from the PRI switch and stabilizer trim wheel.

SUBTASK 27-09-91-860-009

(5) Set the STAB TRIM PRI switch, S272, on the aft area of the control stand, to the NORMAL position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LOOKING AFT FROM FRONT SPAR OF LEFT AND RIGHT STABILIZERS)

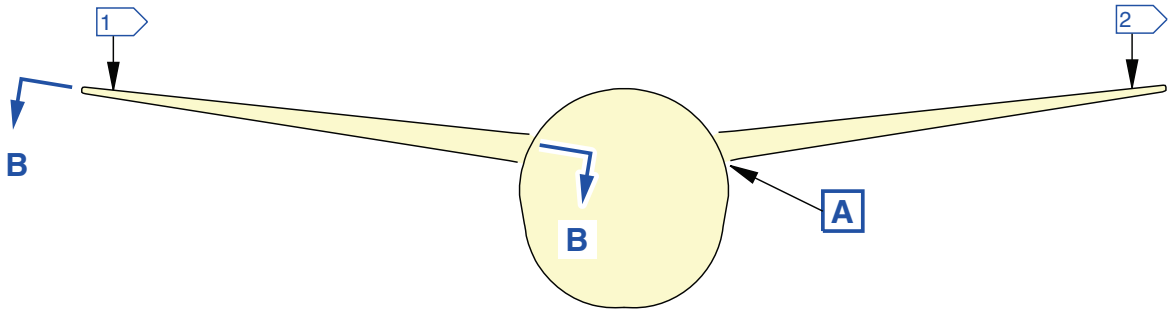
A-A

1. APPLY WEIGHTS ON THE OUTBOARD AREA OF THE RIGHT STABILIZER ABOVE THE FRONT SPAR TO DETERMINE FREEPLAY AT THE LEFT STABILIZER PIVOT HINGE.
2. APPLY WEIGHTS ON THE OUTBOARD AREA OF THE LEFT STABILIZER ABOVE THE FRONT SPAR TO DETERMINE FREEPLAY AT THE RIGHT STABILIZER PIVOT HINGE.
3. APPLY 500 ±20 LBS. NEXT TO THE FUSELAGE, ON TOP OF THE FRONT SPAR (ON LEFT AND RIGHT SIDES) TO DETERMINE THE STABILIZER JACKSCREW FREEPLAY.

3011065 S0000784987_V1

Horizontal Stabilizer Freeplay Overview
Figure 604/27-09-91-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

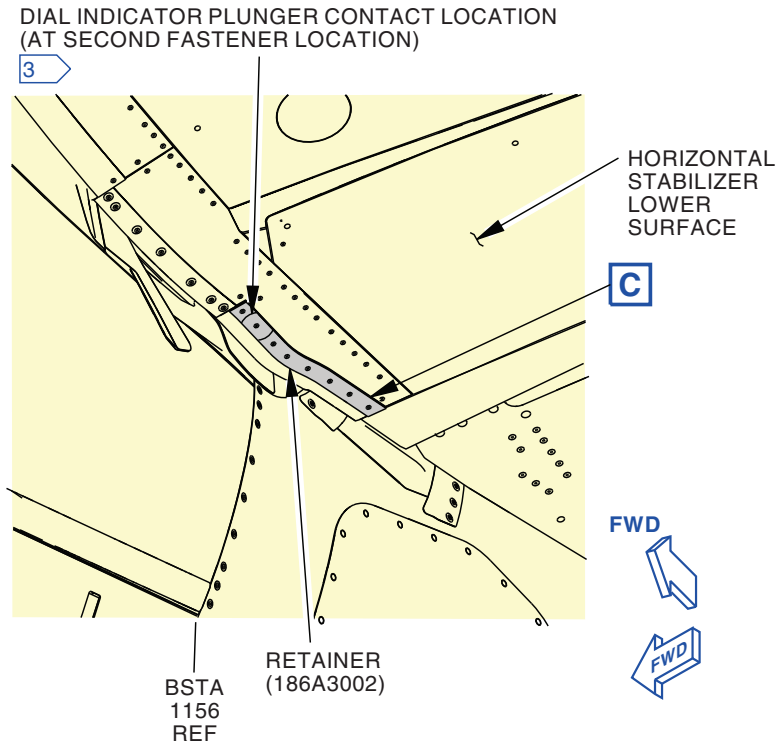


(VIEW IN THE AFT DIRECTION)
(FROM FRONT SPAR OF LEFT AND RIGHT STABILIZERS)

1 APPLY WEIGHTS ON THE OUTBOARD AREA OF THE RIGHT STABILIZER ABOVE THE FRONT SPAR TO DETERMINE FREEPLAY AT THE LEFT STABILIZER PIVOT HINGE.

2 APPLY WEIGHTS ON THE OUTBOARD AREA OF THE LEFT STABILIZER ABOVE THE FRONT SPAR TO DETERMINE FREEPLAY AT THE RIGHT STABILIZER PIVOT HINGE.

3 THE PLUNGER HEAD SHOULD NOT CONTACT THE FASTENER HEAD.



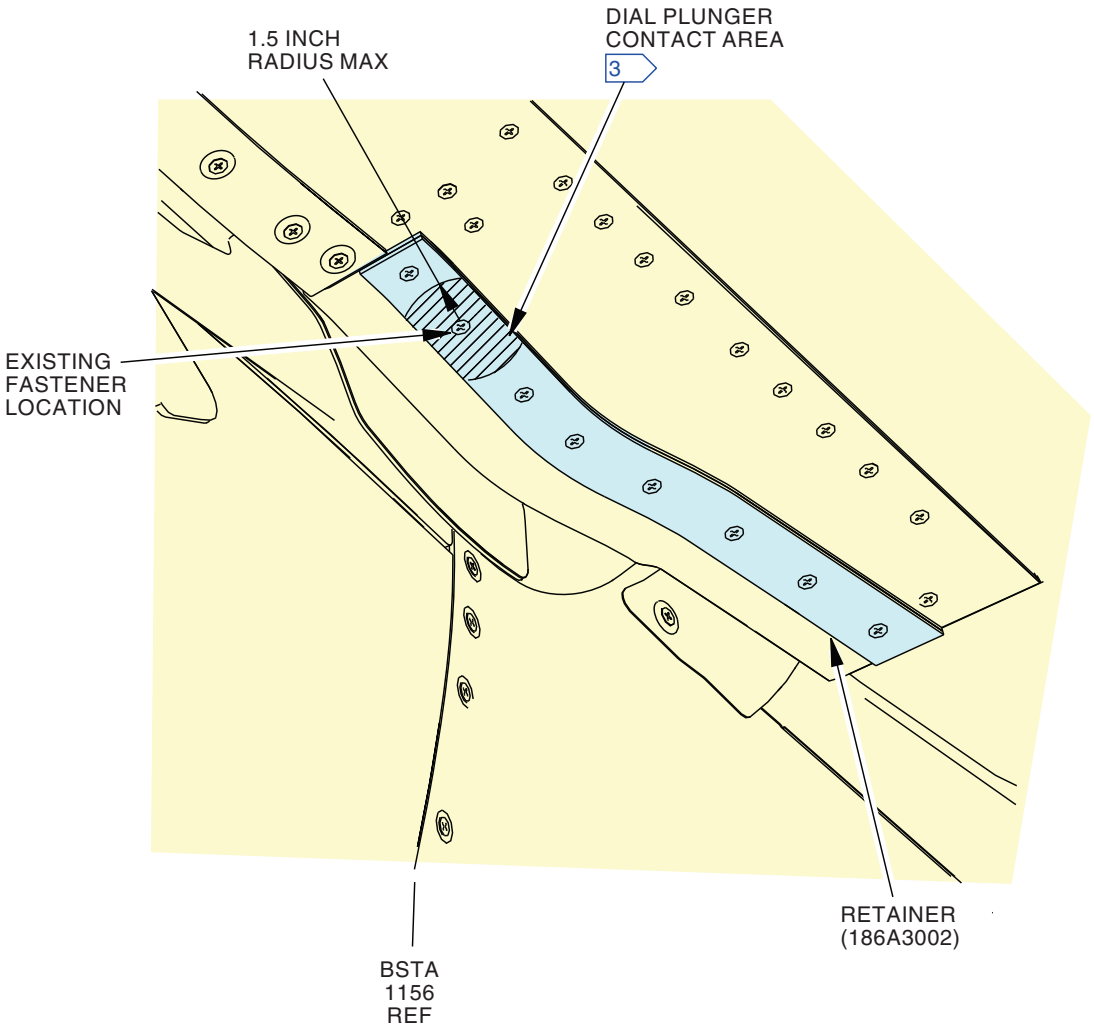
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)
(LEFT PIVOT HINGE INSPECTION IS SHOWN, RIGHT PIVOT HINGE INSPECTION IS OPPOSITE)

A

3011068 S0000784988_V1

Horizontal Stabilizer Pivot Hinge Freeplay Inspection
Figure 605/27-09-91-990-806 (Sheet 1 of 3)

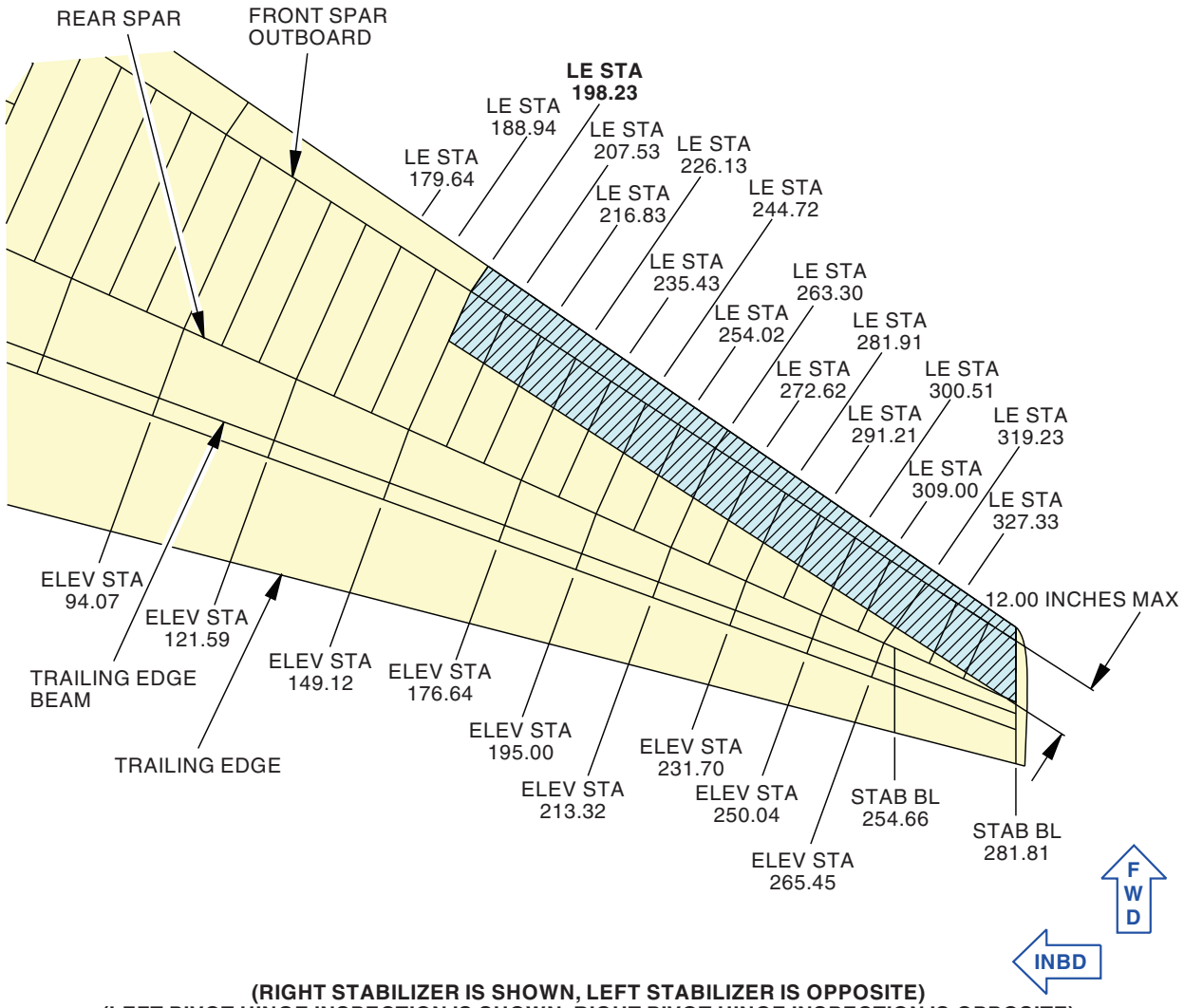
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3016237 S0000789234_V1

Horizontal Stabilizer Pivot Hinge Freeplay Inspection
Figure 605/27-09-91-990-806 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(RIGHT STABILIZER IS SHOWN, LEFT STABILIZER IS OPPOSITE)
(LEFT PIVOT HINGE INSPECTION IS SHOWN, RIGHT PIVOT HINGE INSPECTION IS OPPOSITE)

B-B

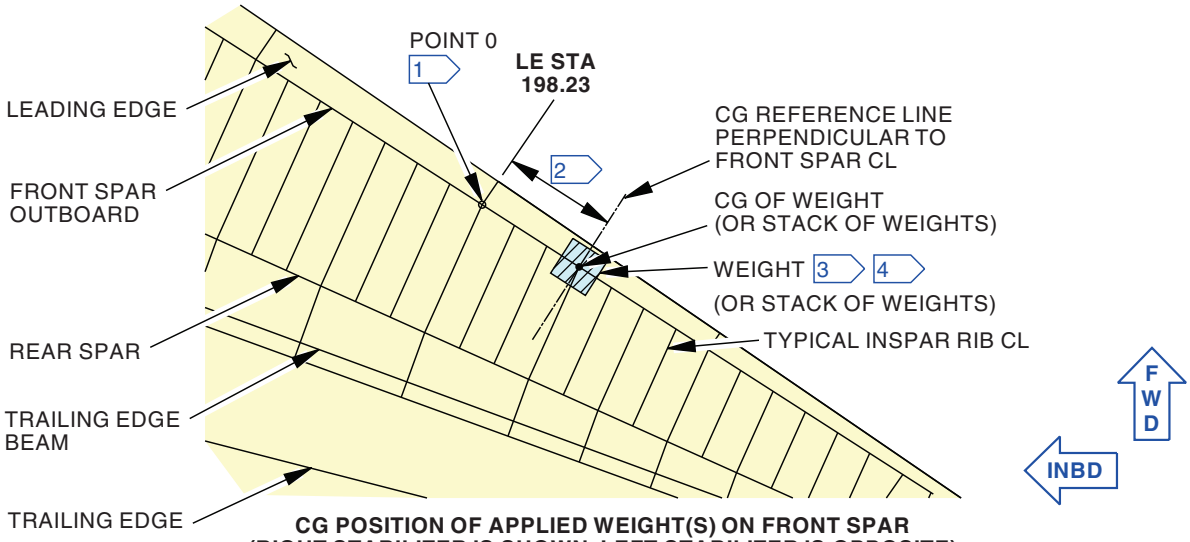
LEGEND:

WEIGHT APPLICATION AREA

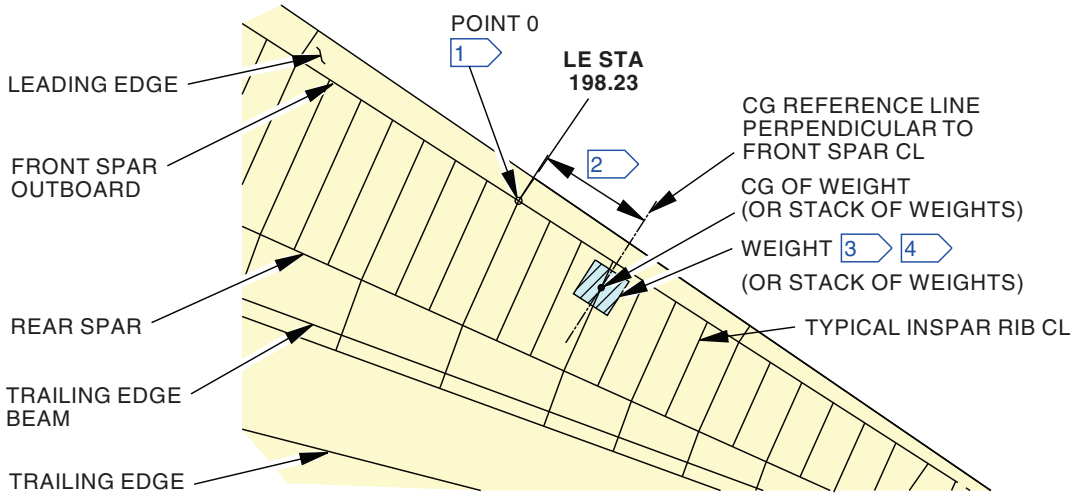
3011071 S0000784990_V1

Horizontal Stabilizer Pivot Hinge Freeplay Inspection
Figure 605/27-09-91-990-806 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CG POSITION OF APPLIED WEIGHT(S) ON FRONT SPAR
(RIGHT STABILIZER IS SHOWN, LEFT STABILIZER IS OPPOSITE)
(LEFT PIVOT HINGE INSPECTION IS SHOWN, RIGHT PIVOT HINGE INSPECTION IS OPPOSITE)



CG POSITION OF APPLIED WEIGHT(S) AFT ON FRONT SPAR ON THE RIB
(RIGHT STABILIZER IS SHOWN, LEFT STABILIZER IS OPPOSITE)
(LEFT PIVOT HINGE INSPECTION IS SHOWN, RIGHT PIVOT HINGE INSPECTION IS OPPOSITE)

LEGEND:

WEIGHT(S) CONTACT AREA

1 POINT 0 IS THE UPPER AFT AND INBOARD CORNER OF THE 186A1210 REMOVABLE LEADING EDGE SKIN.

2 DISTANCE ALONG THE FRONT SPAR CENTERLINE BETWEEN POINT 0 AND THE CENTER OF GRAVITY (CG) OF THE APPLIED WEIGHT (OR STACK OF WEIGHTS).

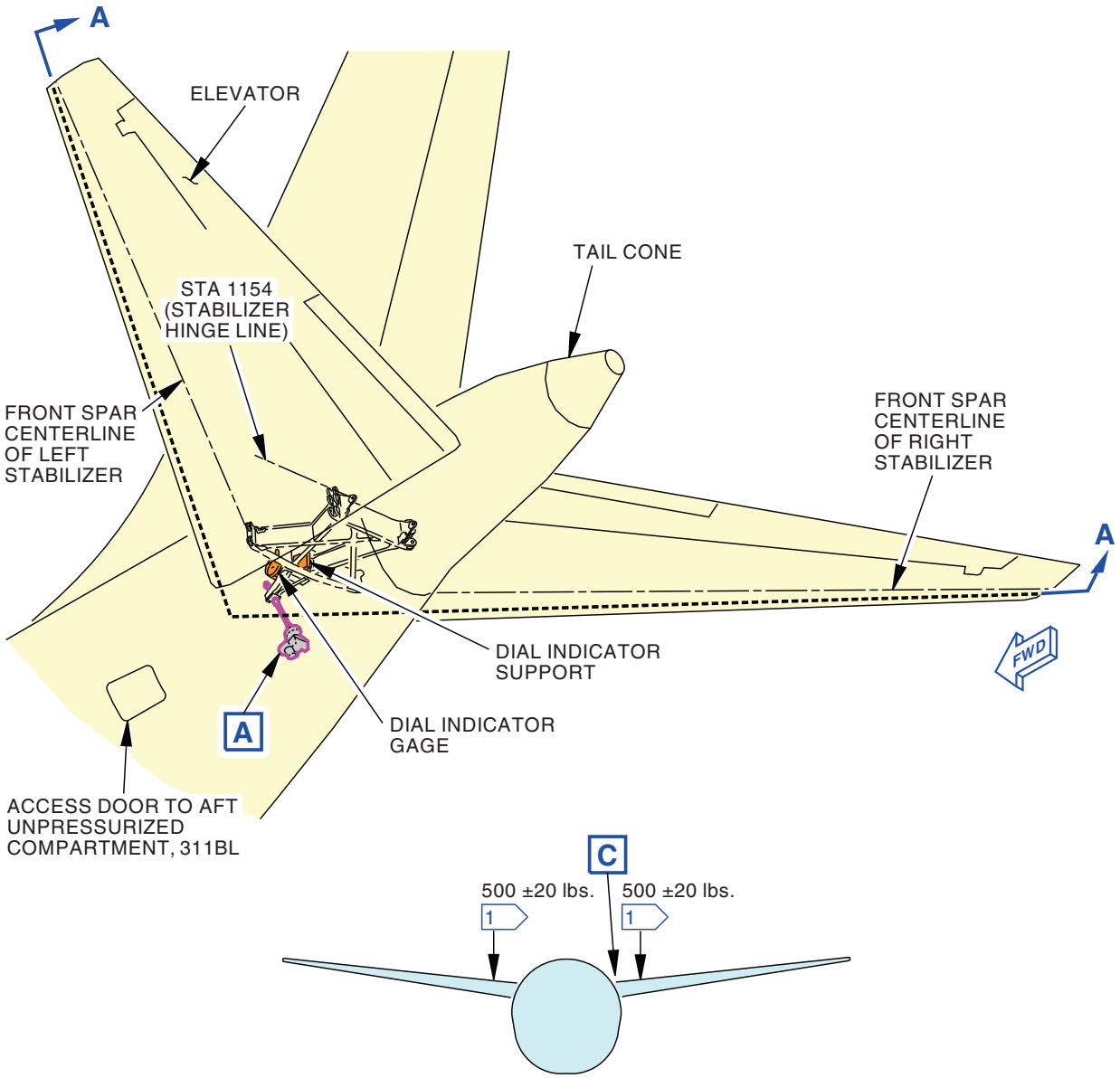
3 A SQUARE SHAPED WEIGHT IS SHOWN. OTHER SHAPED WEIGHTS MAY BE USED. THE USER SHALL CONSIDER THE WEIGHT VARIATIONS AND THE VARIATIONS IN DETERMINING THE CG, WHEN DEMONSTRATING THAT THE TOTAL REVERSE LOAD REQUIREMENT IS MET

4 THE CONTACT AREA OF THE WEIGHTS (OR STACK OF WEIGHTS) APPLIED SHOULD RESULT IN A PRESSURE LESS THAN 4 PSI AVERAGE.

3011070 S0000784991_V1

Weight Application Requirements and Calculating Total Reverse Load
Figure 606/27-09-91-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LOOKING AFT FROM FRONT SPAR OF LEFT AND RIGHT STABILIZERS)

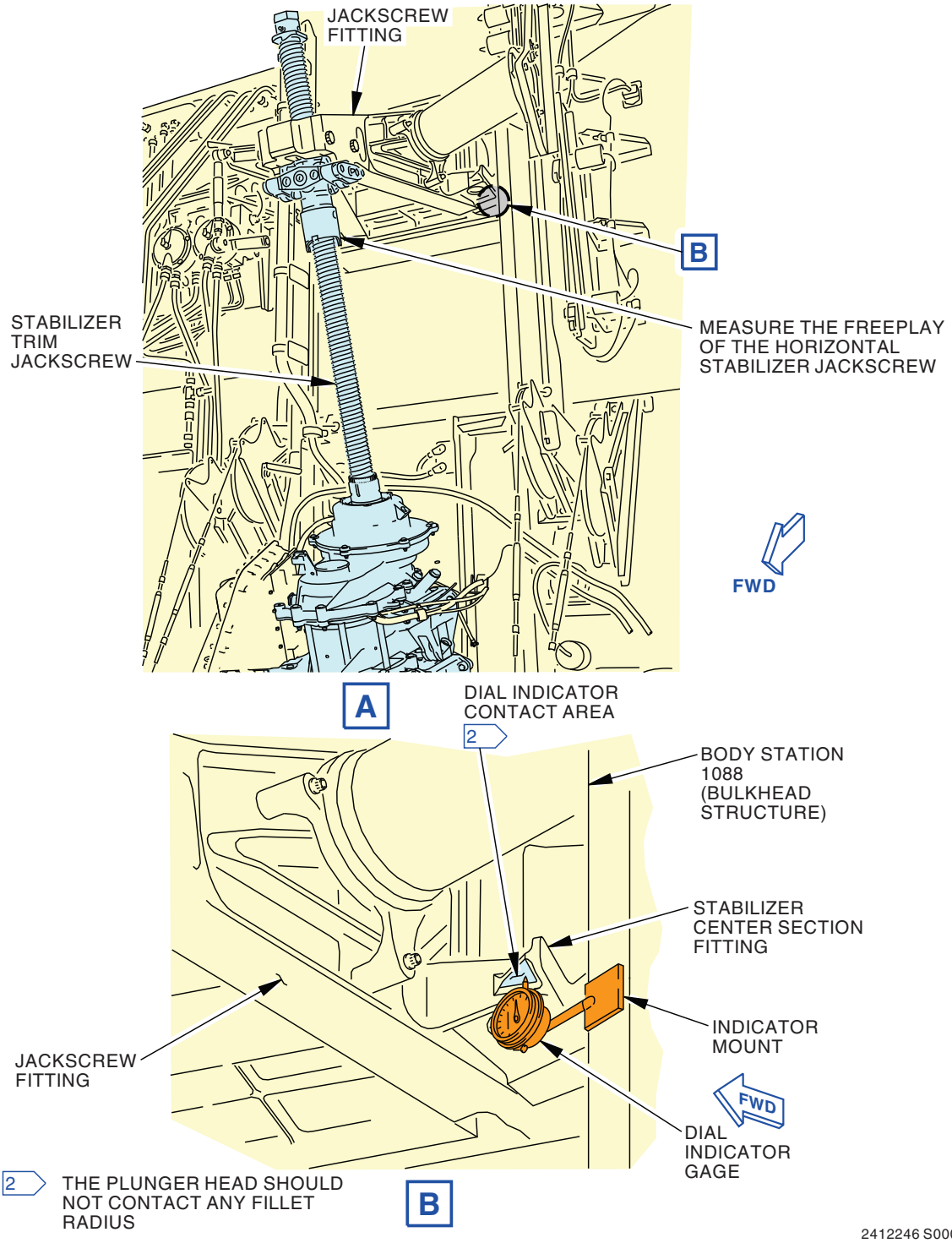
A-A

1 APPLY 500 ±20 LBS. NEXT TO THE FUSELAGE, ON TOP OF THE FRONT SPAR (ON LEFT AND RIGHT SIDES) TO DETERMINE THE STABILIZER JACKSCREW FREEPLAY.

2412245 S00061530288_V2

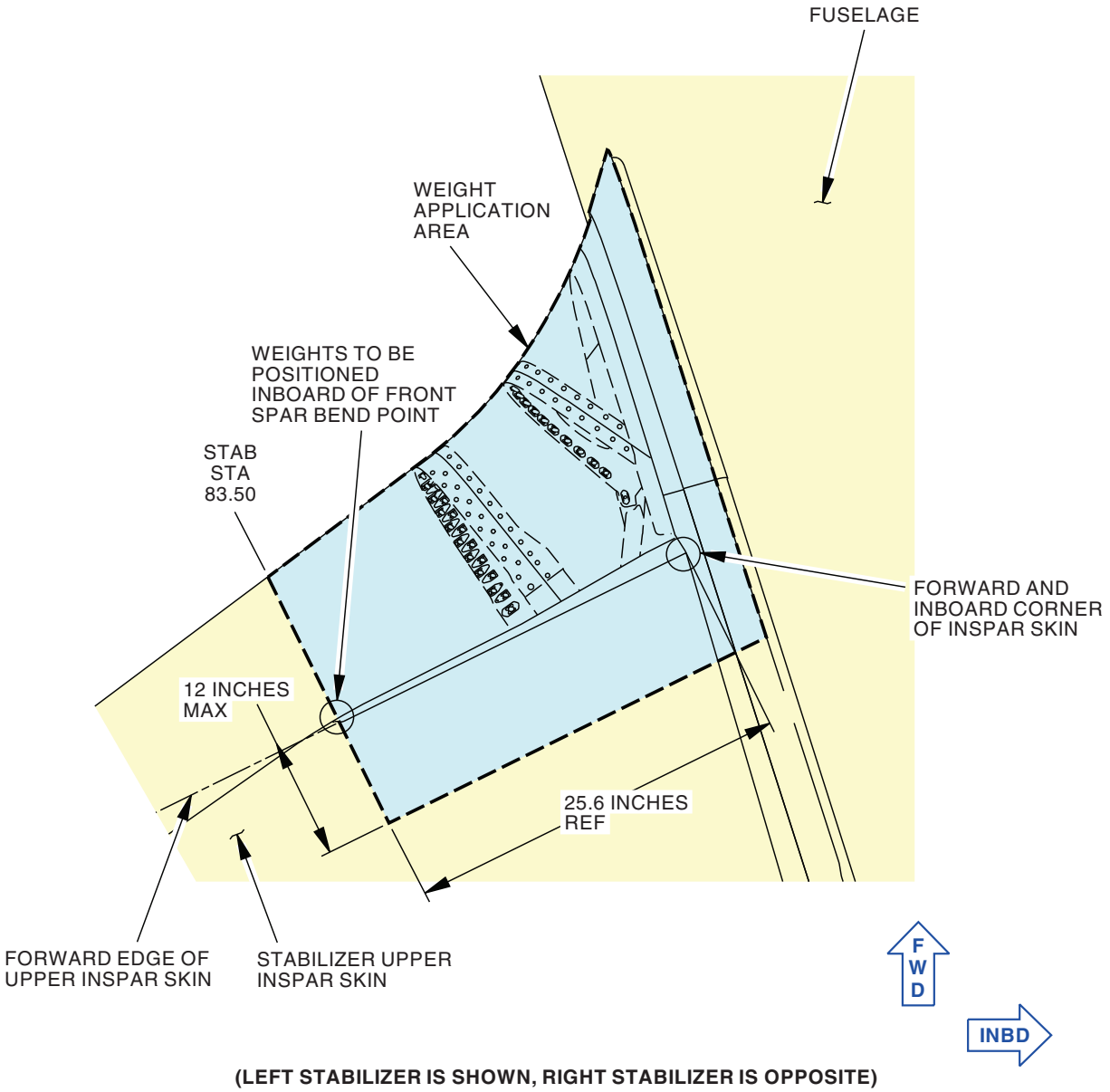
Horizontal Stabilizer Jackscrew Freeplay Inspection
Figure 607/27-09-91-990-808 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Horizontal Stabilizer Jackscrew Freeplay Inspection
Figure 607/27-09-91-990-808 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LEFT STABILIZER IS SHOWN, RIGHT STABILIZER IS OPPOSITE)

C

3011075 S0000784993_V2

Horizontal Stabilizer Jackscrew Freeplay Inspection
Figure 607/27-09-91-990-808 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TOTAL REVERSE LOAD TABLE							
Weight Locations	(A) Enter the weight in lbs.	(B) Enter the distance in inches	(C)	(D) Multiply (B) by (C)	(E)	(F) Add (E) to (D)	Reverse Load in Pounds Multiply (A) by (F)
1	50	6	x 0.016	0.096	+ 3.068	3.164	158.200
2	50	18	x 0.016	0.288	+ 3.068	3.356	167.800
3	50	30	x 0.016	0.480	+ 3.068	3.548	177.400
4	50	42	x 0.016	0.672	+ 3.068	3.740	187.000
5	50	54	x 0.016	0.864	+ 3.068	3.932	196.600
6	50	66	x 0.016	1.056	+ 3.068	4.124	206.200
7	50	78	x 0.016	1.248	+ 3.068	4.316	215.800
8	50	90	x 0.016	1.440	+ 3.068	4.508	225.400
9	50	102	x 0.016	1.632	+ 3.068	4.700	235.000
10	50	114	x 0.016	1.824	+ 3.068	4.892	244.600
11	50	126	x 0.016	2.016	+ 3.068	5.084	254.200
12	50	138	x 0.016	2.208	+ 3.068	5.276	263.800

REFERENCE TOTAL WEIGHT (lbs) =	600

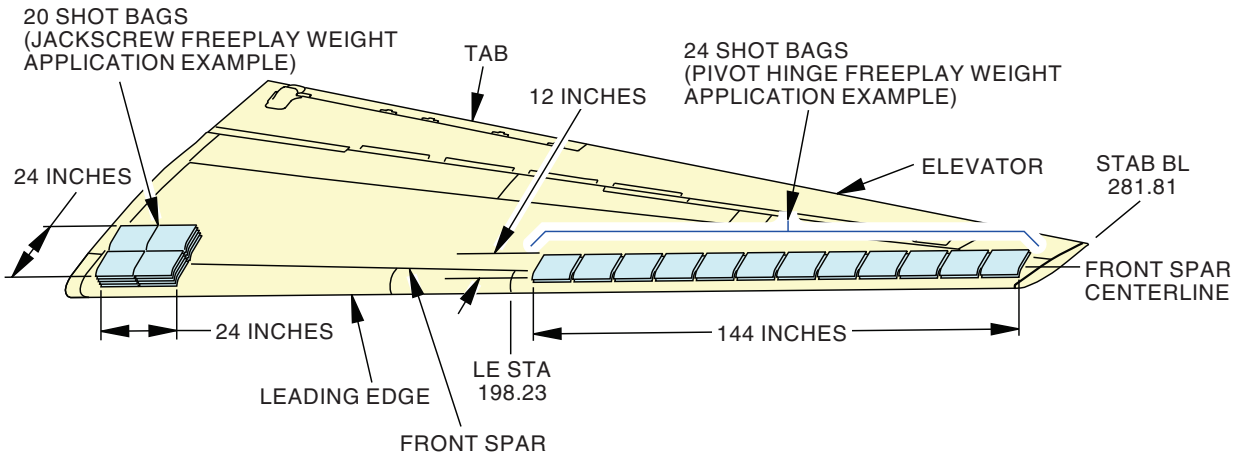
TOTAL REVERSE LOAD (lbs) (ADD ALL VALUES IN REVERSE LOAD COLUMN TO GET TOTAL REVERSE LOAD) =	2532.000

EXAMPLE 1: TOTAL REVERSE LOAD CALCULATION

3011076 S0000784994_V1

Total Reverse Load Examples and Weight Application
Figure 608/27-09-91-990-809 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



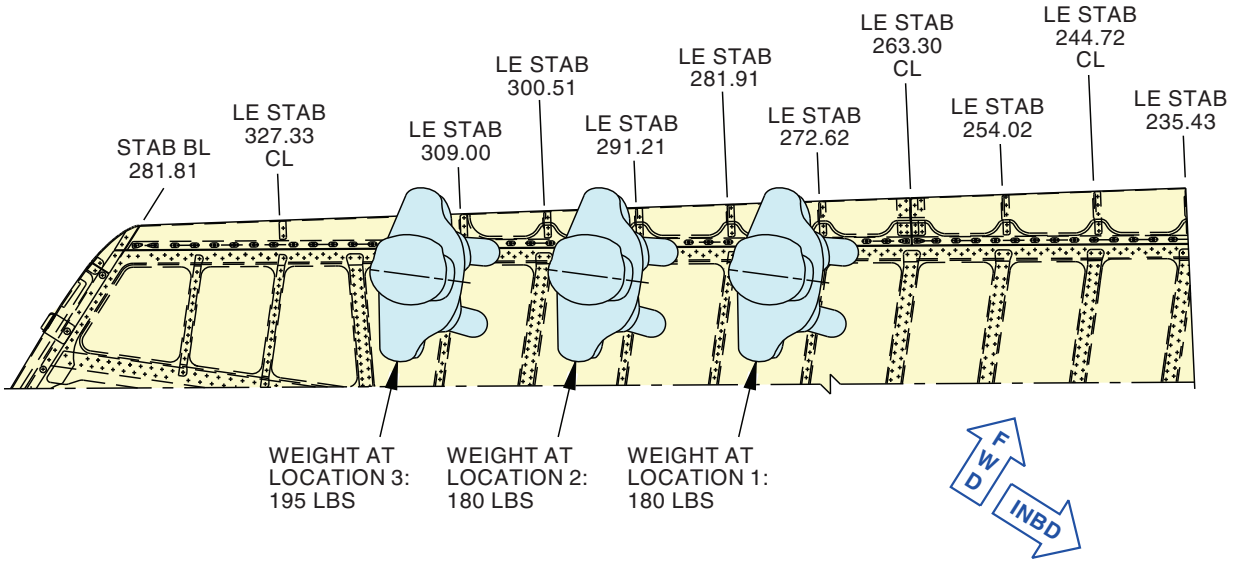
NOMINAL SHOTBAG SIZE: 12 INCHES BY 12 INCHES
NOMINAL SHOTBAG WEIGHT: 25 POUNDS

EXAMPLE 2: WEIGHT APPLICATION USING SHOT BAGS

3011078 S0000784996_V1

Total Reverse Load Examples and Weight Application
Figure 608/27-09-91-990-809 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Weight Locations	Weight of person in (lbs)	Distance in inches	Reverse Load in (lbs)
1	180	74	765.4
2	180	92	817.2
3	195	110	941.5
TOTAL REVERSE LOAD: 2524.0			

EXAMPLE 3: WEIGHT APPLICATION USING PERSONS

3011080 S0000784997_V1

Total Reverse Load Examples and Weight Application
Figure 608/27-09-91-990-809 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-09-91-200-805

6. Spoiler Inspection

(Figure 609)

A. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
STD-1238	Indicator - Dial
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator

C. Location Zones

Zone	Area
552	Left Wing - Spoiler No. 6
562	Left Wing - Spoiler No. 5
563	Left Wing - Spoiler No. 4
564	Left Wing - Spoiler No. 3
565	Left Wing - Spoiler No. 2
566	Left Wing - Spoiler No. 1
652	Right Wing - Spoiler No. 7
662	Right Wing - Spoiler No. 8
663	Right Wing - Spoiler No. 9
664	Right Wing - Spoiler No. 10
665	Right Wing - Spoiler No. 11
666	Right Wing - Spoiler No. 12

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Check

SUBTASK 27-09-91-860-005

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE TRAILING EDGE FLAPS, AND THE LEADING EDGE FLAPS AND SLATS. THESE SURFACES WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Extend the trailing edge flaps to the FLAP DOWN position with primary control. Do this task: Extend the Trailing Edge Flaps, TASK 27-51-00-860-803.

NOTE: You can do a check of the spoilers with hydraulic power on or off.

SUBTASK 27-09-91-480-002

(2) Attach a mount, STD-1279 to the spoiler trailing edge.

(a) Adjust the dial indicator, STD-1238 to zero.

E. Spoiler Inspection

SUBTASK 27-09-91-220-010

(1) Do a check of the freeplay for spoilers No. 1, 2, 3, 4, 5, 8, 9, 10, 11 and 12:

CAUTION PUT SUFFICIENT PADDING BETWEEN THE FORCE SCALE, AND THE CONTROL SURFACE TO PREVENT DAMAGE TO THE CONTROL SURFACE. THE FORCE SCALE CAN CAUSE DAMAGE TO THE CONTROL SURFACE.

(a) Use a force gauge, COM-1557 and apply approximately 21 lbf (93.41 N), 0.5 in. (12.70 mm) forward of the spoiler trailing edge and directly opposite of the power control actuator.

NOTE: The up force of approximately 21 lbf (93.41 N) consists of the spoiler panel weight of 16 lb (7.26 kg) plus 5 lbf (22.24 N) push force.

(b) Measure the freeplay on the inboard edge of the spoiler not more than 0.5 in. (12.70 mm) from the trailing edge.

(c) Make sure the spoiler freeplay is not more than 0.1 in. (2.54 mm).

SUBTASK 27-09-91-220-011

(2) Do a check of the freeplay for spoilers No. 6 and 7 (graphite/composite):

CAUTION PUT SUFFICIENT PADDING BETWEEN THE FORCE SCALE, AND THE CONTROL SURFACE TO PREVENT DAMAGE TO THE CONTROL SURFACE. THE FORCE SCALE CAN CAUSE DAMAGE TO THE CONTROL SURFACE.

(a) Use a force gauge, COM-1557 and apply an up force of approximately 32 lbf (142.34 N), 0.5 in. (12.70 mm) forward of the spoiler trailing edge and at the middle between the two spoiler power control actuators.

NOTE: The up force of approximately 32 lbf (142.34 N) consists of the spoiler panel weight of 27 lb (12.25 kg) plus 5 lbf (22.24 N) push force.

(b) Measure the freeplay on the inboard edge of the spoiler not more than 0.5 in. (12.70 mm) from the trailing edge.

(c) Make sure the spoiler freeplay is not more than 0.2 in. (5.08 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-09-91-080-003

(1) Remove the dial indicator, STD-1238.

SUBTASK 27-09-91-080-004

(2) Remove the mount, STD-1279.

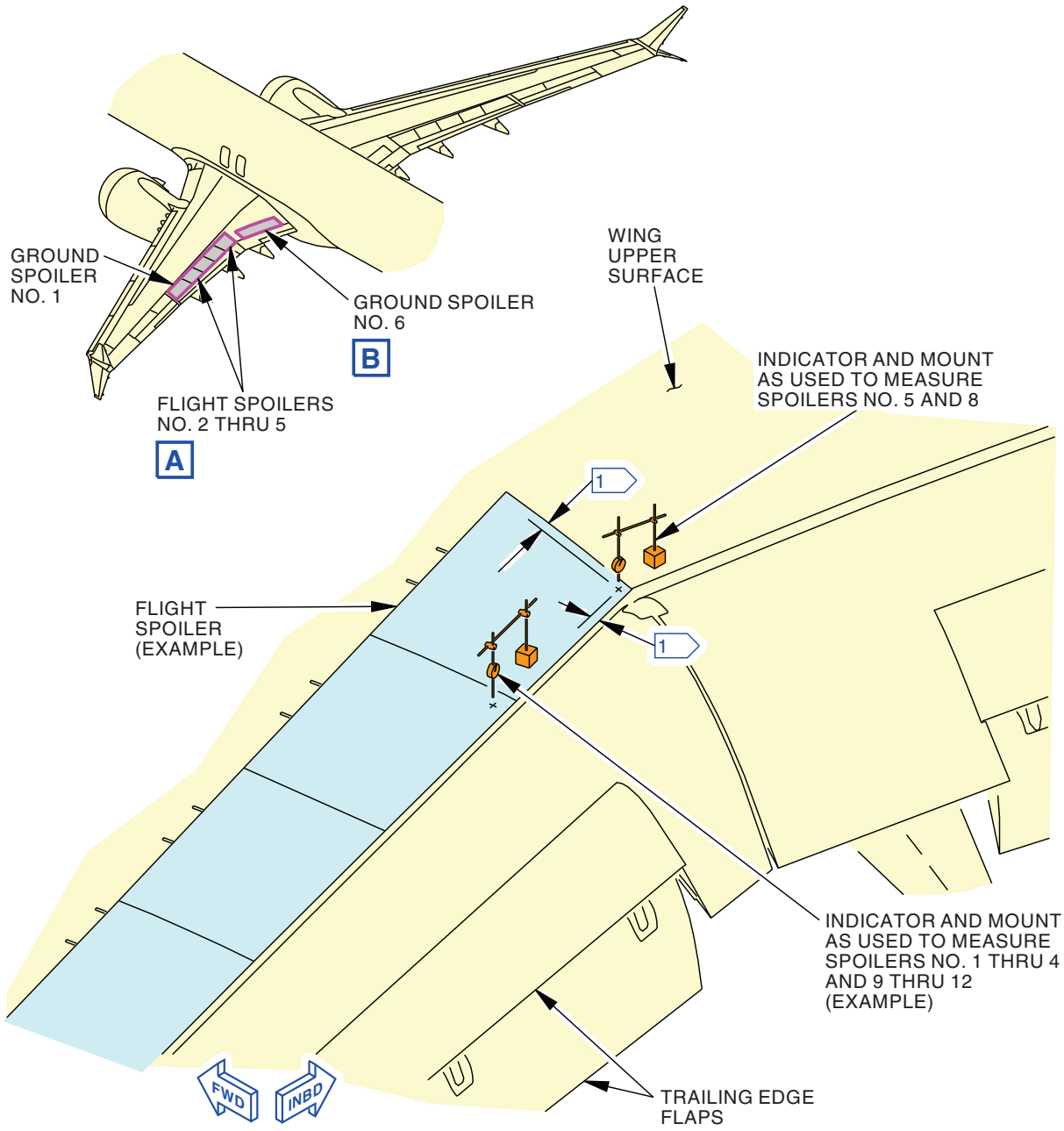
SUBTASK 27-09-91-860-006

WARNING MOVE ALL PERSONS AND EQUIPMENT AWAY FROM THE TRAILING EDGE FLAPS, AND THE LEADING EDGE FLAPS AND SLATS. THESE SURFACES WILL MOVE AND CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(3) Retract the trailing edge flaps with primary control. Do this task: Retract the Trailing Edge Flaps, TASK 27-51-00-860-804.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT SPOILERS NO. 2 THRU 5 FREEPLAY CHECK
(FLIGHT SPOILERS NO. 8 THRU 11 ARE OPPOSITE)

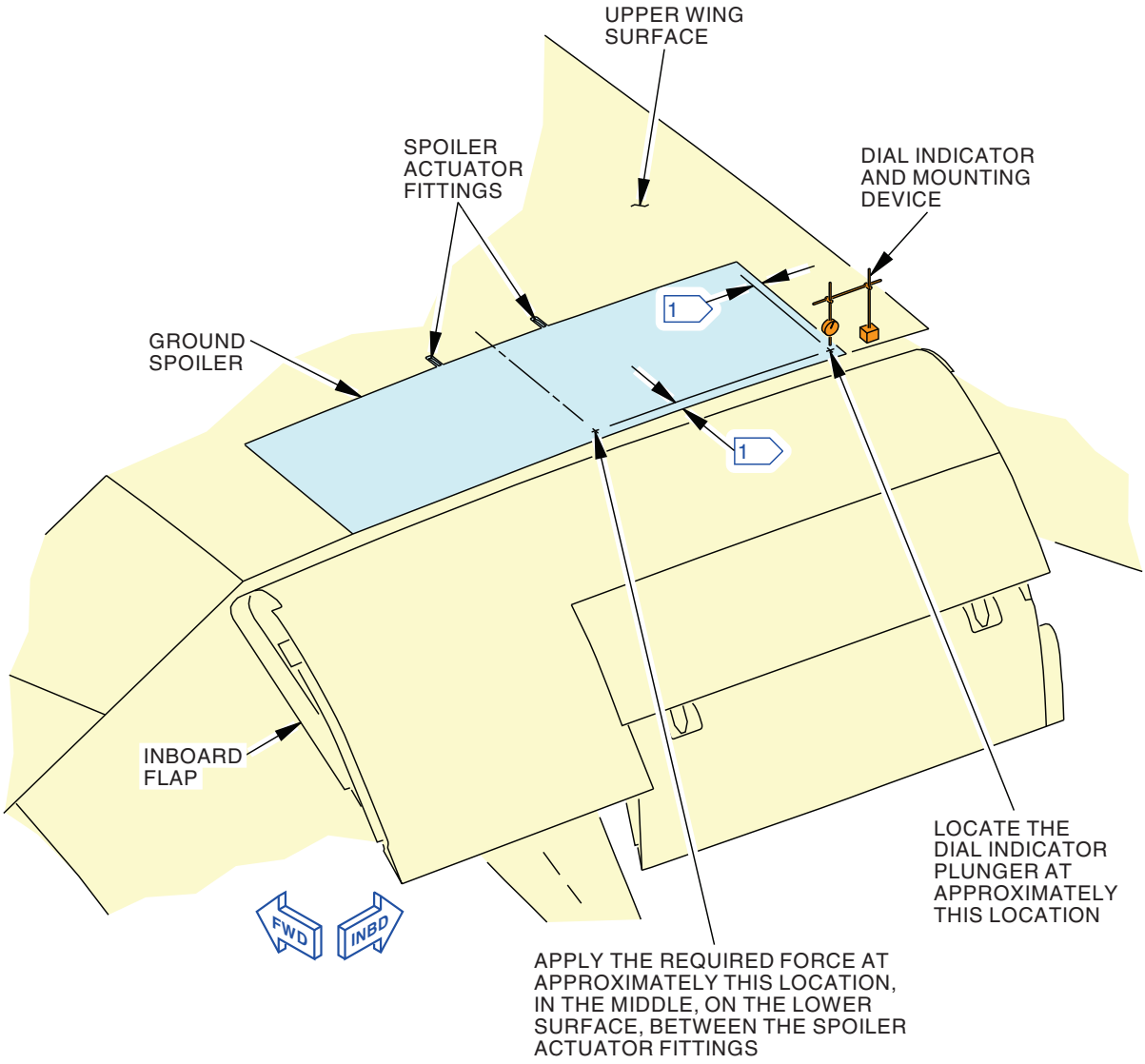
A

1 INDICATOR PLUNGER LOCATION TYPICAL ON EACH SPOILER 0.5 INCH (12.5 mm)

2412247 S00061530291_V1

Flight and Ground Spoiler Freeplay Check
Figure 609/27-09-91-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GROUND SPOILER NO. 6 FREEPLAY CHECK
(GROUND SPOILER NO. 7 IS OPPOSITE)

B

2412248 S00061530292_V1

Flight and Ground Spoiler Freeplay Check
Figure 609/27-09-91-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON AND AILERON TRIM CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Pressure from the aileron hydraulic systems A and B - Deactivation.

(2) Pressure to the aileron hydraulic systems A and B - Activation.

TASK 27-11-00-860-801

2. Pressure from the Aileron Hydraulic Systems A and B - Deactivation

(Figure 201)

A. General

(1) This procedure removes pressure from the aileron hydraulic systems A and B before you do maintenance on the aileron control system.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

E. Pressure from the Aileron Hydraulic Systems A and B - Deactivation

WARNING

MAKE SURE THAT THERE IS NO PRESSURE IN THE HYDRAULIC SYSTEM. PRESSURE IN THE HYDRAULIC SYSTEM CAN MOVE THE CONTROL SURFACES, CABLES, LANDING GEARS, AND LANDING GEAR DOORS QUICKLY. THEIR MOVEMENT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

SUBTASK 27-11-00-860-001

(1) Remove pressure from the aileron hydraulic systems A and B.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Set the FLT CONTROL A and B switches on the forward overhead panel, P5, to the OFF position.

(b) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

(c) Set the ELEC 2, HYD PUMPS A and ELEC 1, HYD PUMPS B switches on the forward overhead panel, P5, to the OFF position.

(d) Operate the rudder pedals to decrease the remaining hydraulic system pressure.

SUBTASK 27-11-00-860-104

(2) If it is necessary to open a hydraulic line, release pressure from the reservoir for the applicable hydraulic system (TASK 29-09-00-860-802).

SUBTASK 27-11-00-480-001

(3) Install the lock, SPL-1675, on the flight controls shutoff valves for systems A and B.

(a) Disconnect the electrical connectors from the shutoff valves.

NOTE: The flight controls shutoff valve is part of the flight controls hydraulic module. The hydraulic module for hydraulic system A is found in the left main wheel well. The hydraulic module for hydraulic system B is found in the right main wheel well.

(b) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(c) If there was corrosion, refer to (SWPM 20-60-03) to correct the problem.

(d) Make sure that the manual override levers on the shutoff valves are moved to position 2.

(e) Install the lock, SPL-1675, on each shutoff valve.

1) Make sure to install the detent pin.

F. Aileron Control System - Tryout

NOTE: This tryout is to make sure that the Aileron Control System is in a zero energy state.

SUBTASK 27-11-00-010-015

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-11-00-410-011

(2) Make sure that the lock, SPL-1675, is correctly installed on each flight controls shutoff valve.

SUBTASK 27-11-00-710-001

(3) Rotate the captain's or first officer's control wheel through full travel in each direction from the neutral position.

NOTE: The force required to move the aileron control surfaces increases significantly when the wheel rotation has exceeded 7 degrees.

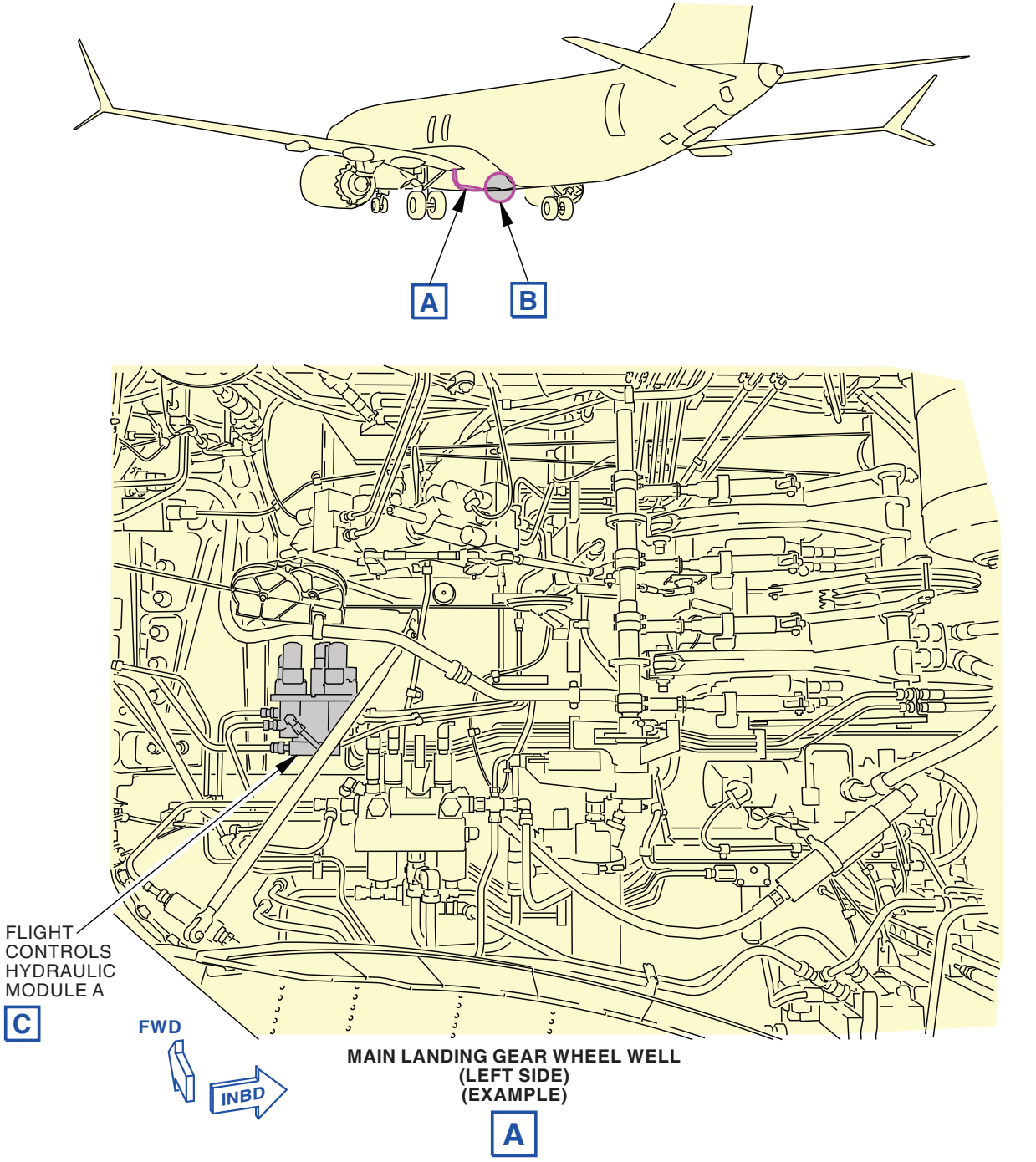
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-040-001

(4) Install a DO NOT OPERATE tag, STD-858, on the captain's and first officer's control wheel.

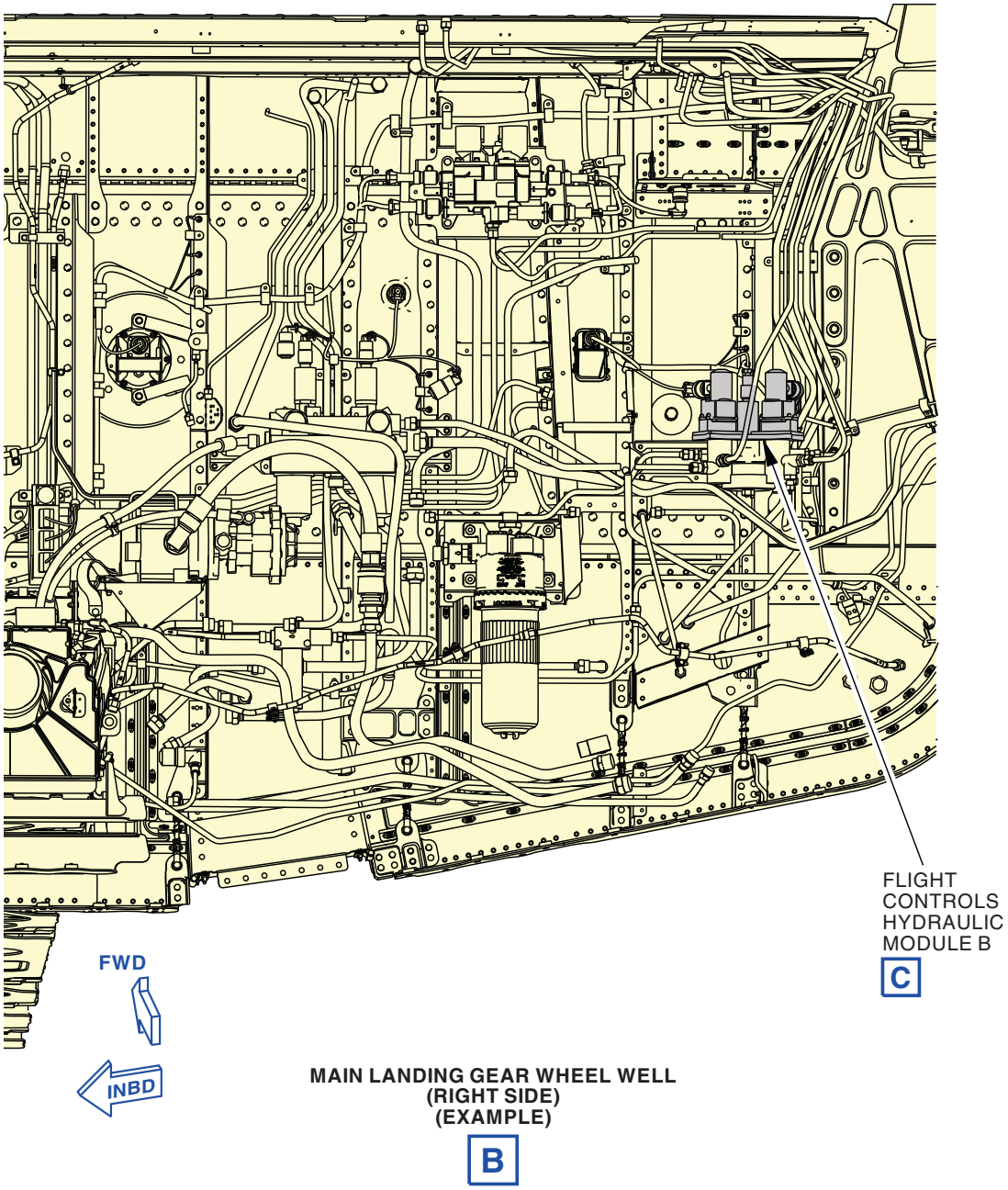
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Hydraulic Module Lock Installation
Figure 201/27-11-00-990-801 (Sheet 1 of 4)

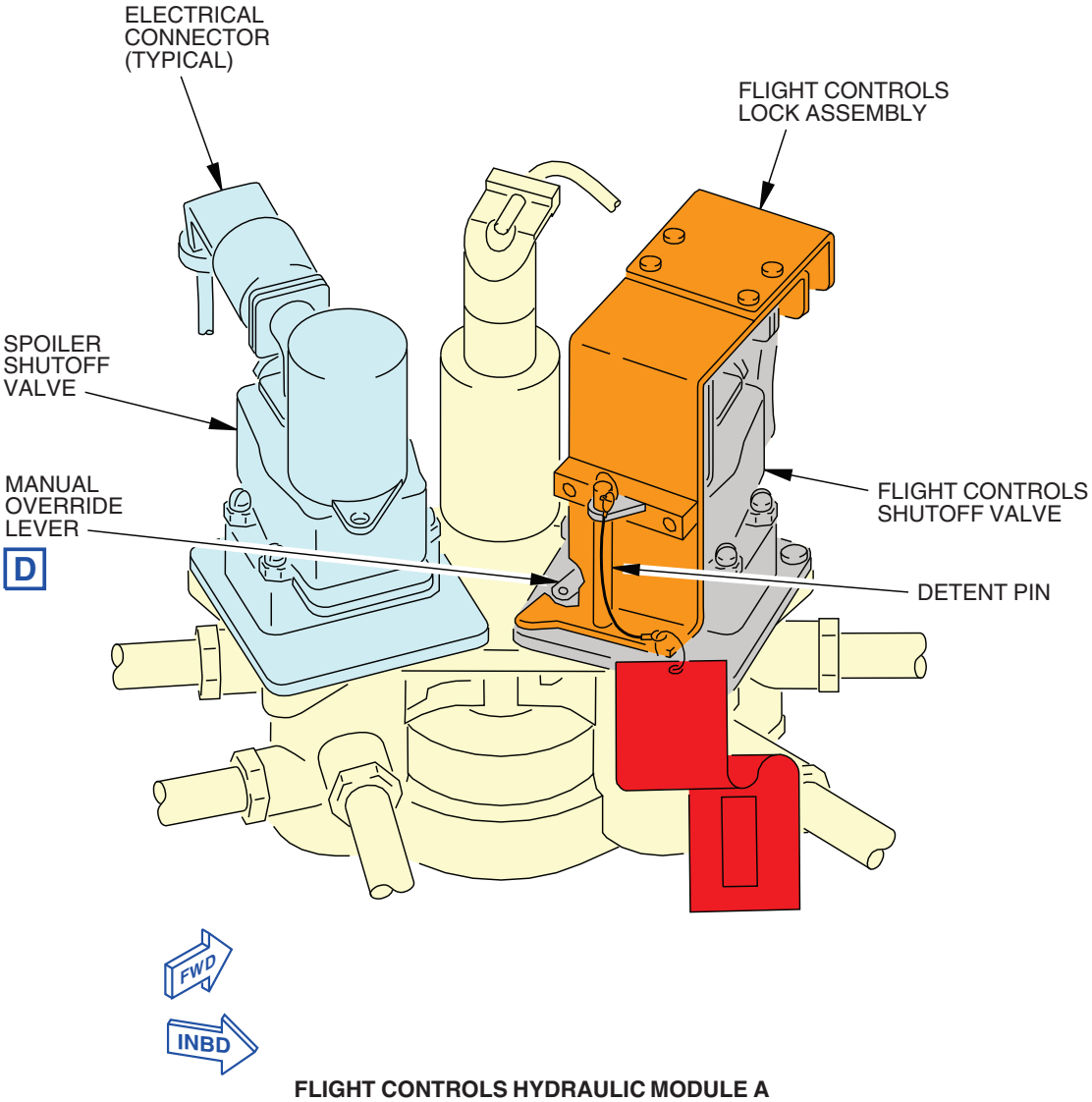
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412250 S00061530298_V3

Flight Controls Hydraulic Module Lock Installation
Figure 201/27-11-00-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



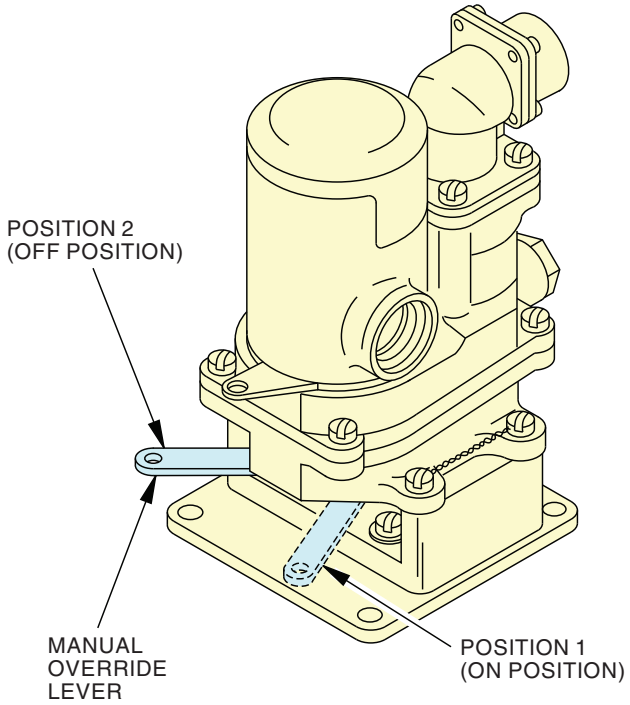
FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2412251 S00061530299_V6

Flight Controls Hydraulic Module Lock Installation
Figure 201/27-11-00-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



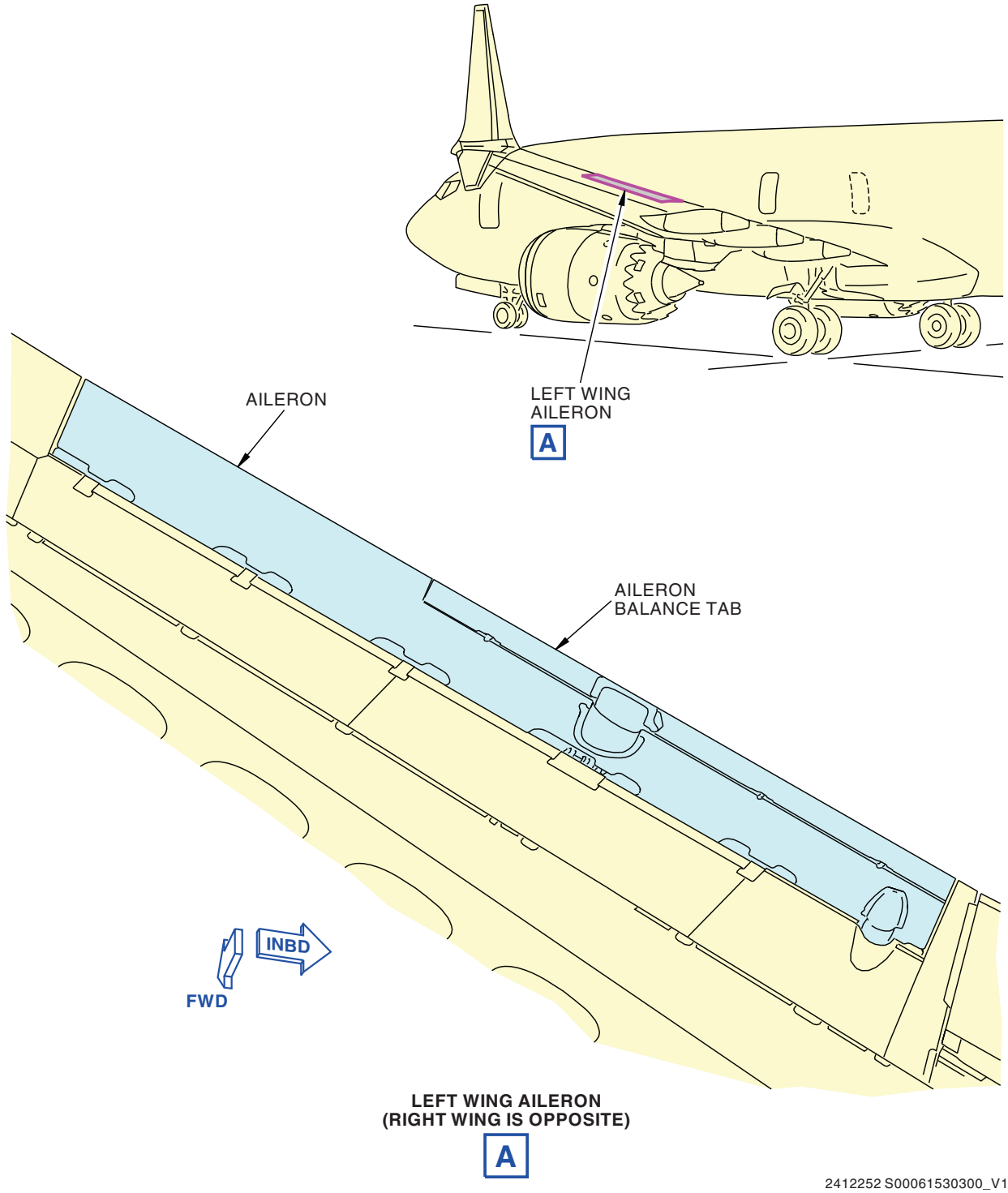
**MANUAL OVERRIDE LEVER POSITION
(EXAMPLE)**

D

2844536 S0000662328_V3

**Flight Controls Hydraulic Module Lock Installation
Figure 201/27-11-00-990-801 (Sheet 4 of 4)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron
Figure 202/27-11-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-860-802

3. Pressure to the Aileron Hydraulic Systems A and B - Activation

(Figure 201)

A. General

(1) This task will put the aileron hydraulic systems A and B back to the condition before pressure removal.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-09-31-710-801	Flight Controls Shutoff Valve Test (P/B 501)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

F. Pressure to the Aileron Hydraulic Systems A and B Activation

SUBTASK 27-11-00-080-001

(1) Remove the lock, SPL-1675, from the flight controls shutoff valves for systems A and B.

(a) Remove the lock, SPL-1675, from the shutoff valves.

1) Make sure that the detent pin is removed first from the lock, SPL-1675.

(b) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

1) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If there was corrosion, correct the problem (SWPM 20-60-03).

(c) Apply D5026NS or ZC-026 compound, G50171, to the connectors (SWPM 20-60-03).

(d) Install electrical connectors on the shutoff valves.

(e) Move the manual override levers on the shutoff valves to position 1.

SUBTASK 27-11-00-860-002

(2) Set the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the ON position.

SUBTASK 27-11-00-440-002

(3) Remove the DO NOT OPERATE tag, STD-858, from the captain's and first officer's control wheel.

SUBTASK 27-11-00-860-003

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-11-00-610-001

(5) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-11-00-700-003

(6) Do this task: Flight Controls Shutoff Valve Test, TASK 27-09-31-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Adjustment of the aileron cables
(2) Adjustment of the pogo and Power Control Unit (PCU)
(3) Adjustment of the aileron balance tab (Recommended Method)
(4) Adjustment of the aileron balance tab (Alternate Method)
(5) Adjustment of the aileron feel and centering unit
(6) Check of the aileron system adjustment
(7) Test of the aileron surface friction and motion
(8) Test of the manual reversion control friction
(9) Test of the aileron response (operation)
(10) Test of the control wheel centering check
(11) Test of the aileron control wheel force
(12) Test of the aileron trim response
(13) Operational test of the transfer mechanism
(14) Functional test of the transfer mechanism
(15) Test of the control wheel travel stop
(16) Test of the control wheel travel interference
(17) Test of the aileron balance tab position
(18) Test of the aileron travel
(19) Functional test of the aileron PCU pogos
(20) Flight test to find and correct asymmetrical flight control forces

TASK 27-11-00-820-801

2. Aileron Cables Adjustment

(Figure 501, Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Before you do cable adjustment, make sure that the outside ambient temperature is stable for the last hour.
(a) The temperature change must not exceed ±5°F or ±2.8°C.
(b) The temperature must not change at a rate more than ±5°F or ±2.8°C per hour while the cables are adjusted.
(c) It is recommended to do the cable adjustments in a controlled environment such as a hangar.
(d) All temperature readings will be taken as ambient outside but adjacent to the applicable area of the airframe.
(e) The cable tension values will not be correct when there are temperature differences along the cable run.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(2) It is recommended that airplanes supplied with the cables that contain higher zinc coating (Grade D or Grade K) be re-tensioned 3 to 4 months into service. This will bring the tension of the control cables within the desired range. After the cables are re-tensioned, the high zinc cables (Grade D or Grade K) will continue to relax at a rate that is equivalent to the Grade B cables. To identify the grade of the cable, refer to the table below:

Table 501/27-11-00-993-802 Control Cable Grades and Diameters

CONTROL CABLE DIAMETER (TERMINAL SIZE REF)	CONSTRUCTION FIRST DASH NUMBER *[1]									
	GRADE A					GRADE B		GRADE D		GRADE K
	ZINC			TIN OVER ZINC		ZINC	TIN OVER ZINC	ZINC	TIN OVER ZINC	ZINC
	7 X 7	7 X 7 (RT)	7 X 19	7 X 7	7 X 7 (RT)	7 X 19	7 X 19	7 X 19	7 X 19	7 X 19
.094 (3/32)	3B	3G*[2]	3A	3D	3E *[2]	- — —	- — —	- — —	- — —	- — —
.125 (1/8)	- — —	- — —	4A	- — —	- — —	4F	4C	4J*[3]	4H*[3]	4K*[3]
.188 (3/16)	- — —	- — —	- — —	- — —	- — —	- — —	- — —	6J*[4]	- — —	- — —

*[1] First dash number in the part number of the cable. Example: BACC2C-3D-01085-CE is Grade A, Tin Over Zinc, 7 x 7.

*[2] Code 3E and 3G are specified only where there is a specific requirement for REDUCED TOLERANCE (RT) percent elongation wire rope, 7 X 7 RT, per BMS7-265.

*[3] Only Grade D and Grade K .125 (1/8) cables (4J, 4H and 4K) are high zinc cables.

*[4] It is not necessary to adjust the Grade D 0.188 (3/16) cables (6J) 3 to 4 months after installation.

(3) It is recommended that cable assemblies on each side of a cable loop be replaced at the same time using the same cable grade material in order to minimize asymmetric relaxation.

(4) Make sure that you only adjust the applicable aileron cable or cables.

B. References

Reference	Title
20-10-10-400-801	Turnbuckle Lock Installation (P/B 401)
27-09-14-860-802	Cable Control - Tension Measurement (P/B 201)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
COM-1554	Tensiometer - Cable, High Tension (200 lbs and above) Part #: 102-03108 Supplier: 21844 Part #: 102-03120 Supplier: 21844 Part #: ACM-300 Supplier: 13331 Part #: ACM-600 Supplier: 13331 Part #: ACX-500-1 Supplier: 13331 Part #: T5-8008-106-00 Supplier: 0N8U4 Part #: T60-1001-C9-1A Supplier: 0N8U4
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1671	Straight Edge - Control Wheel Part #: C27097-1 Supplier: 81205 Part #: SE27-0001 Supplier: 81205

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

F. Prepare for the Adjustment

SUBTASK 27-11-00-480-028

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-00-860-004

(2) Remove pressure from the aileron hydraulic system (TASK 27-11-00-860-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-480-003

(3) Use the high tension cable tensiometer, COM-1554, or low tension cable tensiometer, COM-1553, to measure the cable tension (TASK 27-09-14-860-802).

NOTE: If the tension measurement is in the limit and new control cables are not installed, the following adjustment procedure is not necessary.

SUBTASK 27-11-00-010-019

(4) To get access to cables AA and AB or cables ACBA and ACBB, do these steps:

(a) Open this access door:

Number	Name/Location
112A	Forward Access Door

(b) Remove these access panels (TASK 53-14-01-020-801):

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

G. Aileron Transfer Cables ACBA and ACBB Adjustment

SUBTASK 27-11-00-820-002

WARNING

MAKE SURE THAT THERE IS NO PRESSURE IN THE HYDRAULIC SYSTEM. PRESSURE IN THE HYDRAULIC SYSTEM CAN MOVE THE CONTROL SURFACES, CABLES, LANDING GEARS, AND LANDING GEAR DOORS QUICKLY. THEIR MOVEMENT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) If you install new ACBA1, ACBA2, ACBB1, and ACBB2 cables, do these steps to adjust the cables (Figure 502):

(a) Install the straight edge, SPL-1671, across the top of the two control wheels.

(b) Tighten cables ACBA2 and ACBB2 to 140 lbf (623 N) - 175 lbf (778 N).

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.50 in. (12.70 mm) or less from the straight edge, SPL-1671.

(c) Remove the straight edge, SPL-1671.

(d) Operate the control wheels ± 90 degrees 25 times.

(e) Decrease the cable tension to 0 lbf (0 N) - 20 lbf (89 N).

(f) Repeat these steps for cables ACBA1 and ACBB1.

(g) Install the straight edge, SPL-1671, across the upper ends of both control wheels.

(h) Tighten cables ACBA2 and ACBB2 to the specified cable rigging tension for ACBA2 and ACBB2 (TASK 27-09-14-860-802).

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(i) Tighten cables ACBA1 and ACBB1 to the specified cable rigging tension for ACBA1 and ACBB1 (TASK 27-09-14-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(j) Remove the straight edge, SPL-1671.

(k) Operate the control wheels ± 90 degrees 10 times.

(l) Install the straight edge, SPL-1671, across the top of the two control wheels.

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(m) Make sure that the cable tension is still within the specified cable rigging tension for cables ACBA1, ACBA2, ACBB1, and ACBB2 (TASK 27-09-14-860-802).

(n) Remove the straight edge, SPL-1671.

(o) Do not adjust cables ACBA1, ACBA2, ACBB1, and ACBB2 after they have been tightened to the specified cable rigging tension (TASK 27-09-14-860-802).

SUBTASK 27-11-00-820-003

(2) If the ACBA1, ACBA2, ACBB1, and ACBB2 cables are not new, do these steps to adjust the cables:

(a) Install the straight edge, SPL-1671, across the upper ends of both control wheels.

(b) Tighten cables ACBA2 and ACBB2 to the specified cable rigging tension for ACBA2 and ACBB2 (TASK 27-09-14-860-802).

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(c) Tighten cables ACBA1 and ACBB1 to the specified cable rigging tension for ACBA1 and ACBB1 (TASK 27-09-14-860-802).

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(d) Remove the straight edge, SPL-1671.

(e) Operate the control wheels ± 90 degrees 10 times.

(f) Install the straight edge, SPL-1671, across the top of the two control wheels.

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(g) Make sure that the cable tension is still within the specified cable rigging tension for cables ACBA1, ACBA2, ACBB1 and ACBB2 (TASK 27-09-14-860-802).

(h) Remove the straight edge, SPL-1671.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Do not adjust cables ACBA1, ACBA2, ACBB1, and ACBB2 after they have been tightened to the specified cable rigging tension (TASK 27-09-14-860-802).

SUBTASK 27-11-00-480-006

(3) Install the turnbuckle clip locks (TASK 20-10-10-400-801).

H. Aileron Body Cables AA and AB Adjustment

SUBTASK 27-11-00-860-109

(1) Disconnect the upper and lower pogos for each Power Control Unit (PCU) (TASK 27-11-00-820-802).

SUBTASK 27-11-00-860-007

(2) Make sure that the captain's and the first officer's control wheels are in the neutral position:

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize hydraulic systems A and B (TASK 29-11-00-860-801).

(b) Operate the aileron trim switches, on the P8 control stand panel, until you can freely install and remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron centering mechanism (Figure 501).

(c) Make sure that the rig pin A/S-15 is removed.

(d) Remove power from hydraulic systems A and B (TASK 29-11-00-860-805).

SUBTASK 27-11-00-210-001

(3) Make sure that the centering cam follower roller is firmly in the cam detent (Figure 505).

SUBTASK 27-11-00-820-004

WARNING MAKE SURE THAT THERE IS NO PRESSURE IN THE HYDRAULIC SYSTEM. PRESSURE IN THE HYDRAULIC SYSTEM CAN MOVE THE CONTROL SURFACES, CABLES, LANDING GEARS, AND LANDING GEAR DOORS QUICKLY. THEIR MOVEMENT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(4) If you install new cables AA and AB, do these steps to adjust the cables (Figure 502):

(a) Tighten the turnbuckles on the cables to get the tension to 230 lbf (1023.1 N) - 250 lbf (1112.1 N).

(b) Operate the captain's control wheel 25 times.

1) Do not turn the control wheel more than 100 degrees from the neutral position.

(c) Reduce the tension in the cables to get the specified cable rigging tension for cables AA and AB (TASK 27-09-14-860-802).

(d) Operate the captain's control wheel 10 times.

(e) Put the captain's control wheel back to the neutral position.

(f) Install the straight edge, SPL-1671, across the top of the two control wheels.

1) Make sure that both ends of the captain’s control wheel and one end of the first officer’s control wheel are in contact with the straight edge, SPL-1671.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(g) Make sure that the cable tension is still within the specified cable rigging tension for cables AA and AB (TASK 27-09-14-860-802).

(h) Make sure that you can freely install and remove rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron centering mechanism.

(i) Make sure that rig pin A/S-15 is removed.

(j) Remove the straight edge, SPL-1671.

SUBTASK 27-11-00-820-005

(5) If the cables AA and AB are not new, do these steps to adjust the cables:

(a) Tighten the turnbuckles on the cables to get the specified cable rigging tension for cables AA and AB (TASK 27-09-14-860-802).

(b) Operate the captain's control wheel 10 times.

(c) Put the captain's control wheel back to the neutral position.

(d) Install the straight edge, SPL-1671, across the top of the two control wheels.

1) Make sure that both ends of the captain's control wheel and one end of the first officer's control wheel are in contact with the straight edge, SPL-1671.

2) Make sure that the remaining end of the first officer's control wheel is 0.20 in. (5.08 mm) or less from the straight edge, SPL-1671.

(e) Make sure that the cable tension is still within the specified cable rigging tension for cables AA and AB (TASK 27-09-14-860-802).

(f) Make sure that you can freely install and remove rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron centering mechanism.

(g) Make sure that rig pin A/S-15 is removed.

(h) Remove the straight edge, SPL-1671.

SUBTASK 27-11-00-420-007

(6) Connect the upper and lower pogos for each PCU (TASK 27-11-00-820-802).

SUBTASK 27-11-00-420-001

(7) Install the turnbuckle clip locks (TASK 20-10-10-400-801).

I. Aileron Wing Cables ABSA and ABSB Adjustment

SUBTASK 27-11-00-820-007

(1) Make sure that the pogos and power control unit is correctly adjusted before you do the wing cables adjustment (TASK 27-11-00-820-802).

SUBTASK 27-11-00-860-130

(2) Make sure that the captain's and the first officer's control wheels are in the neutral position:

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Pressurize hydraulic systems A and B (TASK 29-11-00-860-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Operate the aileron trim switches, on the P8 control stand panel, until you can freely install and remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron centering mechanism (Figure 501).

(c) Make sure that the rig pin A/S-15 is removed.

(d) Remove power from hydraulic systems A and B (TASK 29-11-00-860-805).

(e) Make sure that you can freely install and remove rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum.

(f) Make sure that the rig pin A/S-4 is removed.

SUBTASK 27-11-00-820-008

WARNING

MAKE SURE THAT THERE IS NO PRESSURE IN THE HYDRAULIC SYSTEM. PRESSURE IN THE HYDRAULIC SYSTEM CAN MOVE THE CONTROL SURFACES, CABLES, LANDING GEARS, AND LANDING GEAR DOORS QUICKLY. THEIR MOVEMENT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(3) If you install new cables ABSA and ABSB, do these steps to adjust the cables (Figure 502):

(a) Tighten the turnbuckles on the cables to get the tension to 420 lbf (1868 N) - 440 lbf (1957 N).

(b) Operate the captain's control wheel 25 times.

1) Do not turn the control wheel more than 100 degrees from the neutral position.

(c) Reduce the tension in the cables to get the specified cable rigging tension for cables ABSA and ABSB (TASK 27-09-14-860-802).

(d) Operate the captain's control wheel 10 times.

(e) Put the captain's control wheel back to the neutral position.

(f) Make sure that the cable tension is still within the specified cable rigging tension for the cables ABSA and ABSB (TASK 27-09-14-860-802).

(g) Make sure that you can freely install and remove rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum.

(h) Make sure that rig pin A/S-4 is removed.

SUBTASK 27-11-00-820-009

(4) If the cables ABSA and ABSB are not new, do these steps to adjust cables ABSA and ABSB:

(a) Tighten the turnbuckles until you get the specified cable rigging tension for cables ABSA and ABSB (TASK 27-09-14-860-802).

(b) Operate the captain's control wheel 10 times.

(c) Put the captain's control wheel back to the neutral position.

(d) Make sure that the cable tension is still within the specified cable rigging tension for cables ABSA and ABSB (TASK 27-09-14-860-802).

(e) Make sure that you can freely install and remove rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum.

(f) Make sure that rig pin A/S-4 is removed.

SUBTASK 27-11-00-860-013

(5) Make sure that the aileron is in the rig position (Figure 506).

(a) Make sure that the aft-most outboard edge of the aileron upper surface panel is 0.28 ±0.03 in. (7.11 ±0.76 mm) below the aft-most upper surface of the fixed trailing edge.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) If the aileron is not in the rig position, adjust the turnbuckles for each cable.

SUBTASK 27-11-00-860-014

(6) Make sure that the cable tension is still within the specified cable rigging tension for cables ABSA and ABSB (TASK 27-09-14-860-802).

SUBTASK 27-11-00-860-107

(7) Make sure that you can freely install and remove rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum.

(a) Make sure that rig pin A/S-4 is removed.

SUBTASK 27-11-00-420-002

(8) Install the turnbuckle clip locks (TASK 20-10-10-400-801).

SUBTASK 27-11-00-860-108

(9) Make sure that the gaps between the aileron bus drum and the screws are 0.05 ±0.01 in. (1.27 ±0.25 mm) (View A-A, Figure 503).

NOTE: The screws are adjacent to the hole for rig pin A/S-4.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-410-002

(1) To close access to cables AA and AB or cables ACBA and ACBB, do these steps:

(a) Install these access panels (TASK 53-14-01-420-801):

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

(b) Close this access door:

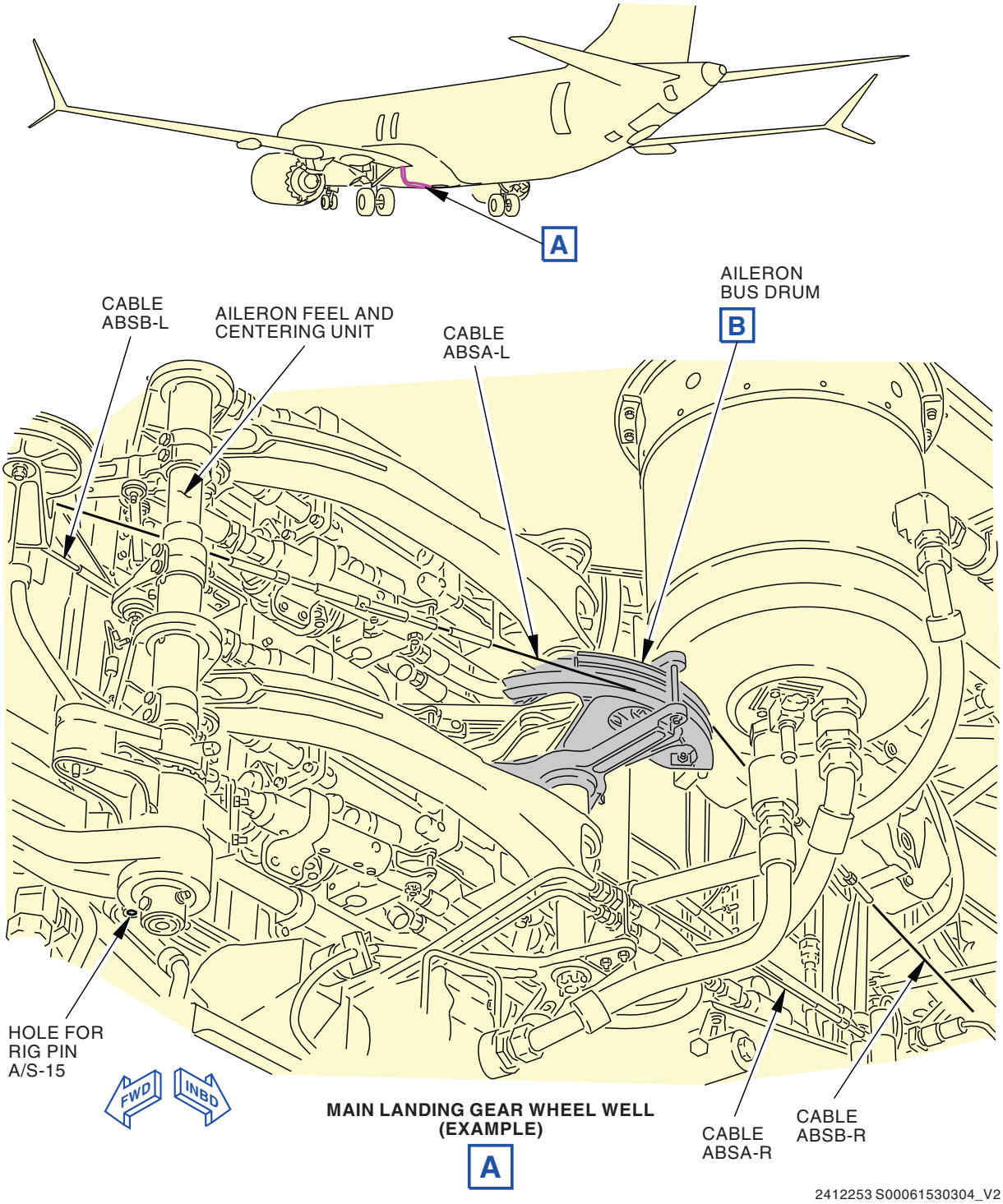
Number	Name/Location
112A	Forward Access Door

SUBTASK 27-11-00-860-015

(2) Pressurize the aileron hydraulic system (TASK 27-11-00-860-802).

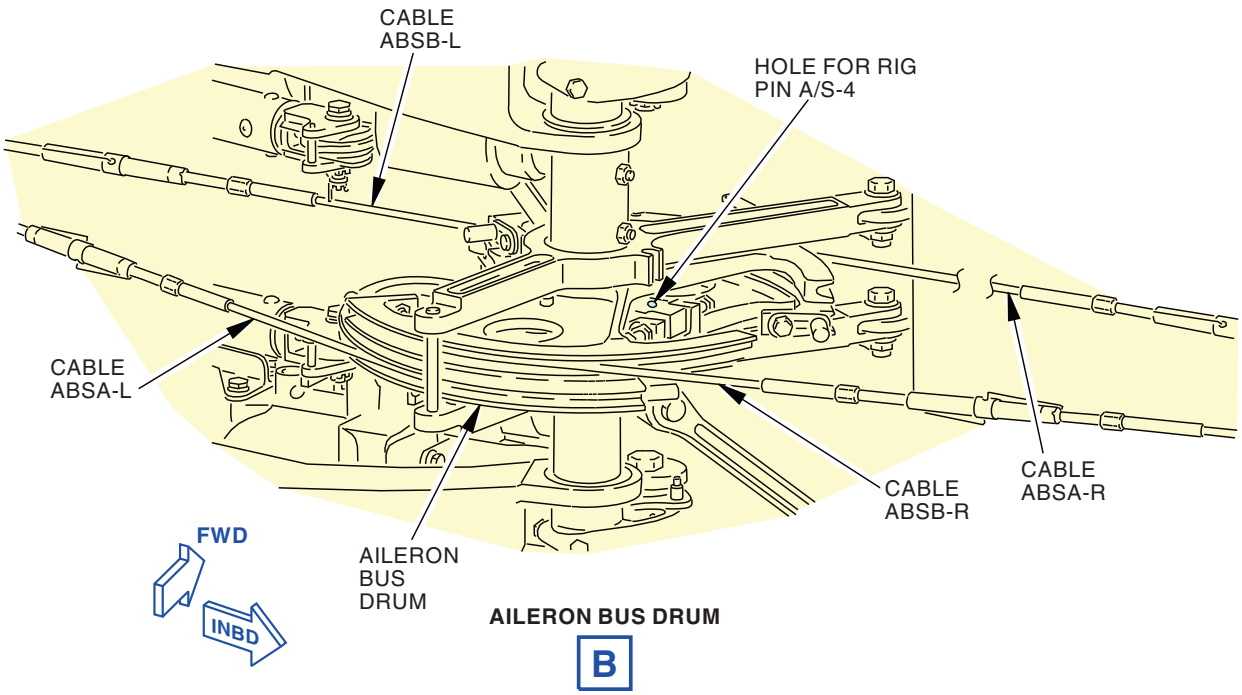
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rig Pin Location
Figure 501/27-11-00-990-803 (Sheet 1 of 2)

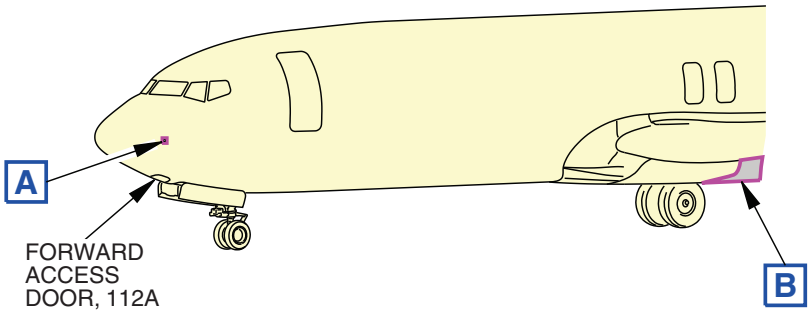
ECCN 9E991 BOEING PROPRIETARY - See title page for details

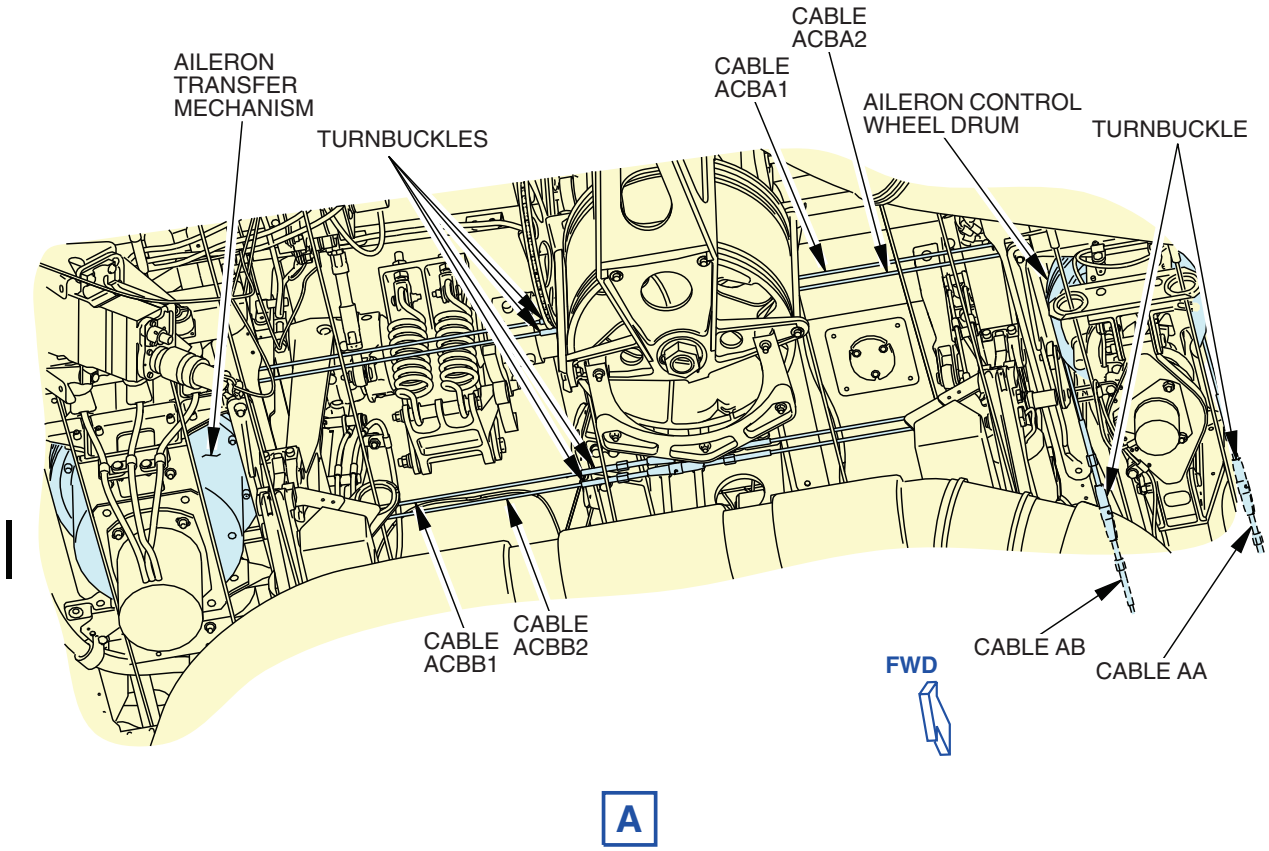


2412254 S00061530305_V1

Rig Pin Location
Figure 501/27-11-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

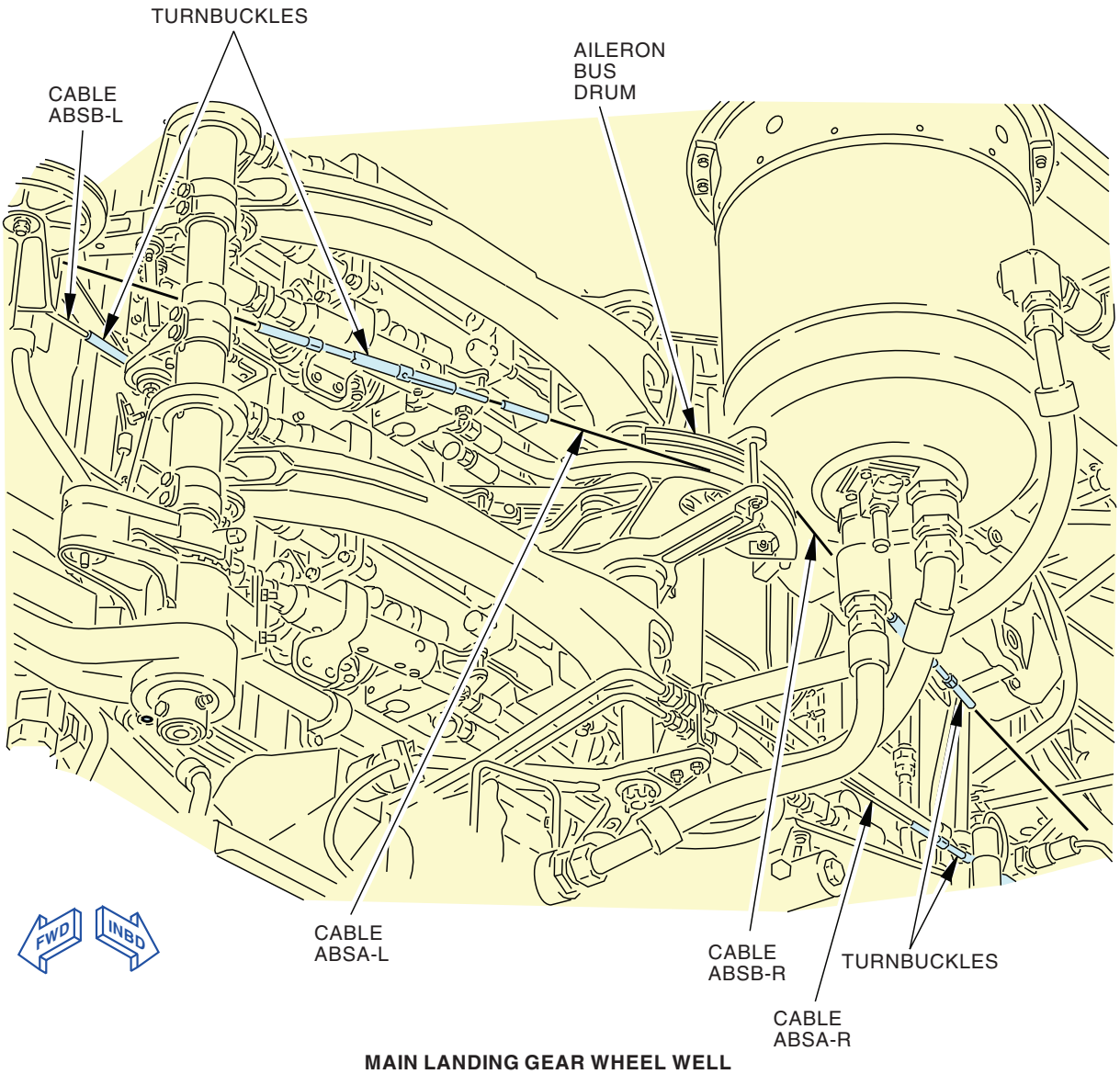




2412256 S00061530307_V4

Aileron Cable Adjustment
Figure 502/27-11-00-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR WHEEL WELL
(EXAMPLE)

B

2412257 S00061530308_V2

Aileron Cable Adjustment
Figure 502/27-11-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-820-802

3. Pogo and Power Control Unit (PCU) Adjustment

(Figure 503 and Figure 506)

A. References

Reference	Title
27-11-71 P/B 401	AILERON POWER CONTROL UNIT - REMOVAL/INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G02436	Lockwire - MS20995NC40, Monel - 0.040 Inch (1.016 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Adjustment

SUBTASK 27-11-00-480-009

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-00-860-016

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize hydraulic systems A and B (TASK 29-11-00-860-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-860-017

(3) Operate the aileron trim switches, on the P8 control stand panel, until you can freely install and remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim actuator (Figure 501).

SUBTASK 27-11-00-860-018

(4) Remove power from hydraulic systems A and B (TASK 29-11-00-860-805).

F. Pogo and the Power Control Unit (PCU) Adjustment

SUBTASK 27-11-00-860-019

(1) Make sure that you can freely install and remove the rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum (Figure 501).

SUBTASK 27-11-00-820-010

(2) If you cannot easily install and remove the rig pin A/S-4, adjust the pogos [1] in the specified sequence:

NOTE: When adjustment of the four pogos is necessary, make sure that you follow these instructions in the specified sequence. Incorrect adjustment of the upper (secondary) and lower (primary) pogos on system A or B Power Control Unit (PCU) when adjusted in the opposite directions will cause the incorrect condition of 'free fit' of rig pin A/S 4.

NOTE: Do not adjust the primary pogos without first removing the secondary pogos. If primary pogos are adjusted the secondary pogos must be re-rigged.

(a) Disconnect the upper (secondary) pogo [1] from system B PCU and the upper (secondary) pogo [1] from system A PCU.

1) Remove the pins [8], nuts [7], washers [6], bushings [5], washers [4] and bolts [3] to disconnect the two pogos [1] from the PCUs.

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Pressurize hydraulic system A and B (TASK 29-11-00-860-801).

(c) Operate the control wheels ± 90 degrees 10 times.

1) Operate the captain’s control wheel carefully and smoothly.

(d) Make sure that the captain's control wheel is the center of the neutral position, do this task: Control Wheel Centering Check, TASK 27-11-00-700-804.

(e) Adjust the lower (primary) pogos [1] on system A PCU and on system B PCU:

1) Loosen the locknut [2] on the each pogo [1].

2) Turn the shaft in the barrel and left clevis to adjust the length of the pogo [1] until you can freely install and remove rig pin A/S-4 in the aileron bus drum.

a) Make sure that the PCU stops on the aileron bus drum are not in contact.

b) Make sure that the drain holes on the pogo [1] are down.

3) Tighten the locknut [2] by hand.

4) Make sure that you can freely install and remove rig pin A/S-4.

a) If it is necessary, adjust the lower (primary) pogos [1] on system A PCU and on system B PCU again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Remove power from hydraulic systems A and B (TASK 29-11-00-860-805).

(g) Connect the upper (secondary) pogo [1] to system A PCU.

1) Adjust the upper (secondary) pogo [1] on system A PCU:

a) Loosen the locknut [2] on the pogo [1].

b) Turn the shaft in the barrel and left clevis to adjust the length of the upper (secondary) pogo [1] until you can easily install the bolt [3].

<1> Make sure that the drain holes on the pogo [1] are down.

c) Tighten the locknut [2] by hand.

d) Make sure that the bolt [3] fits freely in its hole.

<1> If it is necessary, adjust the upper (secondary) pogo [1] on system A PCU again.

2) Install the bolt [3] with the washer [4], bushing [5], washer [6], nut [7] and pin [8] to connect the upper (secondary) pogo [1] to the system A PCU.

(h) Connect the upper (secondary) pogo [1] to system B PCU.

1) Adjust the upper (secondary) pogo [1] on system B PCU:

a) Turn the shaft in the barrel and left clevis to adjust the length of the upper (secondary) pogo [1] until you can easily install the bolt [3] in the PCU lever.

<1> Make sure that the drain holes on the pogo [1] are down.

b) Tighten the locknut [2] by hand.

c) Make sure that the bolt [3] fits freely in the hole.

<1> Adjust the upper (secondary) pogo [1] on system B again if it is necessary.

2) Install the bolt [3] with washer [4], bushing [5], washer [6], nut [7] and pin [8] to connect the upper (secondary) pogo [1] to the system B PCU.

SUBTASK 27-11-00-420-004

(3) Tighten the locknut [2] on the each pogo [1] to 160 ±10 in-lb (18.1 ±1.2 N·m).

SUBTASK 27-11-00-860-126

(4) Make sure that the left pogo clevis bushing flanges are pointed up, and the drain hole in the pogo barrel is pointed down.

SUBTASK 27-11-00-420-005

(5) Install MS20995NC40 lockwire, G02436, on the locknut [2] of the each pogo [1].

SUBTASK 27-11-00-860-115

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Pressurize hydraulic systems A and B (TASK 29-11-00-860-801).

SUBTASK 27-11-00-820-025

(7) Operate the aileron trim switches, on the P8 control stand panel, until you can freely install and remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim actuator (Figure 501).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-860-116

(8) Make sure that the captain's control wheel is the center of the neutral position.

(a) Operate the captain’s control wheel carefully and smoothly.

(b) Jiggle the control wheel about neutral to make sure that it is centered.

SUBTASK 27-11-00-480-010

(9) Make sure that you can freely install and remove the rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum.

NOTE: Do not adjust the primary pogos without first removing the secondary pogos. If primary pogos are adjusted the secondary pogos must be re-rigged.

(a) If it is necessary, adjust the pogos [1] in the same specified sequence as above again.

SUBTASK 27-11-00-860-110

(10) Do these steps at the same time:

(a) Make sure that the valve arms on the upper (secondary) and lower (primary) pogos [1] for each PCU are at the center between the stops (View C and View E, Figure 503).

1) If the valve arms are not at the center between the stops, adjust the four pogos [1] in the same specified sequence as above.

(b) Make sure that the gaps between the lug on the top aileron bus drum and the screws [12] on the bottom aileron bus drum are 0.05 ±0.01 in. (1.27 ±0.25 mm) (View A-A, Figure 503).

1) If the gap is out of tolerance, do these steps:

a) Remove the MS20995NC40 lockwire, G02436.

b) Loosen the nut [13].

c) Install and remove rig pin A/S-4 in the aileron bus drum.

d) Remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

e) Adjust the screw [12] until the gap is 0.05 ±0.01 in. (1.27 ±0.25 mm).

f) Install the MS20995NC40 lockwire, G02436, on the two screws [12] and nuts [13].

2) If the conditions above are applicable and valve arms on the lower (primary) pogos [1] or PCU are still not centered, do this step:

a) Replace the PCU (PAGEBLOCK 27-11-71/401).

3) If the conditions above are applicable and valve arms on the upper (secondary) pogos [1] or PCU are still not centered, do these steps:

a) Disconnect the upper (secondary) pogo [1] from system B PCU and the upper (secondary) pogo [1] from system A PCU.

<1> If the conditions above are applicable and valve arms on the upper (secondary) pogos [1] and PCU are centered, do this step:

<a> Adjust the upper (secondary) pogos [1] in the same specified sequence as above again.

<2> If the conditions above are applicable and valve arms on the upper (secondary) pogos [1] and PCU are still not centered, do this step:

<a> Replace the PCU (PAGEBLOCK 27-11-71/401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-720-009

(1) Put the captain's control wheel back to the neutral position (TASK 27-11-00-700-804).

SUBTASK 27-11-00-080-003

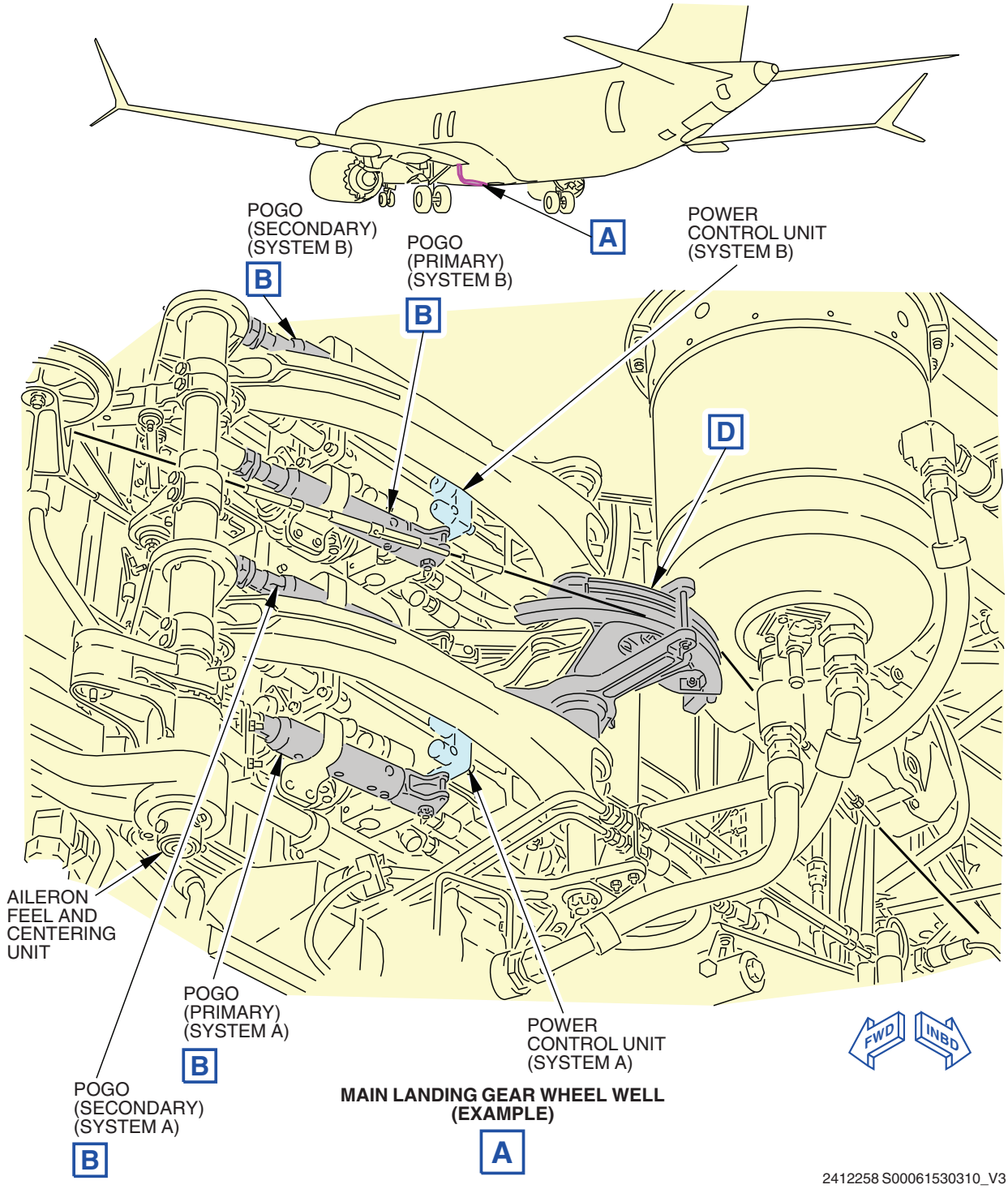
(2) Remove rig pins A/S-15 and A/S-4.

SUBTASK 27-11-00-860-127

(3) If applicable, remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

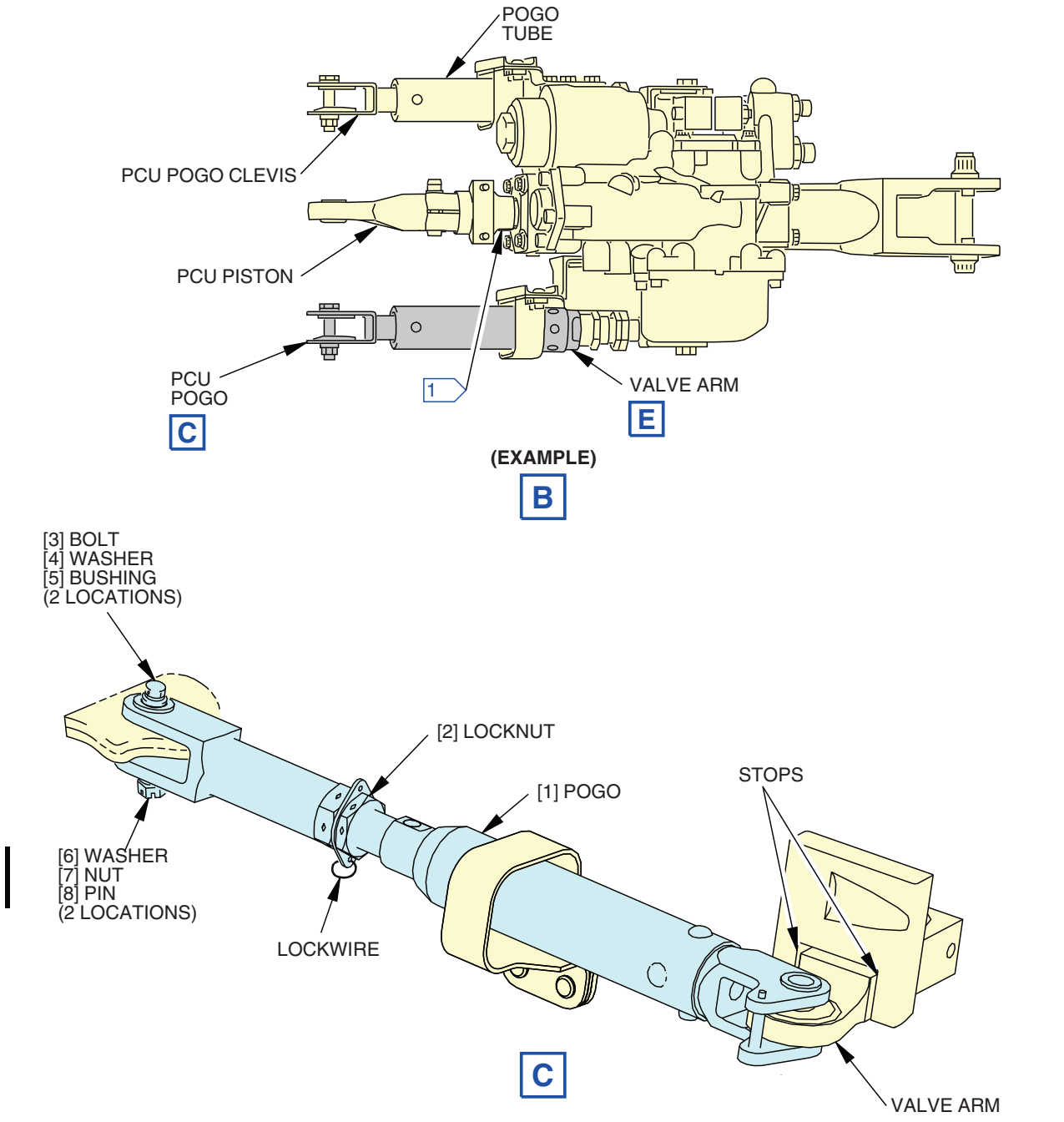
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



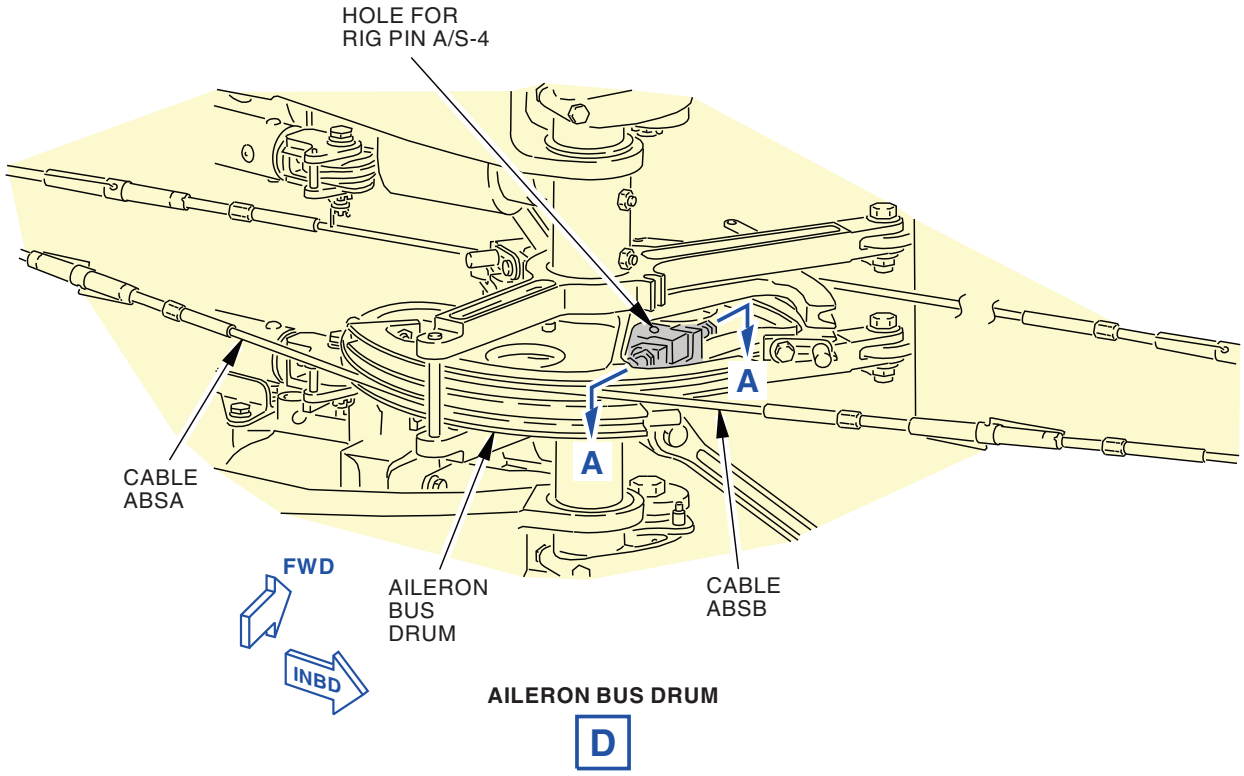
Aileron Pogo and Power Control Unit Adjustment
Figure 503/27-11-00-990-805 (Sheet 1 of 4)

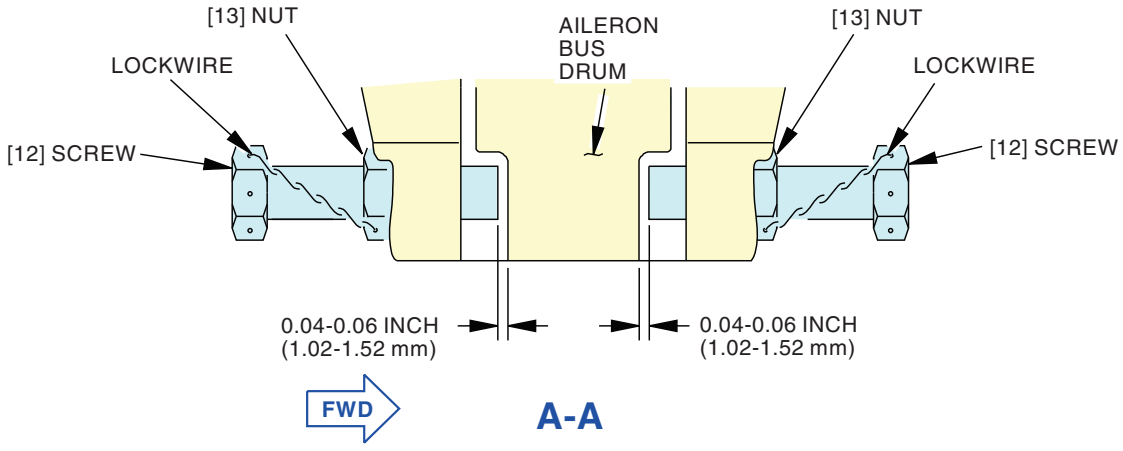
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Aileron Pogo and Power Control Unit Adjustment
Figure 503/27-11-00-990-805 (Sheet 2 of 4)

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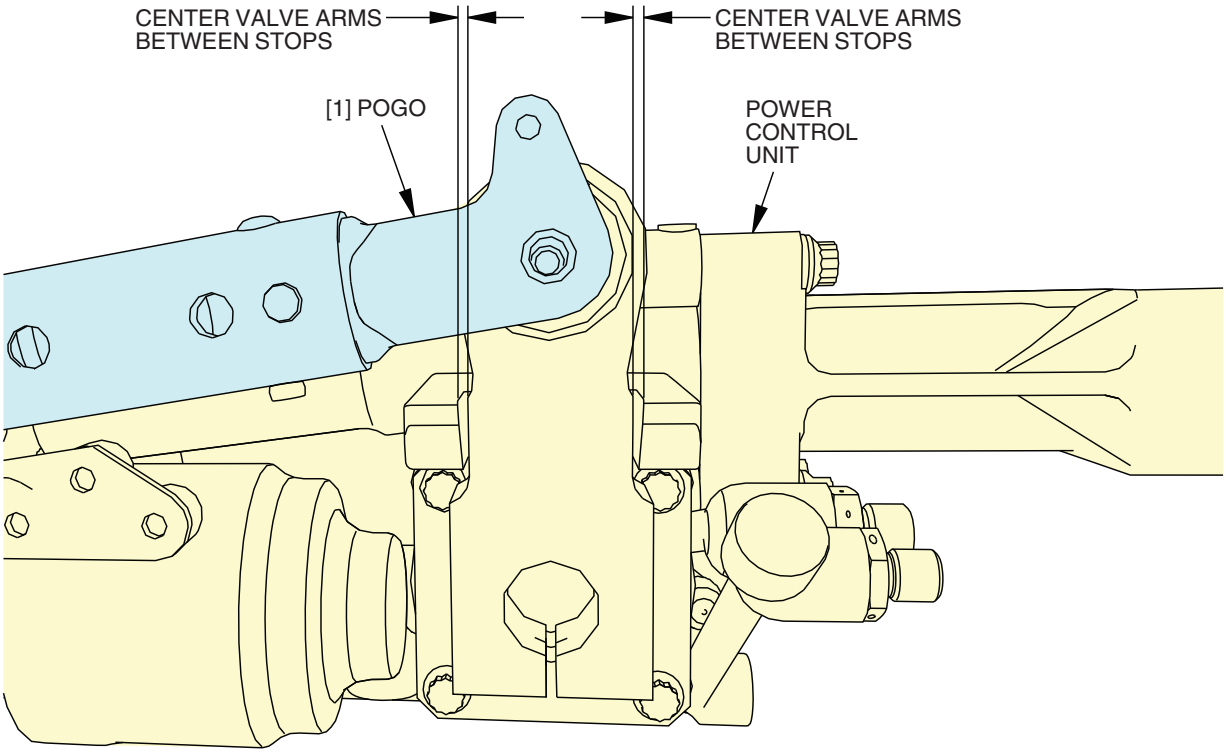




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Aileron Pogo and Power Control Unit Adjustment
Figure 503/27-11-00-990-805 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aileron Pogo and Power Control Unit Adjustment
Figure 503/27-11-00-990-805 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-820-803

4. Aileron Balance Tab Adjustment (Recommended Method)

(Figure 504)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1682	Protractor - Aileron Tab Alignment Part #: F80222-11 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

D. Prepare for the Adjustment

SUBTASK 27-11-00-860-020

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

E. Aileron Balance Tab Adjustment (Recommended Method)

NOTE: You can adjust the aileron balance tab with the aileron in the faired position or with the aileron at the downrig position.

NOTE: Measurements can also be taken with the protractor, SPL-1682, if it is necessary.

SUBTASK 27-11-00-820-013

(1) Do these steps to do a check of the aileron balance tab [31] with the aileron [32] in the faired position:

(a) Align the lower surface of the aileron trailing edge flush with the lower trailing edge surface of the adjacent fixed wing structure to ±0.03 in. (0.76 mm) at the outboard end of the aileron.

(b) Make sure that the outboard lower trailing edge surface of the aileron balance tab [31] aligns with the lower trailing edge surface of the aileron [32] to ±0.03 in. (0.76 mm).

1) If this condition is not met, adjust the aileron balance tab [31].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-220-001

(2) Do these steps to do a check of the aileron balance tab [31] with the aileron [32] in the downrig position (Figure 506):

NOTE: The aileron rig position is measured 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, inboard corner of the fixed lower panel, and 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, outboard corner of the aileron. At this location, the distance between the lower surface of the fixed lower panel and the lower surface of the aileron should be 0.25 ±0.03 in. (6.35 ±0.76 mm).

(a) Measure the vertical distance between the lower surface of the aileron balance tab [31] to the lower surface of the aileron [32] at the outboard end of the aileron balance tab [31] (dimension A) (Figure 504):

1) Measure the distance from the lower aft most points of the aileron balance tab [31] and aileron [32].

2) Make sure that the dimension A is 0.078 ±0.070 in. (1.98 ±1.78 mm).

(b) If the dimension A is out of tolerance, adjust the aileron balance tab [31].

SUBTASK 27-11-00-820-014

(3) Do these steps to adjust the aileron balance tab [31]:

(a) Move the aileron trailing edge to the down position.

(b) Remove the bolts [45] that attach the forward removable fairing [46] to the aileron.

(c) Remove the bolts [43] that attach the aft removable fairing [44] to the aileron balance tab [31].

(d) Loosen the jam nuts [47] and jam nuts [49] on the aileron tab control rods [33].

1) You must adjust the two aileron tab control rods [33] the same quantity.

(e) Do these steps if a coarse adjustment is necessary:

1) Remove the pins [42], nuts [41], washers [40], bushings [39], washers [38], and bolts [37] to disconnect the aileron tab control rods [33] from the aileron balance tab [31].

2) Turn the rod ends on the tab control rods [33] for adjustment.

NOTE: One-half turn of the rod end changes the position of the tab trailing edge 0.067 in. (1.70 mm). One-half turn of the rod end and barrel together changes the position of the tab trailing edge 0.080 in. (2.03 mm).

a) To raise the trailing edge of the aileron balance tab [31], lengthen the aileron tab control rods [33].

b) To lower the trailing edge of the aileron balance tab [31], shorten the aileron tab control rods [33].

3) Connect the aileron tab control rods [33] to the aileron balance tab [31].

a) Install the bushings [39], washers [38], washers [40], bolts [37], and nuts [41].

b) If there is binding between the aileron tab control rods [33] and the bolt [37], adjust the length of the aileron tab control rods [33].

c) Make sure there is a minimum of 0.015 in. (0.38 mm) gap between the bushing [39] and the washer [38].

d) Tighten the nuts [41] to 65 ±15 in-lb (7.3 ±1.7 N·m).

e) Install the pins [42].

(f) Do these steps if fine adjustment is necessary:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Turn the barrels [48] on the aileron tab control rods [33].

NOTE: One turn of the barrel, with the rod end fixed, changes the position of the tab trailing edge 0.027 in. (0.69 mm).

a) To raise the trailing edge of the aileron balance tab [31], lengthen the aileron tab control rods [33].

b) To lower the trailing edge of the aileron balance tab [31], shorten the aileron tab control rod [33].

2) Make sure that the threads are visible in the inspection holes in the barrel [48].

(g) Tighten the jam nuts [47] to 120 ±20 in-lb (13.6 ±2.3 N·m).

(h) Tighten the jam nuts [49] to 90 ±10 in-lb (10.2 ±1.2 N·m).

(i) Install the bolts [43] that attach the aft removable fairing [44] to the aileron balance tab [31].

(j) Install the bolts [45] that attach the forward removable fairing [46] to the aileron.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-080-019

(1) If used to take measurements, remove the protractor, SPL-1682.

SUBTASK 27-11-00-440-001

(2) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-00-700-001

(3) Do this task: Flight Test to Find and Correct Asymmetrical Flight Control Forces, TASK 27-11-00-750-801.

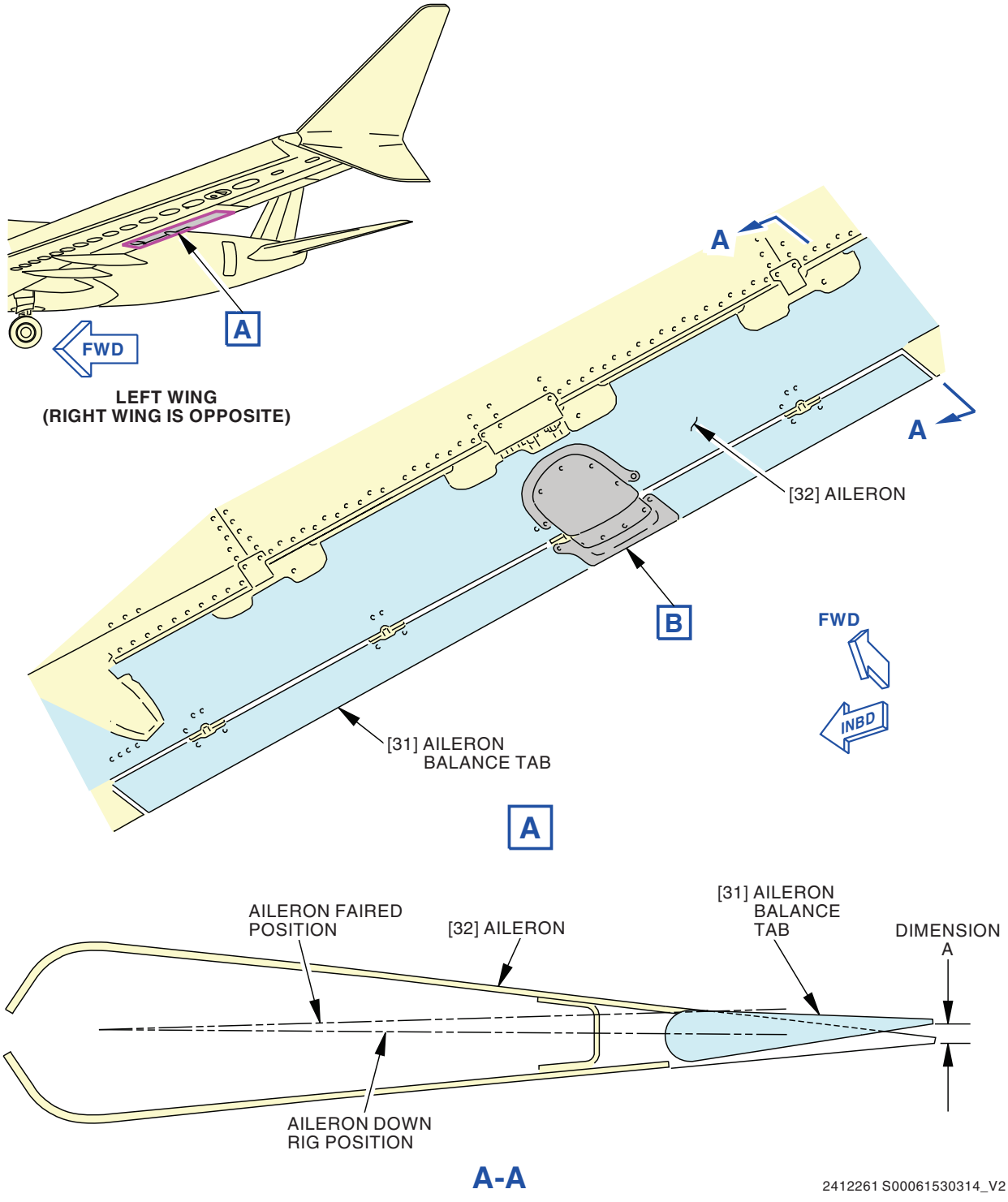
NOTE: For each aileron tab, if both control rods are adjusted, then flight test is necessary.

If only one rod is adjusted, and you can verify that the remaining rod had not been replaced or adjusted since the last flight test, then flight test is not necessary.

If only one rod is adjusted, and you can not verify that the remaining rod had not been replaced or adjusted since the last flight test, then flight test is necessary.

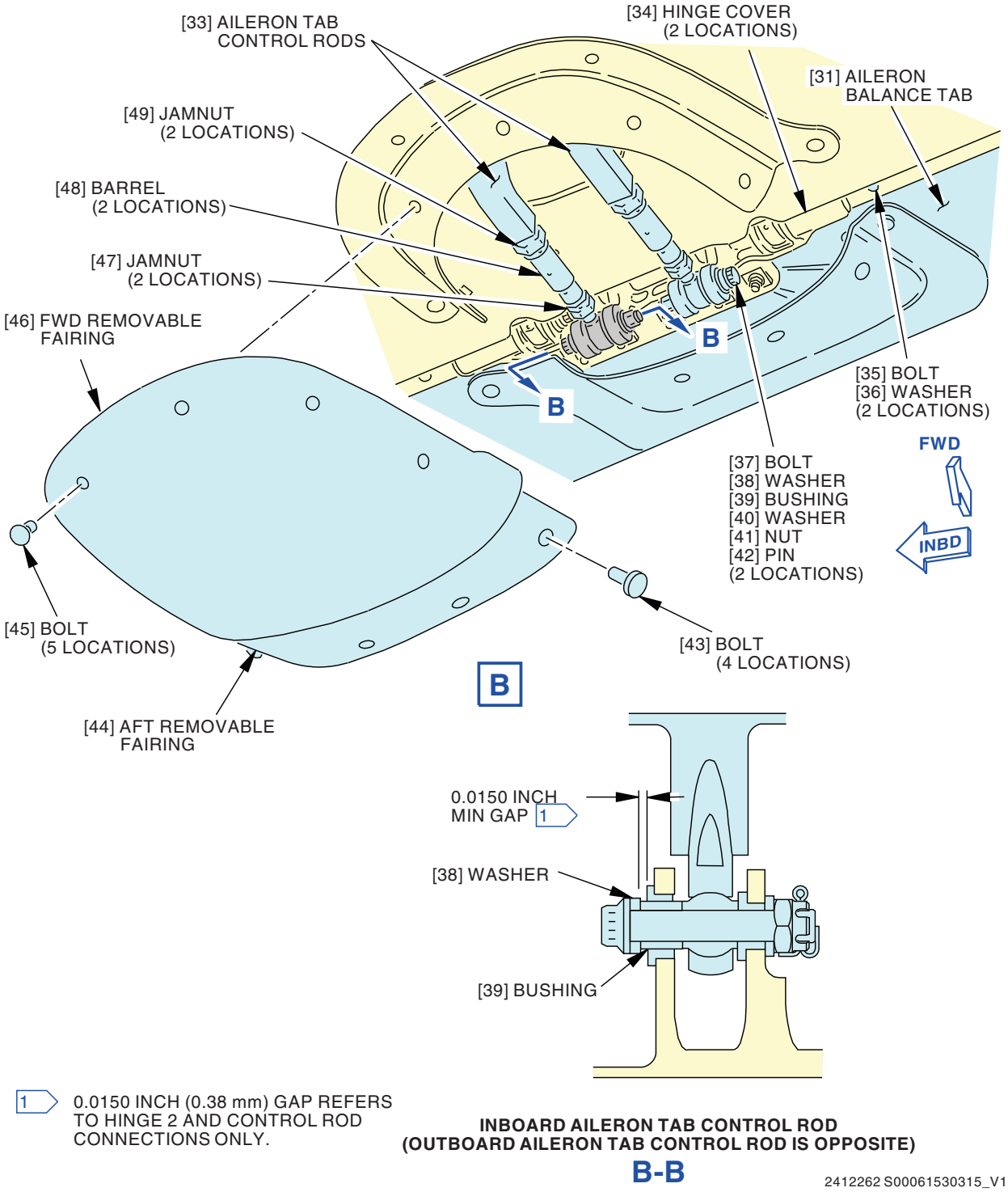
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Tab Adjustment
Figure 504/27-11-00-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Tab Adjustment
Figure 504/27-11-00-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-820-805

5. Aileron Feel and Centering Unit Adjustment

(Figure 505)

A. General

(1) The aileron feel and centering unit is referred to as the "centering unit" in this task.

B. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Adjustment

SUBTASK 27-11-00-480-012

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-00-860-025

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THE CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR CAN MOVE SUDDENLY WITH THE PRESSURIZATION OF A HYDRAULIC SYSTEM. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-11-00-820-017

(3) Make sure that the springs [63] on the feel and centering unit [61] are not loose.

(a) If the springs [63] are loose, tighten the nuts [65] on the eyebolts [66] until you see 4 to 5 threads beyond the nuts [65].

(b) Do not stretch the springs [63] from free length.

F. Aileron Feel and Centering Unit Adjustment

SUBTASK 27-11-00-860-026

(1) Shake the control wheels to make sure that the wheels are in the neutral position.

SUBTASK 27-11-00-710-002

(2) Do this task: Control Wheel Centering Check, TASK 27-11-00-700-804.

SUBTASK 27-11-00-970-001

(3) Measure the control wheel breakout force in the clockwise and counterclockwise directions:

(a) Install the adapter, SPL-1674, to the captain's control wheel.

NOTE: If the adapter is not available, put a tangent force to the captain's control wheel at a 6.5 in. (165 mm) radius.

(b) Remove the map holder assembly from the first officer's control wheel.

(c) Install the mount, SPL-1670 and the protractor, SPL-1680, to the first officer's control wheel.

(d) Set the indicator on the protractor, SPL-1680, to zero degrees.

(e) Turn the control wheel clockwise until the aileron starts to move.

1) Using a push/pull scale 0-25 pound (0-11 kilogram), STD-753, make sure that the force to turn the control wheel is not more than 29.0 in-lb (3.3 N·m) or 4.5 lbf (20.0 N).

(f) Release the control wheel.

(g) Turn the control wheel counterclockwise until the aileron starts to move.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Using a push/pull scale 0-25 pound (0-11 kilogram), STD-753, make sure that the force to turn the control wheel is not more than 29.0 in-lb (3.3 N·m) or 4.5 lbf (20.0 N).

SUBTASK 27-11-00-820-018

(4) If the control wheels center properly, and breakout forces are within the limits above, then no adjustment is necessary.

SUBTASK 27-11-00-820-019

(5) If the control wheels are not in the center or the breakout forces are out of tolerance, adjust the tension of the springs [63] on the centering unit [61]:

(a) Hold the eyebolt [66] while you tighten or loosen the nut [65] in 1/2 turn increments.

NOTE: You must tighten the two eyebolts [66] equally. Tighten the eyebolts [66] to increase the spring forces. Loosen the eyebolts [66] to decrease the spring forces.

(b) Adjust the nuts [65] until the force to turn the control wheel is not more than 29.0 in-lb (3.3 N·m) or 4.5 lbf (20.0 N).

(c) Tighten the checknuts [64].

(d) Make sure that the shaft [70] can turn freely.

SUBTASK 27-11-00-750-001

(6) After adjustment of breakout forces above, do the following steps to do a check of the breakout forces and make sure that the control wheel is centered properly:

(a) Turn the control wheel clockwise until the aileron starts to move.

1) Make sure that the force to turn the control wheel is not more than 29.0 in-lb (3.3 N·m) or 4.5 lbf (20.0 N).

(b) Release the control wheel.

1) Make sure that the control wheel goes back to the neutral position in 0.0 ±0.4 units of trim (2.4 degrees).

(c) Turn the control wheel counterclockwise until the aileron starts to move.

1) Make sure that the force to turn the control wheel is not more than 29.0 in-lb (3.3 N·m) or 4.5 lbf (20.0 N).

(d) Release the control wheel.

1) Make sure that the control wheel goes back to the neutral position in 0.0 ±0.4 units of trim (2.4 degrees).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-080-004

(1) Remove the adapter, SPL-1674, from the captain's aileron control wheel.

SUBTASK 27-11-00-080-005

(2) Remove the mount, SPL-1670 and the protractor, SPL-1680, from the first officer's control wheel.

SUBTASK 27-11-00-410-006

(3) Install the map holder assembly to the first officer's control wheel.

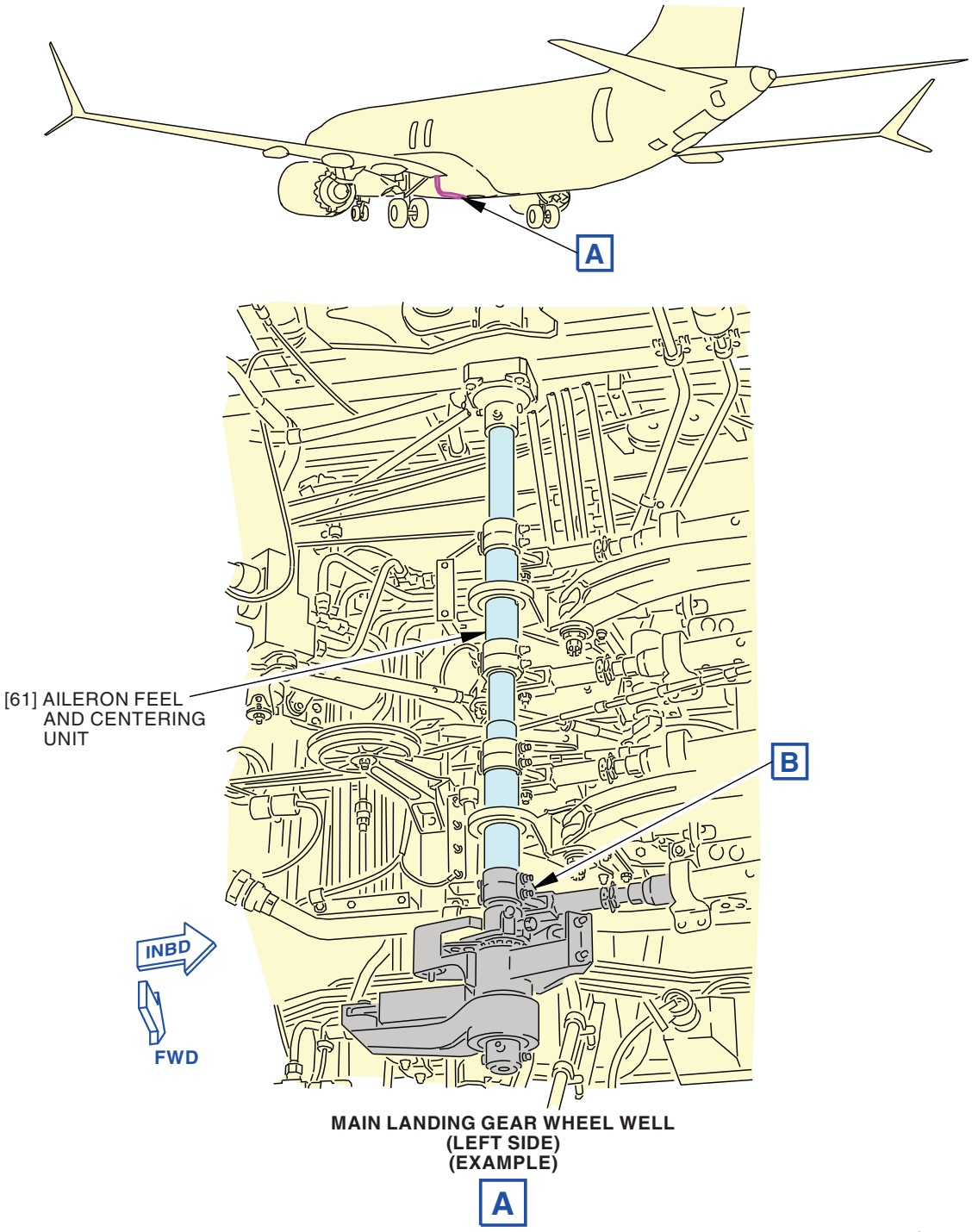
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-080-006

(4) Remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

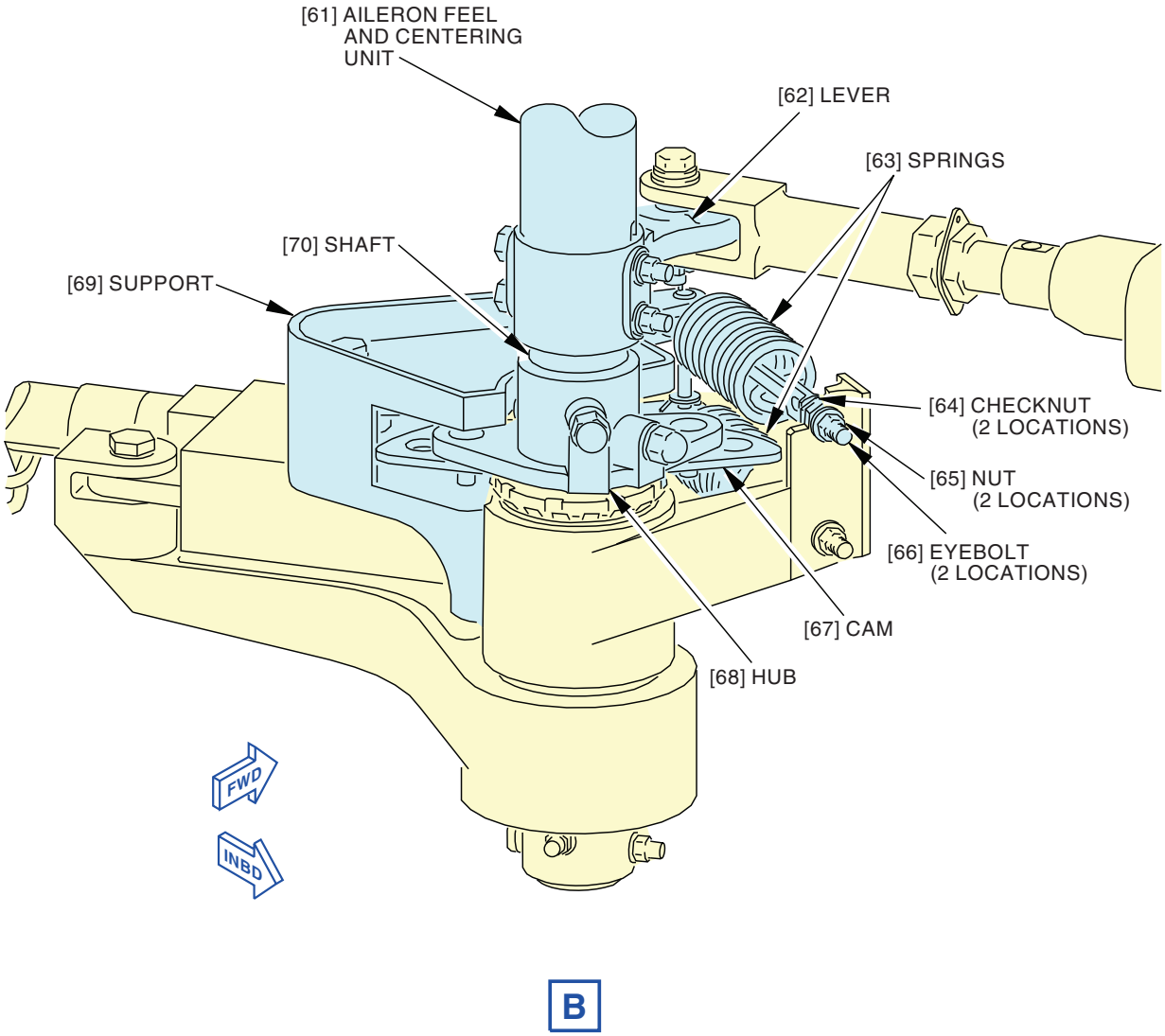
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Aileron Feel and Centering Unit Adjustment
Figure 505/27-11-00-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aileron Feel and Centering Unit Adjustment
Figure 505/27-11-00-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-820-806

6. Do A Check of the Aileron System Adjustment

(Figure 501, Figure 504, Figure 506)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1671	Straight Edge - Control Wheel Part #: C27097-1 Supplier: 81205 Part #: SE27-0001 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Prepare for the Check

SUBTASK 27-11-00-480-013

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-00-010-004

(2) Open this access panel:

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Do a Check of the Aileron System Adjustment

SUBTASK 27-11-00-820-020

(1) Do a check of the aileron system adjustment.

(a) Attach the straight edge, SPL-1671, on the captain's and first officer's control wheels.

(b) Make sure that a rig pin A/S-4, from the rigging pin kit, SPL-1585, can be freely installed in the aileron bus drum (Figure 501).

1) If a rig pin A/S-4 cannot be freely installed, do this task: Pogo and Power Control Unit (PCU) Adjustment, TASK 27-11-00-820-802.

(c) Remove rig pin A/S-4 from the aileron bus drum.

(d) Remove the straight edge, SPL-1671, from the captain's and first officer's control wheels.

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(e) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(f) Operate the aileron trim switches on the control stand panel, P8, until the rig pin A/S-15 from the rigging pin kit, SPL-1585, can be freely installed and removed (Figure 501).

(g) Make sure that the captain's control wheel is the center of the neutral position.

1) Gently move the control wheel in the backlash band.

2) Make sure that the center of the backlash band aligns with the protractor zero position.

(h) Make sure that the index mark on the captain's control wheel aligns with the zero mark on the control column.

(i) Attach the straight edge, SPL-1671, on the captain's and first officer's control wheels.

(j) Make sure that the top ends of the control wheels touch the straight edge, SPL-1671.

(k) Make sure that one end of the first officer's control wheel is no more than 0.20 in. (5.1 mm) from the straight edge, SPL-1671.

1) If the control wheels do not align, adjust the cables ACBA and ACBB, do this task: Aileron Cables Adjustment, TASK 27-11-00-820-801.

(l) Make sure that these rig pins, from the rigging pin kit, SPL-1585, can be freely installed and removed (Figure 501):

NOTE: Install these rig pins at the same time.

1) Rig pin A/S-4 in the aileron bus drum

2) Rig pin A/S-15 in the aileron trim actuator.

(m) If the above rig pins cannot be freely installed and removed, adjust the cables AA and AB, do this task: Aileron Cables Adjustment, TASK 27-11-00-820-801.

(n) Remove the straight edge, SPL-1671, from the captain's and first officer's control wheels.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(o) Make sure that the aileron is in the rig position (Figure 506).

NOTE: The aileron rig position is measured 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, inboard corner of the fixed lower panel, and 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, outboard corner of the aileron. At this location, the distance between the lower surface of the fixed lower panel and the lower surface of the aileron should be 0.25 ±0.03 in. (6.35 ±0.76 mm).

1) If the aileron is not in the rig position, adjust the cables ABSA and ABSB, do this task: Aileron Cables Adjustment, TASK 27-11-00-820-801.

(p) Make sure that the aileron balance tab aligns with the aileron.

1) Measure the distance between the rig position of the aileron balance tab [31] to the aileron lower surface (dimension A) (Figure 504).

NOTE: You must measure from the lower aft most point of the aileron balance tab.

a) At the outboard end of the aileron balance tab, make sure that the dimension A is 0.078 ±0.070 in. (2.0 ±1.8 mm) or 1.07 ±0.25 degrees.

2) If the aileron balance tab does not align with the aileron, do this task: Aileron Balance Tab Adjustment (Recommended Method), TASK 27-11-00-820-803.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-410-003

(1) Close this access panel:

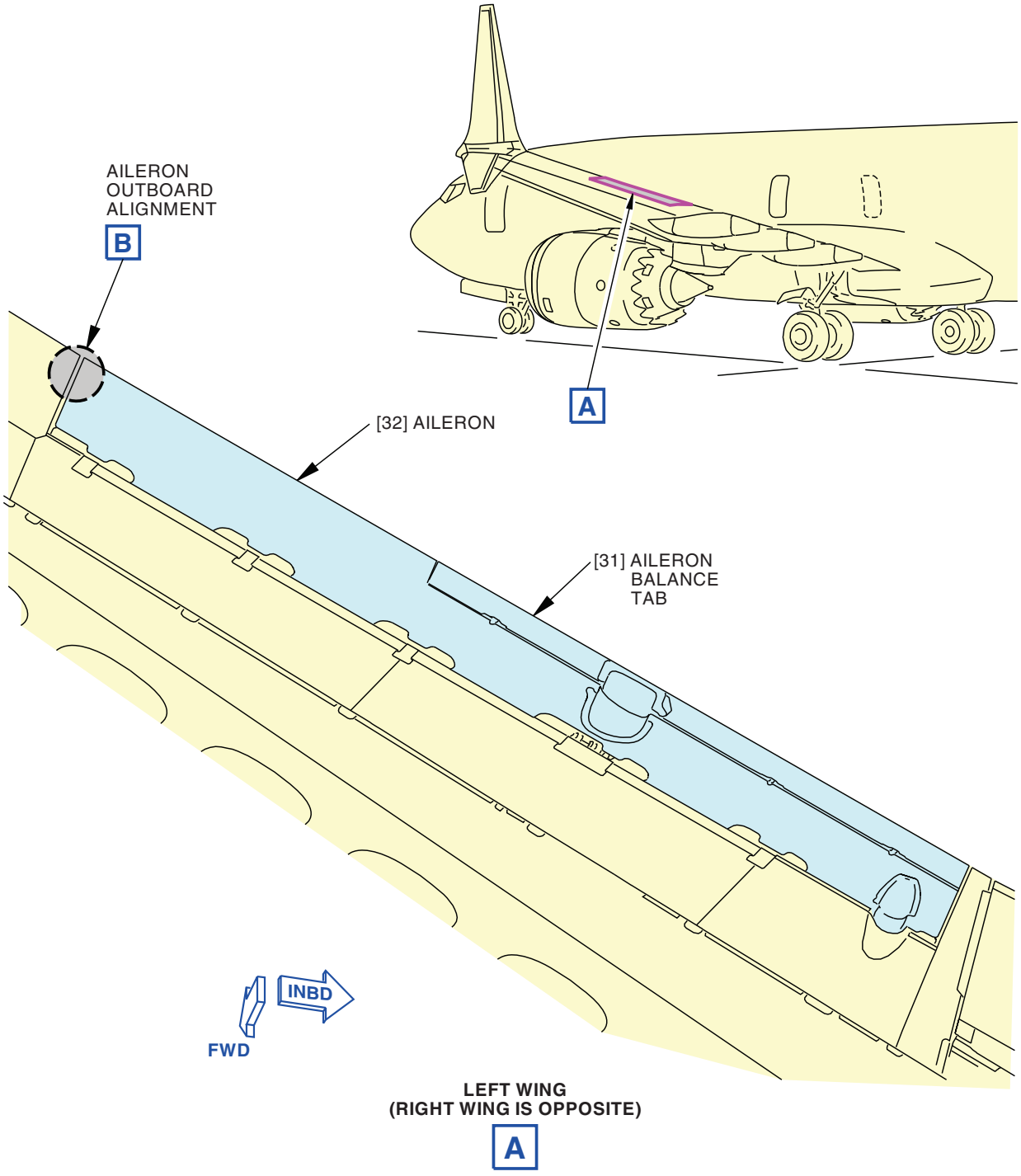
Number	Name/Location
112A	Forward Access Door

SUBTASK 27-11-00-860-028

(2) Remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

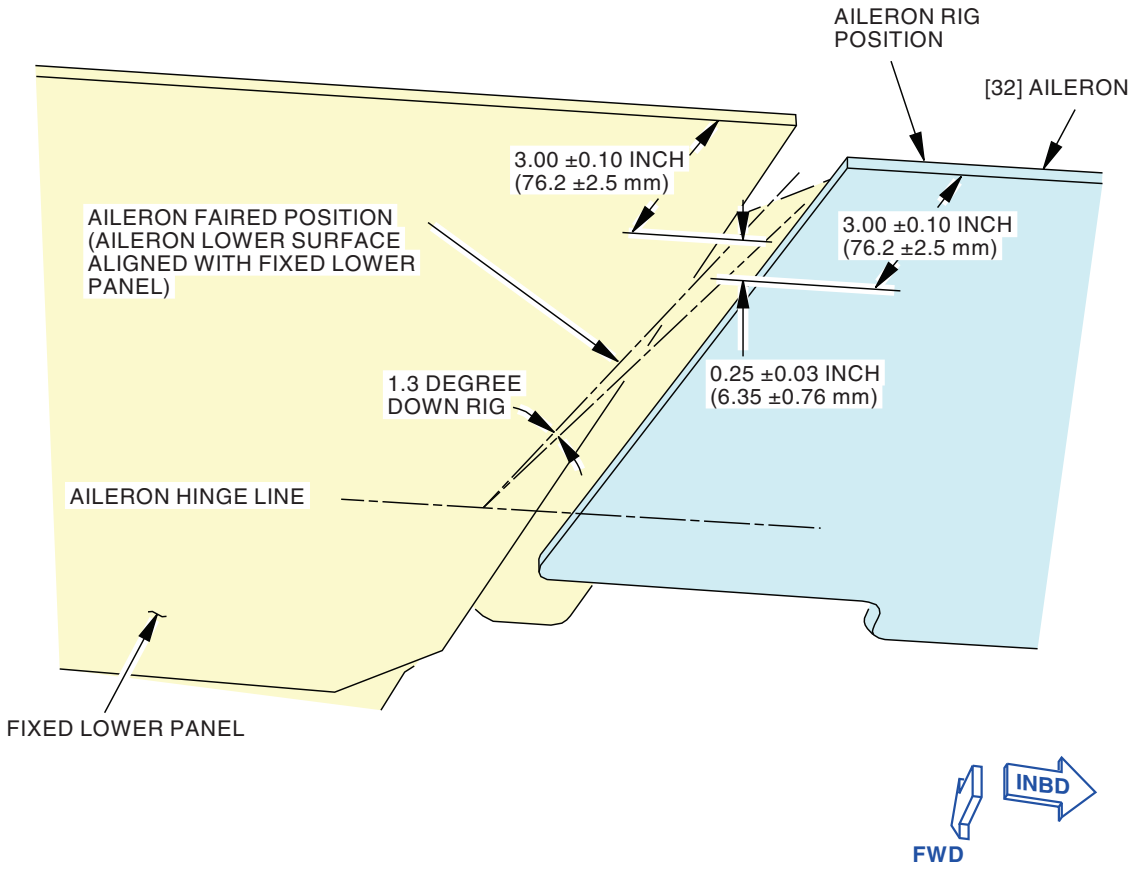
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron System Adjustment
Figure 506/27-11-00-990-809 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON OUTBOARD ALIGNMENT

B

2412269 S00061530325_V4

Aileron System Adjustment
Figure 506/27-11-00-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-700-801

7. Aileron Surface Friction and Motion Test

(Figure 506, Figure 507)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
82	Pin	27-11-21-01-010	PTF ALL

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-11-00-860-029

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-00-020-001

(2) Disconnect the aileron control rod [81] from the aileron [32]. (Figure 507)

(a) Remove the pin [82], nut [83], washer [84], bushing [85], washer [88] and bolt [89] to disconnect the aileron control rod [81] from the aileron [32].

1) Discard the pin [82].

SUBTASK 27-11-00-860-030

(3) Make sure that the Flap Lever is in the up, retracted position.

E. Do a Test of the Aileron Friction and Motion

SUBTASK 27-11-00-720-001

(1) Do a test of the aileron friction and motion:

NOTE: During this test, the force is positive (+) if you need to push the aileron to get it to a specified position. The force is negative (-) if you need to hold the aileron to keep it at a specified position.

(a) Move the aileron surface to the rig position and release (Figure 506).

NOTE: The aileron rig position is measured 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, inboard corner of the fixed lower panel, and 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, outboard corner of the aileron. At this location, the distance between the lower surface of the fixed lower panel and the lower surface of the aileron should be 0.25 ±0.03 in. (6.35 ±0.76 mm).

(b) If the aileron moves away from the rig position, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Hold the aileron at the rig position with a force scale at 8.75 ±0.50 in. (222.3 ±12.8 mm) aft of the hinge centerline and perpendicular to the aileron chord plane (held negative (-) force).
2) Slowly apply or relax the force as it is necessary until the aileron moves up and down from the rig position.
   a) Measure the force that is necessary to move the aileron up (up positive (+) force).
   b) Measure the force that is necessary to move the aileron down (down positive (+) force).
3) Add the up force with the down force.

   NOTE: Add the forces with the same signs. Subtract forces with the opposite signs.
4) Make sure the total of the two values is not more than 5 lbf (22.2 N).

(c) If the aileron does not move away from the rig position, do these steps;
1) Slowly apply a force perpendicular to the aileron chord plane 8.75 ±0.50 in. (222.3 ±12.8 mm) aft of the hinge centerline.
2) Measure the force that is necessary to move the aileron up (up force).
3) Measure the force that is necessary to move the aileron down (down force).
4) Add the up force with the down force.

   NOTE: Add the forces with the same signs. Subtract forces with the opposite signs.
5) Make sure the total of the two forces is not more than 5 lbf (22.2 N).

(d) Measure the maximum positive force necessary to move the aileron to these specified positions:
1) Make sure that the aileron moves smoothly throughout its travel, and that there is no binding at the aileron hinges and seals.
2) Move the aileron from the rig position to the full down position.
3) Move the aileron from the full down position to the full up position.
4) Move the aileron from the full up position to the rig position.

(e) Make sure the force is not more than 13 lbf (57.8 N) for aileron down.

(f) Make sure the force is not more than 3 lbf (13.3 N) aileron up.

(g) Measure the aileron travel before the aileron touches the wing structure at the upper and lower limits of travel.

NOTE: Apply only enough force to maintain contact while you measure the aileron travel.
1) Make sure the aileron travels a minimum of 3.67 in. (93.2 mm) up before the aileron touches the structure.
2) Make sure the aileron travels a minimum of 3.15 in. (80.0 mm) down before the aileron touches the structure.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-020-002

(1) Connect the aileron control rod [81] to the aileron [32].

(a) Install the bushing [85], bolt [89], washer [88], washer [84], nut [83], and new pin [82] to connect the aileron control rod [81] to the aileron.

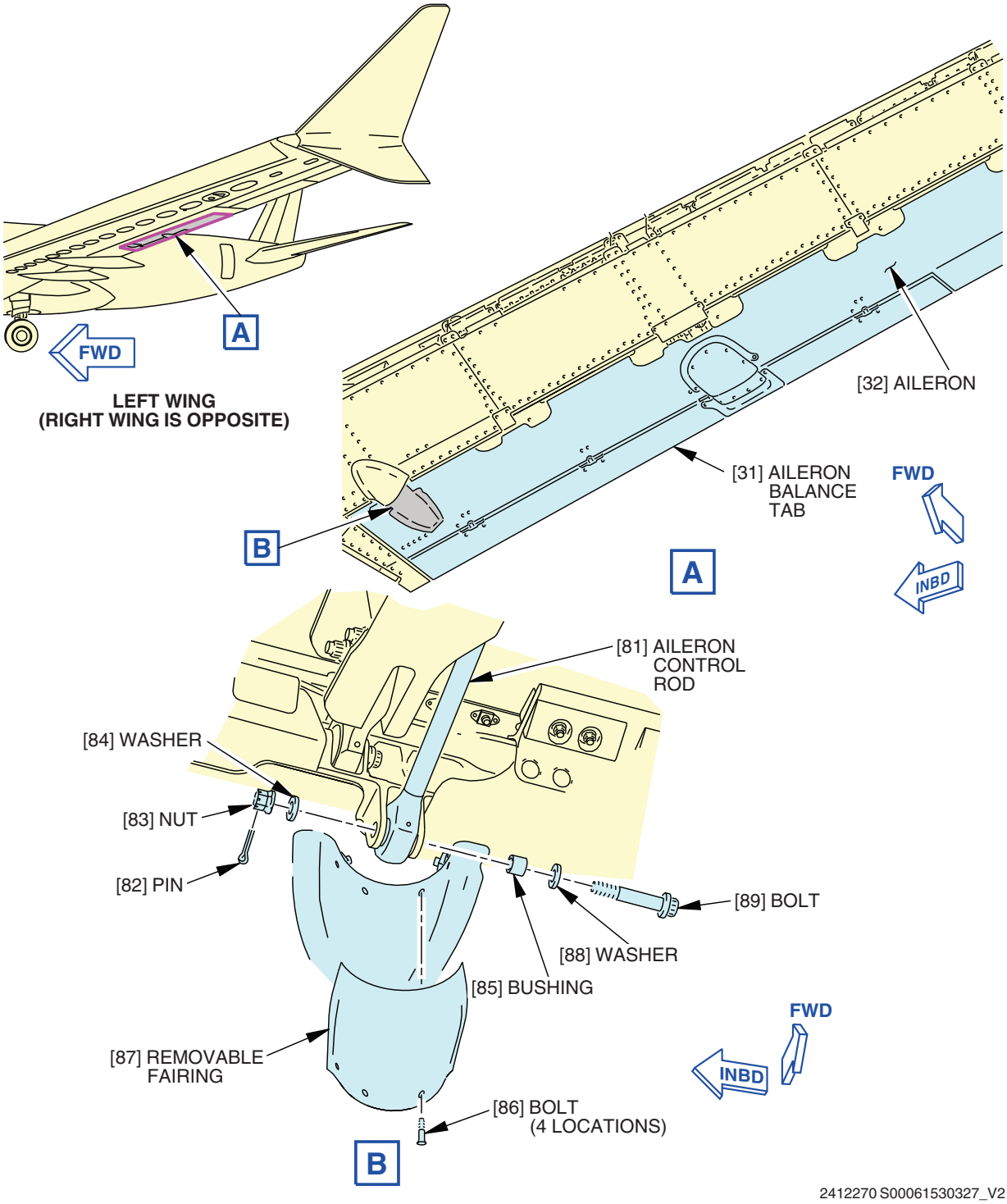
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-860-031

(2) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Surface Friction and Motion Test
Figure 507/27-11-00-990-810

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-700-803

8. Manual Reversion Control Friction Test

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205
STD-1019	Wrench - Torque, 0 to 600 in-lbs (0 to 67.78 N-m), +/-5% accuracy

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

C. Prepare for the Test

SUBTASK 27-11-00-010-010

(1) Remove the map holder assembly from the first officer's control wheel.

SUBTASK 27-11-00-480-015

(2) Install the mount, SPL-1670, and the protractor, SPL-1680, to the first officer's control wheel.

SUBTASK 27-11-00-480-016

(3) Install the adapter, SPL-1674, to the captain's control wheel.

SUBTASK 27-11-00-860-037

(4) Make sure that the aileron trim system is at the center.

SUBTASK 27-11-00-860-038

(5) Set the protractor, SPL-1680, to zero degrees.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Manual Reversion Control Friction Test

SUBTASK 27-11-00-730-001

(1) Do the manual reversion control friction test:

(a) Use the torque wrench, STD-1019, on the adapter, SPL-1674, to measure the force that is necessary to turn the captain's control wheels in the clockwise direction.

NOTE: If the adapter, SPL-1674, is not available, apply a tangent force to the captain's control wheel at 6.5 in. (165 mm) radius.

1) Turn the control wheel in the clockwise direction until the ailerons start to move.
   a) Make sure that the force is 215 in-lb (24.3 N·m) or 33.1 lbf (147.2 N) or less.
2) Continue to turn the control wheel to 20 degrees.
   a) Make sure that the force is 243 in-lb (27.5 N·m) or 37.4 lbf (166.4 N) or less.
3) Continue to turn the control wheel to 40 degrees.
   a) Make sure that the force is 279 in-lb (31.5 N·m) or 42.9 lbf (190.8 N) or less.
4) Continue to turn the control wheel to 75 degrees.
   a) Make sure that the force is 333 in-lb (37.6 N·m) or 51.2 lbf (227.7 N) or less.

(b) Measure the force that is necessary to turn the captain's control wheel in the counterclockwise direction.

NOTE: If the adapter, SPL-1674, is not available, apply a tangent force to the captain's control wheel at 6.5 in. (165 mm) radius.

1) Turn the control wheel in the counterclockwise direction until the ailerons start to move.
   a) Make sure that the force is 215 in-lb (24.3 N·m) or 33.1 lbf (147.2 N) or less.
2) Continue to turn the control wheel to 20 degrees.
   a) Make sure that the force is 243 in-lb (27.5 N·m) or 37.4 lbf (166.4 N) or less.
3) Continue to turn the control wheel to 40 degrees.
   a) Make sure that the force is 279 in-lb (31.5 N·m) or 42.9 lbf (190.8 N) or less.
4) Continue to turn the control wheel to 75 degrees.
   a) Make sure that the force is 333 in-lb (37.6 N·m) or 51.2 lbf (227.7 N) or less.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-080-009

(1) Remove the mount, SPL-1670, and the protractor, SPL-1680, from the first officer's control wheel.

SUBTASK 27-11-00-410-005

(2) Install the map holder assembly to the first officer's control wheel.

SUBTASK 27-11-00-080-010

(3) Remove the adapter, SPL-1674, from the captain's control wheel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-710-801

9. Aileron Response (Operational) Test

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

C. Prepare for the Test

SUBTASK 27-11-00-860-041

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize hydraulic systems A and B (TASK 29-11-00-860-801).

D. Aileron Response (Operational) Test

SUBTASK 27-11-00-710-003

(1) Do the aileron response test:

(a) Turn the aileron control wheel through full travel in each direction from the neutral position.

1) Make sure the ailerons can move smoothly through the full range of travel.

(b) Remove pressure from the hydraulic system A (TASK 29-11-00-860-805).

(c) Turn the aileron control wheel in the clockwise direction from the neutral position.

1) Make sure the trailing edges of the left aileron moves down and the aileron tab moves up.

2) Make sure the trailing edges of the right aileron moves up and the aileron tab moves down.

(d) Turn the aileron control wheel in the counterclockwise direction from the neutral position.

1) Make sure the trailing edges of the left aileron moves up and the aileron tab moves down.

2) Make sure the trailing edges of the right aileron moves down and the aileron tab moves up.

(e) Remove pressure from the hydraulic system B (TASK 29-11-00-860-805).

(f) Pressurize the hydraulic system A (TASK 29-11-00-860-801)..

(g) Turn the aileron control wheel in the clockwise direction from the neutral position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure the trailing edges of the left aileron moves down and the aileron tab moves up.

2) Make sure the trailing edges of the right aileron moves up and the aileron tab moves down.

(h) Turn the aileron control wheel in the counterclockwise direction from the neutral position.

1) Make sure the trailing edges of the left aileron moves up and the aileron tab moves down.

2) Make sure the trailing edges of the right aileron moves down and the aileron tab moves up.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-860-042

(1) Remove pressure from the hydraulic system A (TASK 29-11-00-860-805).

——— END OF TASK ———

TASK 27-11-00-700-804

10. Control Wheel Centering Check

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Check

SUBTASK 27-11-00-860-045

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize hydraulic systems A and B (TASK 29-11-00-860-801).

D. Control Wheel Centering Check

SUBTASK 27-11-00-720-003

(1) Do the centering check:

(a) Turn the captain's control wheel in the clockwise direction from the neutral position (zero units) to 6 units of trim (36 degrees).

(b) Turn the captain's control wheel back to 2 units of trim and release the control wheel.

(c) Make sure the control wheel goes back to the neutral position in 0.0 ±0.4 units of trim (±2.4 degrees).

(d) Turn the captain's control wheel in the counterclockwise direction from the neutral position (zero units) to 6 units of trim (-36 degrees).

(e) Turn the captain's control wheel back to 2 units of trim and release the control wheel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Make sure the control wheel goes back to the neutral position in 0.0 ±0.4 units of trim (±2.4 degrees).

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-860-046

(1) Remove power from hydraulic systems A and B (TASK 29-11-00-860-805).

——— END OF TASK ———

TASK 27-11-00-700-805

11. Aileron Control Wheel Force Test

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-11-00-860-049

(1) Make sure that you can freely install and remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim actuator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-860-050

WARNING	MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THE CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR CAN MOVE SUDDENLY WITH THE PRESSURIZATION OF A HYDRAULIC SYSTEM. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-11-00-020-008

(3) If necessary, remove the map holder assembly from the captain's control wheel.

NOTE: Applicable for P/N F72867-1 of the adapter, SPL-1674.

SUBTASK 27-11-00-480-017

(4) Install the adapter, SPL-1674, to the captain's control wheel.

NOTE: If the adapter is not available, put a tangent force to the captain's control wheel at a 6.5 in. (165 mm) radius.

SUBTASK 27-11-00-010-011

(5) Remove the map holder assembly from the first officer's control wheel.

SUBTASK 27-11-00-480-018

(6) Install the mount, SPL-1670 and the protractor, SPL-1680, to the first officer's control wheel.

E. Do a Test of the Control Wheel Force

SUBTASK 27-11-00-730-002

(1) Do a check of the control wheel force:

(a) Cycle the control wheel CW (Clockwise) and Counter-Clock-Wise (CCW) at least six times and return to center.

(b) Set the indicator on the protractor, SPL-1680, to zero degrees.

(c) Turn the control wheel clockwise until the aileron starts to move.

1) Make sure that the force to turn the control wheel is not more than 29.0 in-lb (3.3 N·m) or 4.5 lbf (20.0 N).

(d) Continue to turn the control wheel clockwise to 5 degrees.

1) Make sure that the force to turn the control wheel is not more than 35.5 in-lb (4.0 N·m) or 5.5 lbf (24.5 N).

(e) Continue to turn the control wheel clockwise to 20 degrees.

1) Make sure that the force to turn the control wheel is not more than 56.0 in-lb (6.3 N·m) or 8.5 lbf (37.8 N).

(f) Continue to turn the control wheel clockwise to 40 degrees.

1) Make sure that the force to turn the control wheel is not more than 89.5 in-lb (10.1 N·m) or 14.0 lbf (62.3 N).

(g) Continue to turn the control wheel clockwise to 75 degrees.

1) Make sure that the force to turn the control wheel is not more than 123.5 in-lb (14.0 N·m) or 19.0 lbf (84.5 N).

(h) Release the control wheel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the control wheel goes back to the neutral position.

(i) Turn the control wheel counterclockwise until the aileron starts to move.

1) Make sure that the force to turn the control wheel is not more than 29.0 in-lb (3.3 N·m) or 4.5 lbf (20.0 N).

(j) Continue to turn the control wheel counterclockwise to 5 degrees.

1) Make sure that the force to turn the control wheel is not more than 35.5 in-lb (4.0 N·m) or 5.5 lbf (24.5 N).

(k) Continue to turn the control wheel counterclockwise to 20 degrees.

1) Make sure that the force to turn the control wheel is not more than 56.0 in-lb (6.3 N·m) or 8.5 lbf (37.8 N).

(l) Continue to turn the control wheel counterclockwise to 40 degrees.

1) Make sure that the force to turn the control wheel is not more than 89.5 in-lb (10.1 N·m) or 14.0 lbf (62.3 N).

(m) Continue to turn the control wheel counterclockwise to 75 degrees.

1) Make sure that the force to turn the control wheel is not more than 123.5 in-lb (14.0 N·m) or 19.0 lbf (84.5 N).

(n) Release the control wheel.

1) Make sure that the control wheel goes back to the neutral position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-080-011

(1) Remove the adapter, SPL-1674, from the captain's aileron control wheel.

SUBTASK 27-11-00-080-012

(2) Remove the mount, SPL-1670 and the protractor, SPL-1680, from the first officer's control wheel.

SUBTASK 27-11-00-410-015

(3) Install the map holder assembly on the captain's control wheel, if it was removed.

SUBTASK 27-11-00-410-007

(4) Install the map holder assembly to the first officer's control wheel.

SUBTASK 27-11-00-080-013

(5) Remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-11-00-730-801

12. Aileron Trim Response Test

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-11-00-860-054

(1) Make sure that you can freely install and remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim actuator.

(2) Remove rig pin A/S-15 from the aileron trim actuator.

SUBTASK 27-11-00-480-019

(3) Install the adapter, SPL-1674, on the captain's control wheel.

NOTE: If the adapter, SPL-1674, is not available, put a tangent force to the captain's control wheel at a 6.5 in. (165.1 mm) radius.

SUBTASK 27-11-00-010-012

(4) Remove the map holder assembly from the first officer's control wheel.

SUBTASK 27-11-00-480-020

(5) Install the mount, SPL-1670, and the protractor, SPL-1680, on the first officer's control wheel.

SUBTASK 27-11-00-860-097

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Pressurize hydraulic systems A and B (TASK 29-11-00-860-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Aileron Trim Response Test

SUBTASK 27-11-00-730-005

(1) Operate the Captain's control wheel through full travel clockwise and counterclockwise a minimum six times.

(a) Turn the captain's control wheel back to the neutral position.

SUBTASK 27-11-00-730-006

(2) Make sure that the pointer decal on the captain's and first officer's control wheels align with the aileron trim decal (zero unit) to a tolerance of 0.00 ±0.02 in. (0.0 ±0.5 mm).

SUBTASK 27-11-00-730-007

(3) At the same time push and hold the aileron trim switches in the LEFT WING DOWN direction until the trim actuator stops.

(a) Make sure that the two control wheels turn in the counterclockwise direction.

SUBTASK 27-11-00-730-008

(4) Record the necessary time for the actuator to move to full LEFT WING DOWN trim.

(a) Make sure the time is less than 30 seconds.

SUBTASK 27-11-00-730-009

(5) Record the captain's wheel position at full LEFT WING DOWN trim.

(a) Make sure the wheel position is at a minimum 9.5 units.

NOTE: When you read the control wheel position, shake the captain's control wheel to make sure that the control system is centered in the trim detent.

SUBTASK 27-11-00-730-017

(6) Manually turn the captain's control wheel to the left stop (105 degrees) counterclockwise.

SUBTASK 27-11-00-730-026

(7) Manually turn and hold the captain's control wheel to the right stop (105 degrees) clockwise.

(a) Make sure that the control wheel moves freely through travel.

SUBTASK 27-11-00-730-027

(8) Release the captain's control wheel.

(a) Make sure that the control wheel goes back to at least 9.5 units counterclockwise position.

NOTE: When you read the control wheel position, shake the captain's control wheel to make sure that the control system is centered in the trim detent.

SUBTASK 27-11-00-730-028

(9) Record the captain's wheel position.

SUBTASK 27-11-00-730-029

(10) At the same time push and hold both aileron trim switches until the captain's and first officer's control wheels align with the aileron trim decal (zero unit) to a tolerance of 0.00 ±0.02 in. (0.0 ±0.5 mm)

(a) Shake the Captain's control wheel to get zero trim units.

SUBTASK 27-11-00-730-030

(11) At the same time push and hold the aileron trim switches in the RIGHT WING DOWN direction until the trim actuator stops.

(a) Make sure that the two control wheels turn in the clockwise direction.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-730-031

(12) Record the necessary time for the actuator to move to full RIGHT WING DOWN trim.

(a) Make sure the time is less than 30 seconds.

SUBTASK 27-11-00-730-032

(13) Record the captain's wheel position at full RIGHT WING DOWN trim.

(a) Make sure the wheel position is at a minimum 9.5 units.

NOTE: When you read the control wheel position, shake the captain's control wheel to make sure that the control system is centered in the trim detent.

SUBTASK 27-11-00-730-033

(14) Manually turn the captain's control wheel to the left stop (105 degrees) clockwise.

SUBTASK 27-11-00-730-034

(15) Manually turn and hold the captain's control wheel to the right stop (105 degrees) counterclockwise.

(a) Make sure that the control wheel moves freely through travel.

SUBTASK 27-11-00-730-035

(16) Release the captain's control wheel.

(a) Make sure that the control wheel goes back to at least 9.5 units clockwise position.

NOTE: When you read the control wheel position, shake the captain's control wheel to make sure that the control system is centered in the trim detent.

SUBTASK 27-11-00-730-036

(17) Record the captain's wheel position.

SUBTASK 27-11-00-730-037

(18) At the same time push and hold the aileron trim switches in the RIGHT WING DOWN direction until the trim actuator stops.

(a) Make sure that the control wheel is at least 9.5 units clockwise position.

NOTE: When you read the control wheel position, shake the captain's control wheel to make sure that the control system is centered in the trim detent.

SUBTASK 27-11-00-730-038

(19) Turn and hold the captain's control wheel at 75 degrees counterclockwise.

SUBTASK 27-11-00-730-039

(20) Record the force to hold the captain's wheel at 75 degrees counterclockwise.

(a) Make sure that the force to hold the control wheel is not more than 137 in-lb (15.5 N·m) or 21.0 lbf (93.4 N).

SUBTASK 27-11-00-730-040

(21) Release the captain's control wheel.

SUBTASK 27-11-00-730-041

(22) At the same time push and hold the aileron trim switches in the LEFT WING DOWN direction until the trim actuator stops.

(a) Make sure that the control wheel is at least 9.5 units counterclockwise position.

NOTE: When you read the control wheel position, shake the captain's control wheel to make sure that the control system is centered in the trim detent.

SUBTASK 27-11-00-730-042

(23) Turn and hold the captain's control wheel at 75 degrees clockwise.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-730-043

(24) Record the force to hold the captain's wheel at 75 degrees clockwise.

(a) Make sure that the force to hold the control wheel is not more than 137 in-lb (15.5 N·m) or 21.0 lbf (93.4 N).

SUBTASK 27-11-00-730-044

(25) Release the captain's control wheel.

SUBTASK 27-11-00-730-045

(26) At the same time push and hold both aileron trim switches in the RIGHT WING DOWN direction until the captain's and first officer's control wheels align with the aileron trim decal (zero unit) to a tolerance of 0.00 ±0.02 in. (0.0 ±0.5 mm)

(a) Shake the Captain's control wheel to get zero trim units.

SUBTASK 27-11-00-730-046

(27) At the same time push and hold the FORWARD aileron trim switch in the LEFT WING DOWN direction for a minimum of 3 seconds.

(28) At the same time push and hold the FORWARD aileron trim switch in the RIGHT WING DOWN direction for a minimum of 3 seconds.

(a) Make sure that the control wheels do not move.

SUBTASK 27-11-00-730-047

(29) At the same time push and hold the AFT aileron trim switch in the LEFT WING DOWN direction for a minimum of 3 seconds.

(30) At the same time push and hold the AFT aileron trim switch in the RIGHT WING DOWN direction for a minimum of 3 seconds.

(a) Make sure that the control wheels do not move.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-860-055

(1) Remove power from hydraulic systems A and B (TASK 29-11-00-860-805).

SUBTASK 27-11-00-080-014

(2) Remove the adapter, SPL-1674, from the captain's aileron control wheel.

SUBTASK 27-11-00-080-015

(3) Remove the mount, SPL-1670, and the protractor, SPL-1680, from the first officer's control wheel.

SUBTASK 27-11-00-410-008

(4) Install the map holder assembly to the first officer's control wheel.

——— END OF TASK ———

TASK 27-11-00-710-802

13. Transfer Mechanism Operational Test

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 27-11-00-860-058

> **WARNING** MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THE CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR CAN MOVE SUDDENLY WITH THE PRESSURIZATION OF A HYDRAULIC SYSTEM. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-11-00-210-002

(2) Make sure the FLT CONTROL A and B switches, on the forward overhead panel, P5, are ON.

SUBTASK 27-11-00-860-059

(3) Make sure that the aileron trim control system is at the neutral position.

D. Transfer Mechanism Operational Test

SUBTASK 27-11-00-710-004

(1) Do the operational test of the transfer mechanism:

(a) Operate the two aileron trim switches to put the aileron control wheels to the neutral (zero unit) position.

(b) Shake the control wheel to center the aileron system.

(c) Turn the captain's control wheel fully clockwise and hold.

(d) Turn the first officer's control wheel counterclockwise through neutral and beyond 10 units left aileron, and return the control wheel to full clockwise.

1) After the initial breakout, the force required to continue wheel motion should increase steadily and smoothly.

Verify that there is no binding or sudden increase in the force required to turn the wheel.

(e) Turn the captain's control wheel fully counterclockwise and hold.

(f) Turn the first officer's control wheel clockwise through neutral and beyond 10 units right aileron, and return the control wheel to full counterclockwise.

1) After the initial breakout, the force required to continue wheel motion should increase steadily and smoothly.

Verify that there is no binding or sudden increase in the force required to turn the wheel.

(g) Return the captain's control wheel to neutral.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-860-060

(1) Remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-11-00-700-806

14. Transfer Mechanism Functional Test

(Figure 508)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Test

SUBTASK 27-11-00-860-063

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-860-098

(2) Make sure that the FLT CONTROL A and B switches, on the P5 forward overhead panel, are in the ON position.

SUBTASK 27-11-00-860-064

(3) Make sure that you can freely install and remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim actuator.

SUBTASK 27-11-00-010-013

(4) Remove the map holder assembly from the captain's control wheel:

(a) Remove the three screws that hold the map holder assembly to the control wheel.

NOTE: The three screws will be used to install the torque adapter in the next subtask.

(b) Remove the map holder assembly.

SUBTASK 27-11-00-480-021

(5) Install the torque adapter, SPL-1674, on the captain's control wheel.

NOTE: If the adapter, SPL-1674, is not available, apply a tangent force to the captain's control wheel at 6.5 in. (165 mm) radius.

(a) Use the three screws from the map holder assembly to install the torque adapter.

SUBTASK 27-11-00-860-100

(6) Do one of the steps that follow:

(a) Use the MAX Display System (MDS) to view the captain and first officer's control column positions during this test.

1) Follow the steps on the MDS:

a) MAINT

b) MAINT DATA PGS

c) 27 FLIGHT CONTROL - DISPLAY

d) CAPT CTRL WHL and SCE FO CTRL WHL.

(b) Install the protractor, SPL-1680, to the captain's control wheel.

E. Transfer Mechanism Functional Test

SUBTASK 27-11-00-720-004

(1) Do the functional test of the transfer mechanism:

(a) Operate the two aileron trim switches, to put the aileron control wheels to the neutral (zero unit) position.

(b) Shake the control wheel to center the aileron system.

(c) Hold the first officer's control wheel at the neutral position.

(d) Turn the captain's control wheel clockwise from the neutral position.

1) Make sure that the breakout force is 240 in-lb (27 N·m) - 436 in-lb (49 N·m) or 37 lbf (165 N) - 67 lbf (298 N).

NOTE: The “breakout” force occurs when the captain's control wheel starts to move with the first officer's control wheel hold in the neutral position.

2) Make sure that the force at 75 degrees is 330 in-lb (37 N·m) - 676 in-lb (76 N·m) or 51 lbf (227 N) - 104 lbf (463 N).

(e) Turn the captain's control wheel counterclockwise from the neutral position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the breakout force is 240 in-lb (27 N·m) - 436 in-lb (49 N·m) or 37 lbf (165 N) - 67 lbf (298 N).

NOTE: The “breakout” force occurs when the captain's control wheel starts to move with the first officer's control wheel hold in the neutral position.

2) Make sure that the force at 75 degrees is 330 in-lb (37 N·m) - 676 in-lb (76 N·m) or 51 lbf (227 N) - 104 lbf (463 N).

(f) Turn and hold the first officer's control wheel clockwise at 80 degrees from the neutral position.

(g) Turn the captain's control wheel counterclockwise, until the transfer mechanism stops touch.

1) Make sure that the captain's control wheel turns a minimum of 138 degrees from the starting position (80 degrees clockwise).

2) Make sure that the transfer mechanism can move freely, when you turn the control wheel.

(h) Turn and hold the first officer's control wheel counterclockwise at 80 degrees from the neutral position.

(i) Turn the captain's control wheel clockwise, until the transfer mechanism stops touch.

1) Make sure that the captain's control wheel turns a minimum of 138 degrees from the starting position (80 degrees counterclockwise).

2) Make sure that the transfer mechanism can move freely when you turn the control wheel.

(j) Turn and hold the first officer's control wheel clockwise at 80 degrees from the neutral position.

(k) Turn the captain's control wheel counterclockwise 130 degrees from the starting position (80 degrees clockwise).

1) Make sure that the maximum force to turn the captain's control wheel is 900 in-lb (102 N·m) or 138 lbf (614 N).

(l) Turn and hold the first officer's control wheel counterclockwise at 80 degrees from the neutral position.

(m) Turn the captain's control wheel clockwise 130 degrees from the starting position (80 degrees counterclockwise).

1) Make sure that the maximum force to turn the captain's control wheel is 900 in-lb (102 N·m) or 138 lbf (614 N).

(n) Return both control wheels to neutral.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-860-065

(1) Remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-11-00-080-016

(2) If applicable, remove the protractor, SPL-1680, from the captain's control wheel.

SUBTASK 27-11-00-000-001

(3) Remove the torque adapter, SPL-1674, from the captain's control wheel:

(a) Remove the three screws that hold the torque adapter to the control wheel.

(b) Remove the torque adapter, SPL-1674.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-410-009

(4) Install the map holder assembly to the captain's control wheel:

(a) Put the map holder assembly in place on the captain's control wheel.

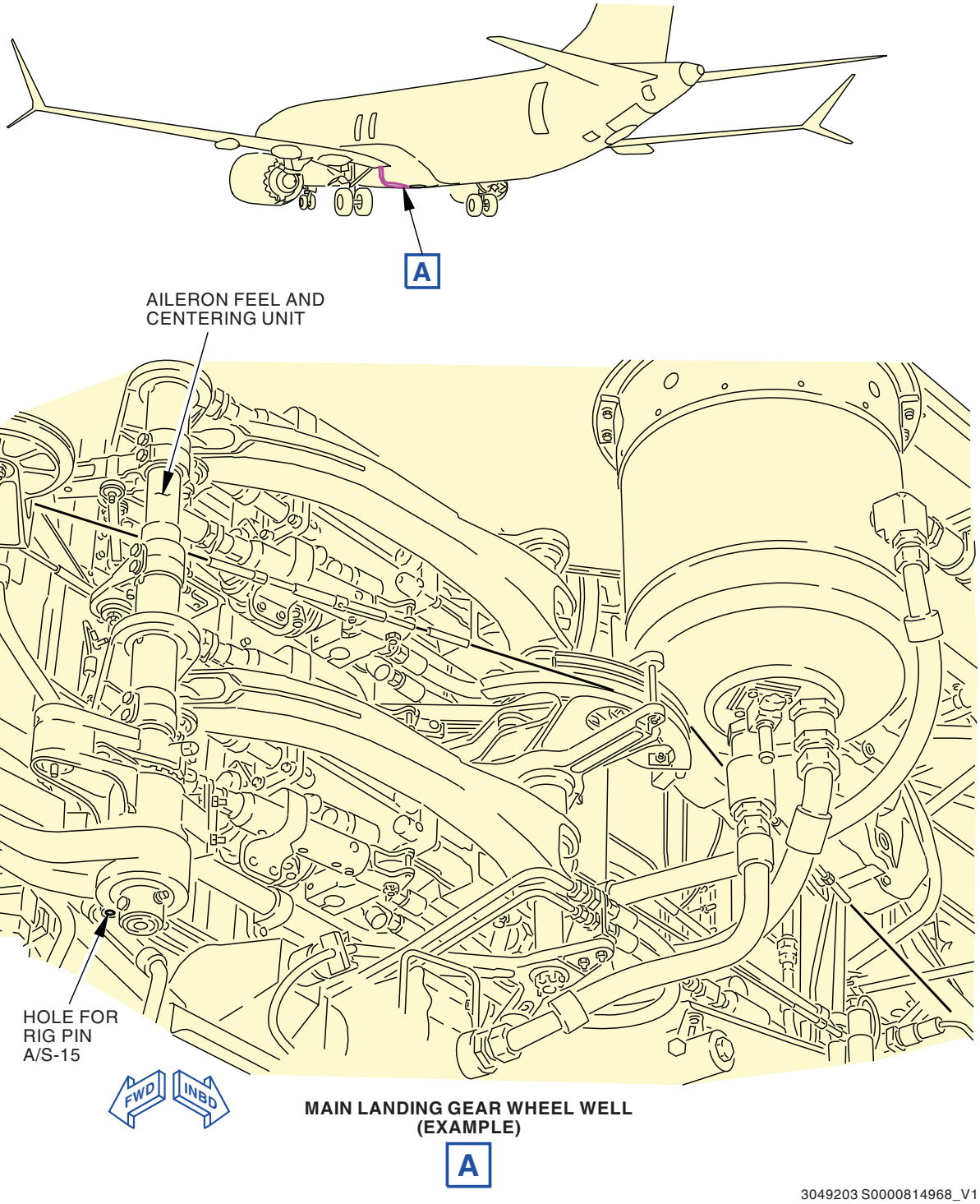
(b) Install the three screws that hold the map holder assembly.

SUBTASK 27-11-00-860-099

(5) Make sure that the FLT CONTROL A and B switches, on the P5 forward overhead panel, are in the ON position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rig Pin Location
Figure 508/27-11-00-990-814

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-700-807

15. Control Wheel Travel Stop Test

NOTE: This procedure is a scheduled maintenance task.

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Do a Test of the Control Wheel Travel Stop

SUBTASK 27-11-00-020-005

(1) Remove the map holder assembly from the first officer's control wheel.

SUBTASK 27-11-00-480-022

(2) Install the mount, SPL-1670 and the protractor, SPL-1680, on the first officer's control wheel.

(a) Set the indicator on the protractor, SPL-1680, to zero degree.

SUBTASK 27-11-00-720-005

(3) Do the test of the control wheel travel stop:

(a) Turn the captain's control wheel clockwise until the control wheel stops.

1) Make sure the first officer's control wheel turns a minimum of 105 degrees.

NOTE: A minimum of 105 degrees control wheel travel is required for the captain's control wheel to touch the forward stop. This large deflection can be necessary for certain flight conditions in manual reversion.

The system reaches the aft quadrant stop at approximately 90 degrees. Additional force will be necessary to stretch the control cable past 90 degrees until the control wheel stops.

(b) Turn the captain's control wheel counterclockwise until the control wheel stops.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure the first officer's control wheel moves freely and turns a minimum of 105 degrees.

NOTE: Turn the control wheel 105 degrees clockwise to make sure the minimum travel requirements of the control wheel are met and that there are no obstructions in the control column and control wheel installations. A minimum of 105 degrees control wheel travel is required for the captain's control wheel to touch the forward stop. This large deflection can be necessary for certain flight conditions in manual reversion. The system reaches the aft stops at approximately 90 degrees. Apply additional force to turn the control wheel more than 90-95 degrees. The control cables stretch slightly when the control wheels are turned beyond 90 degrees.

SUBTASK 27-11-00-080-017

(4) Remove the protractor, SPL-1680 and the mount, SPL-1670, from the first officer's control wheel.

SUBTASK 27-11-00-420-003

(5) Install the map holder assembly to the first officer's control wheel.

——— END OF TASK ———

TASK 27-11-00-700-808

16. Control Wheel Travel Interference Test

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Control Wheel Travel Interference Test

SUBTASK 27-11-00-700-002

(1) Do the test of the control wheel travel interference:

(a) Hold the control column full forward and do the following steps:

1) Turn the control wheel to the full left (counterclockwise) position.

a) Make sure that you make a slow and deliberate movement of the control wheel.

NOTE: If you move the control wheel too quickly, the binding will not be felt.

2) Make sure that the control wheel travel is smooth through the full turn.

3) Turn the control wheel to the center position.

4) Turn the control wheel to the full right (clockwise) position.

a) Make sure that you make a slow and deliberate movement of the control wheel.

NOTE: If you move the control wheel too quickly, the binding will not be felt.

5) Make sure that the control wheel travel is smooth through the full turn.

6) Turn the control wheel to the center position.

(b) Hold the control column full aft and do the following steps:

1) Turn the control wheel to the full left (counterclockwise) position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Make sure that you make a slow and deliberate movement of the control wheel.

NOTE: If you move the control wheel too quickly, the binding will not be felt.

2) Make sure that the control wheel travel is smooth through the full turn.

3) Turn the control wheel to the center position.

4) Turn the control wheel to the full right (clockwise) position.

a) Make sure that you make a slow and deliberate movement of the control wheel.

NOTE: If you move the control wheel too quickly, the binding will not be felt.

5) Make sure that the control wheel travel is smooth through the full turn.

6) Turn the control wheel to the center position.

7) Return the control column to the neutral position.

——— END OF TASK ———

TASK 27-11-00-700-809

17. Aileron Balance Tab Position Test

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

B. Aileron Balance Tab Position Test

SUBTASK 27-11-00-720-006

(1) Do a test of the position of the aileron LEFT balance tab:

NOTE: For all aileron tab measurements in this test, measure the distances from the LOWER surface of the aileron tab at the tip of the trailing edge to the LOWER surface of the aileron at the tip of the trailing edge.

(a) Set the aileron at the rig position (Figure 506).

NOTE: The aileron rig position is measured 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, inboard corner of the fixed lower panel, and 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, outboard corner of the aileron. At this location, the distance between the lower surface of the fixed lower panel and the lower surface of the aileron should be 0.25 ±0.03 in. (6.35 ±0.76 mm).

(b) Measure the vertical distance from the tip of the outboard trailing edge of the aileron tab to the tip of the adjacent aileron trailing edge.

1) Make sure the distance is 0.078 ±0.070 in. (2.0 ±1.8 mm).

(c) Manually move the aileron DOWN so that the tip of the outboard lower trailing edge of the aileron is 3.39 ±0.10 in. (86.1 ±2.5 mm) below the tip of the lower trailing edge of the adjacent fixed trailing edge surface.

(d) Measure the distance from the tip of the lower surface of the aileron tab to the tip of the lower surface of the adjacent aileron trailing edge.

1) Make sure the aileron balance tab moves UP 0.88 ±0.10 in. (22.4 ±2.6 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Manually move the aileron UP so that the tip of the outboard lower trailing edge of the aileron is 4.59 ±0.10 in. (116.6 ±2.6 mm) above the tip of the lower trailing edge of the adjacent fixed trailing edge surface.

(f) Measure the distance from the tip of the lower surface of the aileron tab to the tip of the lower surface of the adjacent aileron trailing edge.

1) Make sure the aileron balance tab moves DOWN 1.14 ±0.10 in. (29.0 ±2.6 mm).

SUBTASK 27-11-00-720-007

(2) Do a test of the position of the aileron RIGHT balance tab:

NOTE: For all aileron tab measurements in this test, measure the distances from the LOWER surface of the aileron tab at the tip of the trailing edge to the LOWER surface of the aileron at the tip of the trailing edge.

(a) Set the aileron at the rig position (Figure 506).

NOTE: The aileron rig position is measured 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, inboard corner of the fixed lower panel, and 3.0 ±0.1 in. (76.2 ±2.5 mm) forward of the aft, outboard corner of the aileron. At this location, the distance between the lower surface of the fixed lower panel and the lower surface of the aileron should be 0.25 ±0.03 in. (6.35 ±0.76 mm).

(b) Measure the vertical distance from the tip of the outboard trailing edge of the aileron tab to the tip of the adjacent aileron trailing edge.

1) Make sure the distance is 0.078 ±0.070 in. (2.0 ±1.8 mm).

(c) Manually move the aileron DOWN so that the tip of the outboard lower trailing edge of the aileron is 3.39 ±0.10 in. (86.1 ±2.5 mm) below the tip of the lower trailing edge of the adjacent fixed trailing edge surface.

(d) Measure the distance from the tip of the lower surface of the aileron tab to the tip of the lower surface of the adjacent aileron trailing edge.

1) Make sure the aileron balance tab moves UP 0.88 ±0.10 in. (22.4 ±2.5 mm).

(e) Manually move the aileron UP so that the tip of the outboard lower trailing edge of the aileron is 4.59 ±0.10 in. (116.6 ±2.6 mm) above the tip of the lower trailing edge of the adjacent fixed trailing edge surface.

(f) Measure the distance from the tip of the lower surface of the aileron tab to the tip of the lower surface of the adjacent aileron trailing edge.

1) Make sure the aileron balance tab moves DOWN 1.14 ±0.10 in. (29.0 ±2.5 mm).

——— END OF TASK ———

TASK 27-11-00-700-810

18. Aileron Travel Test

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

C. Prepare for the Test

SUBTASK 27-11-00-860-074

WARNING MAKE SURE THAT PERSONNEL AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THE CONTROL SURFACES, THE THRUST REVERSERS AND THE LANDING GEAR CAN MOVE SUDDENLY WITH THE PRESSURIZATION OF A HYDRAULIC SYSTEM. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-11-00-010-014

(2) Remove the map holder assembly from the captain's aileron control wheel.

SUBTASK 27-11-00-480-023

(3) Install the mount, SPL-1670 and the protractor, SPL-1680 to the captain's aileron control wheel.

D. Aileron Travel Test

SUBTASK 27-11-00-730-004

(1) Do a test of the aileron travel:

(a) Operate the captain's aileron control wheel through at least 6 full cycles then return the wheel to neutral position.

(b) Turn the captain's aileron control wheel more than 87 degrees counterclockwise from the neutral position and hold.

(c) Measure and record the distance between the outboard lower trailing edge of the aileron and the lower trailing edge of the adjacent fixed trailing structure for both the left and right ailerons.

1) Make sure that the left aileron moves up 4.83 ±0.20 in. (122.7 ±5.1 mm).

2) Make sure that the right aileron moves down 3.63 ±0.20 in. (92.2 ±5.1 mm).

3) Make sure that the aileron balance tab moves in the opposite direction to the aileron.

(d) Turn the captain's aileron control wheel more than 87 degrees clockwise from the neutral position and hold.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Measure and record the distance between the outboard lower trailing edge of the aileron and the lower trailing edge of the adjacent fixed trailing structure for both the left and right ailerons.

1) Make sure that the right aileron moves up 4.83 ±0.20 in. (122.7 ±5.1 mm).

2) Make sure that the left aileron moves down 3.63 ±0.20 in. (92.2 ±5.1 mm).

3) Make sure that the aileron balance tab moves in the opposite direction to the aileron.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-080-018

(1) Remove the mount, SPL-1670 and the protractor, SPL-1680 from the captain's aileron control wheel.

SUBTASK 27-11-00-410-010

(2) Install the map holder assembly to the captain's aileron control wheel.

SUBTASK 27-11-00-860-075

(3) To remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 27-11-00-720-801

19. Aileron Power Control Unit (PCU) Pogos - Functional Test

(Figure 501 and Figure 509)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this task to do a check of the force that is necessary to extend each pogo of the aileron Power Control Unit (PCU).

(2) This task is applicable for the system A and B PCUs.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1749	Lock Equipment - Aileron/Elevator PCU Input Rod (POGO) Part #: C27066-1 Supplier: 81205

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Aileron Power Control Unit (PCU) Pogos Functional Test

SUBTASK 27-11-00-480-026

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 27-11-00-020-003

(2) Make sure that you can freely install and remove rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim actuator (Figure 501).

SUBTASK 27-11-00-860-077

(3) Remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-11-00-280-001

(4) Do a functional test of the system A (lower) pogo.

CAUTION MAKE SURE THAT THE TOOL IS INSTALLED CORRECTLY. IF YOU DO NOT OBEY, DAMAGE TO THE LOCKWIRE CAN OCCUR

(a) Install a lock equipment, SPL-1749, on the system A (lower) aileron PCU piston (Figure 509).

(b) Turn the Captain's control wheel clockwise from the neutral position until system A (lower) pogos extend.

NOTE: Approximately 100 lbf (445 N) force is necessary to turn the control wheel.

(c) Make sure that the pogos extend approximately 0.20 in. (5.1 mm) minimum.

NOTE: Approximately 7 units of wheel required.

(d) Return the Captain's control wheel to the neutral position.

(e) Make sure that all four pogos retract to the original position.

SUBTASK 27-11-00-280-002

(5) Do a functional test of the system B (upper) pogo.

(a) Remove a lock equipment, SPL-1749, from the system A (lower) aileron PCU piston.

CAUTION MAKE SURE THAT THE TOOL IS INSTALLED CORRECTLY. IF YOU DO NOT OBEY, DAMAGE TO THE LOCKWIRE CAN OCCUR

(b) Install the lock equipment, SPL-1749, on the system B (upper) aileron PCU piston (Figure 509).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Turn the Captain's control wheel clockwise from the neutral position until system B (upper) pogos extend.

NOTE: Approximately 100 lbf (445 N) force is necessary to turn the control wheel.

(d) Make sure that the pogos extend approximately 0.20 in. (5.1 mm) minimum.

NOTE: Approximately 7 units of wheel required.

(e) Return the Captain's control wheel to the neutral position.

(f) Make sure that the pogos retract to the original position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-020-006

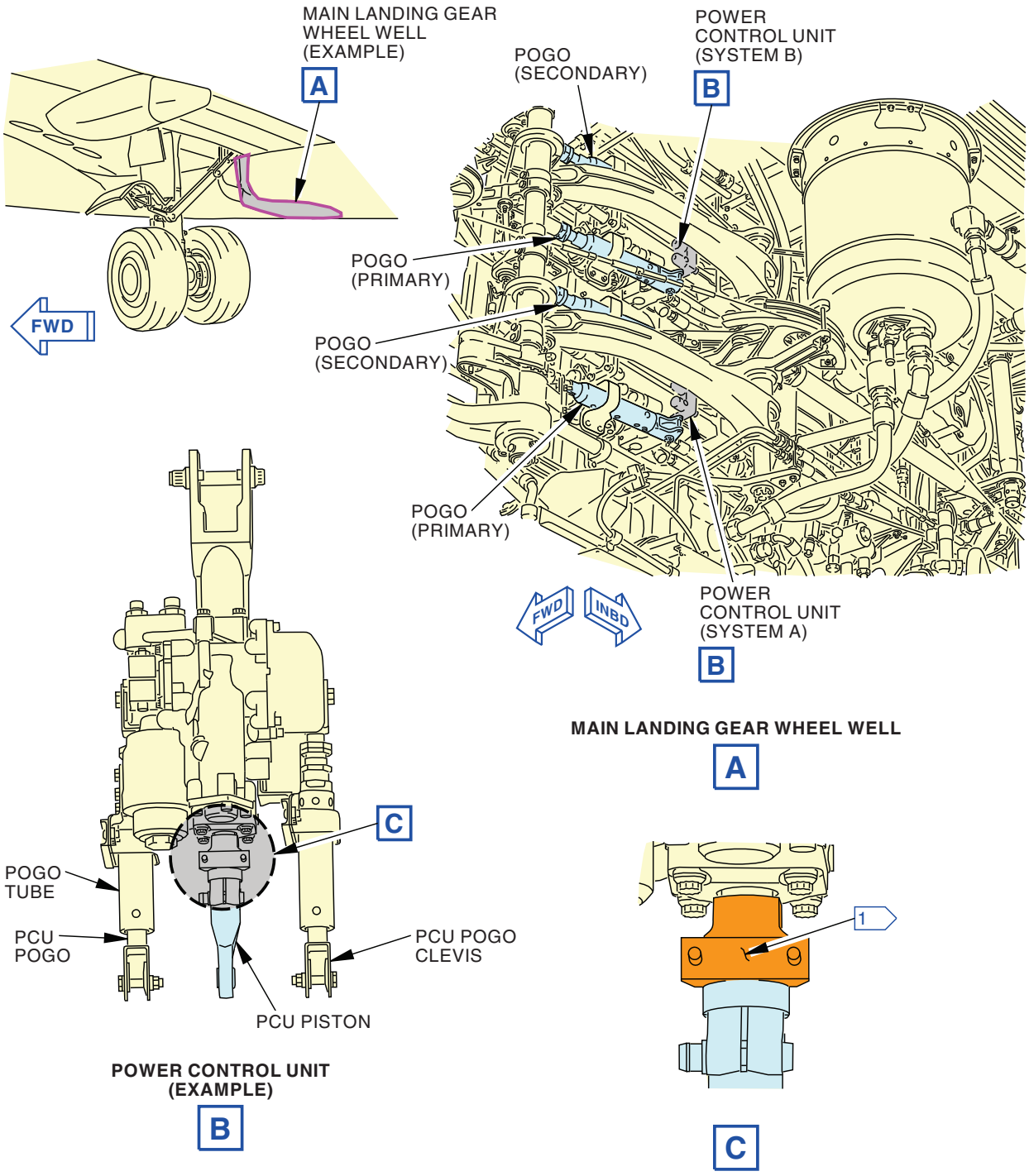
(1) Remove a lock equipment, SPL-1749, from the system B (upper) aileron PCU piston.

SUBTASK 27-11-00-020-007

(2) Do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Power Control Unit (PCU) Pogos Functional Test
Figure 509/27-11-00-990-813

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-750-801

20. Flight Test to Find and Correct Asymmetrical Flight Control Forces

A. General

(1) An aileron system flight test to find and correct asymmetrical flight control forces is necessary after any of the following maintenance procedures:

(a) For the aileron:

1) Installation of a replacement aileron.

NOTE: Replacement of a new aileron, or one that is not the same as the one that was originally on the airplane, requires an aileron system flight test.

NOTE: A flight test is not required if the same ailerons that were removed from the airplane are re-installed without adjustment, modification, or repair.

2) Installation of an aileron that was repaired, repainted, or reworked and was subsequently rebalanced, either by on-jig or calculation method, and required adjustment of the aileron balance adjust weights (SRM 51-60).

NOTE: Any aileron which has been repaired, repainted, or reworked must meet the aerodynamic smoothness requirements of SRM 51-10-01.

3) Removal or replacement of the aileron front spar hinge bearing nearest to the tab control rods.

4) Replacement of any of the following components of the hinge bearing support fitting within the aileron front spar:

a) Installation of a replacement hinge at any of the aileron positions.

b) Installation of a replacement bearing support fitting at any of the aileron positions.

c) Installation of a replacement bearing in the aileron front support hinge at the inboard position, nearest the tab control rods.

NOTE: Installation of new aileron hinge bearings (except the one nearest the tab control rods) does not make a flight test necessary if the original bearing support fittings are installed in their original locations.

5) Installation of a replacement aileron rear spar tab hinge bearing support fitting.

NOTE: Installation of new tab hinge bearings does not make a flight test necessary if the original bearing support fittings are installed in their original locations.

(b) For the aileron tab:

1) Installation of a replacement aileron balance tab.

NOTE: Replacement of a new aileron tab, or one that is not the same as the one that was originally on the airplane, requires an aileron system flight test.

NOTE: A flight test is not required if the same aileron tab that was removed is re-installed without adjustment, modification, or repair.

2) Installation of a replacement front spar hinge of the aileron balance tab.

3) Installation of replacement bushings on the aileron balance tab front spar hinge nearest the aileron tab control rods.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) After an adjustment or replacement of an aileron tab control rod.

NOTE: For each aileron balance tab, if both control rods are adjusted, then flight test is necessary.

NOTE: If only one control rod is adjusted, and you can verify that the remaining rod had not been replaced or adjusted since the last flight test, then flight test is not necessary.

NOTE: If only one control rod is adjusted, and you can not verify that the remaining rod had not been replaced or adjusted since the last flight test, then flight test is necessary.

5) Installation of an aileron tab that was repaired, repainted, or reworked and required adjustment of the aileron tab adjust weights (SRM 51-60).

NOTE: Any aileron tab which has been repaired, repainted, or reworked must meet the aerodynamic smoothness requirements of SRM 51-10-01.

B. References

Reference	Title
27-09-71-820-802	Aileron Balance Tab Adjustment to Equalize Lateral Manual Reversion Forces (P/B 501)
27-09-71-993-801	Table: Lateral Trim Limits (P/B 501)
27-21-00-820-805	Control Rod for Standby Rudder Power Control Unit Adjustment (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1557	Gauge - Force Part #: DG-200 Supplier: 92456 Part #: FDIX 100 Supplier: 0BFD9 Part #: FDIX 50 Supplier: 0BFD9 Part #: FG-60KBM Supplier: 45KA8 Part #: LG-050 Supplier: 92456 Part #: LG-100 Supplier: 92456 Opt Part #: DPP-500G Supplier: 92456 Opt Part #: DPPH-150 Supplier: 92456 Opt Part #: DPPH-200 Supplier: 92456 Opt Part #: DPPH-50 Supplier: 92456 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Test

SUBTASK 27-11-00-820-021

(1) Adjust the applicable system or systems before the flight test if you get these conditions (Table 27-09-71-993-801):

(a) With the gear up and flaps up, the aileron trim is more than 3/4 unit at .78 mach and more than 37,000 feet, 3/4 unit at 250 knots and 10,000 to 17,000 feet.

(b) With the gear up, the aileron trim is more than 1 unit at flaps 1, maneuver speed and 10,000 to 17,000 feet.

(c) With the landing gear extended, the aileron trim is more than 1 unit at flaps 15, maneuver speed and 10,000 to 17,000 feet.

(d) With the landing gear extended, the aileron trim is more than 1.3 units at Vref+5, and 10,000 to 17,000 feet.

(e) The rudder trim is more than 1/2 unit left or 1 unit right at .78 mach and more than 37,000 feet, 3/4 units at 250 knots and 10,000 to 17,000 feet, or 1 unit at any configuration other than gear up and flaps up.

SUBTASK 27-11-00-210-005

(2) Make sure that the airspeed is 250 knots.

SUBTASK 27-11-00-860-078

(3) Make sure that the flight altitude is 10,000 to 17,000 feet.

SUBTASK 27-11-00-860-079

(4) Make sure that the flaps are up.

SUBTASK 27-11-00-860-080

(5) Make sure that all persons are seated.

SUBTASK 27-11-00-860-081

(6) Make sure that the aileron trim is not more than 3/4 unit.

SUBTASK 27-11-00-860-082

(7) Make sure that the rudder trim is not more than 3/4 unit.

F. Flight Test

SUBTASK 27-11-00-860-083

(1) Move the control wheel to the neutral position.

SUBTASK 27-11-00-860-084

(2) Set the SPOILER A and B switches to OFF.

SUBTASK 27-11-00-860-085

(3) Make sure that the FLT CONTROL A and FLT CONTROL B switches are ON.

SUBTASK 27-11-00-860-086

(4) Set the FLT CONTROL A switch to OFF.

NOTE: The airplane may pitch up or down during transition.

SUBTASK 27-11-00-860-087

(5) Carefully move the elevators, ailerons, and rudder.

(a) Make sure that the control forces and operation do not change.

SUBTASK 27-11-00-860-088

(6) Set the FLT CONTROL A switch to ON.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-860-089

(7) Set the FLT CONTROL B switch to OFF.

NOTE: The airplane may pitch up or down during transition.

SUBTASK 27-11-00-860-090

(8) Carefully move the elevators, ailerons, and rudder.

(a) Make sure that the control forces and operation do not change.

SUBTASK 27-11-00-860-091

(9) Set the FLT CONTROL B switch to STBY RUD.

SUBTASK 27-11-00-860-092

(10) Set the FLT CONTROL A switch to OFF.

NOTE: The airplane may pitch up or down during transition.

SUBTASK 27-11-00-280-003

(11) When hydraulics are switched off for the test, apply pitch and roll control input as required to maintain steady level flight.

SUBTASK 27-11-00-750-002

(12) Do these steps to find and correct asymmetrical flight control forces:

(a) Trim the airplane with the rudder to keep wings level flight.

(b) Make sure that the rudder trim does not change by more than 2 units.

1) If the change is more than 2 units, do this task: Control Rod for Standby Rudder Power Control Unit Adjustment, TASK 27-21-00-820-805.

2) If the standby rudder is in adjustment, do this task: Aileron Balance Tab Adjustment (Recommended Method), TASK 27-11-00-820-803.

(c) Make a note if there is a roll tendency.

(d) Turn the control wheel clockwise 30 degrees.

(e) Turn the control wheel counterclockwise 30 degrees.

(f) Make sure that the control wheel forces are symmetrical.

1) If the control wheel forces are not symmetrical by a ratio of 2 to 1 or more, then use the tool to measure the control wheel forces.

a) Use a force gauge, COM-1557, to push down on the right handgrip knob of the control wheel, until the wheel is turned 30 degrees clockwise.

b) Measure the control wheel force.

c) Use a force gauge, COM-1557, to push down on the left handgrip knob of the control wheel, until the wheel is turned 30 degrees counterclockwise.

d) Measure the control wheel force.

e) If the control wheel force in one direction is 2 or more times the force in the other direction, do this task: Aileron Balance Tab Adjustment to Equalize Lateral Manual Reversion Forces, TASK 27-09-71-820-802.

f) If the control wheel force in one direction is less than 2 times the force in the other direction, no action is necessary.

SUBTASK 27-11-00-860-093

(13) Set the FLT CONTROL A switch to STBY RUD.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-00-860-094

(14) Set the FLT CONTROL B switch to OFF.

SUBTASK 27-11-00-750-003

(15) Do all the steps to find and correct asymmetrical flight control forces again.

SUBTASK 27-11-00-860-095

(16) Set the SPOILER A and B, FLT CONTROL A and B switches back to ON.

SUBTASK 27-11-00-860-096

(17) Set the Yaw Damper switch, on the forward overhead panel, to the ON position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON AND AILERON TRIM CONTROL SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has these inspection tasks:

(1) Aileron Mechanical Components from the Control Wheel to the Aileron PCU's.

(2) Aileron Mechanical Components from the Aileron PCU to the Aileron and the Flight Spoiler Mechanical Control Path.

TASK 27-11-00-210-801

2. Aileron Mechanical Components Inspection - Forward components

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Aileron Mechanical Components Inspection

SUBTASK 27-11-00-010-007

(1) To get access to the aileron mechanical components, open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-11-00-210-006

(2) Do a general visual inspection of the forward aileron mechanical components to include the following:

(a) Control wheels

(b) Aileron control drum

(c) Bus drums.

D. Put the Airplane Back to Its Usual Condition

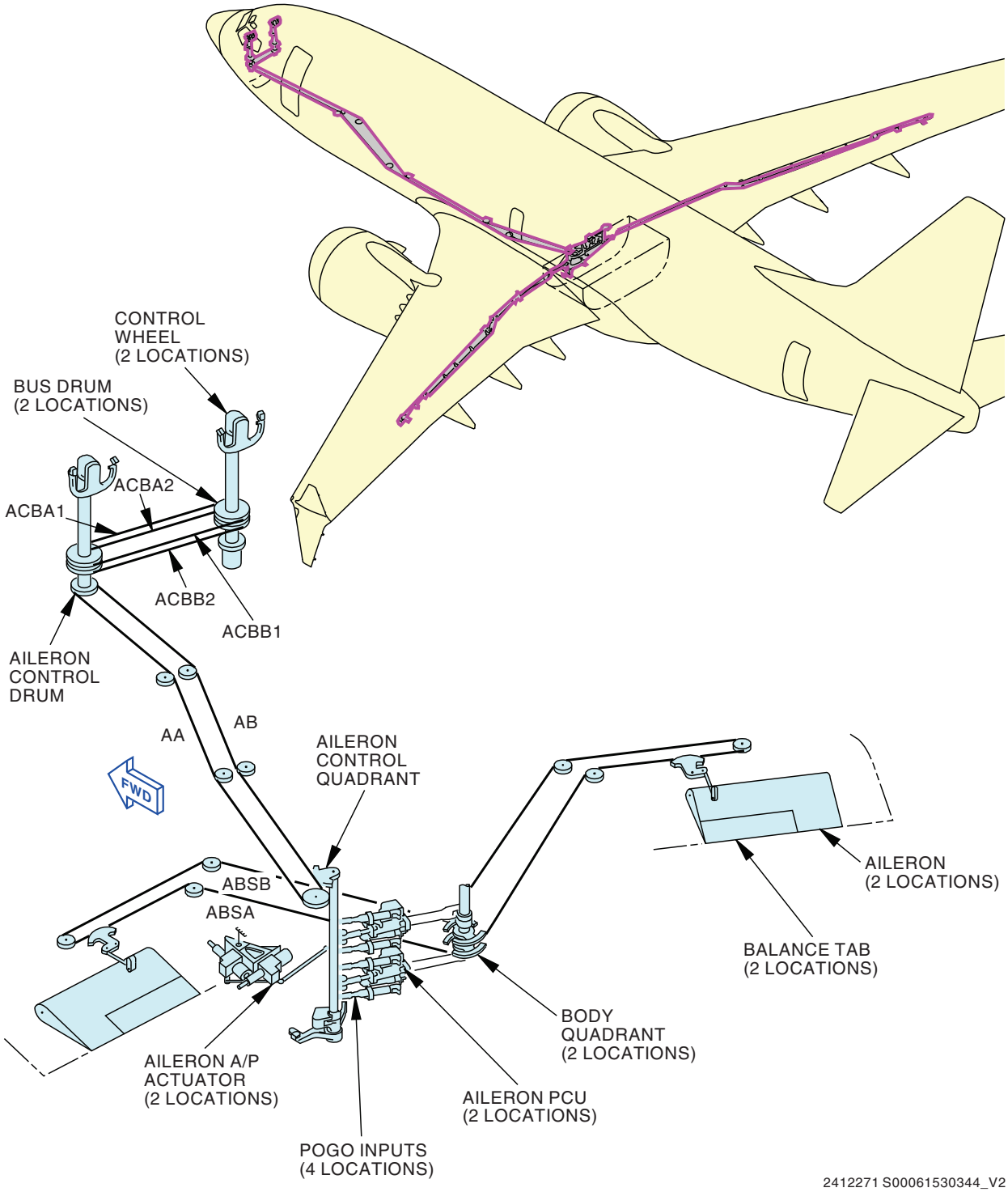
SUBTASK 27-11-00-410-004

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

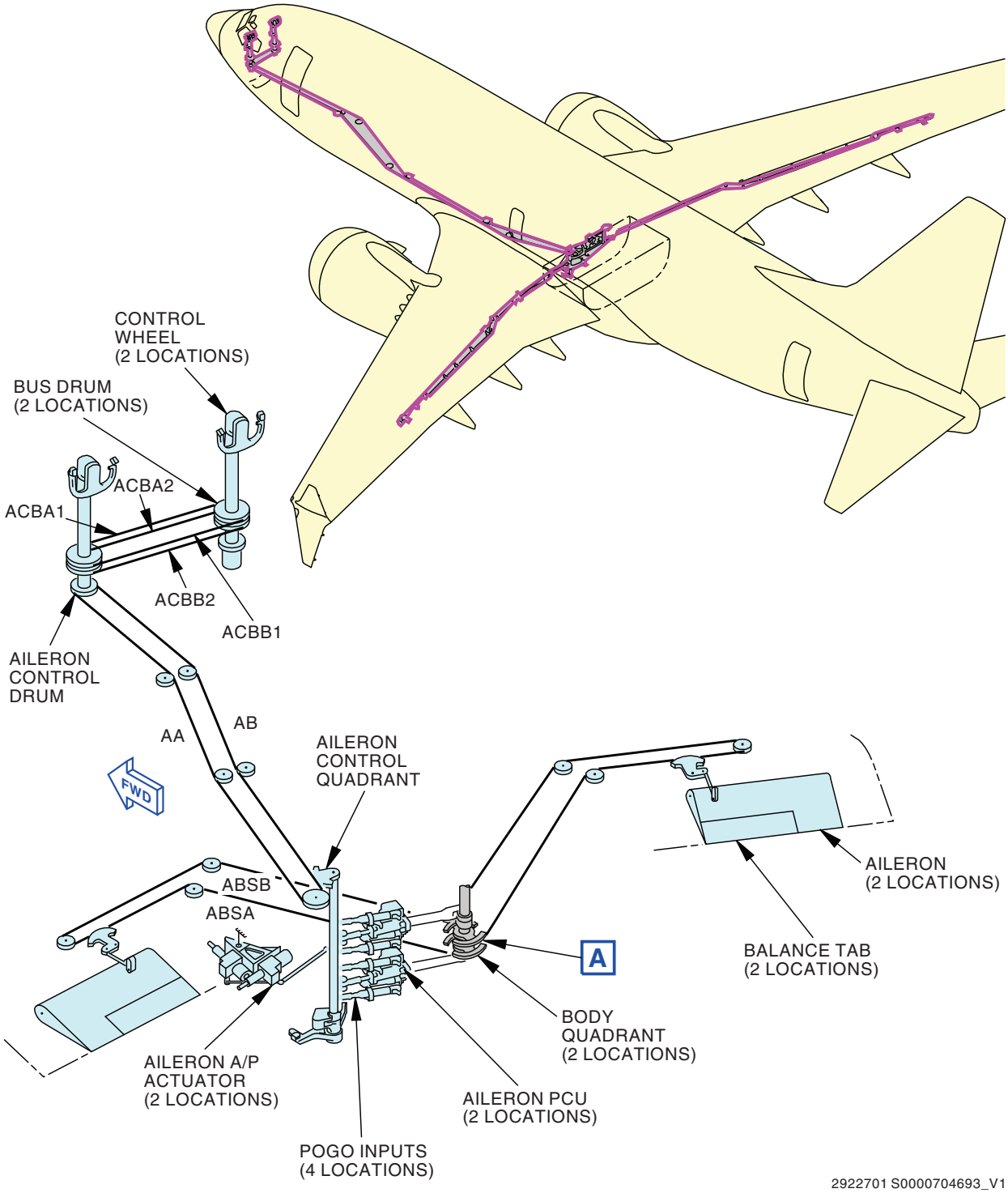
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



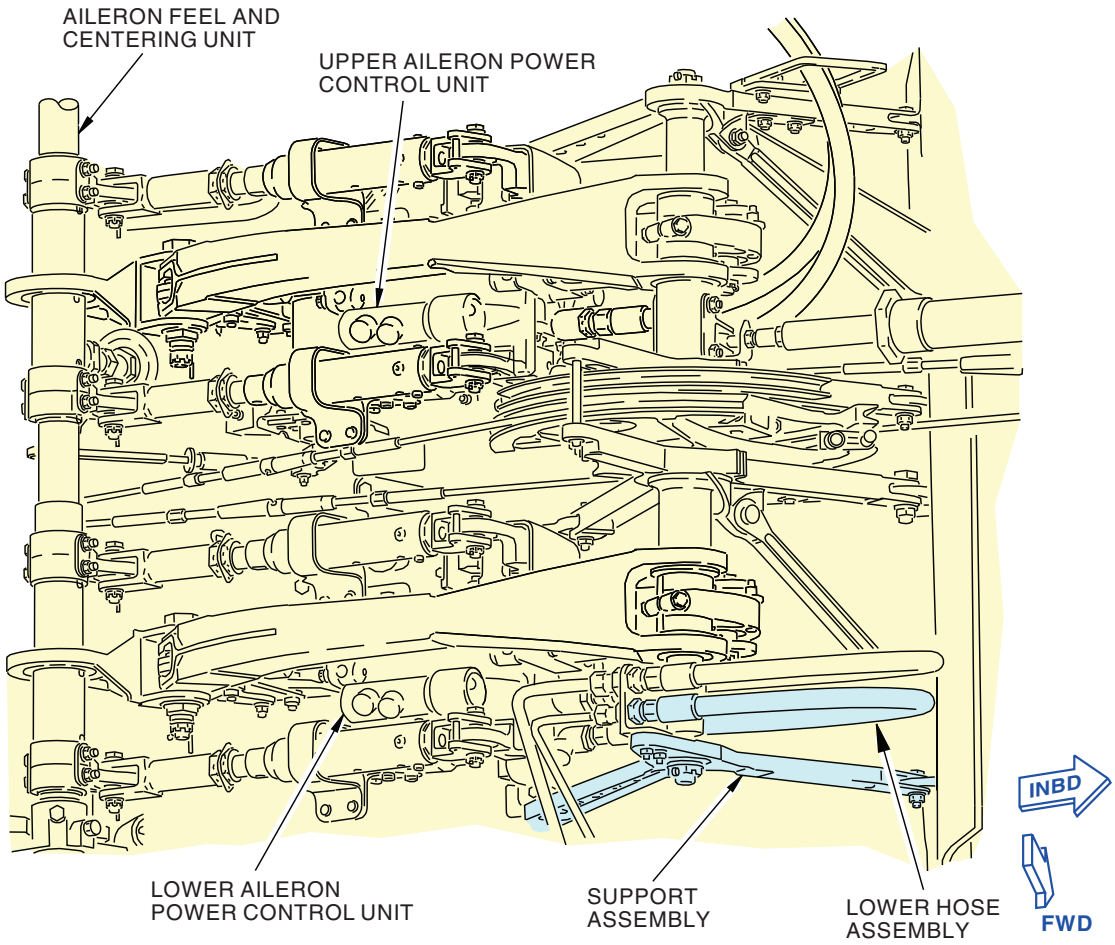
Aileron and Aileron Trim Control System
Figure 601/27-11-00-990-811 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron and Aileron Trim Control System
Figure 601/27-11-00-990-811 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aileron and Aileron Trim Control System
Figure 601/27-11-00-990-811 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-00-210-802

3. Aileron Mechanical Components Inspection - From the Aileron PCUs to the Aileron

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
57-61-02-000-802	Aileron Lower Hinge Covers Removal (P/B 401)
57-61-02-400-802	Aileron Lower Hinge Covers Installation (P/B 401)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

C. Access Panels

Number	Name/Location
572BB	Lower Aileron, Actuator Rod Fairing - WBL 427.00
672BB	Lower Aileron, Actuator Rod Fairing - WBL 427.00

D. Aileron Mechanical Components Inspection

SUBTASK 27-11-00-860-101

(1) Extend the Trailing Edge (TE) flaps (TASK 27-51-00-860-803).

SUBTASK 27-11-00-010-008

(2) For the left wing, remove this access panel:

(TASK 57-61-02-000-802)

Number	Name/Location
572BB	Lower Aileron, Actuator Rod Fairing - WBL 427.00

SUBTASK 27-11-00-010-017

(3) For the right wing, remove this access panel:

(TASK 57-61-02-000-802)

Number	Name/Location
672BB	Lower Aileron, Actuator Rod Fairing - WBL 427.00

SUBTASK 27-11-00-210-007

(4) Do a general visual inspection of all aileron mechanical components from the aileron Power Control Unit (PCU)s to the aileron including the following:

(a) Aileron PCUs

(b) Pogo inputs.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-29-1129

SUBTASK 27-11-00-210-009

(5) Examine the support assembly and lower hose assembly:

(a) Look for any chafing damage on the support assembly from the lower hose assembly.

1) If damage is found, the allowable limits are listed below:

a) Location - Any location along the support assembly

b) Depth - less than 0.125 in. (3.175 mm)

c) Length - 0.80 in. (20.32 mm) maximum

d) Width - equal to the width of the flange on the support assembly

e) If the support assembly is out of tolerance, repair or replace the part.

(b) Look for any chafing to the lower hose assembly where contact with the support assembly can occur.

1) Replace the lower hose assembly if chafing is found.

PTF ALL

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-00-410-013

(1) For the left wing, install this access panel:

(TASK 57-61-02-400-802)

Number	Name/Location
572BB	Lower Aileron, Actuator Rod Fairing - WBL 427.00

SUBTASK 27-11-00-410-014

(2) For the right wing, install this access panel:

(TASK 57-61-02-400-802)

Number	Name/Location
672BB	Lower Aileron, Actuator Rod Fairing - WBL 427.00

SUBTASK 27-11-00-860-102

(3) Retract the TE flaps (TASK 27-51-00-860-804).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron

(2) An installation of the aileron.

B. The aileron has the aileron assembly, aileron balance tab, and four aileron balance panels.

TASK 27-11-11-000-801

2. Aileron Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-21-000-801	Aileron Balance Tab Removal (P/B 401)
27-11-34-000-801	Aileron Tab Control Rods Removal (P/B 401)
57-50-02-000-803	Outboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
57-61-02-000-802	Aileron Lower Hinge Covers Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2042	Sling - Aileron, Removal/Installation Part #: C27055-1 Supplier: 81205

C. Location Zones

Zone	Area
571	Left Wing - Fixed Trailing Edge
572	Left Wing - Aileron
671	Right Wing - Fixed Trailing Edge
672	Right Wing - Aileron

D. Access Panels

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel
572AB	Lower Aileron, Hinge Cover - WBL 423.00
572CB	Lower Aileron, Hinge Cover - WBL 447.00
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00
572FB	Lower Aileron, Hinge Cover - WBL 502.00
572GB	Lower Aileron, Hinge Cover - WBL 528.00
572HB	Lower Aileron, Hinge Cover - WBL 553.00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel
672AB	Lower Aileron, Hinge Cover - WBL 423.00
672CB	Lower Aileron, Hinge Cover - WBL 447.00
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00
672FB	Lower Aileron, Hinge Cover - WBL 502.00
672GB	Lower Aileron, Hinge Cover - WBL 528.00
672HB	Lower Aileron, Hinge Cover - WBL 553.00

E. Prepare for the Removal

SUBTASK 27-11-11-860-005

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-11-860-003

(2) Remove these access panels found on the lower panel of the outboard fixed Trailing Edge (TE):

(a) On the left wing, remove these access panels:

(TASK 57-50-02-000-803)

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(b) On the right wing, remove these access panels:

(TASK 57-50-02-000-803)

Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-11-860-004

(3) Remove these aileron access panels found on the lower wing surface adjacent to the Leading Edge (LE) of the aileron:

(a) On the left wing, remove these access panels:

(TASK 57-61-02-000-802)

Number	Name/Location
572AB	Lower Aileron, Hinge Cover - WBL 423.00
572CB	Lower Aileron, Hinge Cover - WBL 447.00
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00
572FB	Lower Aileron, Hinge Cover - WBL 502.00
572GB	Lower Aileron, Hinge Cover - WBL 528.00
572HB	Lower Aileron, Hinge Cover - WBL 553.00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the right wing, remove these access panels:

(TASK 57-61-02-000-802)

Number	Name/Location
672AB	Lower Aileron, Hinge Cover - WBL 423.00
672CB	Lower Aileron, Hinge Cover - WBL 447.00
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00
672FB	Lower Aileron, Hinge Cover - WBL 502.00
672GB	Lower Aileron, Hinge Cover - WBL 528.00
672HB	Lower Aileron, Hinge Cover - WBL 553.00

F. Aileron Removal

SUBTASK 27-11-11-020-008

(1) Remove the aileron [1] with the sling, SPL-2042 (Figure 401).

(a) Install the sling, SPL-2042, on the aileron [1].

1) At hinge 2, do these steps (View C, Figure 402):

a) Remove the bolts [35], washers [14], radial seal retainer [37], and radial seal [39].

b) Remove the bolts [29], washers [14], radial seal retainer [33], and radial seal [31].

c) Remove and discard the pin [17].

d) Remove and keep the nut [18], washer [19], bushing [20], washer [21], and bolt [22].

2) At hinge 5, do these steps (View E, Figure 402):

a) If installed, remove the bolts [61], washers [14], and radial seal retainer [62].

b) Remove the bolts [52], washers [14], radial seal retainer [53], and radial seal [54].

c) Remove and discard the pin [55].

d) Remove and keep the nut [56], washer [57], bushing [58], washer [59], and bolt [60].

3) Attach the outboard C-bracket to the outboard clevis with the ball lock pin.

4) Attach the inboard C-bracket to the inboard clevis with the ball lock pin.

5) Put the sling in position around the TE of the aileron [1].

6) Attach the inboard clevis to hinge 5 with the attached ball lock pin.

7) Attach the outboard clevis to hinge 2 with the attached ball lock pin.

8) Lift the sling with a cable and hook to hold the aileron [1].

SUBTASK 27-11-11-020-009

(2) Remove the aileron [1] with the sling, SPL-2042, attached.

(a) At hinge 6, do these steps (View F, Figure 402):

1) Remove the bolts [73] that attach the removable fairing [74] to the aileron [1].

2) Remove the removable fairing [74].

3) Remove and discard the pin [77].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Remove and keep the nut [78], washer [76], bushing [75], washer [72], and bolt [71] to disconnect the aileron control rod [67] from the aileron [1].

5) Remove the nut [82] and washer [83] to disconnect the bonding jumper [84] from the structure.

(b) At hinge 1, do these steps (View B, Figure 402):

1) Remove the bolts [23], washers [14], radial seal retainer [24], and radial seal [25].

2) If installed, remove the bolts [15], washers [14], and radial seal retainer [16].

3) Remove the nut [11] and washer [12] to disconnect the bonding jumper [13] from the structure.

(c) At hinge 3, do these steps (View C, Figure 402):

1) Remove the bolts [36], washers [14], radial seal retainer [38], and radial seal [40].

2) Remove the bolts [30], washers [14], radial seal retainer [34], and radial seal [32].

(d) At hinge 4, do these steps (View D, Figure 402):

1) Remove the bolts [86], washers [14], radial seal retainer [85], and radial seal [87].

2) Remove the bolts [43], washers [14], radial seal retainer [44], and radial seal [42].

3) Remove and discard the pins [45].

4) Remove and keep the nuts [46], washers [47], bushings [48], washers [49], and bolts [50] to disconnect the tab control rods [41] from the forward fittings.

(e) Remove the hinge seals [7] from the balance panels [5] (View G, Figure 402).

1) Remove the bolts [4], bolts [6], bolts [27], bolts [8] with collars [9], bolts [88], washers [89], nuts [90], and balance weights [26].

2) Remove the bolts [3] and balance weights [28].

3) Remove the hinge seals [7] from the balance panels [5].

4) Remove and discard the pins [63].

5) Remove and keep the nuts [64], washers [65] (under the nut), and bolts [66] to remove the aileron [1] from the aileron balance panels [5].

NOTE: The washer under the bolt is bonded to the hinge half.

(f) At hinge 6, do these steps:

1) Remove and discard the pin [81].

2) Remove and keep the nut [80], washer [79], bushing [68], washer [69], and bolt [70].

(g) At hinges 1, 3 and 4, do these steps:

1) Remove and discard the pin [17].

2) Remove and keep the nut [18], washer [19], bushing [20], washer [21], and bolt [22].

(h) Remove the aileron [1] from the structure.

SUBTASK 27-11-11-020-010

(3) Remove the aileron [1] without the sling, SPL-2042 (Figure 402).

(a) Disconnect the aileron control rod [67] and bonding jumper [84].

1) At hinge 6, do these steps:

a) Remove the bolts [73] that attach the removable fairing [74] to the aileron [1].

b) Remove the removable fairing [74].

c) Remove and discard the pin [77].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) Remove and keep the nut [78], washer [76], bushing [75], washer [72], and bolt [71] to disconnect the aileron control rod [67] from the aileron [1].

e) Remove the nut [82] and washer [83] to disconnect the bonding jumper [84] from the structure.

(b) Remove the radial seals and radial seal retainers to get access to the aileron hinge and control rods connections.

1) At hinge 1, do these steps:

a) Remove the bolts [23], washers [14], radial seal retainer [24], and radial seal [25].

b) If installed, remove the bolts [15], washers [14], radial seal retainer [16].

c) Remove the nut [11] and washer [12] to disconnect the bonding jumper [13] from the structure.

2) At hinge 2, do these steps:

a) Remove the bolts [35], washers [14], radial seal retainer [37], and radial seal [39].

b) Remove the bolts [29], washers [14], radial seal retainer [33], and radial seal [31].

3) At hinge 3, do these steps:

a) Remove the bolts [36], washers [14], radial seal retainer [38], and radial seal [40].

b) Remove the bolts [30], washers [14], radial seal retainer [34], and radial seal [32].

4) At hinge 4, do these steps:

a) Remove the bolts [86], washers [14], radial seal retainer [85], and radial seal [87].

b) Remove the bolts [43], washers [14], radial seal retainer [44], and radial seal [42].

c) Remove and discard the pins [45].

d) Remove and keep the nuts [46], washers [47], bushings [48], washers [49], and bolts [50] to disconnect the tab control rods [41] from the forward fittings.

5) At hinge 5, do these steps:

a) If installed, remove the bolts [61], washers [14], and radial seal retainer [62].

b) Remove the bolts [52], washers [14], radial seal retainer [53], and radial seal [54].

(c) Remove the hinge seals [7] from the balance panels [5].

1) Remove the bolts [4], bolts [6], bolts [27], bolts [8] with collars [9], bolts [88], washers [89], nuts [90], and balance weights [26].

2) Remove the bolts [3] and balance weights [28].

3) Remove the hinge seals [7] from the balance panels [5].

4) Remove and discard the pins [63].

5) Remove and keep the nuts [64], washers [65] (under the nut), and bolts [66] to remove the aileron [1] from the aileron balance panels [5].

NOTE: The washer under the bolt is bonded to the hinge half.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Hold the aileron [1] while you remove aileron [1] from the structure.

1) At hinges 1, 2, 3, and 4, remove and discard the pin [17].

a) Remove and keep the nut [18], washer [19], bushing [20], washer [21], and bolt [22].

2) At hinge 5, remove and discard the pin [55].

a) Remove and keep the nut [56], washer [57], bushing [58], washer [59], and bolt [60].

3) At hinge 6, remove and discard the pin [81].

a) Remove and keep the nut [80], washer [79], bushing [68], washer [69], and bolt [70].

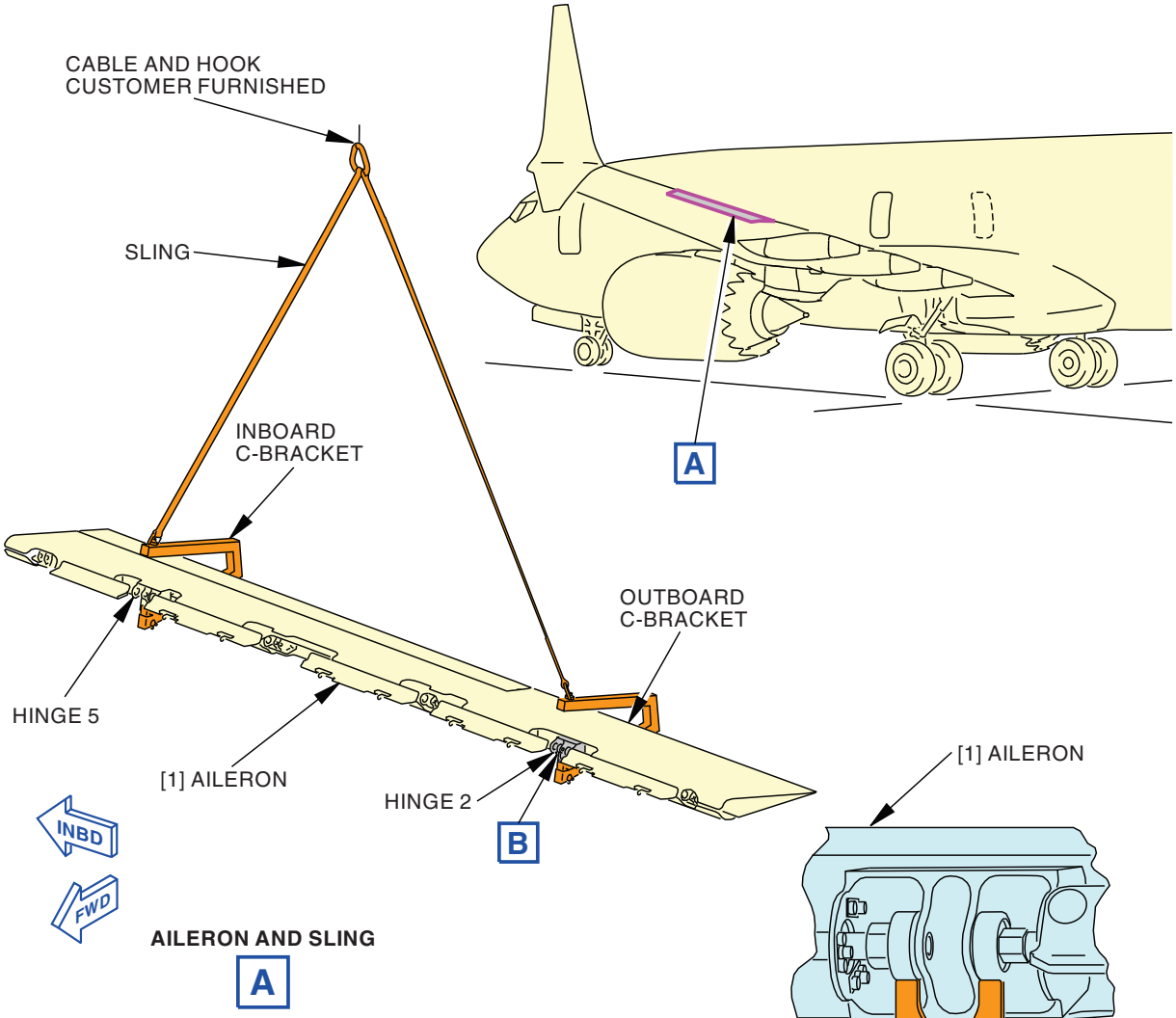
(e) Remove the aileron [1] from the structure.

SUBTASK 27-11-11-020-011

(4) If it is necessary to remove the aileron balance tab [2], do this task: Aileron Balance Tab Removal, TASK 27-11-21-000-801 or Aileron Tab Control Rods Removal, TASK 27-11-34-000-801.

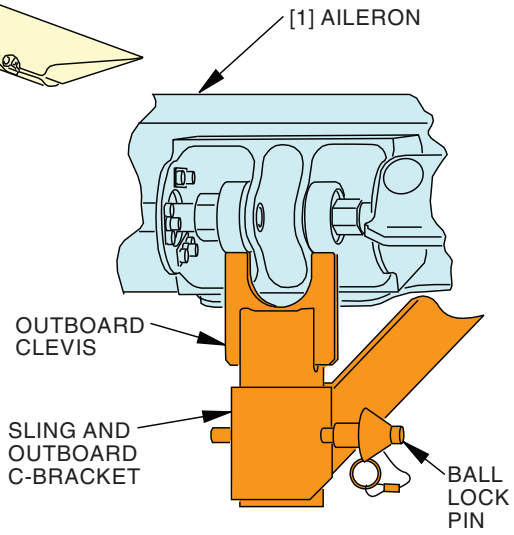
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON AND SLING

A



OUTBOARD HINGE NO. 2

B

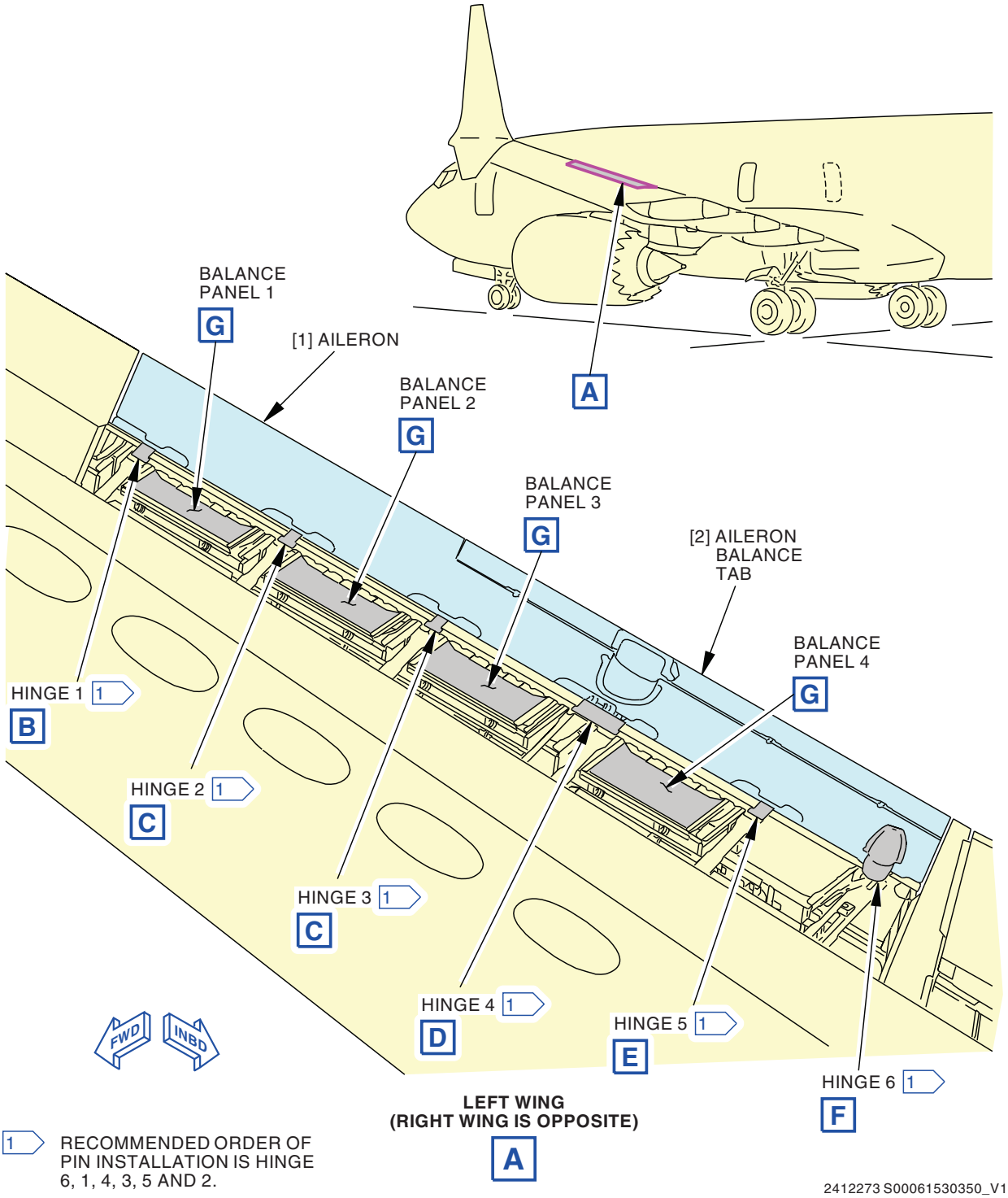
NOTE:

PUT EACH C-BRACKET ASSEMBLY IN THE CORRECT POSITION FOR THE LEFT OR THE RIGHT AILERON. CONNECT EACH CLEVIS ASSEMBLY TO THE C-BRACKET ASSEMBLY WITH A BALL LOCK PIN. PUT THE SLING ASSEMBLY IN POSITION AROUND THE TRAILING EDGE OF THE AILERON. ATTACH THE SLING ASSEMBLY TO THE AILERON HINGES NO. 2 AND 5 WITH THE ATTACHED PINS.

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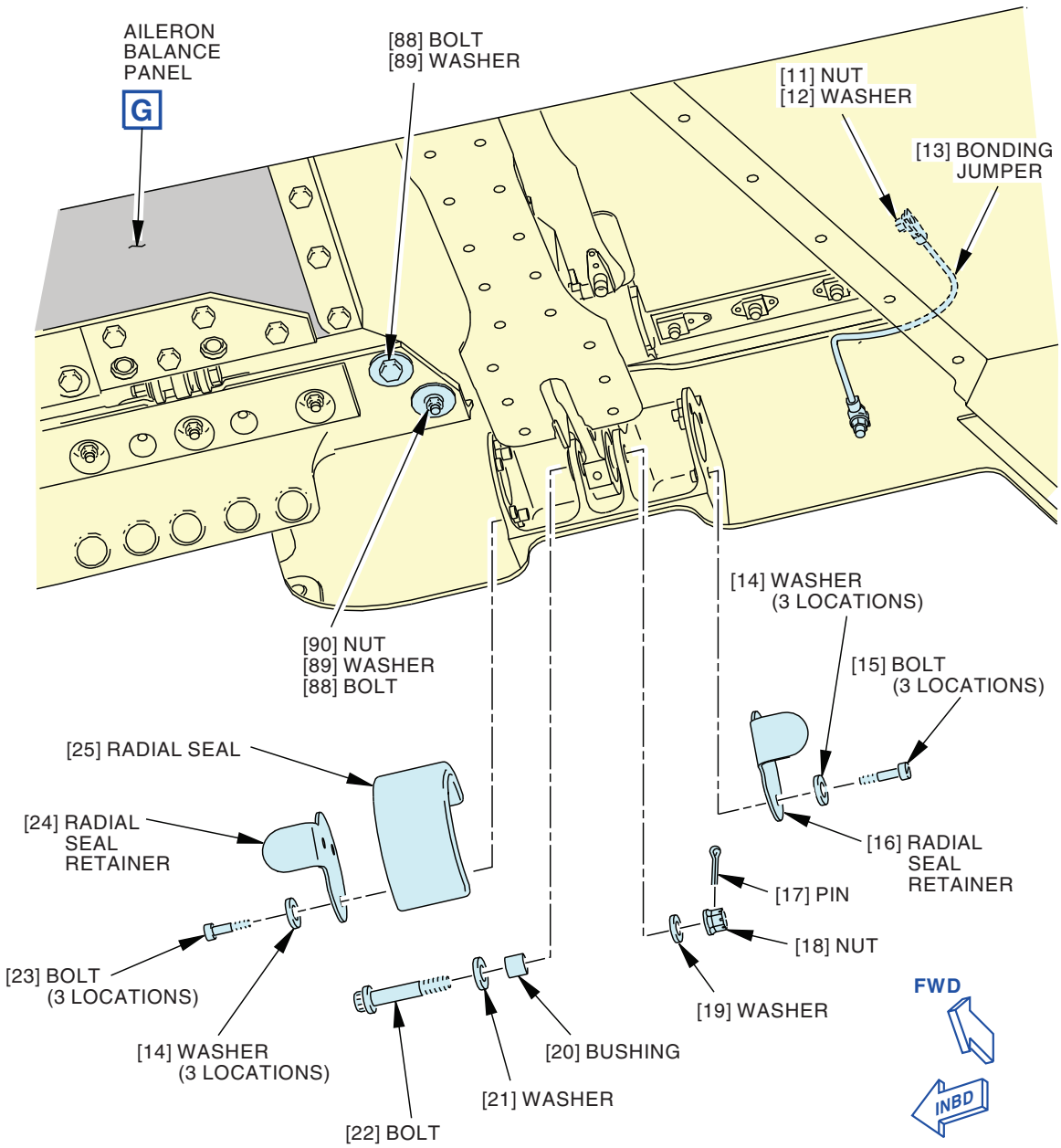
Aileron and Sling Installation
Figure 401/27-11-11-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details



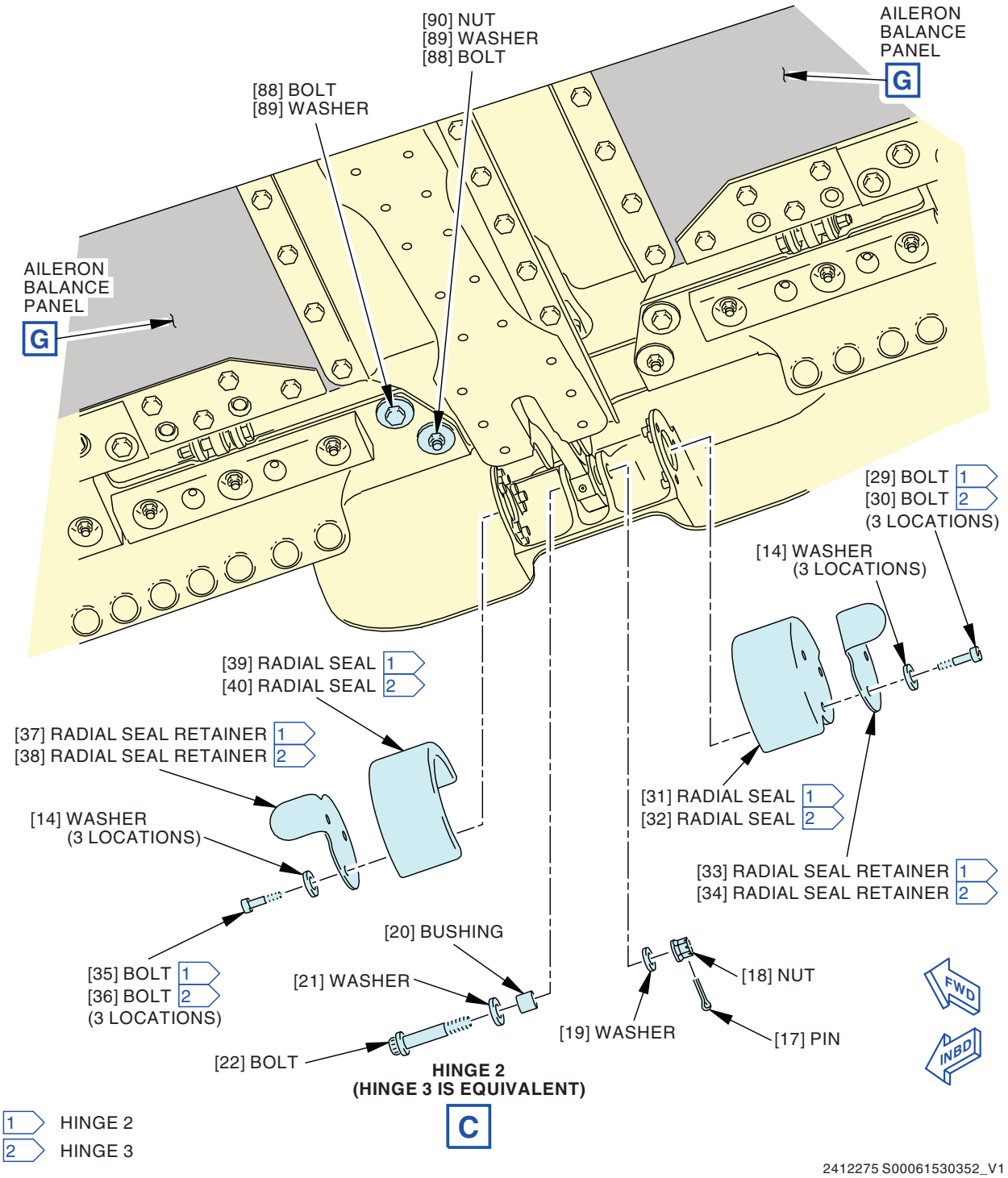
Aileron Installation
Figure 402/27-11-11-990-807 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



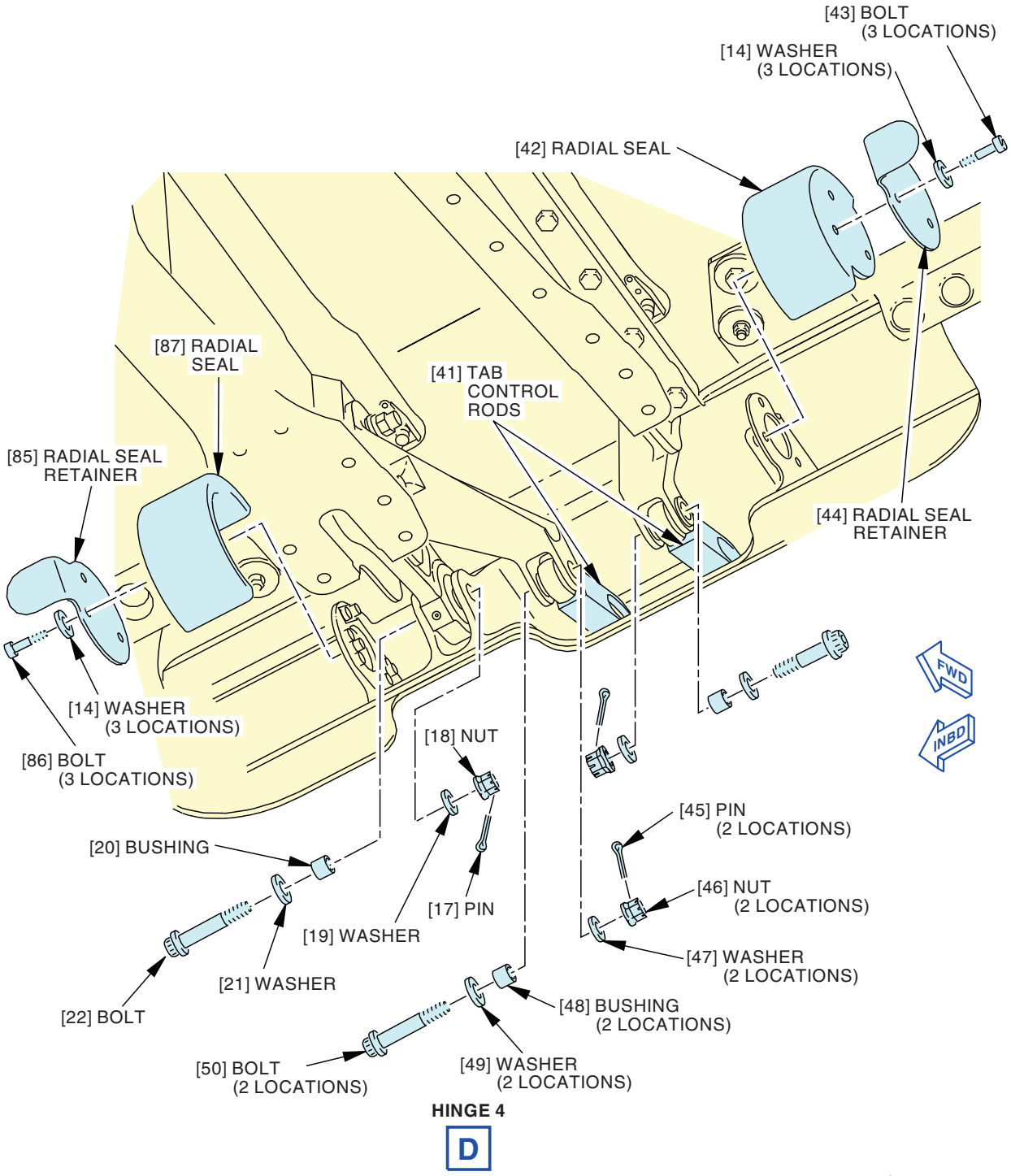
Aileron Installation
Figure 402/27-11-11-990-807 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



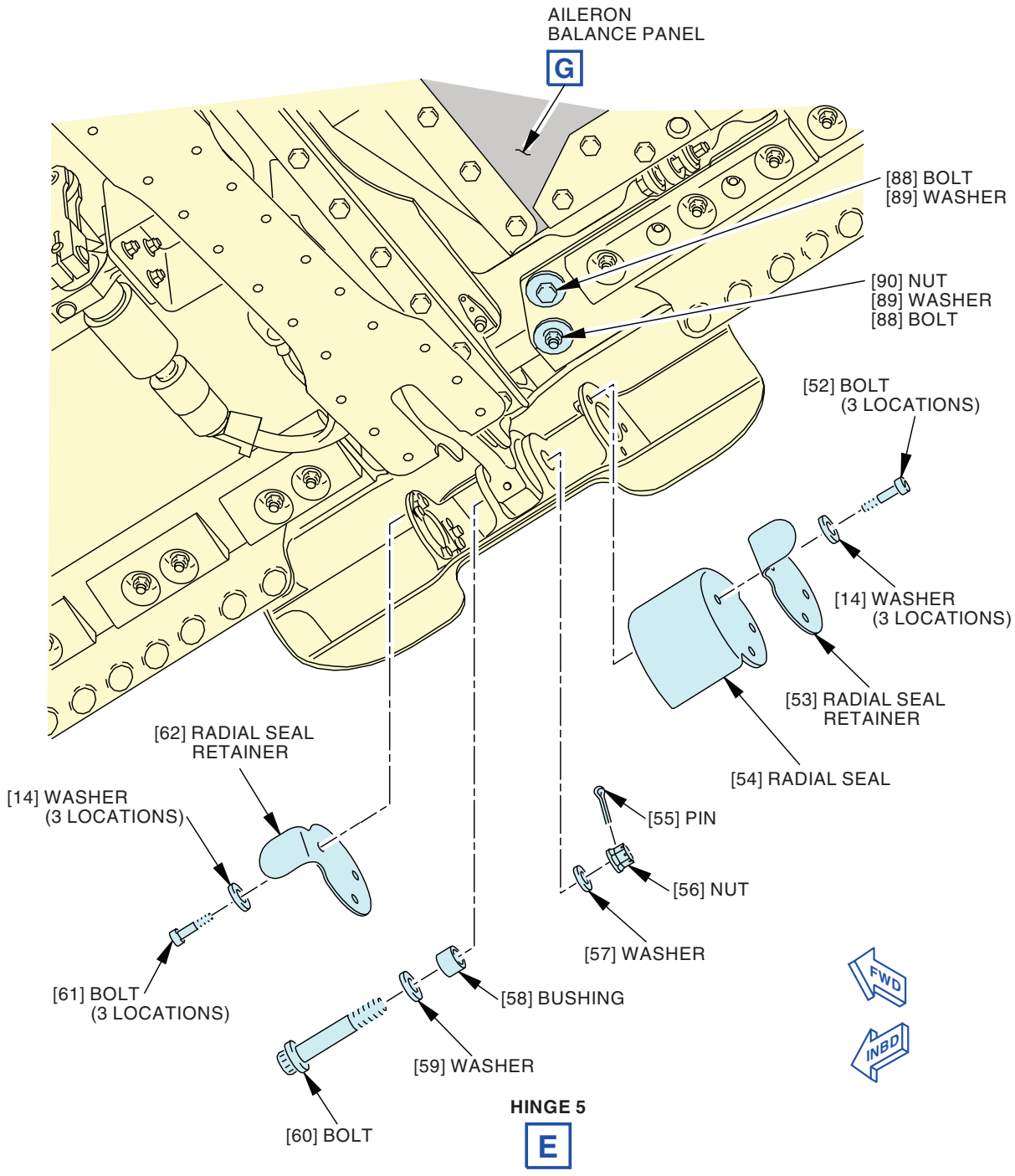
Aileron Installation
Figure 402/27-11-11-990-807 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



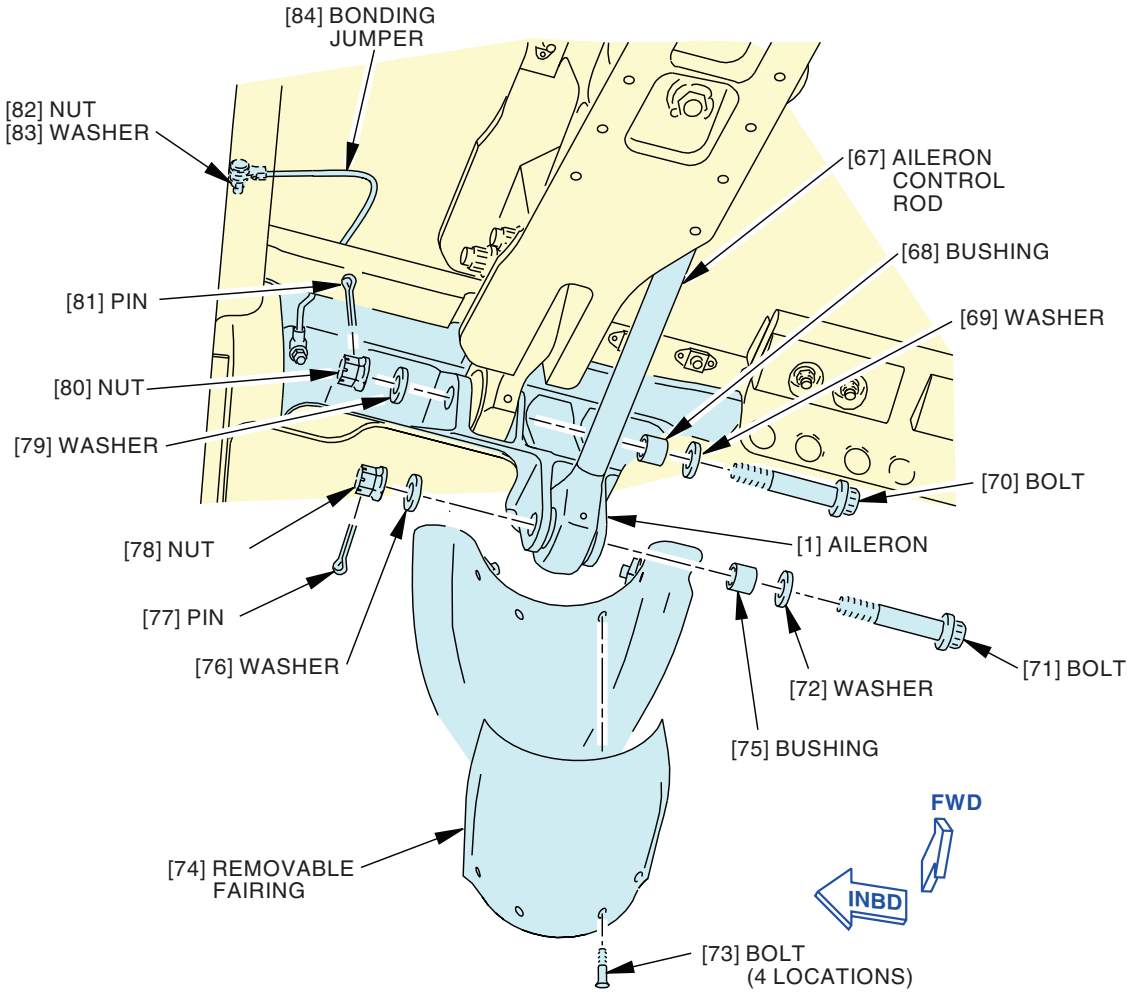
Aileron Installation
Figure 402/27-11-11-990-807 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



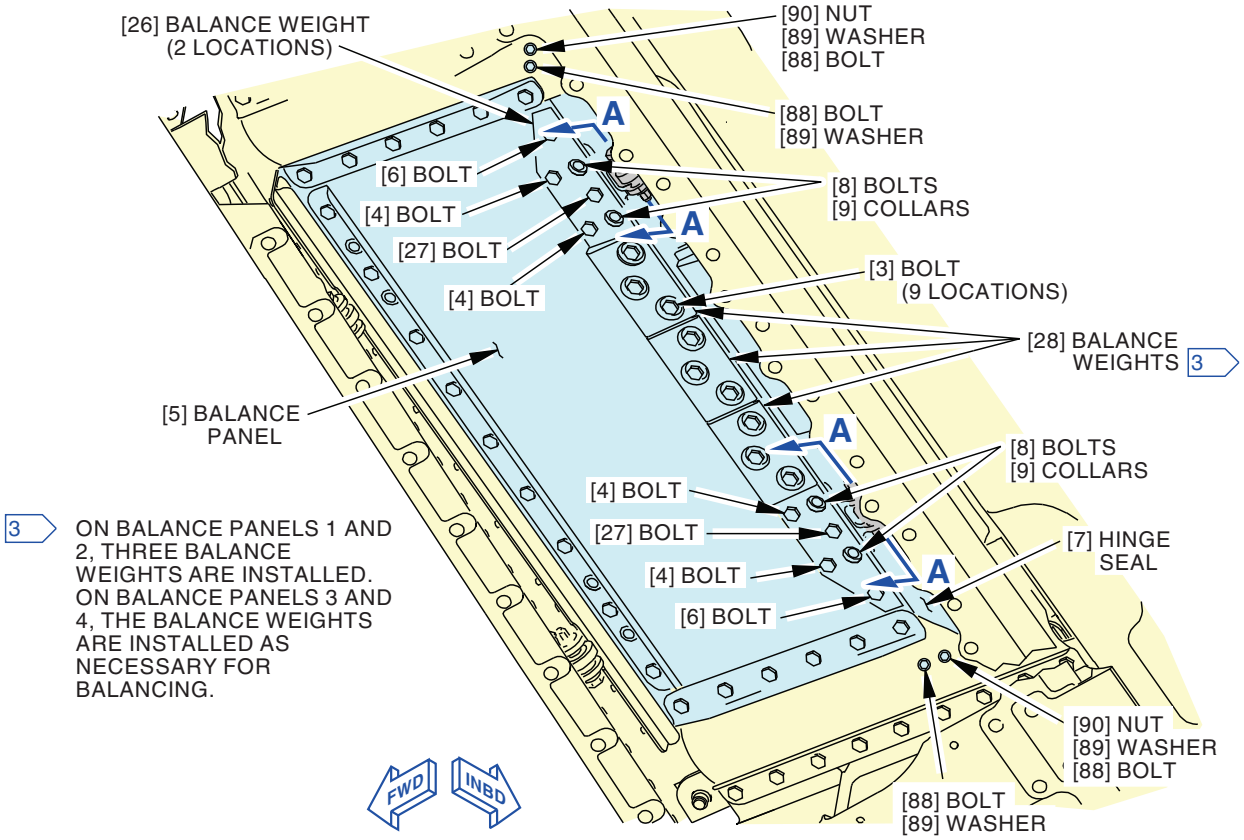
Aileron Installation
Figure 402/27-11-11-990-807 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



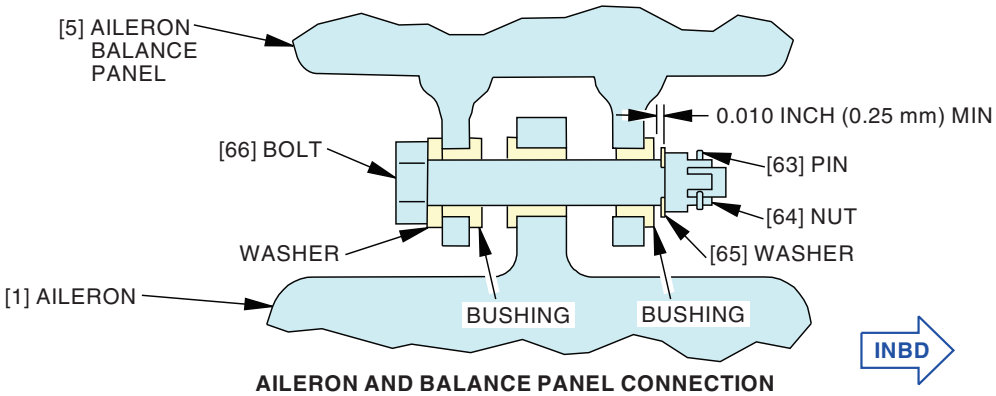
Aileron Installation
Figure 402/27-11-11-990-807 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BALANCE PANEL 1
(BALANCE PANELS 2, 3, AND 4 ARE ALMOST THE SAME)

G



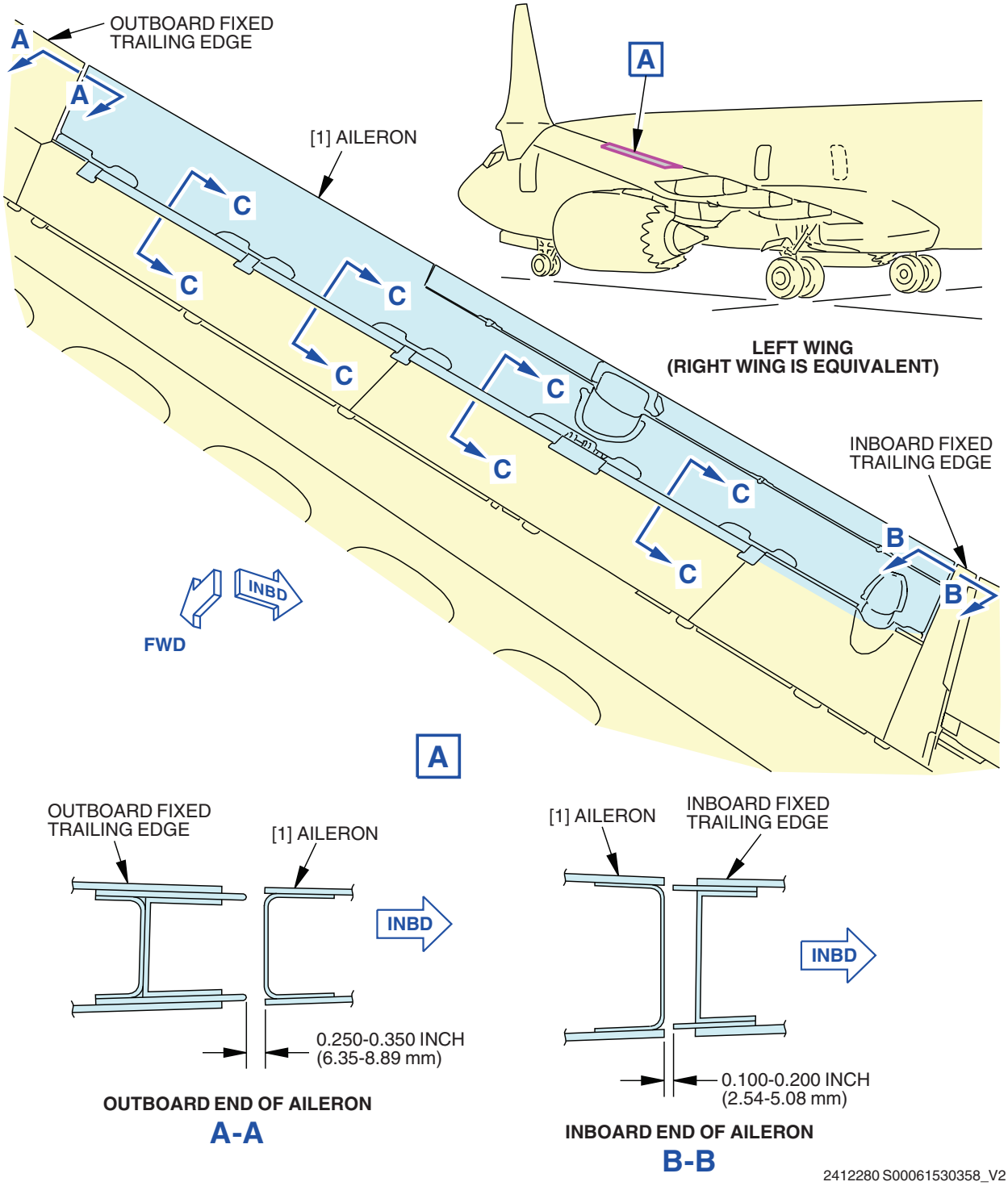
AILERON AND BALANCE PANEL CONNECTION
(OUTBOARD IS SHOWN, INBOARD IS OPPOSITE)
(EXAMPLE)

A-A

2412279 S00061530356_V1

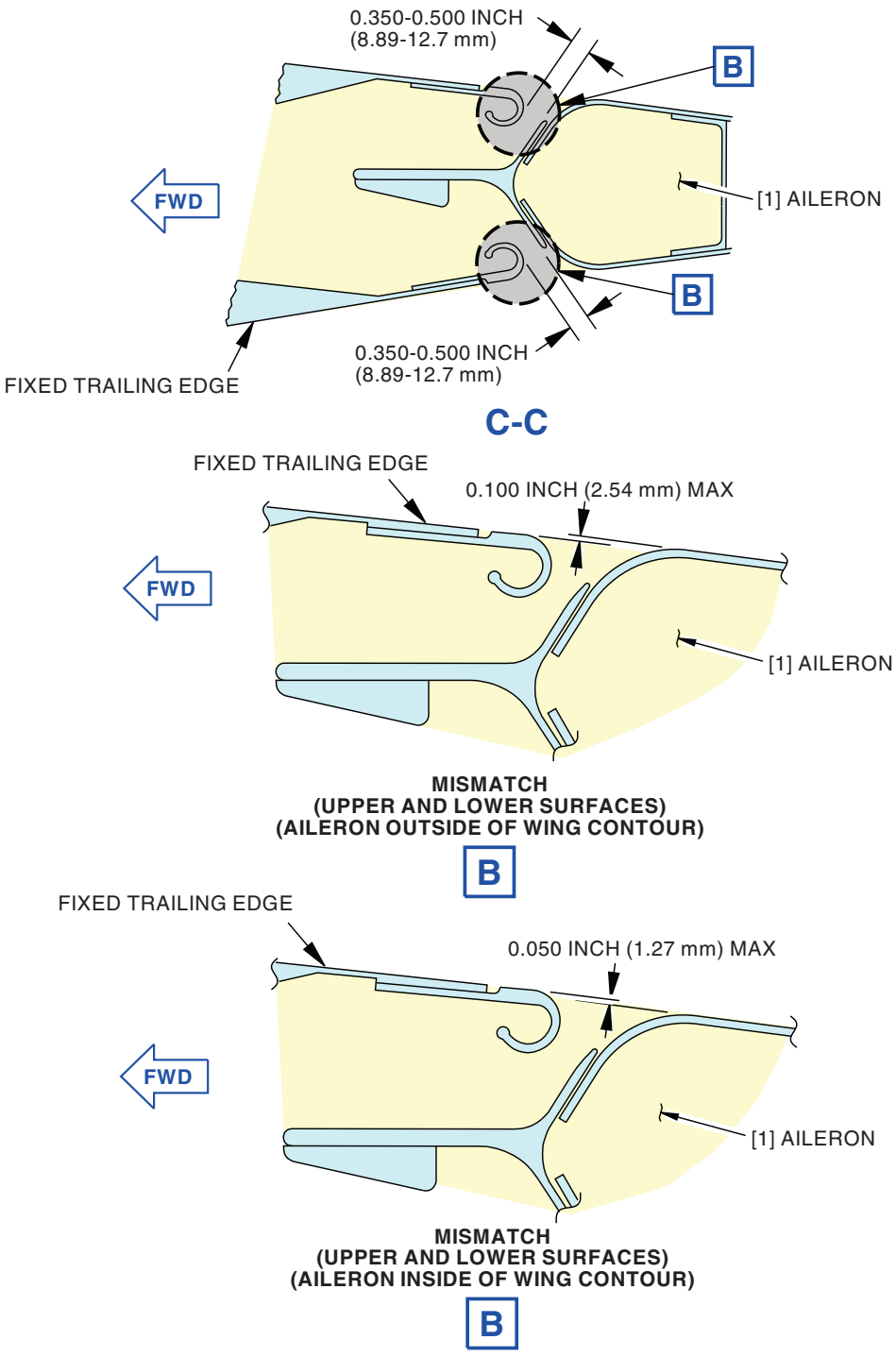
Aileron Installation
Figure 402/27-11-11-990-807 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Clearance and Mismatch Requirements
Figure 403/27-11-11-990-808 (Sheet 1 of 2)

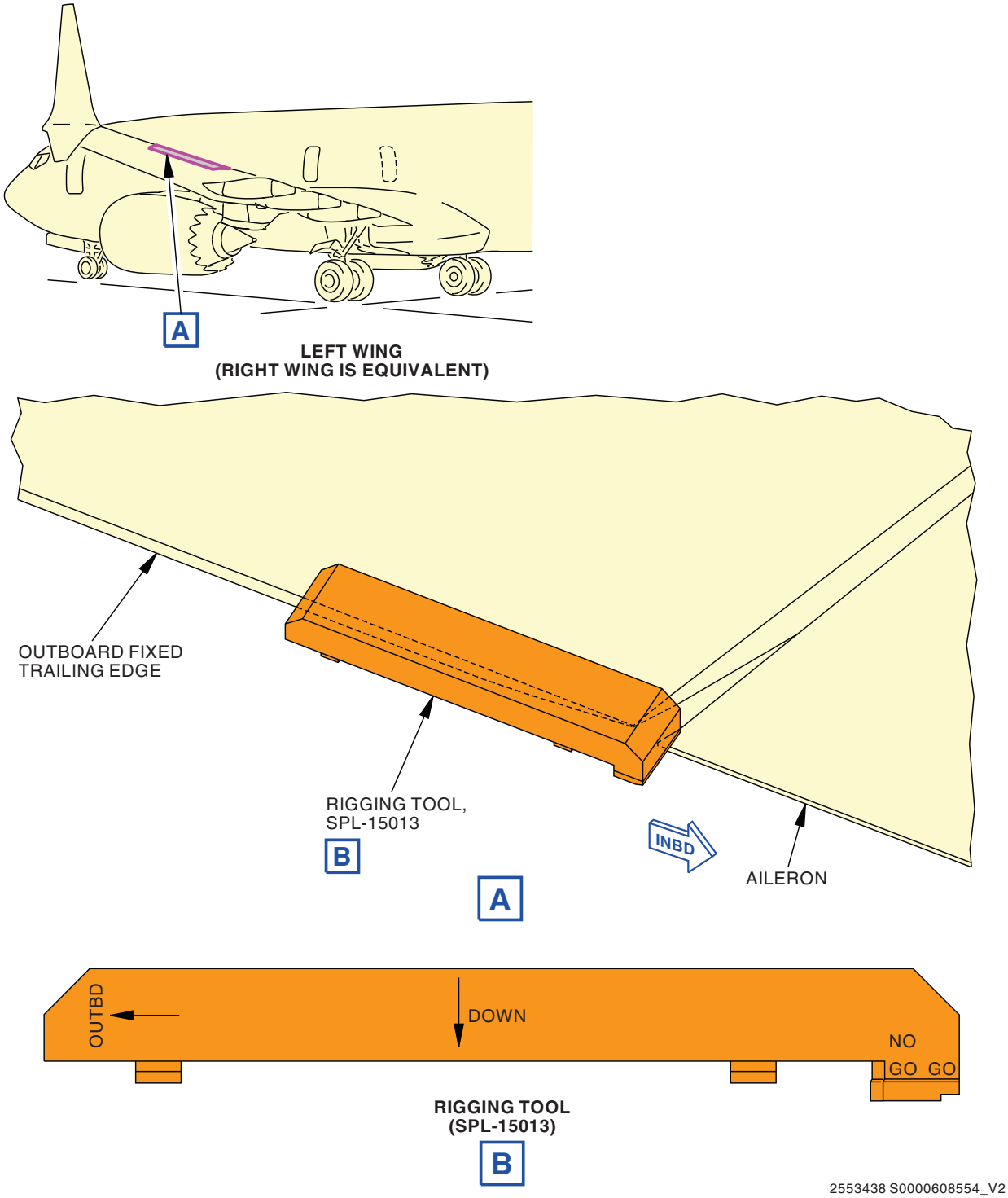
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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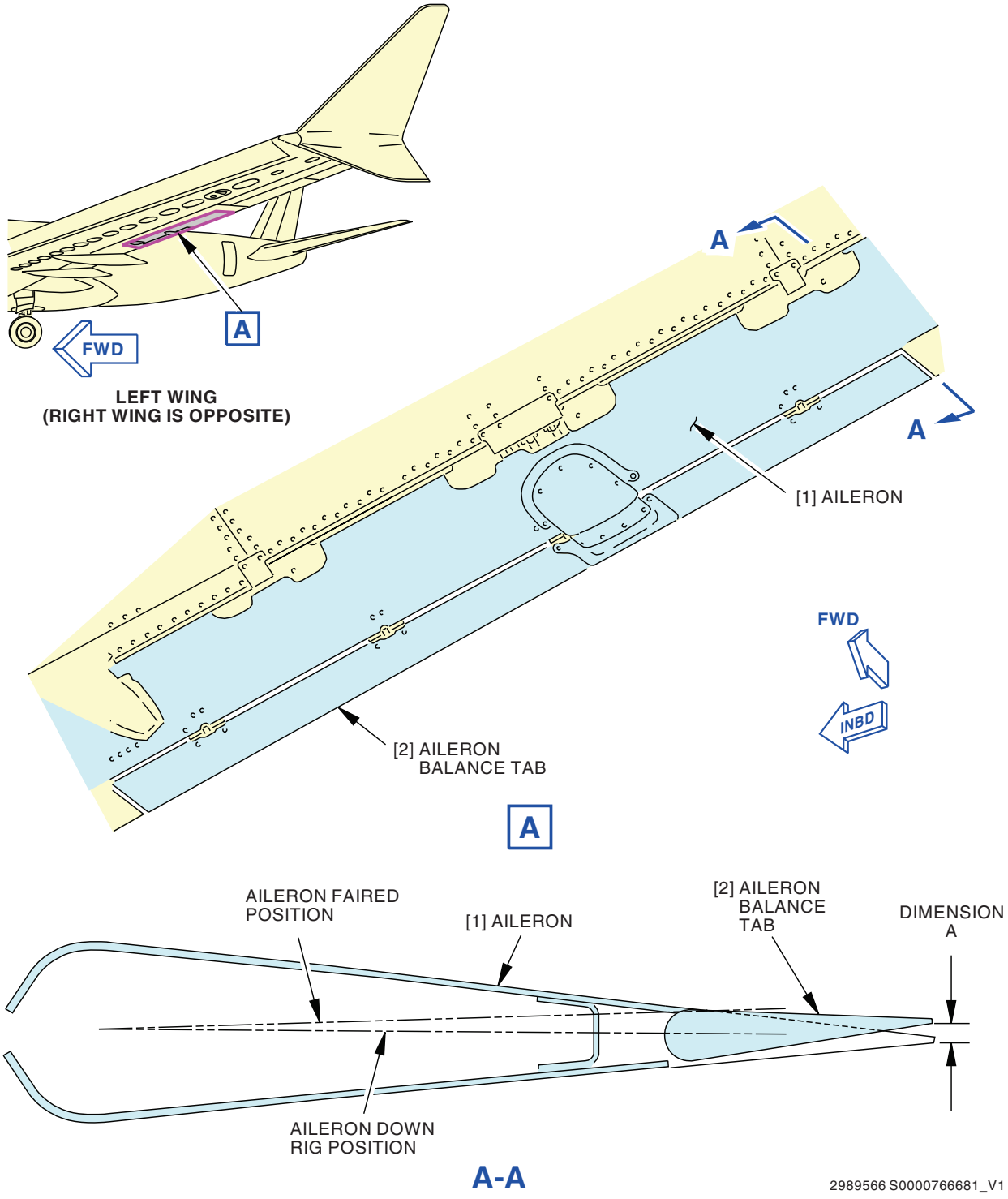
Aileron Clearance and Mismatch Requirements
Figure 403/27-11-11-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Rigging Tool
Figure 404/27-11-11-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Tab Adjustment
Figure 405/27-11-11-990-810

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-11-400-801

3. Aileron Installation

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405)

A. General

(1) A flight test may be necessary after the installation of the component.

(2) Check the requirements in the "General" section of this task: Flight Test to Find and Correct Asymmetrical Flight Control Forces, TASK 27-11-00-750-801.

B. References

Reference	Title
12-22-11-640-801	Aileron Hinge Lubrication (P/B 301)
27-11-00-710-801	Aileron Response (Operational) Test (P/B 501)
27-11-00-750-801	Flight Test to Find and Correct Asymmetrical Flight Control Forces (P/B 501)
27-11-00-820-803	Aileron Balance Tab Adjustment (Recommended Method) (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-11-11-200-801	Aileron Inspection (P/B 601)
27-11-21-400-801	Aileron Balance Tab Installation (P/B 401)
27-11-34-400-801	Aileron Tab Control Rods Installation (P/B 401)
57-50-02-400-803	Outboard Fixed Trailing Edge Lower Panel Installation (P/B 401)
57-61-02-400-802	Aileron Lower Hinge Covers Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2042	Sling - Aileron, Removal/Installation Part #: C27055-1 Supplier: 81205
SPL-15013	Rigging Tool - Aileron Part #: C27096-1 Supplier: 81205
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
D00633	Grease - Aircraft General Purpose	BMS3-33

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Aileron	27-11-11-02-010	PTF ALL
		27-11-11-02-011	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
1 (cont.)		27-11-31-02-010	PTF ALL
		27-11-31-02-013	PTF ALL
		57-61-00-01-175	PTF ALL
		57-61-00-01-180	PTF ALL
2	Aileron balance tab	27-11-21-01-100	PTF ALL
		27-11-21-01-105	PTF ALL

F. Location Zones

Zone	Area
571	Left Wing - Fixed Trailing Edge
572	Left Wing - Aileron
671	Right Wing - Fixed Trailing Edge
672	Right Wing - Aileron

G. Access Panels

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel
572AB	Lower Aileron, Hinge Cover - WBL 423.00
572CB	Lower Aileron, Hinge Cover - WBL 447.00
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00
572FB	Lower Aileron, Hinge Cover - WBL 502.00
572GB	Lower Aileron, Hinge Cover - WBL 528.00
572HB	Lower Aileron, Hinge Cover - WBL 553.00
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel
672AB	Lower Aileron, Hinge Cover - WBL 423.00
672CB	Lower Aileron, Hinge Cover - WBL 447.00
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00
672FB	Lower Aileron, Hinge Cover - WBL 502.00
672GB	Lower Aileron, Hinge Cover - WBL 528.00
672HB	Lower Aileron, Hinge Cover - WBL 553.00

H. Aileron Installation

SUBTASK 27-11-11-220-001

(1) If you install used bolts, bushings, bearings or other components with worn areas, do this task: Aileron Inspection, TASK 27-11-11-200-801.

SUBTASK 27-11-11-420-001

(2) If you removed the aileron balance tab [2], do this task: Aileron Balance Tab Installation, TASK 27-11-21-400-801 or Aileron Tab Control Rods Installation, TASK 27-11-34-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-11-280-001

(3) Make sure that the correct number of balance weights [28] are installed on the lower surface of the aileron balance panels 3 and 4 (SRM PROCEDURE 51-60-03-0G-0).

SUBTASK 27-11-11-820-001

WARNING MAKE SURE THAT YOU HAVE THE AILERONS BALANCED. IF NOT BALANCED, THE AILERONS CAN CAUSE UNWANTED FLUTTER AND INSTABILITY OF THE AIRPLANE. THIS CAN CAUSE AN IN-FLIGHT EMERGENCY. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Balance and stamp the aileron assembly with the new static balance hinge moment if you install a reworked, repaired, repainted, or new aileron assembly (SRM SUBJECT 51-60-01).

SUBTASK 27-11-11-400-001

(5) Install the aileron [1] with the sling, SPL-2042, (Figure 401).

(a) Make sure that the sling, SPL-2042, is installed on the aileron [1].

(b) If the sling, SPL-2042, is not installed, do these steps to install it:

1) Attach the outboard C-bracket to the outboard clevis with the ball lock pin.

2) Attach the inboard C-bracket to the inboard clevis with the ball lock pin.

3) Put the sling in position around the Trailing Edge (TE) of the aileron [1].

4) Attach the inboard clevis to hinge 5 with the attached ball lock pins.

5) Attach the outboard clevis to hinge 2 with the attached ball lock pins.

6) Lift the sling with a cable and hook to hold the aileron [1].

SUBTASK 27-11-11-420-002

(6) Install the aileron [1] on the hinges (Figure 402).

(a) Apply grease, D00633, to the outer diameters of all hinge bolts and bushings and inner diameters of bearings.

(b) Put the aileron [1] in position to install.

(c) Align the hinges with the support fittings.

(d) Connect the aileron [1] at hinge 6.

1) Install the bolt [70], washer [69], bushing [68], washer [79], and nut [80].

2) Tighten the nut [80] to 160 in-lb (18 N·m) - 240 in-lb (27 N·m).

3) Tighten the nut [80] to the lower end of the torque range.

a) Continue to tighten the nut until the next castellation aligns with the hole in the bolt [70].

4) Install the new pin [81].

(e) Connect the aileron [1], in the sequence that follows, at hinges 1, 4 and 3.

1) To install the pin in hinge 4, hold the aileron from below.

2) Install the bolt [22], washer [21], bushing [20], washer [19], and nut [18].

3) Tighten the nut [18] to 50 in-lb (6 N·m) - 80 in-lb (9 N·m).

4) Tighten the nut [18] to the lower end of the torque range.

a) Continue to tighten the nut until the next castellation aligns with the hole in the bolt [22].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) Install the new pin [17].

(f) Remove the sling, SPL-2042, from the aileron [1].

1) Remove the outboard clevis from hinge 2.

2) Remove the inboard clevis from hinge 5.

3) Remove the sling, SPL-2042, from the TE of the aileron [1].

4) Remove the ball lock pin and disconnect the outboard clevis from the outboard C-bracket.

5) Remove the ball lock pin and disconnect the inboard clevis from the inboard C-bracket.

(g) Connect the aileron [1] at hinge 5.

1) Install the bolt [60], washer [59], bushing [58], washer [57], and nut [56].

2) Tighten the nut [56] to 160 in-lb (18 N·m) - 240 in-lb (27 N·m).

3) Tighten the nut [56] to the lower end of the torque range.

a) Continue to tighten the nut until the next castellation aligns with the hole in the bolt [60].

4) Install the new pin [55].

(h) Connect the aileron [1] at hinge 2.

1) Install the bolt [22], washer [21], bushing [20], washer [19], and nut [18].

2) Tighten the nut [18] to 50 in-lb (6 N·m) - 80 in-lb (9 N·m).

3) Tighten the nut [18] to the lower end of the torque range.

a) Continue to tighten the nut until the next castellation aligns with the hole in the bolt [22].

4) Install the new pin [17].

(i) Align the aileron [1] with the balance panels [5].

1) Make sure that the washer under the bolt is bonded to the hinge half.

a) If it is necessary, bond the washer to the hinge half with sealant, A00247.

2) Install the bolts [66], washers [65], and nuts [64].

3) Tighten the nuts [64] to 25 in-lb (3 N·m) - 35 in-lb (4 N·m).

4) Tighten the nuts [64] to the lower end of the torque range.

a) Continue to tighten the nut until the next castellation aligns with the hole in the bolts [66].

5) Make sure that the gap between the bushings and the washers [65] is 0.01 in. (0.25 mm) minimum.

6) Install the new pins [63].

(j) Attach the hinge seals [7] to the aileron balance panels [5].

1) Attach the hinge seals [7] to the aileron balance panels [5].

a) Make sure that the correct number of balance weights [28] are installed on aileron balance panels 3 and 4 (SRM PROCEDURE 51-60-03-0G-0).

2) Apply compound, C00528, to bolts [3], bolts [4], bolts [6], bolts [27], and bolts [88].

3) Install the balance weights [28] with bolts [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Install the balance weights [26] with the bolts [8] with collars [9], bolts [27], bolts [6], bolts [4], bolts [88], washers [89], and nuts [90].

(k) At hinge 4, connect the tab control rods [41] to the forward fittings.

1) Install the bolts [50], washers [49], bushings [48], washers [47], and nuts [46].

2) Tighten the nuts [46] to 50 in-lb (6 N·m) - 80 in-lb (9 N·m).

3) Tighten the nuts [46] to the lower end of the torque range.

a) Continue to tighten the nut until the next castellation aligns with the hole in the bolts [50].

4) Install the new pins [45].

5) At hinge 1, do these steps:

a) If removed, install the radial seal retainer [16], washers [14], and bolts [15].

b) Install the radial seal [25], radial seal retainer [24], washers [14], and bolts [23].

c) Connect the bonding jumper [13] with washer [12] and nut [11].

6) At hinge 2, do these steps:

a) Install the radial seal [31], radial seal retainer [33], washers [14], and bolts [29].

b) Install the radial seal [39], radial seal retainer [37], washers [14], and bolts [35].

7) At hinge 3, do these steps:

a) Install the radial seal [32], radial seal retainer [34], washers [14], and bolts [30].

b) Install the radial seal [40], radial seal retainer [38], washers [14], and bolts [36].

8) At hinge 4, do these steps:

a) Install the radial seal [42], radial seal retainer [44], washers [14], and bolts [43].

b) Install the radial seal [87], radial seal retainer [85], washers [14], and bolts [86].

9) At hinge 5, do these steps:

a) Install the radial seal [54], radial seal retainer [53], washers [14], and bolts [52].

b) If removed, install the radial seal retainer [62], washers [14], and bolts [61].

SUBTASK 27-11-11-640-001

(7) Do this task: Aileron Hinge Lubrication, TASK 12-22-11-640-801.

SUBTASK 27-11-11-420-003

(8) Install the aileron [1] without the sling, SPL-2042.

(a) Install the aileron [1] on the hinges:

1) Apply grease, D00633, to the outer diameters of all hinge bolts and bushings and inner diameters of bearing.

2) Put the aileron [1] in position to install.

3) Align the hinges with the support fittings.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Connect the aileron [1] at hinges 1, 2, 3 and 4.
   a) Install the bolt [22], washer [21], bushing [20], washer [19], and nut [18].
   b) Tighten the nuts nut [18] to 50 in-lb (6 N·m) - 80 in-lb (9 N·m).
   c) Tighten the nut [18] to the lower end of the torque range.
      <1> Continue to tighten the nut until the next castellation aligns with the hole in the bolt [22].
   d) Install the new pin [17].
5) Connect the aileron [1] to hinge 5.
   a) Install the bolt [60], washer [59], bushing [58], washer [57], and nut [56].
   b) Tighten the nut [56] to 160 in-lb (18 N·m) - 240 in-lb (27 N·m).
   c) Tighten the nut [56] to the lower end of the torque range.
      <1> Continue to tighten the nut until the next castellation aligns with the hole in the bolt [60].
   d) Install the new pin [55].
6) Connect the aileron [1] at hinge 6.
   a) Install the bolt [70], washer [69], bushing [68], washer [79], and nut [80].
   b) Tighten the nut [80] to 160 in-lb (18 N·m) - 240 in-lb (27 N·m).
   c) Tighten the nut [80] to the lower end of the torque range.
      <1> Continue to tighten the nut until the next castellation aligns with the hole in the bolt [70].
   d) Install the new pin [81].

(b) Align the aileron [1] with the aileron balance panels [5].
1) Make sure that the washer under the bolt is bonded to the hinge half.
   a) If it is necessary, bond the washer to the hinge half with sealant, A00247.
2) Install the bolts [66], washers [65], and nuts [64].
3) Tighten the nuts [64] to 25 in-lb (3 N·m) - 35 in-lb (4 N·m).
4) Tighten the nuts [64] to the lower end of the torque range.
   a) Continue to tighten the nut until the next castellation aligns with the hole in the bolts [66].
5) Make sure that the gap between the bushings and the washers [65] is 0.01 in. (0.25 mm) minimum.
6) Install the new pins [63].

(c) Attach the hinge seals [7] to the balance panels [5]:
1) Attach the hinge seals [7] to the balance panels [5].
   a) Make sure that the correct number of balance weights [28] are installed on balance panels 3 and 4 (SRM PROCEDURE 51-60-03-0G-0).
2) Apply compound, C00528, to bolts [3], bolts [4], bolts [6], bolts [27], and bolts [88].
3) Install the balance weights [28] with bolts [3].
4) Install the balance weights [26] with the bolts [8] with collars [9], bolts [27], bolts [6], bolts [4], bolts [88], washers [89], and nuts [90].

(d) At hinge 4, connect the tab control rods [41] to the forward fittings.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Install the bolts [50], washers [49], bushings [48], washers [47], and nuts [46].
2) Tighten the nuts [46] to 50 in-lb (6 N·m) - 80 in-lb (9 N·m).
3) Tighten the nuts [46] to the lower end of the torque range.
   a) Continue to tighten the nut until the next castellation aligns with the hole in the bolts [50].
4) Install the pins [45].

(e) At hinge 1, do these steps:
1) If removed, install the radial seal retainer [16], washers [14], and bolts [15].
2) Install the radial seal [25], radial seal retainer [24], washers [14], and bolts [23].
3) Connect the bonding jumper [13] with washer [12] and nut [11].

(f) At hinge 2, do these steps:
1) Install the radial seal [31], radial seal retainer [33], washers [14], and bolts [29].
2) Install the radial seal [39], radial seal retainer [37], washers [14], and bolts [35].

(g) At hinge 3, do these steps:
1) Install the radial seal [32], radial seal retainer [34], washers [14], and bolts [30].
2) Install the radial seal [40], radial seal retainer [38], washers [14], and bolts [36].

(h) At hinge 4, do these steps:
1) Install the radial seal [42], radial seal retainer [44], washers [14], and bolts [43].
2) Install the radial seal [87], radial seal retainer [85], washers [14], and bolts [86].

(i) At hinge 5, do theses steps:
1) Install the radial seal [54], radial seal retainer [53], washers [14], and bolts [52].
2) If removed, install the radial seal retainer [62], washers [14], and bolts [61].

SUBTASK 27-11-11-640-002

(9) Do this task: Aileron Hinge Lubrication, TASK 12-22-11-640-801.

SUBTASK 27-11-11-720-001

(10) Do the aileron surface friction and motion test.
(a) With a push/pull scale 0-25 pound (0-11 kilogram), STD-753, measure the force at 8.75 ±0.25 in. (222.25 ±6.35 mm) aft of the hinge centerline and perpendicular to the aileron chord plane.
(b) Write down the (positive) forces necessary to move each aileron through the neutral position.
(c) Write down the (negative) forces necessary to prevent movement of the aileron through the neutral position.
(d) Add these forces together.
(e) Make sure that the result is not more than 5.0 lbf (22.2 N).
1) If the result is more than 5.0 lbf (22.2 N), do these checks:
   a) Make sure that all the hinge bolts are sufficiently lubricated.
   b) Make sure that the torque values of the hinge bolts are as specified.
(f) Measure the force necessary to move the aileron through full travel in each direction, then back to the neutral position.

NOTE: The movement of the aileron must be free and smooth.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Apply the force perpendicular to the aileron chord plane 8.75 ±0.25 in. (222.25 ±6.35 mm) aft of the aileron hinge line.

2) Make sure that the maximum force is not more than 13 lbf (57.8 N) for aileron down and 3 lbf (13.3 N) for aileron up.

SUBTASK 27-11-11-420-004

(11) At hinge 6, connect the aileron control rod [67] to the aileron [1].

(a) Put the aileron control rod [67] in its position.

(b) Install the bolt [71], washer [72], bushing [75], washer [76], nut [78], and pin [77].

SUBTASK 27-11-11-420-005

(12) At hinge 6, install the removable fairing [74].

(a) Connect the bonding jumper [84] with washer [83] and nut [82].

(b) Put the removable fairing [74] in its position.

(c) Install the bolts [73] to attach the removable fairing [74] to the aileron [1].

SUBTASK 27-11-11-280-002

(13) Use the aileron control wheel to manually operate the aileron through a minimum of two full cycles, to full travel.

(a) Make sure that the aileron can move smoothly and freely.

SUBTASK 27-11-11-220-002

(14) Do a check of the aileron clearances and mismatch.

(a) Make sure that the clearances for the upper and lower surfaces of the aileron and the fixed TE are not out of tolerance.

1) At the support beams for the aileron balance panel, make sure that the clearance is 0.350 in. (8.890 mm) - 0.500 in. (12.700 mm) (View B, Figure 403).

(b) Make sure that the mismatch for the upper and lower surfaces of the aileron and the fixed TE are not out of tolerance (Rigging Tool Method).

1) Set the rigging tool, SPL-15013, on the outboard fixed TE.

2) Make sure that the upper surface of the aileron is between the “GO” and “NO GO” limits.

(c) Make sure that the mismatch for the upper and lower surfaces of the aileron and the fixed TE are not out of tolerance (Straight Edge Method).

1) For the aileron outside of the wing contour, make sure that the maximum mismatch is 0.1 in. (2.5 mm) (View B, Figure 403).

2) For the aileron inside of the wing contour, make sure that the maximum mismatch is 0.05 in. (1.27 mm) (View B, Figure 403).

(d) Make sure that the clearance at the inboard end of the aileron is 0.1 in. (2.5 mm) - 0.2 in. (5.1 mm) (View A, Figure 403).

(e) Make sure that the clearance at the outboard end of the aileron is 0.25 in. (6.4 mm) - 0.35 in. (8.9 mm) (View A, Figure 403).

(f) Make sure that the aileron balance tab [2] aligns with the aileron:

1) Measure the distance between the rig position of the aileron balance tab [2] to the aileron lower surface (dimension A) at these three positions (View A, Figure 405).

NOTE: You must measure from the lower aft most point of the aileron balance tab.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) At the outboard end of the aileron balance tab [2], make sure that the dimension A is 0.06 ±0.07 in. (1.52 ±1.78 mm) or 0.82 ±0.25°arc (0.01 ±0.01 rad).

b) At the middle point along the aileron balance tab [2], make sure that the dimension A is 0.065 ±0.070 in. (1.65 ±1.78 mm) or 0.60 ±0.25°arc (0.01 ±0.00 rad).

c) At the inboard end of the aileron balance tab [2], make sure that the dimension A is 0.070 ±0.070 in. (1.778 ±1.778 mm) or 0.34 ±0.25°arc (0.01 ±0.01 rad).

2) If the aileron balance tab [2] does not align with the aileron, do this task: Aileron Balance Tab Adjustment (Recommended Method), TASK 27-11-00-820-803.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-11-410-001

(1) Install these aileron access panels found on the lower wing surface adjacent to the Leading Edge (LE) of the aileron:

(a) For the left wing, install these access panels:

(TASK 57-61-02-400-802)

Number	Name/Location
572AB	Lower Aileron, Hinge Cover - WBL 423.00
572CB	Lower Aileron, Hinge Cover - WBL 447.00
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00
572FB	Lower Aileron, Hinge Cover - WBL 502.00
572GB	Lower Aileron, Hinge Cover - WBL 528.00
572HB	Lower Aileron, Hinge Cover - WBL 553.00

(b) For the right wing, install these access panels:

(TASK 57-61-02-400-802)

Number	Name/Location
672AB	Lower Aileron, Hinge Cover - WBL 423.00
672CB	Lower Aileron, Hinge Cover - WBL 447.00
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00
672FB	Lower Aileron, Hinge Cover - WBL 502.00
672GB	Lower Aileron, Hinge Cover - WBL 528.00
672HB	Lower Aileron, Hinge Cover - WBL 553.00

SUBTASK 27-11-11-410-002

(2) Install these access panels found on the lower panel of the outboard fixed TE.

(a) On the left wing, install these access panels:

(TASK 57-50-02-400-803)

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(b) On the right wing, install these access panels:

(TASK 57-50-02-400-803)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-11-860-002

(3) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-11-710-001

(4) Do this task: Aileron Response (Operational) Test, TASK 27-11-00-710-801.

SUBTASK 27-11-11-720-002

(5) If it is necessary, do this task: Flight Test to Find and Correct Asymmetrical Flight Control Forces, TASK 27-11-00-750-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON - INSPECTION/CHECK

1. General

A. This procedure contains the task to do a check of the aileron hinge.

TASK 27-11-11-200-801

2. Aileron Inspection

(Figure 601)

A. General

(1) This task gives wear limits and repair data for the aileron hinge inspection.

B. References

Reference	Title
27-11-11-000-801	Aileron Removal (P/B 401)
27-11-11-400-801	Aileron Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

E. Procedure

SUBTASK 27-11-11-020-004

(1) Do this task: Aileron Removal, TASK 27-11-11-000-801.

SUBTASK 27-11-11-020-005

(2) Use these tools to do a check for worn parts of the aileron hinge:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-11-11-020-006

(3) Examine the parts of the aileron PCU linkage for worn areas:

(a) Remove the parts you want to examine.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are out of tolerance, replace or repair the parts if it is necessary.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-11-11-993-801

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	0.2500 in 6.3500 mm	0.2505 in 6.3627 mm	0.2525 in 6.4135 mm	0.0005 in 0.0127 mm
#1 BOLT	O.D.	0.2585 in 6.5659 mm	0.2595 in 6.5913 mm	0.2475 in 6.2865 mm	
#2 BUSHING	I.D.	0.3750 in 9.5250 mm	0.3756 in 9.5402 mm	0.3773 in 9.5834 mm	0.0005 in 0.0127 mm
#2 BUSHING	O.D.	0.3740 in 9.4996 mm	0.3745 in 9.5123 mm	0.3728 in 9.4691 mm	
#3 BUSHING	I.D.	0.2500 in 6.3500 mm	0.2505 in 6.3627 mm	0.2525 in 6.4135 mm	0.0005 in 0.0127 mm
#3 BOLT	O.D.	0.2485 in 6.3119 mm	0.2495 in 6.3373 mm	0.2475 in 6.2865 mm	
#4 BEARING	I.D.	0.2495 in 6.3373 mm	0.2500 in 6.3500 mm	0.2520 in 6.4008 mm	0.0000 in 0.0000 mm
#4 BOLT	O.D.	0.2485 in 6.3119 mm	0.2495 in 6.3373 mm	0.2475 in 6.2865 mm	
#5 BUSHING	I.D.	0.3750 in 9.5250 mm	0.3756 in 9.5402 mm	0.3778 in 9.5961 mm	0.0005 in 0.0127 mm
#5 BOLT	O.D.	0.3735 in 9.4869 mm	0.3745 in 9.5123 mm	0.3723 in 9.4564 mm	
#6 BUSHING	I.D.	0.5620 in 14.2748 mm	0.5627 in 14.2926 mm	0.5645 in 14.3383 mm	0.0005 in 0.0127 mm
#6 BUSHING	O.D.	0.5610 in 14.2494 mm	0.5615 in 14.2621 mm	0.5595 in 14.2113 mm	
#7 BUSHING	I.D.	0.3750 in 9.5250 mm	0.3755 in 9.5377 mm	0.3777 in 9.5936 mm	0.0005 in 0.0127 mm
#7 BOLT	O.D.	0.3735 in 9.4869 mm	0.3745 in 9.5123 mm	0.3723 in 9.4564 mm	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-11-11-993-801 (Continued)

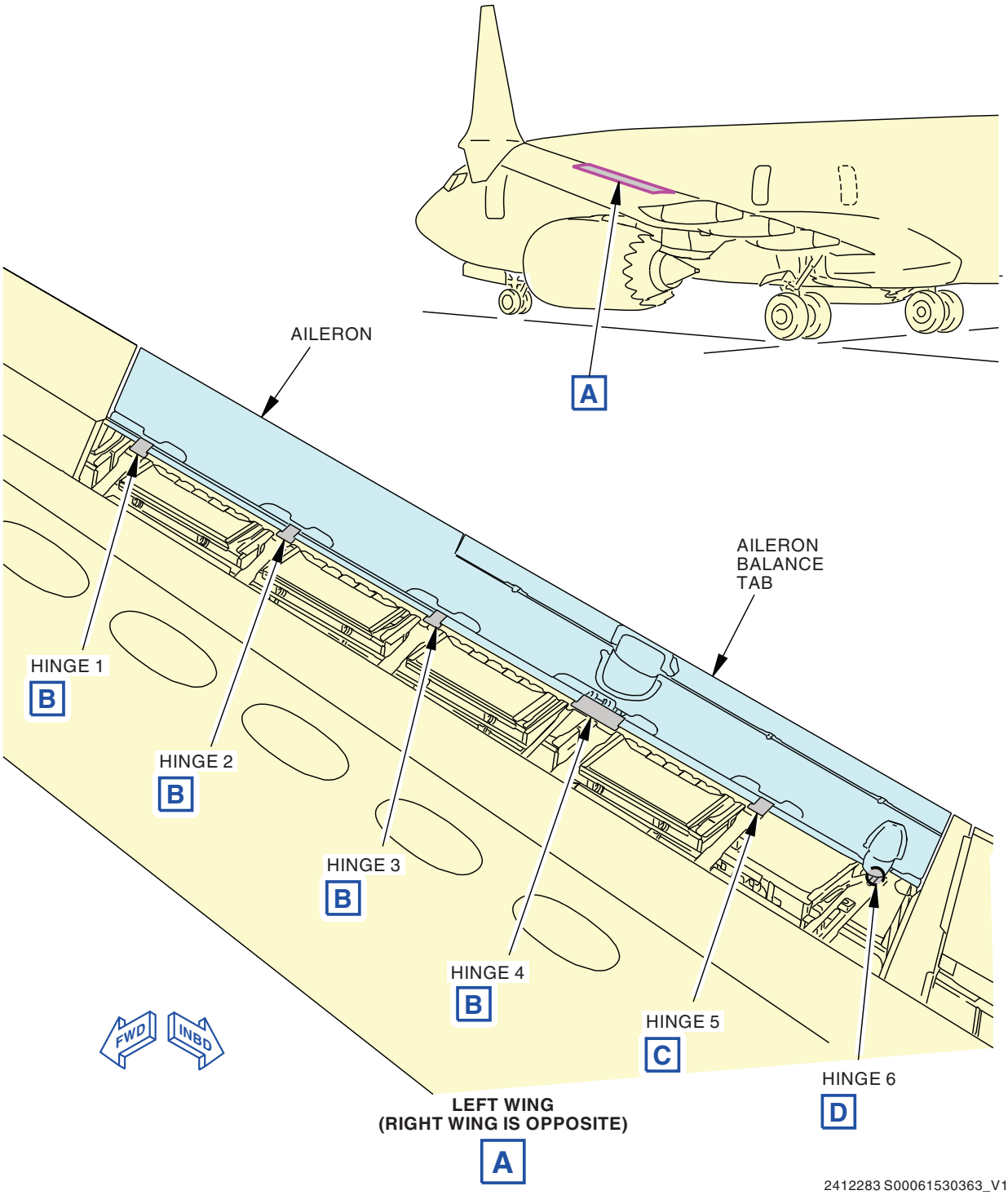
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#8 BEARING	I.D.	0.3745 in 9.5123 mm	0.3750 in 9.5250 mm	0.3772 in 9.5809 mm	0.0000 in
#8 BOLT	O.D.	0.3735 in 9.4869 mm	0.3745 in 9.5123 mm	0.3723 in 9.45642mm	0.0000 mm
#9 BUSHING	I.D.	0.3750 in 9.5250 mm	0.3755 in 9.5377 mm	0.3777 in 9.5936 mm	0.0005 in
#9 BOLT	O.D.	0.3735 in 9.4869 mm	0.3745 in 9.5123 mm	0.3723 in 9.4564 mm	0.0127 mm
#10 BUSHING	I.D.	0.3750 in 9.5250 mm	0.3756 in 9.5402 mm	0.3778 in 9.5961 mm	0.0005 in
#10 BOLT	O.D.	0.3735 in 9.4869 mm	0.3745 in 9.5123 mm	0.3723 in 9.4564 mm	0.0127 mm
#11 BEARING	I.D.	0.3745 in 9.5123 mm	0.3750 in 9.5250 mm	0.3772 in 9.5809 mm	0.0000 in
#11 BOLT	O.D.	0.3735 in 9.4869 mm	0.3745 in 9.5123 mm	0.3723 in 9.4564 mm	0.0000 mm
#12 BUSHING	I.D.	0.5620 in 14.2748 mm	0.5627 in 14.2926 mm	0.5645 in 14.3383 mm	0.0005 in
#12 BUSHING	O.D.	0.5610 in 14.2494 mm	0.5615 in 14.2621 mm	0.5595 in 14.2113 mm	0.0127 mm

SUBTASK 27-11-11-020-007

(4) Do this task: Aileron Installation, TASK 27-11-11-400-801.

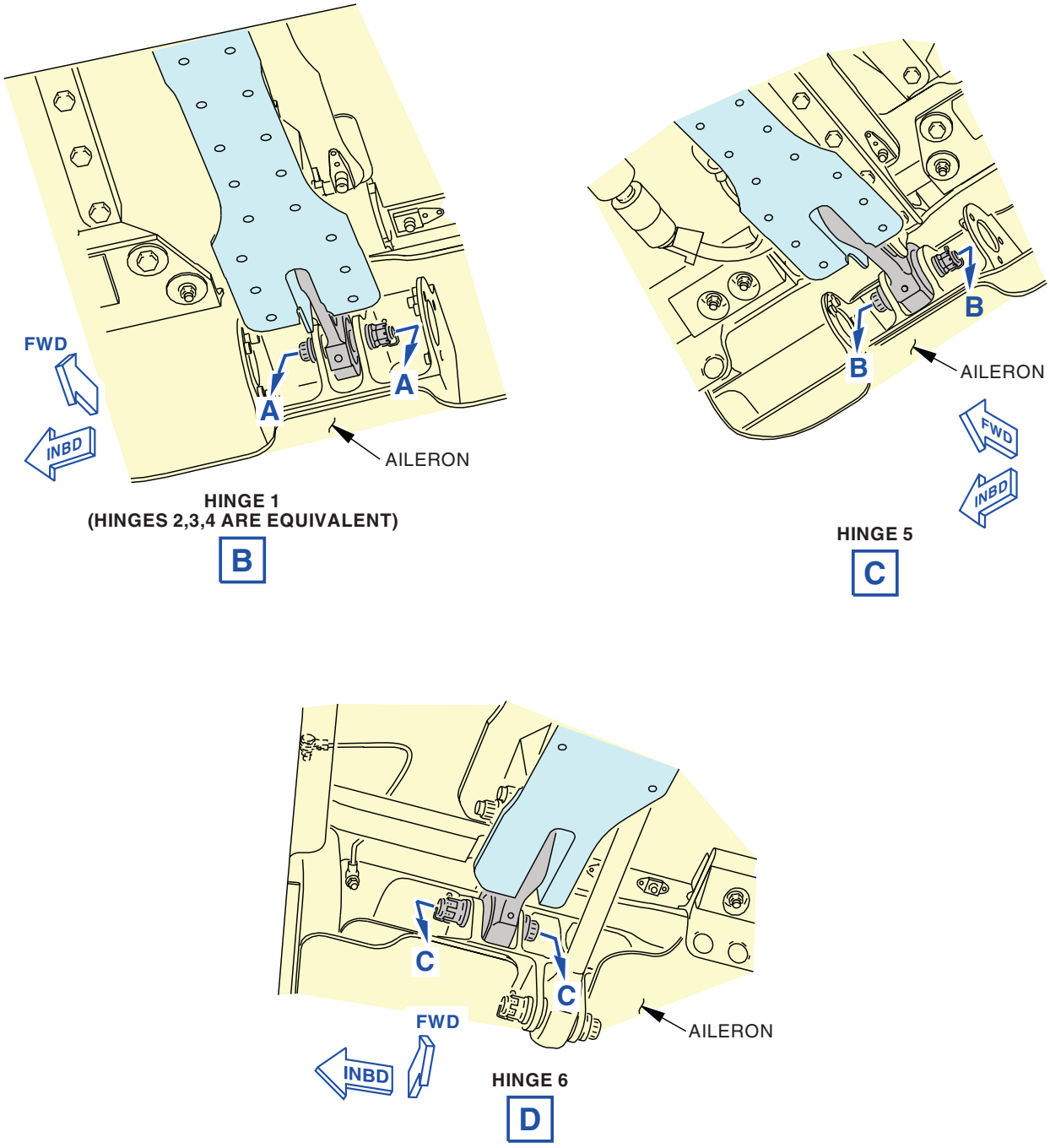
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Wear Limits
Figure 601/27-11-11-990-804 (Sheet 1 of 3)

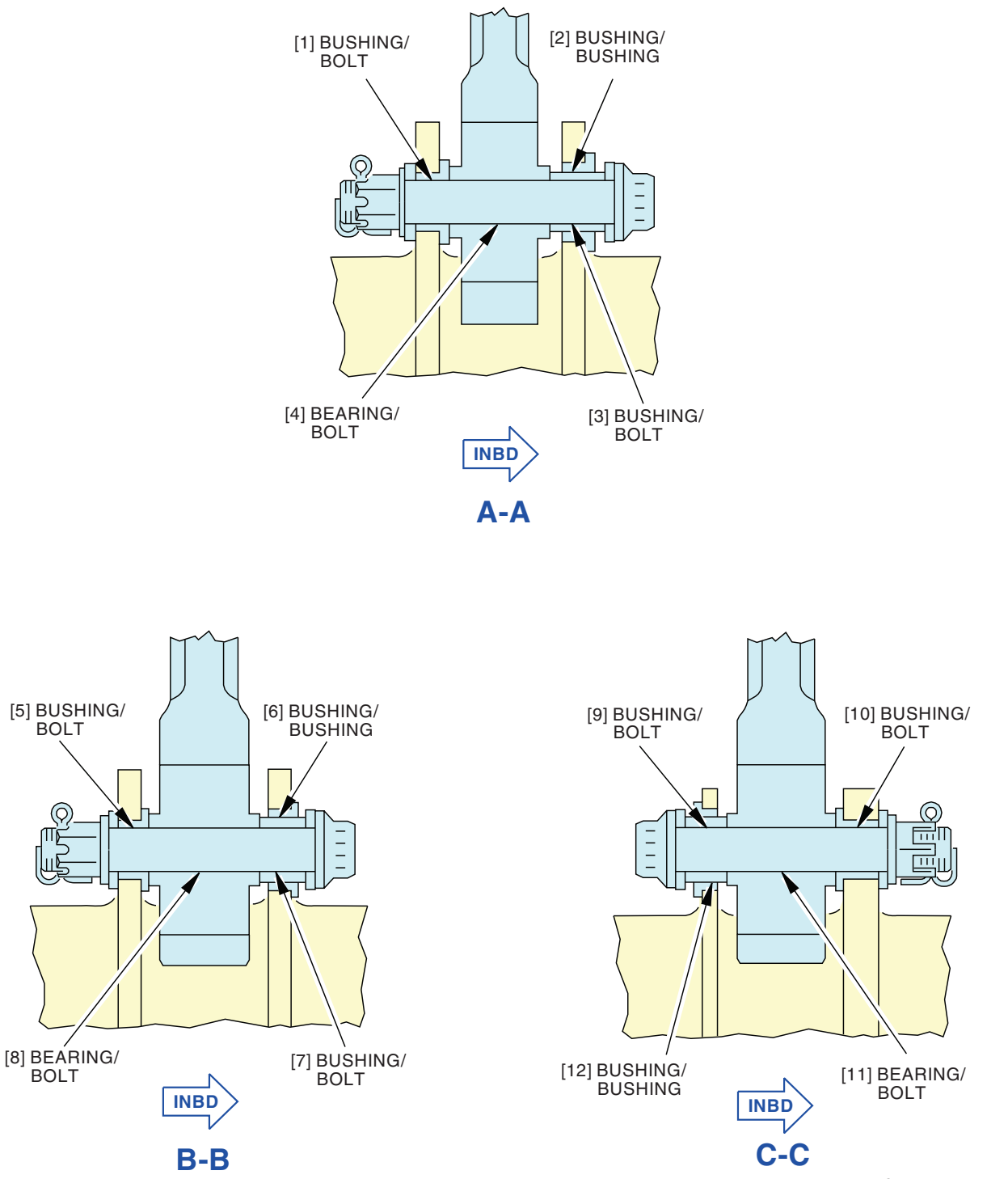
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Aileron Wear Limits
Figure 601/27-11-11-990-804 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Wear Limits
Figure 601/27-11-11-990-804 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON POWER CONTROL UNIT LINKAGE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Aileron Power Control Unit Linkage Removal

(2) Aileron Power Control Unit Linkage Installation.

B. The Aileron Power Control Unit Linkage is referred to as the aileron power control unit linkage (A) [1] or aileron power control unit linkage (B) [2] in this procedure.

C. This procedure is applicable for system A and B linkages.

TASK 27-11-14-000-801

2. Aileron Power Control Unit Linkage Removal

(Figure 401)

A. References

Reference	Title
27-11-71-000-801	Aileron Power Control Unit Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Removal

SUBTASK 27-11-14-480-001

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-14-020-001

(2) Do this task: Aileron Power Control Unit Removal, TASK 27-11-71-000-801.

D. Aileron Power Control Unit Linkage Removal

SUBTASK 27-11-14-020-002

(1) Do these steps to remove the aileron power control unit linkage (A) [1] or aileron power control unit linkage (B) [2]:

(a) Disconnect cable ABSA [22] and cable ABSB [19] at the turnbuckles [3].

(b) Remove the nuts [18], washers [17], washers [16] and bolts [15] to remove the retaining plates [14] from the aileron bus drum [20].

(c) Disengage cable ABSA [22] and cable ABSB [19] from the aileron bus drum [20].

(d) Safety the disconnected cable ABSA [22] and cable ABSB [19] to secure them.

(e) Remove the cotter pin [6] and washers [5] to remove the cable retention pins [4] between both support frame [7] and support frame [21].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Remove the nuts [23] and screws [25] to disconnect the System B upper quadrant assembly from the System A lower quadrant assembly.

(g) For system A, do these steps:

1) Remove the nuts [47], washers [46], washer [45], and bolt [44] to disconnect the diagonal support [39] from the structure.

2) Remove the nuts [34], washers [33], washer [32], and bolt [31] to remove the diagonal support [39] from the support frame [21].

3) Remove the nuts [49] to remove the hydraulic fittings.

4) Remove the nut [30], washer [29], washer [28], bolt [27], nut [38], washer [37], washer [36], and bolt [35] to remove the support frame [21] from the structure.

(h) For system B, do these steps:

1) Remove the nut [68], washer [69], bushing [70], taper shims [67], washer [72], and bolt [71] to disconnect the diagonal support [62] from the structure;

2) Remove the nut [55], washer [54], washer [53], and bolt [52] to remove the diagonal support [62] from the support assembly [56];

3) Remove the nut [30], washer [29], washer [28], bolt [27], nut [66], washer [65], washer [64], and bolt [63] to remove the support frame [7] from the structure.

(i) For system A, do these steps:

1) Remove the nuts [43], washers [42], washers [41], and bolts [40] to disconnect the support assembly [50] from the structure;

2) Remove the aileron power control unit linkage (A) [1] from the airplane.

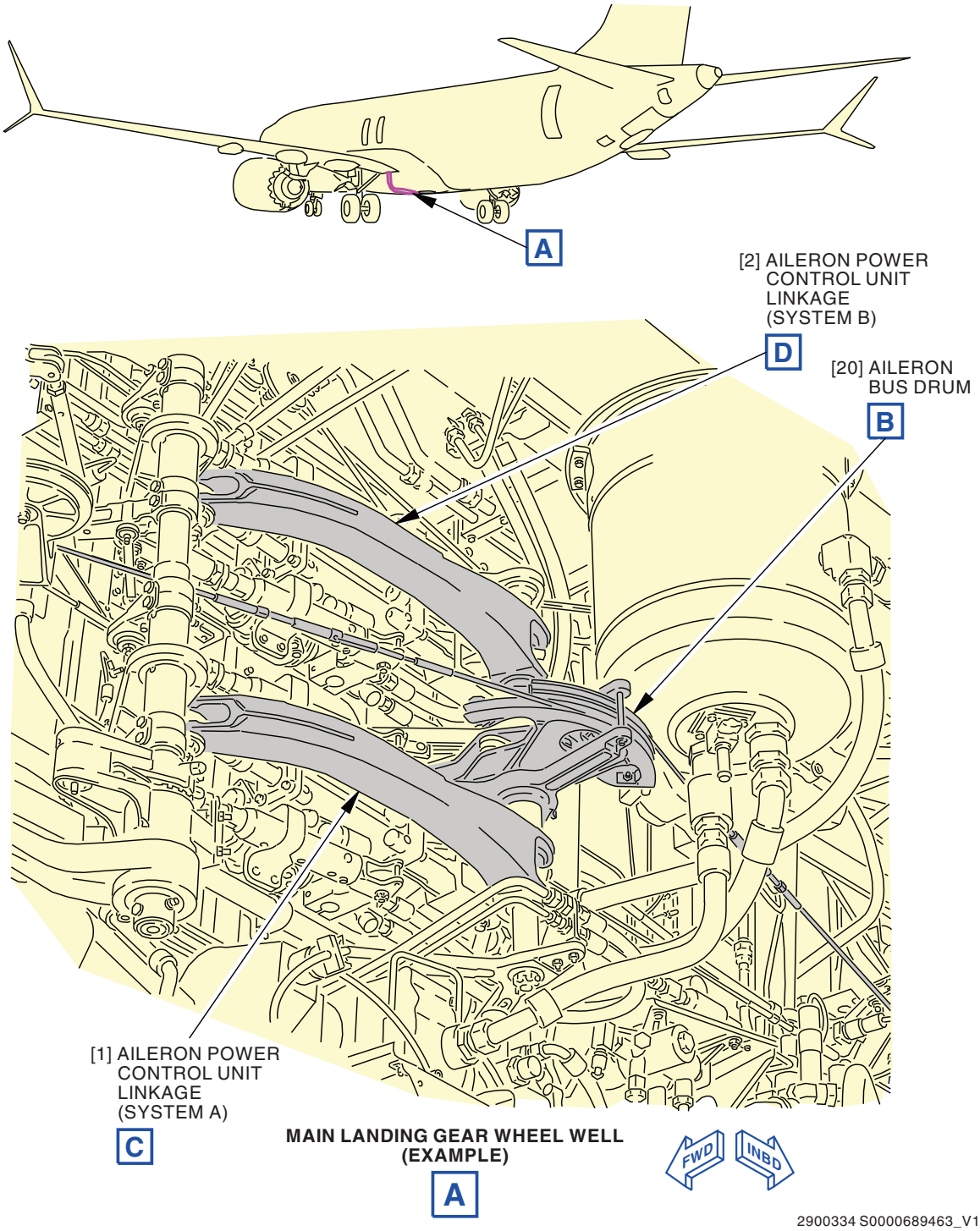
(j) For system B, do these steps:

1) Remove the nuts [43], washers [42], washers [41], and bolts [40] to disconnect the support assemblies [56] from the structure.

2) Remove the aileron power control unit linkage (B) [2] from the airplane.

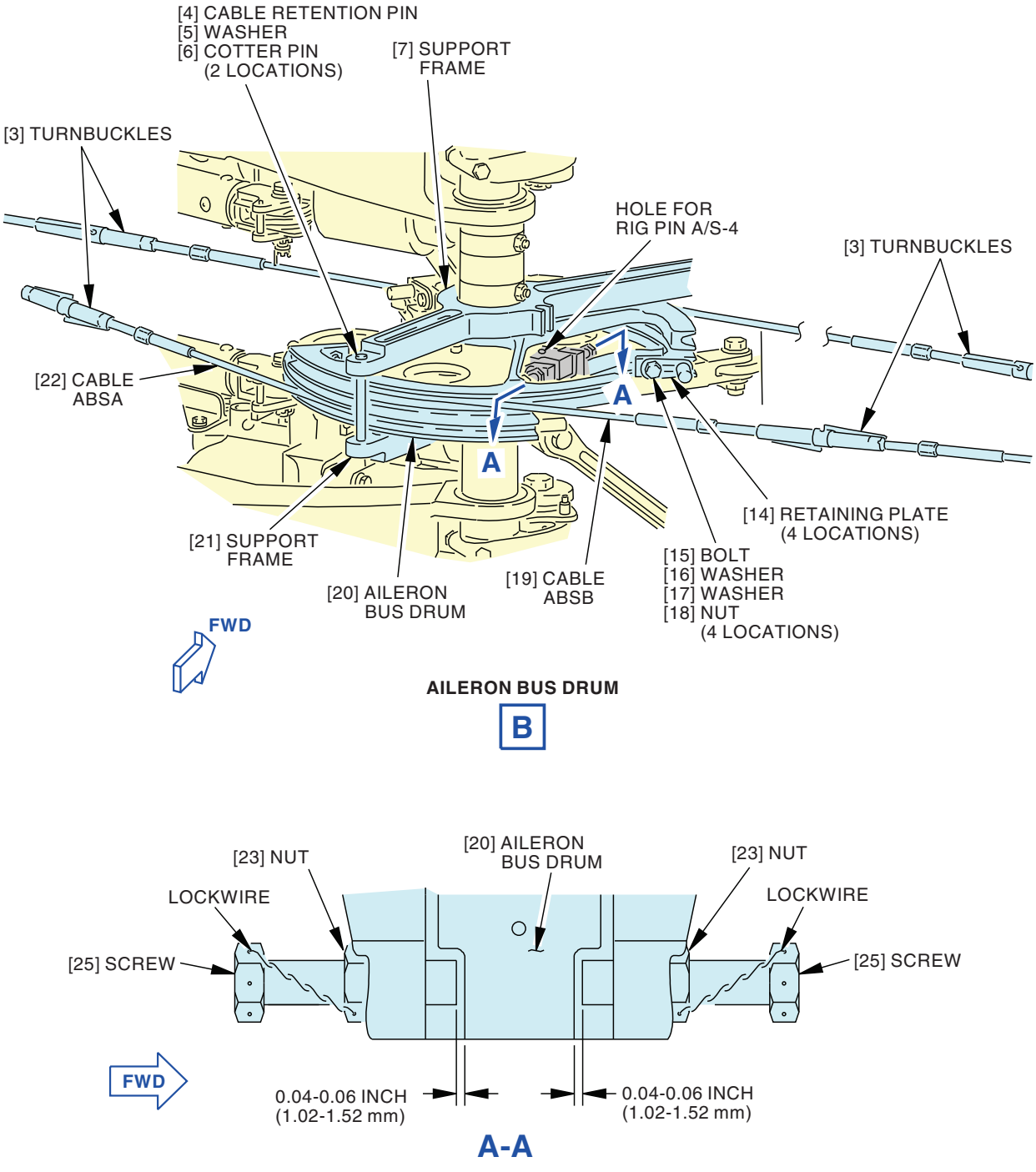
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



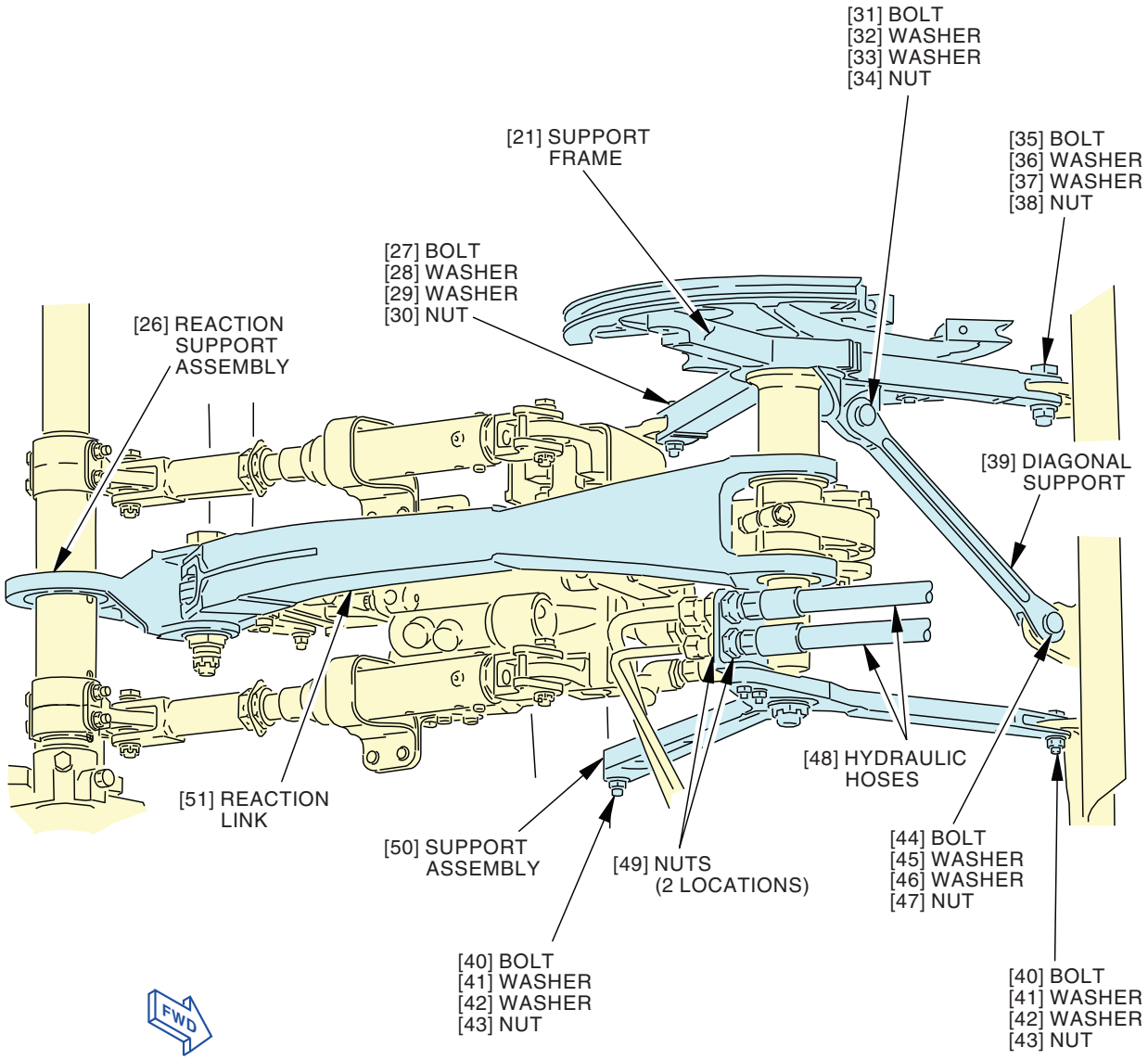
Aileron Power Control Unit Linkage Installation
Figure 401/27-11-14-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Power Control Unit Linkage Installation
Figure 401/27-11-14-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



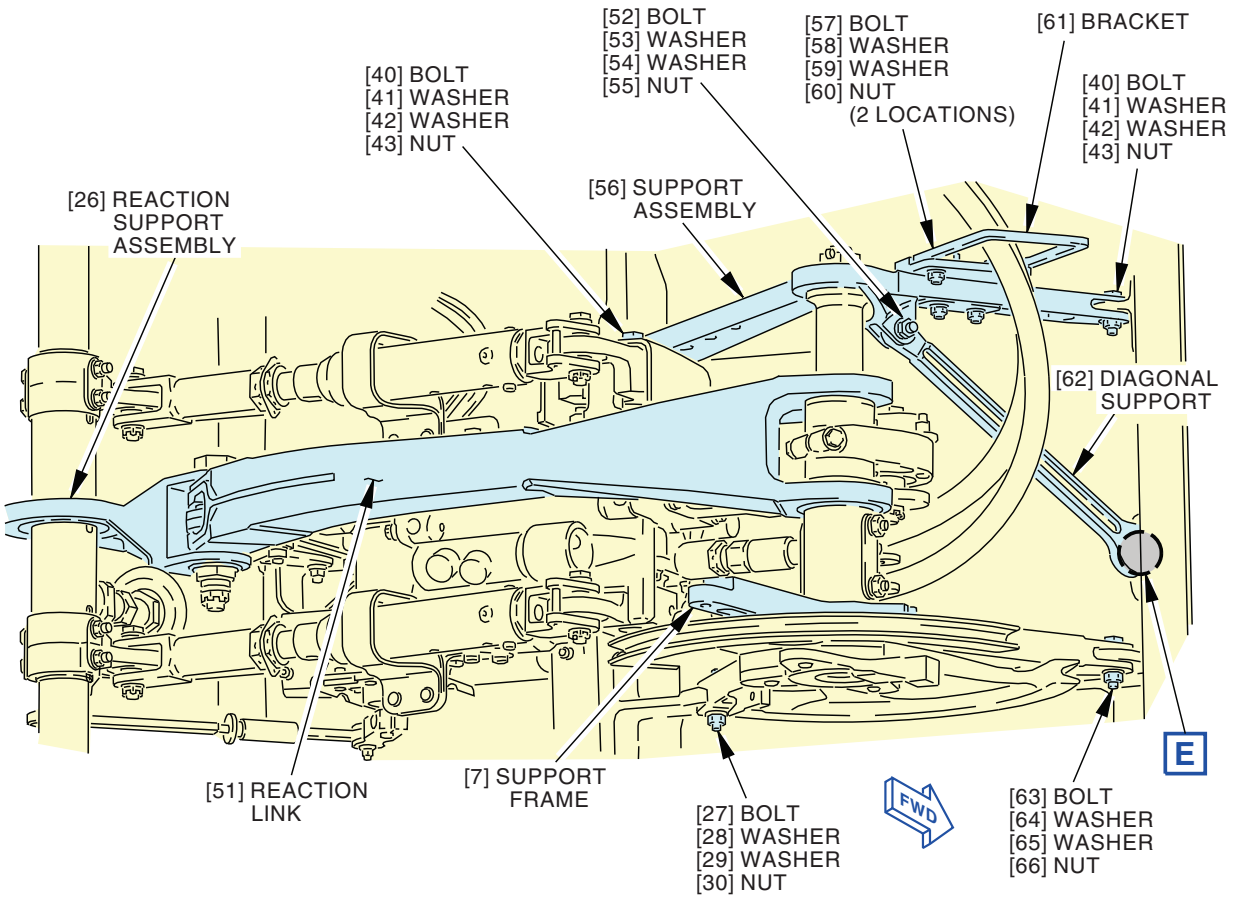
AILERON POWER CONTROL UNIT LINKAGE (SYSTEM A)

C

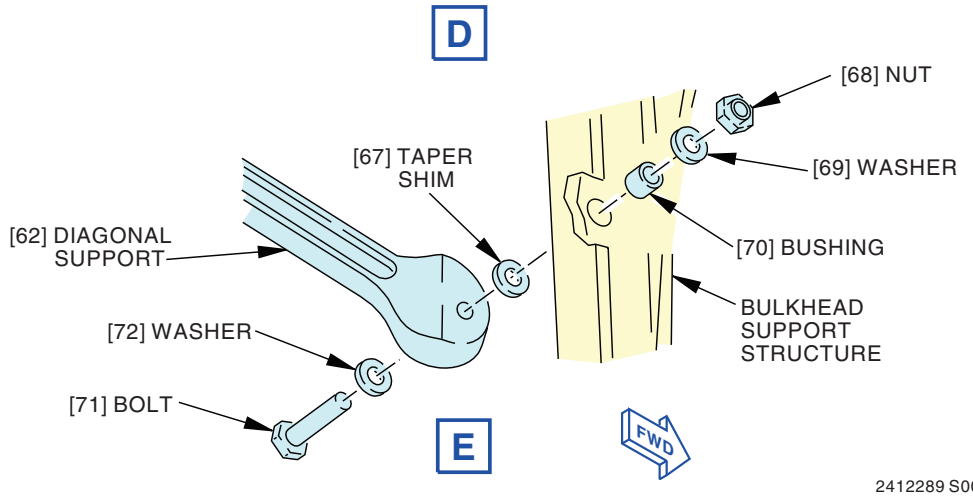
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Aileron Power Control Unit Linkage Installation
Figure 401/27-11-14-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON POWER CONTROL UNIT LINKAGE (SYSTEM B)



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Aileron Power Control Unit Linkage Installation
Figure 401/27-11-14-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-14-400-801

3. Aileron Power Control Unit Linkage Installation

(Figure 401)

A. References

Reference	Title
20-10-93-400-801	Control Cable End Retaining Plate - Installation (P/B 401)
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-11-71-400-801	Aileron Power Control Unit Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G02436	Lockwire - MS20995NC40, Monel - 0.040 Inch (1.016 mm) Diameter	NASM20995

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Aileron Power Control Unit Linkage Installation

SUBTASK 27-11-14-420-001

(1) Install the aileron power control unit linkage (A) [1] or aileron power control unit linkage (B) [2].

(a) Put the aileron power control unit linkage (A) [1] or aileron power control unit linkage (B) [2] in its position.

(b) Apply sealant, A00247, to the bolts [40].

(c) Install the bolts [40], washers [41], washer [42], and nuts [43] to connect the support assembly [50] or support assembly [56] to the structure.

(d) For system A, do these steps:

1) Apply sealant, A00247, to bolt [31], bolt [44], bolts [27], and bolts [35].

2) Install the diagonal support [39] to the support frame [21] with bolt [31], washer [32], washer [33], and nut [34].

3) Connect the diagonal support [39] to the structure with bolt [44], washer [45], washer [46], and nut [47].

4) Install the hydraulic fittings with nuts [49].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) Install the bolt [27], washer [28], washer [29], nut [30], bolt [35], washer [36], washer [37], and nut [38] to connect the support frame [21] to the structure.

(e) For system B, do these steps:

1) Apply sealant, A00247, to bolt [52], bolt [71], bolt [27], bolt [63], taper shims [67], and bushing [70].

2) Install the diagonal support [62] to the support assembly [56] with bolt [52], washer [53], washer [54], and nut [55].

3) Install the diagonal support [62] to the structure with bolt [71], washer [72], taper shims [67], bushing [70], washer [69], and nut [68].

a) Install or remove the taper shim [67] as necessary to decrease the gap to 0.003 in. (0.076 mm) or less, between the diagonal support [62] and bulkhead support structure that it is attached to.

4) Install the bolt [27], washer [28], washer [29], nut [30], bolt [63], washer [64], washer [65], nut [66] to connect the support frame [7] to the structure.

(f) Install the cable retention pins [4], washers [5], and cotter pins [6].

1) Apply a maximum preload of 35 in-lb (4 N·m) during installation.

SUBTASK 27-11-14-420-002

(2) Do this task: Aileron Power Control Unit Installation, TASK 27-11-71-400-801.

SUBTASK 27-11-14-420-003

(3) Install the screws [25] and nuts [23].

(a) Make sure that the gap between the screw [25] and aileron bus drum [20] is 0.04 in. (1.02 mm) - 0.06 in. (1.52 mm).

(b) Install MS20995NC40 lockwire, G02436, with one continuous twist on the two screws [25].

SUBTASK 27-11-14-480-002

(4) Install the rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum [20].

SUBTASK 27-11-14-420-005

(5) Do these steps to connect the cable ABSA [22] and cable ABSB [19].

(a) Connect the cable ABSA [22] and cable ABSB [19] on the aileron bus drum [20].

1) Install the retaining plates [14] with bolts [15], washers [16], washers [17], and nuts [18] (TASK 20-10-93-400-801).

(b) Connect the cable ABSA [22] and cable ABSB [19] at the turnbuckles [3].

(c) Tighten the turnbuckles [3] but keep the cables tension at zero.

(d) Adjust the cable ABSA [22] and cable ABSB [19] (TASK 27-11-00-820-801).

(e) Install the turnbuckles clip locks.

SUBTASK 27-11-14-480-005

(6) Remove the rig pin A/S-4, from the rigging pin kit, SPL-1585, from the aileron bus drum [20].

SUBTASK 27-11-14-710-001

(7) Do this task: Aileron Travel Test, TASK 27-11-00-700-810.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON BALANCE TAB - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of an aileron balance tab

(2) An installation of an aileron balance tab.

B. This procedure is applicable for the left and right aileron balance tab.

TASK 27-11-21-000-801

2. Aileron Balance Tab Removal

(Figure 401)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)

B. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

C. Aileron Balance Tab Removal

SUBTASK 27-11-21-860-001

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-21-020-001

(2) To remove an aileron balance tab [2], do these steps (Figure 401):

(a) Remove the bolts [8] that attach the forward removable fairing [9] to the aileron [1].

1) Remove the forward removable fairing [9].

(b) Remove the bolts [6] that attach the aft removable fairing [7] to the aileron balance tab [2].

1) Remove the aft removable fairing [7].

(c) Remove the pins [12], nuts [11], washers [13], bushings [16], washers [15], and bolts [14] that attach the aileron tab control rods [10].

(d) Turn the trailing edge of the aileron balance tab [2] up until you can see the bolts [3] that attach the hinge covers [5].

(e) Remove the bolts [3] and washers [4].

(f) Remove the hinge covers [5].

(g) Disconnect the bonding jumpers [29] that are adjacent to the tab hinge 1 and 5.

1) Remove the nuts [32], washers [31], and bolts [30] to disconnect the bonding jumpers [29].

(h) To remove an aileron balance tab [2] from the aileron [1]:

1) Do these steps at tab hinge 2:

a) Remove the pin [22], nut [17], washer [21], bushing [18], washer [20], and bolt [19].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<1> Discard the pin [22].

2) Do these steps at hinge 3:

a) Remove the pin [23], nut [28], washer [27], washer [25], bushings [26], and bolt [24].

<1> Discard the pin [23].

3) Do these steps at tab hinge 1, 4, and 5:

a) Remove the pins [37], nuts [38], washers [39], bushings [35], washers [34], and bolts [33].

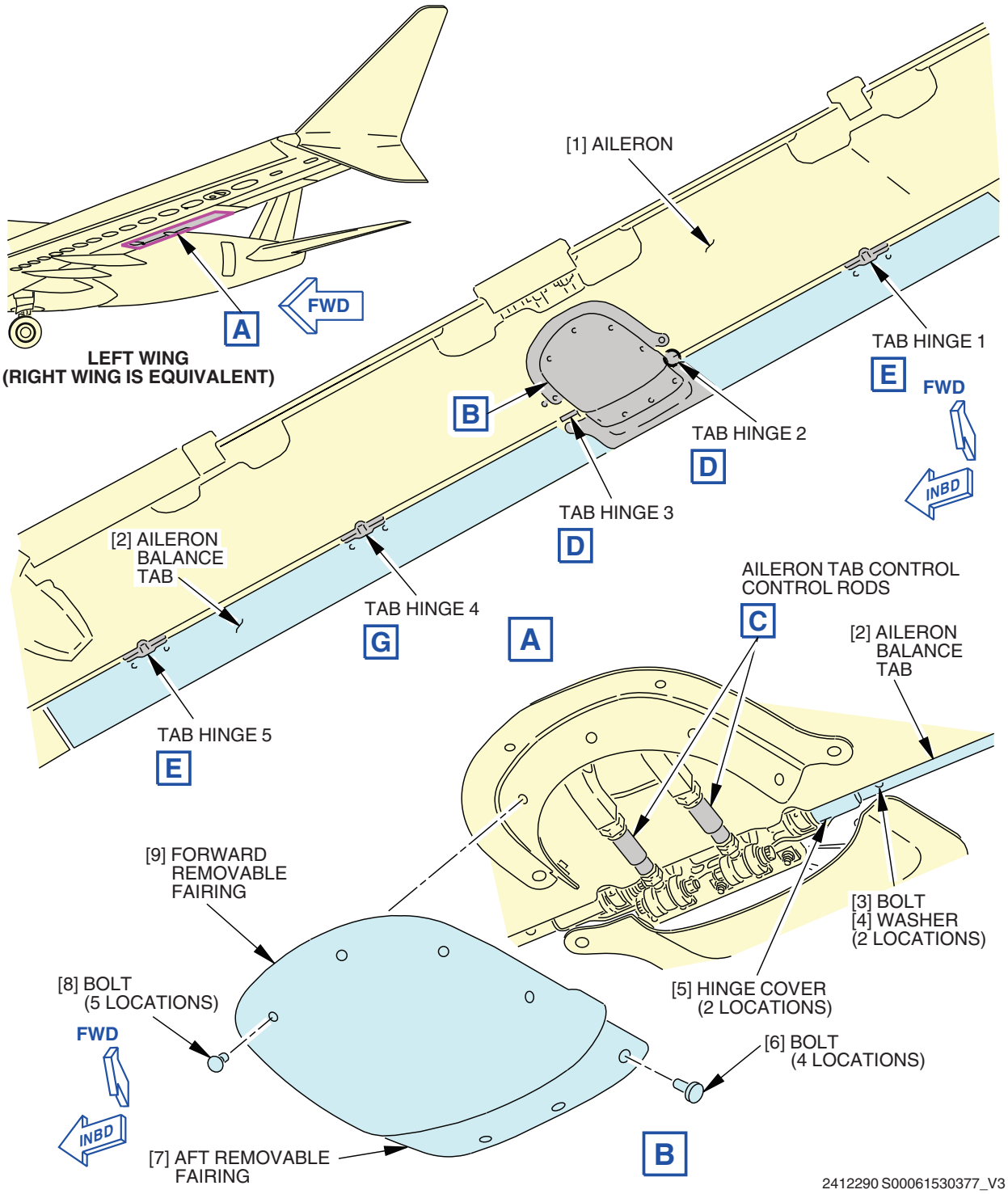
<1> Discard the pins [37].

(i) Carefully remove the aileron balance tab [2] from the hinge assemblies.

(j) Remove the aileron balance tab [2] from the aileron [1].

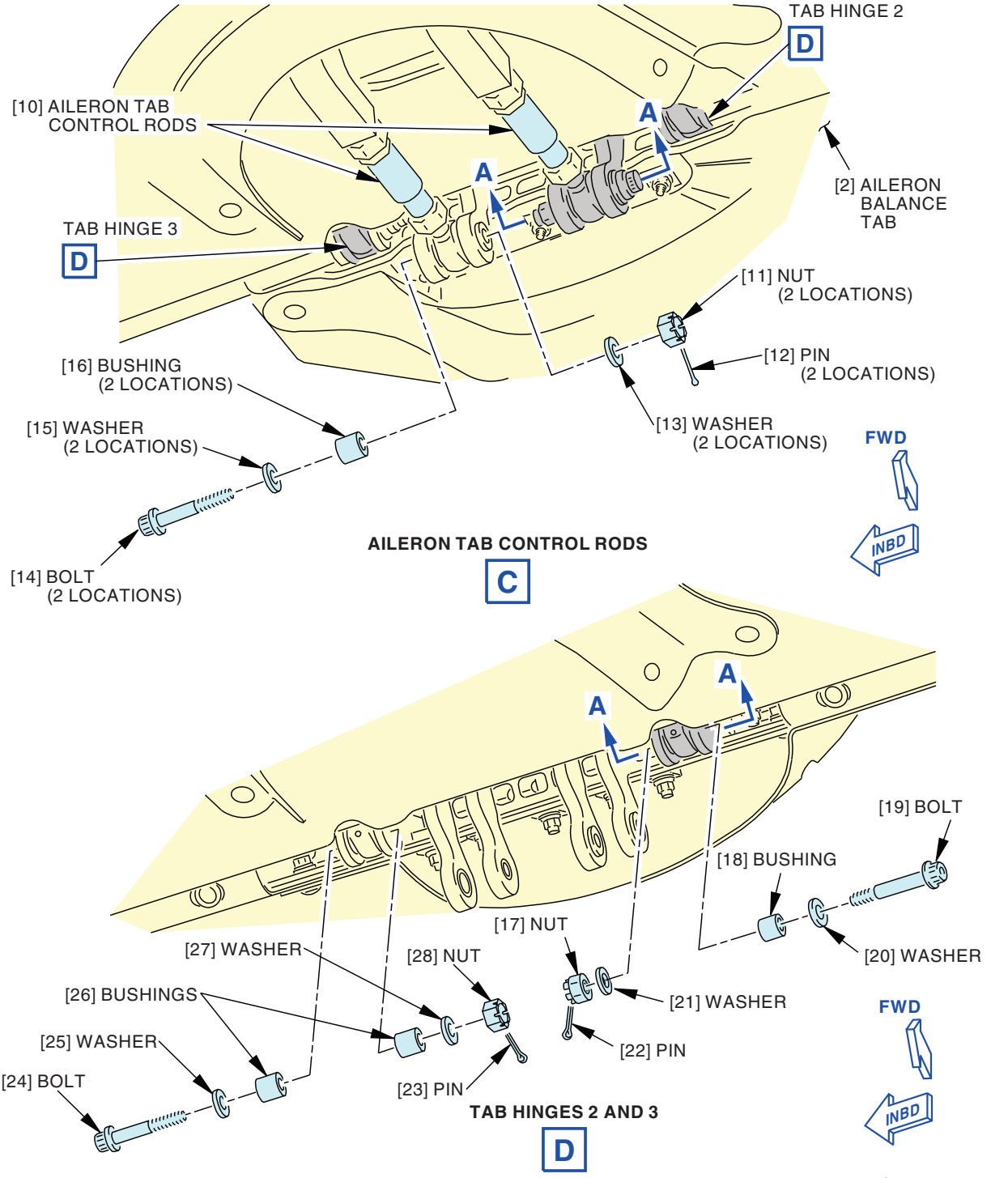
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



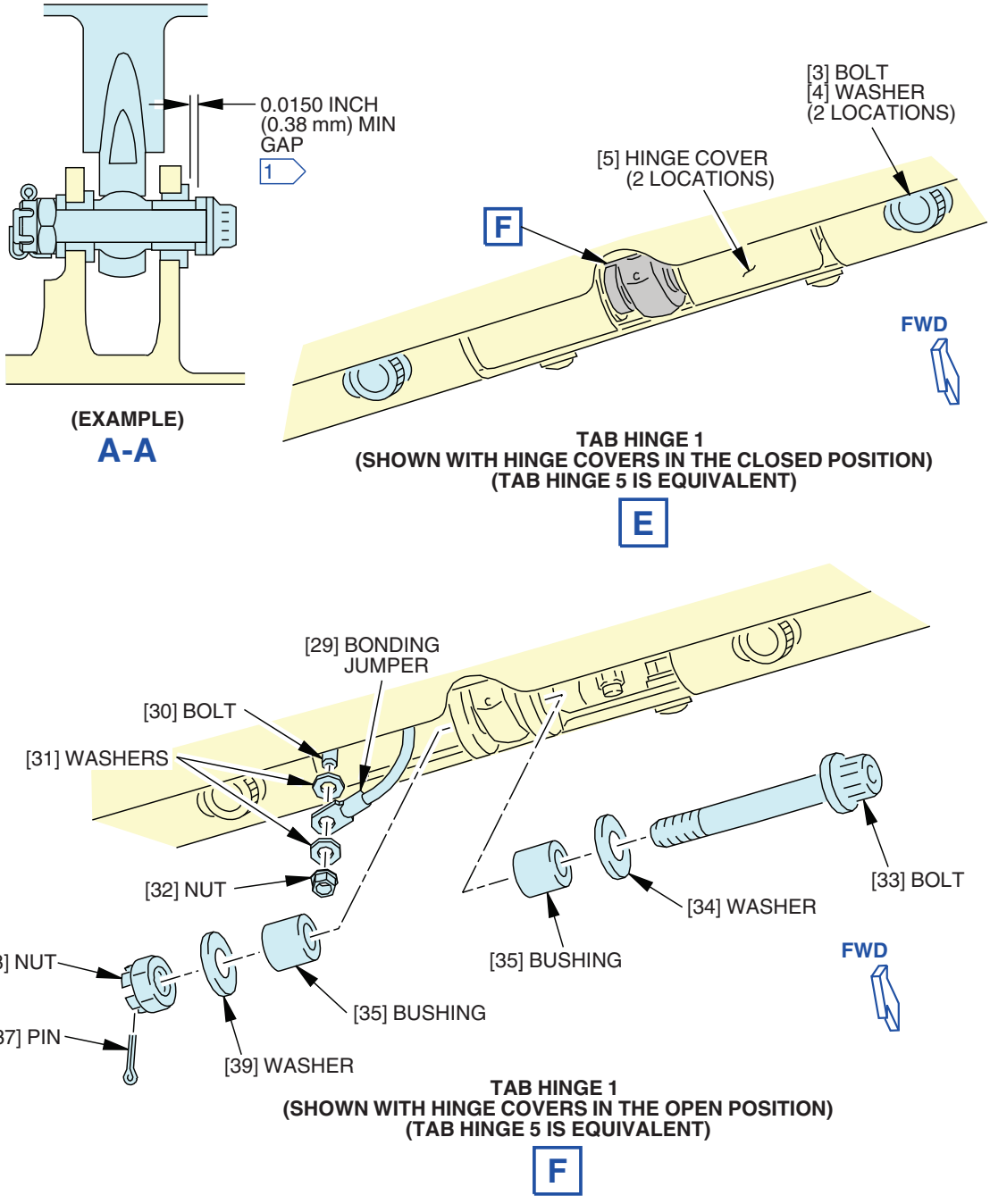
Aileron Balance Tab Installation
Figure 401/27-11-21-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Tab Installation
Figure 401/27-11-21-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

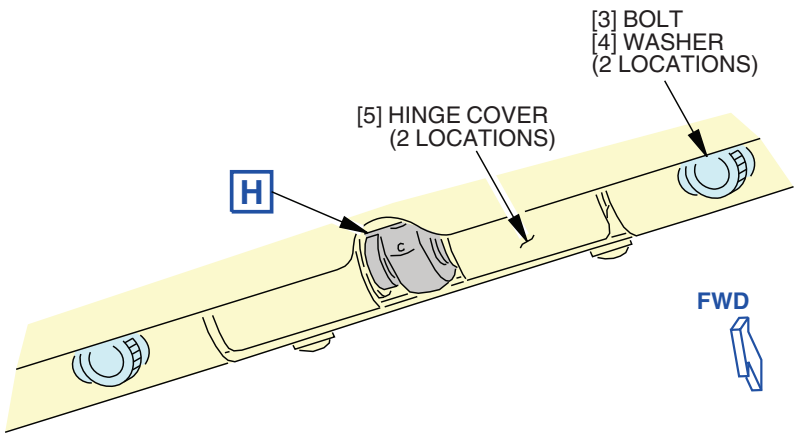


1 0.0150 INCH (0.38 mm) GAP REFERS TO TAB HINGE 2 AND CONTROL ROD CONNECTIONS ONLY.

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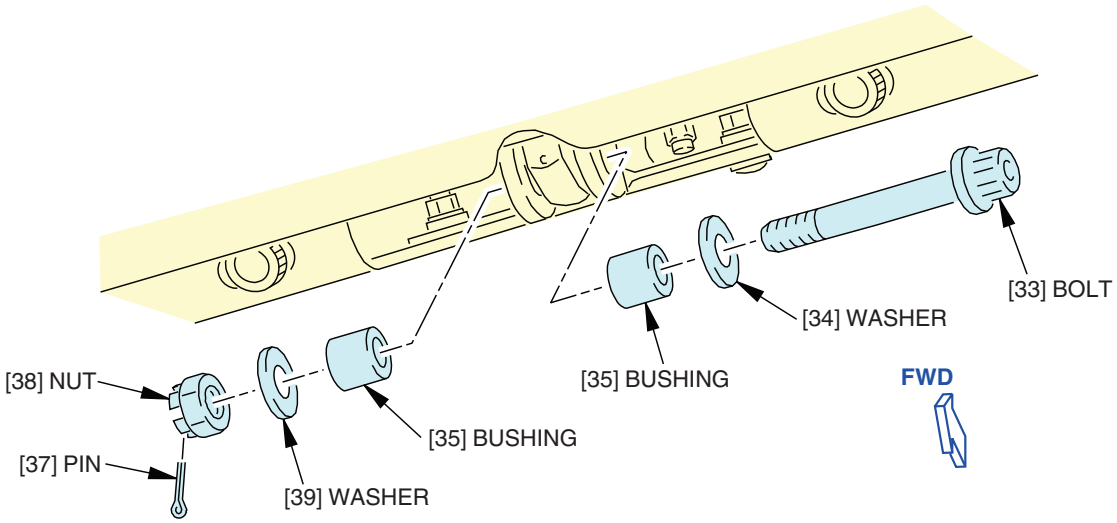
Aileron Balance Tab Installation
Figure 401/27-11-21-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TAB HINGE 4
(SHOWN WITH HINGE COVERS IN THE CLOSED POSITION)

G



TAB HINGE 4
(SHOWN WITH HINGE COVERS IN THE OPEN POSITION)

H

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Aileron Balance Tab Installation
Figure 401/27-11-21-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-21-400-801

3. Aileron Balance Tab Installation

(Figure 401, Figure 402)

A. References

Reference	Title
27-11-00-700-809	Aileron Balance Tab Position Test (P/B 501)
27-11-00-820-803	Aileron Balance Tab Adjustment (Recommended Method) (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-11-21-200-801	Aileron Balance Tab Inspection (P/B 601)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A00767	Sealant - Fuel Tank	BMS5-45
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00130	Alcohol - Isopropyl	TT-I-735
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
D00633	Grease - Aircraft General Purpose	BMS3-33
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Aileron balance tab	27-11-21-01-100	PTF ALL
		27-11-21-01-105	PTF ALL
12	Pin	27-11-21-01-010	PTF ALL
22	Pin	27-11-21-01-060	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
23	Pin	27-11-21-01-060	PTF ALL
37	Pin	27-11-21-01-060	PTF ALL

E. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

F. Aileron Balance Tab Installation

SUBTASK 27-11-21-220-001

(1) If you install used bearings, bushings, and bolts, do this task: Aileron Balance Tab Inspection, TASK 27-11-21-200-801.

SUBTASK 27-11-21-210-001

WARNING MAKE SURE THAT YOU HAVE THE AILERON TABS BALANCED. IF NOT BALANCED, THE AILERON TABS CAN CAUSE UNWANTED FLUTTER AND INSTABILITY OF THE AIRPLANE. THIS CAN CAUSE AN IN-FLIGHT EMERGENCY. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Make sure that the balance panels have the correct number of the tab adjust weights (Figure 402).

(a) If it is necessary, add or remove tab adjust weights (SRM SUBJECT 51-60-03).

SUBTASK 27-11-21-640-001

(3) Lubricate the hinge bearings, bushing [18], bushings [26], bushings [35], bolt [19], bolt [24], and bolt [33] with grease, D00633, except on the threads (Figure 401).

SUBTASK 27-11-21-420-001

(4) To install an aileron balance tab [2] to the aileron [1], do these steps:

(a) Put the aileron balance tab [2] in its position.

(b) Install the aileron balance tab [2] hinge fittings and aileron [1] fittings:

1) Do these steps at tab hinge 1, 4, and 5:

a) Install the bolts [33], washers [34], bushings [35], washers [39], and nuts [38].

b) Tighten the nuts [38] to the lower end of the torque range 65 ±15 in-lb (7.3 ±1.7 N·m).

<1> Continue to tighten the nuts [38] until the first castellation aligns with the hole in the bolts [33].

c) Install the new pins [37].

2) Do these steps at tab hinge 3:

a) Install the bolt [24], washer [25], bushings [26], washer [27], and nut [28].

b) Tighten the nut [28] to the lower end of the torque range 65 ±15 in-lb (7.3 ±1.7 N·m).

<1> Continue to tighten the nut [28] until the first castellation aligns with the hole in the bolt [24].

c) Install the new pin [23].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Do these steps at tab hinge 2:
   a) Install the bolt [19], washer [20], bushing [18], washer [21], and nut [17].
   b) Make sure that there is a minimum of 0.0150 in. (0.38 mm) gap between the washer [20] and bushing [18] (View A-A, Figure 401).
   c) Tighten the nut [17] to the lower end of the torque range 65 ±15 in-lb (7.3 ±1.7 N·m).
      <1> Continue to tighten the nut [17] until the first castellation aligns with the hole in the bolt [19].
   d) Install the new pin [22].

(c) Install the bonding jumper [29] at tab hinge 1 and 5 as follows:
1) Use cotton wiper, G00034, moist in alcohol, B00130, or solvent, B00148, to clean the hole of the bolt [30], washers [31], and bonding jumper [29].
2) Install the bonding jumper [29] through the slot in the hinge cover [5].
3) Install the bolt [30], washers [31], and nut [32] that attach the bonding jumper [29].
4) Measure the electrical bonding resistance between the head of the bolt [30] and aileron balance tab [2] (SWPM 20-20-00).
   a) Use intrinsically safe approved bonding meter, COM-1550.
   b) Make sure that the electrical bonding resistance is 300 000 ohm or less.
5) Measure the electrical bonding resistance between the nut [32] and bonding jumper [29] (SWPM 20-20-00).
   a) Use intrinsically safe approved bonding meter, COM-1550.
   b) Make sure that the electrical bonding resistance is 1.0 ohm (1000 milliohm) or less.
6) Fillet seal the nut [32], washer [31], and terminal of the bonding jumper [29] with sealant, A00247, or sealant, A00767.

(d) To install the hinge covers [5] to close the hinge bolt access areas, do these steps:
1) Install the bolts [3] and washers [4].
2) Make sure that the washers [4] touch the aileron balance tab [2] on a minimum of two edges.
3) Tighten the bolts [3] to 25 ±3 in-lb (2.8 ±0.4 N·m).

(e) Connect the aileron tab control rods [10] to the aileron balance tab [2]:
1) Install the bolts [14], washers [15], bushings [16], washers [13], and nuts [11].
   a) Make sure that there is a minimum of 0.0150 in. (0.38 mm) gap between the washers [15] and bushings [16] (View A-A, Figure 401).
2) Tighten the nuts [11] to 65 ±15 in-lb (7.3 ±1.7 N·m).
3) Install the new pins [12].

SUBTASK 27-11-21-420-002

(5) Install the bolts [6] to attach the aft removable fairing [7] to the aileron balance tab [2].

SUBTASK 27-11-21-420-003

(6) Install the bolts [8] to attach the forward removable fairing [9] to the aileron [1].
(a) Apply sealant, A02315, to the upper edge of the forward removable fairing [9] to get aerodynamic smoothing.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-21-710-001

(7) Operate the aileron control wheel in a full cycle.

(a) Make sure that the aileron balance tab [2] moves freely in the opposite direction of the aileron [1].

SUBTASK 27-11-21-860-002

(8) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-21-820-001

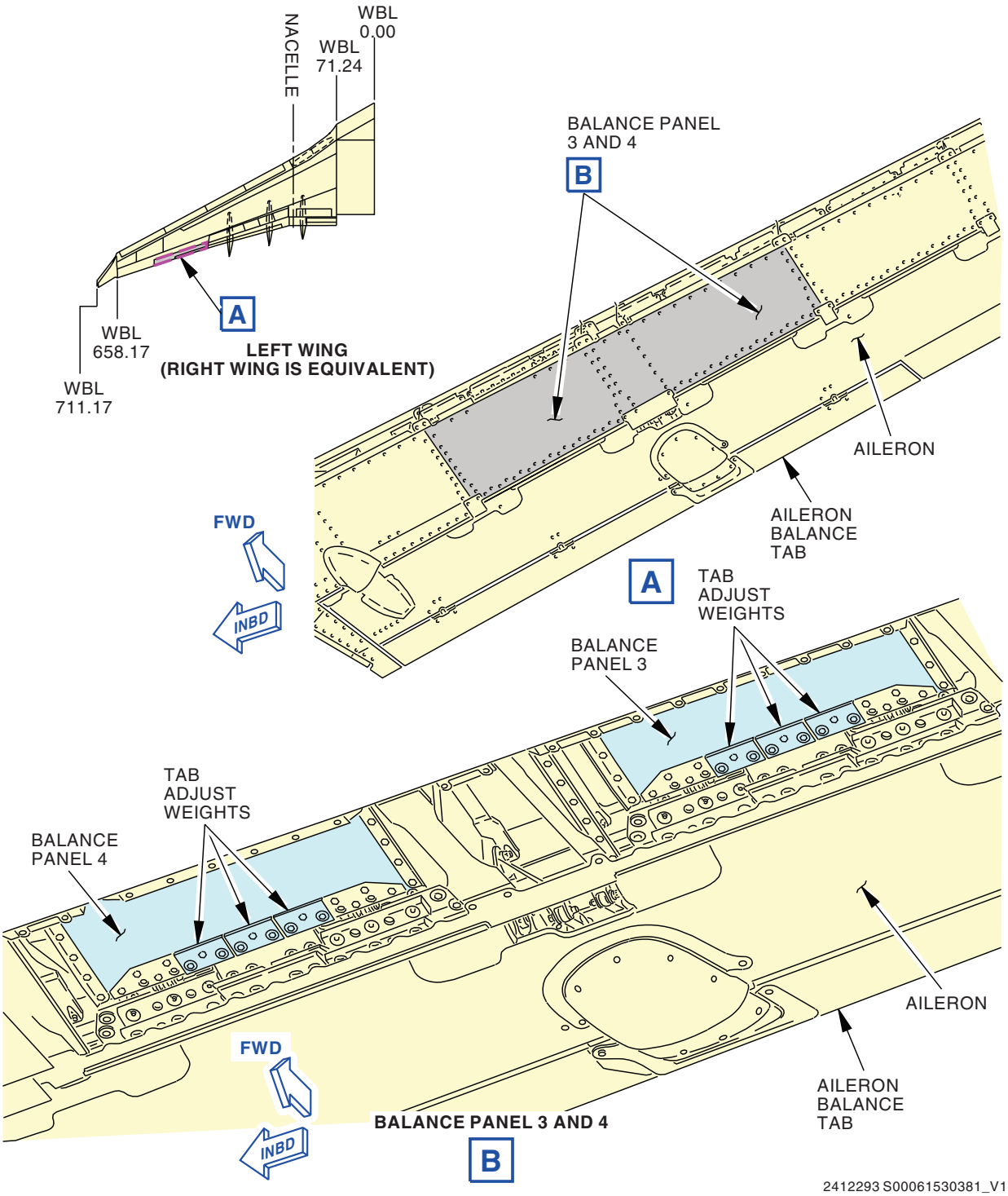
(9) Do this task: Aileron Balance Tab Adjustment (Recommended Method), TASK 27-11-00-820-803.

SUBTASK 27-11-21-730-001

(10) Do this task: Aileron Balance Tab Position Test, TASK 27-11-00-700-809.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Tab Adjust Weights Location
Figure 402/27-11-21-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON BALANCE TAB - INSPECTION/CHECK

1. General

A. This procedure contains the task to do a check of the aileron balance tab.

TASK 27-11-21-200-801

2. Aileron Balance Tab Inspection

(Figure 601)

A. General

(1) This task gives wear limits and repair data for the aileron balance tab inspection.

B. References

Reference	Title
27-11-21-000-801	Aileron Balance Tab Removal (P/B 401)
27-11-21-400-801	Aileron Balance Tab Installation (P/B 401)
27-11-34-000-801	Aileron Tab Control Rods Removal (P/B 401)
27-11-34-400-801	Aileron Tab Control Rods Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

E. Procedure

SUBTASK 27-11-21-020-002

(1) Do this task: Aileron Balance Tab Removal, TASK 27-11-21-000-801 or Aileron Tab Control Rods Removal, TASK 27-11-34-000-801.

SUBTASK 27-11-21-020-003

(2) Use these tools to do a check for wear on the parts of the aileron balance tab:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-11-21-020-004

(3) Examine the parts of the aileron balance tab for worn areas:

(a) Remove the parts you want to do a check.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are out of tolerance, replace or repair the parts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-11-21-993-802

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	0.3750 in. (9.5250 mm)	0.3756 in. (9.5402 mm)	0.3770 in. (9.5758 mm)	0.0030 in. (0.0762 mm)
#1 BUSHING	O.D.	0.3740 in. (9.4996 mm)	0.3745 in. (9.5123 mm)	0.3725 in. (9.4615 mm)	
#2 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.0030 in. (0.0762 mm)
#2 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	
#3 BEARING	I.D.	0.2495 in. (6.3373 mm)	0.2500 in. (6.3500 mm)	0.2510 in. (6.3754 mm)	0.0025 in. (0.0635 mm)
#3 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	
#4 BUSHING	I.D.	0.3750 in. (9.5250 mm)	0.3756 in. (9.5402 mm)	0.3770 in. (9.5758 mm)	0.0030 in. (0.0762 mm)
#4 BUSHING	O.D.	0.3740 in. (9.4996 mm)	0.3745 in. (9.5123 mm)	0.3725 in. (9.4615 mm)	
#5 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.0030 in. (0.0762 mm)
#5 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	
#6 BEARING	I.D.	0.2495 in. (6.3373 mm)	0.2500 in. (6.3500 mm)	0.2510 in. (6.3754 mm)	0.0025 in. (0.0635 mm)
#6 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	
#7 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.0030 in. (0.0762 mm)
#7 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	
#8 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.0030 in. (0.0762 mm)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-11-21-993-802 (Continued)

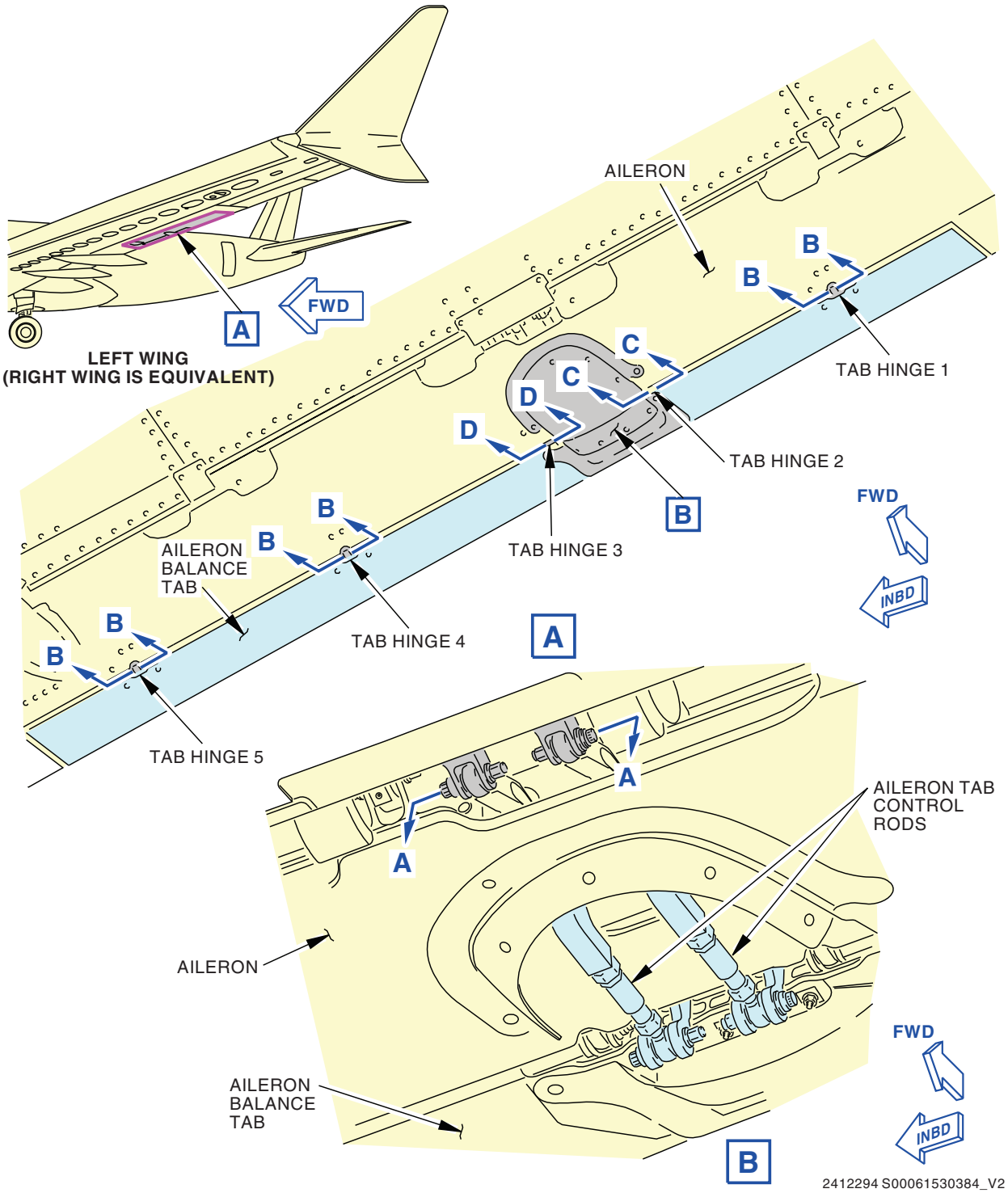
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#8 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	

SUBTASK 27-11-21-020-005

(4) Do this task: Aileron Balance Tab Installation, TASK 27-11-21-400-801 or Aileron Tab Control Rods Installation, TASK 27-11-34-400-801.

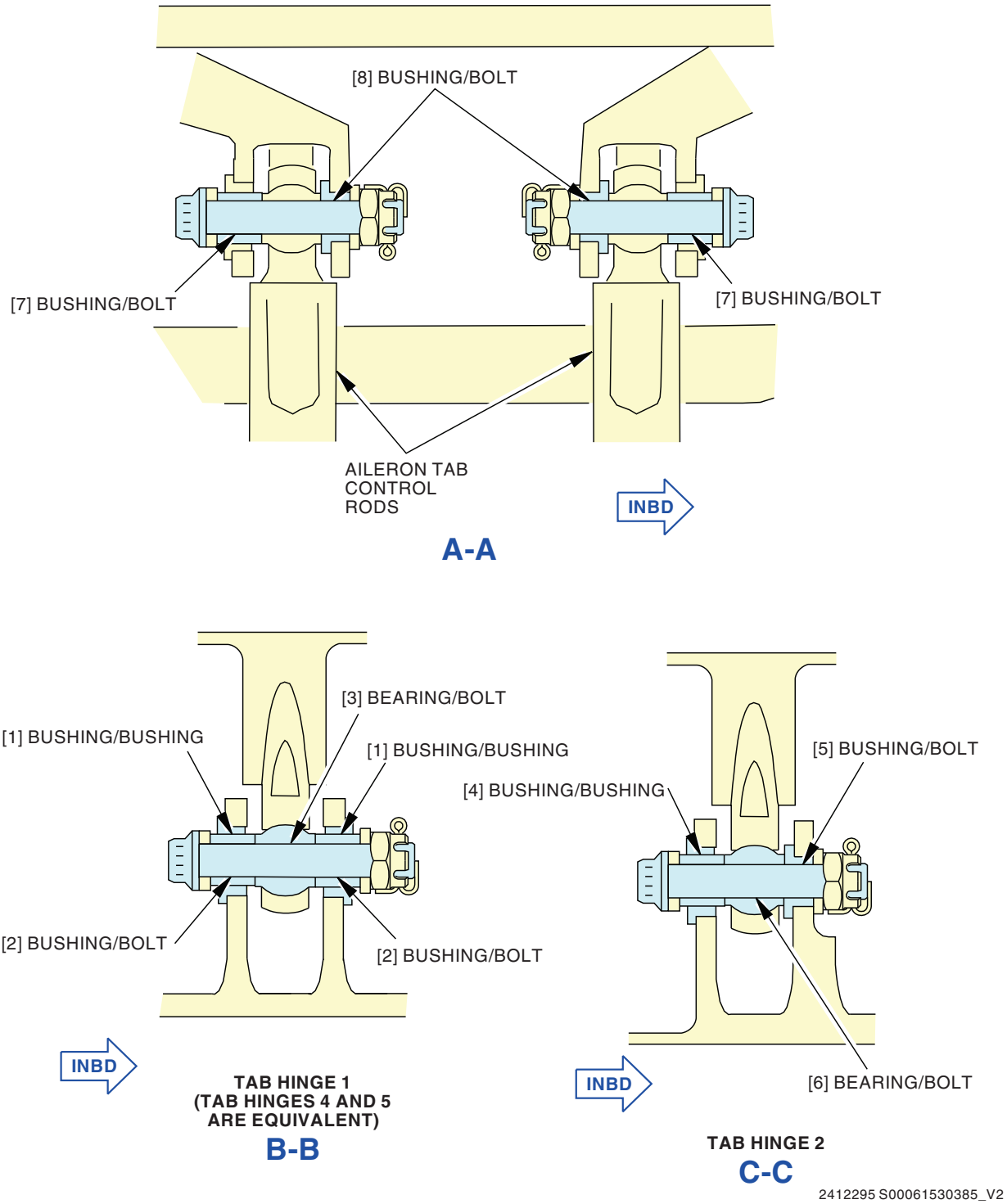
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



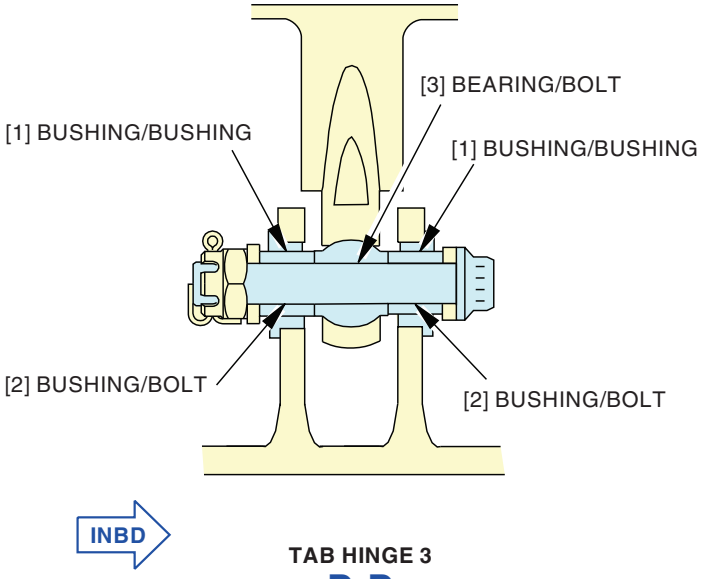
Aileron Balance Tab Wear Limits
Figure 601/27-11-21-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Tab Wear Limits
Figure 601/27-11-21-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TAB HINGE 3
D-D

2884123 S0000684254_V1

Aileron Balance Tab Wear Limits
Figure 601/27-11-21-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON WING QUADRANT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron wing quadrant

(2) An installation of the aileron wing quadrant.

TASK 27-11-24-000-801

2. Aileron Wing Quadrant Removal

(Figure 401)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
57-50-02-000-803	Outboard Fixed Trailing Edge Lower Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

D. Access Panels

Number	Name/Location
571AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel
671AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

E. Prepare for the Removal

SUBTASK 27-11-24-860-001

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-24-480-001

(2) Install the rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-24-010-001

(3) Remove the applicable panel:

(a) On the left wing, remove this access panel:

(TASK 57-50-02-000-803)

Number	Name/Location
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(b) On the right wing, remove this access panel:

(TASK 57-50-02-000-803)

Number	Name/Location
671DB	Lower Outboard Fixed Trailing Edge Access Panel

F. Aileron Wing Quadrant Removal

SUBTASK 27-11-24-020-001

(1) Do these steps to remove the aileron wing quadrant [9]:

(a) Disconnect cable ABSA [2] and cable ABSB [4] at the turnbuckles [1].

(b) Safety the disconnected cable ABSA [2] and cable ABSB [4].

(c) Remove the nuts [39], washers [38], and bolts [37] to remove the cable guard [36].

(d) Remove the nuts [8], washers [7], washers [6], and bolts [5] to remove the cable retaining plates [19] from the aileron wing quadrant [9].

(e) Disengage cable ABSA [2] and cable ABSB [4] from the aileron wing quadrant [9].

(f) Open the applicable access panel to get access to the bolt [23]:

For the left wing, open this access panel:

Number	Name/Location
571AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel

For the right wing, open this access panel:

Number	Name/Location
671AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel

(g) Remove the bolt [31] and washer [32] to disconnect the aileron position transmitter rod [33] from the aileron wing quadrant [9].

(h) Remove the pin [13], nut [14], washer [15], bushing [12], washer [11], and bolt [10] to remove the aileron control rod [17] from the aileron [16]. Discard the pin [13].

(i) Disconnect the aileron wing aileron wing quadrant [9] from the aileron hinge support [18].

1) Remove pin [26], nut [27], washer [25], bushing [24], washer [35], and bolt [23] to disconnect the aileron wing quadrant [9] from the aileron hinge support [18]. Discard the pin [26].

(j) Remove the aileron wing quadrant [9].

(k) If it is necessary, do these steps to disassemble the aileron wing quadrant [9]:

1) Remove the bushing [41], and bearings [40].

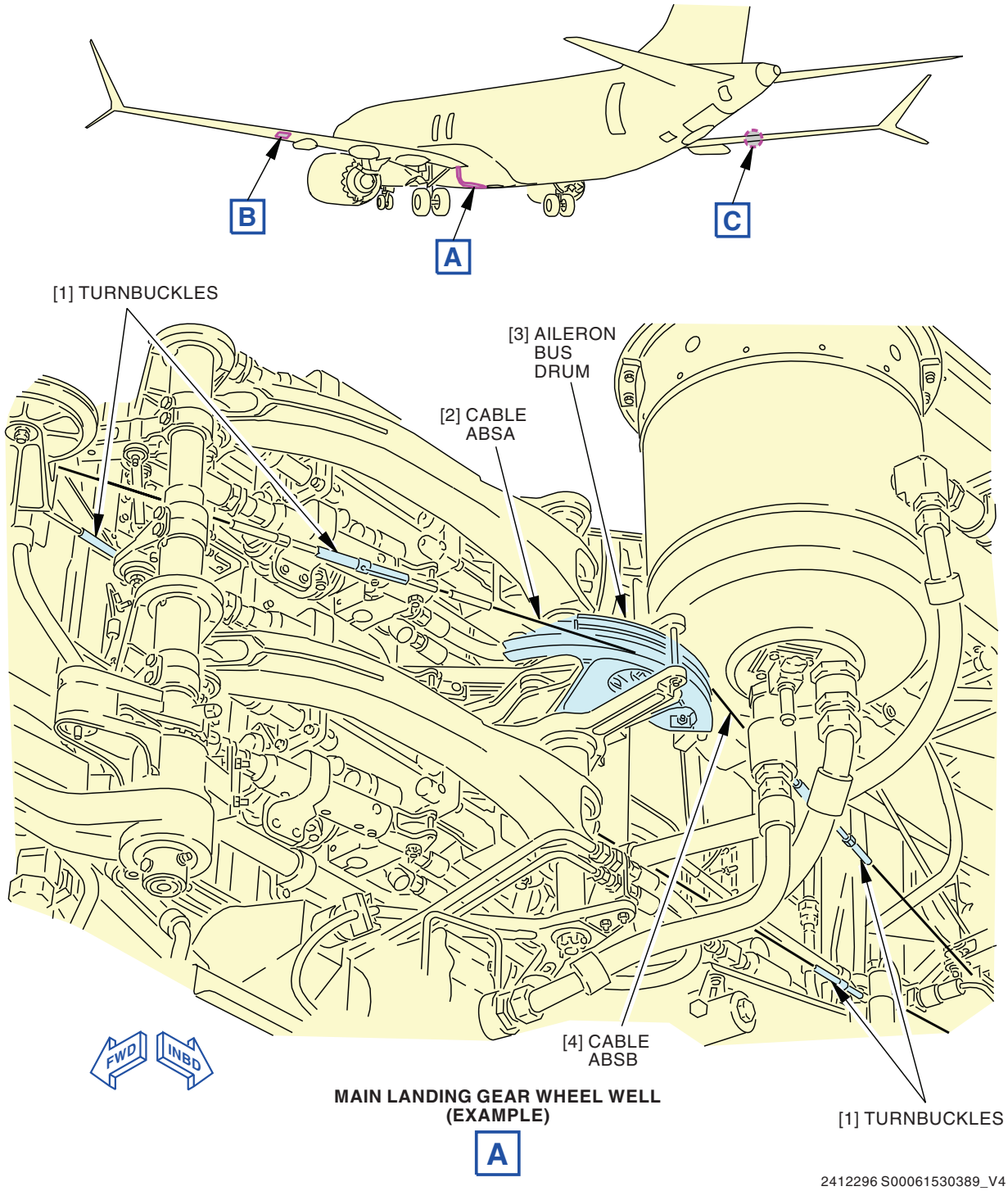
NOTE: The top bearing [40] is swaged to the wing aileron wing quadrant [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Remove the pin [28], nut [29], washer [30], bushing [22], washer [21], and bolt [20] to disconnect the aileron control rod [17] from the aileron wing quadrant [9]. Discard the pin [28].

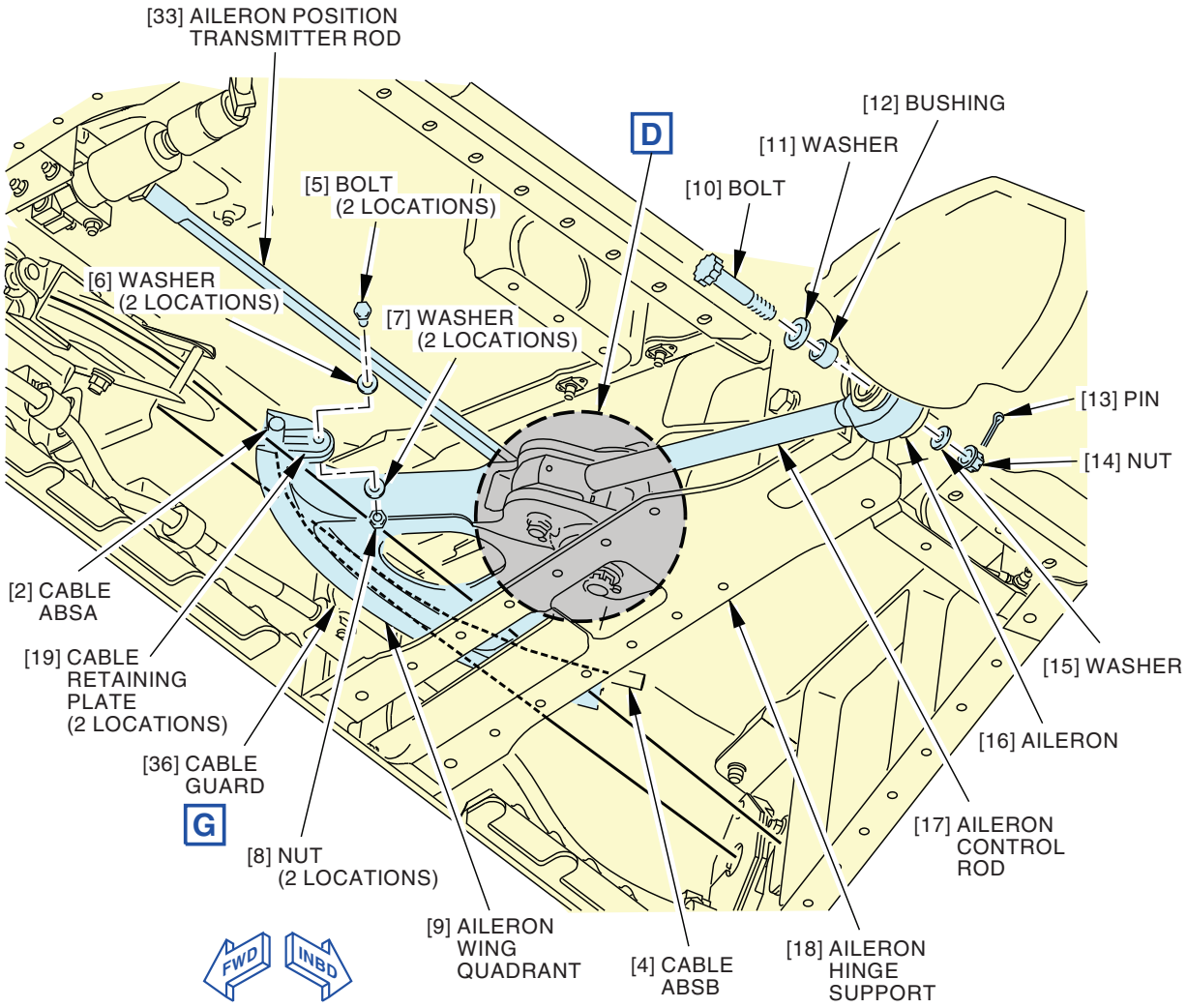
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Wing Quadrant Installation
Figure 401/27-11-24-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



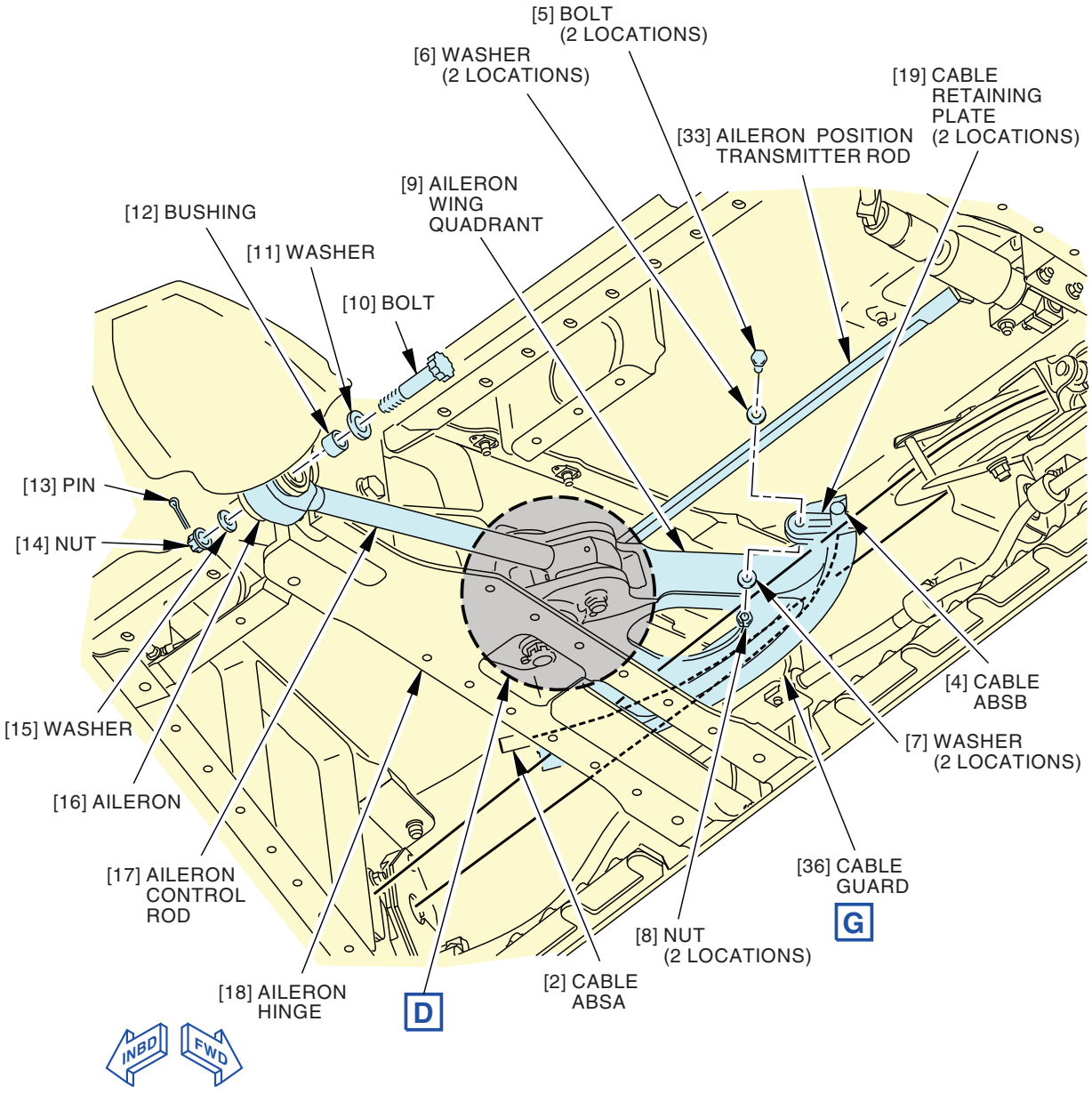
LEFT AILERON WING QUADRANT

B

2412297 S00061530390_V2

Aileron Wing Quadrant Installation
Figure 401/27-11-24-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



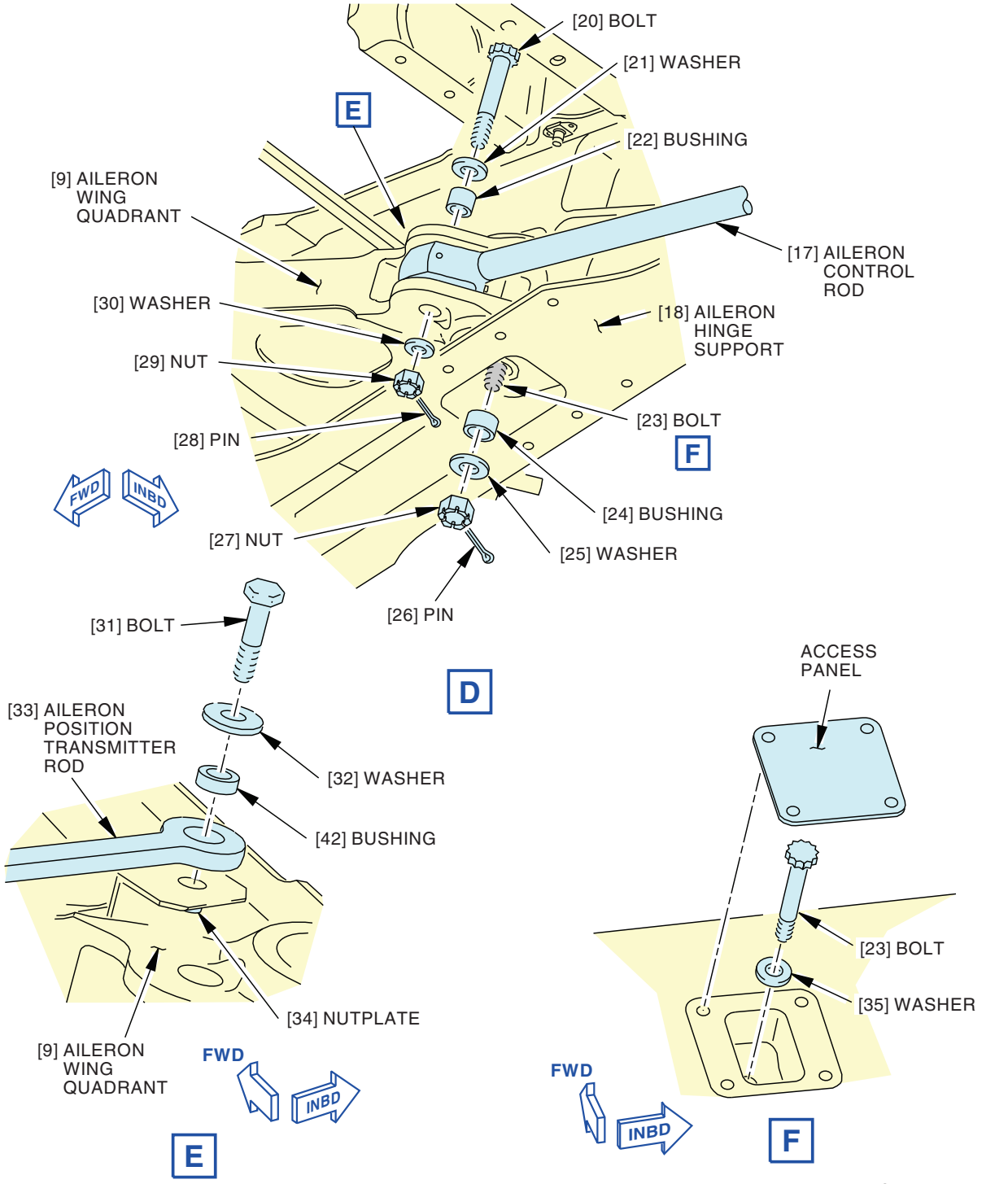
RIGHT AILERON WING QUADRANT

C

2412298 S00061530391_V2

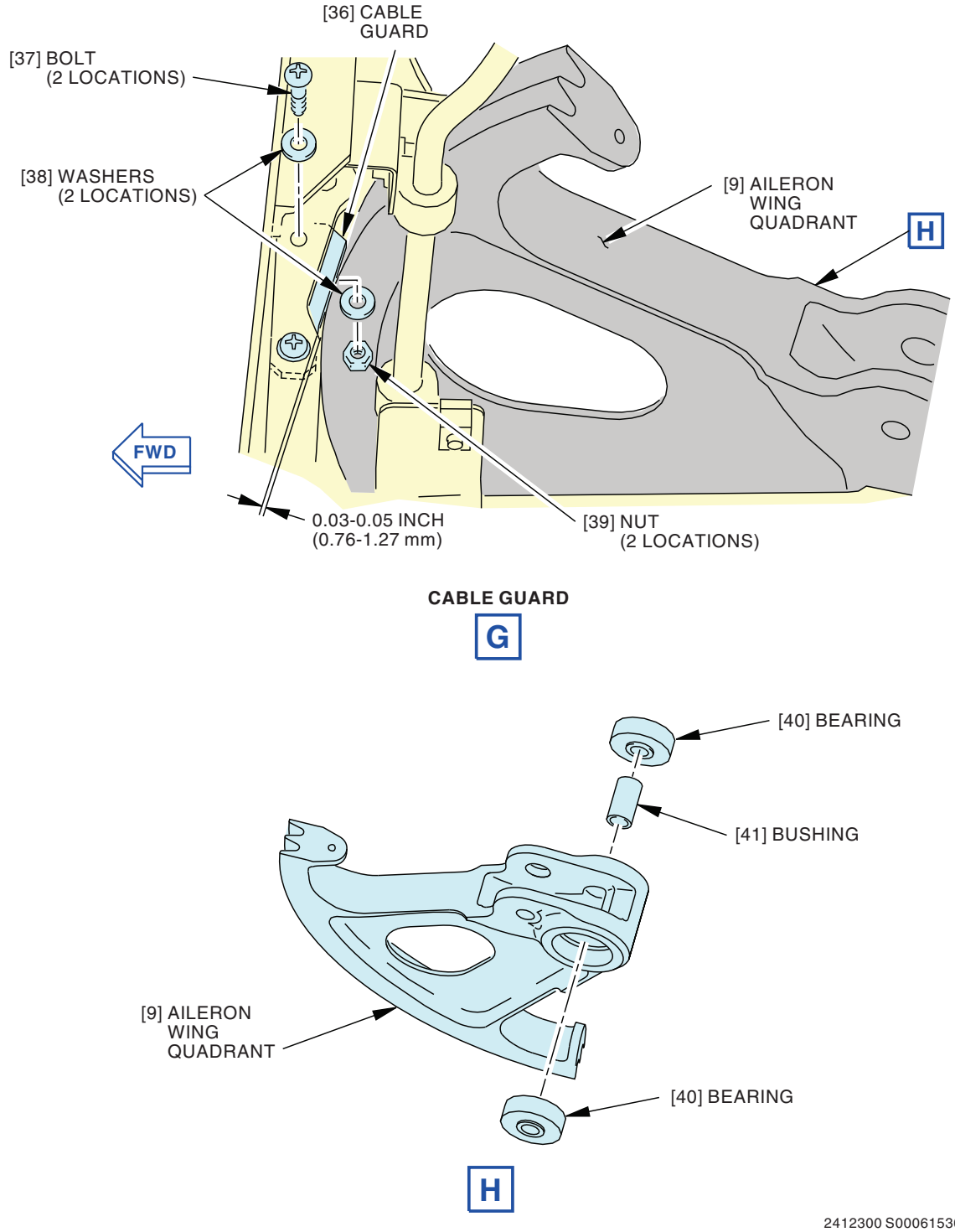
Aileron Wing Quadrant Installation
Figure 401/27-11-24-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Wing Quadrant Installation
Figure 401/27-11-24-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Wing Quadrant Installation
Figure 401/27-11-24-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-24-400-801

3. Aileron Wing Quadrant Installation

(Figure 401)

A. References

Reference	Title
20-10-93-400-801	Control Cable End Retaining Plate - Installation (P/B 401)
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
31-31-31-700-801	Aileron Position Transmitter Installation Test (P/B 401)
57-50-02-400-803	Outboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Aileron wing quadrant	27-11-24-01-085	PTF ALL
		27-11-24-01-090	PTF ALL
13	Pin	27-11-24-01-010	PTF ALL
26	Pin	27-11-24-01-010	PTF ALL
28	Pin	27-11-24-01-125	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

E. Access Panels

Number	Name/Location
571AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel
671AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

F. Aileron Wing Installation

SUBTASK 27-11-24-420-001

(1) Do these steps to install the aileron wing quadrant [9]:

(a) If it is necessary, do these steps to assemble the aileron wing quadrant [9]:

1) Apply sealant, A00247, to the top bearing [40].

2) Roller swage the top bearing [40].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

3) Apply Cor-Ban 27L Compound, G50237, to the bottom bearing [40].

4) Install the bearings [40] and bushing [41] in the aileron wing quadrant [9].

5) Install the bolt [20], washer [21], bushing [22], washer [30], nut [29], and new pin [28] to connect the aileron wing quadrant [9] with the aileron control rod [17].

(b) Do this step to connect the aileron wing quadrant [9] to the aileron hinge support [18]:

1) Install the bolt [23], washer [35], bushing [24], washer [25], nut [27], and new pin [26].

(c) Install the bolt [10], washer [11], bushing [12], washer [15], nut [14], and new pin [13] to connect the aileron control rod [17] to the aileron [16].

(d) Install the bolt [31] and washer [32] to connect the aileron position transmitter rod [33] to the aileron wing quadrant [9].

1) Make sure that the aileron position transmitter rod [33] is 17.01 in. (432.05 mm) from bolt center to bolt center.

2) If it is necessary to adjust the aileron position transmitter rod [33], do this task: Aileron Position Transmitter Installation Test, TASK 31-31-31-700-801.

(e) Put cable ABSA [2] (for the left wing) and cable ABSB [4] (for the right wing) in the bottom groove of the aileron wing quadrant [9].

(f) Put cable ABSB [4] (for the left wing) and cable ABSA [2] (for the right wing) in the top groove of the aileron wing quadrant [9].

(g) Install the cable terminations through the cable retaining plates [19].

(h) Install the cable retaining plates [19] with the bolts [5], washers [6], washers [7], and nuts [8] (TASK 20-10-93-400-801).

(i) Install the cable guard [36] with bolts [37], washers [38], and nuts [39].

1) Adjust the cable guard [36] until the gap between the cable guard [36] and aileron wing quadrant [9] is 0.030 in. (0.762 mm) - 0.05 in. (1.27 mm).

(j) Connect cable ABSA [2] and cable ABSB [4] to the turnbuckles [1], but keep the cable tension at zero.

SUBTASK 27-11-24-820-001

(2) Adjust the tension of cable ABSA [2] and cable ABSB [4] (TASK 27-11-00-820-801).

SUBTASK 27-11-24-860-002

(3) Close these access panels:

(a) On the left wing, close this access panel:

Number	Name/Location
571AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel

(b) On the left wing, install this access panel:

(TASK 57-50-02-400-803)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(c) On the right wing, close this access panel:

Number	Name/Location
671AT	Upper Outboard Fixed Trailing Edge, Quadrant Bolt Access Panel

(d) On the right wing, install this access panel:

(TASK 57-50-02-400-803)

Number	Name/Location
671DB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-24-860-003

(4) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-24-730-001

(5) Do this task: Aileron Travel Test, TASK 27-11-00-700-810.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON BALANCE PANELS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron balance panel

(2) An installation of the aileron balance panel.

B. There are four aileron balance panels on each wing. This procedure is applicable for all the aileron balance panels.

TASK 27-11-31-000-801

2. Aileron Balance Panels Removal

(Figure 401)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
57-50-02-000-803	Outboard Fixed Trailing Edge Lower Panel Removal (P/B 401)

B. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

C. Access Panels

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

D. Prepare for the Removal

SUBTASK 27-11-31-860-001

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-31-010-001

(2) Remove the applicable access panels on the lower wing surface at the balance panel.

(a) For the left wing, remove these access panels:

(TASK 57-50-02-000-803)

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(b) For the right wing, remove these access panels:

(TASK 57-50-02-000-803)

Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

E. Aileron Balance Panels Removal

SUBTASK 27-11-31-020-001

(1) Remove the balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25].

(a) Remove the bolts [1] and bolts [14] to remove the seal retainers [27] and seal retainers [30] from the aerodynamic seals [15].

(b) Write down the location and number of balance weights [11] installed on the balance panel 3 [24] or balance panel 4 [25].

(c) Remove the bolts [32], washers [31], and nuts [33] from the hinge seal [29] and aerodynamic seals [15].

(d) Remove the bolts [12] to remove the balance weights [11] from the hinge seal [29].

(e) Remove bolts [4], bolts [5], bolts [21], and bolts [2] with collars [3] to remove the balance weights [19] and hinge seal [29] from the aileron balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25].

(f) Remove the bolts [16] and washers [17] to remove the seal retainer [18] from the hinge seal [20].

(g) Remove the pins [10], nuts [9], washers [8], washers [7], and bolts [6] to remove the balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25] from the aileron.

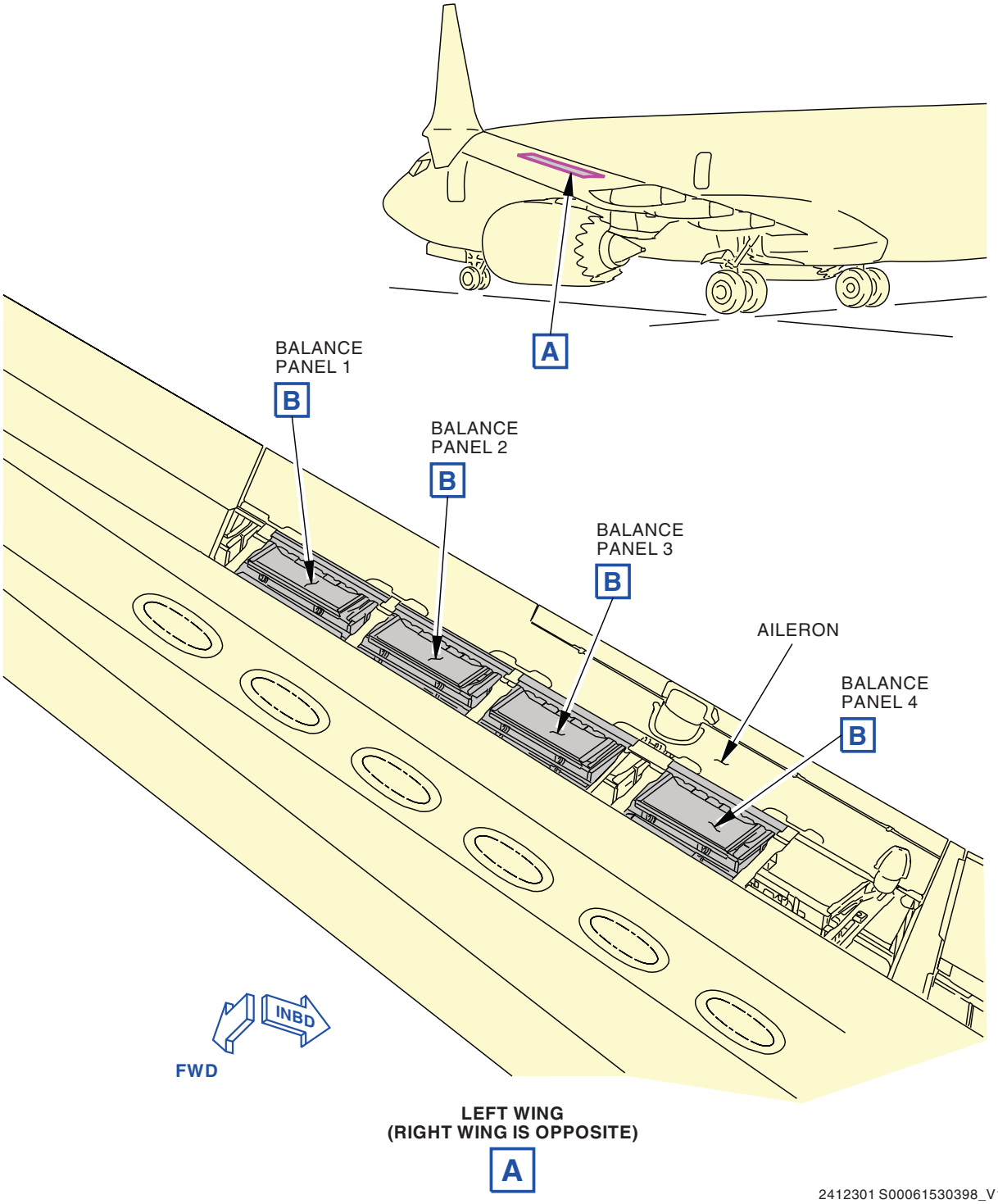
1) Discard the pin [10].

(h) Remove the pins [10], nuts [9], washers [8], washers [7], and bolts [26] to remove the balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25] from the wing structure.

1) Discard the pin [10].

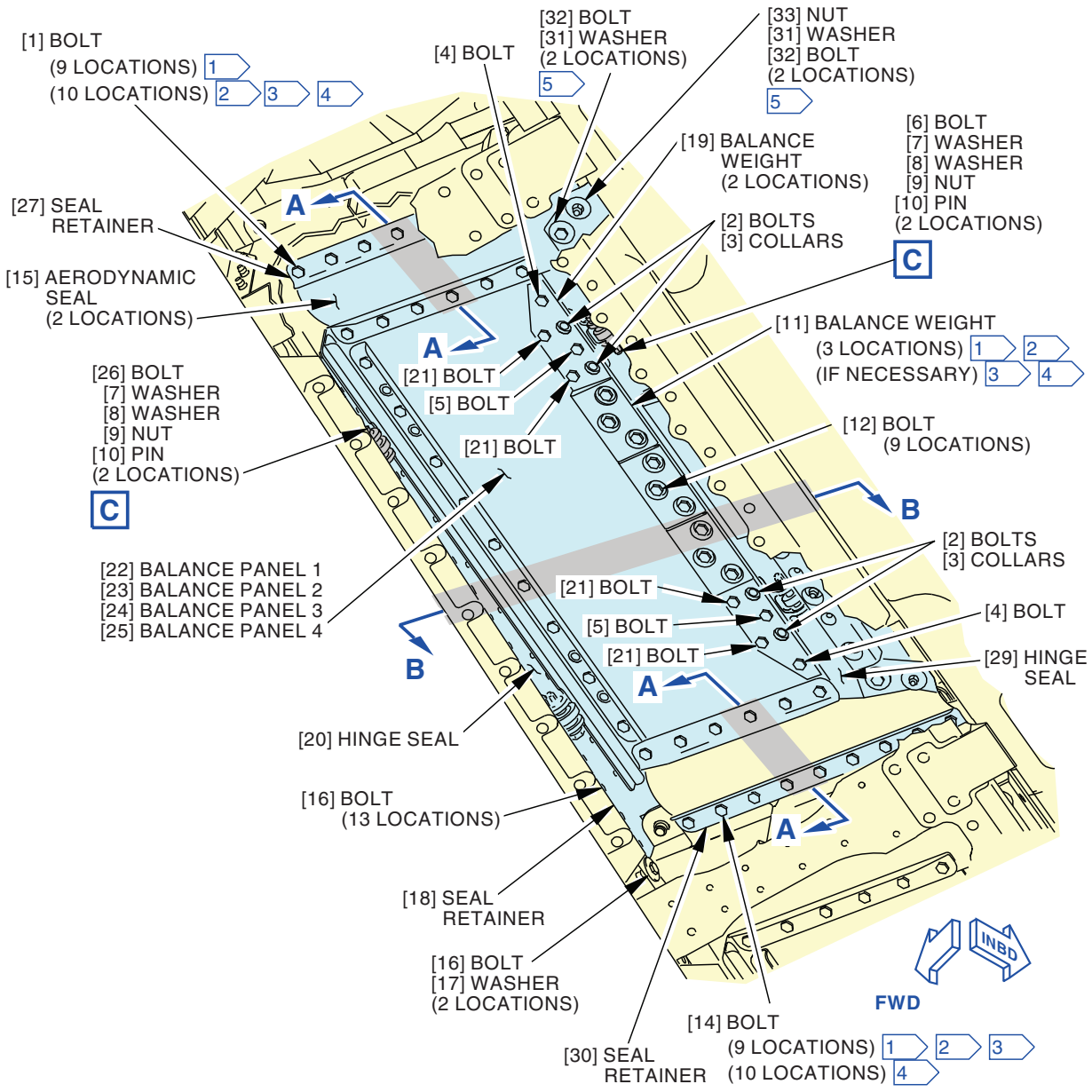
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Panel Installation
Figure 401/27-11-31-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 BALANCE PANEL 1
2 BALANCE PANEL 2
3 BALANCE PANEL 3
4 BALANCE PANEL 4
5 IT IS PERMITTED TO INSTALL THE HINGE SEAL BELOW THE AERODYNAMIC SEAL OR AERODYNAMIC SEAL BELOW THE HINGE SEAL.

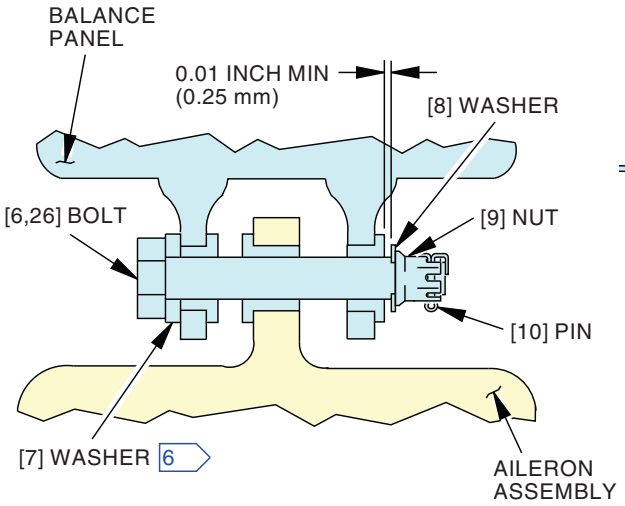
BALANCE PANEL 1
(BALANCE PANELS 2, 3, AND 4 ARE ALMOST THE SAME)

B

2412302 S00061530399_V2

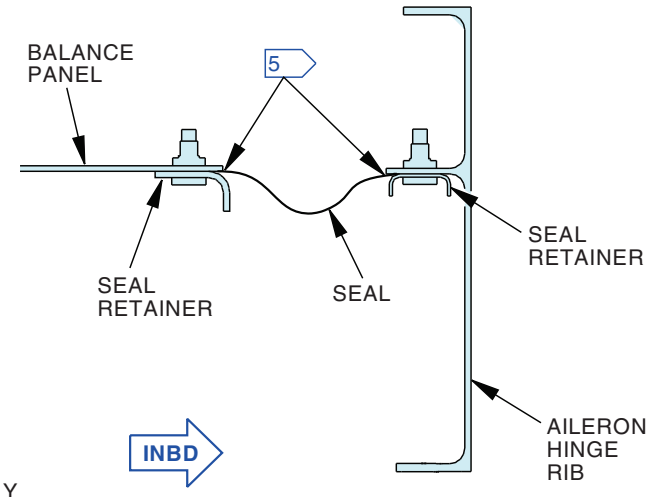
Aileron Balance Panel Installation
Figure 401/27-11-31-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



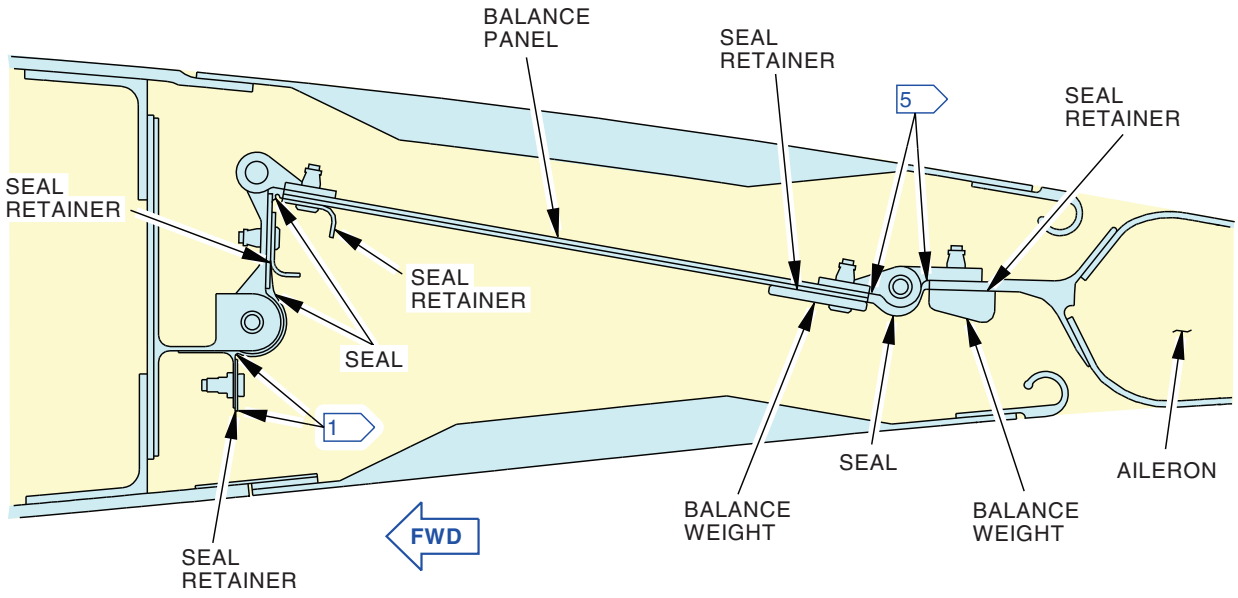
BALANCE PANEL HINGE (EXAMPLE)

C



INBOARD SEAL IS SHOWN (OUTBOARD SEAL IS OPPOSITE)

A-A



(EXAMPLE)

B-B

5 APPLY BMS 3-23 BY BRUSH TO ALL EDGES OF SEAL RETAINERS AND INTERFACE BETWEEN SEAL RETAINERS AND SEALS.

6 BOND WASHER TO HINGE HALF USING BMS 5-95 SEALANT.

2412303 S00061530400_V1

Aileron Balance Panel Installation
Figure 401/27-11-31-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-31-400-801

3. Aileron Balance Panels Installation

(Figure 401)

A. References

Reference	Title
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-11-31-210-801	Aileron Balance Panel Seals Detailed - Inspection (P/B 601)
57-50-02-400-803	Outboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-3911	Brush - Bristle, Medium Nylon

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G00009	Compound - Organic Corrosion Inhibiting	BMS3-23

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Pin	27-11-31-01-069	PTF ALL
		27-31-41-01-110	PTF ALL
22	Balance panel 1	27-11-31-01-519	PTF ALL
		27-11-31-01-522	PTF ALL
23	Balance panel 2	27-11-31-01-366	PTF ALL
		27-11-31-01-369	PTF ALL
24	Balance panel 3	27-11-31-01-213	PTF ALL
		27-11-31-01-216	PTF ALL
25	Balance panel 4	27-11-31-01-063	PTF ALL
		27-11-31-01-066	PTF ALL

E. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

F. Access Panels

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

G. Aileron Balance Panels Installation

SUBTASK 27-11-31-220-001

(1) If you install used bolts, bushings, or other components that have worn areas, do this task: Aileron Balance Panel Seals Detailed - Inspection, TASK 27-11-31-210-801.

SUBTASK 27-11-31-420-001

(2) Install the balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25].

(a) Install the bolts [26], washers [7], washers [8], and nuts [9] to attach the balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25] to the wing structure.

(b) Tighten the nuts [9] to 25 in-lb (3 N·m) - 35 in-lb (4 N·m).

(c) Tighten the nut [9] to the lower end of the torque range.

1) Continue to tighten the nut [9] until the next castellation aligns with the hole in the bolt [26].

2) Make sure that the gap between the bushing face and the washer [8] under the nut [9] is 0.010 in. (0.254 mm) minimum after torque up.

(d) Install the pins [10].

(e) Make sure that the washers [7] is bonded to the hinge half.

1) If it is necessary, bond the washers [7] to the hinge half with sealant, A00247.

(f) Install the bolts [6], washers [7], washers [8], and nuts [9] to attach the aileron balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25] to the aileron.

(g) Tighten the nuts [9] to 25 in-lb (3 N·m) - 35 in-lb (4 N·m).

(h) Tighten the nut [9] to the lower end of the torque range.

1) Continue to tighten the nut [9] until the next castellation aligns with the hole in the bolt [6].

2) Make sure that the gap between the bushing face and the washer [8] under the nut [9] is 0.010 in. (0.254 mm) minimum after torque up.

(i) Install the pins [10].

(j) Make sure that the aerodynamic seals [15], hinge seals [20], and hinge seals [29] are not damaged.

(k) Install the seal retainer [18] to the hinge seal [20] with the bolts [16] and washers [17].

(l) Attach the hinge seal [29] to the balance panel 1 [22], balance panel 2 [23], balance panel 3 [24], or balance panel 4 [25].

(m) Install the balance weights [19] to the balance panel assembly with the bolts [4], bolts [5], bolts [21], and bolts [2] with collars [3].

1) Apply primer, C00259, to the holes of the balance weights [19].

2) Apply compound, C00528, to the bolts [2], bolts [4], bolts [5], and bolts [21].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Tighten the bolts [2], bolts [4], bolts [5], and bolts [21] to 35 in-lb (4 N·m) - 45 in-lb (5 N·m).

(n) Make sure that there are the same number of balance weights [11] on the new balance panel assembly as on the removed balance panel 3 [24] or balance panel 4 [25].

(o) Install the balance weights [11] to the balance panel assembly with bolts [12].

1) Apply primer, C00259, to the holes of the balance weights [11].

2) Apply compound, C00528, to the bolts [12].

3) Tighten the bolt [12] to 35 in-lb (4 N·m) - 45 in-lb (5 N·m).

(p) Install the seal retainers [27] and seal retainer [30] on the aerodynamic seals [15] with the bolts [1] and bolts [14].

SUBTASK 27-11-31-860-004

(3) Apply compound, C00528, to the bolts [32].

SUBTASK 27-11-31-420-002

(4) Install the bolts [32], nuts [33], and washers [31] to attach the hinge seal [29] and aerodynamic seal [15].

NOTE: It is permitted to install the hinge seal below the aerodynamic seal or aerodynamic seal below the hinged seal.

(a) Make sure that one seal is on top of the other across the bolts [32], nuts [33], and washers [31].

SUBTASK 27-11-31-600-001

(5) Use a medium nylon bristle brush, STD-3911, to apply corrosion inhibiting compound, G00009, to all edges of the seal retainer [18], seal retainers [27], and interface between seal retainer [18], seal retainers [27], seal retainer [30], and aerodynamic seals [15], hinge seal [20], and hinge seal [29].

SUBTASK 27-11-31-410-001

(6) Install the applicable access panels on the lower wing surface at the aileron balance panel.

(a) For the left wing, install these access panels:

(TASK 57-50-02-400-803)

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(b) For the right wing, install these access panels:

(TASK 57-50-02-400-803)

Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-31-860-002

(7) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-31-710-001

(8) Do this task: Aileron Travel Test, TASK 27-11-00-700-810.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON BALANCE PANEL - INSPECTION/CHECK

1. General

A. The procedure contains these tasks:

(1) Aileron Balance Panel Inspection

(2) Aileron Balance Panel Seals Detailed Inspection

TASK 27-11-31-220-801

2. Aileron Balance Panel - Inspection

(Figure 601)

A. General

(1) This task gives wear limits and repair data for the aileron balance panel inspection.

B. References

Reference	Title
27-11-31-000-801	Aileron Balance Panels Removal (P/B 401)
27-11-31-400-801	Aileron Balance Panels Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

E. Aileron Balance Panel Inspection

SUBTASK 27-11-31-020-002

(1) Do this task: Aileron Balance Panels Removal, TASK 27-11-31-000-801.

SUBTASK 27-11-31-020-003

(2) Use these tools to do a check for wear on the parts of the aileron balance panel:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-11-31-020-004

(3) Examine the parts of the aileron balance panel for worn areas:

(a) Remove the parts you want to do a check.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are out of tolerance, replace or repair the part.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-11-31-993-801

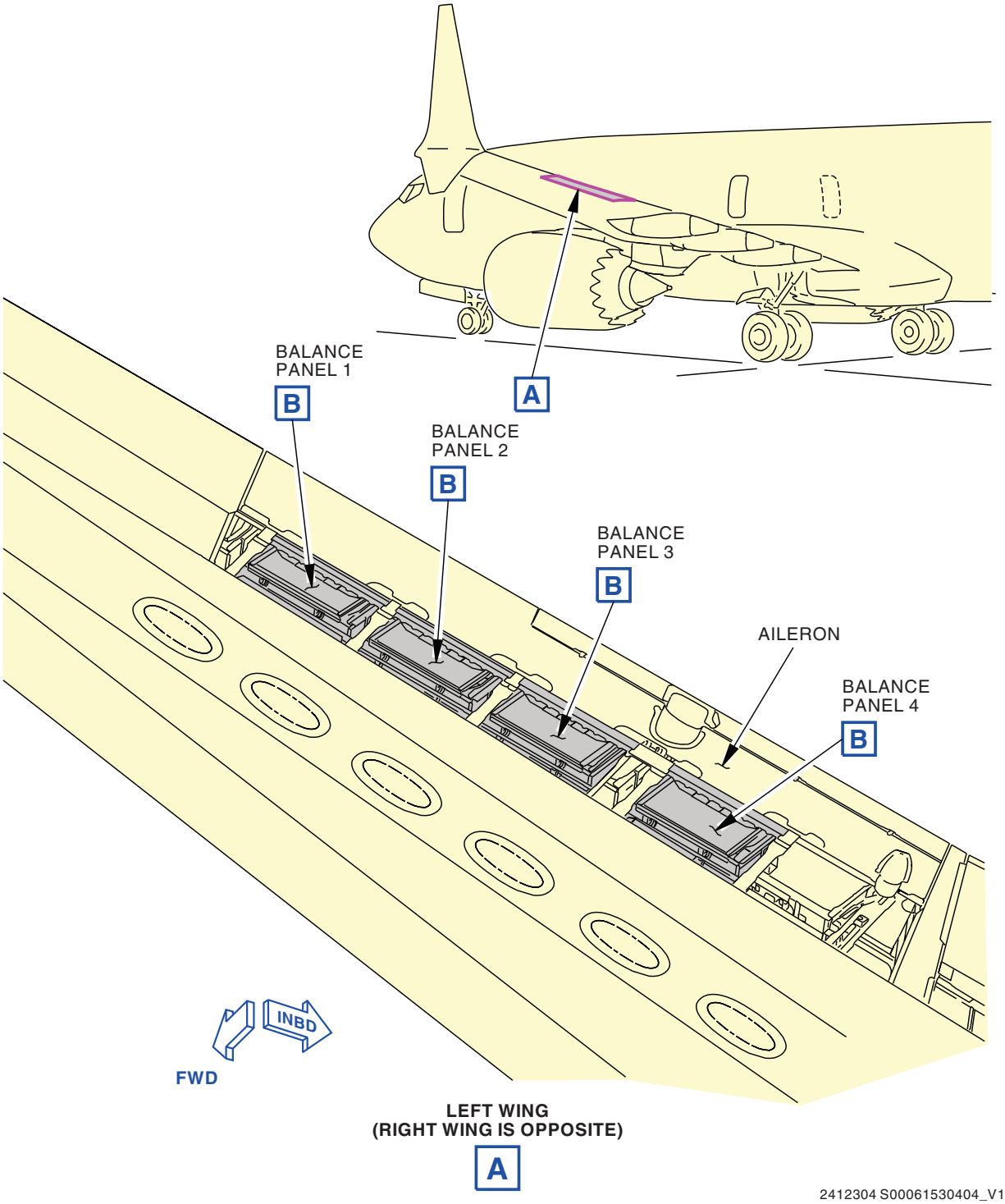
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.0030 in. (0.0762 mm)
#1 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	0.0030 in. (0.0762 mm)
#2 BUSHING	I.D.	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.2525 in. (6.4135 mm)	0.0040 in. (0.1016 mm)
#2 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	0.0040 in. (0.1016 mm)
#3 BUSHING	I.D.	0.2500 in. (6.3500 mm)	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.0030 in. (0.0762 mm)
#3 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	0.0030 in. (0.0762 mm)
#4 BUSHING	I.D.	0.2505 in. (6.3627 mm)	0.2515 in. (6.3881 mm)	0.2525 in. (6.4135 mm)	0.0040 in. (0.1016 mm)
#4 BOLT	O.D.	0.2485 in. (6.3119 mm)	0.2495 in. (6.3373 mm)	0.2475 in. (6.2865 mm)	0.0040 in. (0.1016 mm)

SUBTASK 27-11-31-020-005

(4) Do this task: Aileron Balance Panels Installation, TASK 27-11-31-400-801.

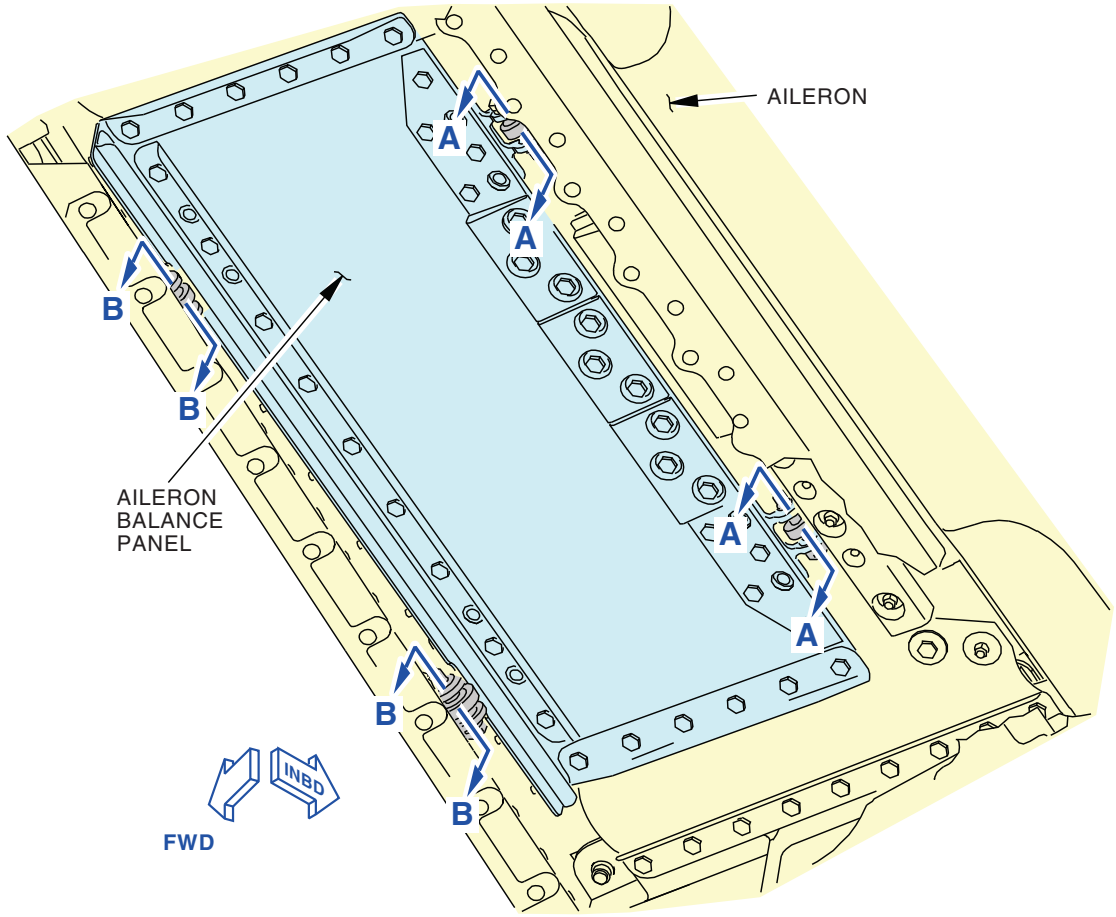
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



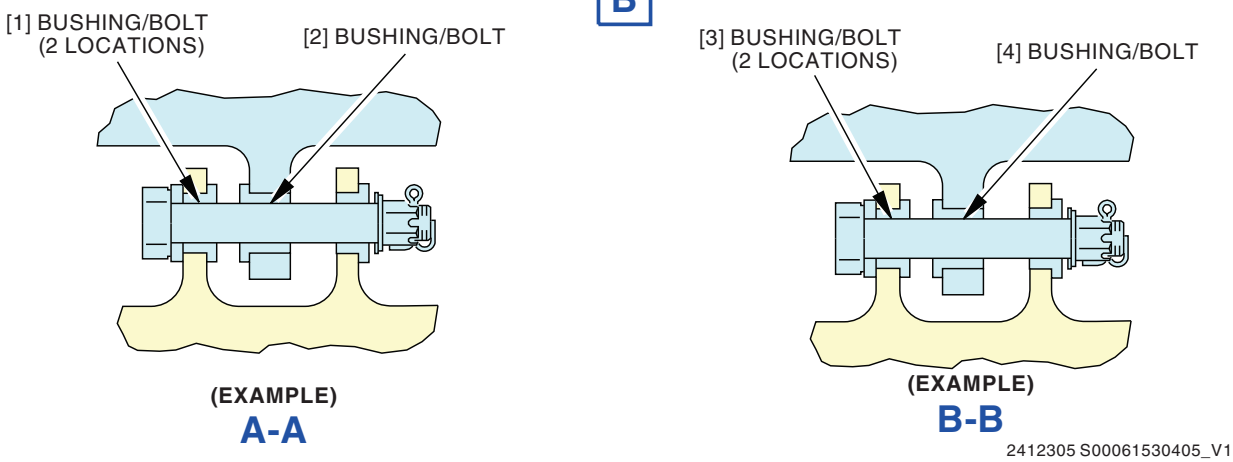
Aileron Balance Panel Wear Limits
Figure 601/27-11-31-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BALANCE PANEL 1
(BALANCE PANELS 2, 3, AND 4 ARE ALMOST THE SAME)

B



(EXAMPLE)
A-A

(EXAMPLE)
B-B

2412305 S00061530405_V1

Aileron Balance Panel Wear Limits
Figure 601/27-11-31-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-31-210-801

3. Aileron Balance Panel Seals Detailed - Inspection

(Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
57-50-02-000-803	Outboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
57-50-02-400-803	Outboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

B. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

C. Access Panels

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

D. Aileron Balance Panel Seals Detailed Inspection

SUBTASK 27-11-31-860-003

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-31-010-002

(2) Remove the applicable access panels on the lower wing surface at the balance panel.

(a) For the left wing, remove these access panels:

(TASK 57-50-02-000-803)

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(b) For the right wing, remove these access panels:

(TASK 57-50-02-000-803)

Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-31-210-001

(3) Do a detailed inspection of the seals around the aileron balance panel.

(a) Make sure that there are no creases, nicks, tears and punctures on the seals.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-31-410-002

(1) Install the applicable access panels on the lower wing surface at the balance panel.

(a) For the left wing, install these access panels:

(TASK 57-50-02-400-803)

Number	Name/Location
571BB	Lower Outboard Fixed Trailing Edge Access Panel
571CB	Lower Outboard Fixed Trailing Edge Access Panel
571DB	Lower Outboard Fixed Trailing Edge Access Panel

(b) For the right wing, install these access panels:

(TASK 57-50-02-400-803)

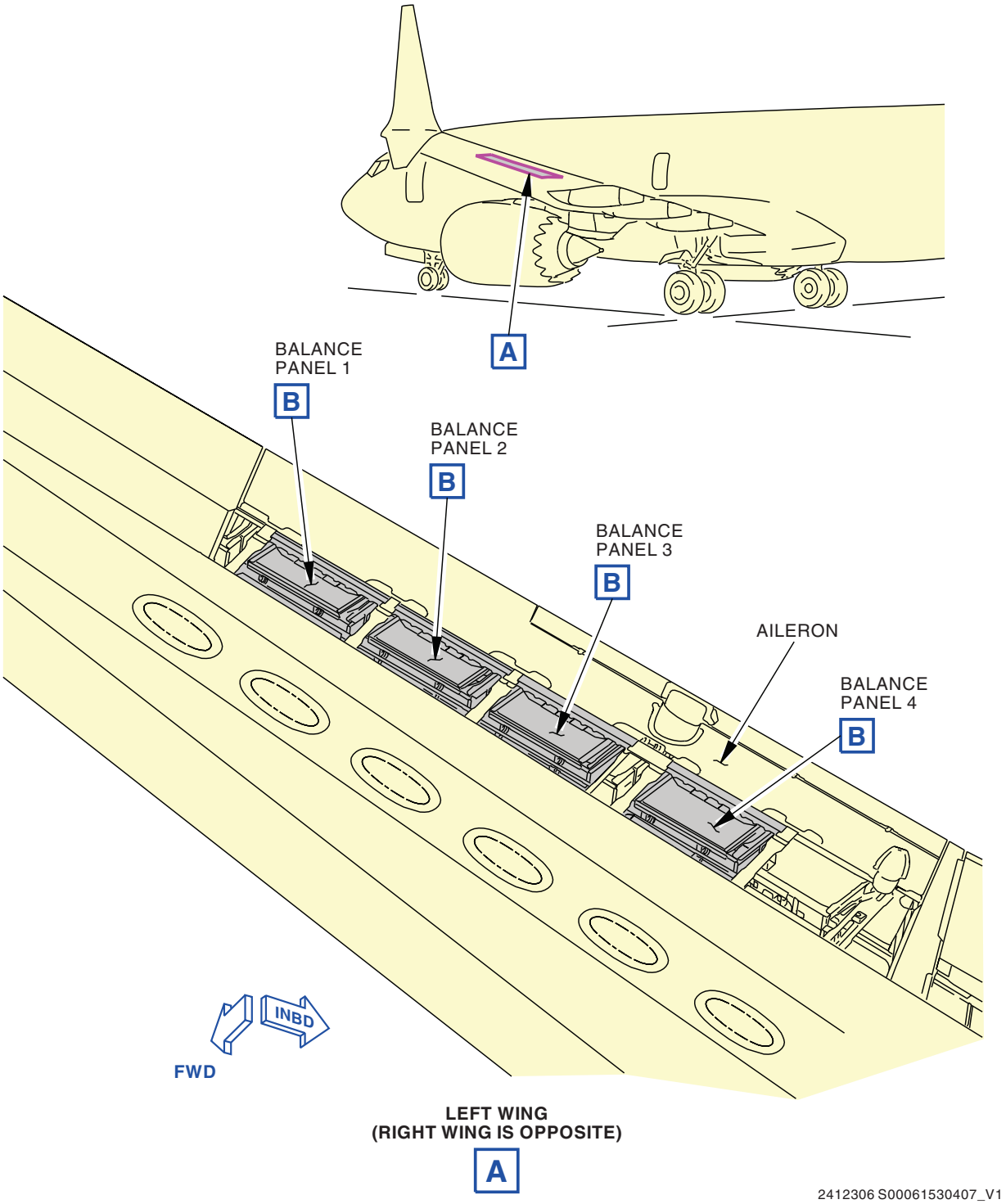
Number	Name/Location
671BB	Lower Outboard Fixed Trailing Edge Access Panel
671CB	Lower Outboard Fixed Trailing Edge Access Panel
671DB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-31-840-001

(2) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

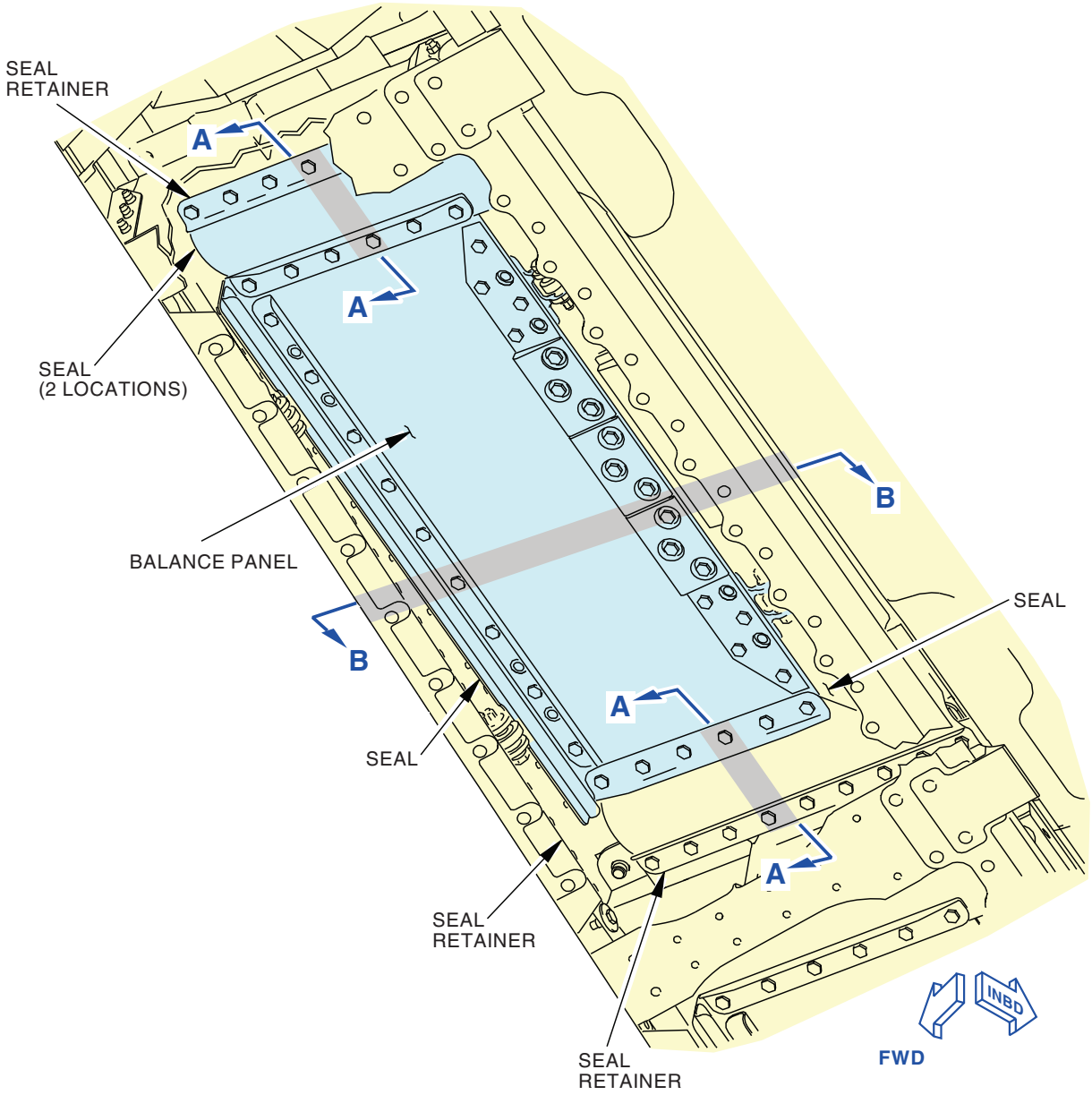
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Balance Panel Seals Detailed Inspection
Figure 602/27-11-31-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



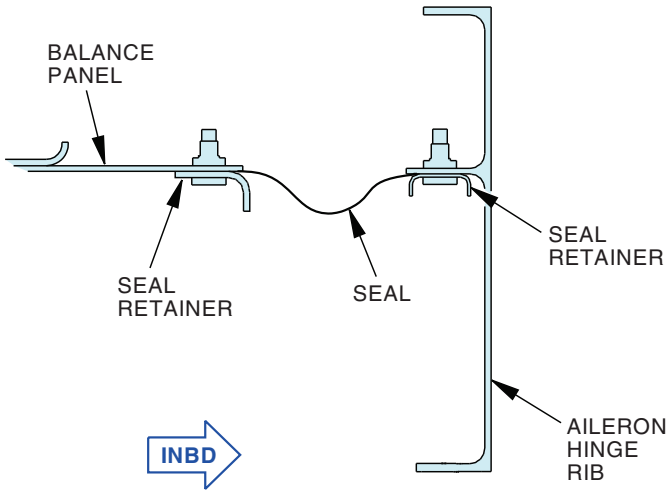
BALANCE PANEL 1
(BALANCE PANELS 2, 3, AND 4 ARE ALMOST THE SAME)

B

2412307 S00061530408_V1

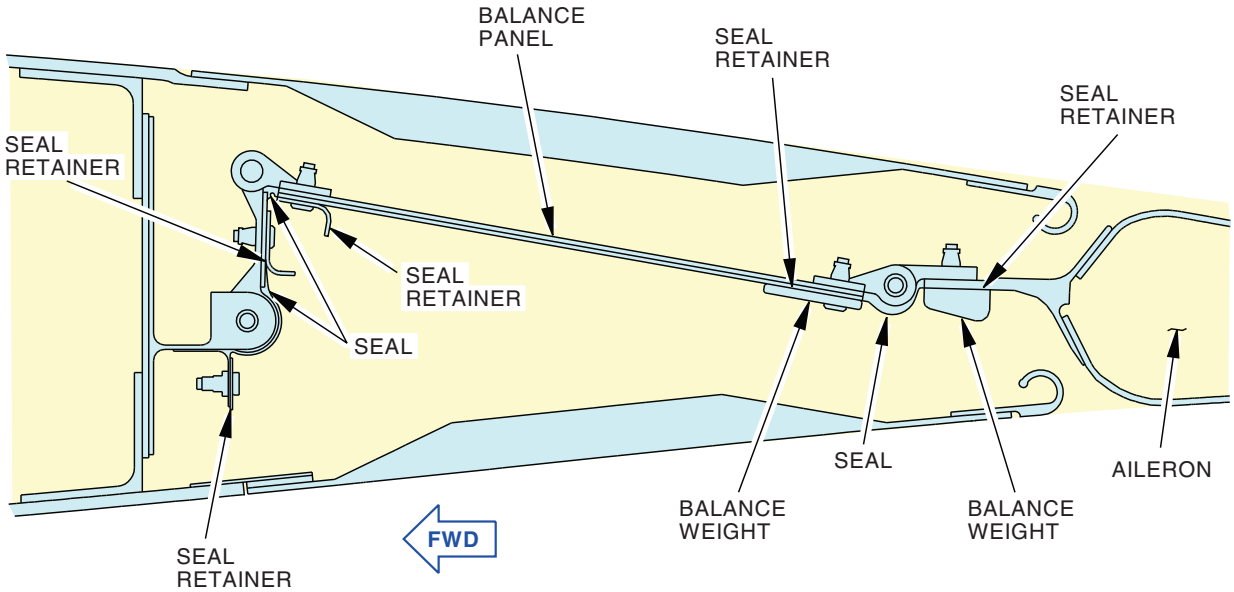
Aileron Balance Panel Seals Detailed Inspection
Figure 602/27-11-31-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



INBOARD SEAL IS SHOWN
(OUTBOARD SEAL IS OPPOSITE)

A-A



(EXAMPLE)

B-B

2412308 S00061530409_V1

Aileron Balance Panel Seals Detailed Inspection
Figure 602/27-11-31-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON TAB CONTROL RODS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of an aileron tab control rods

(2) An installation of the aileron tab control rods.

B. In this procedure the aileron tab control rods are referred to as the "control rods".

TASK 27-11-34-000-801

2. Aileron Tab Control Rods Removal

(Figure 401)

A. References

Reference	Title
57-50-02-000-803	Outboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
57-61-02-000-802	Aileron Lower Hinge Covers Removal (P/B 401)

B. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

C. Access Panels

Number	Name/Location
571CB	Lower Outboard Fixed Trailing Edge Access Panel
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00
671CB	Lower Outboard Fixed Trailing Edge Access Panel
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00

D. Prepare for the Removal

SUBTASK 27-11-34-010-001

(1) On the left wing, remove this access panel:

(TASK 57-50-02-000-803)

Number	Name/Location
571CB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-34-010-002

(2) On the left wing, remove these access panels:

(TASK 57-61-02-000-802)

Number	Name/Location
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00

SUBTASK 27-11-34-010-003

(3) On the right wing, remove this access panel:

(TASK 57-50-02-000-803)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
671CB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-34-010-004

(4) On the right wing, remove these access panels:

(TASK 57-61-02-000-802)

Number	Name/Location
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00

E. Aileron Tab Control Rods Removal

SUBTASK 27-11-34-020-001

(1) Remove the bolts [6] that attach the forward removable fairing [2] to the aileron [1].

(a) Remove the forward removable fairing [2].

SUBTASK 27-11-34-020-002

(2) Remove the bolts [4] that attach the aft removable fairing [3] to the aileron balance tab [5].

(a) Remove the aft removable fairing [3].

SUBTASK 27-11-34-020-003

(3) Remove the aileron tab control rod [17]:

(a) Lift up the aileron Trailing Edge (TE).

(b) Remove the bolts [13] and washers [14] that attach the radial seal [16] to the radial seal retainer [15] to the structure.

(c) Remove the radial seal [16] and radial seal retainer [15].

(d) Remove the pins [12], nuts [11], washers [10], bushings [9], washers [8] and bolts [7] that connect the aileron tab control rod [17] to the forward fittings.

1) Discard the pins [12].

(e) Move the aileron TE down.

(f) Remove the pins [12], nuts [11], washers [10], bushings [18], washers [8], and bolts [19] that connect the aileron tab control rod [17] to the tab mast.

1) Discard the pins [12].

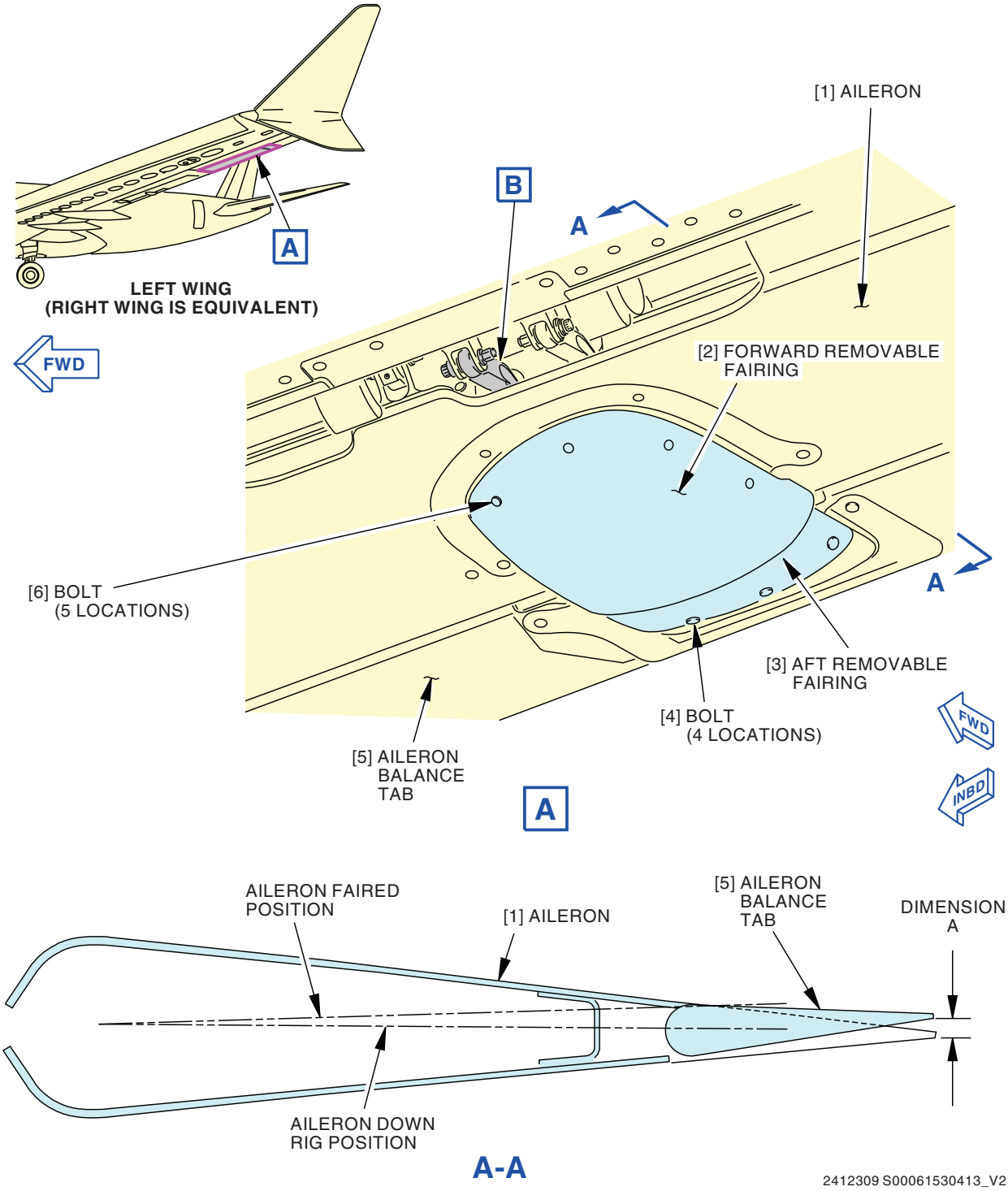
(g) Remove the aileron tab control rod [17] from the airplane.

SUBTASK 27-11-34-970-001

(4) Write down the length of the aileron tab control rod [17].

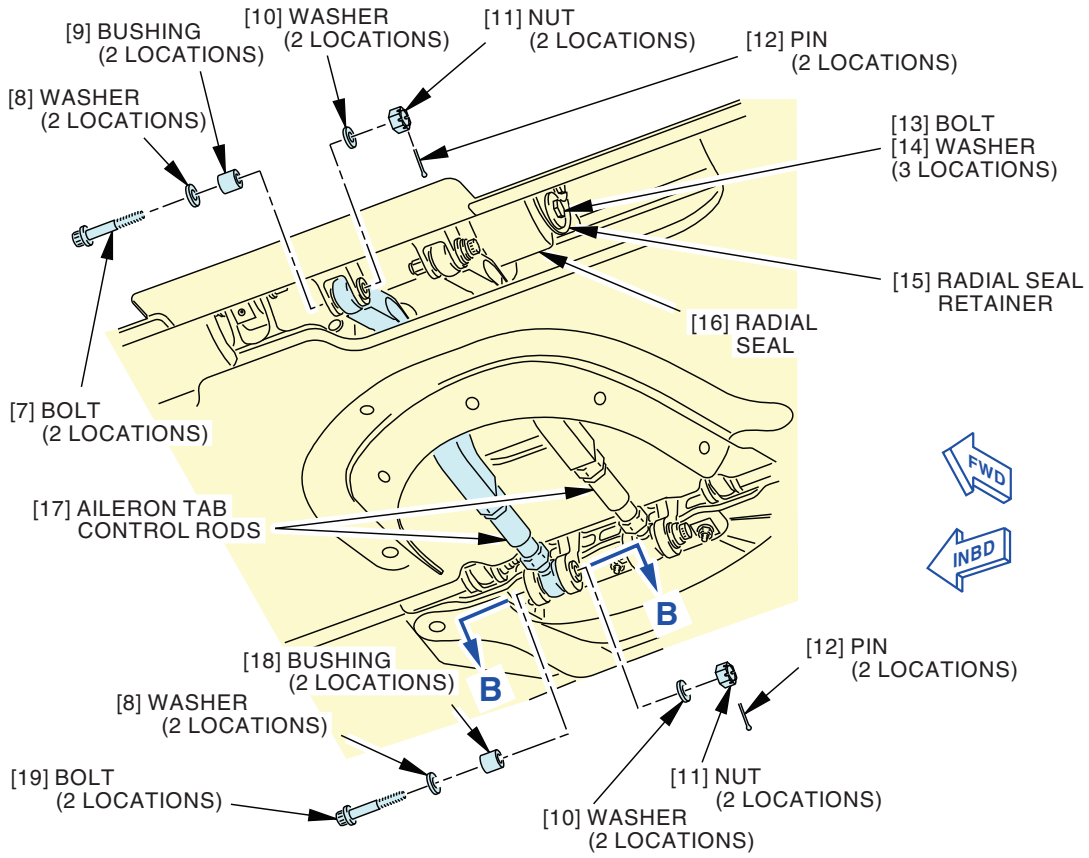
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



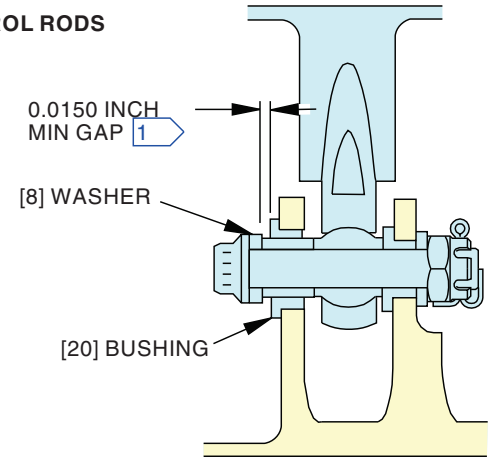
Aileron Tab Control Rods Installation
Figure 401/27-11-34-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AILERON TAB CONTROL RODS

B



1 0.0150 INCH (0.38 mm) GAP REFERS TO HINGE 2 AND CONTROL ROD CONNECTIONS ONLY.

INBOARD AILERON TAB CONTROL ROD
(OUTBOARD AILERON TAB CONTROL ROD IS OPPOSITE)

B-B

2412310 S00061530414_V2

Aileron Tab Control Rods Installation
Figure 401/27-11-34-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-34-400-801

3. Aileron Tab Control Rods Installation

(Figure 401)

A. References

Reference	Title
27-11-00-750-801	Flight Test to Find and Correct Asymmetrical Flight Control Forces (P/B 501)
27-11-00-820-803	Aileron Balance Tab Adjustment (Recommended Method) (P/B 501)
27-11-34-200-801	Aileron Tab Control Rods Inspection (P/B 601)
57-50-02-400-803	Outboard Fixed Trailing Edge Lower Panel Installation (P/B 401)
57-61-02-400-802	Aileron Lower Hinge Covers Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	Pin	27-11-21-01-010	PTF ALL
17	Aileron tab control rod	27-11-21-01-040	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
572	Left Wing - Aileron
672	Right Wing - Aileron

D. Access Panels

Number	Name/Location
571CB	Lower Outboard Fixed Trailing Edge Access Panel
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00
671CB	Lower Outboard Fixed Trailing Edge Access Panel
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00

E. Prepare for installation

SUBTASK 27-11-34-220-001

(1) If you install used bolts, bushings, bearings or other components with worn areas, do this task: Aileron Tab Control Rods Inspection, TASK 27-11-34-200-801.

F. Aileron Tab Control Rods Installation

SUBTASK 27-11-34-420-001

(1) Install the aileron tab control rod [17]:

(a) Lift up the aileron Trailing Edge (TE).

(b) Put the two aileron tab control rods [17] in position, align the rod ends with the attach fittings.

NOTE: The length of the aileron tab control rods must be the same as those written down after the removal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Install the bolts [7], washers [8], bushings [9], washers [10], and nuts [11] to connect the aileron tab control rod [17] to the forward fittings.

1) Tighten the nut [11] to 50 in-lb (6 N·m) - 80 in-lb (9 N·m) until the next nut castellation aligns with the hole in the bolt.

2) Install the pins [12].

(d) Install the radial seal [16] and radial seal retainer [15] to the structure with the bolts [13] and washers [14].

(e) Move the aileron TE down.

(f) Connect the aileron tab control rod [17] to the tab mast with the bolts [19], washers [8], bushings [18], washers [10], and nuts [11].

1) If there is binding between the aileron tab control rods [17] and the bolts [19], adjust the length of the aileron tab control rods [17].

NOTE: You must adjust the aileron tab control rods the same amount.

2) Make sure there is a minimum of 0.0150 in. (0.38 mm) gap between the bushing [20] and washer [8].

3) Tighten the nuts [11] to 50 in-lb (6 N·m) - 80 in-lb (9 N·m).

4) Install the new pins [12].

SUBTASK 27-11-34-220-003

(2) Make sure that the distance from the tab TE to the extension of the aileron bottom surface (Dim. A) is correct (View A-A, Figure 401), (TASK 27-11-00-820-803).

SUBTASK 27-11-34-410-001

(3) On the left wing, install this access panel:

(TASK 57-50-02-400-803)

Number	Name/Location
571CB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-34-410-002

(4) On the left wing, install these access panels:

(TASK 57-61-02-400-802)

Number	Name/Location
572DB	Lower Aileron, Hinge Cover - WBL 469.00
572EB	Lower Aileron, Hinge Cover - WBL 481.00

SUBTASK 27-11-34-410-003

(5) On the right wing, install this access panel:

(TASK 57-50-02-400-803)

Number	Name/Location
671CB	Lower Outboard Fixed Trailing Edge Access Panel

SUBTASK 27-11-34-410-004

(6) On the right wing, install these access panels:

(TASK 57-61-02-400-802)

Number	Name/Location
672DB	Lower Aileron, Hinge Cover - WBL 469.00
672EB	Lower Aileron, Hinge Cover - WBL 481.00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-34-420-002

(7) Install the bolts [4] to attach the aft removable fairing [3] to the aileron balance tab [5].

SUBTASK 27-11-34-420-003

(8) Install the bolts [6] to attach the forward removable fairing [2] to the aileron [1].

SUBTASK 27-11-34-710-001

(9) Move the aileron control wheel through full travel.

(a) Make sure that the aileron tab moves freely.

SUBTASK 27-11-34-820-001

(10) Do this task: Flight Test to Find and Correct Asymmetrical Flight Control Forces, TASK 27-11-00-750-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON TAB CONTROL RODS - INSPECTION/CHECK

1. General

A. This procedure contains the task to do an inspection of the aileron tab control rods.

TASK 27-11-34-200-801

2. Aileron Tab Control Rods Inspection

(Figure 601)

A. General

(1) This task gives wear limits and repair data for the aileron tab control rods inspection.

B. References

Reference	Title
27-11-21-000-801	Aileron Balance Tab Removal (P/B 401)
27-11-21-400-801	Aileron Balance Tab Installation (P/B 401)
27-11-34-000-801	Aileron Tab Control Rods Removal (P/B 401)
27-11-34-400-801	Aileron Tab Control Rods Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
572	Left Wing - Aileron
672	Right Wing - Aileron

E. Procedure

SUBTASK 27-11-34-020-004

(1) Do this task: Aileron Balance Tab Removal, TASK 27-11-21-000-801 or Aileron Tab Control Rods Removal, TASK 27-11-34-000-801.

SUBTASK 27-11-34-020-005

(2) Use these tools to do a check for wear on the parts of the aileron tab control rods:

(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 27-11-34-020-006

(3) Examine the parts of the aileron tab control rods for worn areas:

(a) Remove the parts you want to do a check.

(b) Measure the parts with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096 or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions.

(d) If the parts are out of tolerance, replace or repair the parts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 601/27-11-34-993-801

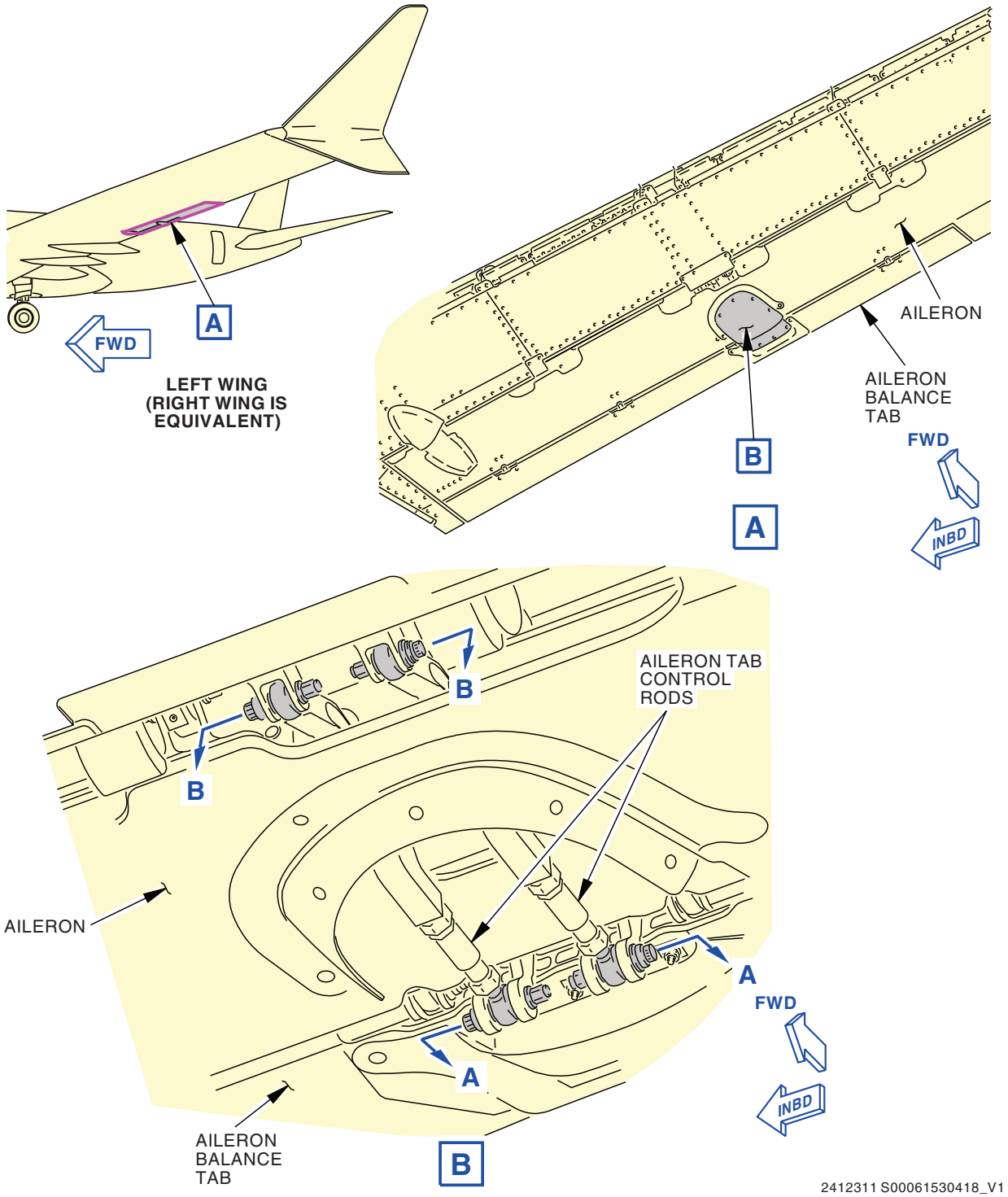
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
#1 ROD END BEARING	I.D.	0.2495 in 6.3373 mm	0.2500 in 6.3500 mm	0.2520 in 6.4008 mm	0.0000 in 0.0000 mm
#1 BOLT	O.D.	0.2485 in 6.3119 mm	0.2495 in 6.3373 mm	0.2475 in 6.2865 mm	
#2 BUSHING	I.D.	0.3750 in 9.5250 mm	0.3756 in 9.5402 mm	0.3773 in 9.5834 mm	0.0005 in 0.0127 mm
#2 BUSHING	O.D.	0.3740 in 9.4996 mm	0.3745 in 9.5123 mm	0.3728 in 9.4691 mm	
#3 BUSHING	I.D.	0.2500 in 6.3500 mm	0.2505 in 6.3627 mm	0.2525 in 6.4135 mm	0.0005 in 0.0127 mm
#3 BOLT	O.D.	0.2485 in 6.3119 mm	0.2495 in 6.3373 mm	0.2475 in 6.2865 mm	
#4 BUSHING	I.D.	0.2500 in 6.3500 mm	0.2505 in 6.3627 mm	0.2525 in 6.4135 mm	0.0005 in 0.0127 mm
#4 BOLT	O.D.	0.2485 in 6.3119 mm	0.2495 in 6.3373 mm	0.2475 in 6.2865 mm	

SUBTASK 27-11-34-020-007

(4) Do this task: Aileron Balance Tab Installation, TASK 27-11-21-400-801 or Aileron Tab Control Rods Installation, TASK 27-11-34-400-801.

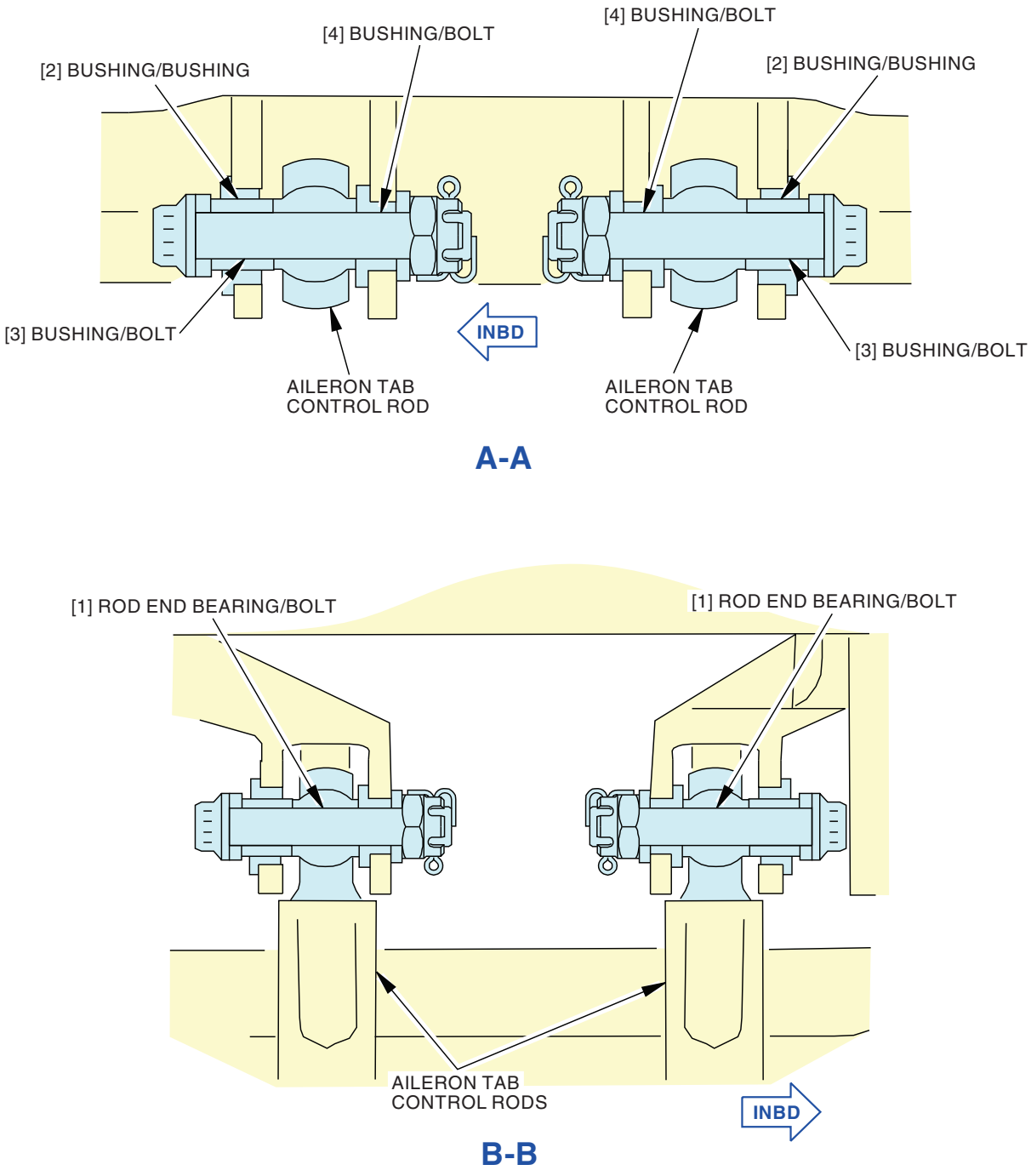
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Tab Control Rods Wear Limits
Figure 601/27-11-34-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Tab Control Rods Wear Limits
Figure 601/27-11-34-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON CONTROL WHEEL - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Aileron Control Wheel Removal

(2) Aileron Control Wheel Installation

B. This procedure is applicable for the captain's and first officer's aileron control wheel [3].

TASK 27-11-41-000-801

2. Aileron Control Wheel Removal

A. References

Reference	Title
22-11-41-000-801	Autopilot Disengage Switch Removal (P/B 401)
23-51-04-000-801	Push-To-Talk (PTT) Control Wheel Switch Removal (P/B 401)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
27-41-97-000-801	Stabilizer Trim Control Switch Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 27-11-41-860-001

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-41-860-004

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-11-41-860-002

(3) Make sure the aileron control wheels [3] are in the neutral position.

SUBTASK 27-11-41-480-002

(4) Install the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron feel and centering unit [21] (Figure 401).

E. Aileron Control Wheel Removal

SUBTASK 27-11-41-020-001

(1) Do these steps to remove the aileron control wheel [3] (Figure 402):

(a) Remove the screws [5] that attach the medallion [6] to the aileron control wheel [3].

(b) Remove the medallion [6] from the aileron control wheel [3].

(c) Remove the pin [10], nut [4], washer [8], washer [7], spacer [9], and bolt [11]. Discard the pin [10].

(d) Write down and make a mark of the Jiffy junction modules that are on top of the terminal cover [16].

(e) Disconnect the autopilot Jiffy junction modules (TASK 22-11-41-000-801).

(f) Disconnect the PTT Jiffy junction modules, similar to autopilot.

(g) Remove the screws [12] and washers [13] to remove the terminal cover [16].

(h) Remove the screws [19], washers [18], and washers [20] to disconnect the wires from the terminal block [17].

(i) Write down and make a mark of the configuration of the identified color wires.

(j) Remove the terminal block [17] from the hub of the wheel.

(k) Carefully remove the aileron control wheel [3] from the control column.

(l) Make a mark of the positions of the aileron control wheel [3], the gear, and the sleeve assembly [1].

(m) Make a mark on the wire bundle to identify the clamp [15] location in the control wheel hub.

(n) Remove the screw [14] to remove the clamp [15] from the aileron control wheel [3].

(o) Carefully disengage the wire bundle from the aileron control wheel [3].

SUBTASK 27-11-41-020-002

(2) If you want to remove the stabilizer trim control switch, do this task: Stabilizer Trim Control Switch Removal, TASK 27-41-97-000-801.

SUBTASK 27-11-41-020-003

(3) If you want to remove the autopilot disengage switch, do this task: Autopilot Disengage Switch Removal, TASK 22-11-41-000-801.

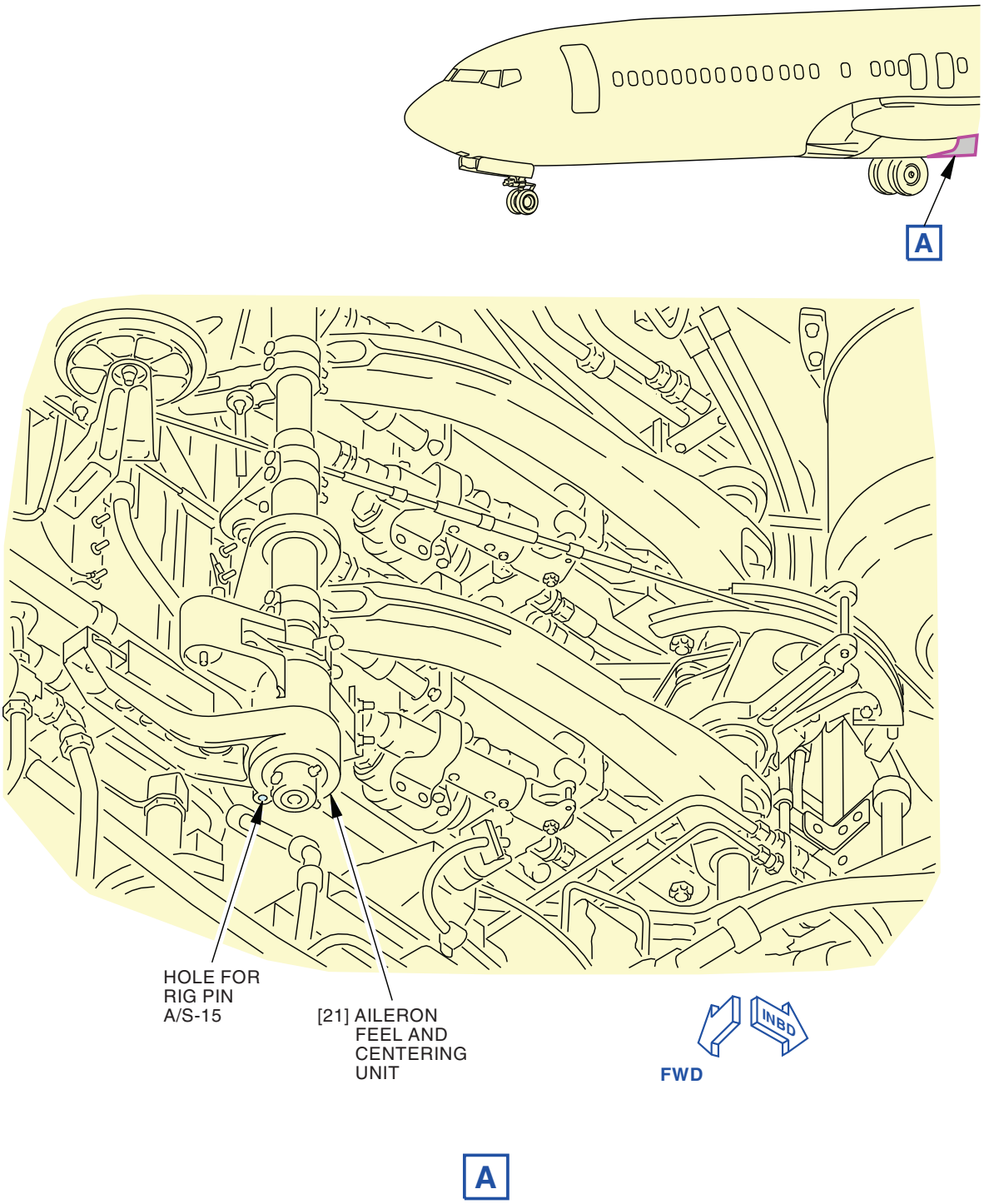
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-41-020-004

(4) If you want to remove the control wheel PTT switch, do this task: Push-To-Talk (PTT) Control Wheel Switch Removal, TASK 23-51-04-000-801.

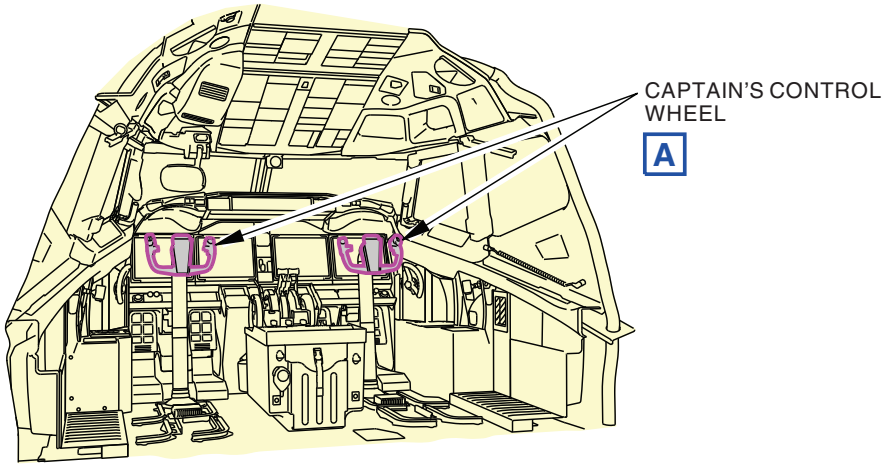
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

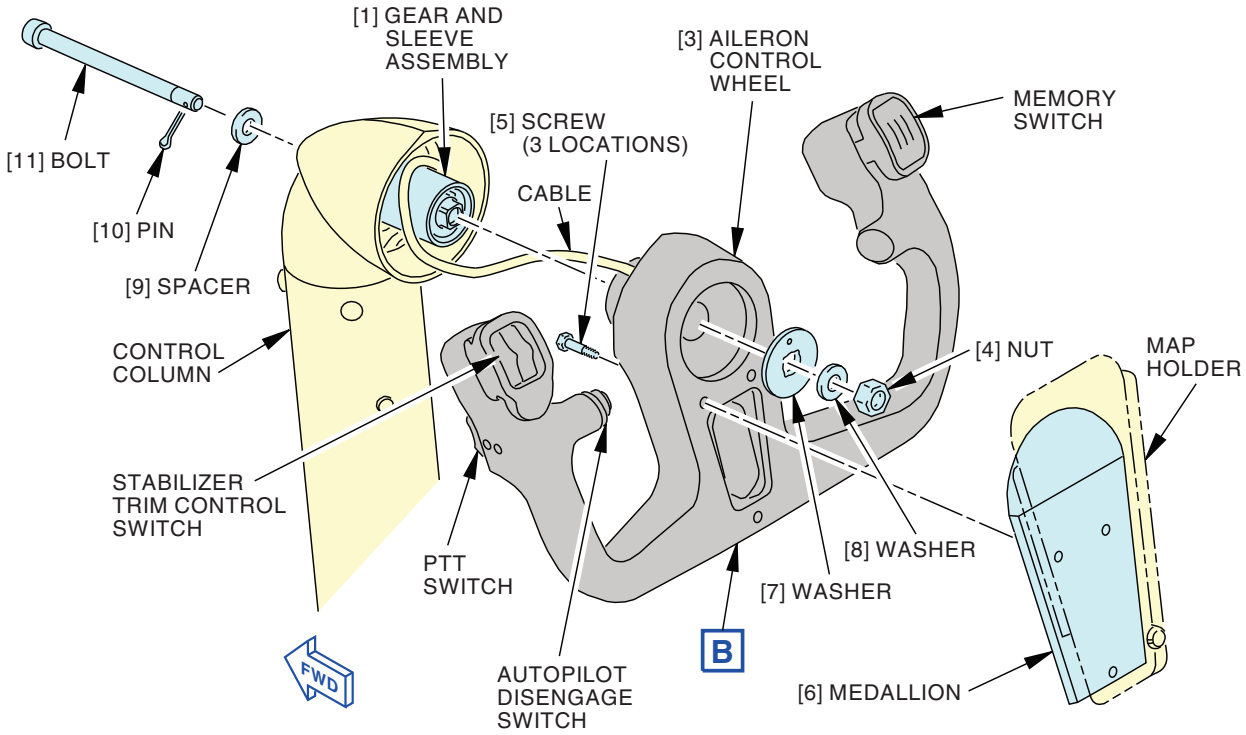


Rig Pin Installation
Figure 401/27-11-41-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT



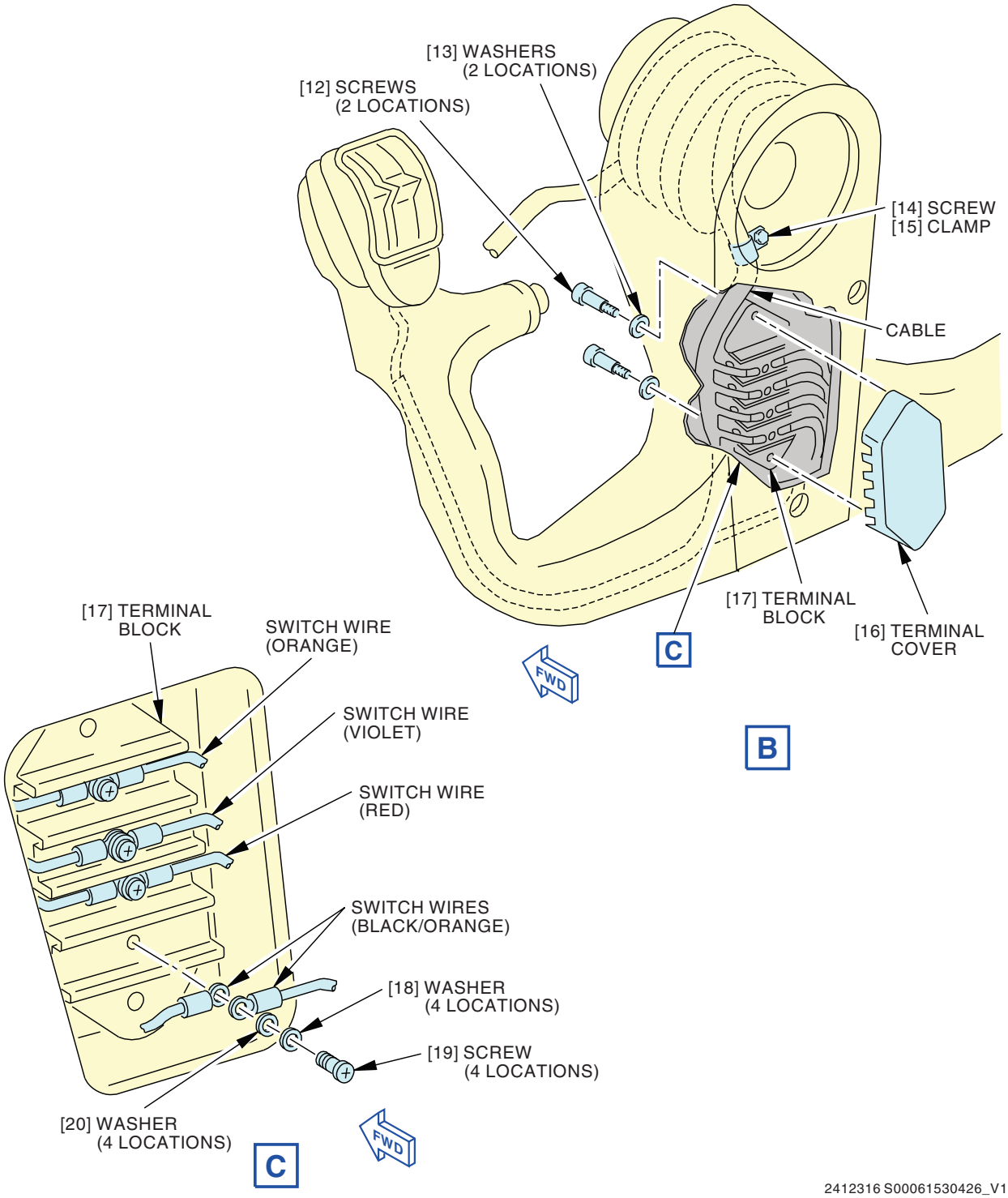
CAPTAIN'S CONTROL WHEEL
(FIRST OFFICER'S CONTROL WHEEL IS OPPOSITE)

A

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Aileron Control Wheel Installation
Figure 402/27-11-41-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Control Wheel Installation
Figure 402/27-11-41-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-41-400-801

3. Aileron Control Wheel Installation

A. References

Reference	Title
22-11-41-400-801	Autopilot Disengage Switch Installation (P/B 401)
23-51-04-400-801	Push To Talk (PTT) Control Wheel Switch Installation (P/B 401)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
27-41-97-400-801	Stabilizer Trim Control Switch Installation (P/B 401)
27-41-97-700-801	Stabilizer Trim Control Switch Test (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-17798	Multimeter - Digital, Insulation Resistance Part #: 1004-745 Supplier: U0146 Part #: 1507 Supplier: 4U744 Part #: 1587 FC Supplier: 4U744 Part #: MIT410/2 Supplier: U0146 Opt Part #: 1587 Supplier: 89536
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1670	Mount - Control Wheel Protractor Part #: C27060-22 Supplier: 81205 Part #: F72790 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205
SPL-1671	Straight Edge - Control Wheel Part #: C27097-1 Supplier: 81205 Part #: SE27-0001 Supplier: 81205
SPL-1674	Assembly - Adapter, Control Wheel, Torque and Force Test Part #: C27060-22 Supplier: 81205 Opt Part #: C27060-1 Supplier: 81205 Opt Part #: F72867-1 Supplier: 81205
SPL-1680	Protractor Assembly - Control Column Part #: A27021-29 Supplier: 81205 Part #: G76002-42 Supplier: 81205 Opt Part #: 4MIT65B80307-1 Supplier: 81205 Opt Part #: F52485-500 Supplier: 81205 Opt Part #: G76002-19 Supplier: 81205 Opt Part #: G76002-37 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G50550	Compound - Electrical Insulating (Dow Corning 5)	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Aileron control wheel	27-11-41-01-210	PTF ALL
		27-11-41-01-215	PTF ALL
10	Pin	27-11-41-01-045	PTF ALL

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Aileron Control Wheel Installation

SUBTASK 27-11-41-020-005

(1) If you removed the stabilizer trim control switch, do this task: Stabilizer Trim Control Switch Installation, TASK 27-41-97-400-801.

SUBTASK 27-11-41-020-006

(2) If you removed the autopilot disengage switch, do this task: Autopilot Disengage Switch Installation, TASK 22-11-41-400-801.

SUBTASK 27-11-41-020-007

(3) If you removed the control wheel PTT switch, do this task: Push To Talk (PTT) Control Wheel Switch Installation, TASK 23-51-04-400-801.

SUBTASK 27-11-41-420-001

(4) Do these steps to install the aileron control wheel [3] (Figure 402):

(a) Install the terminal block [17] in the hub of the wheel.

(b) Install the screws [19], washers [18], and washers [20] to connect the terminals of the switch wires to the terminal block [17].

(c) Make sure the terminals of the switch wires are installed correctly.

(d) Attach the terminal cover [16] to the aileron control wheel [3] with washers [13] and screws [12].

(e) Reconnect the autopilot Jiffy junction modules (TASK 22-11-41-400-801).

(f) Reconnect the PTT Jiffy junction modules, similar to autopilot.

(g) Apply a small amount of the compound, G50550 to the wire bundle and the inside of the hub of the control wheel splines.

(h) Use a wire leader to put the electrical wires in the aileron control wheel [3].

(i) Put a clamp [15] at the mark on the wire bundle.

CAUTION

MAKE SURE THAT THE WIRE BUNDLE LENGTH IS CORRECT BETWEEN THE AILERON CONTROL COLUMN AND THE AILERON CONTROL WHEEL. THE AILERON CONTROL WHEEL CANNOT MOVE FREELY IF THE WIRE BUNDLE IS NOT CORRECT. THIS CAN CAUSE DAMAGE TO THE WIRE BUNDLE.

(j) Make sure the length of the wire bundle is approximately 27 in. (686 mm) between the clamp [15] in the aileron control wheel [3] and the hole in the housing.

(k) Make sure the aileron control wheel [3] moves an equal distance each direction from the neutral position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(l) Install the clamp [15] with the screw [14] to hold the wire bundle to the aileron control wheel [3].

(m) Wind the wire bundle four times around the control wheel hub (Figure 402 (Sheet 2)).

1) As you look forward into the aileron control wheel [3] upper cavity do this step:

a) For the captain's aileron control wheel [3], wind the wire bundle counterclockwise.

b) For the first officer's aileron control wheel [3], wind the wire bundle clockwise .

(n) Engage the splines on the aileron control wheel [3] to the control column at the identified positions.

(o) Install the bolt [11], spacer [9], washer [7], washer [8], and nut [4].

(p) Tighten the nut [4] 85 in-lb (10 N·m)-140 in-lb (16 N·m) until the next nut castellation aligns with the hole in the bolt.

(q) Install the new pin [10].

SUBTASK 27-11-41-720-001

(5) Do these steps to check the operation of the aileron control wheel [3]:

(a) Make sure the captain's and the first officer's aileron control wheels [3] are in the neutral position.

(b) Put the straight edge, SPL-1671 across the top end of the aileron control wheels.

1) Make sure that one end of the first officer's aileron control wheel [3] is not farther than 0.20 in. (5.08 mm) from the straightedge.

(c) Remove the straight edge, SPL-1671 from the aileron control wheels [3].

(d) Install the adapter, SPL-1674 to the captain's aileron control wheel [3].

(e) Remove the map holder assembly from the first officer's control wheel.

(f) Install the mount, SPL-1670 and the protractor, SPL-1680 to the first officer's aileron control wheel [3].

(g) Remove the rig pin A/S-15, rigging pin kit, SPL-1585, from the aileron feel and centering unit (Figure 401).

CAUTION

DO NOT LET THE AILERON CONTROL WHEEL TURN MORE THAN 110 DEGREES IN EACH DIRECTION FROM NEUTRAL. IF THE AILERON CONTROL WHEEL TURNS MORE THAN 110 DEGREES DAMAGE TO THE WIRE BUNDLE CAN OCCUR.

(h) Make sure the aileron control wheel [3] can turn 108 degrees in each direction from the neutral position.

(i) Make slow and careful inputs to the aileron control wheel [3] one direction at a time.

(j) Make sure the aileron control wheel [3] moves freely and smoothly.

(k) Remove the adapter, SPL-1674 from the captain's aileron control wheel [3].

(l) Remove the mount, SPL-1670 and the protractor, SPL-1680 from the first officer's aileron control wheel [3].

(m) Install the map holder assembly to the first officer's control wheel.

SUBTASK 27-11-41-720-002

(6) Do a test of the electrical circuits for continuity changed during the replacement of the aileron control wheel [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-41-720-003

(7) Use the insulation resistance multimeter, COM-17798, to do a test of the terminal block [17] while you turn the aileron control wheel [3] from stop to stop.

(a) Do a test on the terminal block for resistance to ground, or wire to wire for a minimum insulation value of 100 Megaohms.

SUBTASK 27-11-41-420-002

(8) Do these steps to install the medallion [6].

(a) Install the medallion [6] on the aileron control wheel [3].

(b) Install the screws [5] that attach the medallion [6] to the aileron control wheel [3].

SUBTASK 27-11-41-860-005

(9) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
E	7	C01208	STALL WARN ASYM MODE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 27-11-41-700-001

(10) Do the Installation Test of the autopilot switch (TASK 22-11-41-400-801).

SUBTASK 27-11-41-700-002

(11) Do the Installation Test of the PPT switch (TASK 23-51-04-400-801).

SUBTASK 27-11-41-700-003

(12) Do this task: Stabilizer Trim Control Switch Test, TASK 27-41-97-700-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-41-860-003

(1) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON CONTROL WHEEL DRUM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron control wheel drum

(2) An installation of the aileron control wheel drum.

B. The aileron control wheel drum is referred to as the "control wheel drum assembly" in this procedure.

C. The control wheel drum assembly is installed at the base of the captain's control column.

TASK 27-11-51-000-801

2. Aileron Control Wheel Drum Removal

(Figure 401)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

E. Prepare for the Removal

SUBTASK 27-11-51-010-001

(1) Open this access door:

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-51-010-002

(2) If necessary, remove these access panels to access the aileron control wheel drum (TASK 53-14-01-020-801).

(a) Remove these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-11-51-860-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-51-860-002

(4) Put the captain's and first officer's control wheels in the neutral position.

SUBTASK 27-11-51-480-001

(5) Install the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim mechanism.

F. Aileron Control Wheel Drum Removal

SUBTASK 27-11-51-020-001

(1) Disconnect the cable ACBA1 [1], cable ACBA2 [21], cable ACBB1 [2], and cable ACBB2 [22] at the cable turnbuckles [3].

(a) Attach identification tags on the disconnected cables for easier assembly.

SUBTASK 27-11-51-020-002

(2) Disconnect the cable AA [4] and cable AB [6] at the cable turnbuckles [5].

(a) Attach identification tags on the disconnected cables for easier assembly.

SUBTASK 27-11-51-020-005

(3) Disconnect brake control cable LGB1A [39] and cable LGB1B [40] at the turnbuckles.

(a) Attach identification tags on the disconnected cables for easier assembly.

SUBTASK 27-11-51-020-004

(4) Remove the aileron control wheel drum assembly [19]:

(a) Remove the screw [35] and clamp [37] that attach the rudder pedal adjustment crank to the aileron control wheel drum assembly [19].

(b) Disconnect the electrical connector [15], electrical connector [30], and electrical connector [34].

(c) Remove the screws [31], washers [32], and nuts [33] to remove the wiring harness [38] from the bracket on the aileron control wheel drum assembly [19].

(d) Remove the screw [24] and clamp [26].

(e) Remove the nuts [14], washers [13], and bolts [12].

(f) Remove the nuts [11], washers [10], and bolts [9].

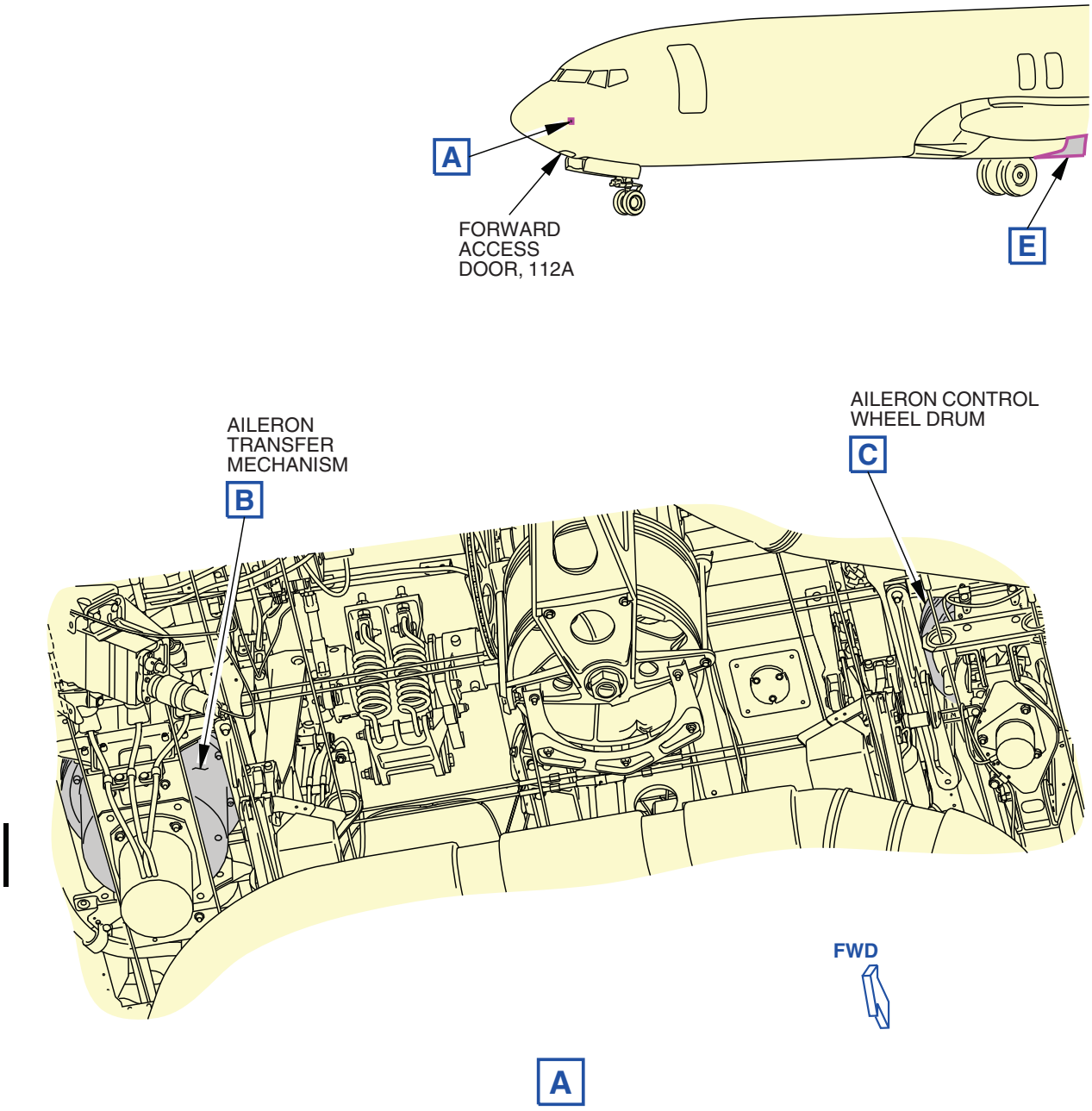
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Move the aileron control wheel drum assembly [19] down while, at the same time, you put the cable ACBA1 [1], cable ACBA2 [21], cable ACBB1 [2], and cable ACBB2 [22] through the openings.

(h) Remove the control wheel drum assembly [19] from the airplane structure.

——— END OF TASK ———

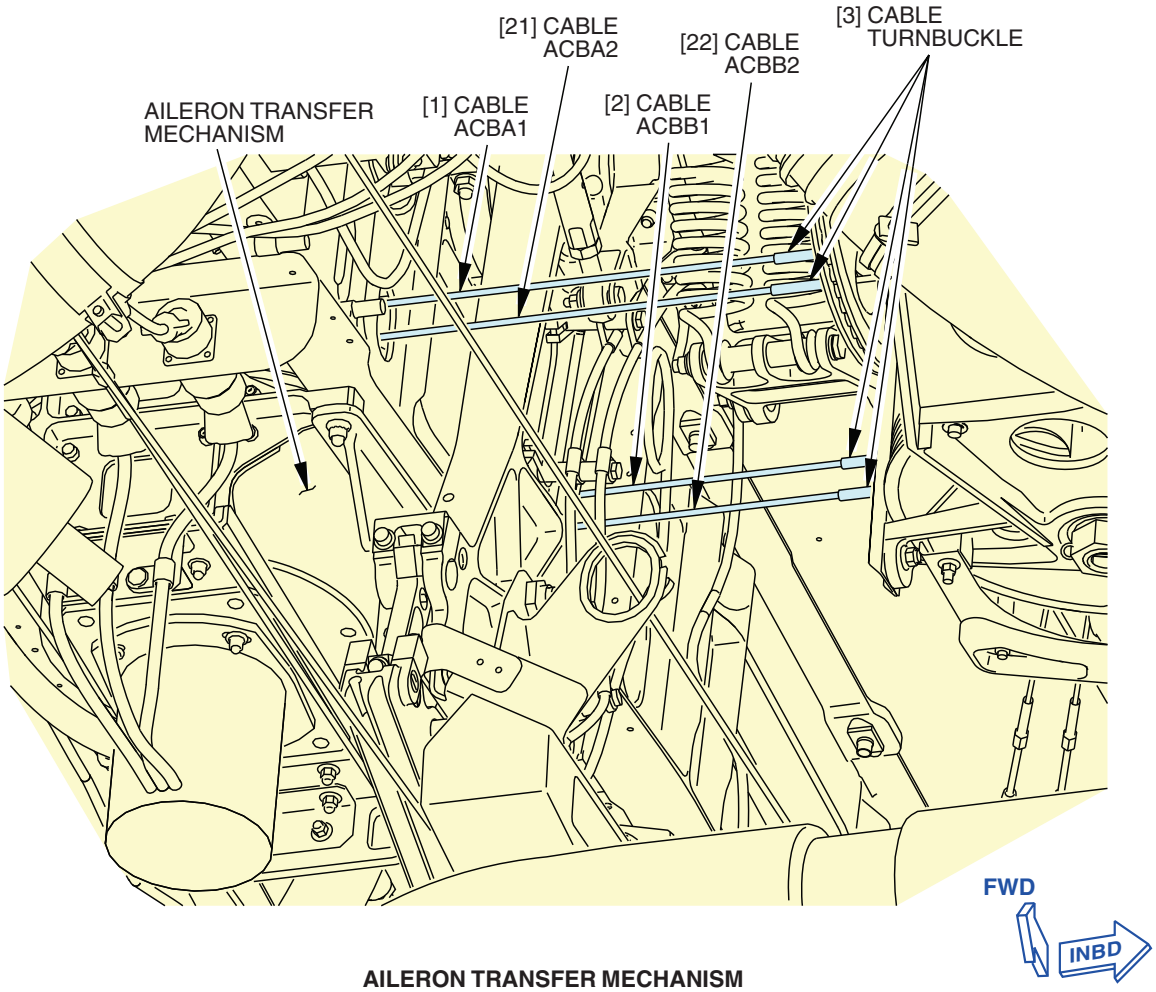
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412317 S00061530431_V2

Aileron Control Wheel Drum Installation
Figure 401/27-11-51-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



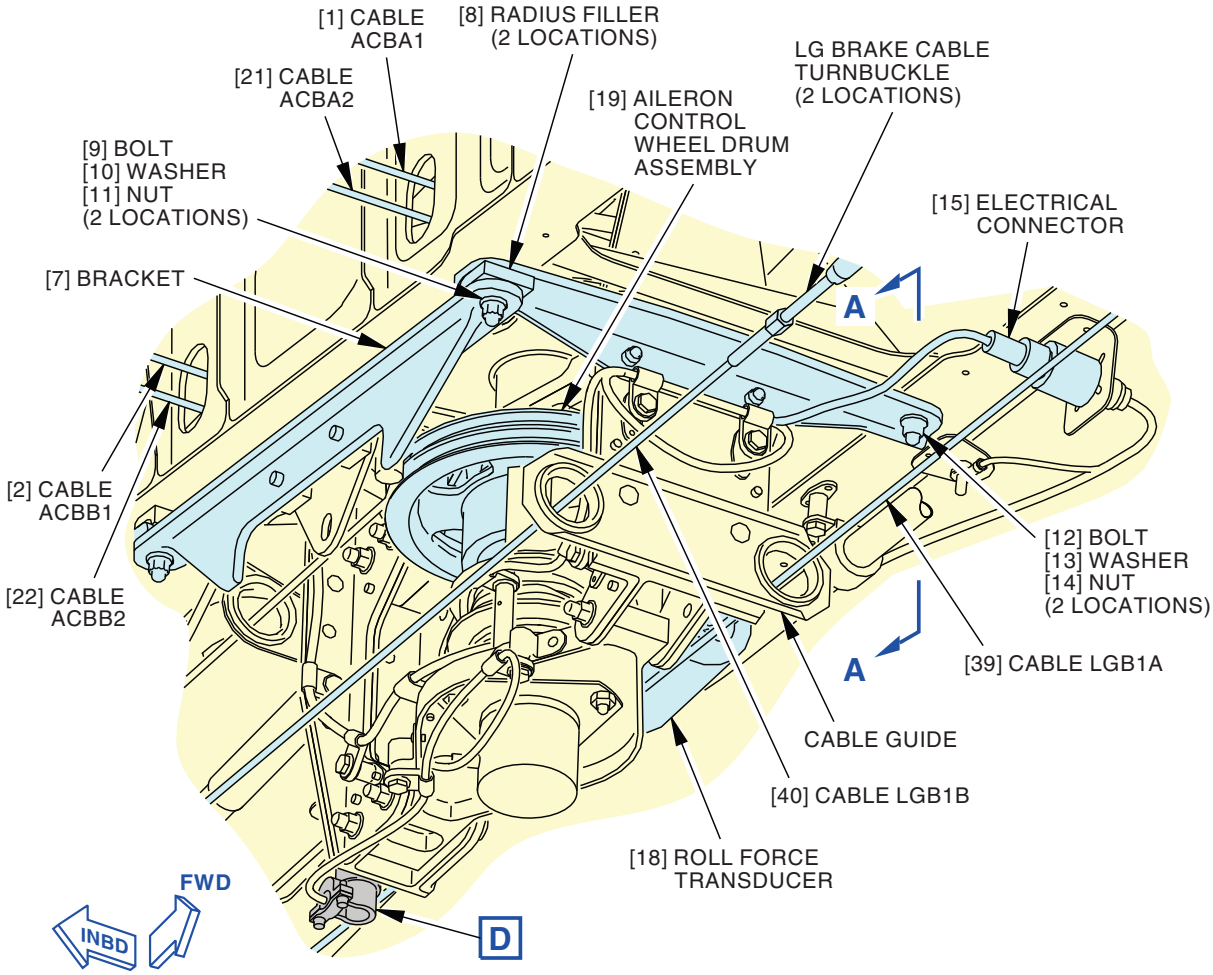
AILERON TRANSFER MECHANISM

B

2412318 S00061530432_V2

Aileron Control Wheel Drum Installation
Figure 401/27-11-51-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



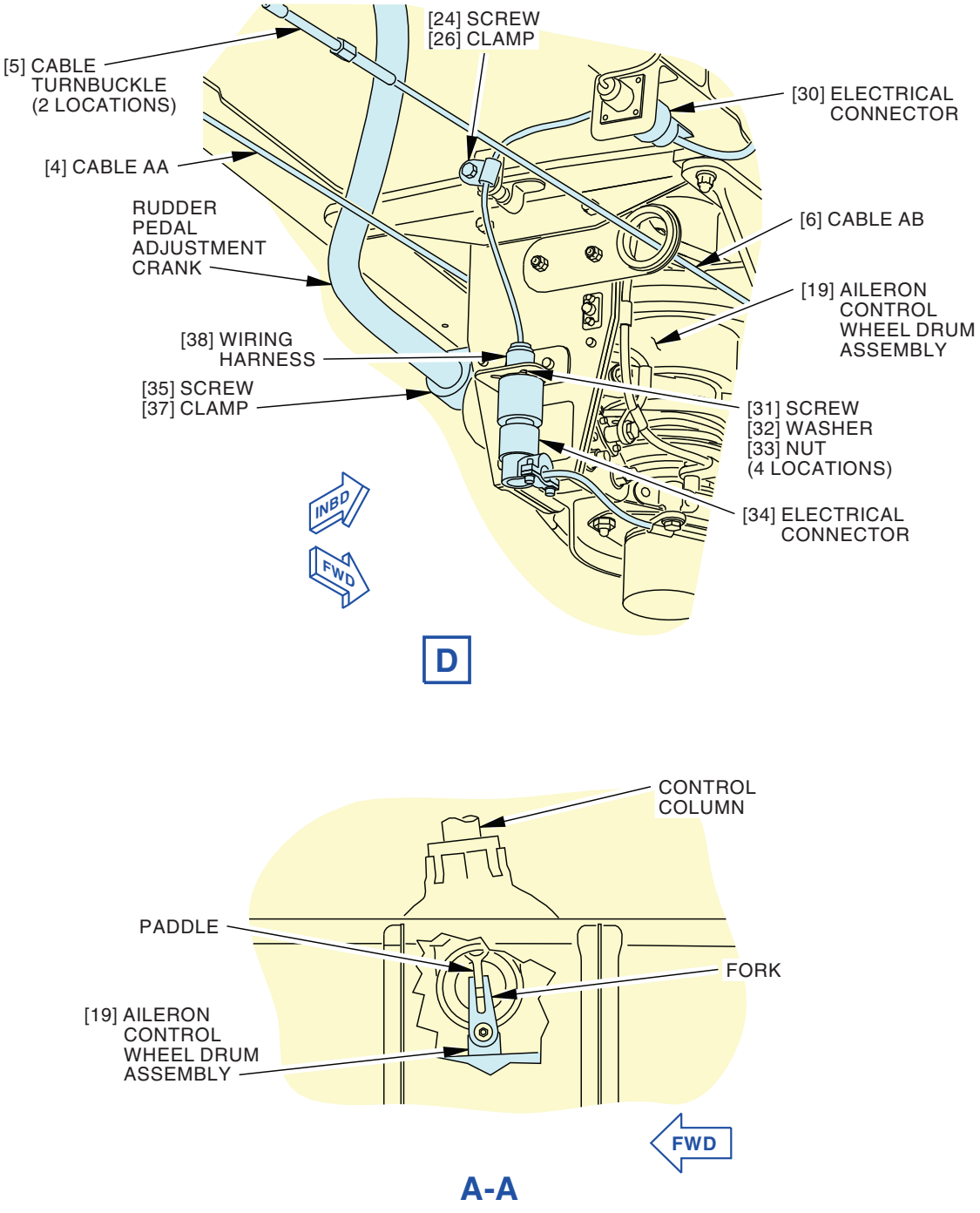
AILERON CONTROL WHEEL DRUM

C

2412319 S00061530433_V4

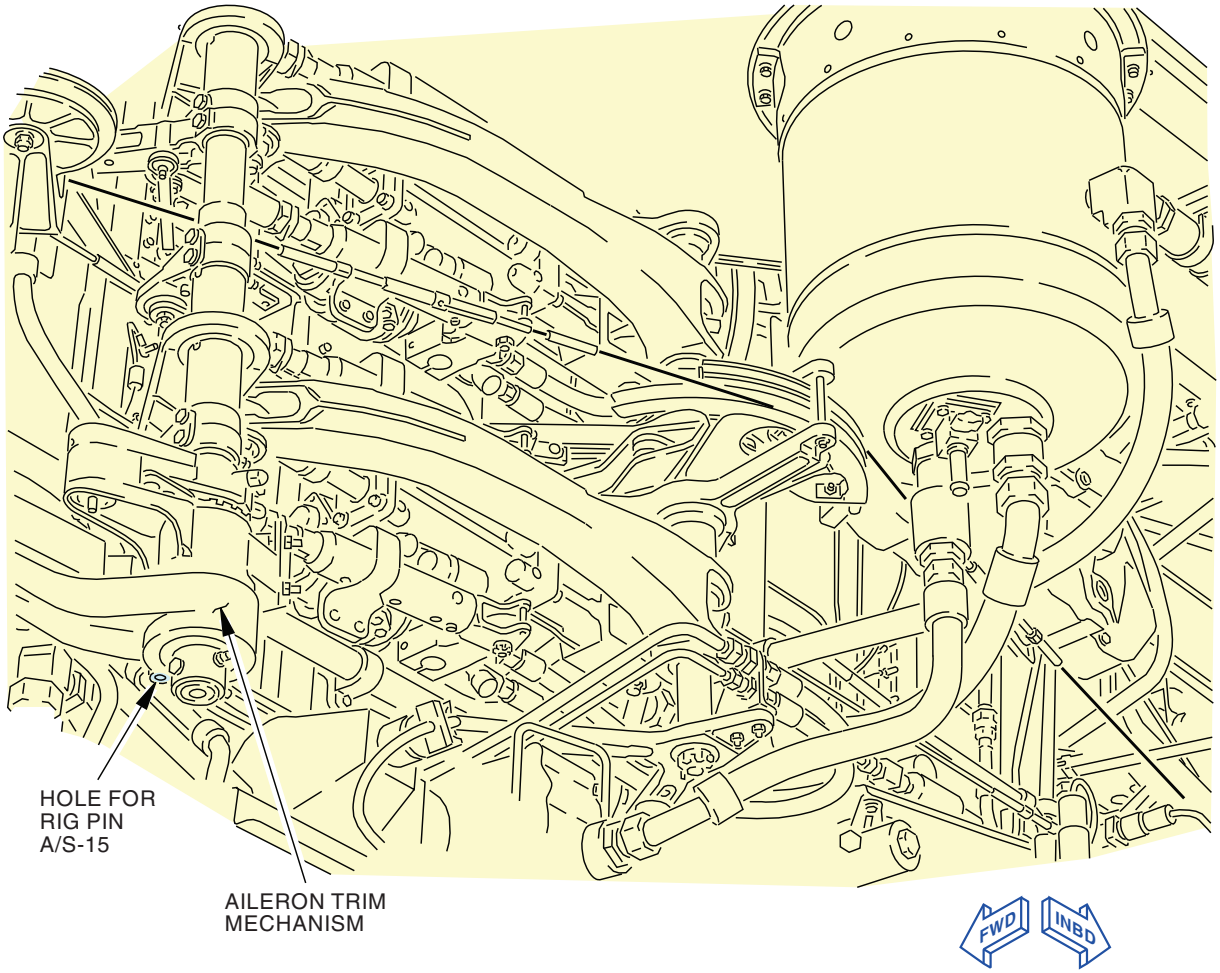
Aileron Control Wheel Drum Installation
Figure 401/27-11-51-990-801 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Control Wheel Drum Installation
Figure 401/27-11-51-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412321 S00061530435_V2

Aileron Control Wheel Drum Installation
Figure 401/27-11-51-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-51-400-801

3. Aileron Control Wheel Drum Installation

(Figure 401)

A. References

Reference	Title
22-11-11-740-801	Roll Force Transducer Test (P/B 501)
27-11-00-700-803	Manual Reversion Control Friction Test (P/B 501)
27-11-00-700-804	Control Wheel Centering Check (P/B 501)
27-11-00-700-805	Aileron Control Wheel Force Test (P/B 501)
27-11-00-700-807	Control Wheel Travel Stop Test (P/B 501)
27-11-00-710-801	Aileron Response (Operational) Test (P/B 501)
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
31-31-32-820-801	Control Wheel Position Sensor Installation Test (P/B 401)
32-41-00-820-801	Hydraulic Brake System Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00113	Lubricant - Solid Film, Liquid Dispersed	BMS3-8
D50108	Lubricant - Solid Film, Air Cured, Corrosion Inhibiting	MIL-L-23398
G50063	Lubricant - PTFE Release Agent - Miller-Stephenson MS-122XD (Replaces MS-122DF)	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
19	Control wheel drum assembly	27-11-51-01-030	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
112A	Forward Access Door
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

G. Aileron Control Wheel Drum Installation

SUBTASK 27-11-51-210-001

WARNING MAKE SURE THAT IT IS THE SELF-LOCKING SHEAR NUT AND NOT THE TENSION TYPE SELF-LOCKING NUT THAT CONNECTS THE FORK TO THE CONTROL WHEEL DRUM. IF AN INCORRECT NUT IS INSTALLED, THE CAPTAIN'S CONTROL COLUMN CAN BECOME DISENGAGED FROM THE FORK. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Make sure that the fork is connected to the control wheel drum with correct self-locking shear nut.

NOTE: You must see a completed bolt chamfer at the end of the nut.

SUBTASK 27-11-51-420-001

(2) Install the aileron control wheel drum assembly [19]:

(a) Apply MS-122XD release agent, G50063, to the paddle.

(b) Apply solid film lubricant, D00113, or lubricant, D50108, to the fork.

(c) Put the bolts [9] and bolts [12] into position on the airplane structure.

(d) Make sure that the radius fillers [8] are bonded to the bracket [7].

(e) Lift the aileron control wheel drum assembly [19] into position while making sure that the fork and paddle engage.

(f) Install the washers [10] and nuts [11] on the bolts [9].

(g) Install the washers [13] and nuts [14] on the bolts [12].

(h) Put the ends of cable ACBA1 [1], cable ACBA2 [21], cable ACBB1 [2], and cable ACBB2 [22] through the openings.

(i) Install the screw [24] and clamp [26].

(j) Attach the wiring harness [38] to the aileron control wheel drum assembly [19] with screws [31], washers [32], and nuts [33].

(k) Connect the electrical connector [15], electrical connector [30], and electrical connector [34].

(l) Install the screw [35] and clamp [37] to attach the rudder pedal adjustment crank to the aileron control wheel drum assembly [19].

SUBTASK 27-11-51-420-003

(3) To connect the cable AA [4] and cable AB [6] at the cable turnbuckles [5], do these steps:

(a) Remove the identification tags from the disconnected cables.

(b) Connect the cable AA [4] and cable AB [6] at the cable turnbuckles [5].

(c) Tighten the cable turnbuckles [5] to remove slack but keep the cable tension at zero.

SUBTASK 27-11-51-420-004

(4) Connect the cable ACBA1 [1], cable ACBA2 [21], cable ACBB1 [2], and cable ACBB2 [22] at the cable turnbuckles [3]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the identification tags from the disconnected cables.

(b) Connect the cable ACBA1 [1], cable ACBA2 [21], cable ACBB1 [2], and cable ACBB2 [22] at the cable turnbuckles [3].

(c) Tighten the cable turnbuckles [3] to remove slack but keep the cable tension at zero.

SUBTASK 27-11-51-420-005

(5) To connect brake control cable LGB1A [39] and cable LGB1B [40] at the LG brake cable turnbuckles, do these steps:

(a) Remove the identification tags from the disconnected cables.

(b) Connect the cable LGB1A [39] and cable LGB1B [40] at the cable turnbuckles.

(c) Tighten the brake cable turnbuckles to remove slack but keep the cable tension at zero.

SUBTASK 27-11-51-800-001

(6) Make sure that the captain's and first officer's control wheels are in the neutral position.

SUBTASK 27-11-51-080-001

(7) Remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, from the aileron trim mechanism.

SUBTASK 27-11-51-860-003

(8) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-51-820-001

(9) Adjust the cables ACBA and ACBB and cables AA and AB (TASK 27-11-00-820-801).

SUBTASK 27-11-51-820-002

(10) Adjust the brake control cables (Hydraulic Brake System Adjustment, TASK 32-41-00-820-801):

(a) Do the steps to adjust brake cables LGB1A and LGB1B.

(b) Adjust the right side brake control cables LGB2A and LGB2B at the same time.

(c) Brake system retest is not required after rigging.

H. Aileron Control Wheel Drum Installation Test

SUBTASK 27-11-51-730-001

(1) Do this task: Manual Reversion Control Friction Test, TASK 27-11-00-700-803.

SUBTASK 27-11-51-710-001

(2) Do this task: Aileron Response (Operational) Test, TASK 27-11-00-710-801.

SUBTASK 27-11-51-720-002

(3) Do this task: Control Wheel Centering Check, TASK 27-11-00-700-804.

SUBTASK 27-11-51-720-003

(4) Do this task: Aileron Control Wheel Force Test, TASK 27-11-00-700-805.

SUBTASK 27-11-51-720-004

(5) Do this task: Control Wheel Travel Stop Test, TASK 27-11-00-700-807.

SUBTASK 27-11-51-710-002

(6) Do this task: Roll Force Transducer Test, TASK 22-11-11-740-801.

SUBTASK 27-11-51-710-003

(7) Do this task: Control Wheel Position Sensor Installation Test, TASK 31-31-32-820-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-51-410-003

(1) Install these access panels if they were removed to gain access to the aileron control wheel drum (TASK 53-14-01-420-801).

(a) Install these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-11-51-410-001

(2) Close this access door:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON TRIM ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron trim actuator

(2) An installation of the aileron trim actuator.

TASK 27-11-54-000-801

2. Aileron Trim Actuator Removal

(Figure 401)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 27-11-54-860-005

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

SUBTASK 27-11-54-860-001

(2) Put the captain's and first officer's control wheels in the neutral position.

SUBTASK 27-11-54-860-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-54-480-001

(4) Install the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim mechanism.

E. Aileron Trim Actuator Removal

SUBTASK 27-11-54-020-001

(1) Disconnect the electrical connector from the aileron trim actuator [1].

(a) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(b) If there was corrosion, correct the problem (SWPM 20-60-03).

SUBTASK 27-11-54-020-002

(2) Remove the aileron trim actuator [1] from the aileron trim mechanism:

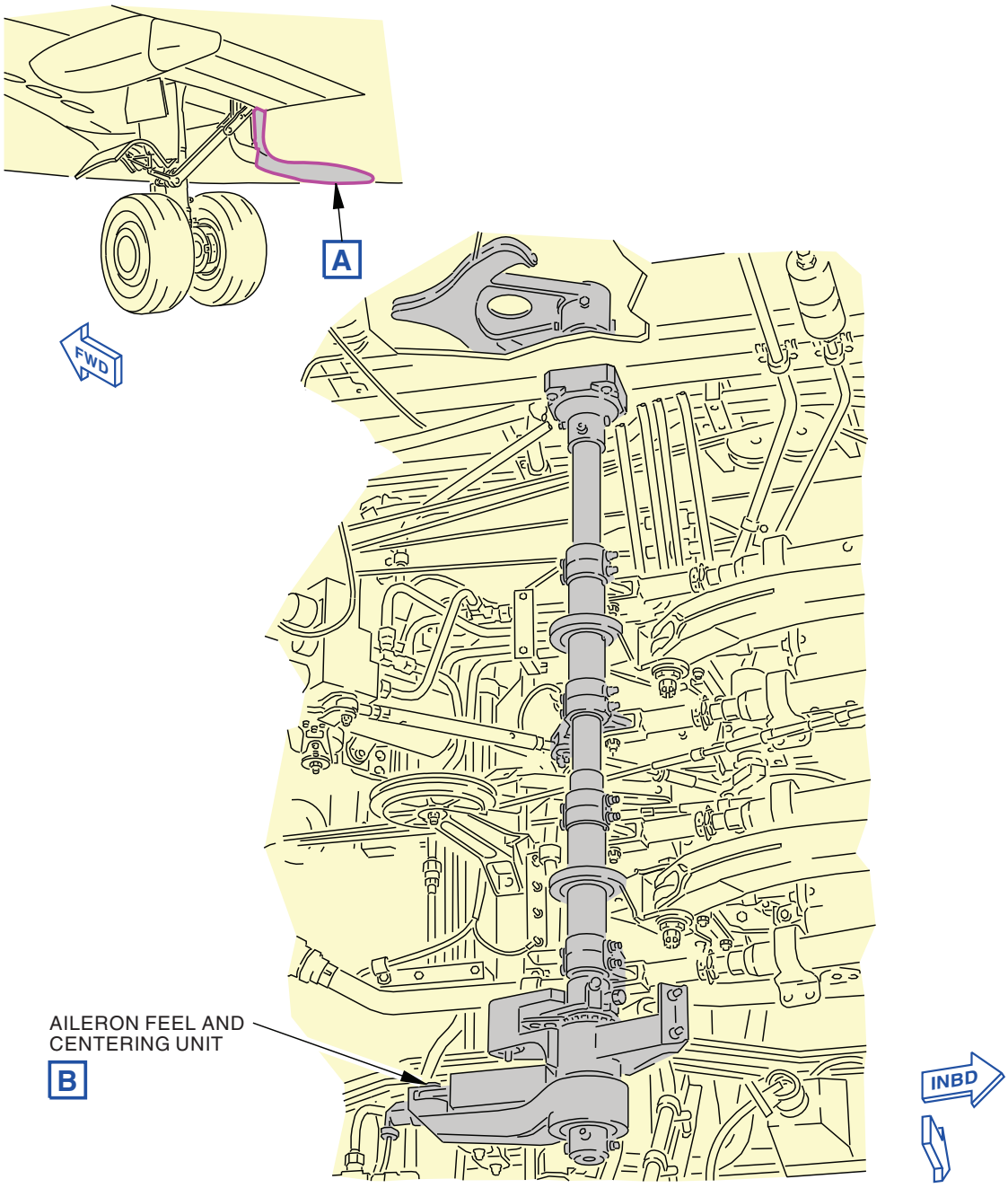
(a) Remove the nut [11], washer [9], bushing [10], washer [8], and bolt [7].

(b) Remove the nut [2], washer [4], bushing [3], washer [5], and bolt [6].

(c) Remove the aileron trim actuator [1].

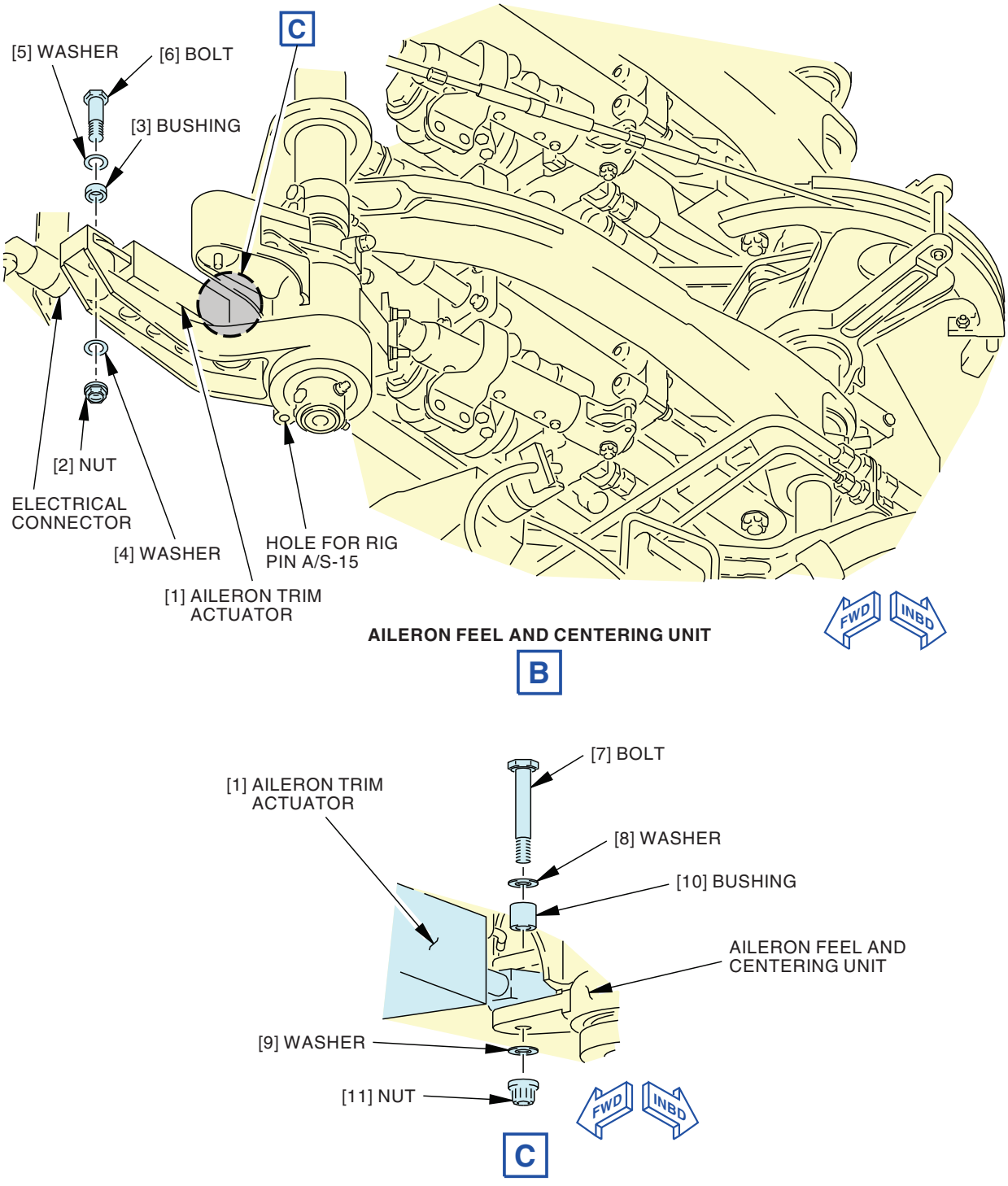
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Trim Actuator Installation
Figure 401/27-11-54-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Trim Actuator Installation
Figure 401/27-11-54-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-54-400-801

3. Aileron Trim Actuator Installation

(Figure 401)

A. References

Reference	Title
27-11-00-730-801	Aileron Trim Response Test (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Aileron trim actuator	27-11-54-01-045	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Aileron Trim Actuator Installation

SUBTASK 27-11-54-420-001

(1) Install the aileron trim actuator [1]:

(a) Make sure that the rig pin A/S-15, from the rigging pin kit, SPL-1585, is installed in the aileron trim mechanism.

(b) Put the aileron trim actuator [1] in position.

(c) Install the bolt [6], washer [5], bushing [3], washer [4], and nut [2].

(d) Install the bolt [7], washer [8], bushing [10], washer [9], and nut [11].

SUBTASK 27-11-54-420-002

(2) Do the following steps to connect the electrical connector:

(a) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

2) If there was corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the connector (SWPM 20-60-03).

(c) Connect the electrical connector to the aileron trim actuator [1].

SUBTASK 27-11-54-860-003

(3) Make sure that the captain's and first officer's control wheels are in the neutral position.

SUBTASK 27-11-54-080-001

(4) Remove the rig pin A/S-15, from the rigging pin kit, SPL-1585, from the aileron trim mechanism.

SUBTASK 27-11-54-860-006

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

SUBTASK 27-11-54-860-004

(6) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-54-710-001

(7) Do this task: Aileron Trim Response Test, TASK 27-11-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON TRANSFER MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron transfer mechanism.

(2) An installation of the aileron transfer mechanism.

B. The aileron transfer mechanism is installed at the base of the first officer's control column.

TASK 27-11-61-000-801

2. Aileron Transfer Mechanism Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
112A	Forward Access Door
114AC	Fwd Nose Wheel Well Upper Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 27-11-61-860-004

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-11-61-010-001

(2) To get access to the aileron transfer mechanism, do this task:

(a) Open these access panels:

Number	Name/Location
112A	Forward Access Door
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-11-61-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-61-860-001

(4) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-61-860-002

(5) Put the aileron control wheels in the neutral position.

(a) Make sure that the aileron trim switches are at the zero trim position.

NOTE: The aileron trim switches are installed on the aft electronic control panel, P8 in the flight compartment.

SUBTASK 27-11-61-480-002

(6) Install the rig pins A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drums.

F. Aileron Transfer Mechanism Removal

SUBTASK 27-11-61-020-001

(1) Disconnect the cable ACBA1 [1], cable ACBA2 [2], cable ACBB1 [3] and cable ACBB2 [4] at the cable turnbuckles [5].

(a) Attach identification tags on the disconnected cables for easier assembly.

SUBTASK 27-11-61-020-005

(2) Disconnect the LG brake control cable LGB2A [25] and cable LGB2B [26] at the LG brake cable turnbuckles.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Attach identification tags on the disconnected cables for easier assembly.

SUBTASK 27-11-61-020-004

(3) To remove the aileron position transducer [7] from the aileron transfer mechanism [13] do these steps:

(a) Disconnect the three electrical connectors [11].

(b) Remove the washers [9] and nuts [10].

(c) Remove the screw [28], screw [29], washers [30], clamps [27] and nuts [31].

(d) Remove the aileron position transducer [7] from the aileron transfer mechanism [13].

SUBTASK 27-11-61-020-003

(4) Remove the aileron transfer mechanism [13]:

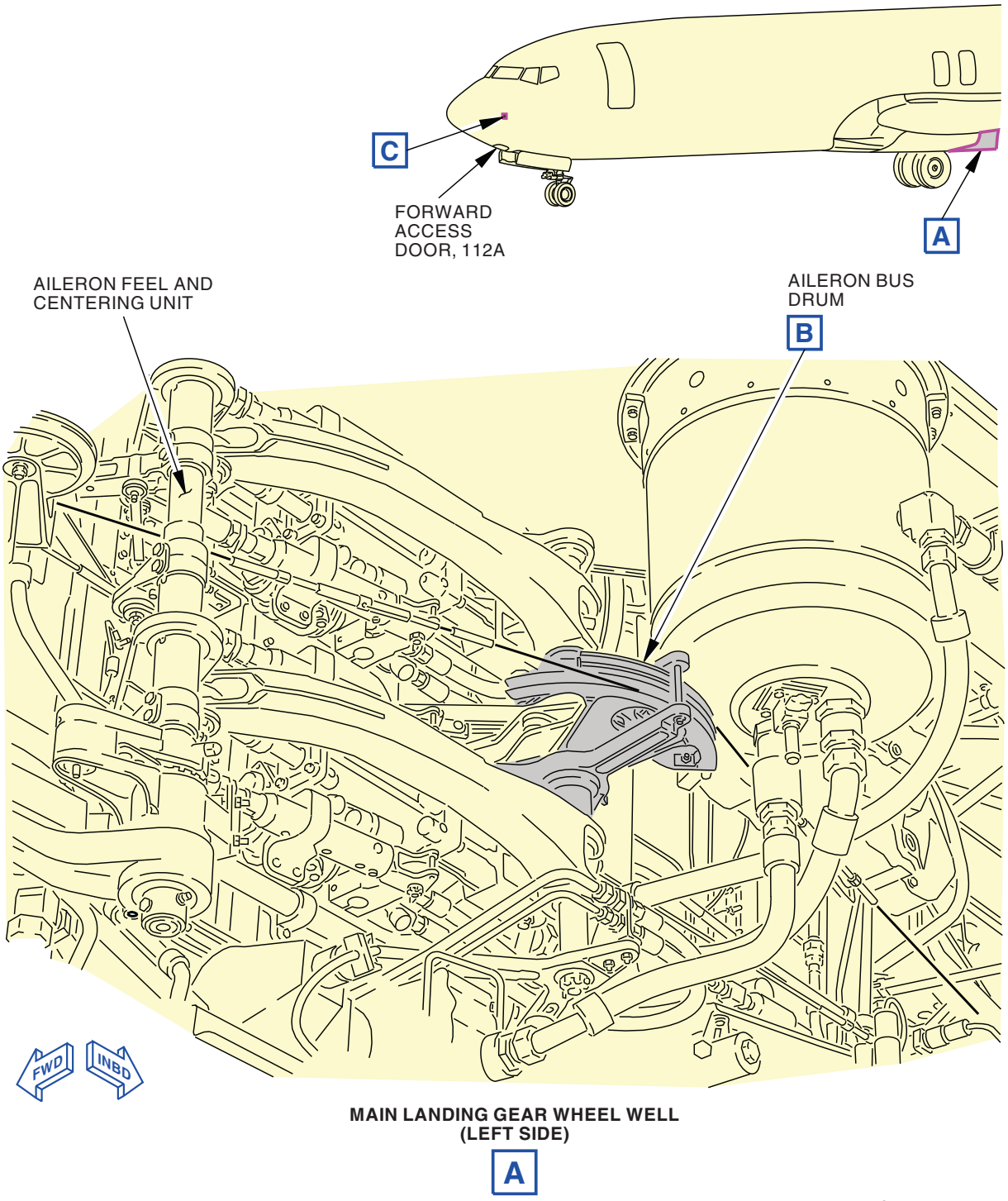
(a) Remove screw [17] and cable clamp [18] attached to the aileron transfer mechanism [13].

(b) Remove the nuts [24], washers [23], and bolts [22] that attach the aileron transfer mechanism [13] to the airplane structure.

(c) Move the aileron transfer mechanism [13] down while you put the cable ACBA1 [1], cable ACBA2 [2], cable ACBB1 [3] and cable ACBB2 [4] through the openings.

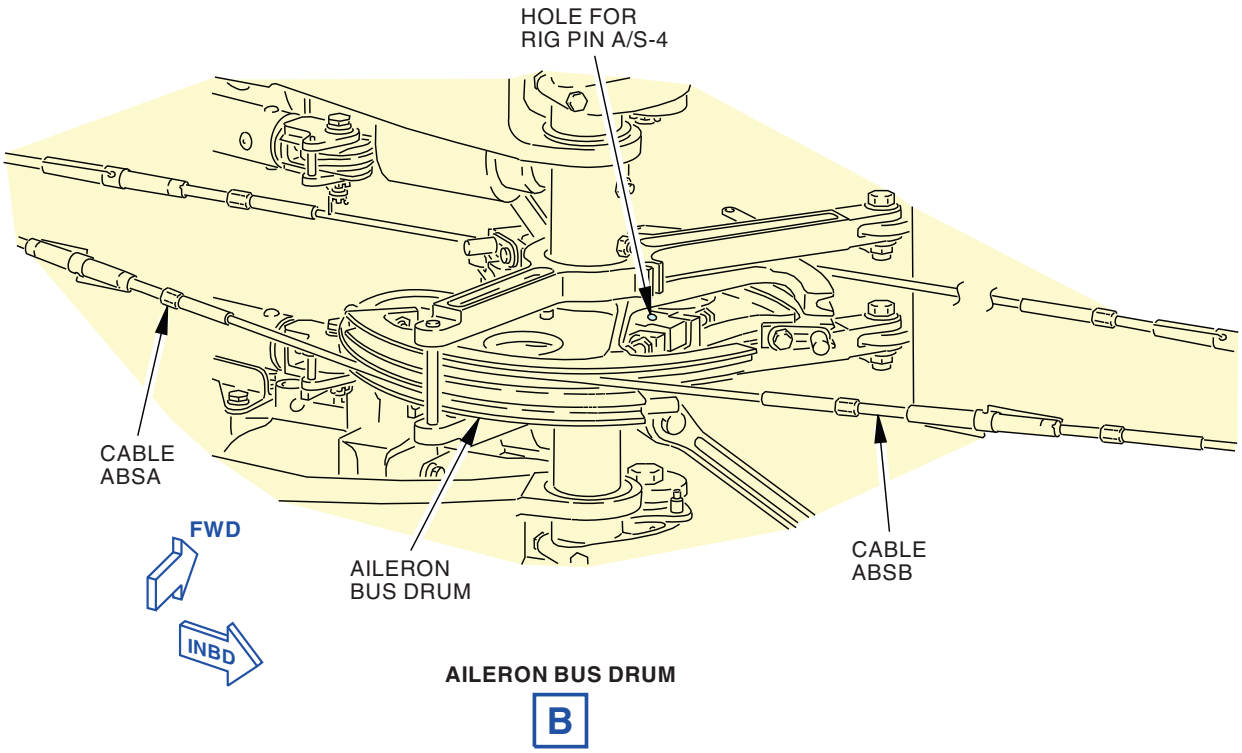
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Transfer Mechanism Installation
Figure 401/27-11-61-990-801 (Sheet 1 of 4)

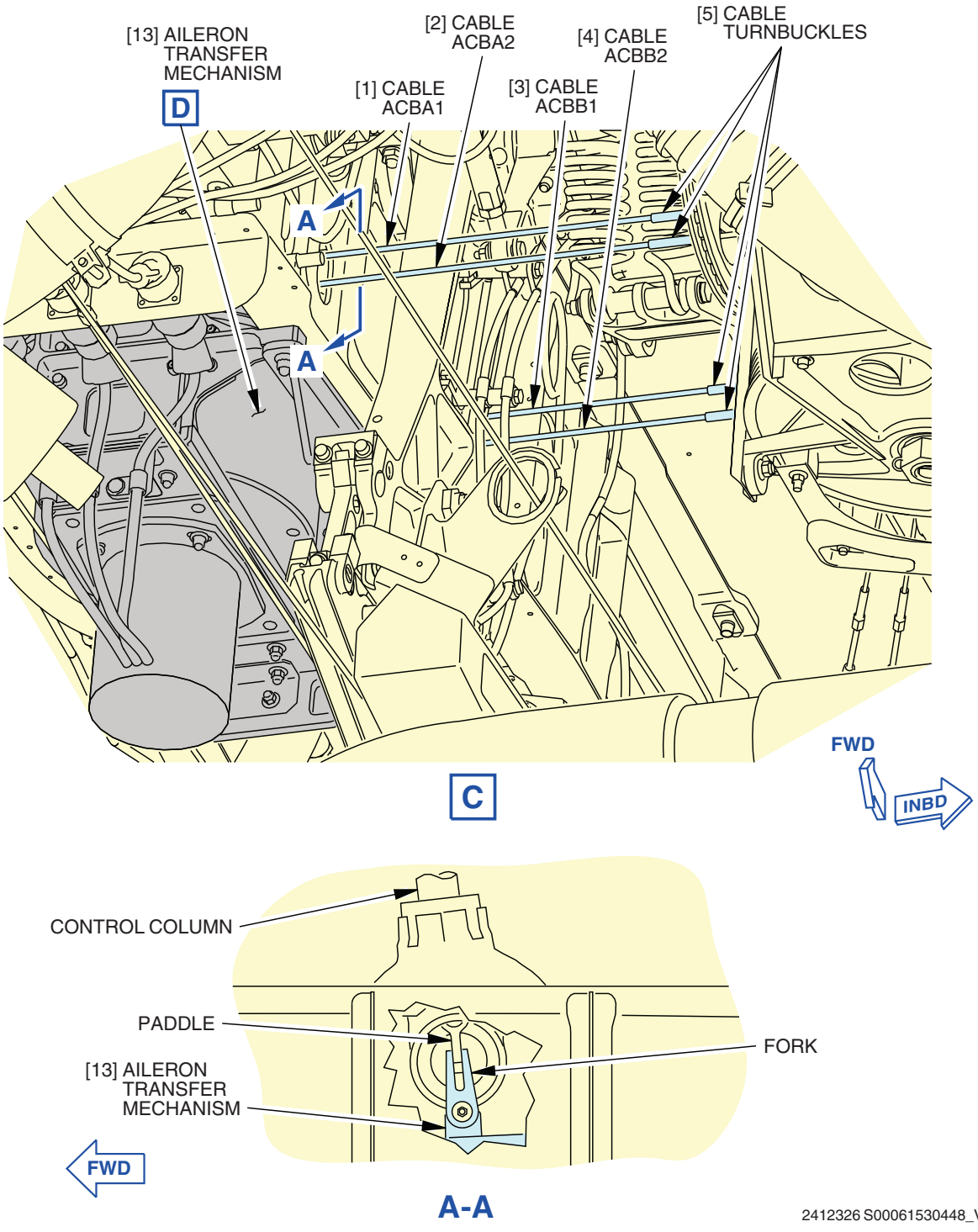
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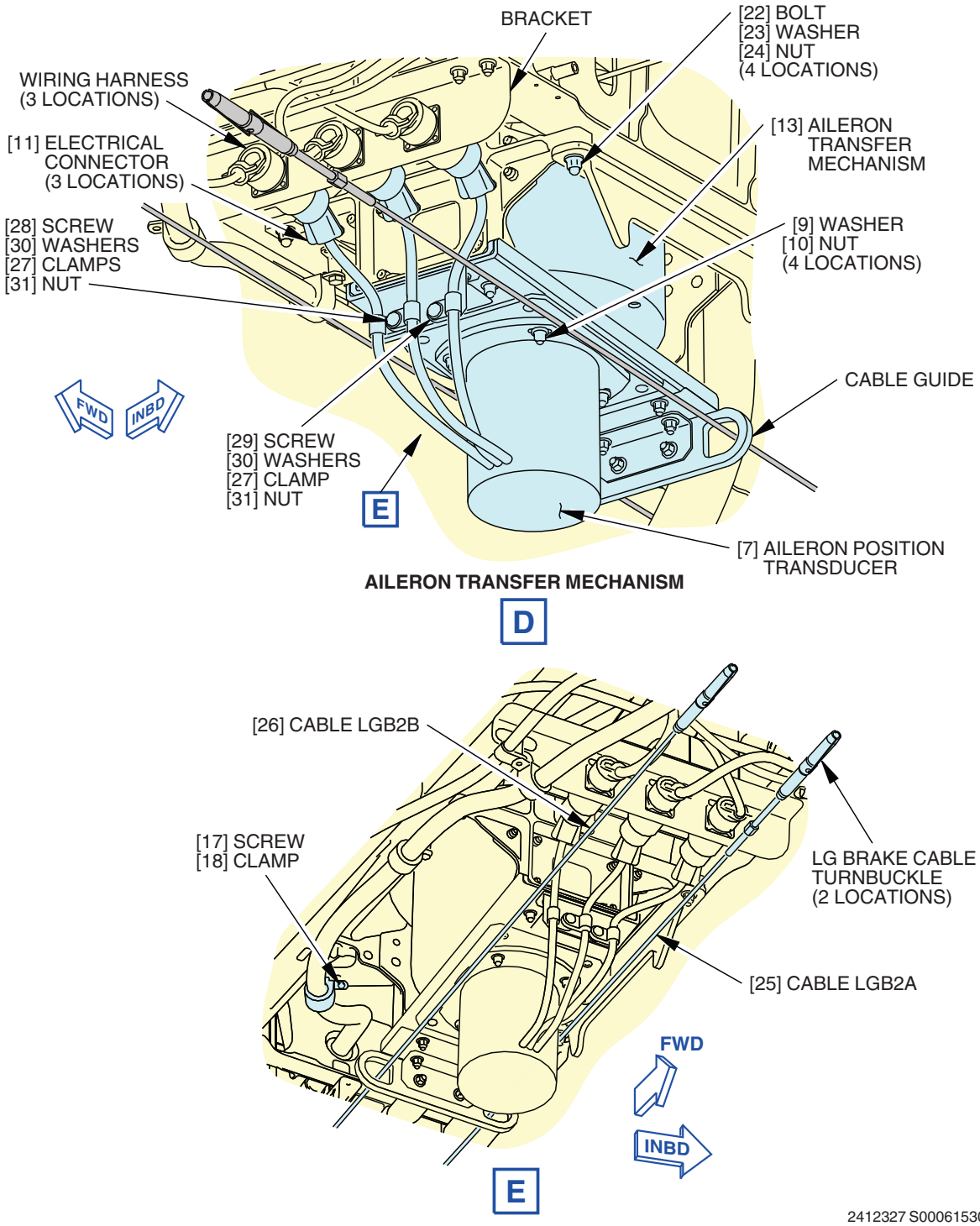
Aileron Transfer Mechanism Installation
Figure 401/27-11-61-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Transfer Mechanism Installation
Figure 401/27-11-61-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Transfer Mechanism Installation
Figure 401/27-11-61-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-61-400-801

3. Aileron Transfer Mechanism Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-11-00-700-803	Manual Reversion Control Friction Test (P/B 501)
27-11-00-700-804	Control Wheel Centering Check (P/B 501)
27-11-00-700-805	Aileron Control Wheel Force Test (P/B 501)
27-11-00-700-806	Transfer Mechanism Functional Test (P/B 501)
27-11-00-700-807	Control Wheel Travel Stop Test (P/B 501)
27-11-00-710-801	Aileron Response (Operational) Test (P/B 501)
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-41-00-820-801	Hydraulic Brake System Adjustment (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00113	Lubricant - Solid Film, Liquid Dispersed	BMS3-8
D00633	Grease - Aircraft General Purpose	BMS3-33
D50108	Lubricant - Solid Film, Air Cured, Corrosion Inhibiting	MIL-L-23398
G50063	Lubricant - PTFE Release Agent - Miller-Stephenson MS-122XD (Replaces MS-122DF)	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Aileron transfer mechanism	27-11-61-01-070	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
112A	Forward Access Door
114AC	Fwd Nose Wheel Well Upper Access Panel

G. Aileron Transfer Mechanism Installation

SUBTASK 27-11-61-420-001

(1) Install the aileron transfer mechanism [13]:

(a) Apply MS-122XD release agent, G50063, to the paddle.

(b) Apply solid film lubricant, D00113, or lubricant, D50108, to the fork.

(c) Put the bolts [22] into position on the airplane structure.

(d) Lift the aileron transfer mechanism [13] into position while you make sure that the fork and paddle engage.

(e) Install the washers [23] and nuts [24] on the bolts [22] to attach the aileron transfer mechanism [13] to the airplane structure.

(f) Put the ends of the cable ACBA1 [1], cable ACBA2 [2], cable ACBB1 [3] and cable ACBB2 [4] through the openings.

(g) Install the screw [17] and clamp [18] to connect the rudder pedal adjustment cable to the aileron transfer mechanism [13].

SUBTASK 27-11-61-420-002

(2) Install the aileron position transducer [7] on the aileron transfer mechanism [13]:

(a) Put a thin film of grease, D00633, on the splines of the aileron position transducer [7].

(b) Put the aileron position transducer [7] in position.

(c) Install the washers [9] and nuts [10].

(d) Install the screw [28], screw [29], washers [30], clamps [27], and nuts [31].

(e) Connect the three electrical connectors [11].

SUBTASK 27-11-61-420-003

(3) Connect the cable ACBA1 [1], cable ACBA2 [2], cable ACBB1 [3] and cable ACBB2 [4] to the cable turnbuckles [5].

(a) Remove the identification tags from the disconnected cables.

(b) Connect the cable ACBA1 [1], cable ACBA2 [2], cable ACBB1 [3] and cable ACBB2 [4] to the cable turnbuckles [5].

(c) Tighten the cable turnbuckles [5] to remove slack but keep the cable tension at zero.

SUBTASK 27-11-61-420-004

(4) To connect brake control cable LGB2A [25] and cable LGB2B [26] at the LG brake cable turnbuckles, do these steps:

(a) Remove the identification tags from the disconnected cables.

(b) Connect the cable LGB2A [25] and cable LGB2B [26] at the cable turnbuckles.

(c) Tighten the brake cable turnbuckles to remove slack but keep the cable tension at zero.

SUBTASK 27-11-61-080-002

(5) Remove the rig pins A/S-4, from the rigging pin kit, SPL-1585, from the aileron bus drums.

SUBTASK 27-11-61-820-001

(6) Adjust the cable ACBA and cable ACBB (TASK 27-11-00-820-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-61-820-002

(7) Adjust the brake control cables LGB2A and LGB2B (Hydraulic Brake System Adjustment, TASK 32-41-00-820-801):

(a) Do the steps to adjust brake cables LGB2A and LGB2B.

(b) Adjust the left side brake control cables LGB1A and LGB1B at the same time.

(c) Brake system retest is not required after rigging.

H. Aileron Transfer Mechanism Installation Test

SUBTASK 27-11-61-860-003

(1) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-61-730-002

(2) Do this task: Manual Reversion Control Friction Test, TASK 27-11-00-700-803.

SUBTASK 27-11-61-710-001

(3) Do this task: Aileron Response (Operational) Test, TASK 27-11-00-710-801.

SUBTASK 27-11-61-720-001

(4) Do this task: Control Wheel Centering Check, TASK 27-11-00-700-804.

SUBTASK 27-11-61-730-003

(5) Do this task: Aileron Control Wheel Force Test, TASK 27-11-00-700-805.

SUBTASK 27-11-61-720-002

(6) Do this task: Transfer Mechanism Functional Test, TASK 27-11-00-700-806.

SUBTASK 27-11-61-730-004

(7) Do this task: Control Wheel Travel Stop Test, TASK 27-11-00-700-807.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 27-11-61-860-005

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

SUBTASK 27-11-61-410-001

(2) Close these access panels:

Number	Name/Location
112A	Forward Access Door
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 27-11-61-080-003

(3) Remove the downlock pins from the landing gear (Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON TRANSFER MECHANISM - INSPECTION/CHECK

1. General

A. This procedure has this inspection task:

(1) Aileron Transfer Mechanism

TASK 27-11-61-210-801

2. Aileron Transfer Mechanism Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Aileron Transfer Mechanism Inspection

SUBTASK 27-11-61-010-002

(1) Get access to the aileron transfer mechanism:

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-11-61-210-001

(2) Do a general visual inspection of the aileron transfer mechanism.

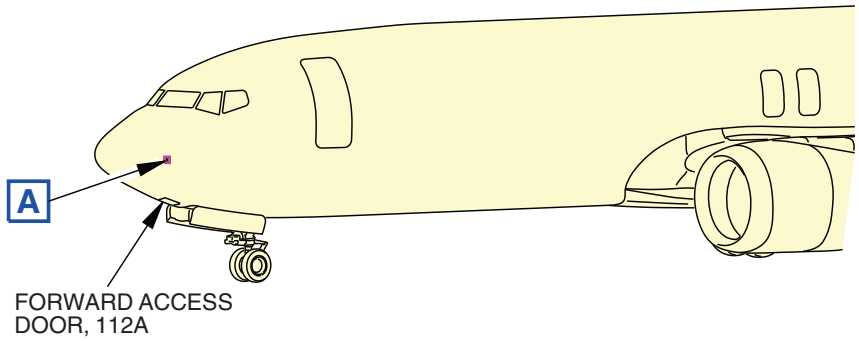
SUBTASK 27-11-61-410-002

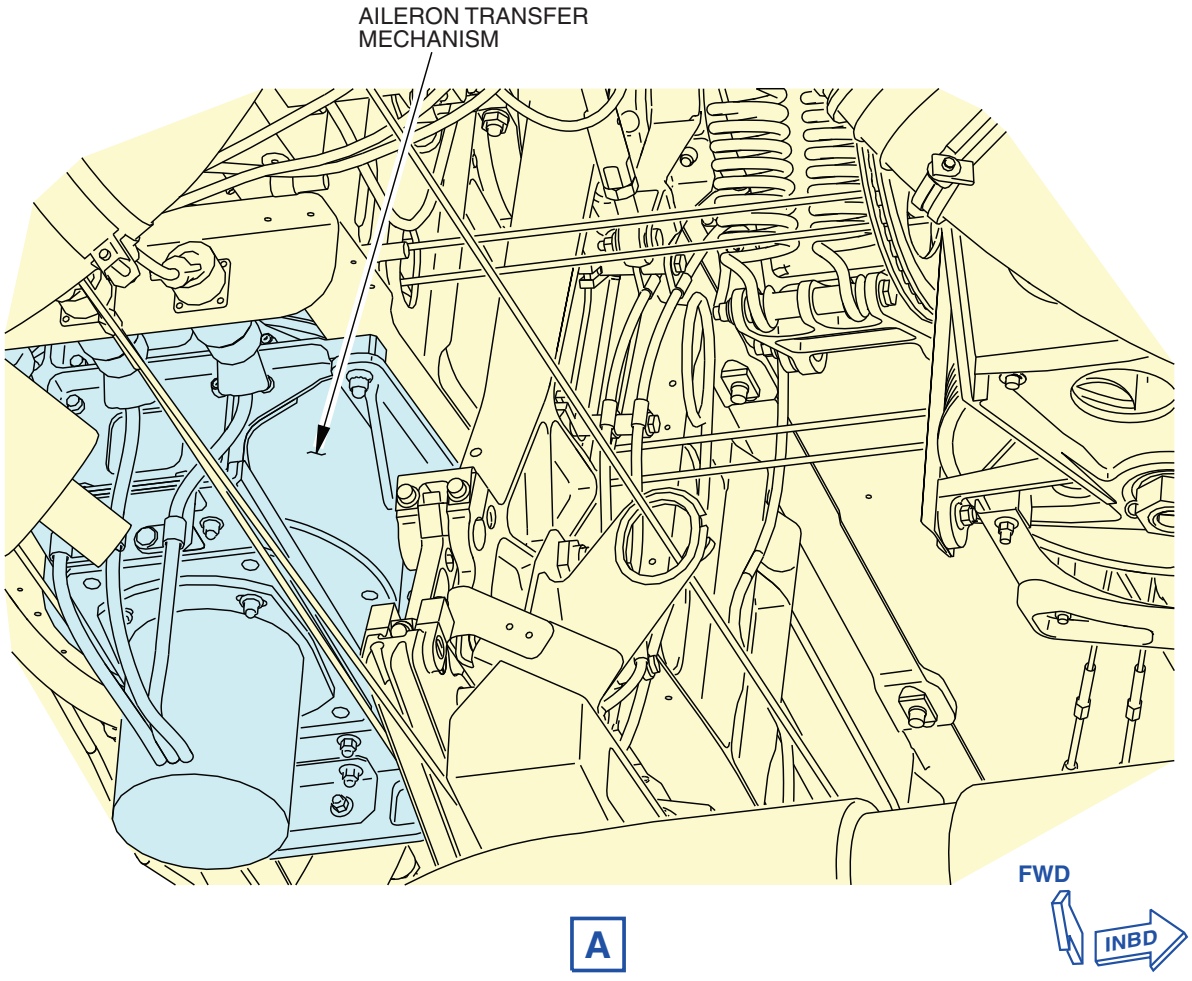
(3) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Aileron Transfer Mechanism Inspection
Figure 601/27-11-61-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON POWER CONTROL UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron power control unit

(2) An installation of the aileron power control unit.

B. The aileron power control unit is referred to as the "power control unit" in this procedure.

C. The two power control units supply hydraulic power to the aileron. The power control unit at the bottom is connected to hydraulic system A. The power control unit at the top is connected to hydraulic system B.

D. This procedure is applicable for the power control units that are connected to hydraulic systems A and B.

TASK 27-11-71-000-801

2. Aileron Power Control Unit Removal

(Figure 401)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 27-11-71-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-71-860-001

(2) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-71-860-008

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

E. Aileron Power Control Unit Linkage Removal

SUBTASK 27-11-71-020-001

(1) Disconnect hydraulic lines [4] at the inlet ports and hydraulic lines [40] at the outlet ports of the power control unit [1]. Use a oil resistant container, STD-203 to contain any hydraulic fluid leakage. Discard the packing [2] and packing [6].

(a) Install the caps and plugs on hydraulic lines [4], hydraulic lines [40] and ports.

SUBTASK 27-11-71-020-002

(2) Remove the power control unit [1]:

(a) Remove the pins [13], nuts [12], washers [11], bushings [10], washers [9], and bolts [8] to disconnect the pogos [7] from the power control unit [1]. Discard the pins [13].

(b) Remove the bolts [30], washers [31], bolts [32] and washers [33] to disconnect the bracket assemblies from the power control unit [1].

(c) Remove the pin [23], nut [22], washer [21], nut [24], washers [25], bushings [26], washers [28], and bolt [14] to disconnect the power control unit [1] from the reaction support assembly [27]. Discard the pin [23].

(d) Remove the nut [20], washer [19], washers [17], spacer [16] and bolt [15] to disconnect the safety link [18] from the reaction support assembly [27].

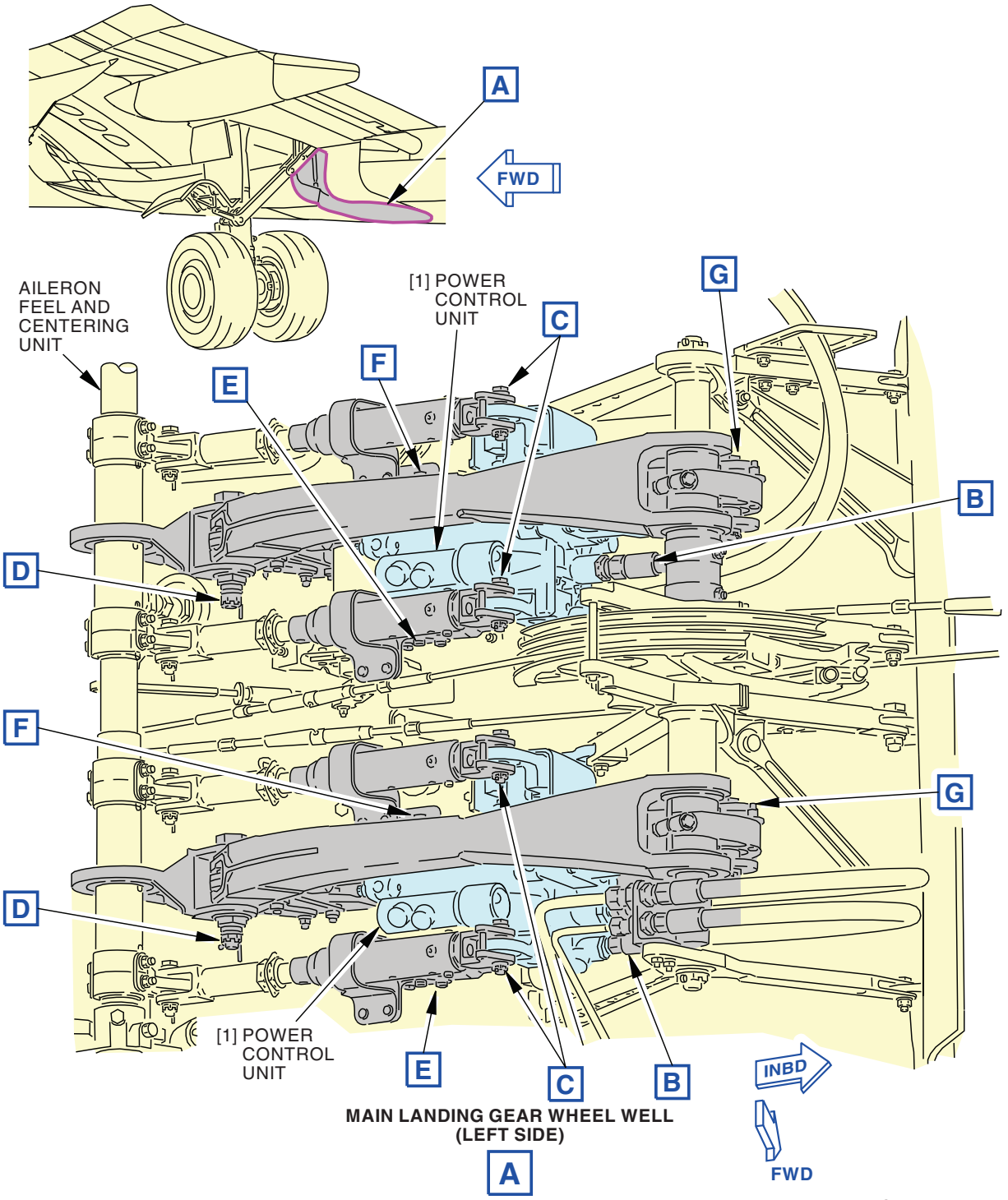
(e) Move the reaction link [29] aft to get access to the power control unit [1].

(f) Remove the pin [37], nut [38], washer [39], washer [35], and bolt [34] to disconnect the power control unit [1] from the output lever assembly [36]. Discard the pin [37].

(g) Remove the power control unit [1].

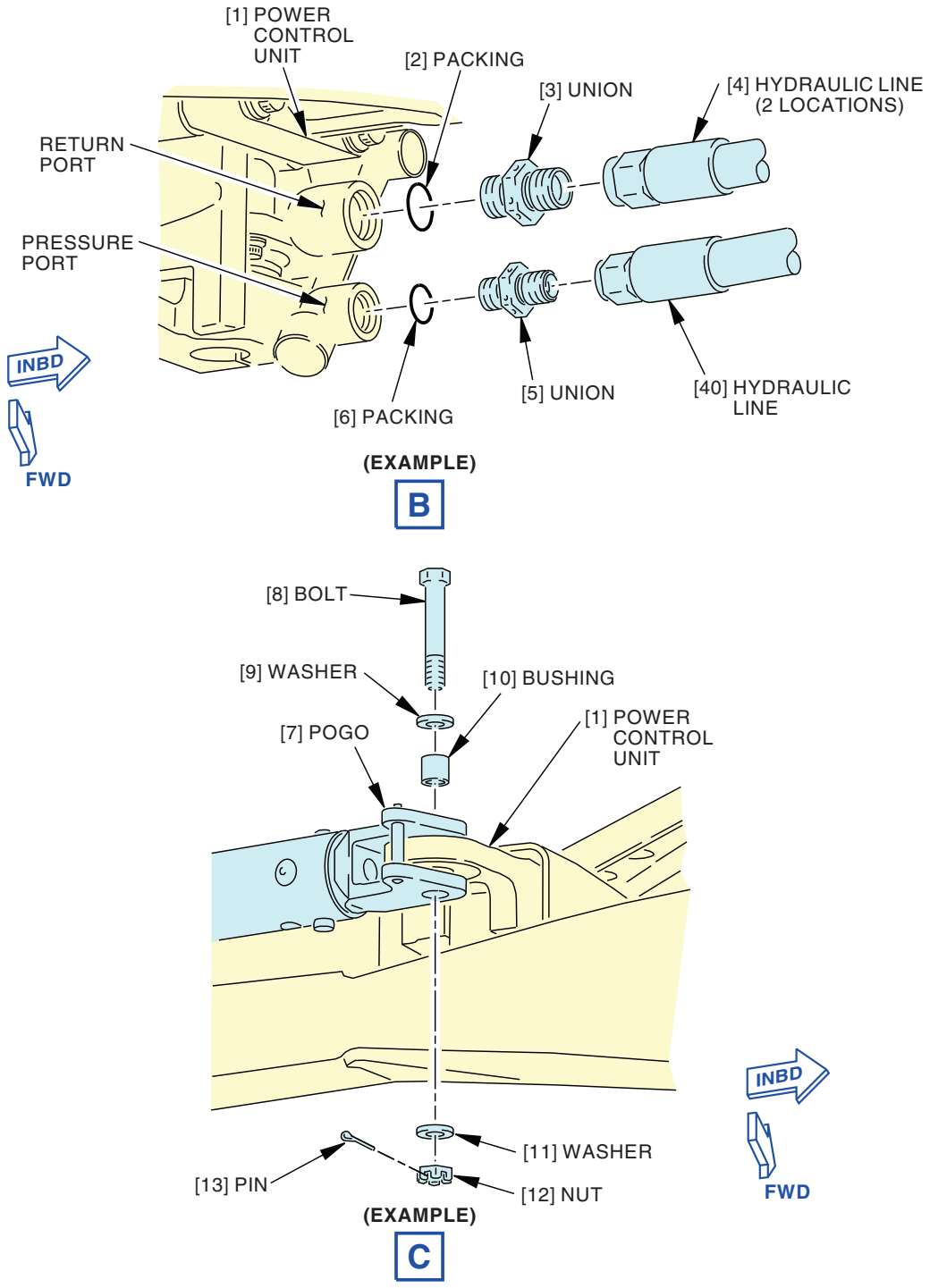
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



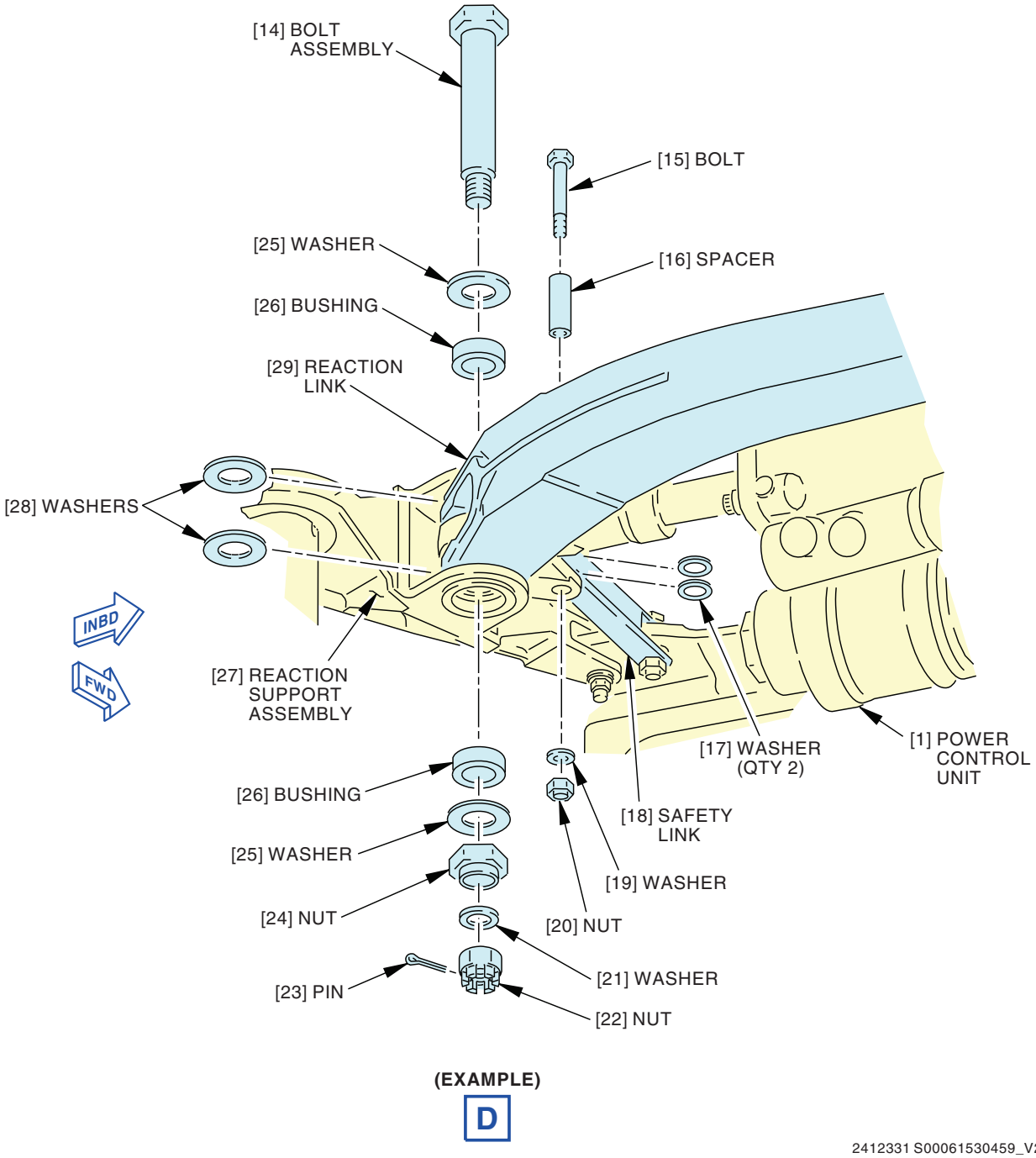
Aileron Power Control Unit Installation
Figure 401/27-11-71-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



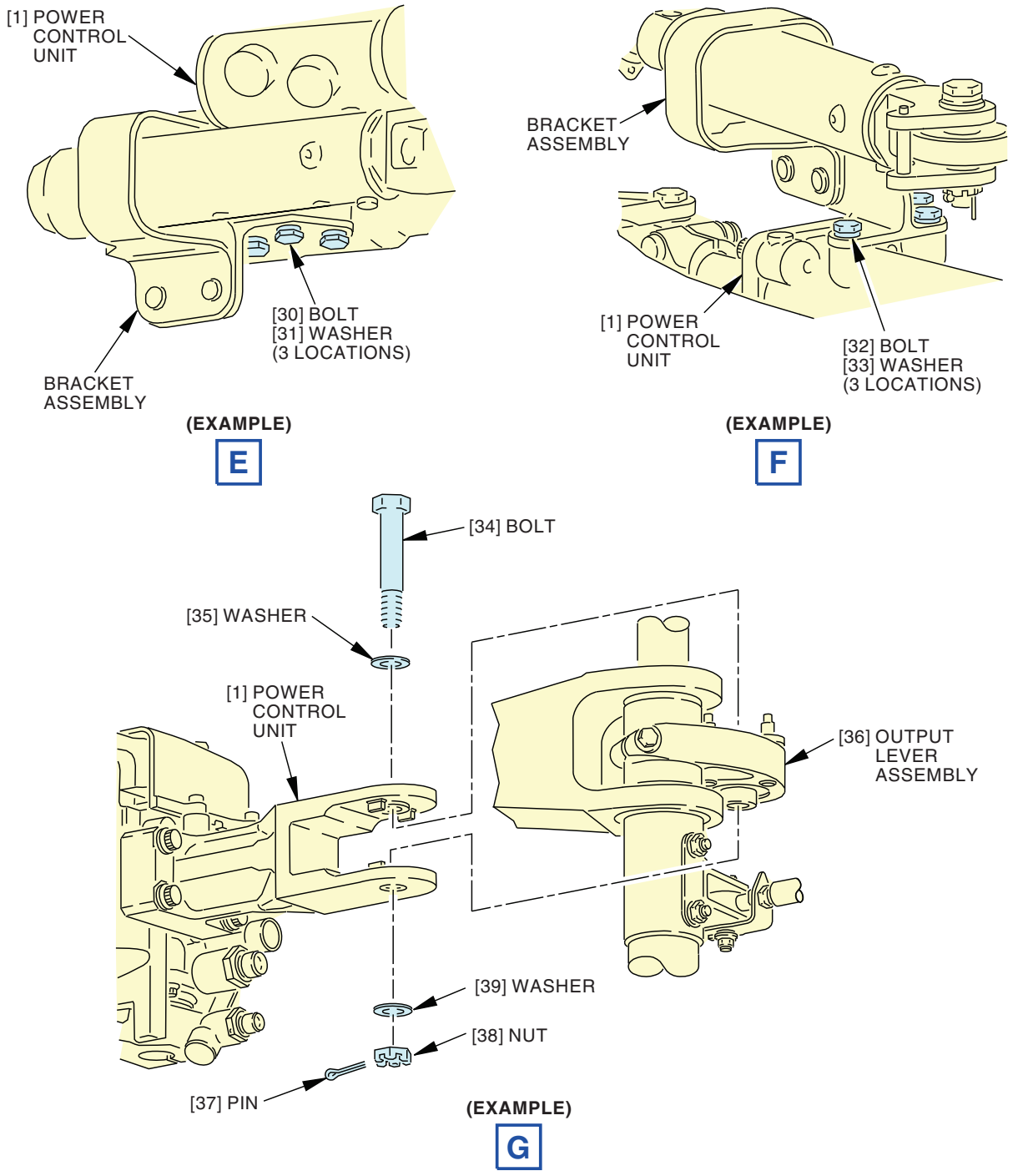
Aileron Power Control Unit Installation
Figure 401/27-11-71-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Power Control Unit Installation
Figure 401/27-11-71-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aileron Power Control Unit Installation
Figure 401/27-11-71-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-71-400-801

3. Aileron Power Control Unit Installation

(Figure 401)

A. References

Reference	Title
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-710-801	Aileron Response (Operational) Test (P/B 501)
27-11-00-820-802	Pogo and Power Control Unit (PCU) Adjustment (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Power control unit	27-11-71-01-430	PTF ALL
2	Packing	27-11-71-10-530	PTF ALL
6	Packing	27-11-71-10-535	PTF ALL
13	Pin	27-11-71-01-115	PTF ALL
23	Pin	27-11-71-01-280	PTF ALL
37	Pin	27-11-71-01-330	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Aileron Power Control Unit Installation

SUBTASK 27-11-71-420-001

(1) If they are not installed, install the union [3], union [5], new packing [2], and new packing [6].

(a) Apply hydraulic fluid, D00153, to the new packing [2] and new packing [6].

(b) Install the union [3], new packing [2], union [5], and new packing [6] in the ports on the power control unit [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Tighten the union [3] to 170 ±9 in-lb (19.2 ±1.0 N·m).

2) Tighten the union [5] to 110 ±6 in-lb (12.4 ±0.7 N·m).

SUBTASK 27-11-71-860-002

(2) Make sure the power control unit [1] is filled with hydraulic fluid, D00153.

SUBTASK 27-11-71-820-001

(3) Remove the oil resistant container, STD-203.

SUBTASK 27-11-71-420-002

(4) Install the power control unit [1]:

(a) Put the power control unit [1] in its position.

(b) Install the bolt [34], washer [35], washer [39], and nut [38] to connect the power control unit [1] to the output lever assembly [36].

1) Coat the mating surfaces with grease, D00633, prior to assembly.

(c) Tighten the nut [38] to the lower end of the torque range 300 in-lb (33.9 N·m) - 410 in-lb (46.3 N·m).

1) Continue to tighten the nut [38] until the next castellation aligns with the hole in the bolt [34].

(d) Install the new pin [37].

(e) Move the reaction link [29] to put the power control unit [1] between the fork end of the reaction link [29].

(f) Connect the power control unit [1] to the reaction support assembly [27].

1) Install the bolt [14], washers [25], bushings [26], washers [28], and nut [24].

a) Coat the mating surfaces with grease, D00633, prior to assembly.

2) Tighten the nut [24] to 300 in-lb (33.9 N·m) - 350 in-lb (39.5 N·m).

3) Install the washer [21] and nut [22].

4) Tighten the nut [22] to the lower end of the torque range 170 in-lb (19.2 N·m) - 250 in-lb (28.2 N·m).

a) Continue to tighten the nut [22] until the next castellation aligns with the hole in the bolt [14].

5) Install the new pin [23].

(g) Apply sealant, A00247, to the bolt [15] and spacer [16] .

(h) Install the bolt [15], spacer [16], washers [17], washer [19], and nut [20] to connect the safety link [18] to the reaction support assembly [27].

(i) Connect the bracket assemblies to the power control unit [1] with bolts [30], washers [31], bolts [32], and washers [33].

(j) Apply grease, D00633, to the bolts [8] and nuts [12].

(k) Connect the pogos [7] to the power control unit [1] with bolts [8], washers [9], bushings [10], washers [11], and nuts [12].

1) Install the bolt [8] with the bolt head up.

(l) Tighten the nuts [12] to the lower end of the torque range 60 in-lb (6.8 N·m) - 90 in-lb (10.2 N·m).

1) Continue to tighten the nuts [12] until the next castellation aligns with the hole in the bolts [8].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Install the new pin [13].

SUBTASK 27-11-71-020-003

(5) Remove the caps and plugs from hydraulic lines [4], hydraulic lines [40], and the ports of the power control unit [1].

SUBTASK 27-11-71-420-003

(6) Connect hydraulic lines [4] to the inlet ports and hydraulic lines [40] to the outlet ports of the power control unit [1].

(a) Tighten the B-nuts on hydraulic lines [4] to 170 ±9 in-lb (19.2 ±1.0 N·m).

(b) Tighten the B-nuts on hydraulic lines [40] to 110 ±6 in-lb (12.4 ±0.7 N·m).

(c) Make sure that there is a minimum clearance of 0.75 in. (19.05 mm) between the flexible hose and the adjacent structure after you tighten the flexible hose.

SUBTASK 27-11-71-860-004

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) To pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 27-11-71-860-005

(8) Operate the aileron control wheel to bleed the system.

SUBTASK 27-11-71-790-001

(9) Examine the connections of the power control unit [1] for leaks.

SUBTASK 27-11-71-860-006

(10) To remove power from hydraulic systems A and B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 27-11-71-860-009

(11) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

SUBTASK 27-11-71-860-007

(12) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-71-820-002

(13) Do this task: Pogo and Power Control Unit (PCU) Adjustment, TASK 27-11-00-820-802.

SUBTASK 27-11-71-710-001

(14) Do this task: Aileron Response (Operational) Test, TASK 27-11-00-710-801.

SUBTASK 27-11-71-730-001

(15) Do this task: Aileron Travel Test, TASK 27-11-00-700-810.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON FEEL AND CENTERING UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the aileron feel and centering unit

(2) An installation of the aileron feel and centering unit.

(3) A removal of the aileron feel and centering unit cam follower bearing.

(4) An installation of the aileron feel and centering unit cam follower bearing.

B. The aileron feel and centering unit is referred to as the "centering unit" in this procedure.

TASK 27-11-81-000-801

2. Aileron Feel and Centering Unit - Removal

(Figure 401 or Figure 402)

A. References

Reference	Title
25-22-00-000-801	Passenger Seat Removal (P/B 401)
25-27-15-000-801	Carpet Removal (P/B 401)
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1671	Straight Edge - Control Wheel Part #: C27097-1 Supplier: 81205 Part #: SE27-0001 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Removal

SUBTASK 27-11-81-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-11-81-860-001

(2) Do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-81-480-002

(3) Install the straight edge, SPL-1671, across the top ends of the two control wheels to keep the control wheels in the neutral position.

SUBTASK 27-11-81-210-001

(4) Make sure that the aileron trim switches are at the zero trim position.

NOTE: The aileron trim switches are installed, on the P8 aft electronic control panel, in the flight compartment.

SUBTASK 27-11-81-480-003

(5) Install the rig pin A/S-15, from the rigging pin kit, SPL-1585, in the aileron trim mechanism (Figure 401).

SUBTASK 27-11-81-480-004

(6) Install the rig pin A/S-4, from the rigging pin kit, SPL-1585, in the aileron bus drum (Figure 401).

SUBTASK 27-11-81-020-001

(7) Above the left main wheel well, do this task: Passenger Seat Removal, TASK 25-22-00-000-801.

SUBTASK 27-11-81-020-002

(8) Above the left main wheel well, do this task: Carpet Removal, TASK 25-27-15-000-801.

SUBTASK 27-11-81-020-003

(9) Above the left main wheel well, do this task: Passenger Cabin Floor Panel Removal, TASK 53-21-00-000-801.

SUBTASK 27-11-81-860-006

(10) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

E. Aileron Feel and Centering Unit Removal

SUBTASK 27-11-81-020-004

(1) Remove the aileron feel and centering unit [11] (Figure 402).

(a) Remove the electrical connector from the aileron trim actuator [15].

(b) Loosen the cable turnbuckle [3] for the cable AB [1] and cable AA [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the pin [32], nuts [61], washers [33], nuts [34], washers [35], bushings [36], washers [37], and bolt assembly [27] from the reaction links [29].

NOTE: The four bushings in the reaction support assemblies are free floating and concealed by the washers above and below them. They can be easily misplaced or lost during disassembly.

1) Discard the pin [32].

(d) Make sure that the four concealed bushings [36] are captured on the disassembly and retained for reassembly of the new support assembly [30] to the reaction link [29].

(e) Remove the nut [41], washer [42], washer [44], and bolt [45] to disconnect the autopilot actuator rod [43] from the aileron feel and centering unit [11].

(f) Record the position and orientation of the aileron position sensor rod [38] before removal as an aid for installation.

(g) Remove the bolt [46], washer [39], and nut [40] to remove the aileron position sensor rod [38] from the aileron feel and centering unit [11].

(h) Remove the pin [25], nuts [24], washers [23], bushings [21], washers [20], and bolts [19] to disconnect the pogos [22] from the lever assemblies [26].

1) Discard the pin [25].

(i) Disconnect the aileron feel and centering unit [11] from the airplane structure.

1) Remove the nut [4], washer [5], bushing [6], washer [7], and bolt [8].

2) Remove the nuts [16], washers [17], washers [31], bushings [28], washers [13], and bolts [14].

(j) Remove the aileron control quadrant [47] from the shaft:

1) Remove the cable AA [2] and cable AB [1] from the aileron control quadrant [47]:

a) Disconnect the left cable AA [2] and cable AB [1] at the cable turnbuckle [3].

b) Remove the nuts [59], washers [58], washers [62] (if applicable), and bolt [57] to remove the retaining plates [60] from the aileron control quadrant [47].

c) Remove the cable AA [2] and cable AB [1] from the aileron control quadrant [47].

d) Secure the disconnected cable AA [2] and cable AB [1] with clamps.

e) Attach identification tags to the disconnected cables for easier assembly.

2) Remove the nut [51], washer [50], washer [49], and bolt [48] to disconnect the aileron control quadrant [47] from the shaft.

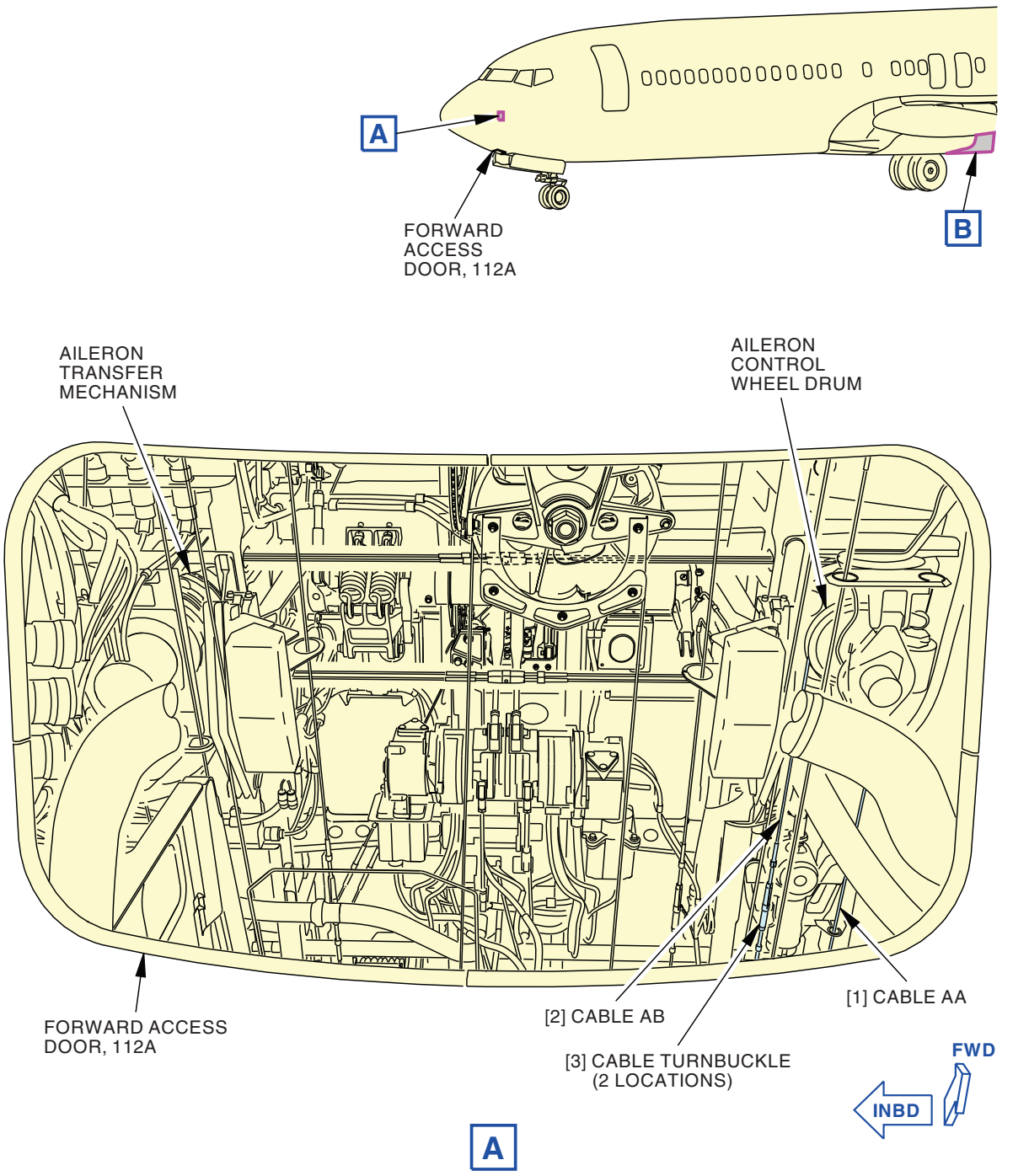
3) Remove the aileron control quadrant [47] from the shaft.

(k) Remove the nuts [56], washers [55], washer [53], and bolts [52] to disconnect the bearing housing [54] from the pressure deck.

(l) Carefully lower and remove the aileron feel and centering unit [11].

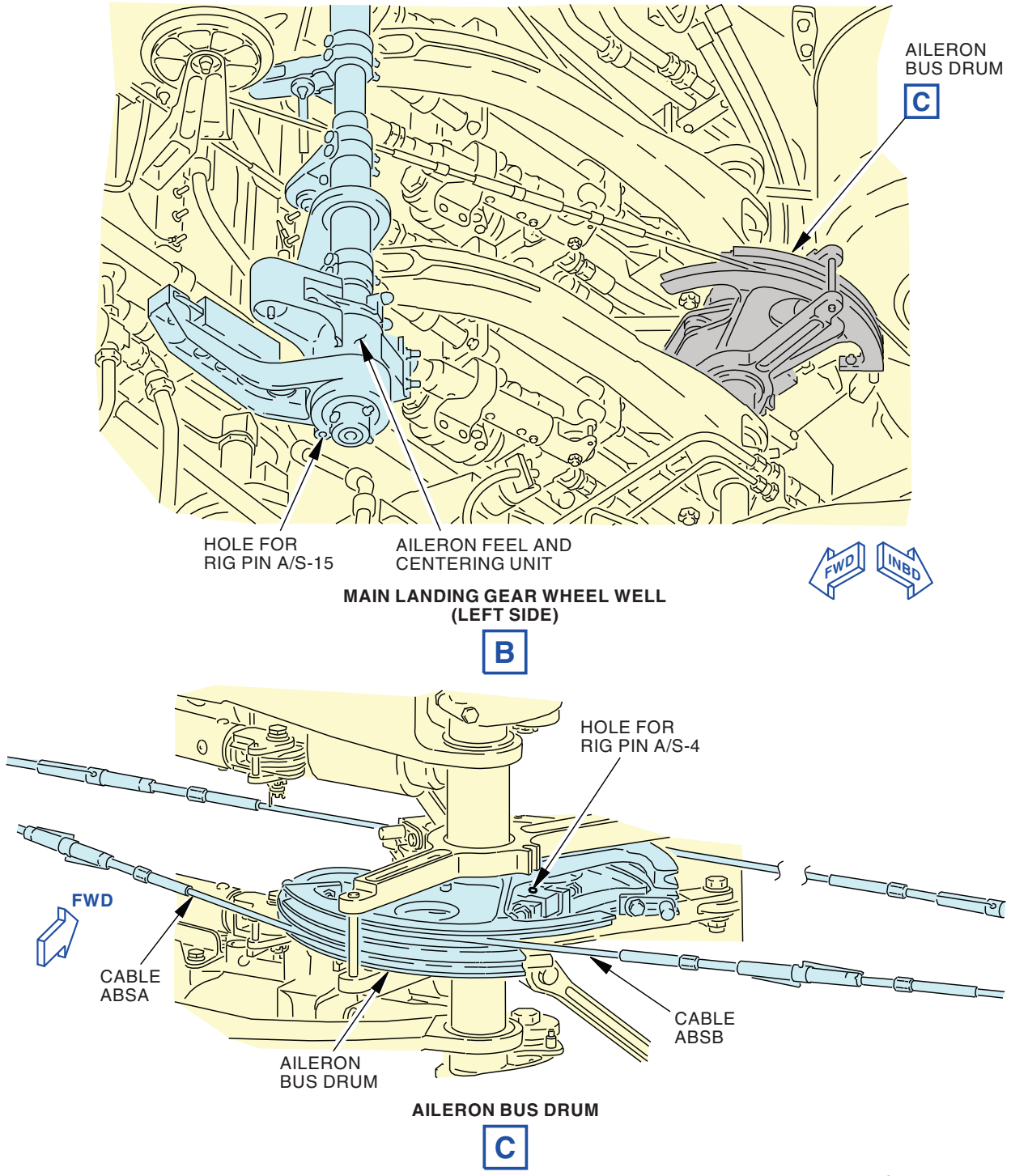
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



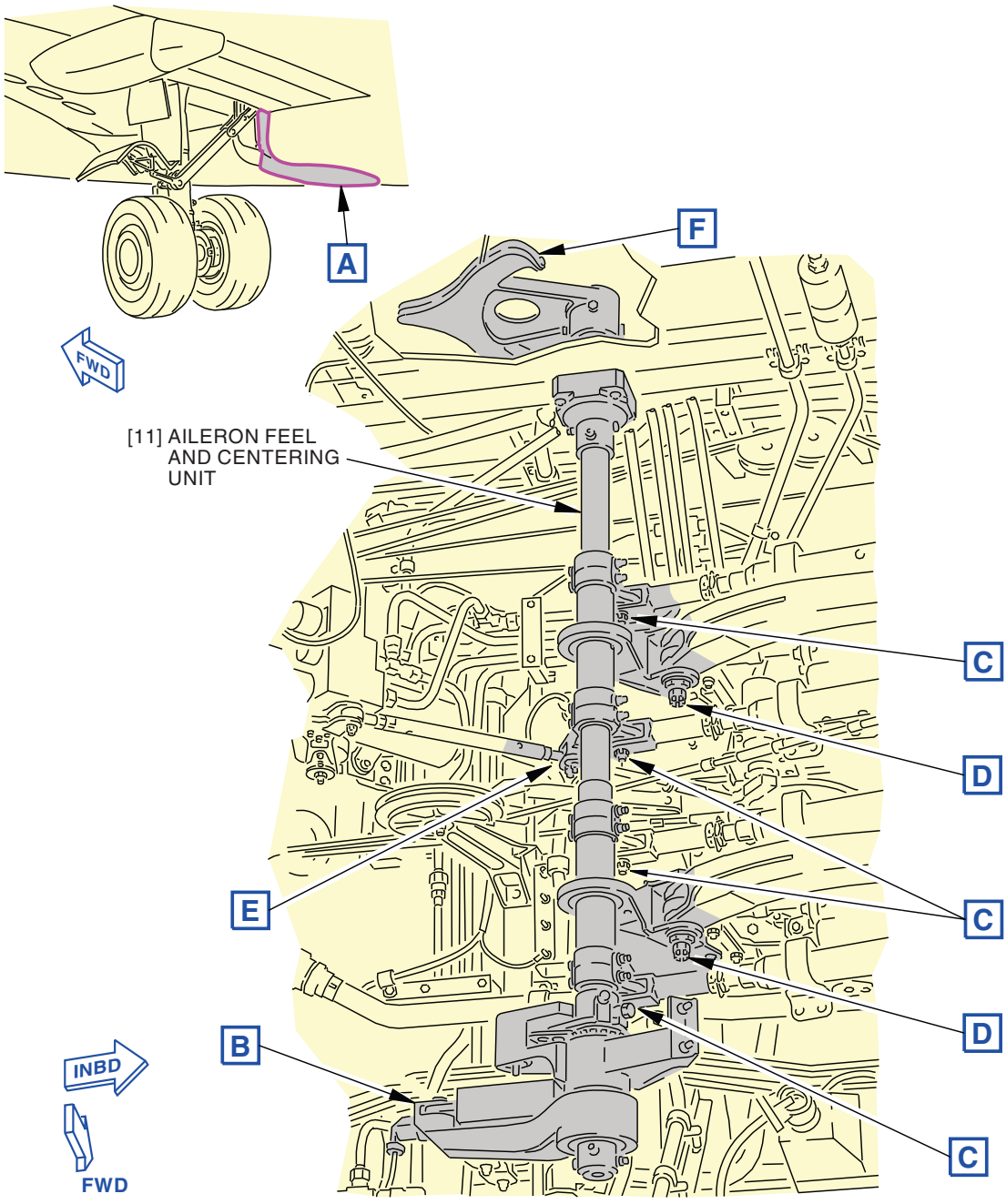
Rig Pin and Turnbuckle Location
Figure 401/27-11-81-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rig Pin and Turnbuckle Location
Figure 401/27-11-81-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



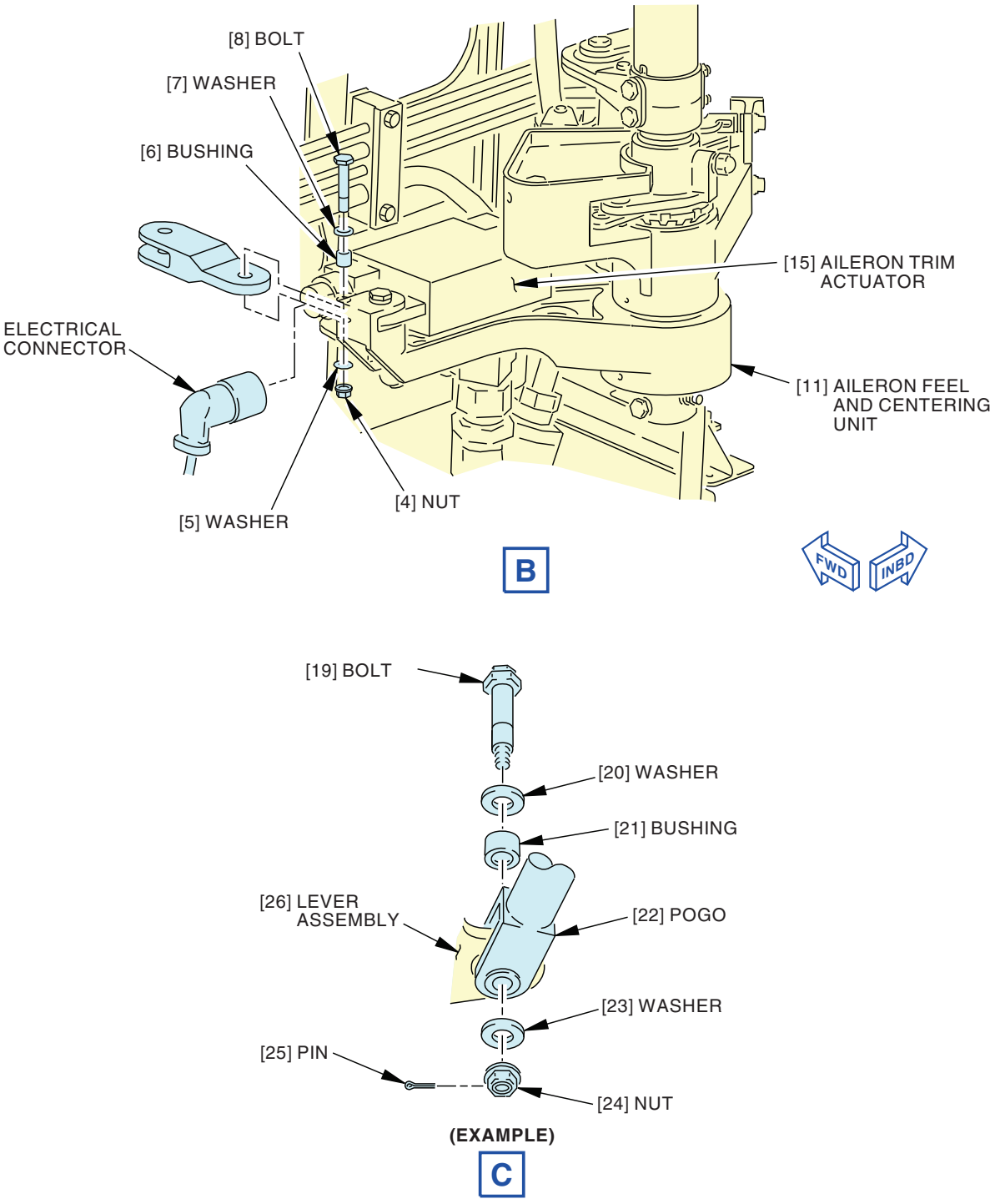
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)

A

2412335 S00061530467_V1

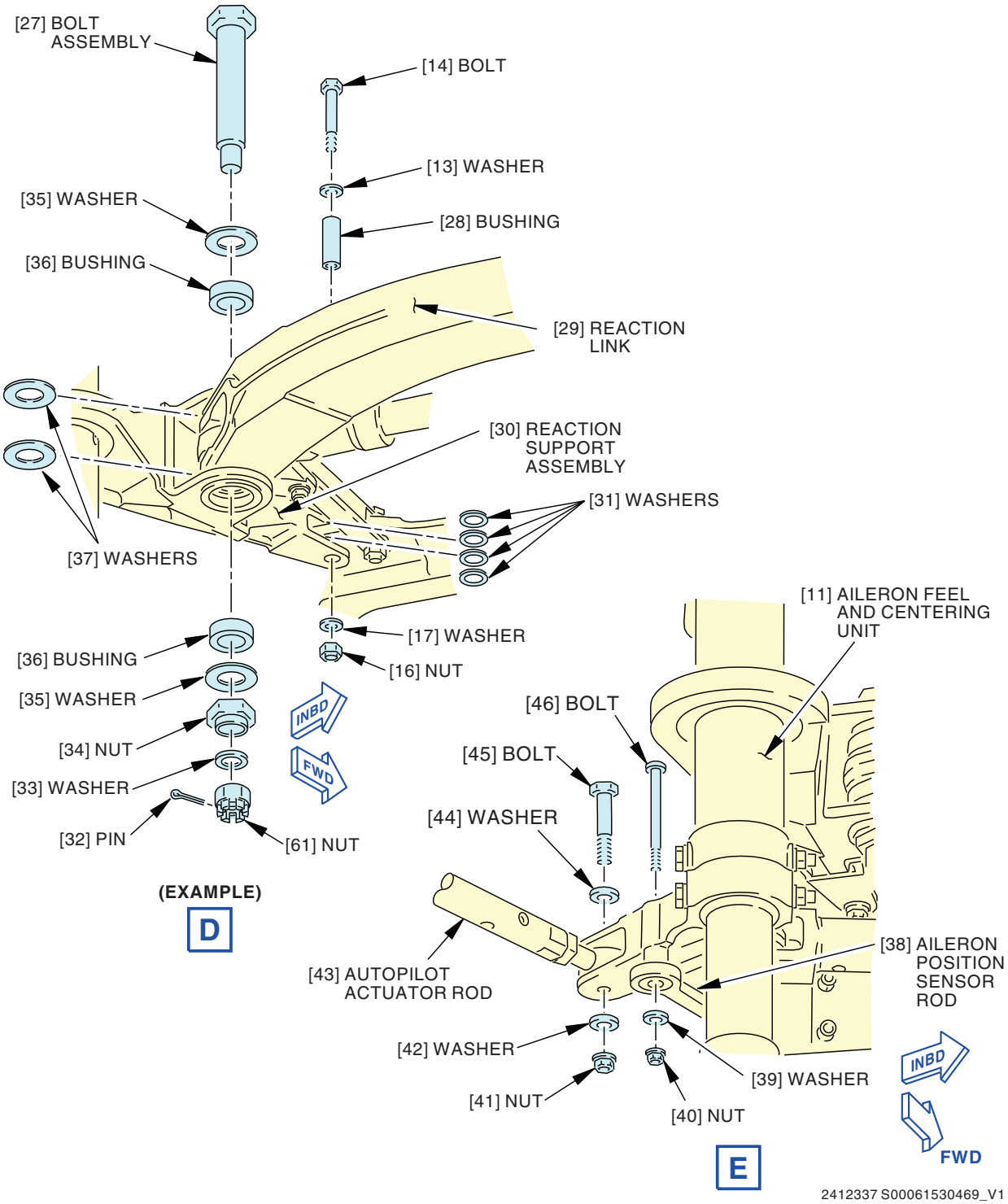
Aileron Feel and Centering Unit Installation
Figure 402/27-11-81-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



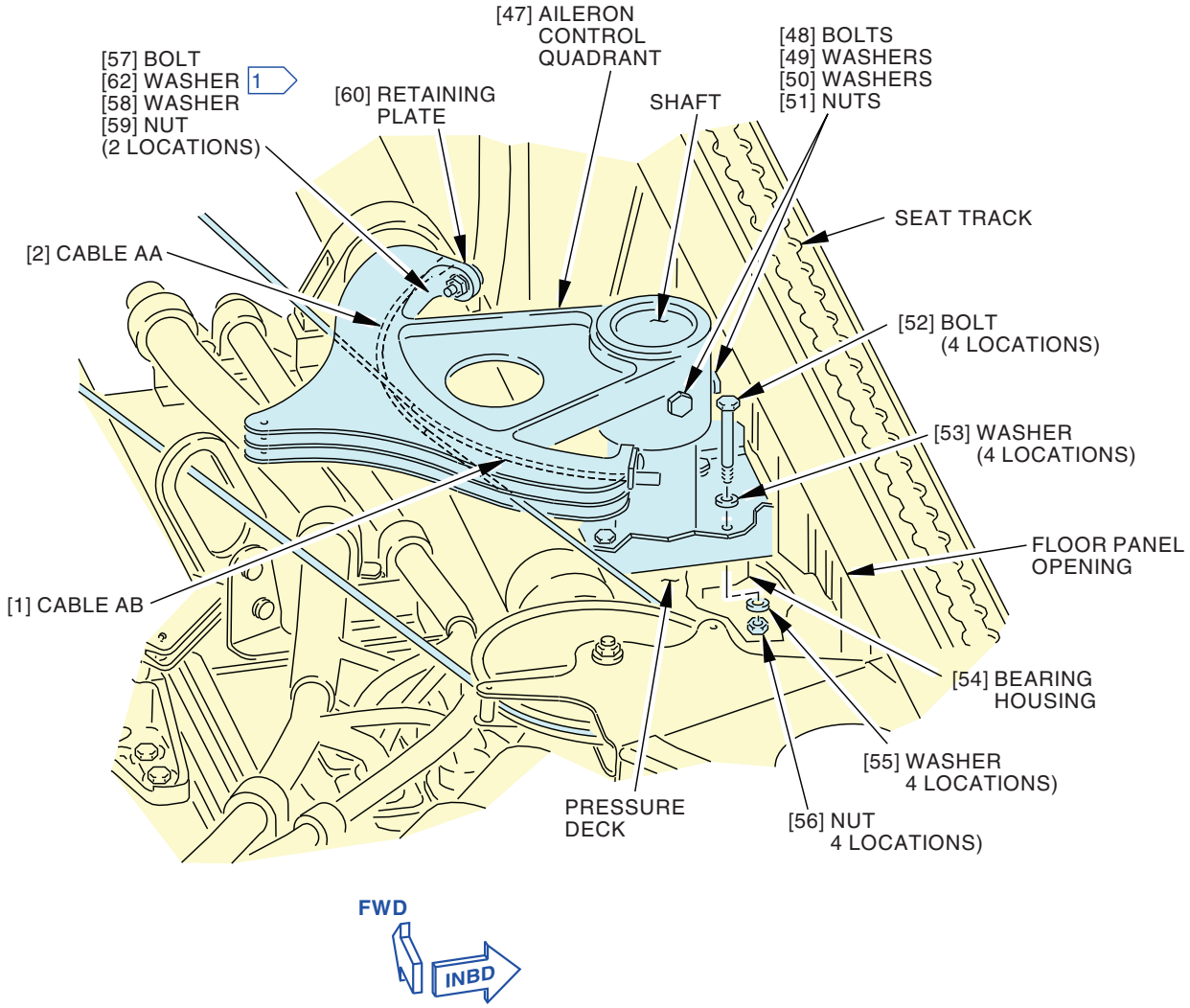
Aileron Feel and Centering Unit Installation
Figure 402/27-11-81-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aileron Feel and Centering Unit Installation
Figure 402/27-11-81-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 IF APPLICABLE

2412338 S00061530470_V2

Aileron Feel and Centering Unit Installation
Figure 402/27-11-81-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-81-400-801

3. Aileron Feel and Centering Unit Installation

(Figure 401 or Figure 402)

A. References

Reference	Title
20-10-93-400-801	Control Cable End Retaining Plate - Installation (P/B 401)
22-11-25-820-801	Autopilot Aileron Actuator Adjustment (P/B 501)
22-11-29-820-801	Aileron Position Sensor Adjustment (P/B 501)
25-22-00-400-801	Passenger Seat Installation (P/B 401)
25-27-15-400-801	Carpet - Installation (P/B 401)
27-11-00-700-803	Manual Reversion Control Friction Test (P/B 501)
27-11-00-700-805	Aileron Control Wheel Force Test (P/B 501)
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-820-801	Aileron Cables Adjustment (P/B 501)
27-11-00-820-802	Pogo and Power Control Unit (PCU) Adjustment (P/B 501)
27-11-00-820-805	Aileron Feel and Centering Unit Adjustment (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1671	Straight Edge - Control Wheel Part #: C27097-1 Supplier: 81205 Part #: SE27-0001 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D50180	Grease - Aircraft General Purpose (AeroShell Grease 33)	BMS3-33, MIL-PRF-23827

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Aileron Feel and Centering Unit Installation

SUBTASK 27-11-81-420-001

(1) Install the aileron feel and centering unit [11] (Figure 402).

(a) If the aileron control quadrant [47] is attached to the aileron feel and centering unit [11], remove the aileron control quadrant [47].

1) Remove and keep the nuts [51], washers [50], washers [49], and bolts [48].

(b) Apply AeroShell Grease 33, D50180, to the bearings (View A, Figure 402).

(c) Make sure that the spherical race of the bearing housing [54] does not bind in the outer housing.

(d) Apply a fillet seal between the bearing housing [54] and pressure deck with sealant, A00247 (TASK 51-22-00-390-806).

(e) Put the aileron feel and centering unit [11] in position with the shaft through the pressure deck.

(f) Install the bolts [52], washers [53], washers [55], and nuts [56].

1) Tighten the nuts [56] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

(g) Attach the reaction support assemblies [30] to the structure (View D, Figure 402).

1) Install the bolts [14], washers [13], bushings [28], washers [31], washers [17], and nuts [16].

a) Install the washers [31] between the lug and clevis.

NOTE: There must be a minimum of one washer on each side of the lug.

(h) Attach the aileron feel and centering unit [11] to the structure (View B, Figure 402).

1) Install the bolt [8], washer [7], bushing [6], washer [5], and nut [4].

2) Connect the electrical connector to the aileron trim actuator [15].

(i) Connect the reaction link [29] to the reaction support assembly [30].

1) Install the bolt assemblies [27], washers [35], bushings [36], washers [37], and nuts [34].

2) Tighten the nuts [34] to 300 in-lb (33.9 N·m) - 350 in-lb (39.5 N·m).

3) Install the washers [33], nuts [61], and new pin [32].

(j) Install the aileron control quadrant [47] on the shaft.

1) If it is necessary, apply sealant, A00247, on the disk of the aileron control quadrant [47].

2) Engage the aileron control quadrant [47] on the shaft.

a) Make sure that the tip of the aileron control quadrant [47] points in the outboard direction.

3) Install the bolts [48], washers [49], washers [50], and nuts [51].

(k) Connect the pogos [22] to the aileron feel and centering unit [11] with the bolts [19], washers [20], bushings [21], washers [23], nuts [24], and new pin [25].

1) Make sure that you install the bolts [19] and bushings [21] with the head up.

(l) Make sure that the top and bottom pogos [22] on each power control unit is at the center between the stops.

1) If it is necessary to adjust the pogo [22], do this task: Pogo and Power Control Unit (PCU) Adjustment, TASK 27-11-00-820-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Connect the autopilot actuator rod [43] with the bolt [45], washer [44], washer [42], and nut [41].

CAUTION

POSITION AND ORIENT AILERON POSITION SENSOR ROD AS RECORDED IN REMOVAL. UPON INSTALLATION OF THE AILERON POSITION SENSOR, PLACE THE AILERON POSITION SENSOR ROD AND CRANK SUCH THAT THEY ARE POINTED FORWARD RELATIVE TO THE FUSELAGE. AN INCORRECT INSTALLATION OF THE ROD CAN CAUSE DAMAGE TO EQUIPMENT WHEN AUTOPILOT IS ENGAGED.

(n) Connect the aileron position sensor rod [38] with bolt [46], washer [39], and nut [40] (Figure 402).

1) Adjust the aileron position sensor (TASK 22-11-29-820-801).

NOTE: Adjustment of the aileron position sensor is necessary for it to operate correctly.

SUBTASK 27-11-81-420-002

CAUTION

MAKE SURE THAT THE ROUTING OF CABLES AA AND AB IS CORRECT. IF THE ROUTING IS INCORRECT, IT CAN INCREASE FRICTION WHICH CAN CAUSE DAMAGE TO THE CABLES.

(2) Connect the cable AA [2] and cable AB [1] to the aileron control quadrant [47].

(a) Remove the identification tags from the cable AA [2] and cable AB [1] (Figure 402).

(b) Remove the clamps from the cable AA [2] and cable AB [1].

(c) Install the retaining plate [60] with bolts [57], washers [62] (if applicable), washers [58], and nuts [59] (TASK 20-10-93-400-801).

SUBTASK 27-11-81-420-003

(3) Connect the cable AA [2] and cable AB [1] to the cable turnbuckles [3].

(a) Tighten the cable turnbuckles [3] but keep the cable tension at zero.

SUBTASK 27-11-81-820-001

(4) Adjust the tension of the cable AA [2] and cable AB [1], do this task: Aileron Cables Adjustment, TASK 27-11-00-820-801.

SUBTASK 27-11-81-080-001

(5) Remove the straight edge, SPL-1671, from the two aileron control wheels.

SUBTASK 27-11-81-480-005

(6) Make sure that the rig pin A/S-15, from the rigging pin kit, SPL-1585, goes through the two walls of the aileron trim mechanism.

SUBTASK 27-11-81-080-002

(7) Remove the rig pin A/S-4, from the rigging pin kit, SPL-1585, from the aileron bus drum.

SUBTASK 27-11-81-400-001

(8) Do this task: Passenger Cabin Floor Panel Installation, TASK 53-21-00-400-801.

SUBTASK 27-11-81-400-002

(9) Do this task: Carpet - Installation, TASK 25-27-15-400-801.

SUBTASK 27-11-81-400-003

(10) Do this task: Passenger Seat Installation, TASK 25-22-00-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-81-860-002

(11) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-81-820-002

(12) Do this task: Aileron Feel and Centering Unit Adjustment, TASK 27-11-00-820-805.

SUBTASK 27-11-81-820-003

(13) Do this task: Pogo and Power Control Unit (PCU) Adjustment, TASK 27-11-00-820-802.

SUBTASK 27-11-81-860-003

(14) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

SUBTASK 27-11-81-730-001

(15) Do this task: Aileron Control Wheel Force Test, TASK 27-11-00-700-805.

SUBTASK 27-11-81-730-002

(16) Do this task: Manual Reversion Control Friction Test, TASK 27-11-00-700-803.

SUBTASK 27-11-81-730-003

(17) Do this task: Aileron Travel Test, TASK 27-11-00-700-810.

SUBTASK 27-11-81-820-004

(18) Do this task: Autopilot Aileron Actuator Adjustment, TASK 22-11-25-820-801.

SUBTASK 27-11-81-820-005

(19) Do this task: Aileron Position Sensor Adjustment, TASK 22-11-29-820-801.

——— END OF TASK ———

TASK 27-11-81-000-802

4. Cam Follower Bearing Removal

(Figure 403)

A. References

Reference	Title
27-11-00-860-801	Pressure from the Aileron Hydraulic Systems A and B - Deactivation (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

C. Prepare for Removal

SUBTASK 27-11-81-490-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gears, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-81-864-001

(2) Make sure that hydraulic pressure is removed.

(a) If it is necessary, do this task: Pressure from the Aileron Hydraulic Systems A and B - Deactivation, TASK 27-11-00-860-801.

SUBTASK 27-11-81-860-004

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

D. Cam Follower Bearing Removal

SUBTASK 27-11-81-030-001

(1) Loosen the nuts [70] to relieve the tension on the springs [72].

NOTE: Record the number of times you turn the nuts [70]. This will make it easier to adjust the centering unit after installation.

SUBTASK 27-11-81-020-005

(2) Remove the cotter pins [75] from the pin [73].

(a) Discard the cotter pins [75].

SUBTASK 27-11-81-020-006

(3) Remove the pin [73] and washers [74] from the arm [85].

SUBTASK 27-11-81-030-002

(4) Disconnect the springs [72] from the eye bolts [71].

SUBTASK 27-11-81-020-007

(5) Remove the cotter pin [83], bolt [80], washer [81], washers [87], washer [88] (if installed), washer [89], and nut [82] from the arm [85] and support [86].

(a) Discard the cotter pin [83].

SUBTASK 27-11-81-020-008

(6) Remove the arm [85] from the support [86].

SUBTASK 27-11-81-020-009

(7) Remove the cotter pin [79], bolt [76], washer [77], and nut [78] from the arm [85].

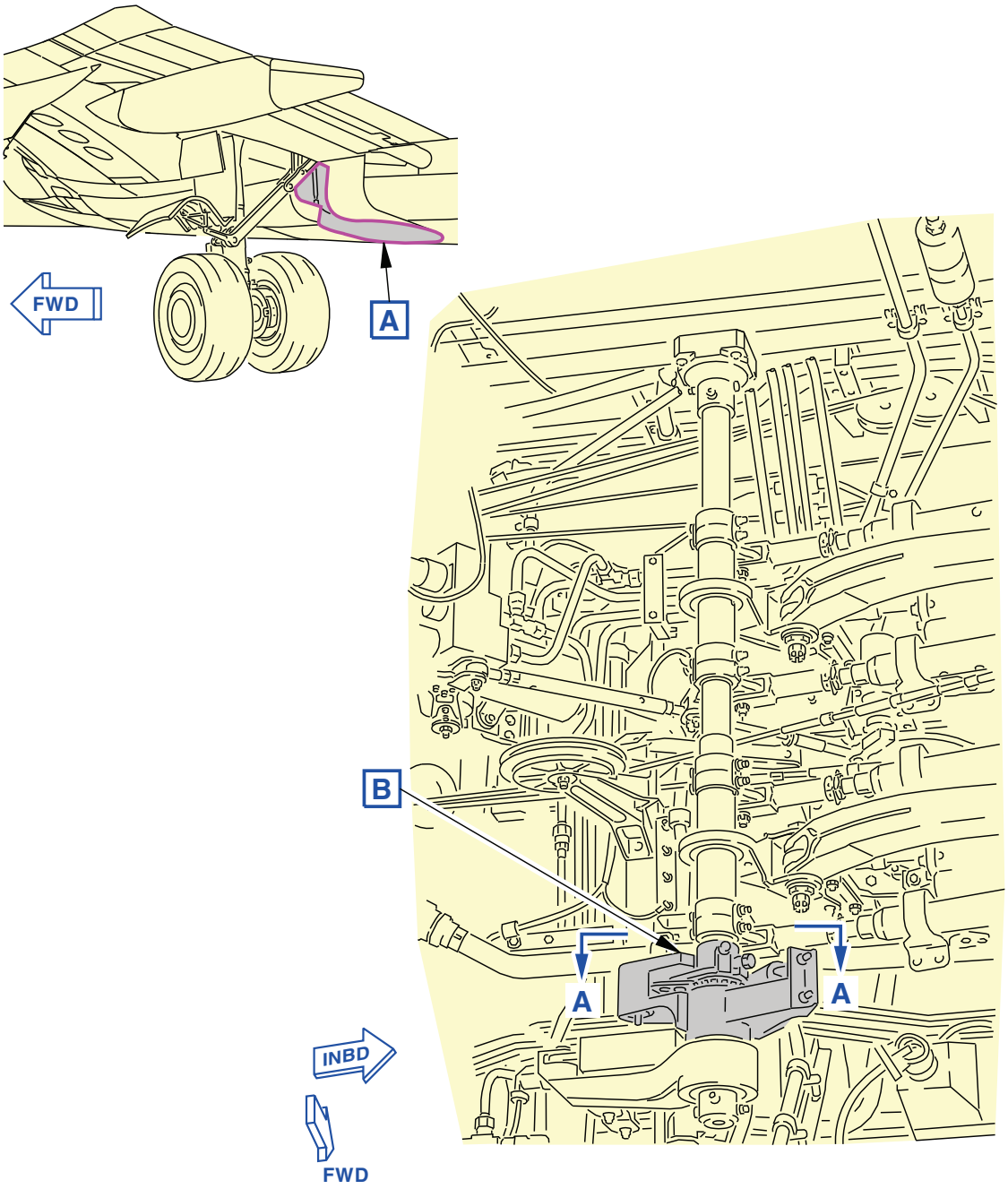
(a) Discard the cotter pin [79].

SUBTASK 27-11-81-020-010

(8) Remove the bearing [84] from the arm [85].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



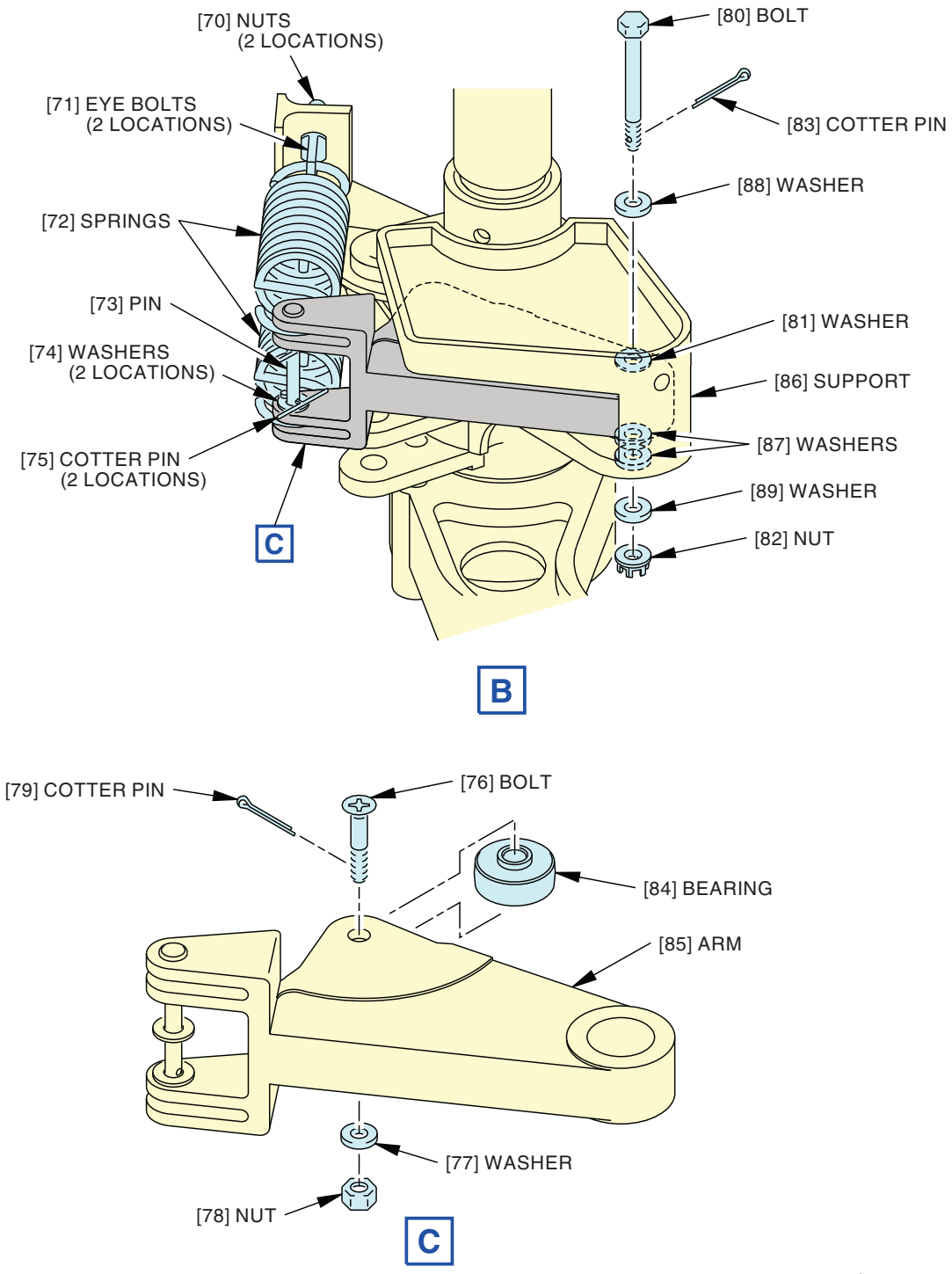
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)

A

2412339 S00061530473_V2

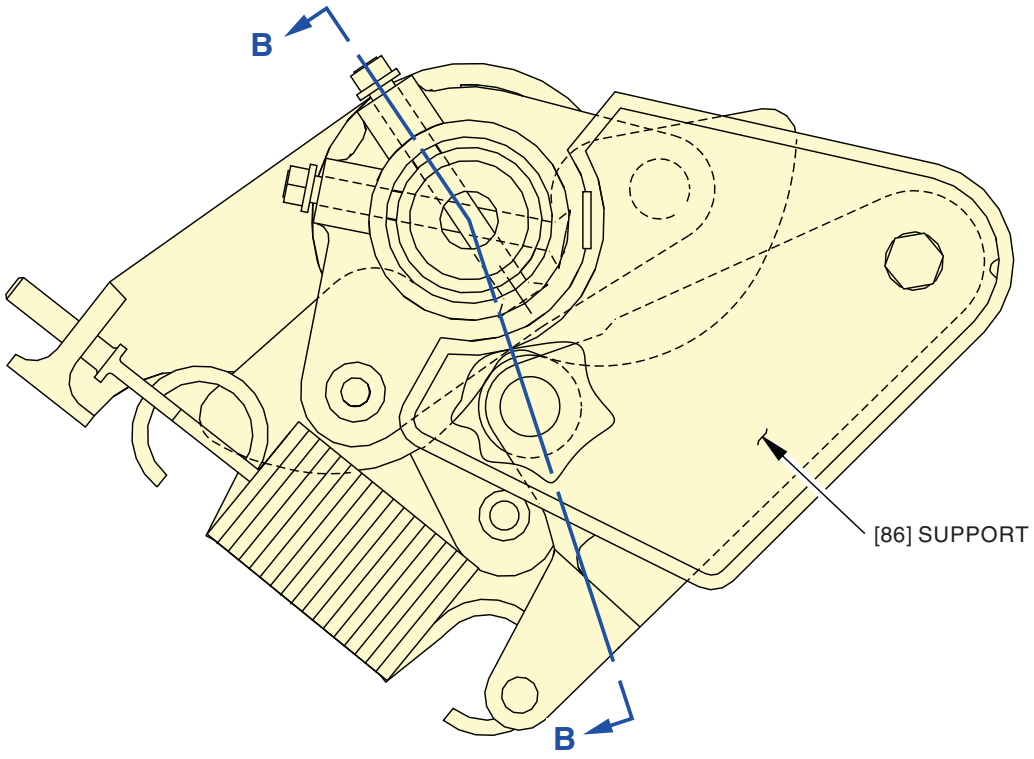
Cam Follower Bearing Installation
Figure 403/27-11-81-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

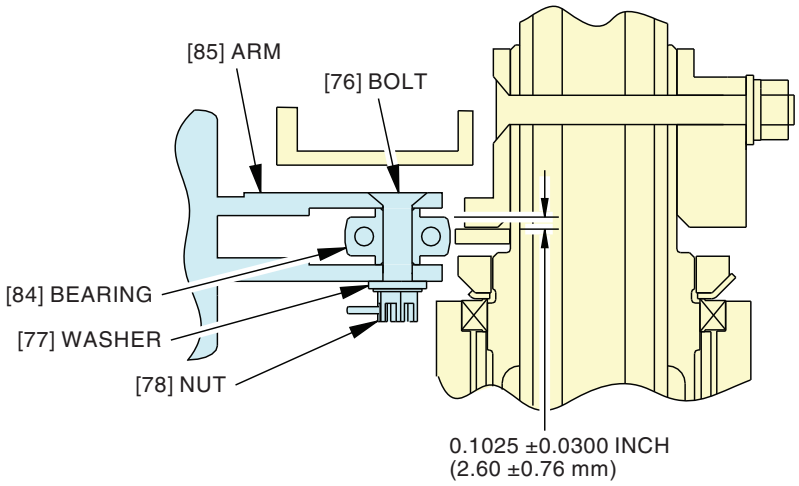


Cam Follower Bearing Installation
Figure 403/27-11-81-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

3054984 S0000819556_V1

Cam Follower Bearing Installation
Figure 403/27-11-81-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-81-400-802

5. Cam Follower Bearing Installation

(Figure 403)

A. References

Reference	Title
27-11-00-700-803	Manual Reversion Control Friction Test (P/B 501)
27-11-00-700-805	Aileron Control Wheel Force Test (P/B 501)
27-11-00-700-810	Aileron Travel Test (P/B 501)
27-11-00-820-805	Aileron Feel and Centering Unit Adjustment (P/B 501)
27-11-00-860-802	Pressure to the Aileron Hydraulic Systems A and B - Activation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

C. Installation of the Cam Follower Bearing

SUBTASK 27-11-81-420-004

(1) Put the bearing [84] in place in the arm [85] and install the bolt [76], washer [77], nut [78], and new cotter pin [79].

SUBTASK 27-11-81-420-005

(2) Install the arm [85] in the support [86].

(a) Put the arm [85] in its position in the support [86].

(b) Put the washer [88], washer [81], and washers [87] between the arm [85] and support [86] as necessary to achieve the cam roller dimension of 0.1025 ±0.0300 in. (2.6 ±0.8 mm) (Figure 403, View B-B).

(c) Install the bolt [80], washer [89], and nut [82].

1) If necessary, install the unused washers [81], washers [87], and washers [88] under the head of the bolt [80] to align the cotter pin [83] with the hole in the end of the bolt [80].

(d) Install the new cotter pin [83].

SUBTASK 27-11-81-430-001

(3) Connect the springs [72] to the eye bolts [71].

SUBTASK 27-11-81-420-006

(4) With the springs [72] in place, install the pin [73], washers [74], and new cotter pins [75] on the arm [85] so that the springs [72] are connected to the pin [73].

SUBTASK 27-11-81-430-002

(5) Tighten the nuts [70] to the pre-removal tension.

SUBTASK 27-11-81-212-001

(6) Make sure that the bearing [84] is in the center of the cam in the support [86].

SUBTASK 27-11-81-863-001

(7) Do this task: Pressure to the Aileron Hydraulic Systems A and B - Activation, TASK 27-11-00-860-802.

SUBTASK 27-11-81-820-006

(8) Do this task: Aileron Feel and Centering Unit Adjustment, TASK 27-11-00-820-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-81-860-005

(9) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL

SUBTASK 27-11-81-720-001

(10) Do this task: Aileron Control Wheel Force Test, TASK 27-11-00-700-805.

SUBTASK 27-11-81-720-002

(11) Do this task: Manual Reversion Control Friction Test, TASK 27-11-00-700-803.

SUBTASK 27-11-81-720-003

(12) Do this task: Aileron Travel Test, TASK 27-11-00-700-810.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON FEEL AND CENTERING UNIT - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Aileron Feel and Centering Unit Springs Detailed Inspection

TASK 27-11-81-210-801

2. Aileron Feel and Centering Unit Springs Detailed Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
27-11-81-000-802	Cam Follower Bearing Removal (P/B 401)
27-11-81-400-802	Cam Follower Bearing Installation (P/B 401)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Aileron Feel and Centering Unit Springs Detailed Inspection

SUBTASK 27-11-81-210-002

(1) Do a detailed inspection of the springs on the aileron feel and centering unit.

(a) Examine the springs for correct installation.

(b) Examine the springs for damage.

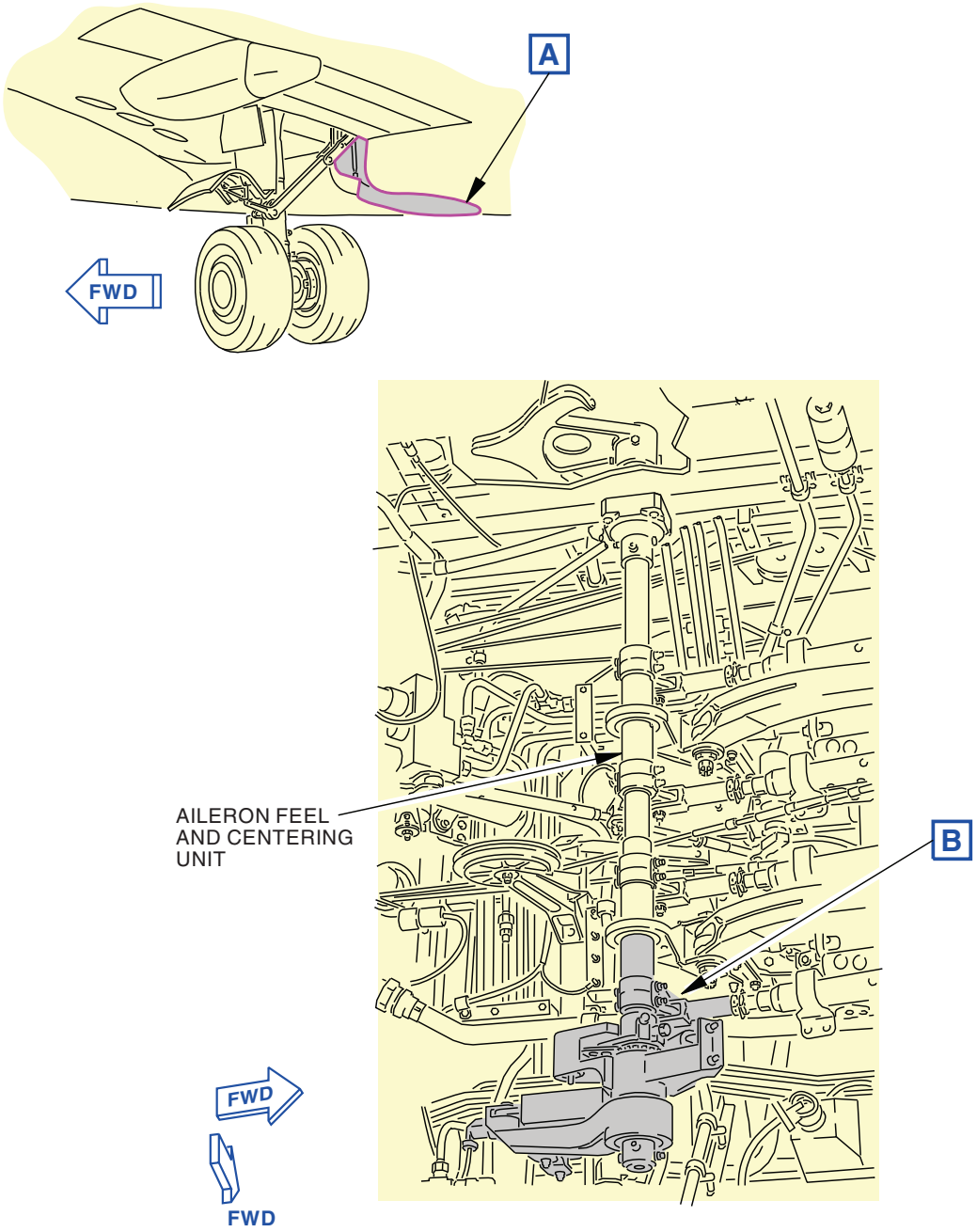
1) If one or both springs are found with damage, worn or broken, do these tasks:

a) Cam Follower Bearing Removal, TASK 27-11-81-000-802.

b) Cam Follower Bearing Installation, TASK 27-11-81-400-802.

——— END OF TASK ———

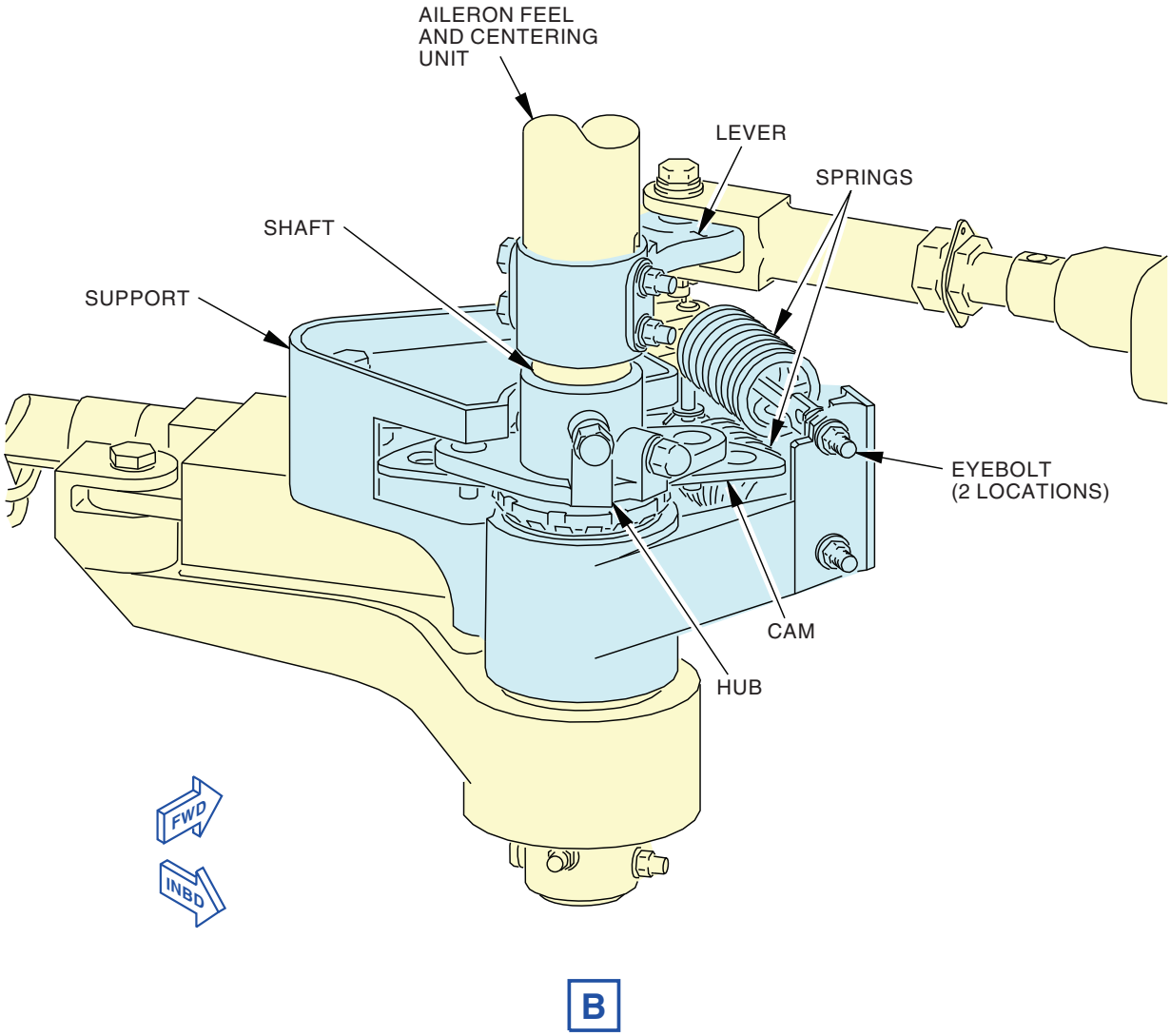
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aileron Feel and Centering Unit Spring Visual Inspection
Figure 601/27-11-81-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412342 S00061530479_V1

Aileron Feel and Centering Unit Spring Visual Inspection
Figure 601/27-11-81-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AILERON/RUDDER TRIM CONTROL MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Aileron and Rudder Trim Control Module Removal

(2) Aileron and Rudder Trim Control Module Installation

(3) Trim Control Switches Removal

(4) Trim Control Switches Installation

(5) Rudder Trim Indicator Removal

(6) Rudder Trim Indicator Installation

B. The Aileron and Rudder Trim Control Module is referred to as the "trim control module assembly [22]" in this procedure.

TASK 27-11-94-000-801

2. Aileron and Rudder Trim Control Module Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-11-94-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

C. Aileron and Rudder Trim Control Module Removal

SUBTASK 27-11-94-020-001

(1) Do these steps to remove the trim control module assembly [22]:

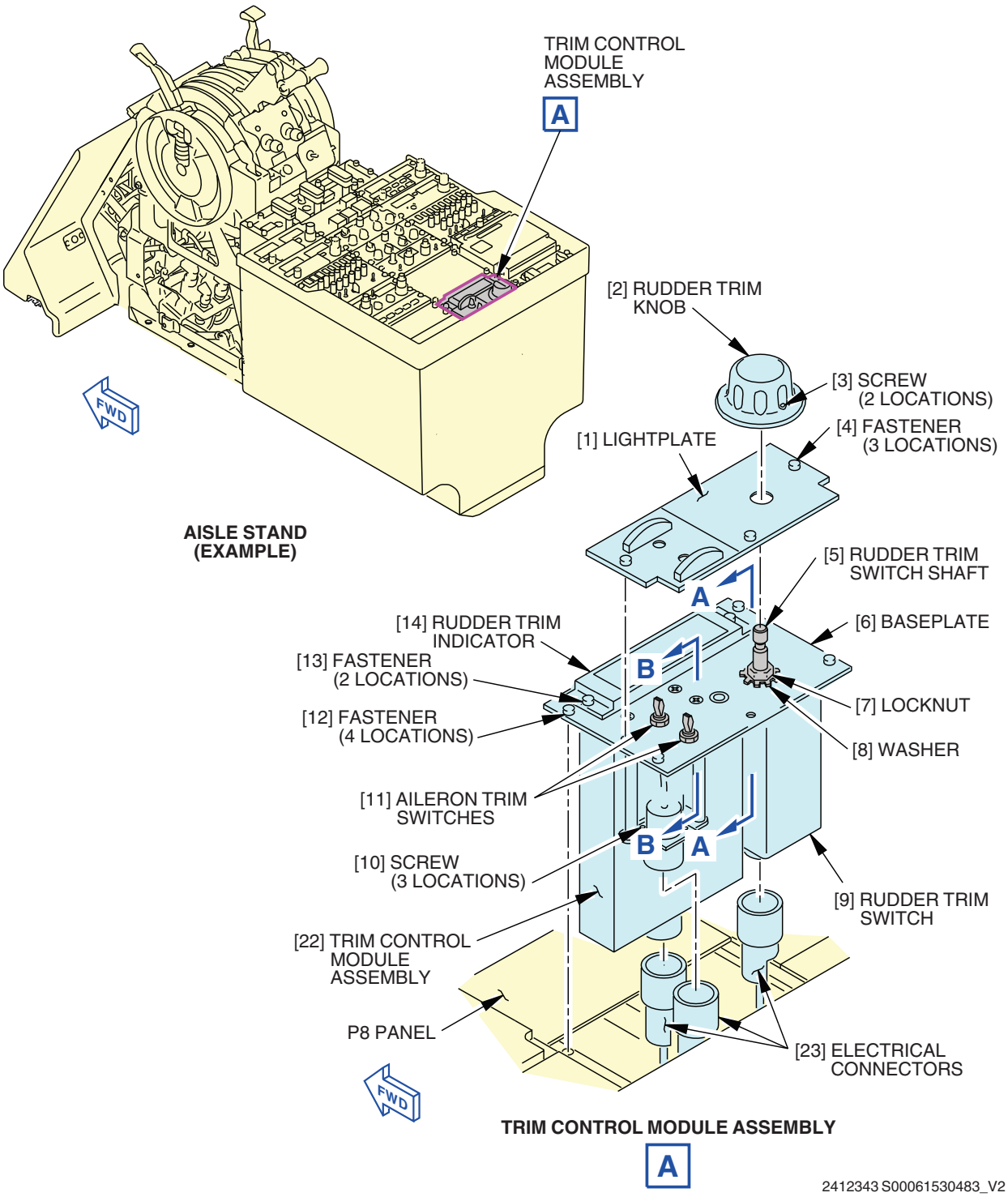
(a) Remove the four fasteners [12] from the trim control module assembly [22].

(b) Lift the trim control module assembly [22] clear of the P8 panel.

(c) Disconnect the three electrical connectors [23] from the trim control module assembly [22].

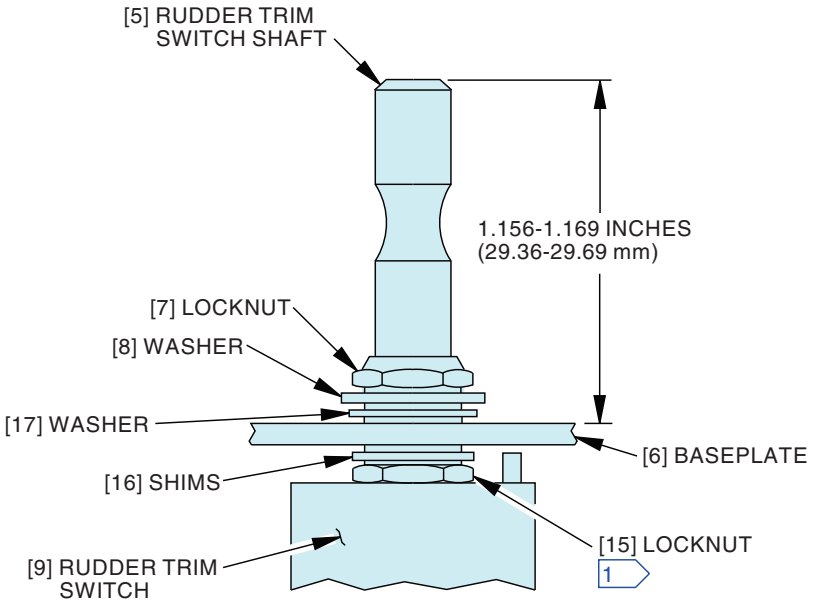
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

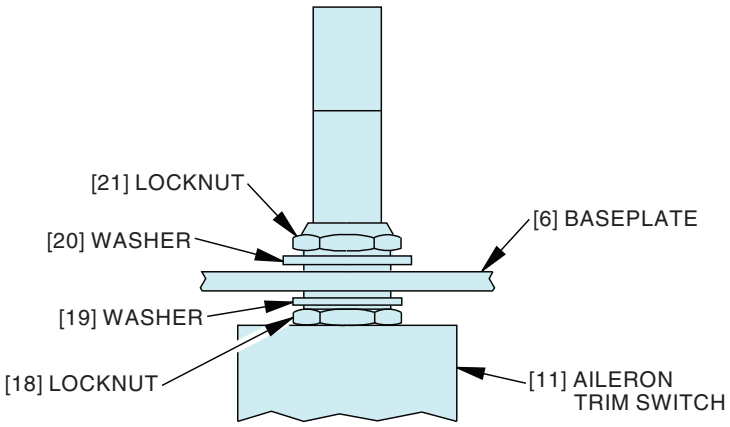


Aileron/Rudder Trim Control Module
Figure 401/27-11-94-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

NOTE:

1 DO NOT CHANGE THIS LOCKNUT SETTING DURING THE SWITCH INSTALLATION OR ADJUSTMENT. IT IS POSSIBLE THAT THE SWITCH WILL NOT ALIGN IF YOU CHANGE THE LOCKNUT SETTING.

2412344 S00061530484_V1

Aileron/Rudder Trim Control Module
Figure 401/27-11-94-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-94-400-801

3. Aileron and Rudder Trim Control Module Installation

(Figure 401)

A. References

Reference	Title
27-11-00-730-801	Aileron Trim Response Test (P/B 501)
27-21-00-700-807	Rudder Trim System Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	Trim control module assembly	27-11-94-01-030	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Aileron and Rudder Trim Control Module Installation

SUBTASK 27-11-94-420-001

(1) Do these steps to install the trim control module assembly [22]:

(a) Connect the electrical connectors [23] to the trim control module assembly [22].

(b) Put the trim control module assembly [22] on the P8 panel.

(c) Install the fasteners [12].

SUBTASK 27-11-94-860-002

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	18	C00786	FLIGHT CONTROL TRIM CONT AIL
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

SUBTASK 27-11-94-730-001

(3) Do this task: Aileron Trim Response Test, TASK 27-11-00-730-801.

SUBTASK 27-11-94-710-001

(4) Do this task: Rudder Trim System Test, TASK 27-21-00-700-807.

——— END OF TASK ———

TASK 27-11-94-000-802

4. Trim Control Switches Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Prepare for the Removal

SUBTASK 27-11-94-020-002

(1) Do this task: Aileron and Rudder Trim Control Module Removal, TASK 27-11-94-000-801.

C. Trim Control Switches Removal

SUBTASK 27-11-94-020-003

(1) Do these steps to remove the rudder trim knob [2]:

(a) Loosen the screws [3] from the rudder trim knob [2].

(b) Remove the rudder trim knob [2] from the rudder trim switch shaft [5].

SUBTASK 27-11-94-020-004

(2) Remove the fasteners [4] to remove the lightplate [1].

SUBTASK 27-11-94-020-005

(3) Do these steps to remove the rudder trim switch [9]:

(a) Bend the tab down on the washer [8].

(b) Remove the locknut [7], washer [8], and washer [17] from the rudder trim switch shaft [5].

(c) Remove the rudder trim switch [9] from behind the baseplate [6].

(d) Remove the shims [16] from the rudder trim switch shaft [5].

1) Make sure the locknut [15] is installed on the rudder trim switch shaft [5].

NOTE: The locknut [15] is installed on the rudder trim switch shaft [5] with the correct torque value. For correct operation of the rudder trim switch [9], do not move the locknut [15].

SUBTASK 27-11-94-020-006

(4) Do these steps to remove the aileron trim switches [11]:

(a) Remove locknuts [21] and washers [20] from the aileron trim switches [11].

(b) Remove the aileron trim switches [11] from behind the baseplate [6].

(c) Remove the washers [19] and locknuts [18] from the aileron trim switches [11].

(d) Disconnect the electrical wires from the aileron trim switches [11].

——— END OF TASK ———

TASK 27-11-94-400-802

5. Trim Control Switches Installation

(Figure 401)

A. References

Reference	Title
27-11-00-730-801	Aileron Trim Response Test (P/B 501)
27-21-00-700-807	Rudder Trim System Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Rudder trim switch	27-11-94-01-020	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right

D. Trim Control Switches Installation

SUBTASK 27-11-94-420-002

(1) Do these steps to install the aileron trim switches [11]:

(a) Connect electrical wires on the aileron trim switches [11].

(b) Install locknuts [18] and washers [19] on the aileron trim switches [11].

(c) Put the aileron trim switches [11] up through the baseplate [6].

(d) Install the washers [20] and locknuts [21] on the aileron trim switches [11].

SUBTASK 27-11-94-420-003

(2) Do these steps to install the rudder trim switch [9]:

(a) Install the shims [16] on the rudder trim switch [9].

1) Use the necessary number of shims [16] to get a dimension of 1.156 in. (29.362 mm)-1.169 in. (29.693 mm) from the top of the baseplate [6] to the end of the rudder trim switch shaft [5].

(b) Put the rudder trim switch [9] up through the baseplate [6].

(c) Install the washer [17], washer [8], and locknut [7] on the rudder trim switch shaft [5].

(d) Tighten the locknut [7] to 36 in-lb (4 N·m)-40 in-lb (5 N·m).

(e) Bend the tab on the washer [8] over the flat of the locknut [7].

NOTE: Small deformation of the star washer [8] is permitted.

SUBTASK 27-11-94-420-004

(3) Install the lightplate [1] with fasteners [4].

SUBTASK 27-11-94-420-005

(4) Do these steps to install the rudder trim knob [2]:

(a) Make sure the surfaces between the rudder trim knob [2] and the lightplate [1] are free of contaminates.

(b) Put the rudder trim knob [2] fully down on the rudder trim switch shaft [5].

(c) Make sure the rudder trim knob [2] is 0.010 in. (0.254 mm)–0.050 in. (1.270 mm) from the top of the lightplate [1].

(d) Install the set screws [3] on the rudder trim knob [2].

(e) Tighten the set screws [3] on the rudder trim knob [2].

SUBTASK 27-11-94-420-006

(5) Do this task: Aileron and Rudder Trim Control Module Installation, TASK 27-11-94-400-801.

SUBTASK 27-11-94-730-002

(6) If you replaced the rudder trim switch [9], do this task: Rudder Trim System Test, TASK 27-21-00-700-807.

SUBTASK 27-11-94-730-003

(7) If you replaced the aileron trim switches [11], do this task: Aileron Trim Response Test, TASK 27-11-00-730-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-11-94-000-803

6. Rudder Trim Indicator Removal

(Figure 401)

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 27-11-94-860-003

(1) Make sure that the rudder is in the neutral position.

SUBTASK 27-11-94-860-004

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

C. Rudder Trim Indicator Removal

SUBTASK 27-11-94-020-007

(1) Do these tasks to remove the rudder trim indicator [14]:

(a) Remove the fasteners [13] that connect the rudder trim indicator [14] to the baseplate [6].

(b) Disconnect the electrical connector from the rudder trim indicator [14].

(c) Remove the rudder trim indicator [14] from the baseplate [6].

——— END OF TASK ———

TASK 27-11-94-400-803

7. Rudder Trim Indicator Installation

(Figure 401)

A. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	Rudder trim indicator	27-11-94-01-020	PTF ALL

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Rudder Trim Indicator Installation

SUBTASK 27-11-94-420-007

(1) Do these steps to install the rudder trim indicator [14]:

(a) Connect the electrical connector to the rudder trim indicator [14].

(b) Put the rudder trim indicator [14] on the baseplate [6].

(c) Install the fasteners [13] to attach the rudder trim indicator [14] to the baseplate [6].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-11-94-860-005

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

SUBTASK 27-11-94-280-001

(3) Do this check to make sure that the rudder trim indicator [14] is in the null position.

(a) With the rudder in the neutral position, make sure the rudder trim indicator [14] is at the zero position.

SUBTASK 27-11-94-820-001

(4) If the rudder trim indicator [14] is not at the null position, do these steps to adjust the rudder trim indicator [14]:

(a) Remove the fasteners [13] that attach the rudder trim indicator [14] to the baseplate [6].

NOTE: Do not disconnect the electrical connector.

(b) Turn the adjustment screw on the bottom of the rudder trim indicator [14] until the indicator is at 0.0 ±0.1 unit.

(c) Install the rudder trim indicator [14] to the baseplate [6] with the fasteners [13].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER AND RUDDER TRIM CONTROL SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Rudder Hydraulic System A, B, or Standby Pressurization

(a) Use this task to pressurize either hydraulic system A or B, both hydraulic systems A and B, or to pressurize the standby hydraulic system.

(2) Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization

(3) Pressure from the Rudder Hydraulic Systems A, B and Standby - Deactivation.

(4) Pressure to the Rudder Hydraulic Systems A, B, and Standby - Activation.

B. Hydraulic systems A, B, and standby must have pressure removed before maintenance is done on the rudder control system. This is to prevent injury to persons because of accidental operation of the control system while you do maintenance. You must be careful to put maintenance stands and equipment away from control surface travel.

C. Electric pump 1 pressurizes the hydraulic system B and electric pump 2 pressurizes the hydraulic system A. Electric pump 1 heat exchanger is installed in the fuel tank 2. Electric pump 2 heat exchanger is installed in the fuel tank 1.

D. This procedure operates the electric hydraulic system A or B pumps to pressurize the hydraulic systems A and B. Pressurization of the hydraulic systems with the engine-driven hydraulic pumps is not included in this procedure.

TASK 27-21-00-800-801

2. Rudder Hydraulic System A, B, or Standby Pressurization

(Figure 201)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

C. Procedure

NOTE: This procedure supply the steps to pressurize either hydraulic system A or B, both systems A and B, or to pressurize the standby hydraulic system.

SUBTASK 27-21-00-860-001

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-002

CAUTION

DO NOT OPERATE SYSTEM A OR B HYDRAULIC PUMPS FOR MORE THAN TWO MINUTES WITHOUT FUEL IN THE FUEL TANKS. THE NO. 1 (FOR SYSTEM A) AND NO. 2 (FOR SYSTEM B) FUEL TANKS MUST HAVE A MINIMUM OF 250 GALLONS (1675 POUNDS/760 KILOGRAMS) OF FUEL IN THEM. IF THERE IS NOT SUFFICIENT FUEL IN THE FUEL TANKS, THE HYDRAULIC FLUID WILL BECOME TOO HOT.

(2) Make sure the No. 1 and No. 2 fuel tanks each have 250 gal (946 l) (1675 lb (760 kg)) of fuel in them.

SUBTASK 27-21-00-210-001

(3) Make sure the hydraulic reservoirs are full. To fill the reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-21-00-860-003

(4) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-21-00-860-004

(5) Set the parking brake.

SUBTASK 27-21-00-480-001

WARNING

MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 27-21-00-860-005

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Pressurize the rudder hydraulic system A, B or Standby (Figure 201):

(a) Set the ELEC 2, HYD PUMPS A or ELEC 1, HYD PUMPS B switch on the forward overhead panel to the ON position.

NOTE: Intermittent system A or B electric pump operation must be kept to a limit of five starts in a 5 minutes period. After the fifth start, operate the pump for more than 5 minutes or turn the pump off for a minimum of 5 minutes.

(b) For the hydraulic system A pressurization, set the FLT CONTROL A switch on the forward overhead panel to the ON position.

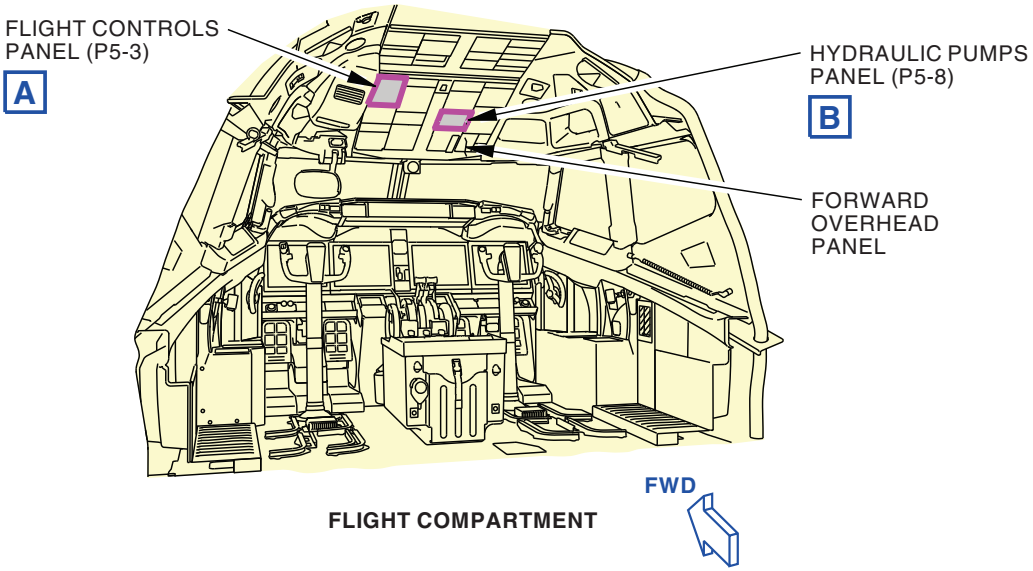
(c) For the hydraulic system B pressurization, set the FLT CONTROL B switch on the forward overhead panel to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

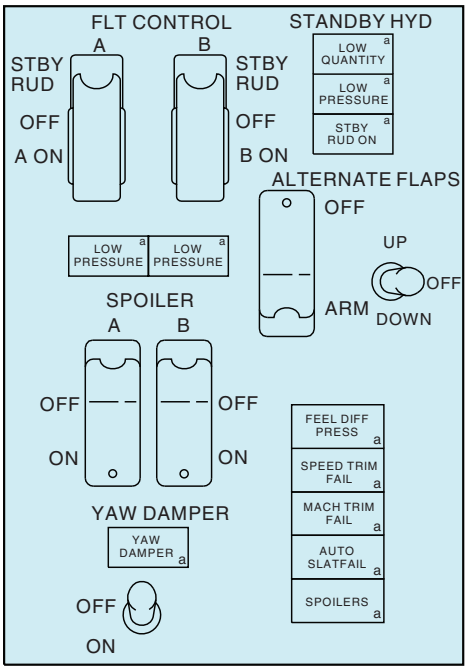
(d) For the standby hydraulic system, set the FLT CONTROL A or B switch on the forward overhead panel to the STDBY RUD position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

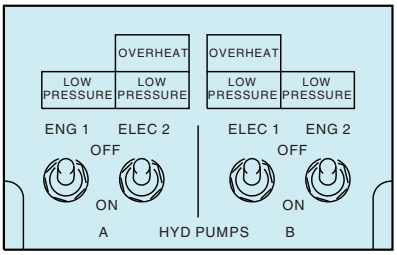


FLIGHT COMPARTMENT



FLIGHT CONTROLS PANEL (P5-3)

A



HYDRAULIC PUMPS PANEL (P5-8)

B

2412345 S00061530494_V1

Rudder Power Control Switches
Figure 201/27-21-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-840-801

3. Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization

(Figure 201)

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

B. Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization

SUBTASK 27-21-00-860-006

(1) Set the ELEC 2, HYD PUMPS A and the ELEC 1, HYD PUMPS B switches, on the forward overhead panel, to the OFF position.

SUBTASK 27-21-00-860-007

(2) Make sure that the FLT CONTROL A and B switches, on the forward overhead panel, are in the ON position.

——— END OF TASK ———

TASK 27-21-00-800-802

4. Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation

(Figure 201, Figure 202, Figure 203)

A. General

(1) This procedure removes pressure from the rudder hydraulic systems A, B, and Standby before you do maintenance on the rudder and rudder trim control system.

B. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge
325	Vertical Fin - Rudder

E. Rudder Hydraulic Systems A, B, and Standby Pressure Deactivation

SUBTASK 27-21-00-860-008

(1) Set the FLT CONTROL A and B switches, on the forward overhead panel, to the OFF position.

SUBTASK 27-21-00-860-009

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-21-00-860-010

(3) Set the ELEC 2, HYD PUMPS A and ELEC 1, HYD PUMPS B switches, on the forward overhead panel, to the OFF position.

SUBTASK 27-21-00-870-001

(4) Operate the rudder pedals or the control column to decrease the remaining hydraulic system pressure.

SUBTASK 27-21-00-860-011

(5) If it is necessary to open a hydraulic line, you must release pressure from the reservoir for the applicable hydraulic system, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 27-21-00-480-002

(6) Install the flight controls ground locks to the flight controls shutoff valves (Figure 202, Figure 203):

(a) Disconnect the electrical connector from the flight controls shutoff valve at the flight controls hydraulic modules A and B.

(b) Disconnect the electrical connector from the standby flight controls shutoff valve at the standby flight controls hydraulic module.

(c) After you disconnect the electrical connector(s), examine the connector(s) for corrosion.

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

(d) If there was corrosion, correct the problem (SWPM 20-60-03).

(e) Put the manual override lever at the flight control A and B shutoff valves in the position 2 (OFF position) and leave the standby valve in the position 1 (OFF position).

NOTE: In normal condition, the flight control A and B shutoff valves are in position 1 (ON position) and standby valve is in position 1 (OFF position).

(f) Install the lock, SPL-1675, on the three flight controls shutoff valves.

1) Install the detent pins.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Rudder and Rudder Trim Control System Tryout

NOTE: This tryout is to make sure the rudder and rudder trim control system is in a zero energy state.

SUBTASK 27-21-00-710-001

(1) In the flight deck, push the Captain's or First Officer's rudder pedals to their stops, left and right.

(a) Make sure that the rudder control surface does not move when you continue to push the rudder pedals.

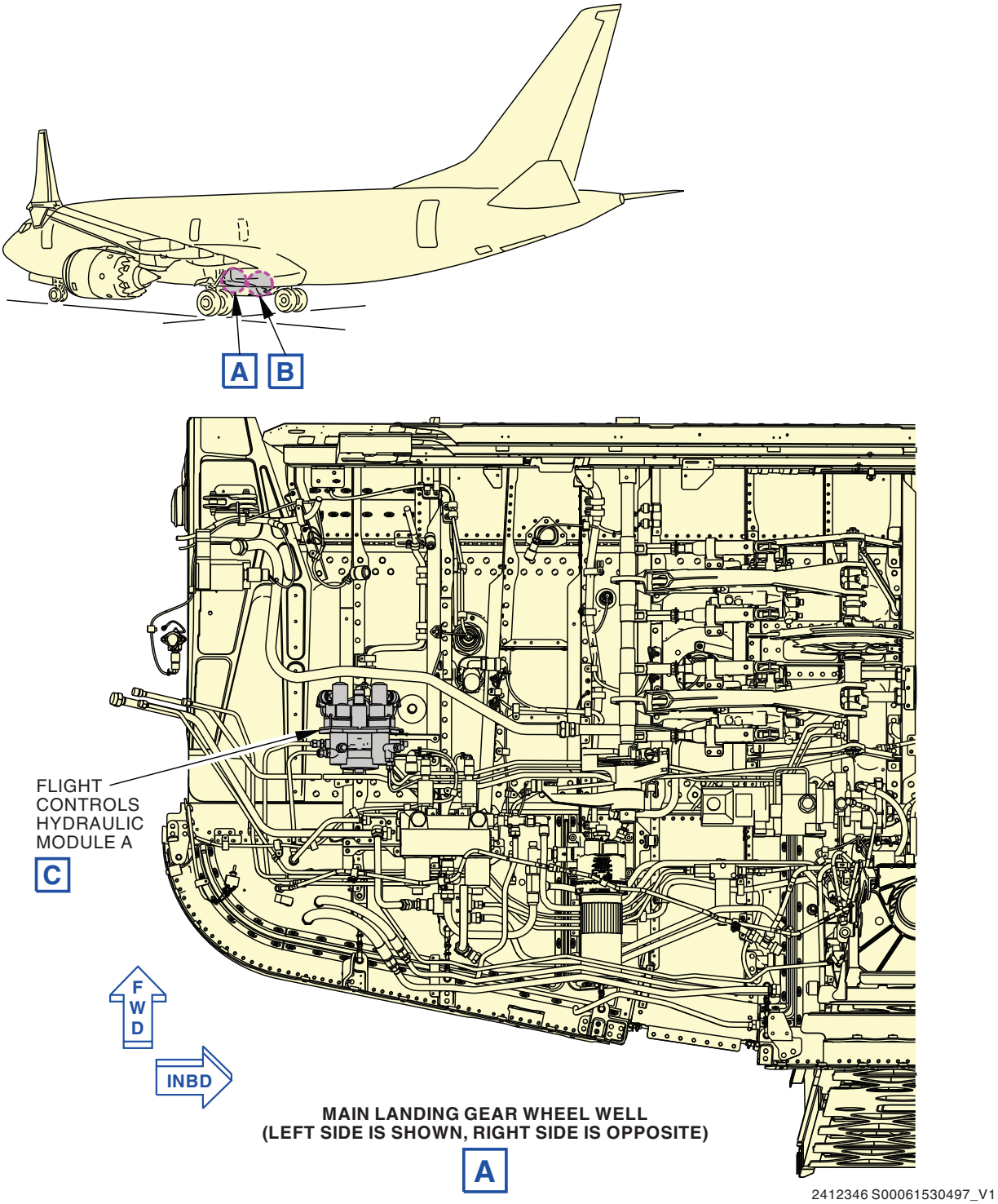
SUBTASK 27-21-00-710-002

(2) Operate the rudder trim knob, on the P8 control stand panel, to simulate the rudder function.

(a) Make sure that the rudder control surface does not move.

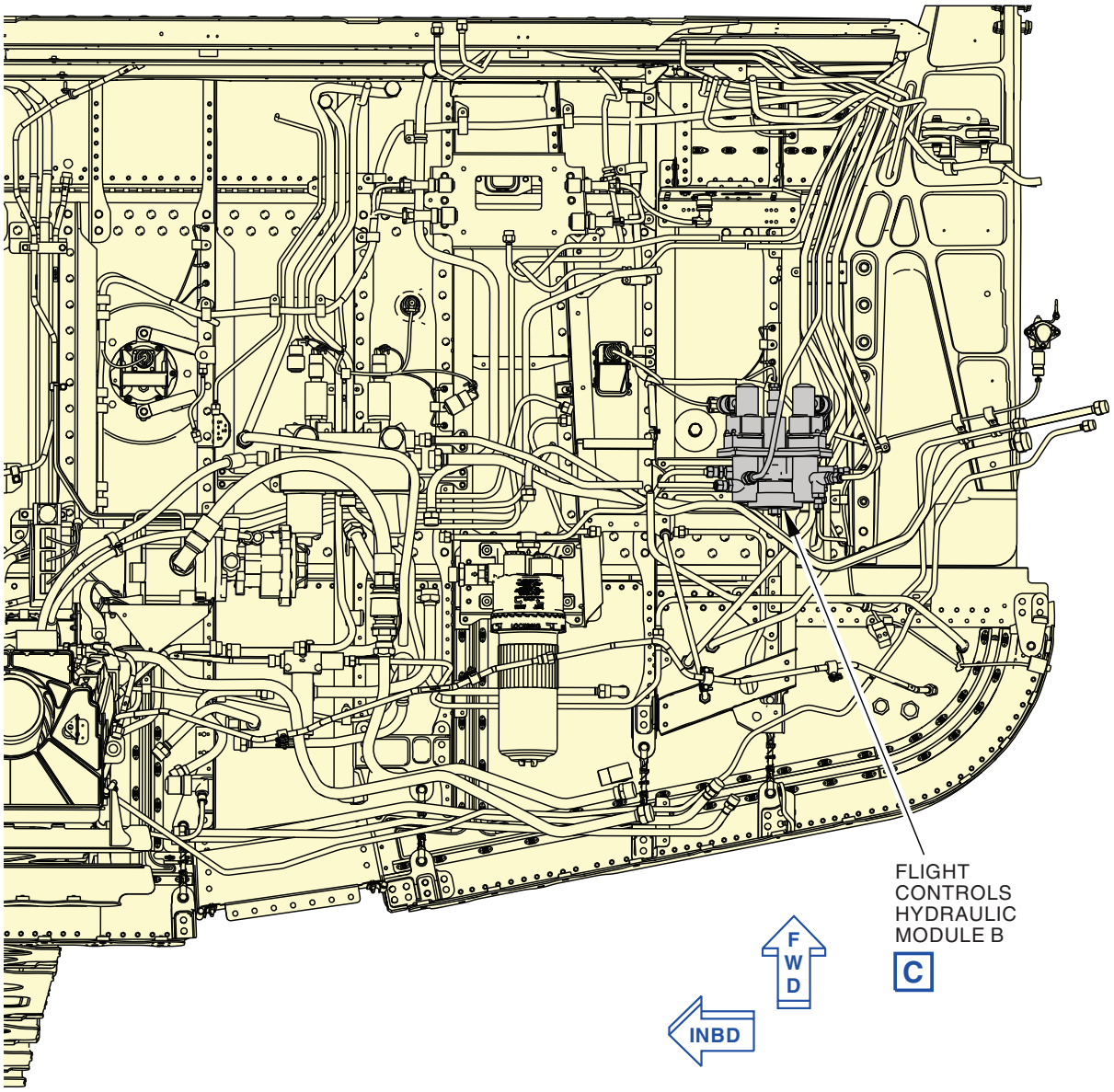
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flight Controls Hydraulic Module Lock Installation
Figure 202/27-21-00-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



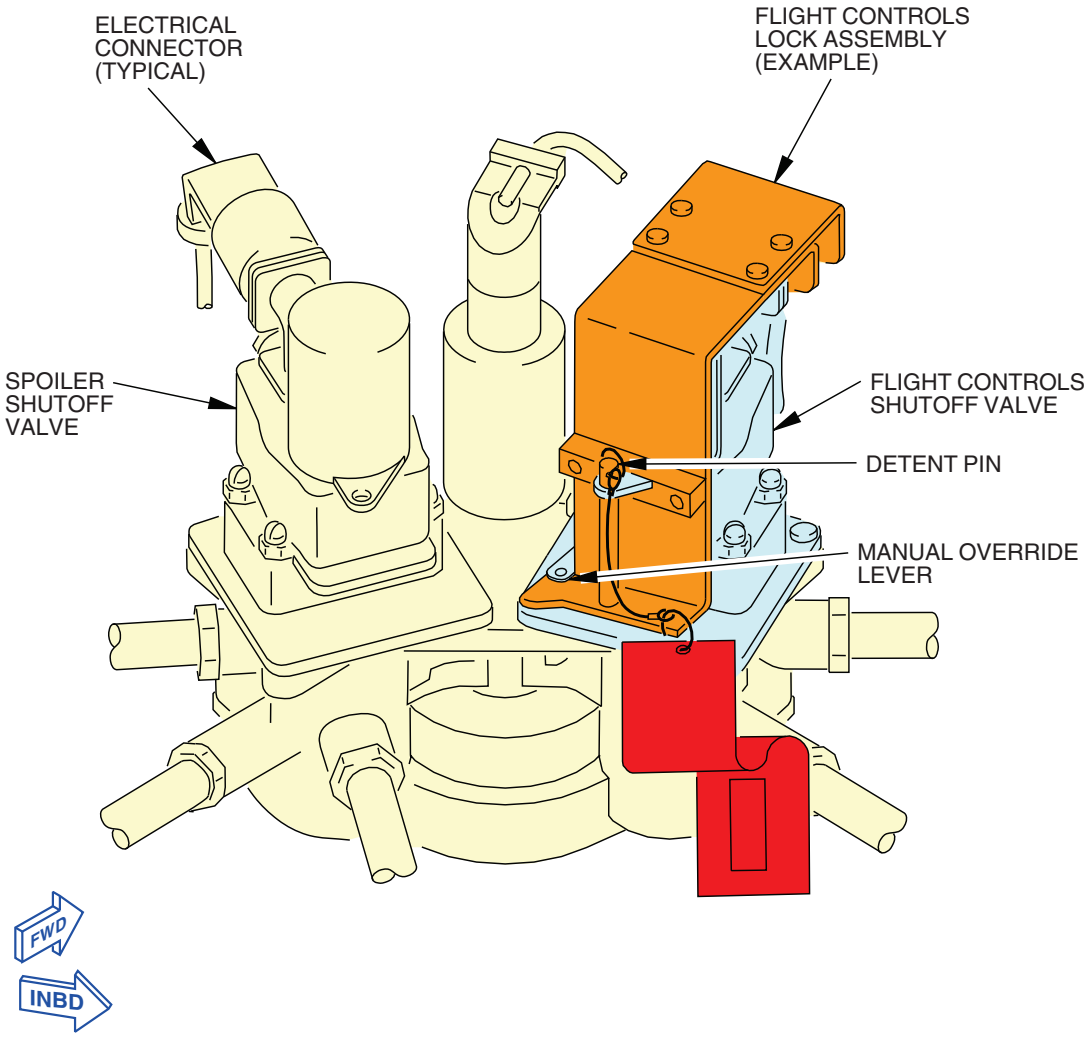
MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE IS SHOWN, LEFT SIDE IS OPPOSITE)

B

2412347 S00061530498_V1

Flight Controls Hydraulic Module Lock Installation
Figure 202/27-21-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



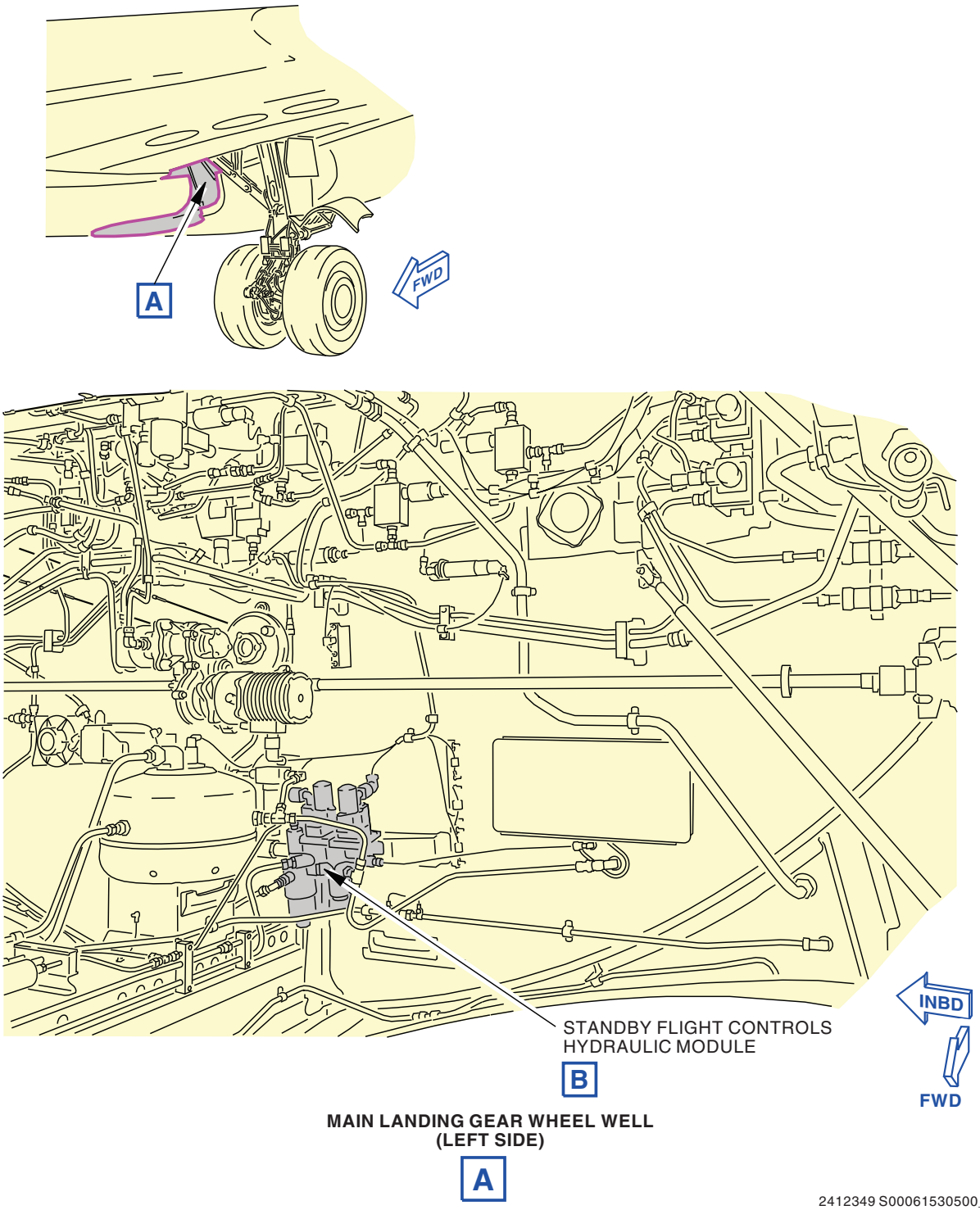
FLIGHT CONTROLS HYDRAULIC MODULE A
(MODULE B IS EQUIVALENT)

C

2412348 S00061530499_V4

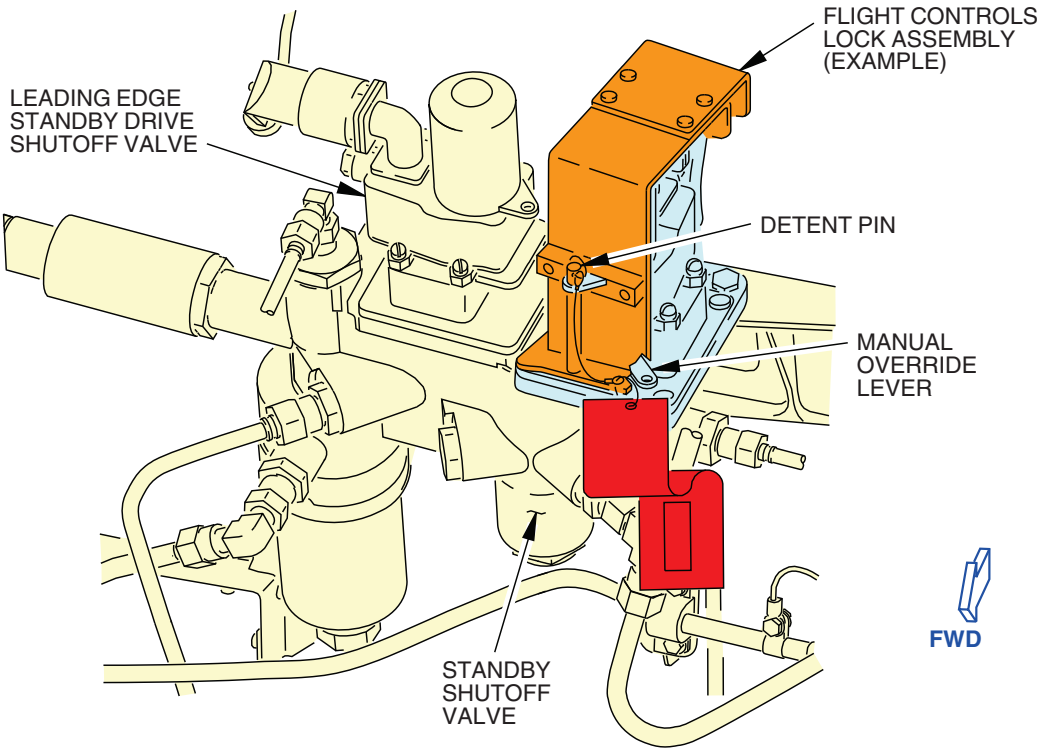
Flight Controls Hydraulic Module Lock Installation
Figure 202/27-21-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Standby Flight Controls Hydraulic Module Lock Installation
Figure 203/27-21-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STANDBY FLIGHT CONTROLS HYDRAULIC MODULE

B

2412350 S00061530501_V4

Standby Flight Controls Hydraulic Module Lock Installation
Figure 203/27-21-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-840-802

5. Pressure to the Rudder Systems A, B, and Standby - Activation

(Figure 201, Figure 202, Figure 203)

A. General

(1) This task will put the rudder hydraulic systems A, B, and Standby back to the condition before pressure removal.

B. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
27-09-31-710-801	Flight Controls Shutoff Valve Test (P/B 501)
29-21-00-700-801	Operational Test of the Standby Hydraulic System (P/B 501)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1675	Lock Assembly - Flight Controls (used on valve p/n 10-60581-2 and 10-60581-3) Part #: F80049-83 Supplier: 81205 Opt Part #: F80049-56 Supplier: 81205 Opt Part #: F80049-65 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50

F. Pressure to the Rudder Systems A, B, and Standby Activation

SUBTASK 27-21-00-080-002

(1) Remove the flight control ground locks from the flight control shutoff valves (Figure 202, Figure 203):

(a) Remove the detent pins.

(b) Remove the lock, SPL-1675, from the three flight controls shutoff valves.

(c) Put the manual override lever at the three flight controls shutoff valves A and B in the position 1 (ON position).

(d) If you did not examine the electrical connector(s) at the removal step, do these steps:

NOTE: The inspection of the electrical connector(s) can be done after the removal step or before the installation step of the electrical connector(s). This is an operator option.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Before you connect the electrical connector(s), examine the connector(s) for corrosion.

2) If there was corrosion, correct the problem (SWPM 20-60-03).

(e) Apply D5026NS or ZC-026 compound, G50171, to the connector (SWPM 20-60-03).

(f) Connect the electrical connector to the flight controls shutoff valve at the flight controls hydraulic modules A and B.

(g) Connect the electrical connector to the flight controls shutoff valve at the standby flight controls hydraulic module.

SUBTASK 27-21-00-860-012

(2) Set the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the ON position.

SUBTASK 27-21-00-860-013

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-21-00-860-014

(4) If it is necessary to service the hydraulic reservoirs, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

SUBTASK 27-21-00-700-011

(5) Do this task: Flight Controls Shutoff Valve Test, TASK 27-09-31-710-801.

SUBTASK 27-21-00-710-006

(6) Do this task: Operational Test of the Standby Hydraulic System, TASK 29-21-00-700-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER AND RUDDER TRIM CONTROL SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Rudder Forward Quadrant Adjustment - Use this task to adjust the rudder forward quadrant.

(2) Rudder Pedals Adjustment - Use this task to adjust the rudder pedals.

(3) Rudder Control Cables RA and RB Adjustment - Use this task to adjust the rudder control cables. This task also includes steps to adjust the rudder forward quadrant and rudder pedals.

(4) Control Rod for Main Rudder Power Control Unit Adjustment - Use this task to adjust the control rod for the main rudder power control unit.

(5) Control Rod for Standby Rudder Power Control Unit Adjustment - Use this test to adjust the control rod for the standby rudder power control unit.

(6) Rudder Pedal Adjustment and Limit Travel Test - Use this test make sure the rudder pedals can move through their full travel and the pedal adjustment crank operates correctly.

(7) Rudder Hinge Bearing and PCU Freeplay - Use this task to do a check of the freeplay of the rudder control system.

(8) Rudder Centering Test - Use this test to make sure the rudder is in its center position.

(9) Rudder Pedal Forces Test - Use this test to make sure the rudder pedals operate with the correct force.

(10) Rudder Travel Test - Use this test to make sure the rudder operates correctly.

(11) Standby Rudder Power Control Unit Deadband Test - Use this test to make sure the standby rudder power control unit is in the correct rigged position.

(12) Rudder Trim System Test - Use this test to make sure the rudder system operates correctly.

(13) Wheel to Rudder Interconnect System Test - Use this test to make sure the wheel to rudder function of the standby rudder power control unit operates correctly.

(14) Rudder Hinge Friction Test - Use this task to do a check of the rudder hinge for excessive friction.

(15) Main Rudder PCU Internal Hydraulic Leakage Test - Use this task to do a check for excessive internal hydraulic leakage in the main rudder PCU.

(16) Rudder Override Operational Test - Use this task to verify that the override mechanisms on the torque tube are operational.

(17) Rudder Main PCU Force Fight Monitor Test - Use this task to verify that the STBY RUD ON light on the overhead panel P5-3 is functional.

TASK 27-21-00-820-801

2. Rudder Forward Quadrant Adjustment

(Figure 501, Figure 502)

A. General

(1) Use this test to make sure the rudder forward quadrant is adjusted correctly. After you adjust the rudder forward quadrant, you must re-adjust the rudder pedals and control cables.

B. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
27-09-14-860-802	Cable Control - Tension Measurement (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
32-51-00-820-805	Nose Wheel Steering System Adjustment (P/B 501)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

F. Prepare for the Test

SUBTASK 27-21-00-860-015

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-00-010-001

(2) Open this access door:

Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-00-010-002

(3) Open these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
114BW	Forward Nose Wheel Well Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

G. Rudder Forward Quadrant Adjustment

SUBTASK 27-21-00-710-003

(1) Do the rudder forward quadrant adjustment:

(a) Do these steps to set the jackshaft yoke to the middle position:

NOTE: The middle position of the jackshaft yoke is where the number of turns of the shaft from the full forward and full aft adjustment are the same.

1) Turn the rudder pedal adjustment crank to move the rudder pedals full forward.

2) Turn the rudder pedal adjustment crank to move the rudder pedals full aft.

a) Record the number of turns.

3) Turn the rudder pedal adjustment crank half the recorded number of turns forward.

(b) Install the rig pin R-3 from the rigging pin kit, SPL-1585, in the forward hole in the captain's rudder forward quadrant.

(c) Install the rig pin R-4 from the rigging pin kit, SPL-1585, in the forward hole in the first officer's rudder forward quadrant.

(d) If you can not install rig pins R-3 or R-4, adjust the bus rod [6]:

1) Turn the turnbuckles to release the tension of rudder control cables RA and RB.

2) Remove the pin [5], nut [4], washer [3], spacers [2], bolt [1] to disconnect the bus rod [6] from rudder forward quadrant.

a) Discard the pin [5].

3) Install the rig pin R-3 from the rigging pin kit, SPL-1585, in the forward hole in the captain's rudder forward quadrant.

4) Install the rig pin R-4 from the rigging pin kit, SPL-1585, in the forward hole in the first officer's rudder forward quadrant.

5) Adjust the bus rod [6] until you can install the bolt [1] through the rudder forward quadrant.

6) Install the spacers [2], washer [3], nut [4], and new new pin [5].

7) If necessary, adjust the brake pushrod [12] to the nose wheel steering. To adjust them, do this task: Nose Wheel Steering System Adjustment, TASK 32-51-00-820-805.

8) Make sure that the control cables RA and RB are at the correct tension (TASK 27-09-14-860-802).

(e) Make sure that the clearance between the cable guard blocks and the rudder forward quadrants is 0.030 in. (0.76 mm) - 0.050 in. (1.27 mm) through full travel.

(f) Remove the rig pins R-3 and R-4.

(g) Remove the nut [45], washer [49], washer [48], and bolt assembly [47] to disconnect the standby rudder PCU standby rudder PCU control rod [34] from the rudder control torque tube [33] (Figure 502).

(h) For each main rudder PCU control rod do this task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Remove the nut [45], washer [44], washer [43] and bolt assembly [42] to disconnect the two main rudder PCU control rod [36] from the rudder control torque tube [33] (Figure 502).

(i) Move the captain's rudder forward quadrant until you can install the rig pin R-3 in the aft rig pin hole.

(j) Make sure that there is no clearance between the captain's forward quadrant and quadrant stop. Add or subtract shims as needed to bring the clearance to zero.

(k) Remove the rig pin R-3.

(l) Move the first officer's rudder forward quadrant until you can install the rig pin R-4 in the aft rig pin hole.

(m) Make sure that there is no clearance between the first officer's forward quadrant and quadrant stop. Add or subtract shims as needed to bring the clearance to zero.

(n) Remove the rig pin R-4.

(o) Connect the main rudder PCU control rod [36] to the center crank [41].

1) Install the bolt assembly [42] with the bolt head end up and washer [43].

2) Tighten the outer bolt of the bolt assembly [42] to 70 in-lb (7.9 N·m) - 85 in-lb (9.6 N·m).

3) Install the washer [44] and nut [45].

CAUTION DO NOT TIGHTEN THE OUTER BOLT OF THE BOLT ASSEMBLY [42] AFTER YOU TIGHTEN THE NUT [45]. DAMAGE TO EQUIPMENT CAN OCCUR.

4) Tighten the nut [45] to 25 in-lb (2.8 N·m) - 30 in-lb (3.4 N·m).

(p) Install the bolt assembly [47] with the bolt head end up, washer [48] (under bolt), washer [49] (under nut), and nut [50] to connect the standby rudder PCU control rod [34] to the upper crank [46].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-820-001

(1) Do this task: Rudder Pedals Adjustment, TASK 27-21-00-820-802.

SUBTASK 27-21-00-820-002

(2) Do this task: Control Rod for Main Rudder Power Control Unit Adjustment, TASK 27-21-00-820-804.

SUBTASK 27-21-00-410-001

(3) Close these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-010-003

(4) Close this access door:

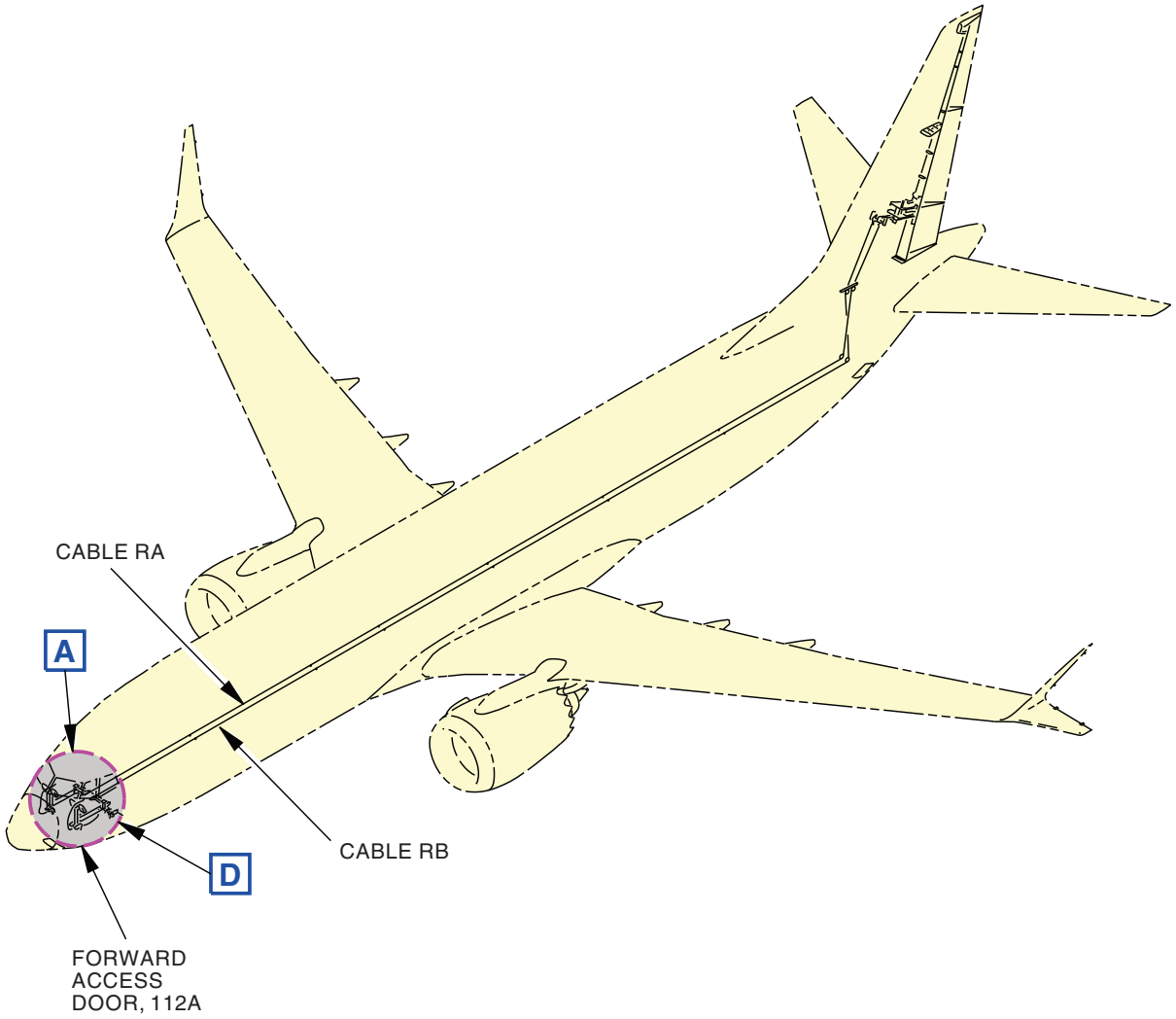
Number	Name/Location
112A	Forward Access Door

SUBTASK 27-21-00-860-016

(5) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

——— END OF TASK ———

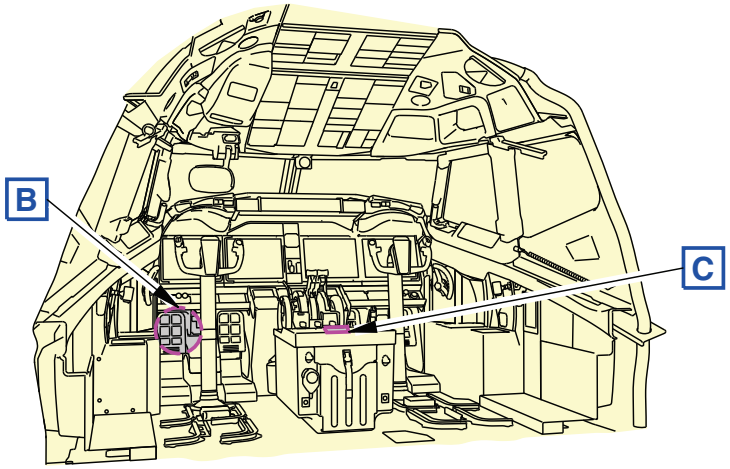
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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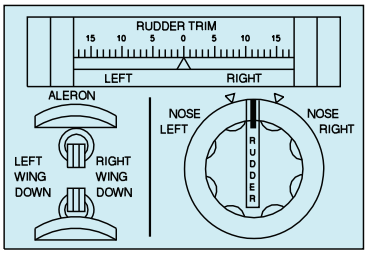
Rudder Control System Forward Components
Figure 501/27-21-00-990-804 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



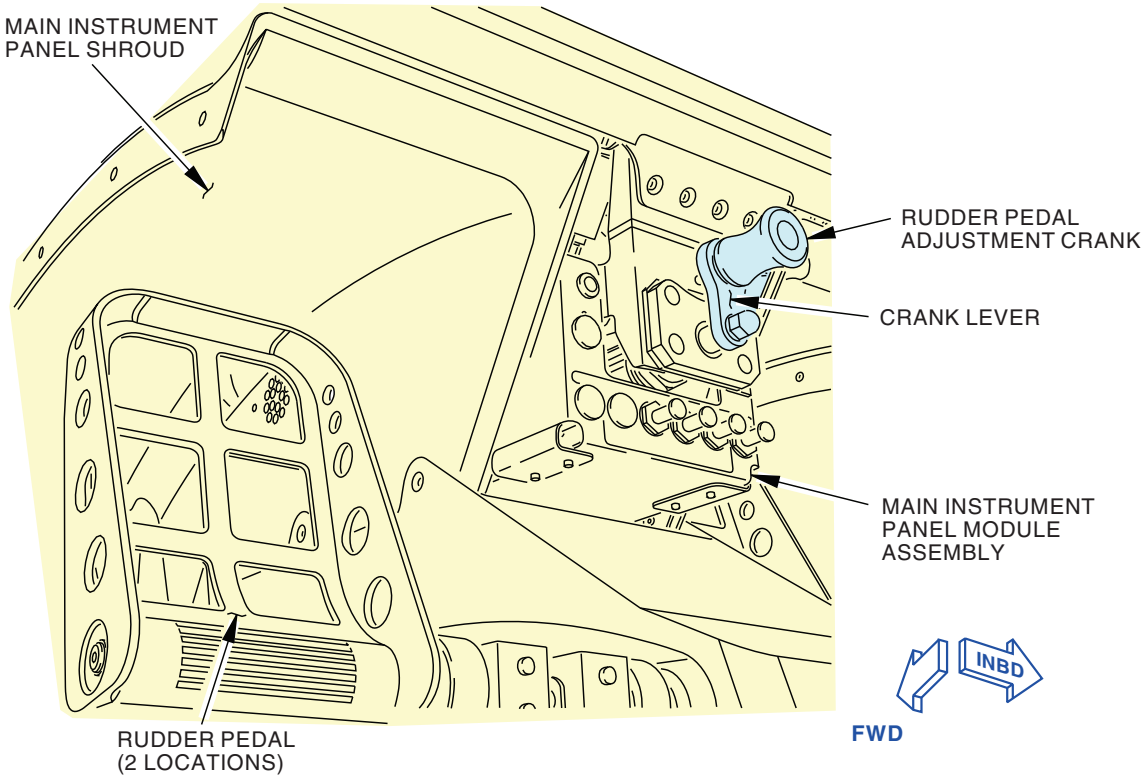
FLIGHT COMPARTMENT

A



AILERON AND RUDDER TRIM MODULE (P8)

C



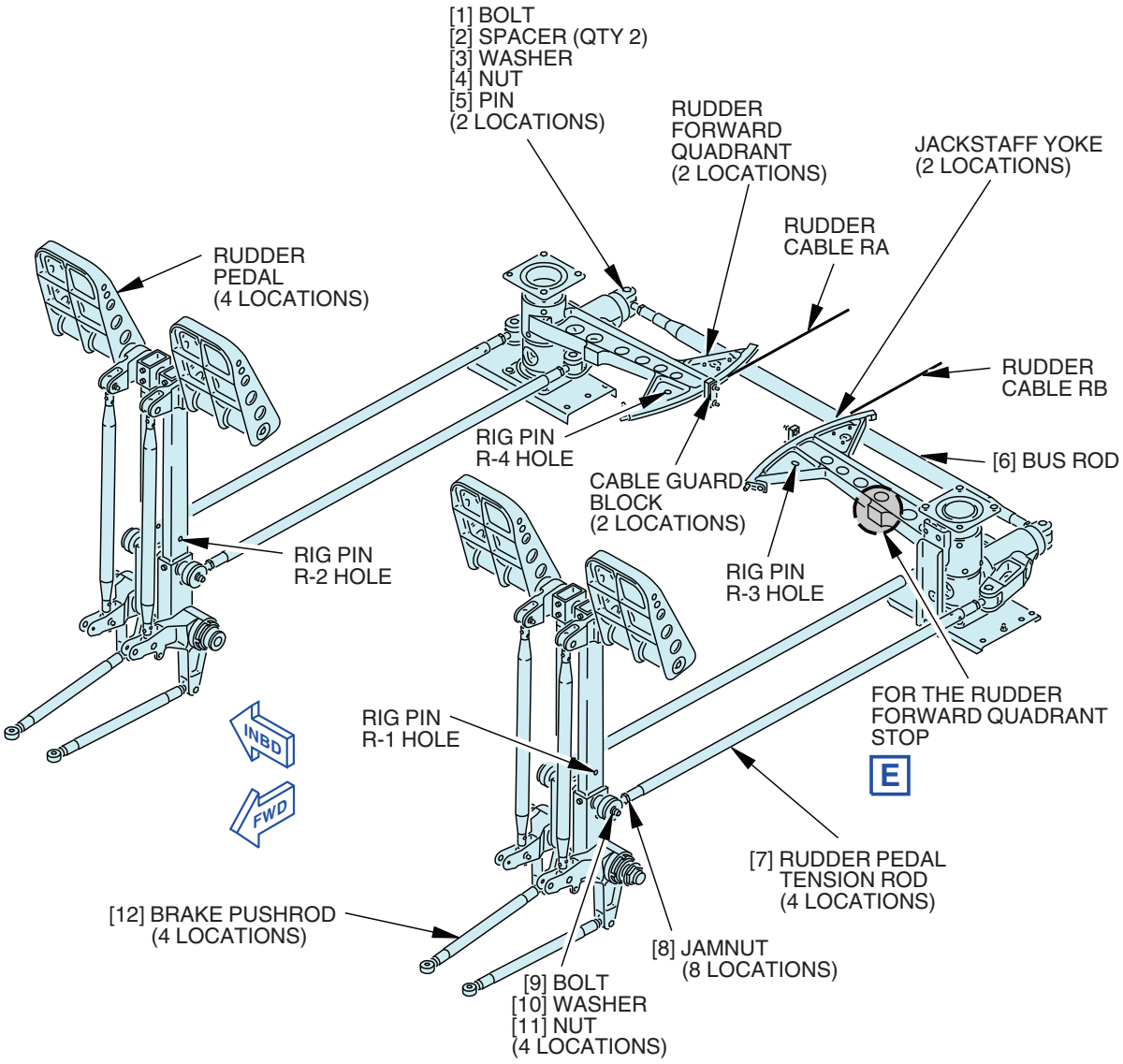
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

B

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Rudder Control System Forward Components
Figure 501/27-21-00-990-804 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



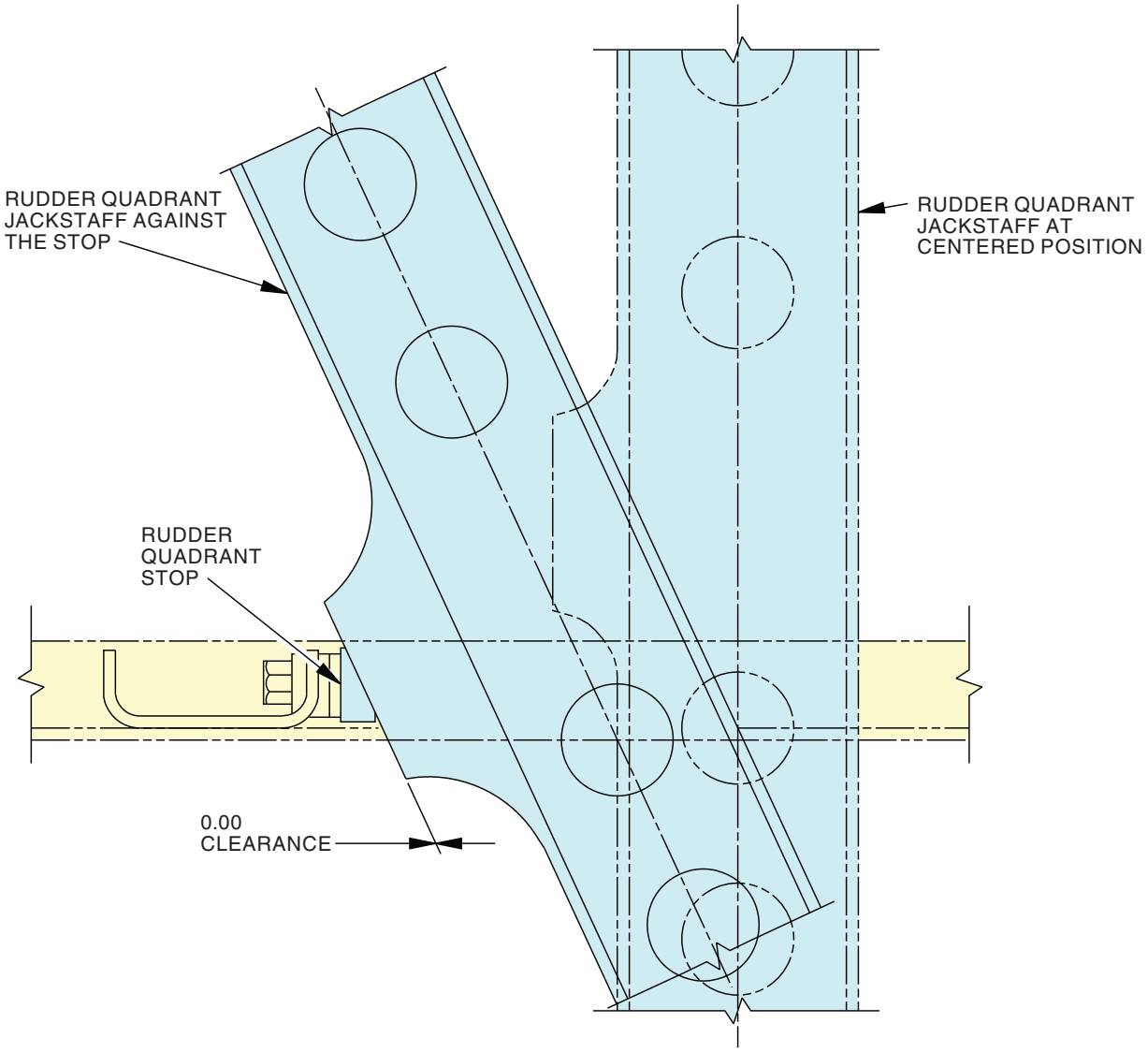
PEDAL SHAFTS AND FORWARD QUADRANTS

D

2412353 S00061530507_V2

Rudder Control System Forward Components
Figure 501/27-21-00-990-804 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RUDDER QUADRANT STOP

E

2412354 S00061530508_V1

Rudder Control System Forward Components
Figure 501/27-21-00-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-820-802

3. Rudder Pedals Adjustment

(Figure 501)

A. General

(1) Use this task to adjust the rudder pedals to the center position.

B. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel
324CL	Vertical Fin, Access

F. Prepare for the Test

SUBTASK 27-21-00-860-017

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-00-010-004

(2) Open these access doors:

Number	Name/Location
112A	Forward Access Door
324CL	Vertical Fin, Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-010-005

(3) Open these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

G. Rudder Pedals Adjustment

SUBTASK 27-21-00-710-004

(1) Do the rudder pedals adjustment:

(a) Install the rig pin R-5 from the rigging pin kit, SPL-1585, in the rudder feel and centering unit (Figure 502).

(b) Install the rig pin R-3 from the rigging pin kit, SPL-1585, in the captain's rudder forward quadrant (Figure 501).

(c) Install the rig pin R-4 from the rigging pin kit, SPL-1585, in the first officer's rudder forward quadrant (Figure 501).

(d) Do these steps to set the jackshaft yoke to the middle position:

NOTE: The middle position of the jackshaft yoke is where the number of turns of the shaft from the full forward and full aft adjustment are the same.

1) Turn the rudder pedal adjustment crank to move the rudder pedals full forward.

2) Turn the rudder pedal adjustment crank to move the rudder pedals full aft.

a) Record the number of turns.

3) Turn the rudder pedal adjustment crank half the recorded number of turns forward.

(e) Make sure that you can install these rig pins from the rigging pin kit, SPL-1585:

1) Rig pin R-1 on the captain's rudder pedals.

2) Rig pin R-2 on the first officer's rudder pedals.

(f) If you can not install the rig pins R-1 and R-2 freely, then do these steps:

1) Determine which pedal arm is out-of-line with the rigging pin hole.

a) If you have problem with installing rig pin R-1, then adjust the captain's side.

b) If you have problem with installing rig pin R-2, then adjust the first officer's side.

c) If you have problem with installing both rig pins R-1 and R-2, then adjust both sides.

2) Loosen both jamnut [8] on the rudder pedal tension rod [7].

3) Remove the bolt [9], washer [10] and nut [11] to disconnect the rudder pedal tension rod [7] at the forward end.

4) Adjust the length of rudder pedal tension rod [7] until you can install rig pin R-1 or R-2 easily.

a) Turn the rod end or rod body of the rudder pedal tension rod [7] as necessary to adjust the length.

b) Keep the lube fitting on the rudder pedal tension rod [7] down.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) Install the bolt [9], washer [10] and nut [11] to connect the rudder pedal tension rod [7].

6) Tighten the jamnut [8].

7) Make sure that the rod ends cover at least 50 percent of the inspection hole at the two ends of the rudder pedal tension rod [7].

(g) Remove the rig pins R-1, R-2, R-3, R-4, and R-5.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-410-002

(1) Close these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

SUBTASK 27-21-00-940-001

(2) Close these access doors:

Number	Name/Location
112A	Forward Access Door
324CL	Vertical Fin, Access

SUBTASK 27-21-00-860-018

(3) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

——— END OF TASK ———

TASK 27-21-00-820-803

4. Rudder Control Cables RA and RB Adjustment

(Figure 501 and Figure 502)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this task to adjust the rudder control cables RA and RB. This task includes steps to adjust the rudder forward quadrant and rudder pedals.

(2) Before you do cable adjustment, permit a minimum of one hour at a constant ambient temperature of ± 5 degrees F (± 2.8 degrees C) for the airframe temperature to become stable.

(a) It is recommended to do the cable adjustments in a controlled environment such as a hangar.

NOTE: Direct sunlight on the airframe will cause the cable readings to be inaccurate.

(b) All temperature readings will be taken as ambient outside but adjacent to the applicable area of the airframe.

(c) The cable tension values will not be correct when there are temperature differences along the cable run.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) It is recommended that airplanes supplied with the cables that contain higher zinc coating (Grade D or Grade K) be re-tensioned 3 to 4 months into service. This will bring the tension of the control cables within the desired range. After the cables are re-tensioned, the high zinc cables (Grade D or Grade K) will continue to relax at a rate that is equivalent to the Grade B cables. To identify the grade of the cable, refer to the table below:

Table 501/27-21-00-993-803 Control Cable Grades and Diameters

CONTROL CABLE DIAMETER (TERMINAL SIZE REF)	CONSTRUCTION FIRST DASH NUMBER *[1]									
	GRADE A					GRADE B		GRADE D		GRADE K
	ZINC			TIN OVER ZINC		ZINC	TIN OVER ZINC	ZINC	TIN OVER ZINC	ZINC
	7 X 7	7 X 7 (RT)	7 X 19	7 X 7	7 X 7 (RT)	7 X 19	7 X 19	7 X 19	7 X 19	7 X 19
.094 (3/32)	3B	3G*[2]	3A	3D	3E *[2]	- — —	- — —	- — —	- — —	- — —
.125 (1/8)	- — —	- — —	4A	- — —	- — —	4F	4C	4J*[3]	4H*[3]	4K*[3]
.188 (3/16)	- — —	- — —	- — —	- — —	- — —	- — —	- — —	6J*[4]	- — —	- — —

*[1] First dash number in the part number of the cable. Example: BACC2C-3D-01085-CE is Grade A, Tin Over Zinc, 7 x 7.

*[2] Code 3E and 3G are specified only where there is a specific requirement for REDUCED TOLERANCE (RT) percent elongation wire rope, 7 X 7 RT, per BMS7-265.

*[3] Only Grade D and Grade K .125 (1/8) cables (4J, 4H and 4K) are high zinc cables.

*[4] It is not necessary to adjust the Grade D 0.188 (3/16) cables (6J) 3 to 4 months after installation.

(e) It is recommended that cable assemblies on each side of a cable loop be replaced at the same time using the same cable grade of material in order to minimize asymmetric relaxation.

B. References

Reference	Title
27-09-14-860-802	Cable Control - Tension Measurement (P/B 201)
27-09-17-860-801	Location of the Flight Control Cables Air Pressure Seals (P/B 201)
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-00-840-802	Pressure to the Rudder Systems A, B, and Standby - Activation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
COM-1554	Tensiometer - Cable, High Tension (200 lbs and above) Part #: 102-03108 Supplier: 21844 Part #: 102-03120 Supplier: 21844 Part #: ACM-300 Supplier: 13331 Part #: ACM-600 Supplier: 13331 Part #: ACX-500-1 Supplier: 13331 Part #: T5-8008-106-00 Supplier: 0N8U4 Part #: T60-1001-C9-1A Supplier: 0N8U4
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
112A	Forward Access Door
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel
311BL	Stabilizer Trim Access Door
324AL	Vertical Fin, Aft Fin Access Door
324CL	Vertical Fin, Access

F. Prepare for the Adjustment

SUBTASK 27-21-00-860-019

(1) Make sure that the rudder trim indicator is at the zero unit of trim.

SUBTASK 27-21-00-730-001

(2) Do this task: Rudder Pedal Adjustment and Limit Travel Test, TASK 27-21-00-700-801.

SUBTASK 27-21-00-860-020

(3) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-00-860-022

(4) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-010-006

(5) Open these access doors:

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door
324AL	Vertical Fin, Aft Fin Access Door
324CL	Vertical Fin, Access

SUBTASK 27-21-00-480-003

(6) Use a low tension cable tensiometer, COM-1553, or high tension cable tensiometer, COM-1554, to measure the cable tension (TASK 27-09-14-860-802).

NOTE: If the tension measurement is in the limit and new control cables are not installed, the following adjustment procedure is not necessary.

SUBTASK 27-21-00-010-007

(7) If cable adjustment is required, open these access panels:

(Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

G. Rudder Control Cables RA and RB Adjustment

SUBTASK 27-21-00-820-003

(1) If you install new rudder control cables RA and RB, do these steps to stretch the cables (Figure 502):

(a) Operate the rudder trim indicator until you can install the rig pin R-5 in the rudder feel and centering unit easily.

(b) Adjust the turnbuckles in the cables RA and RB to get a tension of 300 lbf (1334 N) ± 20 lbf (89 N) each.

1) Make sure that you can install the rig pin R-3 from the rigging pin kit, SPL-1585, freely in the captain's rudder forward quadrant.

(c) Remove the rig pins R-3 and R-5.

(d) Adjust the pressure seals at the aft pressure bulkhead with the control cables RA and RB, do this task: Location of the Flight Control Cables Air Pressure Seals, TASK 27-09-17-860-801.

(e) Make sure that the cables RA and RB do not touch the pulley or quadrant flanges for the total cable travel.

(f) Make sure that the cables RA and RB are in less than 2 degrees of the plane of the pulley or the quadrant.

(g) Make sure that you install the cables through the grommets in the structure.

(h) Make sure that the cables do not move from the neutral rig position.

(i) Make sure that the pulleys turn freely with sufficient clearance with the guards.

(j) Remove the nut [50], washer [49], washer [48], and bolt assembly [47] to disconnect the standby rudder PCU control rod [34] from the rudder control torque tube [33].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(k) Remove the nut [45], washer [44], washer [43], and bolt assembly [42] to disconnect the main rudder PCU control rod [36] from the rudder control torque tube [33].

(l) Move the rudder pedals 25 times through full travel.

(m) Connect the main rudder PCU control rod [36] to the center crank [41].

1) Install the bolt assembly [42] with the bolt head end up and washer [43].

2) Torque the outer bolt of the bolt assembly [42] to 70 in-lb (7.9 N·m) - 85 in-lb (9.6 N·m).

3) Install the washer [44] and nut [45].

CAUTION DO NOT TIGHTEN THE OUTER BOLT OF THE BOLT ASSEMBLY [42] AFTER YOU TIGHTEN THE NUT [45]. DAMAGE TO EQUIPMENT CAN OCCUR.

4) Torque the nut [45] to 25 in-lb (2.8 N·m) - 30 in-lb (3.4 N·m).

(n) Install the bolt assembly [47] with the bolt head end up, washer [48] (under bolt), washer [49] (under nut), and nut [50] to connect the standby rudder PCU control rod [34] to the upper crank [46].

SUBTASK 27-21-00-820-004

(2) Adjust the control cables RA and RB to their correct cable rigging tension.

(a) Use a low tension cable tensiometer, COM-1553, or high tension cable tensiometer, COM-1554, to measure the cable tension.

(b) Install the rig pin R-5 from the rigging pin kit, SPL-1585, in the rudder feel and centering unit.

(c) Adjust the turnbuckles in the cables RA and RB to get the cable tensions (TASK 27-09-14-860-802).

NOTE: Adjust the tensions close to the maximum values in the table. The cables will stretch over a period of time.

(d) Make sure that you can install these rig pins from the rigging pin kit, SPL-1585 (Figure 501):

1) Rig pin R-3 in the captain's rudder forward quadrant.

2) Rig pin R-4 in the first officer's rudder forward quadrant.

(e) If you can not install the rig pins R-3 and R-4 freely, do these steps:

1) Turn the turnbuckles to release the tension of the control cables RA and RB.

2) Remove the pin [5], nut [4], washer [3], spacer [2], and bolt [1] to disconnect one end of the bus rod [6] from rudder forward quadrant.

a) Discard the pin [5].

3) Install the rig pins R-3 in the captain's rudder forward quadrant.

4) Install the rig pin R-4 in the first officer's rudder forward quadrant.

5) Adjust the bus rod [6] until you can install the bolt [1] easily.

6) Install the bolt [1], spacer [2], washer [3], nut [4], and new pin [5] to connect the bus rod [6] to the rudder forward quadrant.

7) Adjust the turnbuckles on the control cables RA and RB to get the correct tensions (TASK 27-09-14-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Do these steps to adjust the captain's and first officer's pedals to the neutral adjustment position:

NOTE: The middle position of the rudder pedals is where the number of turns of the shaft from the full forward and full aft adjustment are the same.

1) Turn the rudder pedal adjustment crank to move the rudder pedals full forward.
2) Turn the rudder pedal adjustment crank to move the rudder pedals full aft.
   a) Record the number of turns.
3) Turn the rudder pedal adjustment crank half the number of turns forward.

(g) Install these rig pins from the rigging pin kit, SPL-1585:

1) Rig pin R-1 on the captain's rudder pedals.
2) Rig pin R-2 on the first officer's rudder pedals.

(h) If you can not install the rig pin R-1 or R-2 freely, do these steps:

1) Loosen the jamnut [8] from the applicable rudder pedal tension rod [7].

NOTE: If you can not install rig pin R-1, adjust the captain's rudder pedal tension rods. If you can not install the rig pin R-2, adjust the first officer's rudder pedal tension rods. You are only required to adjust the applicable rudder pedal tension rod that is out of rig.

2) Remove the nut [11], washer [10], bolt [9] from the forward end of the rudder pedal tension rod [7].
3) Adjust the rudder pedal tension rod [7] until you can install the applicable rig pin freely.
   a) Make sure that you keep the lube fitting on the rudder pedal tension rod [7] in the down direction.
4) Install the bolt [9], washer [10], and nut [11].
5) Tighten the jamnut [8].

(i) Do a check on the rudder forward quadrant bus rod [6] and rudder pedal tension rod [7].

1) Make sure that the rod ends cover at least 50 percent of the inspection holes at two end of the rods (10 locations).

(j) Remove the rig pins R-1, R-2, R-3, R-4, and R-5.

(k) Operate the rudder pedals through one full travel cycle.

1) Make sure that the control cables RA and RB are at the correct tensions (TASK 27-09-14-860-802).

(l) Install the locking clips on the turnbuckles.

H. Put the Airplane Back To Its Usual Condition

SUBTASK 27-21-00-010-008

(1) Close these access panels:

(Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel
114BW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-010-009

(2) Close these access doors:

Number	Name/Location
112A	Forward Access Door
311BL	Stabilizer Trim Access Door
324AL	Vertical Fin, Aft Fin Access Door
324CL	Vertical Fin, Access

SUBTASK 27-21-00-860-023

(3) Do this task: Pressure to the Rudder Systems A, B, and Standby - Activation, TASK 27-21-00-840-802.

——— END OF TASK ———

TASK 27-21-00-820-804

5. Control Rod for Main Rudder Power Control Unit Adjustment

(Figure 501, Figure 502)

A. General

(1) Use this task to adjust the control rod for the main rudder Power Control Unit (PCU).

B. References

Reference	Title
27-11-94	AILERON/RUDDER TRIM CONTROL MODULE
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
324CL	Vertical Fin, Access
324DL	Trailing Edge Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Adjustment

SUBTASK 27-21-00-840-001

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
C	7	C00285	YAW DAMPER AC
D	6	C01354	YAW DAMPER 2 DC
D	7	C00286	YAW DAMPER 1 DC

SUBTASK 27-21-00-010-010

(2) To get access to the main PCU, open this access panel:

Number	Name/Location
324DL	Trailing Edge Access

SUBTASK 27-21-00-010-011

(3) To get access to the rudder feel and centering unit, open this access panel:

Number	Name/Location
324CL	Vertical Fin, Access

SUBTASK 27-21-00-820-005

(4) Manually hold the rudder in the neutral position over the index plate.

SUBTASK 27-21-00-840-002

(5) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

SUBTASK 27-21-00-860-025

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) Pressurize the rudder hydraulic systems A and B, do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

(a) Move the rudder pedals to bleed the main rudder PCU until the movement of the rudder is smooth.

SUBTASK 27-21-00-980-001

(7) Make sure that the rudder pedals are in the neutral position.

(a) If it is necessary, tap on the rudder pedals lightly to put the rudder pedals to the neutral position.

SUBTASK 27-21-00-480-004

(8) Install rig pin R-5 from the rigging pin kit, SPL-1585, in the rudder feel and centering unit.

(a) Make sure that you can install the rig pin R-5 freely.

1) If you cannot install the rig pin freely, adjust the rudder trim actuator:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Jiggle and release the feel and centering unit input arm to make sure that the system is centered.

b) Move the rudder trim knob until rig pin R-5 can be installed freely.

(9) Make sure that the rudder trim indicator is at 0.00 +/- 0.25 unit of trim.

(a) If indicator is not at 0.00 +/- 0.25 unit of trim, adjust the trim indicator per AILERON/RUDDER TRIM CONTROL MODULE, SUBJECT 27-11-94.

SUBTASK 27-21-00-860-026

(10) Set the FLT CONTROL A and B switches to the OFF position.

SUBTASK 27-21-00-480-005

(11) Install the rig pins R-3, and R-4 (Figure 501).

(a) If only one rig pin can be inserted freely, adjust the bus rod, do this task: Rudder Forward Quadrant Adjustment, TASK 27-21-00-820-801.

(b) If both rig pins cannot be installed freely, adjust the control cables RA and RB, do this task: Rudder Control Cables RA and RB Adjustment, TASK 27-21-00-820-803.

(12) Install the rig pins R-1 and R-2.

(a) If you cannot install the rig pins freely, adjust the pedal tension rods, do this task: Rudder Pedals Adjustment, TASK 27-21-00-820-802.

SUBTASK 27-21-00-080-003

(13) Remove the rig pins R-1, R-2, R-3, R-4, and R-5.

G. Control Rods for the Main PCU Adjustment

SUBTASK 27-21-00-860-027

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Set the FLT CONTROL B switch to the ON position.

SUBTASK 27-21-00-210-002

(2) Make sure that the YAW DAMPER switch, on the P5-3 forward overhead panel, is in the OFF position and the YAW DAMPER light is illuminated.

SUBTASK 27-21-00-860-028

(3) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

NOTE: Do not change the position of the rudder trim knob until the main PCU control rods and the standby PCU control rod have been adjusted.

SUBTASK 27-21-00-820-006

(4) Do these steps to put the rudder trailing edge to the left of the index plate:

NOTE: Always start the adjustment of the control rod with the rudder at this position.

(a) Loosen the clamp nut [35] on the lower main rudder PCU control rod [36] until you can rotate the sleeve [37].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Rotate the sleeve [37] until the rudder trailing edge is between 0.20 in. (5.1 mm) to 0.30 in. (7.6 mm) to the left of the index plate.

NOTE: One full rotation of the sleeve will move the rudder trailing edge approximately 0.18 in. (4.6 mm) linear travel. Rotate the sleeve clockwise (looking aft) will move the rudder trailing edge left. Rotate the sleeve counter-clockwise (looking aft) will move the rudder trailing edge right.

SUBTASK 27-21-00-860-029

(5) Set the FLT CONTROL B switch to the OFF position and FLT CONTROL A switch to the ON position.

SUBTASK 27-21-00-820-007

(6) Do these steps to put the rudder trailing edge to the left of the index plate:

NOTE: Always start the adjustment of the control rod with the rudder at this position.

(a) Loosen the clamp nut [35] on the upper main rudder PCU control rod [36] until you can rotate the sleeve [37].

(b) Rotate the sleeve [37] until the rudder trailing edge is between 0.20 in. (5.1 mm) - 0.30 in. (7.6 mm) to the left of the index plate.

NOTE: One full rotation of the sleeve will move the rudder trailing edge approximately 0.18 in. (4.6 mm) linear travel. Rotate the sleeve clockwise (looking aft) will move the rudder trailing edge left. Rotate the sleeve counter-clockwise (looking aft) will move the rudder trailing edge right.

SUBTASK 27-21-00-820-008

(7) Do this adjustment to the upper main rudder PCU control rod [36]:

(a) Loosen the clamp nut [35] on the upper main rudder PCU control rod [36] until you can rotate the sleeve [37].

(b) Rotate the sleeve [37] until the rudder trailing edge is within the center of the index plate.

1) Make sure that the rudder trailing edge does not go beyond the center of the index plate on to the right side.

a) If this happens, start the adjustment again with the rudder trailing edge positioned to the left of the index plate.

(c) Tighten the clamp nut [35] on the upper main rudder PCU control rod [36].

(d) Do not torque the clamp nut [35] at this time.

(e) Make sure that the rudder trailing edge is still within the center of the index plate.

SUBTASK 27-21-00-860-032

(8) Set the FLT CONTROL A switch to the OFF position and set the FLT CONTROL B switch to the ON position.

SUBTASK 27-21-00-820-009

(9) Do this adjustment to the lower main rudder PCU control rod [36]:

(a) Loosen the clamp nut [35] on the lower main rudder PCU control rod [36] until you can rotate the sleeve [37].

(b) Rotate the sleeve [37] until the rudder trailing edge is within the center of the index plate.

1) Make sure that the rudder trailing edge does not go beyond the center of the index plate on to the right side.

a) If this happens, start the adjustment again with the rudder trailing edge positioned to the left of the index plate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Tighten the clamp nut [35] on the lower main rudder PCU control rod [36].

(d) Do not torque the clamp nut [35] at this time.

(e) Make sure that the rudder trailing edge is still within the center of the index plate.

SUBTASK 27-21-00-820-010

(10) Set the FLT CONTROL A switch to the ON position.

(a) Make sure that the rudder trailing edge is still within the center of the index plate.

SUBTASK 27-21-00-820-011

(11) Set the FLT CONTROL B switch to the OFF position.

(a) Make sure that the rudder trailing edge is still within the center of the index plate.

SUBTASK 27-21-00-820-012

(12) Set the FLT CONTROL A switch to the OFF position and FLT CONTROL B switch to the ON position.

(a) Make sure that the rudder trailing edge is still within the center of the index plate.

SUBTASK 27-21-00-860-033

(13) Set the FLT CONTROL A switch to the ON position.

SUBTASK 27-21-00-420-001

(14) Tighten the clamp nut [35] on the upper and lower main rudder PCU control rod [36].

SUBTASK 27-21-00-860-034

(15) Tap on the rudder pedals lightly to put the rudder pedals to the neutral position.

SUBTASK 27-21-00-860-035

(16) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

SUBTASK 27-21-00-700-001

(17) Verify voltage between pin 18 and pin 20 of connector, D10064, on the E1-1 shelf is between 4.38 VDC to 5.62 VDC.

(a) If the voltage measured is not between 4.38 VDC to 5.62 VDC adjust the main rudder PCU control rod [36].

SUBTASK 27-21-00-820-013

(18) Make sure that you can install and remove rig pin R-5 freely.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-820-014

(1) If it is necessary check the standby rudder PCU, do this task: Control Rod for Standby Rudder Power Control Unit Adjustment, TASK 27-21-00-820-805.

SUBTASK 27-21-00-860-036

(2) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-114

(3) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

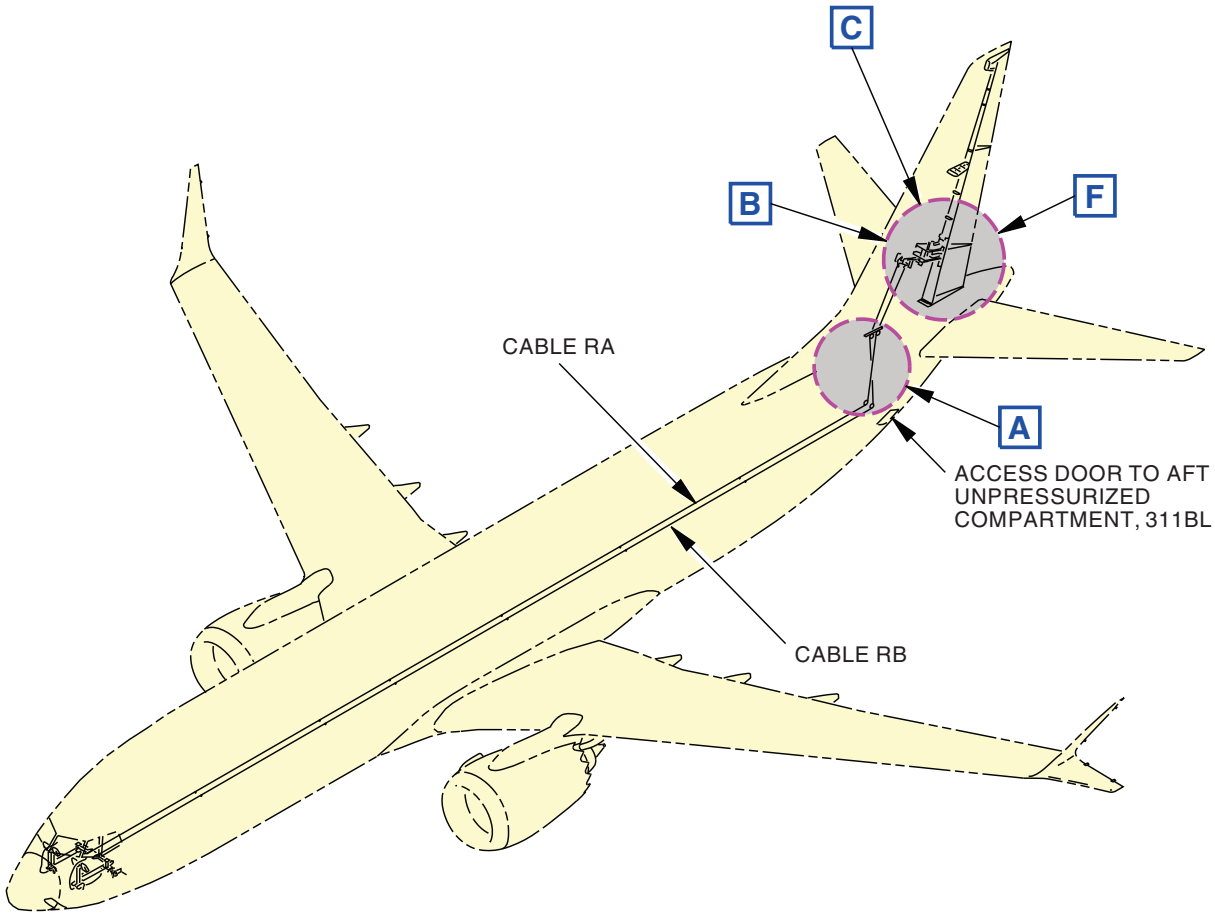
SUBTASK 27-21-00-410-003

(4) Close these access panels:

Number	Name/Location
324CL	Vertical Fin, Access
324DL	Trailing Edge Access

——— END OF TASK ———

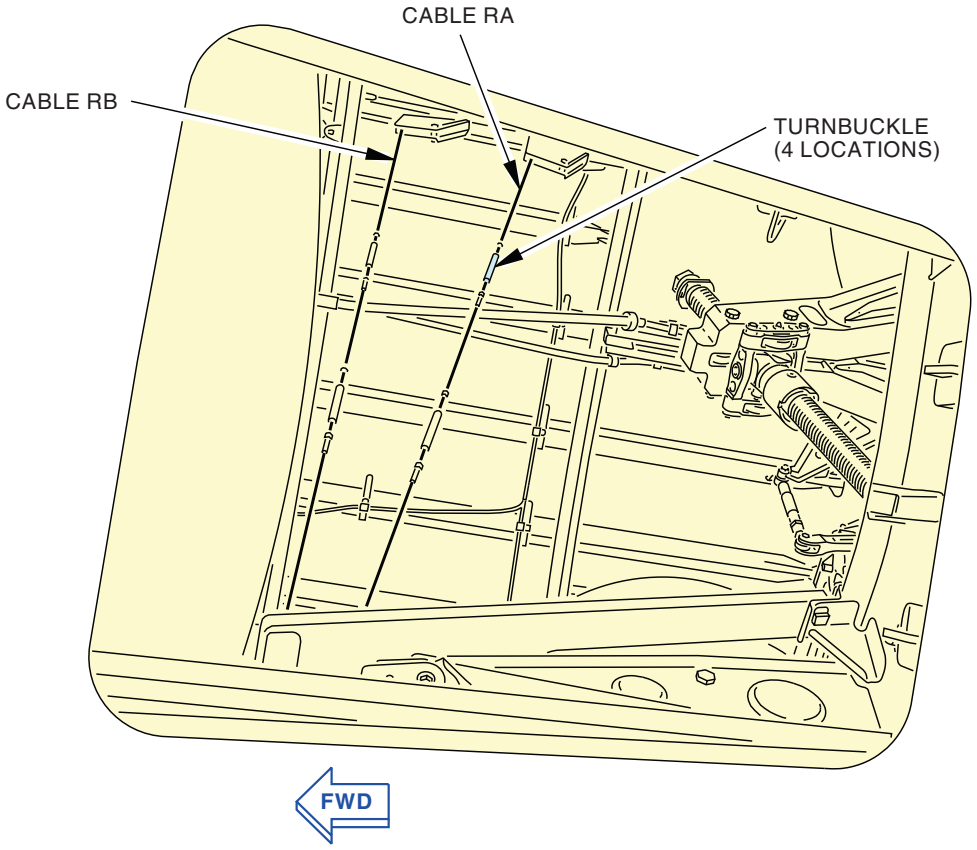
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412355 S00061530512_V1

Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



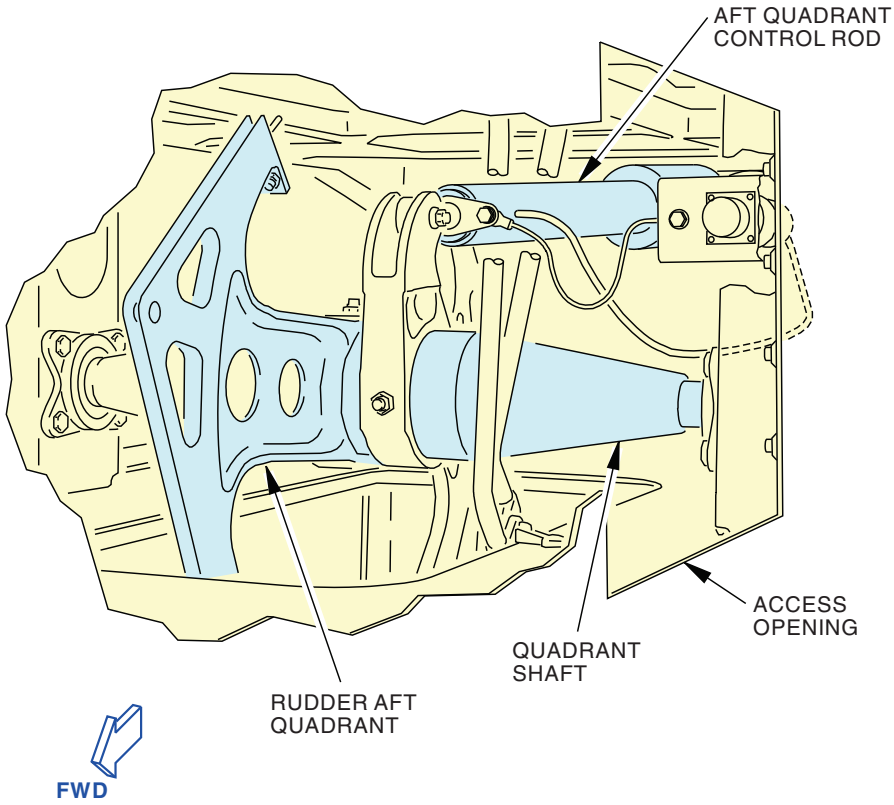
**VIEW THROUGH THE ACCESS DOOR TO
AFT UNPRESSURIZED COMPARTMENT, 311BL**

A

2412356 S00061530513_V1

**Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 2 of 8)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



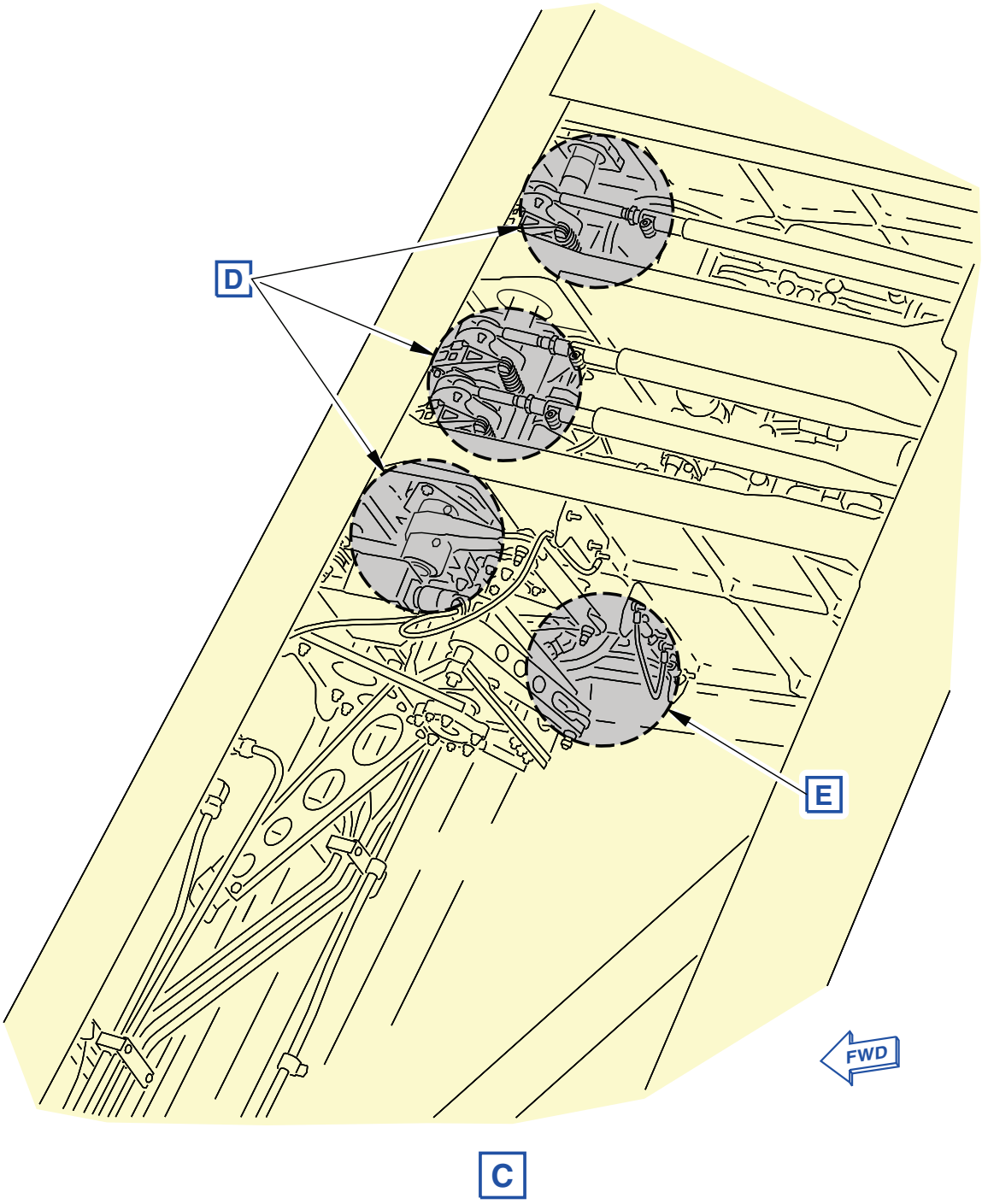
RUDDER AFT QUADRANT

B

2412357 S00061530514_V1

Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 3 of 8)

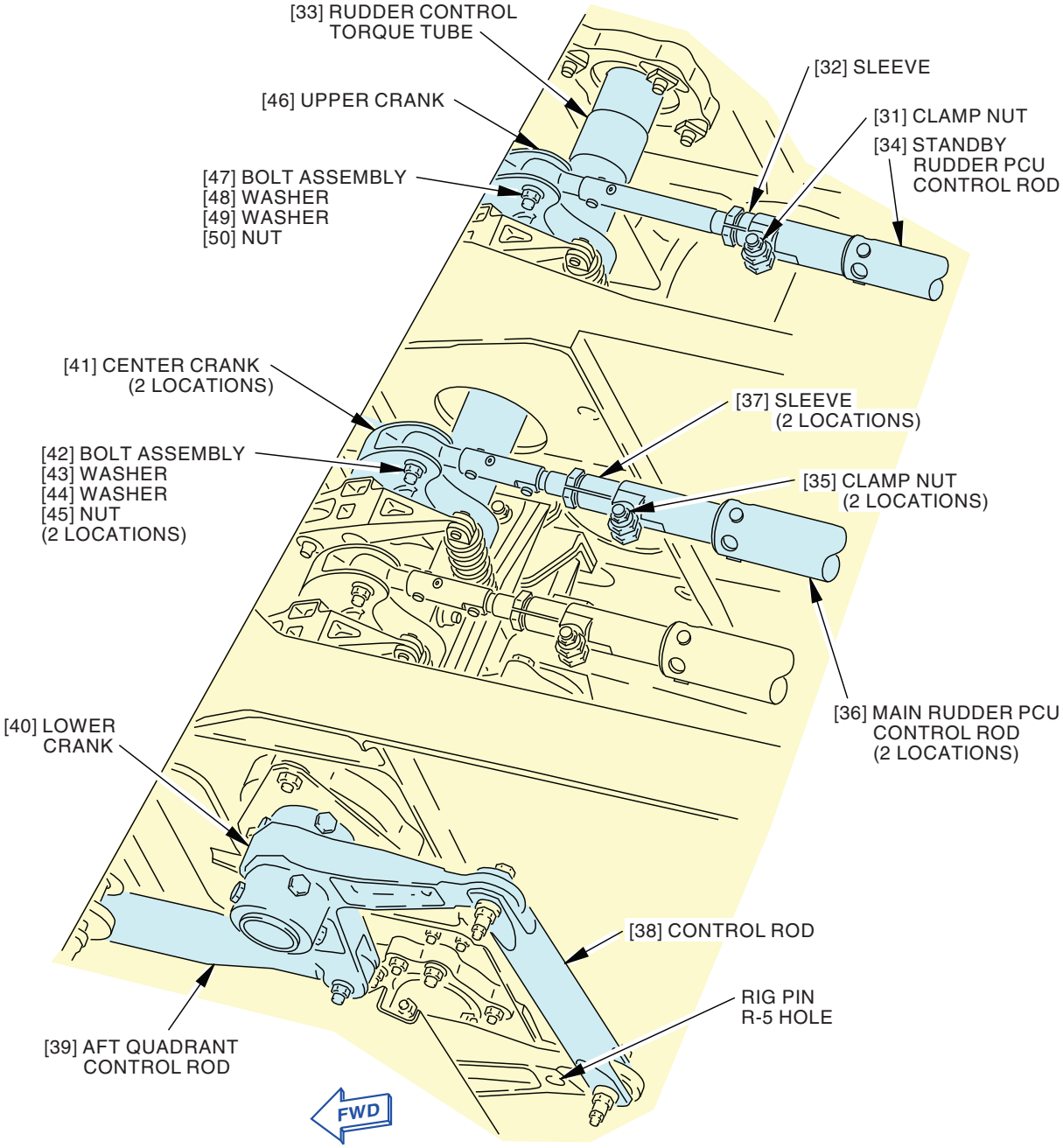
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412358 S00061530515_V1

Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



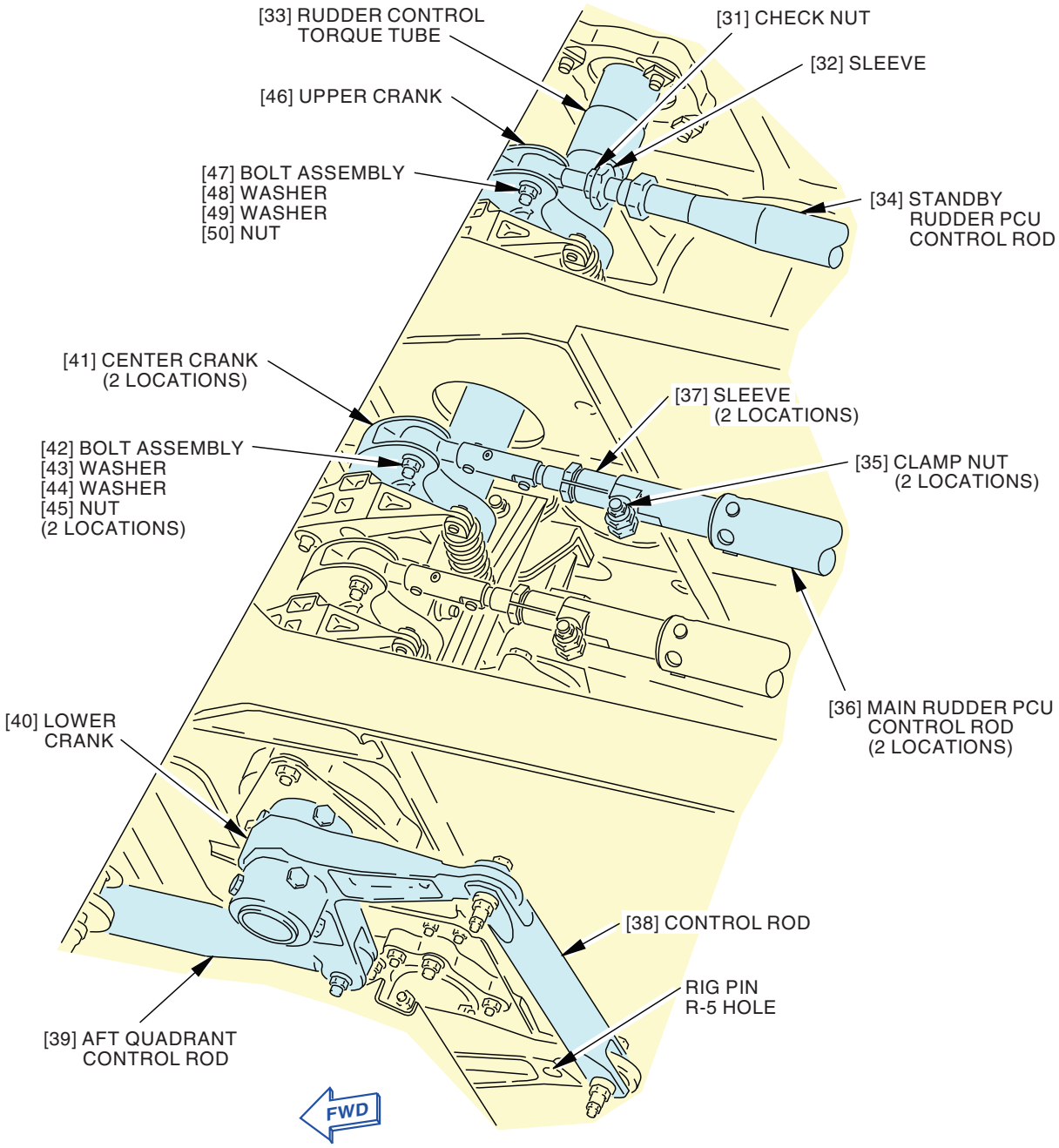
RUDDER CONTROL TORQUE TUBE

D

2412359 S00061530516_V1

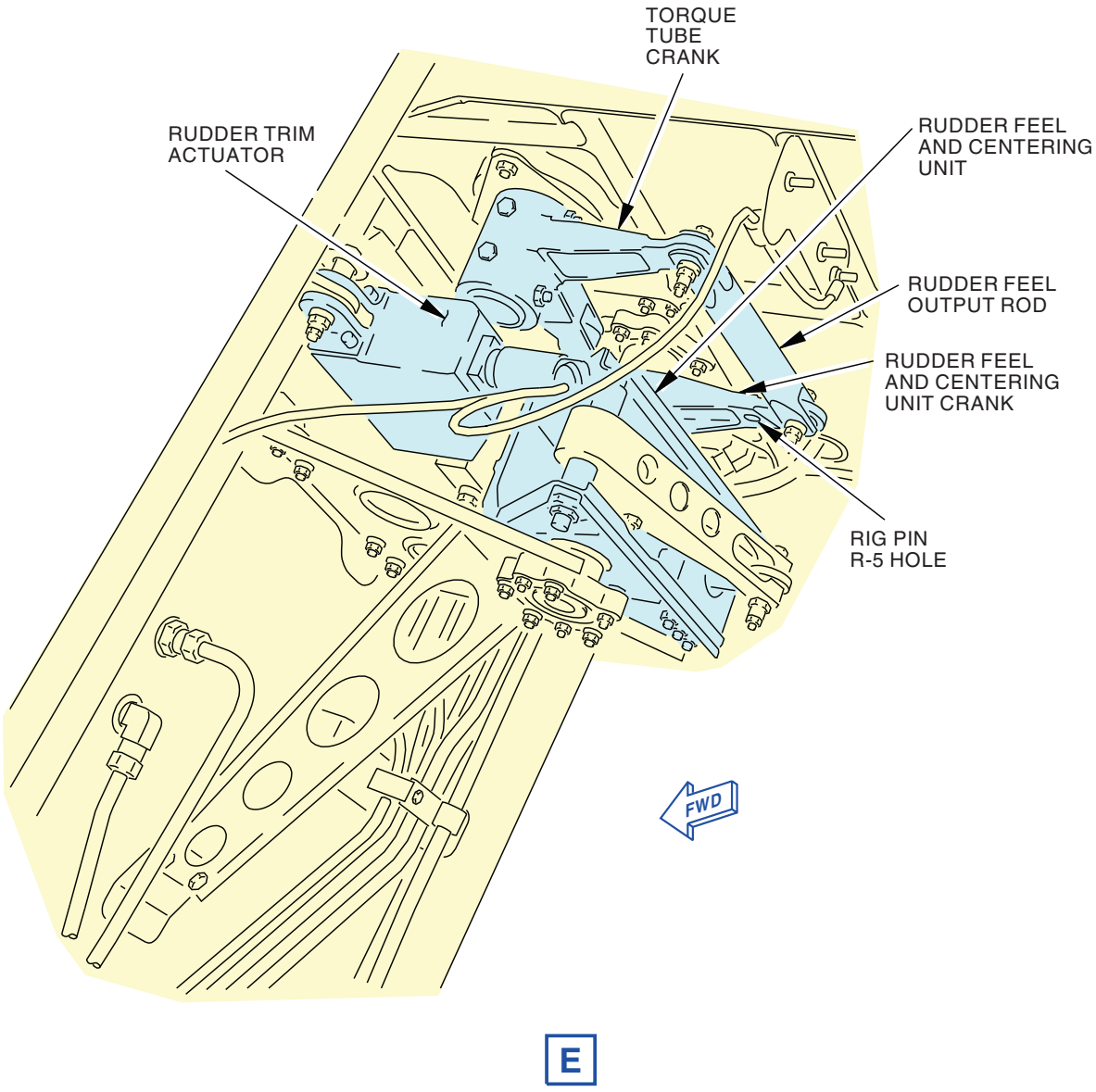
Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 6 of 8)

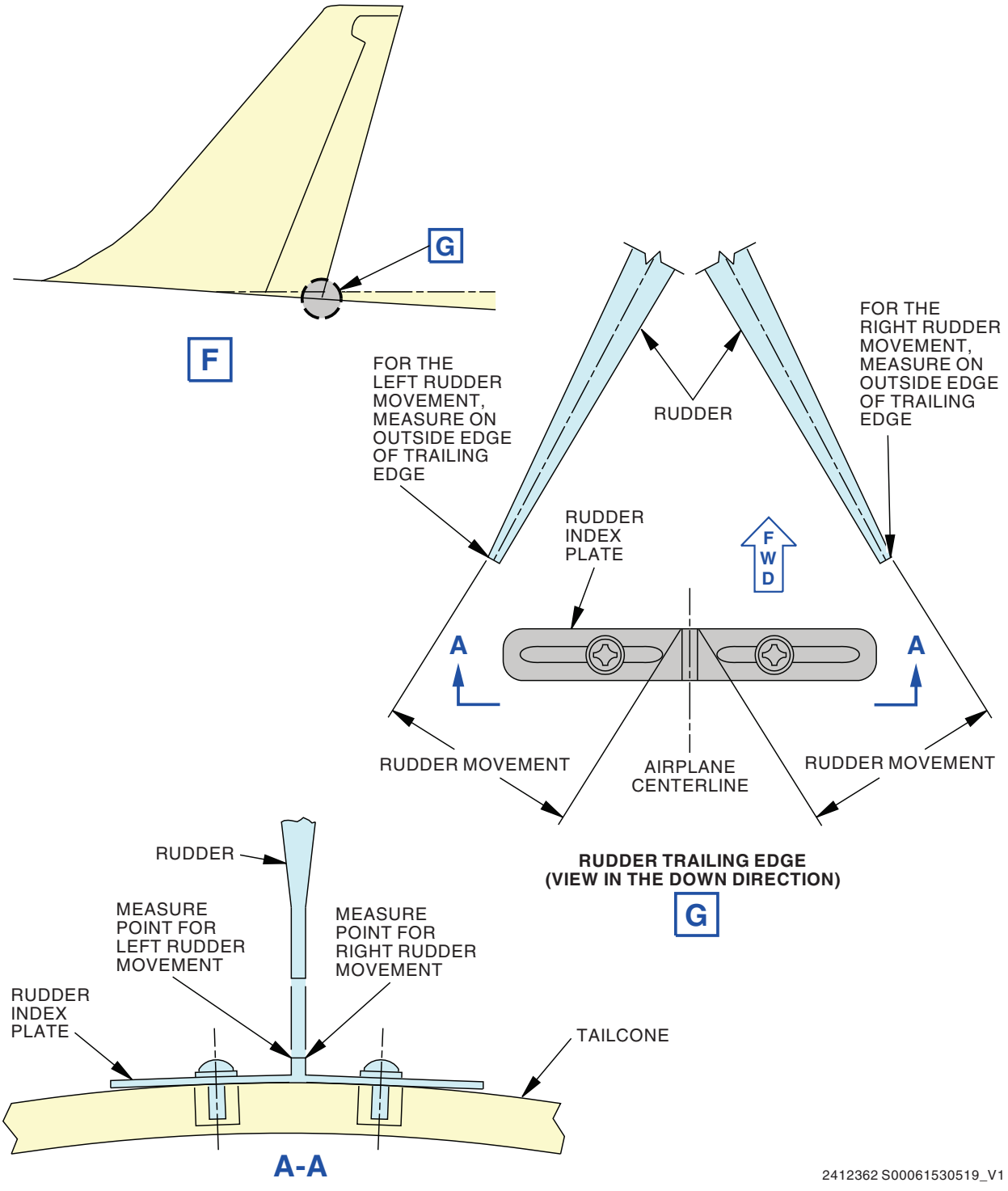
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412361 S00061530518_V1

Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Control System Aft Components
Figure 502/27-21-00-990-805 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-820-805

6. Control Rod for Standby Rudder Power Control Unit Adjustment

(Figure 502)

A. General

(1) Use this task to adjust the control rod for the standby rudder Power Control Unit (PCU). The rudder main PCU must be adjusted before you adjust the standby rudder PCU.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
324CL	Vertical Fin, Access
324DL	Trailing Edge Access

E. Prepare for the Adjustment

SUBTASK 27-21-00-010-012

(1) To get access to the standby rudder PCU, open this access panel:

Number	Name/Location
324DL	Trailing Edge Access

SUBTASK 27-21-00-010-013

(2) To get access to the rudder feel and centering unit, open this access panel:

Number	Name/Location
324CL	Vertical Fin, Access

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-039

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Pressurize the standby rudder hydraulic system, do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-00-860-040

(4) Move the rudder pedals through ten cycles to flush and bleed the standby rudder PCU.

SUBTASK 27-21-00-860-041

(5) Set the FLT CONTROL A and B switches to the ON position.

SUBTASK 27-21-00-860-042

(6) Make sure that the rudder trim indicator is in the neutral position (Figure 501).

SUBTASK 27-21-00-480-006

(7) Make sure that you can install and remove the rig pin R-5 freely with no movement of the rudder feel and centering unit output crank (Figure 502).

SUBTASK 27-21-00-860-043

(8) Align the ADIRS, do this task: Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803.

F. Control Rod of the Standby Rudder PCU Adjustment

SUBTASK 27-21-00-710-005

(1) Adjust the standby rudder PCU control rod [34] of the standby rudder PCU:

(a) Set FLT CONTROL A and B switches to the STBY RUD position.

(b) Set the YAW DAMPER switch, on the P5-3 forward overhead panel, to the ON position.

1) Make sure that the YAW DAMPER light goes out after approximately 2 seconds.

(c) Move the control wheel full right clockwise.

1) Make sure that the rudder moves to the right.

(d) Release the control wheel.

(e) Do these steps to initiate a zero yaw damper command output from the stall Management Yaw Damper No. 2 (SMYD No. 2), M1748:

NOTE: The SMYD No. 2 is installed on the E3-2 electrical equipment rack.

1) Push and release the ON/OFF button on the front BITE display of the SMYD No. 2.

2) Push and release the up arrow or down arrow button until the GROUND TEST shows on the display.

3) Push the YES button to get access to the ground test option.

4) Push and release the up arrow or down arrow button until the SERVO TEST? shows on the display.

5) Push the YES button to start the servo test.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE RUDDER. THE RUDDER CAN MOVE QUICKLY WHEN YOU PUSH THE "YES" BUTTON ON THE SMYD NO. 2. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

6) Push the YES button when the RUDDER CLEARED? shows on the display.
7) Push the YES button when the FLT CTRL A STBY? shows on the display.
8) Push the YES button when the FLT CTRL B STBY? shows on the display.
9) Push the YES button when YAW DMPR ENGAGED? shows on the display.
10) Push the YES button when the ZERO COMMAND? shows on the display.

NOTE: A yaw damper command value and LVDT position will show for 10 seconds each, then the END ZERO COMMAND? will show on the display. You can push the NO button to show these values again.

SUBTASK 27-21-00-820-015

(2) Do this adjustment to the standby rudder PCU control rod [34] (Figure 502):

(a) Loosen the clamp nut [31] on the standby rudder PCU control rod [34] until you can rotate the sleeve [32].

(b) Rotate the sleeve [32] until the rudder trailing edge is within the center of the index plate.

NOTE: One full rotation of the sleeve will move the rudder trailing edge approximately 0.18 in. (4.6 mm) linear travel. Rotate the sleeve clockwise (looking aft) will move the rudder trailing edge left. Rotate the sleeve counter-clockwise (looking aft) will move the rudder trailing edge right.

(c) Tighten the clamp nut [31] on the standby rudder PCU control rod [34].

1) Do not torque the clamp nut [31] at this time.

(d) Make sure that the rudder trailing edge is still within the center of the index plate.

(e) Tighten the clamp nut [31] on the standby rudder PCU control rod [34].

SUBTASK 27-21-00-820-016

(3) Do this adjustment to the standby rudder PCU control rod [34] (Figure 502):

(a) If the rudder trailing edge is greater than 0.50 in. (12.7 mm) from the neutral position, do this coarse adjustment to the standby rudder PCU control rod [34]:

WARNING MAKE SURE THAT YOU SET THESE SWITCHES TO THE OFF POSITION BEFORE YOU DISCONNECT THE CONTROL ROD FROM THE STANDBY RUDDER PCU. THE CONTROL ROD CAN MOVE QUICKLY AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

1) Set the FLT CONTROL A and B switches to the OFF position.
2) Loosen the two checknut [31] on the standby rudder PCU control rod [34].
3) Remove the nut [50], washer [49], washer [48], and bolt assembly [47] to disconnect the standby rudder PCU control rod [34] from the upper crank [46] of the rudder control torque tube [33].
4) Turn the control rod end for half a turn relative to the sleeve [32].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Turn the sleeve [32] and control rod end together for half a turn.

NOTE: The sum of half turn of each is equal to 0.67 in. (17.0 mm) at the rudder trailing edge.

5) Keep number of threads that you can see in the two parts are approximately same.

6) Install the bolt assembly [47] with the bolt head end up, washer [48] (under bolt), washer [49] (under nut), and nut [50] to connect the standby rudder PCU control rod [34] to the upper crank [46].

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE RUDDER. THE RUDDER CAN MOVE QUICKLY WHEN YOU PUSH THE `YES` BUTTON ON THE SMYD NO. 2. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

7) Set the FLT CONTROL A and B switches to the STANDBY RUD position.

(b) If the rudder trailing edge is less than 0.50 in. (12.7 mm) from the neutral position, do this fine adjustment to the standby rudder PCU control rod [34]:

1) Loosen the checknut [31] on the standby rudder PCU control rod [34].

2) Turn the adjustable sleeve [32] until the two sides of the rudder trailing edge are in the width of the rudder index tab.

3) Tighten the two checknut [31].

(c) Do a check for minimum thread engagement of the rod.

1) Make sure that you can see the rod from the inspection hole in the tube.

SUBTASK 27-21-00-820-017

(4) Make sure that you can install and remove the rig pin R-5 freely.

SUBTASK 27-21-00-860-111

(5) Push and release the ON/OFF button to turn off the SMYD BITE display.

SUBTASK 27-21-00-860-112

(6) Set the YAW DAMPER switch, on the P5-3 overhead panel, to the OFF position.

SUBTASK 27-21-00-860-044

(7) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

SUBTASK 27-21-00-820-018

(8) Set the FLT CONTROL A and B switches to the ON position.

(a) Make sure that the STBY RUD ON light is not illuminated after 5 seconds.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-480-007

(1) Remove the rig pin R-5.

SUBTASK 27-21-00-860-045

(2) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-410-004

(3) Close these access panels:

Number	Name/Location
324CL	Vertical Fin, Access
324DL	Trailing Edge Access

——— END OF TASK ———

TASK 27-21-00-700-801

7. Rudder Pedal Adjustment and Limit Travel Test

(Figure 503)

A. General

(1) Use this test to make sure that the rudder pedals can move through their full travel and pedal adjustment crank operates correctly.

(2) It is recommended to measure pedal travel by projecting a line from the pivot point of the pedal normal to the heel rest wear plate at each pedal position and marking the pedal position on a piece of masking tape that applied to the heel rest. Measuring the distance between marks is the measured pedal travel.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Rudder Pedal Adjustment and Limit Travel Test

SUBTASK 27-21-00-420-007

(1) Install the NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism (TASK 32-00-01-480-801).

NOTE: The Nose Landing Gear (NLG) will move when the rudder pedals are moved. To prevent NLG movement, install a lockout pin in the nose wheel steering mechanism. The nose gear steering does not need to be disconnected if the airplane is jacked up such that the nose gear is fully extended.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-046

(2) Do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-00-730-002

(3) Do these steps to set the jackshaft yokes to the middle positions:

NOTE: The middle positions of the jackshaft yokes is where the number of turns of the shaft from the full forward and full aft adjustment are the same.

(a) Turn the rudder pedal adjustment cranks for both the captain and first officer to move the rudder pedals full forward.

(b) Turn the rudder pedal adjustment cranks to move the rudder pedals full aft.

1) Keep a record of number of turns.

(c) Turn the rudder pedal adjustment cranks back half of the number of recorded turns to set the jackshaft yoke to the middle position.

(d) Make sure that you can install rig pins R-1 and R-2 easily.

(e) Remove the rig pins R-1 and R-2.

SUBTASK 27-21-00-730-003

(4) Do the captain's rudder pedal adjustment and the limit travel.

(a) Make sure that the rudder trim indicator is at the zero unit of trim.

NOTE: Rudder in neutral position.

(b) Turn the rudder pedal adjustment crank to move the captain's pedals full forward.

(c) Push the captain's left pedal forward until it touches the stop and hold.

1) Measure the forward travel of the captain's left pedal from the middle position of the pedal adjustment.

2) Make sure that the forward travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).

3) Release the captain's left pedal.

(d) Push the captain's right pedal forward until it touches the stop and hold.

1) Measure the forward travel of the captain's right pedal from the middle position of the pedal adjustment.

2) Make sure that the forward travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).

3) Release the captain's right pedal.

(e) Push forward on both the captain's left and right pedals at the same time.

1) Apply a light load of approximately 2.5 lb (1.1 kg) on each pedal equally and simultaneously.

(f) Turn the rudder pedal adjustment crank to move the captain's pedals to full aft position.

1) Turn the crank at a rate of one revolution (turn) per second.

(g) Make sure that the adjustment crank moves smooth and without any chatter.

(h) Push the captain's left pedal forward until it touches the stop and hold.

1) Measure the aft travel of the captain's right pedal from the middle position of the pedal adjustment.

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2) Make sure that the aft travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).

3) Release the captain's left pedal.

(i) Push the captain's right pedal forward until it touches the stop and hold.

1) Measure the aft travel of the captain's left pedal from the middle position of the pedal adjustment.

2) Make sure that the aft travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).

3) Release the captain's right pedal.

(j) Turn the rudder pedal adjustment crank to move the captain's pedals to full forward position.

NOTE: No load is applied to the rudder pedals during this step.

1) Turn the crank at a rate of one revolution (turn) per second.

(k) Make sure that the adjustment crank only chattered slightly and there is no binding when the crank is turned.

NOTE: The tendency to chatter will increase as the rate of adjustment increases.

SUBTASK 27-21-00-730-004

(5) Do the first officer's rudder pedal adjustment and the limit travel.

(a) Make sure that the rudder trim indicator is at the zero unit of trim.

NOTE: Rudder in neutral position.

(b) Turn the rudder pedal adjustment crank to move the first officer's pedals full forward.

(c) Push the first officer's left pedal forward until it touches the stop and hold.

1) Measure the forward travel of the first officer's left pedal from the middle position of the pedal adjustment.

2) Make sure that the forward travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).

3) Release the first officer's left pedal.

(d) Push the first officer's right pedal forward until it touches the stop and hold.

1) Measure the forward travel of the first officer's right pedal from the middle position of the pedal adjustment.

2) Make sure that the forward travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).

3) Release the first officer's right pedal.

(e) Push on both the first officer's left and right pedals at the same time.

1) Apply a light load of approximately 2.5 lb (1.1 kg) each pedal equally and simultaneously.

(f) Turn the rudder pedal adjustment crank to move the first officer's pedals to full aft position.

1) Turn the crank at a rate of one revolution (turn) per second.

(g) Make sure that the adjustment crank moves smooth and without any chatter.

(h) Push the first officer's left pedal forward until it touches the stop and hold.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Measure the aft travel of the first officer's right pedal from the middle position of the pedal adjustment.
2) Make sure that the aft travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).
3) Release the first officer's left pedal.

(i) Push the first officer's right pedal forward until it touches the stop and hold.

1) Measure the aft travel of the first officer's left pedal from the middle position of the pedal adjustment.
2) Make sure that the aft travel, which measured at the pedal pivot point is at a minimum of 6.04 in. (153.4 mm).
3) Release the first officer's right pedal.

(j) Turn the rudder pedal adjustment crank to move the first officer's pedals to full forward position.

NOTE: No load is applied to the rudder pedals during this step.

1) Turn the crank at a rate of one revolution (turn) per second.

(k) Make sure that the adjustment crank only chatters slightly and there is no binding when the crank is turned.

NOTE: The tendency to chatter will increase as the rate of adjustment increases.

SUBTASK 27-21-00-860-047

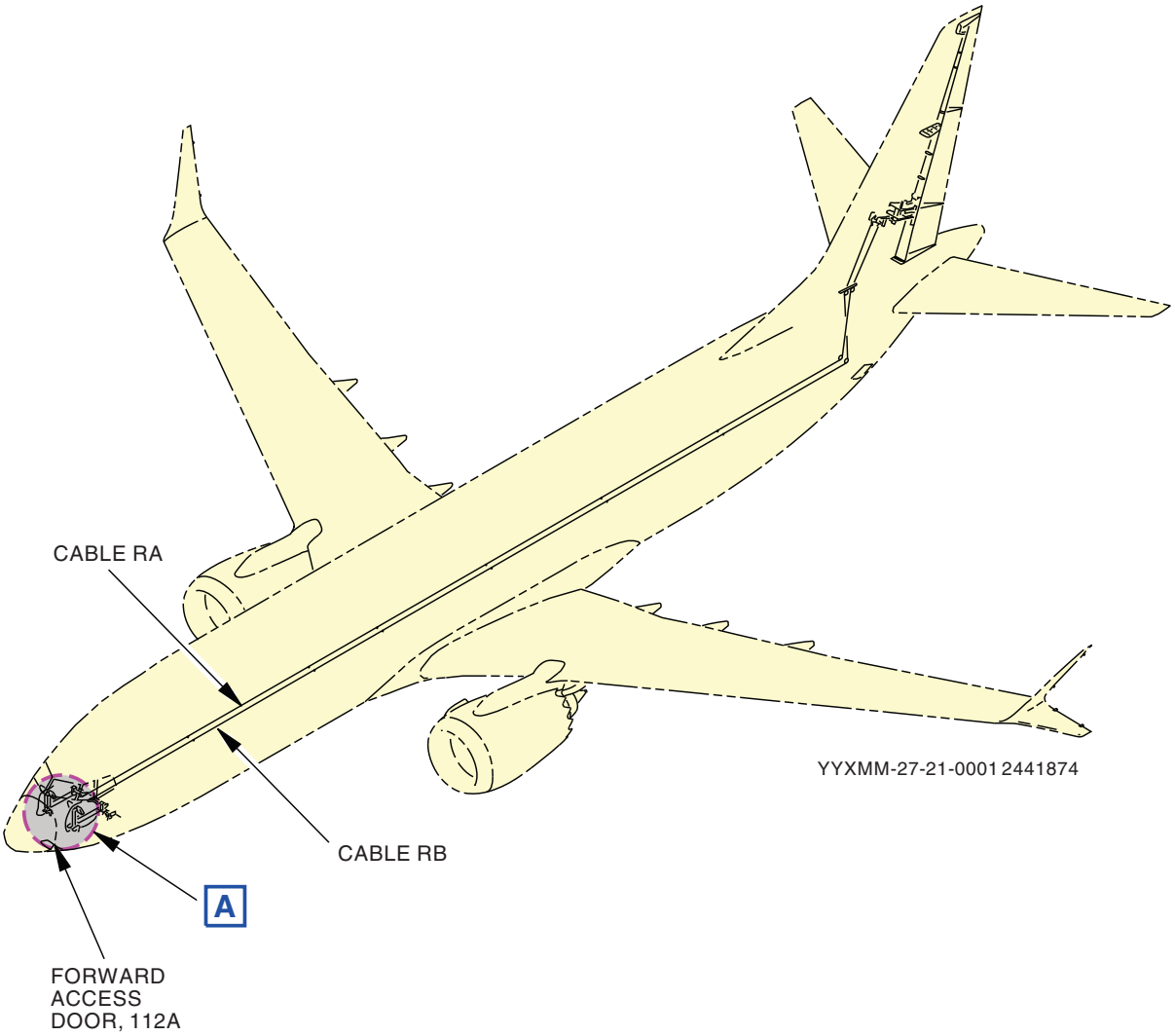
(6) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-020-001

(7) Remove the NLG towing lever pin, SPL-1499, from the nose wheel steering mechanism (TASK 32-00-01-080-801).

——— END OF TASK ———

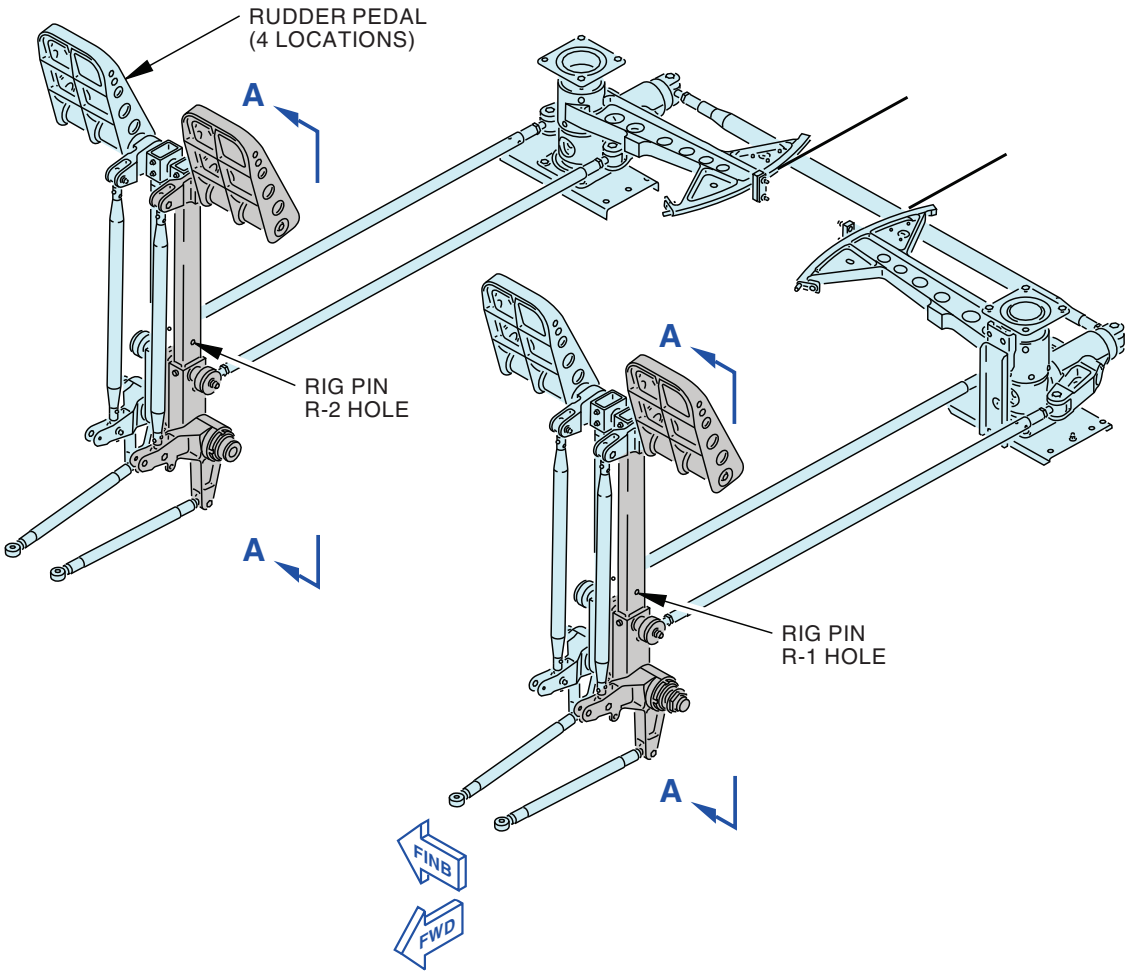
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2412363 S00061530522_V1

Rudder Control System Forward Components
Figure 503/27-21-00-990-806 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



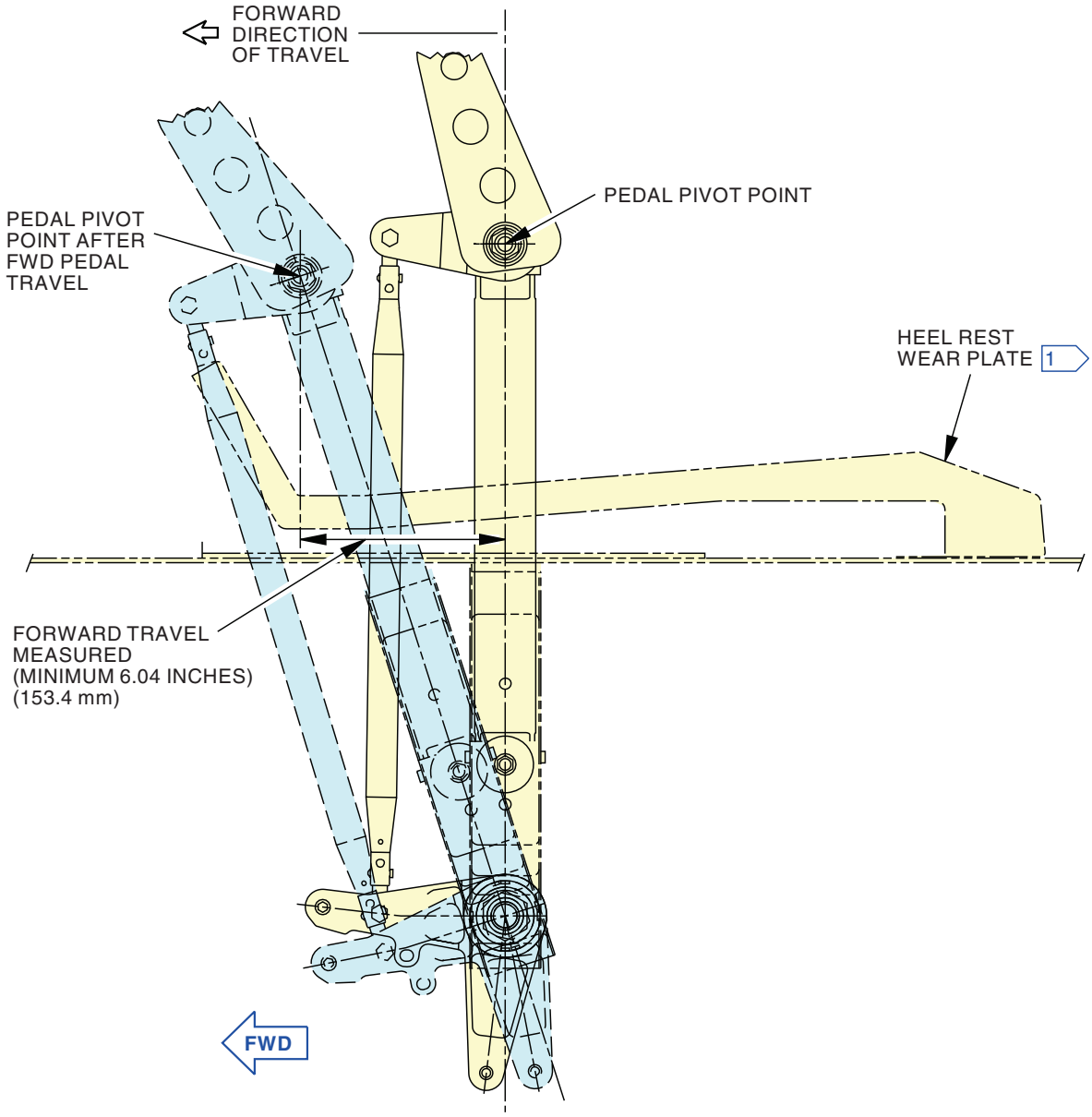
PEDAL SHAFTS AND FORWARD QUADRANTS

A

2412364 S00061530523_V1

Rudder Control System Forward Components
Figure 503/27-21-00-990-806 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

A-A

1 MASKING TAPE CAN BE APPLIED TO THE HEEL REST WEAR PLATE TO MARK THE PEDAL POSITION FOR PEDAL TRAVEL MEASUREMENT.

2412365 S00061530524_V2

Rudder Control System Forward Components
Figure 503/27-21-00-990-806 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-802

8. Rudder Hinge Bearing and PCU Bearing Freeplay

(Figure 504)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this task to check the freeplay of the rudder control system.

(2) This test will have two rudder left position readings. This will verify that the indicator does not move during the test.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy
STD-1238	Indicator - Dial
STD-1278	Block - Loading, 1/8 Inch Thick, 3 Inch by 3 Inch Wood or Fiberglass Block with Pad to Prevent Damage to the Skin.
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

E. Rudder Hinge Bearing and PCU Bearing Freeplay

SUBTASK 27-21-00-420-002

(1) Install the, NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism as given in: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

NOTE: Nose landing gear will move when the rudder pedals are moved. To prevent NLG movement, install a lockout pin in the nose wheel steering mechanism.

Nose gear steering does not need to be disconnected if the airplane is jacked up such that the nose gear is fully extended.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-048

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the rudder hydraulic systems A and B. To pressurize them, do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-00-730-005

(3) Do the rudder hinge bearing and Power Control Unit (PCU) bearing freeplay:

(a) Make sure that the rudder and rudder pedals are in the neutral position.

1) If it is necessary, move the rudder pedals lightly to set the rudder to the neutral position.

(b) Make sure that the rudder trim indicator is at zero units of trim.

(c) Make sure not to move rudder pedals during this test.

(d) Attach the dial indicator, STD-1238, with the mount, STD-1279, at the lower end of the rudder trailing edge.

CAUTION USE A LOADING BLOCK AND DO NOT PUT MORE THAN 50 LBF (222 N) ON THE RUDDER WHEN YOU MOVE THE RUDDER SURFACE BY AN EXTERNAL FORCE. APPLY THE EXTERNAL FORCE APPROXIMATELY BETWEEN 6 TO 8 INCHES (150 TO 200 MILLIMETERS) FROM THE BOTTOM OF THE TRAILING EDGE AND APPROXIMATELY 3 INCHES (80 MILLIMETERS) FORWARD OF THE TRAILING EDGE. IF YOU DO NOT APPLY FORCE AT THIS LOCATION, YOU CAN CAUSE DAMAGE TO THE RUDDER.

(e) Use a block, STD-1278 and push/pull scale 0-25 pound (0-11 kilogram), STD-753, to move the rudder to the left side with a force of 12 lbf (53.4 N) ± 4.0 lbf (17.8 N) and release slowly.

1) Make a record of the rudder position on the dial indicator.

(f) Use a block, STD-1278 and push/pull scale 0-25 pound (0-11 kilogram), STD-753, to move the rudder to the right side with a force of 12 lbf (53.4 N) ± 4.0 lbf (17.8 N) and release slowly.

1) Make a record of the rudder position on the dial indicator.

(g) Make sure that the total left and right freeplay between the left and right positions is not more than 0.05 in. (1.3 mm).

1) If the total left and right freeplay is more than 0.05 in. (1.3 mm), then replace the bearings on the rudder hinge or power control units and repeat this test.

(h) Use a block, STD-1278 and push/pull scale 0-25 pound (0-11 kilogram), STD-753, to move the rudder to the left side with a force of 12 lbf (53.4 N) ± 4.0 lbf (17.8 N) and release slowly.

NOTE: This check compares the two left positions to verify that the dial indicator did not move during the test. If the check shows that the dial indicator had moved, the test has to be repeated.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make a record of the rudder position on the dial indicator.

(i) Make sure that the difference between the two left positions is not more than 0.03 in. (0.8 mm).

1) If the difference is more than 0.03 in. (0.8 mm), then repeat the freeplay check.

(j) Remove the dial indicator, STD-1238, from the rudder.

SUBTASK 27-21-00-860-049

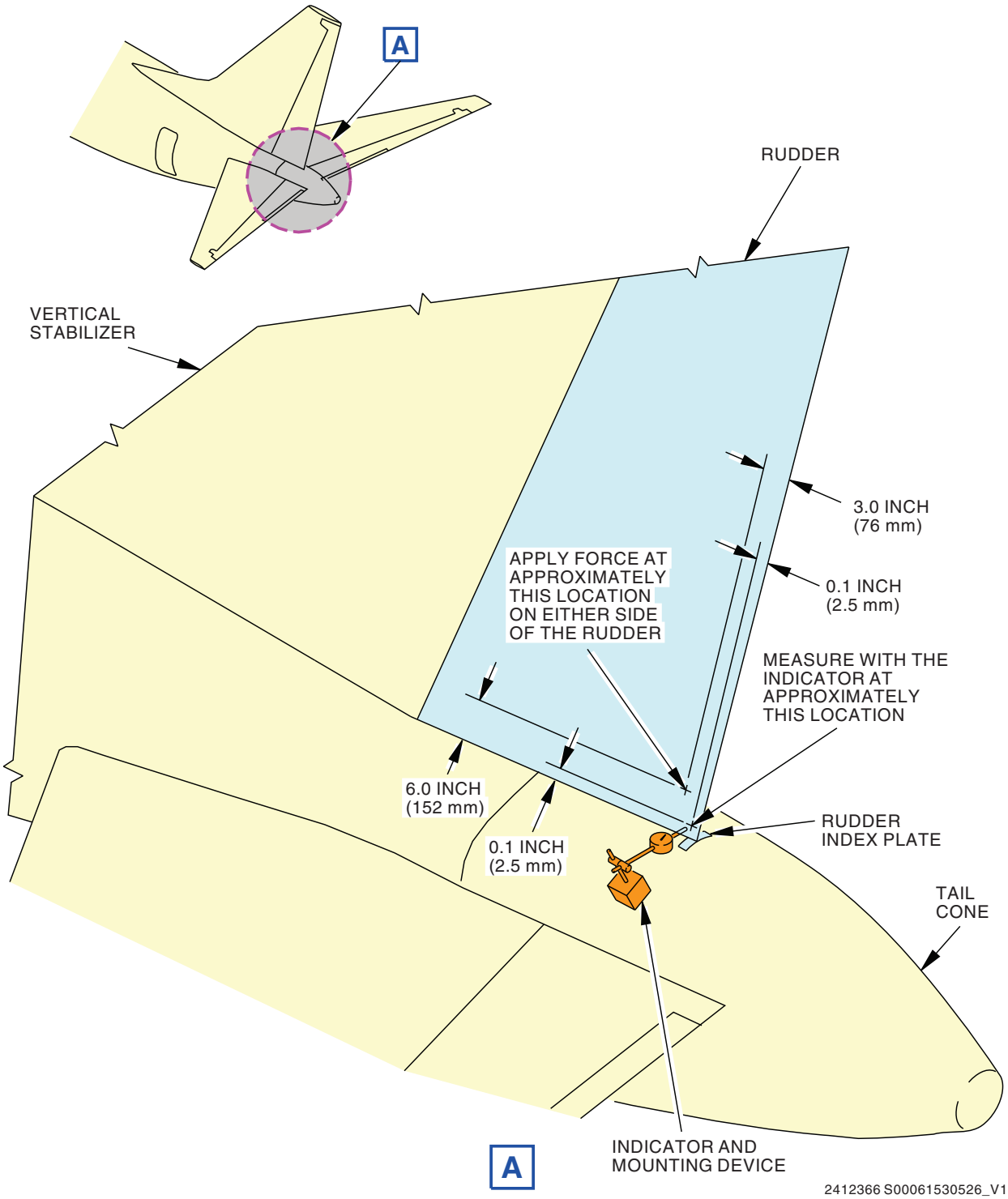
(4) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-860-050

(5) Remove the, NLG towing lever pin, SPL-1499, from the nose wheel steering mechanism as given in: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Hinge and PCU Bearing Freeplay Check
Figure 504/27-21-00-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-803

9. Rudder Centering Test

A. General

(1) Use this test to make sure the rudder is in its center position.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

E. Rudder Centering Test

SUBTASK 27-21-00-420-003

(1) Install the NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism (TASK 32-00-01-480-801).

NOTE: The nose landing gear (NLG) will move when the rudder pedals are moved. To prevent NLG movement, install a lockout pin in the nose wheel steering mechanism.

The nose gear steering does not need to be disconnected if the airplane is jacked up such that the nose gear is fully extended.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-052

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the rudder hydraulic systems A and B (TASK 27-21-00-800-801).

SUBTASK 27-21-00-860-053

(3) Align the ADIRS (TASK 34-21-00-820-802 or TASK 34-21-00-820-801 or TASK 34-21-00-820-803).

SUBTASK 27-21-00-730-006

(4) Do the test of the rudder centering test:

(a) Make sure the rudder is in the neutral position.

1) If it is necessary, move the rudder pedals lightly to set the rudder to the neutral position.

(b) Make sure that the rudder trim indicator is at 0.0 ± 0.25 units of trim.

(c) Move the captain's left rudder pedal approximately 0.5 in. (12.7 mm) forward and release quickly.

1) Make sure the rudder goes back to the neutral position within 0.00 ±0.05 in. (0.00 ±1.27 mm).

(d) Move the captain's right rudder pedal approximately 0.5 in. (12.7 mm) forward and release quickly.

1) Make sure the rudder goes back to the neutral position within 0.00 ±0.05 in. (0.00 ±1.27 mm).

(e) Set FLT CONTROL A and B switches to the STBY RUD position.

1) Make sure that the STANDBY HYD LOW PRESSURE light on the forward overhead panel, P5-3, goes off.

(f) Set the YAW DAMPER switch on the forward overhead panel, P5-3, to the ON position.

(g) Do these steps to initiate a zero yaw damper command output from the stall Management Yaw Damper No. 2 (SMYD No. 2), M1748:

NOTE: The SMYD No. 2 is installed on the E3-2 electrical equipment rack.

1) Push and release the ON/OFF button on the front BITE display of the SMYD No. 2.

2) Push and release the up arrow or down arrow button until the GROUND TEST shows on the display.

3) Push the YES button to get access to the ground test option.

4) Push and release the up arrow or down arrow button until the SERVO TEST? shows on the display.

5) Push the YES button to start the servo test.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF THE RUDDER. THE RUDDER CAN MOVE QUICKLY WHEN YOU PUSH THE `YES` BUTTON ON THE SMYD NO. 2. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

6) Push the YES button when the RUDDER CLEARED? shows on the display.

7) Push the YES button when the FLT CTRL A STBY? shows on the display.

8) Push the YES button when the FLT CTRL B STBY? shows on the display.

9) Push the YES button when YAW DMPR ENGAGED? shows on the display.

10) Push the YES button when the ZERO COMMAND? shows on the display.

NOTE: A yaw damper command value and LVDT position will show for 10 seconds each, then the END ZERO COMMAND? will show on the display. You can push the NO button to show these values again.

(h) Move the captain's left rudder pedal approximately 0.5 in. (12.7 mm) forward and release quickly.

1) Make sure the rudder goes back to the neutral position within 0.00 ±0.05 in. (0.00 ±1.27 mm).

(i) Move the captain's right rudder pedal approximately 0.5 in. (12.7 mm) forward and release quickly.

1) Make sure the rudder goes back to the neutral position within 0.00 ±0.05 in. (0.00 ±1.27 mm).

(j) Push and release the ON/OFF button on the SMYD No. 2.

SUBTASK 27-21-00-860-054

(5) Set the YAW DAMPER switch on the forward overhead panel, P5-3, to the OFF position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-860-055

(1) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-860-056

(2) Remove the NLG towing lever pin, SPL-1499, from the nose wheel steering mechanism (TASK 32-00-01-080-801).

——— END OF TASK ———

TASK 27-21-00-700-804

10. Rudder Pedal Forces Test

(Figure 505)

A. General

(1) Use this test to make sure the rudder pedals operate with the correct forces.

NOTE: To make the measurements, the forces must be applied with the pedal moving away from the neutral position. If the pedal moves more than the specified distances, the test must be repeated and started with the pedal at the neutral position.

(2) You may need two people to do the test. One person can sit in one of the seats and push on the tool. The other person can measure the rudder pedal movement and record the results.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) It may be useful to take three readings on each rudder pedal for better accuracy on the rudder pedal forces test.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1681	Adapter - Force Check, Rudder Pedal Part #: F80212-30 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

E. Rudder Pedal Forces Test

SUBTASK 27-21-00-420-004

(1) Install the NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism (TASK 32-00-01-480-801).

NOTE: Nose landing gear will move when the rudder pedals are moved. To prevent NLG movement, install a lockout pin in the nose wheel steering mechanism.

Nose gear steering does not need to be disconnected if the airplane is jacked up such that the nose gear is fully extended.

SUBTASK 27-21-00-860-058

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the rudder hydraulic systems A and B (TASK 27-21-00-800-801).

SUBTASK 27-21-00-730-007

(3) Do the test of the rudder pedal forces test:

(a) Set the captain's and first officer's rudder pedal adjustment cranks to the middle position (Figure 501).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the rudder pedals are in the neutral position.

(c) Install adapter, SPL-1681, on the captain's left rudder pedal.

1) Push on the adapter, SPL-1681, until the rudder starts to move.

2) Multiply the compression gage value on the adapter, SPL-1681, by a multiplication factor given in (Table 502):

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

Table 502/27-21-00-993-802

Angle from Horizontal	Multiplication Factor
0°	1.0
5°	1.0
10°	0.98
15°	0.97
20°	0.94
25°	0.91
30°	0.87
35°	0.82
40°	0.77
45°	0.71

3) Make sure the pedal force is 18.0 lbf (80.1 N) or less.

4) Push again on the adapter, SPL-1681, until the captain's left rudder pedal is 3.60 ±0.10 in. (91.4 ±2.6 mm) forward from the neutral position.

NOTE: When the rudder pedal reaches 3.60 ±0.10 in. (91.4 ±2.6 mm), DO NOT relax the force on the tool until measurement is taken, or the test will fail. If the 3.60 ±0.10 in. (91.4 ±2.6 mm) travel limit is exceeded, release the pedal to neutral and start again.

NOTE: You can use a ruler or straightedge that is put against the rudder pedal, to measure the rudder pedal movement. It may help to attach a pointer to the pedal, with tape.

5) Multiply the compression gage value on the adapter, SPL-1681 by a multiplication factor given in (Table 502).

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

6) Make sure the pedal force is between 60.0 lbf (266.9 N) to 82.0 lbf (364.8 N).

7) Remove the adapter, SPL-1681, from the captain's left pedal.

(d) Install the adapter, SPL-1681, on the captain's right rudder pedal.

1) Push on the adapter, SPL-1681, until the rudder starts to move.

2) Multiply the compression gage value on the adapter, SPL-1681, by a multiplication factor given in (Table 502).

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Make sure that the pedal force is 18.0 lbf (80.1 N) or less.

4) Push again on the adapter, SPL-1681, until the captain's right rudder pedal is 3.60 ±0.10 in. (91.4 ±2.6 mm) forward from the neutral position.

NOTE: When the rudder pedal reaches 3.60 ±0.10 in. (91.4 ±2.6 mm), DO NOT relax the force on the tool until measurement is taken, or the test will fail. If the 3.60 ±0.10 in. (91.4 ±2.6 mm) travel limit is exceeded, release the pedal to neutral and start again.

5) Multiply the compression gage value on the adapter, SPL-1681, by a multiplication factor given in (Table 502).

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

6) Make sure the pedal force is between 60.0 lbf (266.9 N) to 82.0 lbf (364.8 N).

7) Remove the adapter, SPL-1681, from the captain's right pedal.

(e) Install the adapter, SPL-1681, on the first officer's left rudder pedal.

1) Push on the adapter, SPL-1681, until the rudder starts to move.

2) Multiply the compression gage value on the adapter, SPL-1681, by a multiplication factor given in (Table 502).

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

3) Make sure that the pedal force is 18.0 lbf (80.1 N) or less.

4) Push again on the adapter, SPL-1681, until the first officer's left rudder pedal is 3.60 ±0.10 in. (91.4 ±2.6 mm) forward from the neutral position.

NOTE: When the rudder pedal reaches 3.60 ±0.10 in. (91.4 ±2.6 mm), DO NOT relax the force on the tool until measurement is taken, or the test will fail. If the 3.60 ±0.10 in. (91.4 ±2.6 mm) travel limit is exceeded, release the pedal to neutral and start again.

5) Multiply the compression gage value on the adapter, SPL-1681, by a multiplication factor given in (Table 502).

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

6) Make sure the pedal force is between 60.0 lbf (266.9 N) to 82.0 lbf (364.8 N).

7) Remove the adapter, SPL-1681, from the first officer's left pedal.

(f) Install the adapter, SPL-1681, on the first officer's right rudder pedal.

1) Push on the adapter, SPL-1681, until the rudder starts to move.

2) Multiply the compression gage value on the adapter, SPL-1681, by a multiplication factor given in (Table 502).

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

3) Make sure the pedal force is 18.0 lbf (80.1 N) or less.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Push again on the adapter, SPL-1681, until the first officer's right rudder pedal is 3.60 ±0.10 in. (91.4 ±2.6 mm) forward from the neutral position.

NOTE: When the rudder pedal reaches 3.60 ±0.10 in. (91.4 ±2.6 mm), DO NOT relax the force on the tool until measurement is taken, or the test will fail. If the 3.60 ±0.10 in. (91.4 ±2.6 mm) travel limit is exceeded, release the pedal to neutral and start again.

5) Multiply the compression gage value on the adapter, SPL-1681, by a multiplication factor given in (Table 502).

NOTE: The angle from the horizontal value is also shown on the adapter, SPL-1681.

6) Make sure that the pedal force is between 60.0 lbf (266.9 N) to 82.0 lbf (364.8 N).

7) Remove the adapter, SPL-1681, from the first officer's right pedal.

SUBTASK 27-21-00-860-059

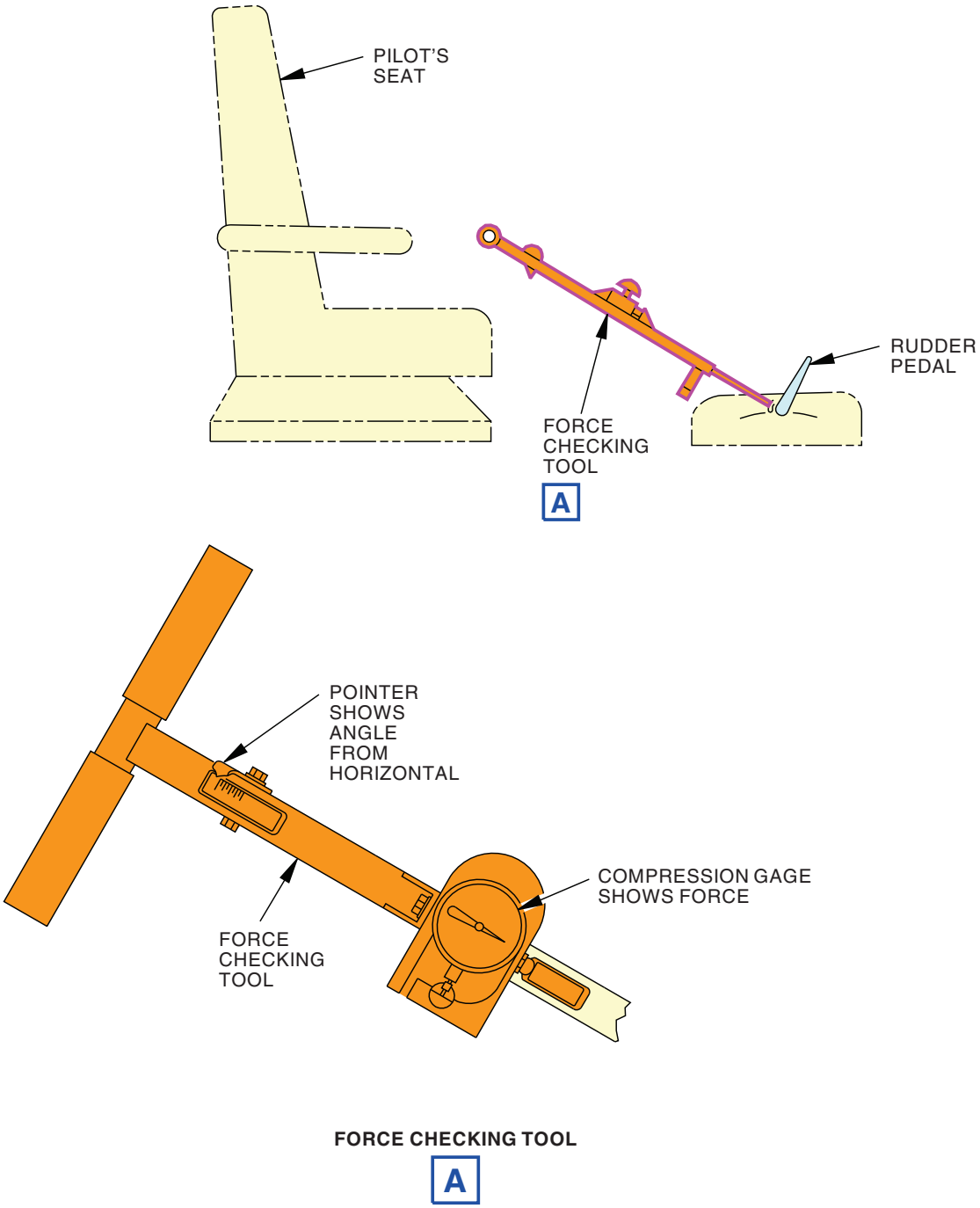
(4) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-860-060

(5) Remove the, NLG towing lever pin, SPL-1499, from the nose wheel steering mechanism (TASK 32-00-01-080-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Pedal Force Checking Tool
Figure 505/27-21-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-805

11. Rudder Travel Test

A. General

(1) Use this test to make sure the rudder operates correctly.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

E. Rudder Travel Test

SUBTASK 27-21-00-420-005

(1) Install the NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism as given in: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

NOTE: Nose landing gear will move when the rudder pedals are moved. To prevent NLG movement, lockout pin in the nose wheel steering mechanism should be installed.

Nose gear steering does not need to be disconnected if the airplane is jacked up such that the nose gear is fully extended.

SUBTASK 27-21-00-860-062

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the rudder hydraulic systems A and B, do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-00-730-008

(3) Do the test of the rudder travel test:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Set the FLT CONTROL A switch to the ON position and FLT CONTROL B switch to the OFF position.

(b) Set the captain's and first officer's rudder pedal adjustment cranks to the middle position (Figure 501).

(c) Make sure the rudder pedals are in the neutral position.

1) If it is necessary, move the rudder pedals lightly to set the rudder to the neutral position.

2) Make sure that the rudder trim indicator is at 0.0 ± 0.25 units of trim.

(d) Push and hold captain's left pedal until it touches the forward quadrant stop.

1) Measure the rudder travel (Figure 507).

2) Make sure that the dimension is 24.28 ±0.92 in. (616.71 ±23.37 mm) (26.00 +/- 1.00 degrees) rudder trailing edge left.

3) Release the captain's left pedal.

(e) Push and hold the captain's right pedal until it touches the forward quadrant stop.

1) Measure the rudder travel (Figure 507).

2) Make sure that the dimension is 24.28 ±0.92 in. (616.71 ±23.37 mm) (26.00 +/- 1.00 degrees) rudder trailing edge right.

3) Release the captain's right pedal.

(f) Make sure that the wire bundle W443 has a minimum clearance of 0.5 in. (12.700 mm) from left forward rudder quadrant full travel motion, while operating the rudder pedals (Figure 506).

(g) Set the FLT CONTROL B switch to the ON position and FLT CONTROL A switch to the OFF position.

(h) Push and hold captain's left pedal until it touches the forward quadrant stop.

1) Measure the rudder travel (Figure 507).

2) Make sure that the dimension is 24.28 ±0.92 in. (616.71 ±23.37 mm) (26.00 +/- 1.00 degrees) rudder trailing edge left.

3) Release the captain's left pedal.

(i) Push and hold the captain's right pedal until it touches the forward quadrant stop.

1) Measure the rudder travel (Figure 507).

2) Make sure that the dimension is 24.28 ±0.92 in. (616.71 ±23.37 mm) (26.00 +/- 1.00 degrees) rudder trailing edge right.

3) Release the captain's right pedal.

(j) Make sure that the wire bundle W443 has a minimum clearance of 0.5 in. (12.700 mm) from left forward rudder quadrant full travel motion, while operating the rudder pedals (Figure 506).

(k) Set the FLT CONTROL B switch to the STBY position.

(l) Push and hold captain's left pedal until it touches the forward quadrant stop.

1) Measure the rudder travel (Figure 507).

2) Make sure that the dimension is 22.62 +0.92 / -0.91 in. (574.55 +23.37 / -23.11 mm) (24.25 +/- 1.00 degrees) rudder trailing edge left.

3) Release the captain's left pedal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(m) Push and hold the captain's right pedal until it touches the forward quadrant stop.

1) Measure the rudder travel (Figure 507).

2) Make sure that the dimension is 22.62 +0.92 / -0.91 in. (574.55 +23.37 / -23.11 mm) (24.25 +/- 1.00 degrees) rudder trailing edge right.

3) Release the captain's right pedal.

(n) Make sure that the wire bundle W443 has a minimum clearance of 0.5 in. (12.700 mm) from left forward rudder quadrant full travel motion, while operating the rudder pedals (Figure 506).

SUBTASK 27-21-00-860-063

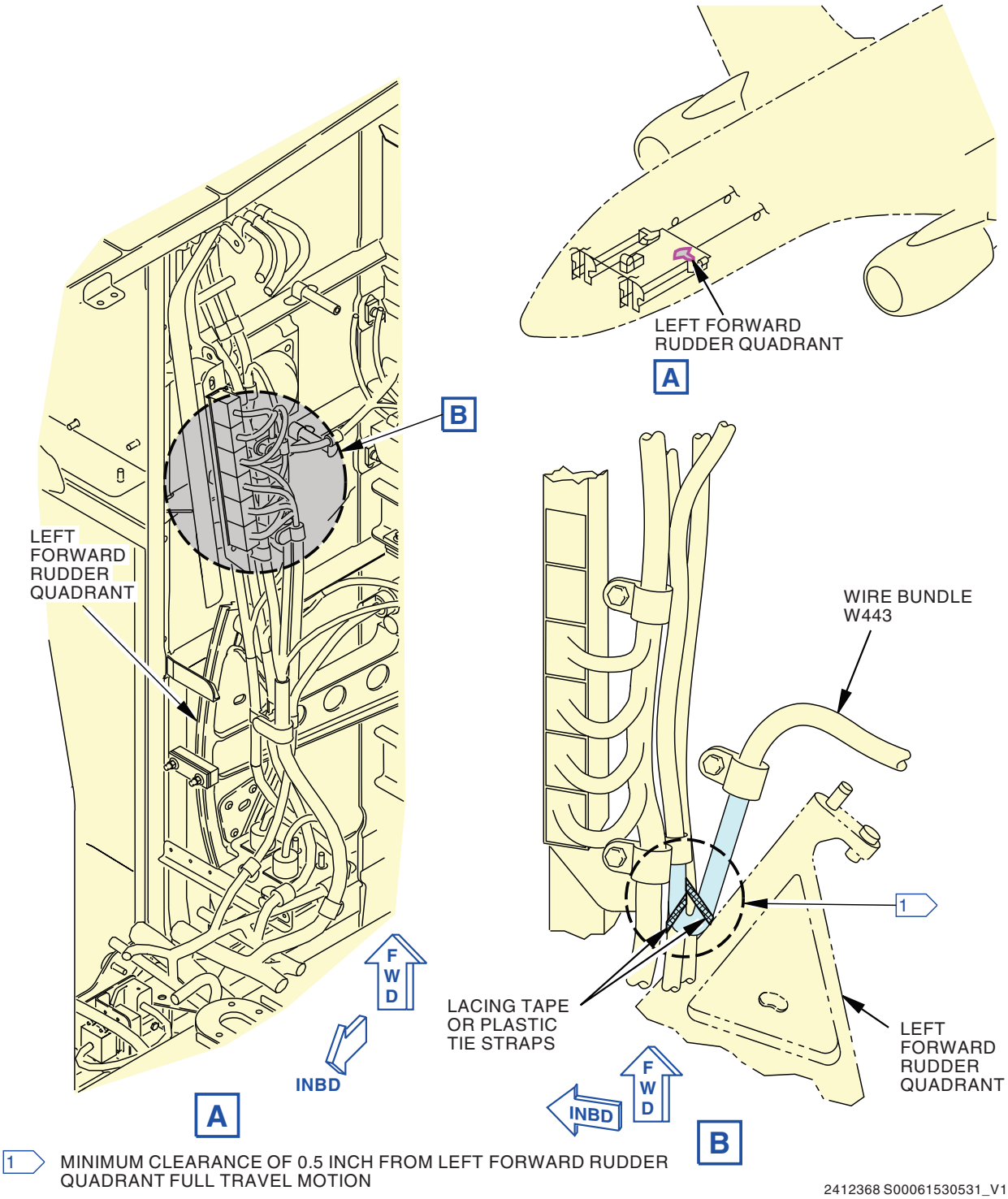
(4) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-080-006

(5) Remove the NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism as given in: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Wire Bundle W443 Clearance
Figure 506/27-21-00-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-806

12. Standby Rudder Power Control Unit Deadband Test

A. General

(1) Use this test to make sure the standby rudder power control unit (PCU) is in the correct rigged position.

(a) Make sure that the YAW DAMPER switch on the P5-3 overhead panel is in the OFF position.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

D. Standby Rudder Power Control Unit Deadband Test

SUBTASK 27-21-00-860-064

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Pressurize the standby rudder hydraulic system (TASK 27-21-00-800-801)..

SUBTASK 27-21-00-860-065

(2) Move the rudder pedals through ten cycles to flush and bleed the standby rudder PCU.

SUBTASK 27-21-00-860-066

(3) Set the FLT CONTROL A and B switches to the ON position.

SUBTASK 27-21-00-730-009

(4) Do the test of the standby rudder PCU deadband test:

(a) Make sure that the rudder is in the neutral position.

1) If necessary, move the rudder pedals lightly to put the rudder to the neutral position.

2) Make sure that the rudder trim indicator is at 0.00 ± 0.25 unit of trim.

(b) Set the FLT CONTROL A and B to the STANDBY RUD position.

(c) Use your hands to push the rudder to the right, then slowly remove the load.

1) Move your hands away from the rudder.

2) Measure the rudder right position from the center of the rudder trailing edge to the centerline of the index plate.

3) Record this measurement as A.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Push the rudder by hands to the left, then slowly remove the load.

1) Move your hands away from the rudder.

2) Measure the rudder left position from the center of the rudder trailing edge to the centerline of the index plate.

3) Record this measurement as B.

(e) Make sure that the difference between A and B is 0.2 in. (5 mm) or less.

1) If the difference is more than 0.2 in. (5 mm), do this task: Control Rod for Standby Rudder Power Control Unit Adjustment, TASK 27-21-00-820-805.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-860-067

(1) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

——— END OF TASK ———

TASK 27-21-00-700-807

13. Rudder Trim System Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to make sure the rudder trim system operates correctly.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the System Test

SUBTASK 27-21-00-420-006

(1) Install the NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism (TASK 32-00-01-480-801).

NOTE: The nose landing gear (NLG) will move when the rudder pedals are moved. To prevent NLG movement, install a lockout pin in the nose wheel steering mechanism.

The nose gear steering does not need to be disconnected if the airplane is jacked up such that the nose gear is fully extended.

SUBTASK 27-21-00-860-069

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize both rudder hydraulic systems A and B (TASK 27-21-00-800-801).

NOTE: It is necessary to pressurize both systems A and B for this task.

F. Rudder Trim System Test

SUBTASK 27-21-00-730-010

(1) Do the test of the rudder trim system:

(a) Make sure that the rudder is in the neutral position.

1) If it is necessary, move the rudder pedals lightly to put the rudder to the neutral position.

2) Make sure the rudder trim indicator is at 0.00 ± 0.25 units of trim.

(b) Move the rudder trim knob to NOSE LEFT until the mark on the rudder trim knob is aligned with the indicator mark on the lightplate under the trim knob.

1) Make sure that the rudder did not move.

(c) Move the rudder trim knob to NOSE RIGHT until the mark on the rudder trim knob is aligned with the indicator mark on the lightplate under the trim knob.

1) Make sure that the rudder did not move.

(d) Move the rudder trim knob to NOSE RIGHT until the rudder trim indicator shows 10.0 ± 0.25 units of trim.

1) Make sure that the rudder moves 10.12 ±1.26 in. (257.0 ±32.1 mm) (10.74 ± 1.35 degrees) to the right side.

(e) Move the rudder trim knob to NOSE RIGHT until the rudder stops.

1) Make sure that the rudder moves 16.89 ±0.93 in. (429.0 ±23.6 mm) (18.03 ± 1.00 degrees) to the right side.

2) Make sure the rudder trim indicator is at least 15 units of trim.

(f) Move the rudder trim knob to NOSE LEFT until the rudder trim indicator shows 10.0 ± 0.25 units of trim.

1) Make sure that the rudder moves 10.12 ±1.26 in. (257.0 ±32.1 mm) (10.74 ± 1.35 degrees) to the left side.

(g) Move the rudder trim knob to NOSE LEFT until the rudder stops.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the rudder moves 16.89 ±0.93 in. (429.0 ±23.6 mm) (18.03 ± 1.00 degrees) to the left side.

2) Make sure the rudder trim indicator is at least 15 units of trim.

(h) Make sure the rudder trim knob operates smoothly and goes back to the center freely.

(i) Make sure that the rudder is in the neutral position.

1) If it is necessary, move the rudder pedals lightly to put the rudder to the neutral position.

2) Make sure the rudder trim indicator is at 0.00 ± 0.25 units of trim.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-860-070

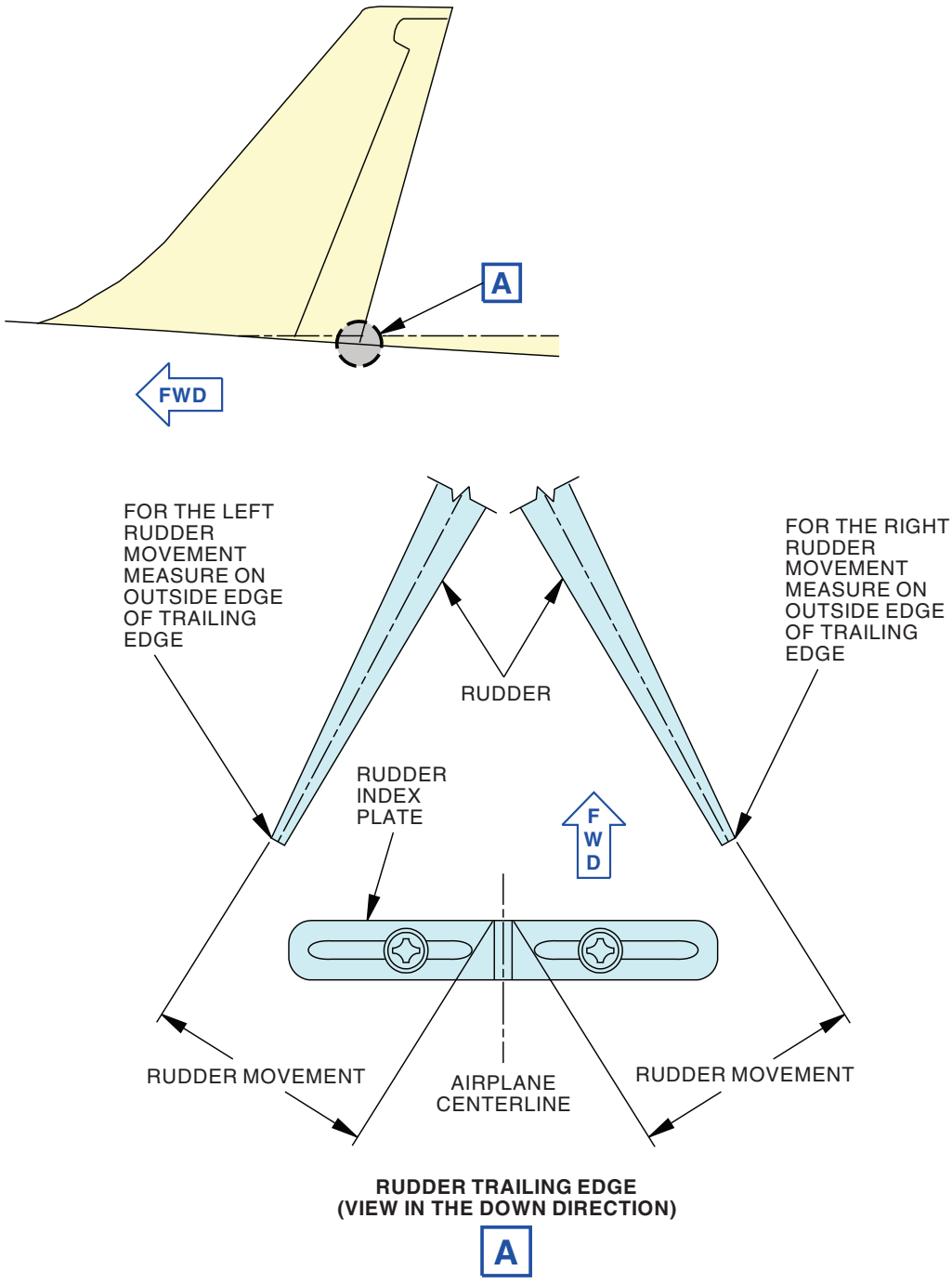
(1) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-860-071

(2) Remove the NLG towing lever pin, SPL-1499, from the nose wheel steering mechanism (TASK 32-00-01-080-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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RUDDER TRIM SYSTEM TEST
Figure 507/27-21-00-990-812

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-808

14. Wheel to Rudder Interconnect System Test

(Figure 508)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to make sure that the wheel to rudder function of the standby rudder Power Control Unit (PCU) operates correctly.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
27-51-00-860-803	Extend the Trailing Edge Flaps (P/B 201)
27-51-00-860-804	Retract the Trailing Edge Flaps (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
325	Vertical Fin - Rudder

D. Prepare for the System Test

SUBTASK 27-21-00-860-072

(1) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-1

Row	Col	Number	Name
E	5	C01009	ADIRU LEFT DC
E	7	C01007	ADIRU LEFT AC

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
B	4	C01207	SMYD-2 SNSR EXC AC
B	5	C01206	SMYD-2 CMPTR DC
C	14	C01008	ADIRU RIGHT AC
C	17	C01010	ADIRU RIGHT DC

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-073

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Pressurize the rudder hydraulic systems A and B (TASK 27-21-00-800-801).

SUBTASK 27-21-00-860-074

(3) Move the flaps to a position other than up position (TASK 27-51-00-860-803).

E. Wheel to Rudder Interconnect System Test

SUBTASK 27-21-00-730-011

(1) Do the test of the wheel to rudder interconnect system:

NOTE: The rudder will move past the center position to the opposite direction when the control wheel returns to the neutral position. To return the rudder to the center position with the control wheel at neutral, you must set the YAW DAMPER switch on the overhead panel to the OFF position then back to the ON position.

(a) Make sure that the Air Data Inertial Reference System (ADIRS) is aligned (TASK 34-21-00-820-802 or TASK 34-21-00-820-801 or TASK 34-21-00-820-803).

(b) Make sure that the rudder is in the neutral position.

1) Make sure that the rudder trim indicator is at 0.0 ± 0.25 unit of trim.

2) If it is necessary, move the rudder pedals lightly to put the rudder to the neutral position.

(c) Set the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the STBY RUD position.

(d) Set the YAW DAMPER switch, on the P5 forward overhead panel, to the ON position.

(e) Slowly turn the Captain's wheel left to 2 units (12°) of wheel and hold.

NOTE: Use aileron trim marker on control column to measure wheel travel.

1) Make sure that the rudder moves 2.47 ±0.25 in. (62.74 ±6.35 mm) (2.4 ± 0.3 degrees) to the left side.

(f) Slowly turn the captain's wheel right to 2 units (12°) of wheel and hold.

NOTE: The 2 units to the right and left are in reference to the neutral position of the captain's control wheel.

1) Make sure that the rudder moves from left to right.

2) Make sure that the rudder moves 2.47 ±0.25 in. (62.74 ±6.35 mm) (2.4 ± 0.3 degrees) to the right side.

(g) Slowly turn the captain's control wheel to the full right and hold.

1) Make sure that the rudder moves 2.47 ±0.25 in. (62.74 ±6.35 mm) (2.4 ± 0.3 degrees) to the right side.

(h) Slowly turn the captain's control wheel to the full left and hold.

NOTE: The rudder should move left 2.47 ±0.25 in. (62.74 ±6.35 mm) before the wheel has reached neutral.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the rudder moves 2.47 ±0.25 in. (62.74 ±6.35 mm) (2.4 ± 0.3 degrees) to the left side.

(i) Slowly turn the captain's control wheel back to the neutral position.

1) Make sure that the rudder moves 2.47 ±0.25 in. (62.74 ±6.35 mm) (2.4 ± 0.3 degrees) and the rudder surface has moved to the right.

2) Put the YAW DAMPER switch, on the P5 panel, in the OFF position.

3) Put the YAW DAMPER switch, on the P5 panel, in the ON position.

NOTE: This will restore the rudder to neutral with the control wheel at neutral.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 27-21-00-860-075

(1) Put the YAW DAMPER switch, on the P5 panel, in the OFF position.

SUBTASK 27-21-00-860-076

(2) Return the flaps to the retracted position (TASK 27-51-00-860-804).

SUBTASK 27-21-00-860-077

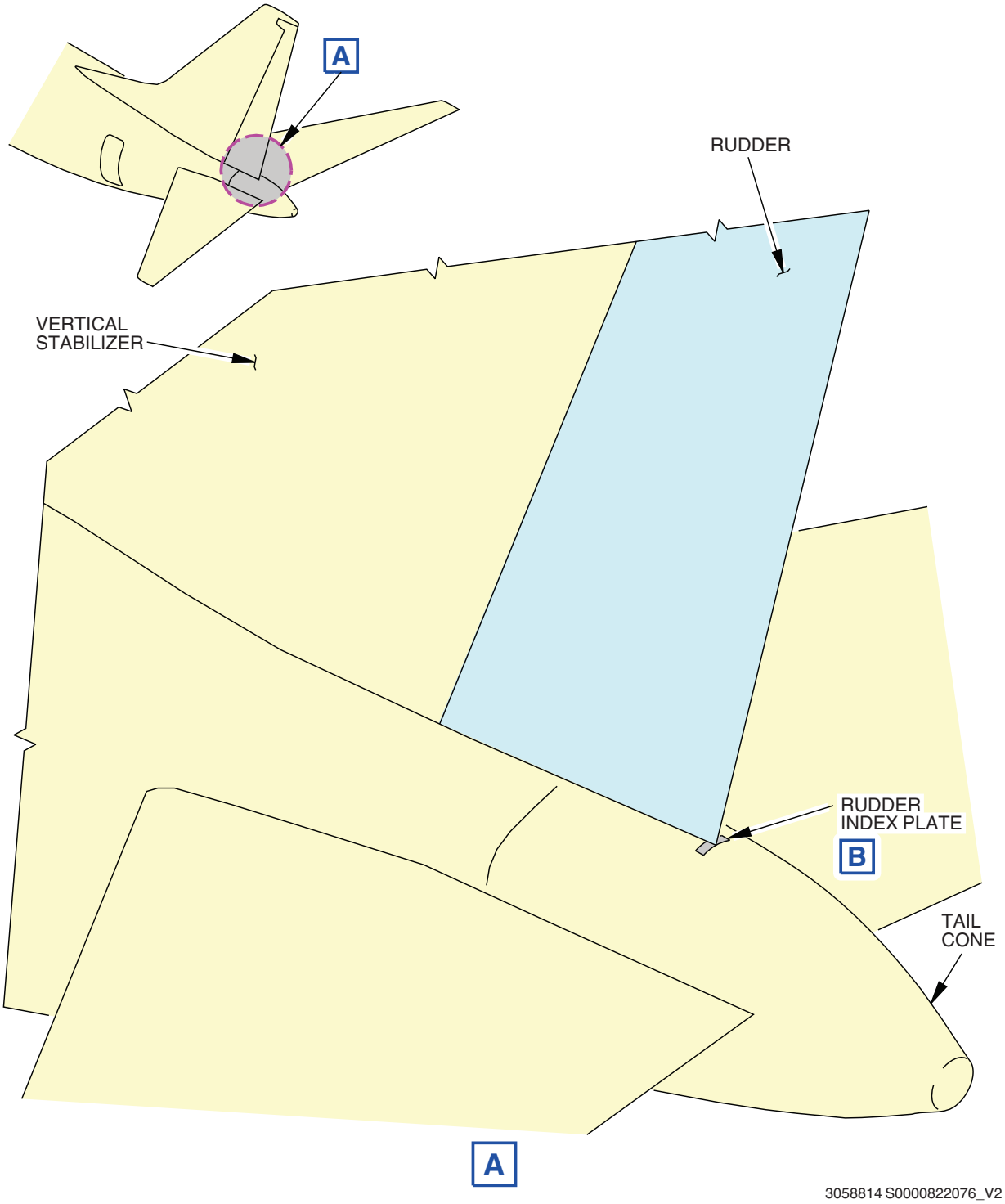
(3) Set the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the ON position.

SUBTASK 27-21-00-860-078

(4) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

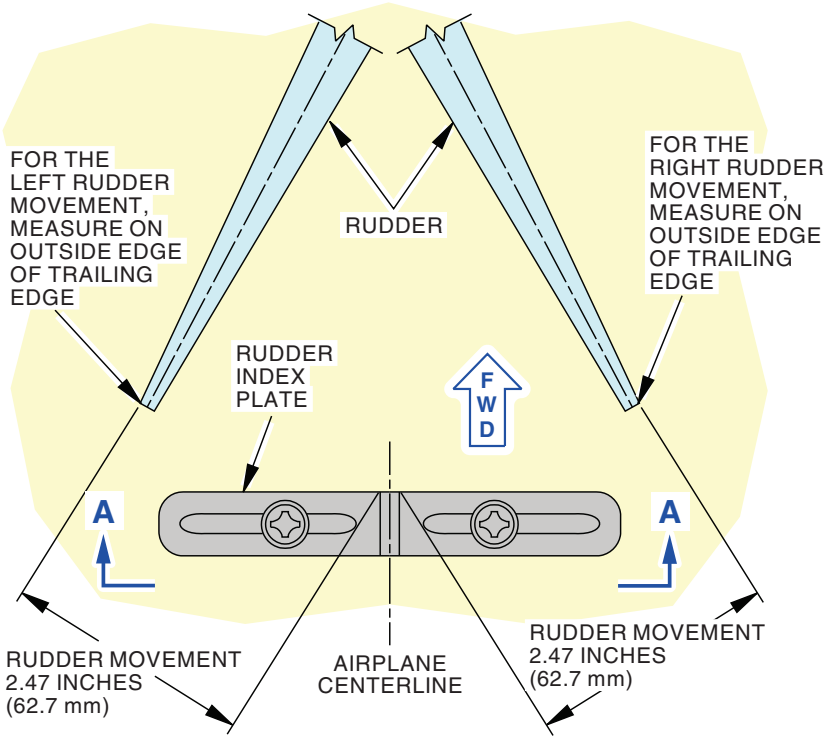
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



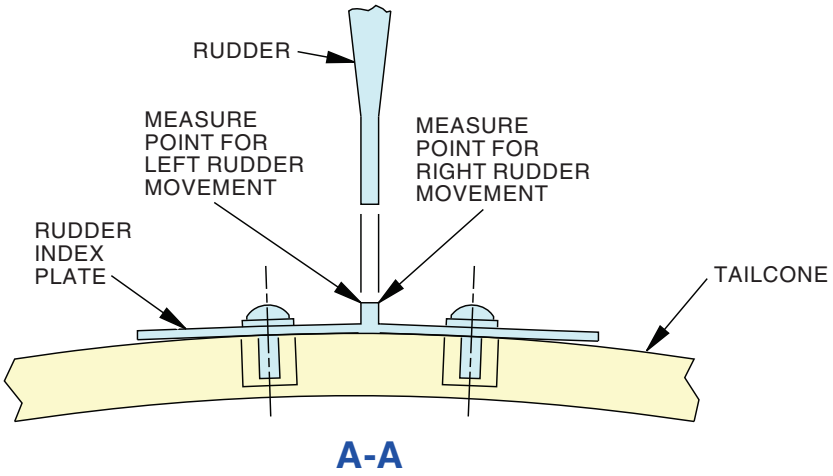
Rudder Control System Aft Components
Figure 508/27-21-00-990-813 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RUDDER TRAILING EDGE
(VIEW IN THE DOWN DIRECTION)

B



A-A

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Rudder Control System Aft Components
Figure 508/27-21-00-990-813 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-809

15. Rudder Hinge Friction Test

(Figure 504)

A. General

(1) Use this test to make sure that the rudder hinge friction is within limit.

B. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-753	Scale - Push/Pull, 0-25 pound (0-11 kilogram) Capacity, 1/4 pound (113 gram) Accuracy
STD-1238	Indicator - Dial
STD-1278	Block - Loading, 1/8 Inch Thick, 3 Inch by 3 Inch Wood or Fiberglass Block with Pad to Prevent Damage to the Skin.
STD-1279	Mount - Device, Mounting, Holds the Dial Indicator

D. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

E. Rudder Hinge Friction Test

SUBTASK 27-21-00-860-079

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU REMOVE OR INSTALL HYDRAULIC POWER. AILERONS, RUDDERS, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU REMOVE HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If hydraulic pressure is powered, do this step to turn off hydraulic power:

(a) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-00-730-012

CAUTION DO NOT USE A LOAD OF MORE THAN 50 POUNDS ON THE RUDDER SURFACE DURING THE TEST. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION	MAKE SURE TO APPLY EXTERNAL FORCE THAT IS APPROXIMATELY AT BETWEEN 6 INCHES (152 MM) AND 18 INCHES (457MM) FROM THE BOTTOM OF THE TRAILING EDGE. IF YOU DO NOT APPLY FORCE AT THIS LOCATION, YOU CAN CAUSE DAMAGE TO THE RUDDER.

(2) Do the rudder hinge friction check:

NOTE: If the dial indicator is not available, you can use a ruler to measure the rudder trailing edge placement in this test.

NOTE: If the test is unsatisfactory, you can remove the seals between the vertical fin and the rudder surface.

(a) Make sure that the rudder and rudder pedals are in the neutral position.

(b) Make sure that the rudder trim indicator is at zero units of trim.

(c) Use your hand to move the rudder slowly through full travel to the left side.

1) Make sure that there is no sign of binding, rough spots, or excessive friction at any point of travel.

a) If binding or excessive friction exist, the bearings can be out of alignment, damaged, or improperly installed, correct this condition and repeat this check.

(d) Use your hand to move the rudder slowly through full travel to the right side.

1) Make sure that there is no sign of binding, rough spots, or excessive friction at any point of travel.

a) If binding or excessive friction exist, the bearings can be out of alignment, damaged, or improperly installed, correct this condition and repeat this check.

(e) Attach the mount, STD-1279, on the tail cone.

(f) Place the pointer of the dial indicator, STD-1238, at the lower aft corner of the rudder trailing edge.

(g) Use a block, STD-1278, and push/pull scale 0-25 pound (0-11 kilogram), STD-753, to move the rudder 2.25 in. (5.72 cm) ± 0.1 in. (0.25 cm) to the left side.

1) Make sure that the force to move the rudder is less than 16.3 lbf (72.5 N).

(h) Use a block, STD-1278, and push/pull scale 0-25 pound (0-11 kilogram), STD-753, to move the rudder 2.25 in. (5.72 cm) ± 0.1 in. (0.25 cm) to the right side.

1) Make sure that the force to move the rudder is less than 16.3 lbf (72.5 N).

(i) Remove the dial indicator, STD-1238, and block, STD-1278, from the rudder.

(j) Remove the mount, STD-1279, from the tailcone.

(k) If the force to move the rudder is greater than 16.3 lbf (72.5 N), do these steps:

1) Disconnect the two ends of the standby Power Control Unit (PCU):

a) Disconnect the two ends of the PCU

b) Move the PCU forward and to one side to let the rudder travel without interference.

2) Disconnect the two ends of the main PCU:

a) Disconnect the two ends of the PCU.

b) Move the PCU forward and to one side to let the rudder travel without interference.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Use your hand to move the rudder slowly through full travel to the left side.
   a) Make sure that there is no sign of binding, rough spots, or excessive friction at any point of travel.
      <1> If binding or excessive friction exist, the bearings can be out of alignment, damaged, or improperly installed, correct this condition and repeat this check.
4) Use your hand to move the rudder slowly through full travel to the right side.
   a) Make sure that there is no sign of binding, rough spots, or excessive friction at any point of travel.
      <1> If binding or excessive friction exist, the bearings can be out of alignment, damaged, or improperly installed, correct this condition and repeat this check.
5) Attach the mount, STD-1279, on the tail cone.
6) Place the pointer of the dial indicator, STD-1238, at the lower aft corner of the rudder trailing edge.
7) Use a block, STD-1278, and push/pull scale 0-25 pound (0-11 kilogram), STD-753, to move the rudder:
   a) Move the rudder 2.25 in. (5.72 cm) ± 0.1 in. (0.25 cm) to the left side.
   b) Measure the force required to move the rudder.
   c) Record this value.
   d) Move the rudder 2.25 in. (5.72 cm) ± 0.1 in. (0.25 cm) to the right side.
   e) Measure the force required to move the rudder.
   f) Record this value.
   g) If the two recorded values are more than 1 lbf (4.4 N), do these steps:
      <1> Correct the faulty condition.
      <2> Connect the two ends of the main PCU.
      <3> Connect the two ends of the standby PCU.
      <4> Do the Rudder Hinge Friction Test again from the start of the test.
   h) If one of the two recorded values is less than 1 lbf (4.4 N), connect the two ends of the main PCU.
8) Use a block, STD-1278, and push/pull scale 0-25 pound (0-11 kilogram), STD-753, to move the rudder:
   a) Move the rudder 2.25 in. (5.72 cm) ± 0.1 in. (0.25 cm) to the right side.
   b) Measure the force required to move the rudder.
   c) Record this value.
   d) Move the rudder 2.25 in. (5.72 cm) ± 0.1 in. (0.25 cm) to the left side.
   e) Measure the force required to move the rudder.
   f) Record this value.
   g) If the two recorded values are more than 11.49 lbf (51.1 N), the friction in the main PCU is too high. Do these steps:
      <1> Correct the faulty condition.
      <2> Connect the two ends of the main PCU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<3> Do the Rudder Hinge Friction Test again from the start of the test.

h) If the two recorded values are less than 11.49 lbf (51.1 N), the friction in the standby PCU is too high. Do these steps:.

<1> Correct the faulty condition.

<2> Connect the two ends of the standby PCU.

<3> Do the Rudder Hinge Friction Test again from the start of the test.

(l) Remove the dial indicator, STD-1238, and block, STD-1278, from the rudder.

(m) Remove the mount, STD-1279, from the tailcone.

——— END OF TASK ———

27-CMR-05

TASK 27-21-00-700-810

16. Main Rudder PCU Internal Hydraulic Leakage Test

(Figure 509)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this task to do a check for excessive internal hydraulic leakage for the main rudder Power Control Unit (PCU).

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

D. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right
324	Vertical Fin - Rear Spar To Trailing Edge

E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
324DL	Trailing Edge Access

F. Prepare for the Main Rudder PCU Check

SUBTASK 27-21-00-840-003

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	9	C01447	FLIGHT CONT RUDDER AUTHORITY LIMITER
D	15	C01628	FORCE FIGHT MONITOR

SUBTASK 27-21-00-840-004

(2) If required, open this access panel:

Number	Name/Location
324DL	Trailing Edge Access

SUBTASK 27-21-00-860-081

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Pressurize the rudder hydraulic systems A and B, do this task: Rudder Hydraulic System A, B, or Standby Pressurization, TASK 27-21-00-800-801.

SUBTASK 27-21-00-860-082

(4) Put the FLT CONTROL A and B switches, on the P5-3 panel, to the ON position.

SUBTASK 27-21-00-860-083

(5) Install the NLG towing lever pin, SPL-1499, in the nose wheel steering mechanism (TASK 32-00-01-480-801).

NOTE: Nose Landing Gear (NLG) will move when the rudder pedals are moved. To prevent NLG movement, install a lockout pin in the nose wheel steering mechanism.

SUBTASK 27-21-00-980-003

(6) Move the rudder pedals through 5 cycles of full travel.

SUBTASK 27-21-00-980-004

(7) Center the rudder pedals and remove feet from the pedals.

SUBTASK 27-21-00-010-014

(8) Get access to the main electronics equipment compartment, do this step:

Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-21-00-700-002

(9) Use a digital voltmeter to make sure that the 4.38 Vdc to 5.62 Vdc between pin 18 and pin 20 of connector D10064, on the E1-1 shelf in the EE bay.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If the voltage is not 4.38 Vdc to 5.62 Vdc, adjust the control rod (TASK 27-21-00-820-804).

SUBTASK 27-21-00-210-003

(10) Make sure that the STBY RUD ON light, on the P5-3 overhead panel, is not illuminated.

SUBTASK 27-21-00-860-084

(11) Put the FLT CONTROL A switch, on the P5-3 panel, to the OFF position.

G. Rudder Main PCU System B Internal Leakage Check in the Extend Direction

NOTE: You will need 3 persons to complete this task.

There is a time limit to complete some steps in this task.

SUBTASK 27-21-00-860-085

(1) Use the Flap/Slat Electronics Unit (FSEU), on the E1-1 shelf in the main equipment center, to command the Rudder Pressure Limiters (RPL) to the "LOW" pressure state.

NOTE: The command will remain active for only 75 seconds after it is set.

(a) Push the ON/OFF button on the front panel of the FSEU, on E1-1 shelf, to turn on the display.

1) The display will show EXISTING FAULTS?.

(b) Select OTHER FUNCTNS? from the MAIN MENU on the FSEU.

1) The display will show SET OUTPUTS?.

(c) Select SET OUTPUTS? from the OTHER FUNCTNS? menu.

1) The display will show RUD PRES LIMIT?.

(d) Select RUD PRES LIMIT? from the SET OUTPUTS? menu.

1) The display will show SET ON?.

(e) Select SET ON? from the RUD PRES LIMIT? menu.

1) The display will show CMD ON, SET OFF?.

NOTE: You only have 75 seconds to complete the next 3 tasks. If you do not finish in 75 seconds, you will have to repeat the test and start with the steps to use the FSEU to command the RPL to the "LOW" state.

SUBTASK 27-21-00-980-005

(2) Push the left rudder pedal until it touches the forward quadrant stop and hold.

SUBTASK 27-21-00-980-006

(3) Manually hold the main rudder PCU lower control rod at location B and pull forward with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

SUBTASK 27-21-00-700-003

(4) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is between 1.87 VDC to 2.62 VDC.

SUBTASK 27-21-00-980-007

(5) Release the main rudder PCU control rod.

SUBTASK 27-21-00-980-008

(6) Release the rudder pedal.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-086

(7) The display screen on the FSEU should show CMD ON, SET OFF?.

(a) Select YES. The display will show CMD OFF, SET ON?.

SUBTASK 27-21-00-980-009

(8) Push the left rudder pedal until it touches the forward quadrant stop and hold.

SUBTASK 27-21-00-980-010

(9) Manually hold the main rudder PCU lower control rod at location B and pull forward with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

SUBTASK 27-21-00-700-004

(10) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is less than 1.87 VDC.

SUBTASK 27-21-00-980-011

(11) Release the main rudder PCU lower control rod.

SUBTASK 27-21-00-980-012

(12) Release the left rudder pedal.

NOTE: The test of system B was completed in the extend direction.

H. Rudder Main PCU System B Internal Leakage Check in the Retract Direction

SUBTASK 27-21-00-860-087

(1) Use the FSEU, on the E1-1 shelf in the main equipment center, to command the RPL to the "LOW" pressure state.

NOTE: The command will remains active for only 75 seconds after it is set.

(a) Push the ON/OFF button on the front panel of the FSEU, on E1-1 shelf, to turn on the display.

1) The display will show EXISTING FAULTS?.

(b) Select OTHER FUNCTNS? from the MAIN MENU on the FSEU.

1) The display will show SET OUTPUTS?.

(c) Select SET OUTPUTS? from the OTHER FUNCTNS? menu.

1) The display will show RUD PRES LIMIT?.

(d) Select RUD PRES LIMIT? from the SET OUTPUTS? menu.

1) The display will show SET ON?.

(e) Select SET ON? from the RUD PRES LIMIT? menu.

1) The display will show CMD ON, SET OFF?.

NOTE: You only have 75 seconds to complete the next 3 tasks. If you do not finish in 75 seconds, you will have to repeat the test and start with the steps to use the FSEU to command the RPL to the "LOW" state.

SUBTASK 27-21-00-980-013

(2) Push the right rudder pedal until it touches the forward quadrant stop and hold.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-980-014

(3) Manually hold the main rudder PCU lower control rod at location B and pull aft with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

SUBTASK 27-21-00-700-005

(4) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is between 7.37 VDC to 8.13 VDC.

SUBTASK 27-21-00-980-015

(5) Release the main rudder PCU control rod.

SUBTASK 27-21-00-980-016

(6) Release the rudder pedal.

SUBTASK 27-21-00-860-088

(7) The display screen on the FSEU should show CMD ON, SET OFF?.

(a) Select YES.

1) The display will show CMD OFF, SET ON?.

SUBTASK 27-21-00-980-017

(8) Push the right rudder pedal until it touches the forward quadrant stop and hold.

SUBTASK 27-21-00-980-018

(9) Manually hold the main rudder PCU lower control rod at location B and pull aft with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

SUBTASK 27-21-00-700-006

(10) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is greater than 8.13 VDC.

SUBTASK 27-21-00-980-019

(11) Release the main rudder PCU lower control rod.

SUBTASK 27-21-00-980-020

(12) Release the right rudder pedal.

NOTE: The test of system B was completed in the retract direction.

I. Rudder Main PCU System A Internal Leakage Check in the Extend Direction

NOTE: You will need 3 persons to complete this task. There is a time limit to complete some steps in this task.

SUBTASK 27-21-00-860-089

(1) Put the FLT CONTROL A switch, on the P5-3 panel, to the ON position.

SUBTASK 27-21-00-860-090

(2) Put the FLT CONTROL B switch, on the P5-3 panel, to the OFF position.

SUBTASK 27-21-00-860-091

(3) Use the FSEU, on the E1-1 shelf in the main equipment center, to command the RPL to the "LOW" pressure state.

NOTE: The command will remains active for only 75 seconds after it is set.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Push the ON/OFF button on the front panel of the FSEU, on E1-1 shelf, to turn on the display.

1) The display will show EXISTING FAULTS?.

(b) Select OTHER FUNCTNS? from the MAIN MENU on the FSEU.

1) The display will show SET OUTPUTS?.

(c) Select SET OUTPUTS? from the OTHER FUNCTNS? menu.

1) The display will show RUD PRES LIMIT?.

(d) Select RUD PRES LIMIT? from the SET OUTPUTS? menu.

1) The display will show SET ON?.

(e) Select SET ON? from the RUD PRES LIMIT? menu.

1) The display will show CMD ON, SET OFF?.

NOTE: You only have 75 seconds to complete the next 3 tasks. If you do not finish in 75 seconds, you will have to repeat the test and start with the steps to use the FSEU to command the RPL to the "LOW" state.

SUBTASK 27-21-00-980-021

(4) Push the left rudder pedal until it touches the forward quadrant stop and hold.

SUBTASK 27-21-00-980-022

(5) Manually hold the main rudder PCU upper control rod at location A and pull forward with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

SUBTASK 27-21-00-700-007

(6) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is between 7.37 VDC to 8.13 VDC.

SUBTASK 27-21-00-980-023

(7) Release the main rudder PCU control rod.

SUBTASK 27-21-00-980-024

(8) Release the rudder pedal.

SUBTASK 27-21-00-860-092

(9) The display screen on the FSEU should show CMD ON, SET OFF?.

(a) Select YES.

1) The display will show CMD OFF, SET ON?.

SUBTASK 27-21-00-980-025

(10) Push the left rudder pedal until it touches the forward quadrant stop and hold.

SUBTASK 27-21-00-980-026

(11) Manually hold the main rudder PCU upper control rod at location A and pull forward with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-700-008

(12) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is greater than 8.13 VDC.

SUBTASK 27-21-00-980-027

(13) Release the main rudder PCU upper control rod.

SUBTASK 27-21-00-980-028

(14) Release the left rudder pedal.

NOTE: The test of system A was completed in the extend direction.

J. Rudder Main PCU System A Internal Leakage Check in the Retract Direction

NOTE: You will need 3 persons to complete this task. There is a time limit to complete some steps in this task.

SUBTASK 27-21-00-860-093

(1) Use the FSEU, on the E1-1 shelf in the main equipment center, to command the RPL to the "LOW" pressure state.

NOTE: The command will remains active for only 75 seconds after it is set.

(a) Push the ON/OFF button on the front panel of the FSEU, on E1-1 shelf, to turn on the display.

1) The display will show EXISTING FAULTS?.

(b) Select OTHER FUNCTNS? from the MAIN MENU on the FSEU.

1) The display will show SET OUTPUTS?.

(c) Select SET OUTPUTS? from the OTHER FUNCTNS? menu.

1) The display will show RUD PRES LIMIT?.

(d) Select RUD PRES LIMIT? from the SET OUTPUTS? menu.

1) The display will show SET ON?.

(e) Select SET ON? from the RUD PRES LIMIT? menu.

1) The display will show CMD ON, SET OFF?.

NOTE: You only have 75 seconds to complete the next 3 tasks. If you do not finish in 75 seconds, you will have to repeat the test and start with the steps to use the FSEU to command the RPL to the "LOW" state.

SUBTASK 27-21-00-980-029

(2) Push the right rudder pedal until it touches the forward quadrant stop and hold.

SUBTASK 27-21-00-980-030

(3) Manually hold the main rudder PCU upper control rod at location A and pull aft with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

SUBTASK 27-21-00-700-009

(4) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is between 1.87 VDC to 2.62 VDC.

SUBTASK 27-21-00-980-031

(5) Release the main rudder PCU control rod.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-980-032

(6) Release the rudder pedal.

SUBTASK 27-21-00-860-094

(7) The display screen on the FSEU should show CMD ON, SET OFF?.

(a) Select YES.

1) The display will show CMD OFF, SET ON?.

SUBTASK 27-21-00-980-033

(8) Push the right rudder pedal until it touches the forward quadrant stop and hold.

SUBTASK 27-21-00-980-034

(9) Manually hold the main rudder PCU upper control rod at location A and pull aft with a force of approximately to 30 lbf (133 N) - 40 lbf (178 N) until it touches the stop and hold (Figure 509).

NOTE: If the control rods are not held against the stops, the voltage reading will differ from the limits in the following step.

SUBTASK 27-21-00-700-010

(10) Make sure that the voltage between pin 18 and pin 20 of connector D10064, on the E1-1 shelf, is less than 1.87 VDC.

SUBTASK 27-21-00-980-035

(11) Release the main rudder PCU upper control rod.

SUBTASK 27-21-00-980-036

(12) Release the right rudder pedal.

NOTE: The test of system A was completed in the retract direction.

SUBTASK 27-21-00-860-095

(13) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-480-008

(14) Remove the NLG towing lever pin, SPL-1499, from the nose wheel steering mechanism (TASK 32-00-01-080-801).

SUBTASK 27-21-00-410-005

(15) If required, close this access panel:

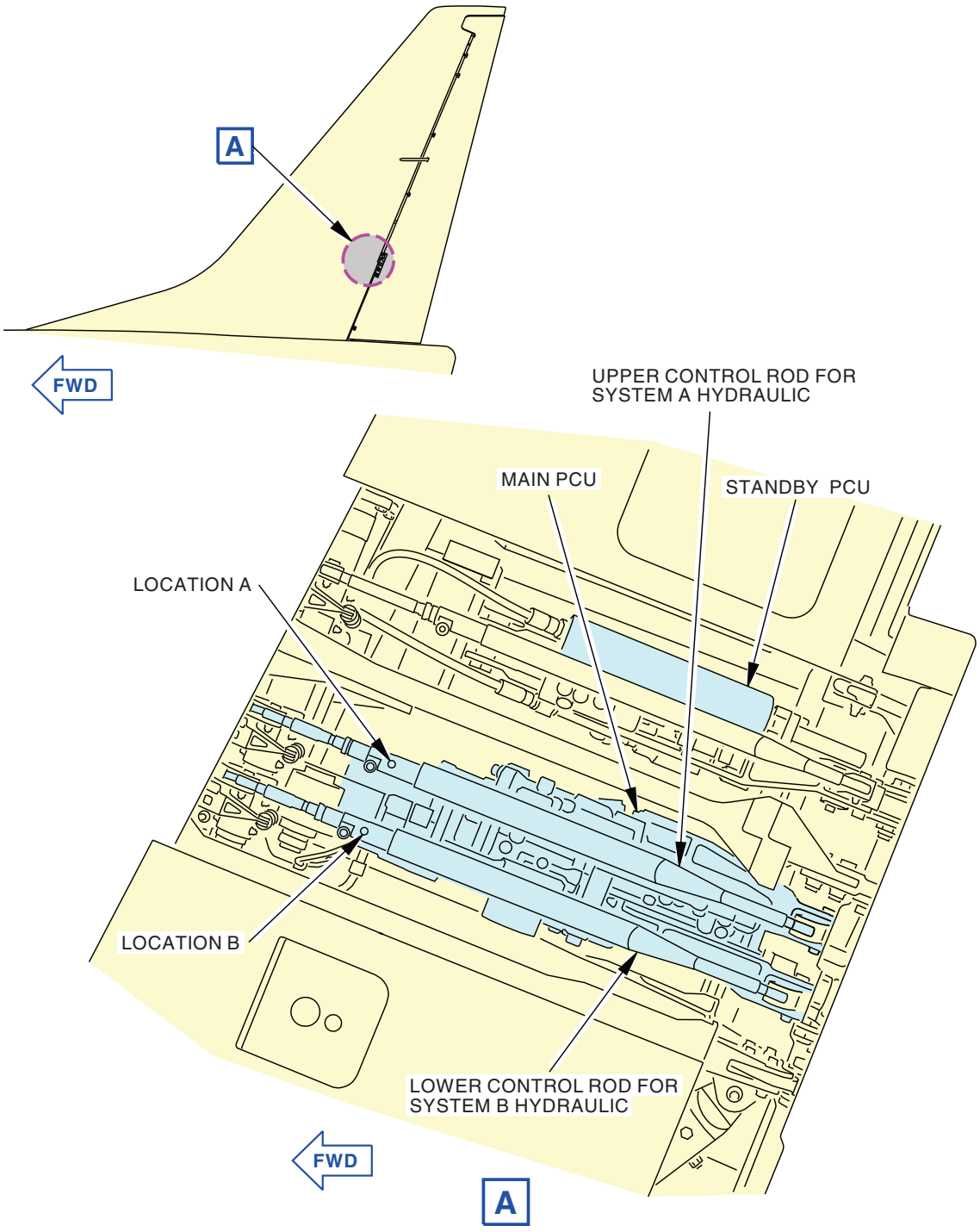
Number	Name/Location
324DL	Trailing Edge Access

Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Main Rudder PCU Internal Leakage Test
Figure 509/27-21-00-990-810

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-811

17. Rudder Override Operational Test

(Figure 510)

NOTE: This procedure is a scheduled maintenance task.

A. General

WARNING MAKE SURE THAT YOU FOLLOW THE TASK STEPS IN THE CORRECT SEQUENCE. IF YOU DO NOT, IT CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Use this test to make sure that the override mechanisms installed on the torque tube are operational.

B. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge

C. Access Panels

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

D. Procedure

SUBTASK 27-21-00-840-005

(1) Put the FLT CONTROL A and B switches, on the P5 panel, to the OFF position.

SUBTASK 27-21-00-010-015

(2) To gain access to the torque tube, do this task:

Open these access panels:

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

SUBTASK 27-21-00-730-013

(3) Do the rudder override operational test:

(a) Make sure that the rudder and rudder pedals are in the neutral position.

(b) Manually hold the rudder aft edge in the centered position.

(c) Observe the torque tube override assemblies and do these steps:

1) Slowly push on the captain's left rudder pedal forward full travel then return the pedal to the neutral position.

NOTE: It will require more force than normal operation for the override mechanism to breakout of the detent position.

2) Make sure that all 3 override assemblies breakout of their detent positions.

3) Make sure that there is no sign of binding or interferences at any point of travel on the three override assemblies.

a) If binding or excessive friction exist, the override assemblies may be damaged or improperly installed. Correct this condition and repeat this check.

(d) Observe the torque tube override assemblies and do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Slowly push on the captain's right rudder pedal forward full travel then return the pedal to the neutral position.

   NOTE: It will require more force than normal operation for the override mechanism to breakout of the detent position.

2) Make sure that all 3 override assemblies breakout of their detent positions.

3) Make sure that there is no sign of binding or interferences at any point of travel on the three override assemblies.

   a) If binding or excessive friction exist, the override assemblies may be damaged or improperly installed. Correct this condition and repeat this check.

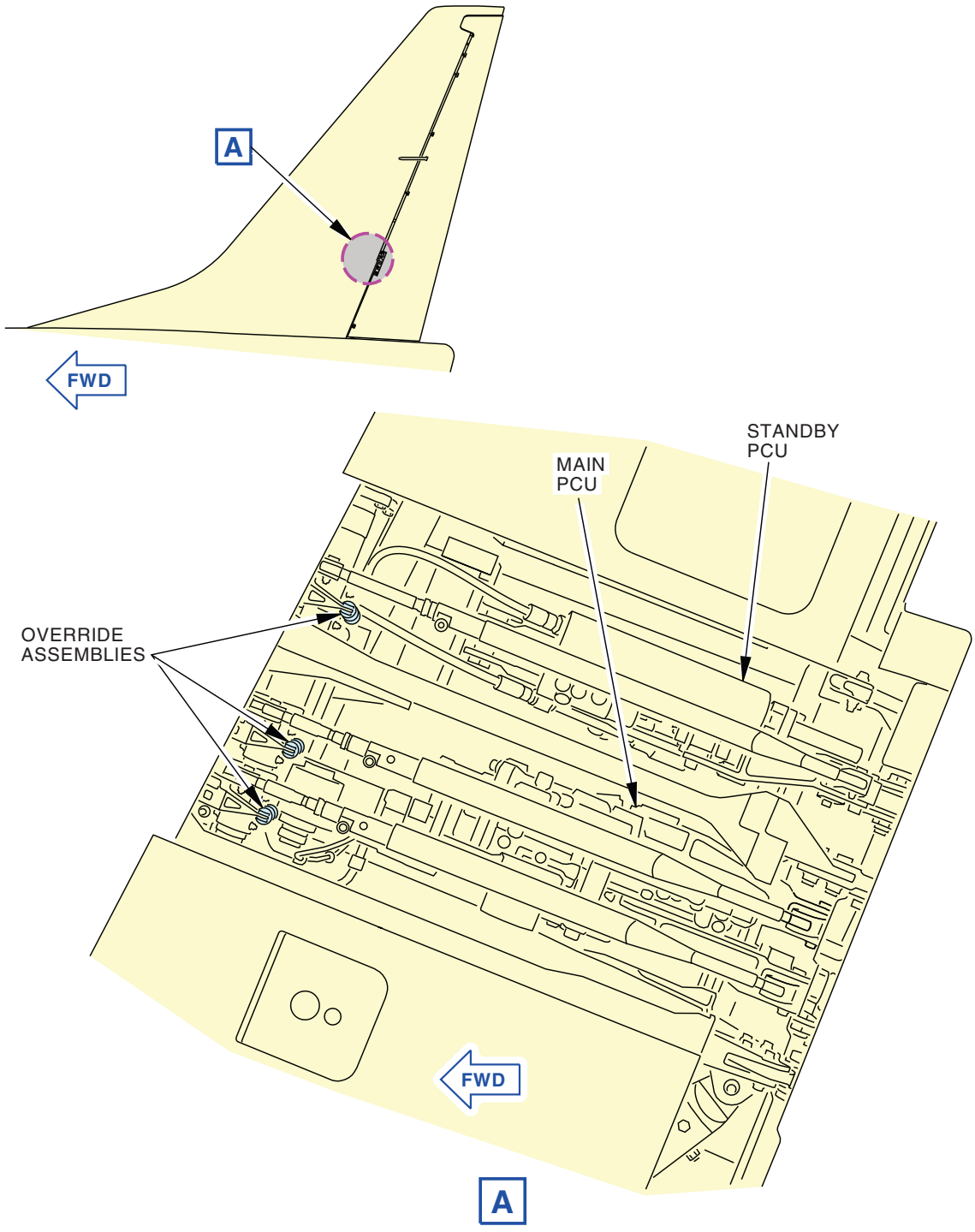
SUBTASK 27-21-00-410-006

(4) Install these access panels:

Number	Name/Location
324BL	Vertical Fin, Trailing Edge Access
324DL	Trailing Edge Access

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Override Operational Test
Figure 510/27-21-00-990-811

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 27-21-00-700-812

18. Rudder Main PCU Force Fight Monitor Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) Use this test to make sure that the STBY RUD ON light on the overhead panel P5-3 is functional.

B. References

Reference	Title
27-21-00-800-801	Rudder Hydraulic System A, B, or Standby Pressurization (P/B 201)
27-21-00-840-801	Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
118	Electrical and Electronics Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
324	Vertical Fin - Rear Spar To Trailing Edge

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
324CL	Vertical Fin, Access
324EL	Vertical Fin, Access

E. Rudder Main PCU Force Fight Monitor Test

SUBTASK 27-21-00-860-098

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. AILERONS, RUDDER, ELEVATORS, FLAPS, SPOILERS, SLATS, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE CLOSED AND IN THE STOWED POSITION OR REMOVED BEFORE YOU EXTEND THE LEADING EDGE FLAPS AND SLATS. THERE IS NOT SUFFICIENT CLEARANCE FOR THE FLAPS AND SLATS TO EXTEND IF THE INBOARD FAN COWL PANEL AND THE INBOARD AND OUTBOARD THRUST REVERSERS ARE IN THE OPEN POSITION. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Pressurize the rudder hydraulic systems A and B (TASK 27-21-00-800-801).

SUBTASK 27-21-00-860-099

(2) Put the FLT CONTROL A and B switches, on the P5 panel, to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-100

(3) Make sure that the airplane is in the air mode (TASK 32-09-00-860-801).

SUBTASK 27-21-00-010-017

(4) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-21-00-860-113

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 27-21-00-410-008

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 27-21-00-860-101

(7) Move the rudder pedals through 5 cycles of full travel.

SUBTASK 27-21-00-210-004

(8) Make sure that the STBY RUD ON light on the overhead panel P5-3 is not on.

SUBTASK 27-21-00-860-102

(9) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-21-00-860-103

(10) Put the FLT CONTROL B switch, on the P5 panel, to the STBY RUD position.

SUBTASK 27-21-00-210-005

(11) Make sure that the STBY RUD ON light on the overhead panel P5-3 is on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-860-104

(12) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	13	C00363	FLIGHT CONT SHUTOFF VALVES FLT CONT
C	13	C01074	FLIGHT CONT SHUTOFF VALVES FLT CONT

SUBTASK 27-21-00-860-105

(13) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD

SUBTASK 27-21-00-010-016

(14) To get access to the rudder main PCU, do this step:

(a) Open these access panels:

Number	Name/Location
324CL	Vertical Fin, Access
324EL	Vertical Fin, Access

SUBTASK 27-21-00-860-106

(15) Put the FLT CONTROL B switch, on the P5 panel, to the ON position.

SUBTASK 27-21-00-730-014

WARNING MAKE SURE THAT YOU FOLLOW THE STEPS IN THIS TASK CORRECT SEQUENCE. IF YOU DO NOT OBEY, INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(16) Do the Force Fight Monitor test:

(a) Make sure that the STBY RUD ON light and STANDBY HYD LOW PRESS light on the overhead panel P5-3 are off.

(b) Move the rudder pedals through 2 cycles of full travel.

NOTE: This will relieve any standby hydraulic system residual pressure.

1) If it is necessary, jiggle the rudder pedals to put the pedals to the neutral position.

(c) Manually hold the rudder main PCU upper control rod at location A (Figure 509) and pull aft with sufficient force to break out the input override (approximately 30 lbf (133 N)).

1) Hold the control rod in this position for at least 7 seconds, then release.

(d) Make sure that the STBY RUD ON light and STANDBY HYD LOW PRESS light on the overhead panel P5-3 are on.

NOTE: The standby hydraulic pumps will not activate.

(e) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD

(f) Make sure that the STANDBY HYD LOW PRESS light on the overhead panel P5-3 is off.

NOTE: The standby hydraulic pumps will activate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Open this circuit breaker on the P6-2 panel for at least 2 seconds, then close:

1) Open and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

(h) Make sure that the STBY RUD ON light on the overhead panel P5-3 is off.

(i) Put the FLT CONTROL B switch, on the P5 panel, to the STBY RUD position.

(j) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD

(k) Put the FLT CONTROL B switch, on the P5 panel, to the ON position.

(l) Move the rudder pedals through 2 cycles of full travel.

1) If it is necessary, jiggle the rudder pedals to put the pedals to the neutral position.

(m) Manually hold the rudder main PCU upper control rod at location A (Figure 509) and pull forward with sufficient force to break out the input override (approximately 30 lbf (133 N)).

1) Hold the control rod in this position for at least 7 seconds, then release.

(n) Make sure that the STBY RUD ON light and STANDBY HYD LOW PRESS light on the overhead panel P5-3 are on.

NOTE: STANDBY HYD LOW PRESS light will not come on until hydraulic pressure has bled off (approximately 5 minutes).

NOTE: The standby hydraulic pumps will not activate.

(o) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	11	C00362	FLIGHT CONT SHUTOFF VALVES STBY RUD

(p) Make sure that the STANDBY HYD LOW PRESS light on the overhead panel P5-3 is off.

NOTE: The standby hydraulic pumps will activate.

(q) Open this circuit breaker on the P6-2 panel for at least 2 seconds, then close:

1) Open and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	15	C01628	FORCE FIGHT MONITOR

(r) Make sure that the STBY RUD ON light on the overhead panel P5-3 is off.

SUBTASK 27-21-00-860-107

(17) Do this task: Put the Rudder Hydraulic Systems A, B, and Standby Back to the Condition Before the Pressurization, TASK 27-21-00-840-801.

SUBTASK 27-21-00-860-108

(18) Put the airplane back to the ground mode (TASK 32-09-00-860-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 27-21-00-410-007

(19) Install these access panels:

Number	Name/Location
324CL	Vertical Fin, Access
324EL	Vertical Fin, Access

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the rudder

(2) An installation of the rudder.

TASK 27-21-11-000-801

2. Rudder Removal

(Figure 401, Figure 402)

A. References

Reference	Title
27-21-00-800-802	Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation (P/B 201)
27-21-24-000-801	Standby Rudder Power Control Unit Removal (P/B 401)
27-21-91-000-801	Main Rudder Power Control Unit Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2033	Sling Equipment - Rudder Removal/Installation Part #: C27035-33 Supplier: 81205

C. Location Zones

Zone	Area
324	Vertical Fin - Rear Spar To Trailing Edge
325	Vertical Fin - Rudder

D. Access Panels

Number	Name/Location
324AAL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ABL	Panel Assy - Trailing Edge, Beam Seal, Vertical Fin
324ACL	Rudder Nose Cover/Rudder Hinge No. 9 At Rudder Station 276.24
324ADL	Rudder Nose Cover/Rudder Hinge No. 8 At Rudder Station 257.93
324AEL	Rudder Nose Cover/Rudder Hinge No. 7 At Rudder Station 239.61
324AFL	Rudder Nose Cover/Rudder Hinge No. 6 At Rudder Station 184.67
324AHL	Rudder Nose Fairing/Rudder Hinge No. 5 At Rudder Station 129.74
324AJL	Rudder Nose Fairing/Rudder Hinge No. 4 At Rudder Station 74.80
324ALL	Rudder Nose Cover/Rudder Hinge No. 1 At Rudder Station 6.96
324QL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324SL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324UL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324XL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ZL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 27-21-11-860-001

(1) Do this task: Pressure from the Rudder Hydraulic Systems A, B, and Standby - Deactivation, TASK 27-21-00-800-802.

SUBTASK 27-21-11-020-001

(2) Remove the seals and seal attachment plates from each side of the fin at the rudder balance weight.

SUBTASK 27-21-11-010-001

(3) Remove these access panels:

Number	Name/Location
324AAL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ABL	Panel Assy - Trailing Edge, Beam Seal, Vertical Fin
324ACL	Rudder Nose Cover/Rudder Hinge No. 9 At Rudder Station 276.24
324ADL	Rudder Nose Cover/Rudder Hinge No. 8 At Rudder Station 257.93
324AEL	Rudder Nose Cover/Rudder Hinge No. 7 At Rudder Station 239.61
324AFL	Rudder Nose Cover/Rudder Hinge No. 6 At Rudder Station 184.67
324AHL	Rudder Nose Fairing/Rudder Hinge No. 5 At Rudder Station 129.74
324AJL	Rudder Nose Fairing/Rudder Hinge No. 4 At Rudder Station 74.80
324ALL	Rudder Nose Cover/Rudder Hinge No. 1 At Rudder Station 6.96
324QL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324SL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324UL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324XL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin
324ZL	Panel Assy - Trailing Edge, Beam Seal, Vert Fin

SUBTASK 27-21-11-860-002

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	16	C00788	FLIGHT CONTROL TRIM IND RUD
D	19	C00787	FLIGHT CONTROL TRIM CONT RUD

F. Rudder Removal

SUBTASK 27-21-11-480-001

(1) Do these steps to install the sling equipment, SPL-2033 (Figure 401):

NOTE: If you use the crane, do this step when you have enough room for the crane.

(a) Attach the master link to a load cell with a minimum capacity of 500 lb (227 kg).

(b) Install the hoist fittings on the rudder [1] with a ¼ in. (6.4 mm) drive socket.

(c) Install four ropes to make sure that the rudder does not rotate and hit the airplane.

(d) Attach the sling to the hoist fittings installed on the rudder [1].

SUBTASK 27-21-11-020-002

(2) Remove the nut [46], washers [45], and bolt [44] to disconnect the bonding jumper [47] from the vertical fin at hinge 9 (Figure 402).

SUBTASK 27-21-11-020-003

(3) Disconnect the control rod of the rudder position transmitter from the vertical fin:

(a) Remove the bolt [23], washer [24], and washer [25].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Disconnect the control rod.

SUBTASK 27-21-11-020-004

CAUTION REMOVE THE CONTROL ROD FROM THE PCU BEFORE YOU DISCONNECT THE PCU FROM THE RUDDER. IF YOU DO NOT REMOVE THE CONTROL ROD FIRST, DAMAGE TO THE CONTROL ROD CAN OCCUR DURING THE CHECK.

(4) Do these steps to disconnect the rudder main power control unit (PCU) from the rudder [1]:

(a) Disconnect and remove the control rods [51] from the main PCU (TASK 27-21-91-000-801).

(b) Remove nut [15], washer [14], nut [13], washer [12], bushing [11], bolt [10], and bolt [9].

SUBTASK 27-21-11-020-005

(5) Do these steps to disconnect the rudder standby power control unit (PCU) from the rudder [1]:

(a) Disconnect and remove the control rod [52] from the standby PCU (TASK 27-21-24-000-801).

(b) Remove nut [29], washer [28], bushing [50], washer [27], and bolt [26].

SUBTASK 27-21-11-020-006

(6) Move the rudder [1] to the neutral position.

SUBTASK 27-21-11-020-007

WARNING DO NOT LIFT THE RUDDER WITH A FORCE THAT IS MORE THAN 257-LBS. IF YOU USE A FORCE THAT IS MORE THAN 257-LBS, IT CAN CAUSE DAMAGE TO THE AIRPLANE AND INJURY TO PERSONNEL.

(7) Lift the rudder [1] until the load is removed from its hinges.

SUBTASK 27-21-11-020-008

WARNING MAKE SURE THAT THE CRANE WILL HOLD THE RUDDER IN A STABLE POSITION WHEN YOU REMOVE THE HINGE BOLTS. ACCIDENTAL RUDDER MOVEMENT WHEN YOU DISCONNECT IT FROM THE AIRPLANE CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(8) Remove the hinge bolts from the hinges:

(a) At hinges 1, 4, 5, 6, and 8, remove the pins [35], nuts [34], washers [33], bushings [32], washers [31], and bolt [30] from the hinges.

1) Discard the pins [35].

(b) At hinge 7, remove the pin [43], nut [42], washer [41], washer [39], bushing [38], washer [37], and bolt [36] from the hinge.

1) Discard the pin [43].

(c) At hinge 2, remove the pin [2], nut [3], washer [4], spacer [5], bushing [6], washer [7], and bolt [8] from the hinge.

1) Discard the pin [2].

(d) At hinge 3, remove the pin [22], nut [21], washer [20], adjustment bushing [19], bushing [18], washer [17], and bolt [16] from the hinge.

1) Discard the pin [22].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) At hinge 9, remove the pin [35], nut [34], washer [33], bushing [32], washer [31], and bolt [30] from the hinge.

1) Discard the pin [35].

SUBTASK 27-21-11-020-009

CAUTION

YOU MUST MOVE THE RUDDER UP BEFORE YOU REMOVE IT FROM THE VERTICAL FIN. IF YOU DO NOT MOVE THE RUDDER UP, YOU CAN CAUSE DAMAGE TO THE RUDDER TIP.

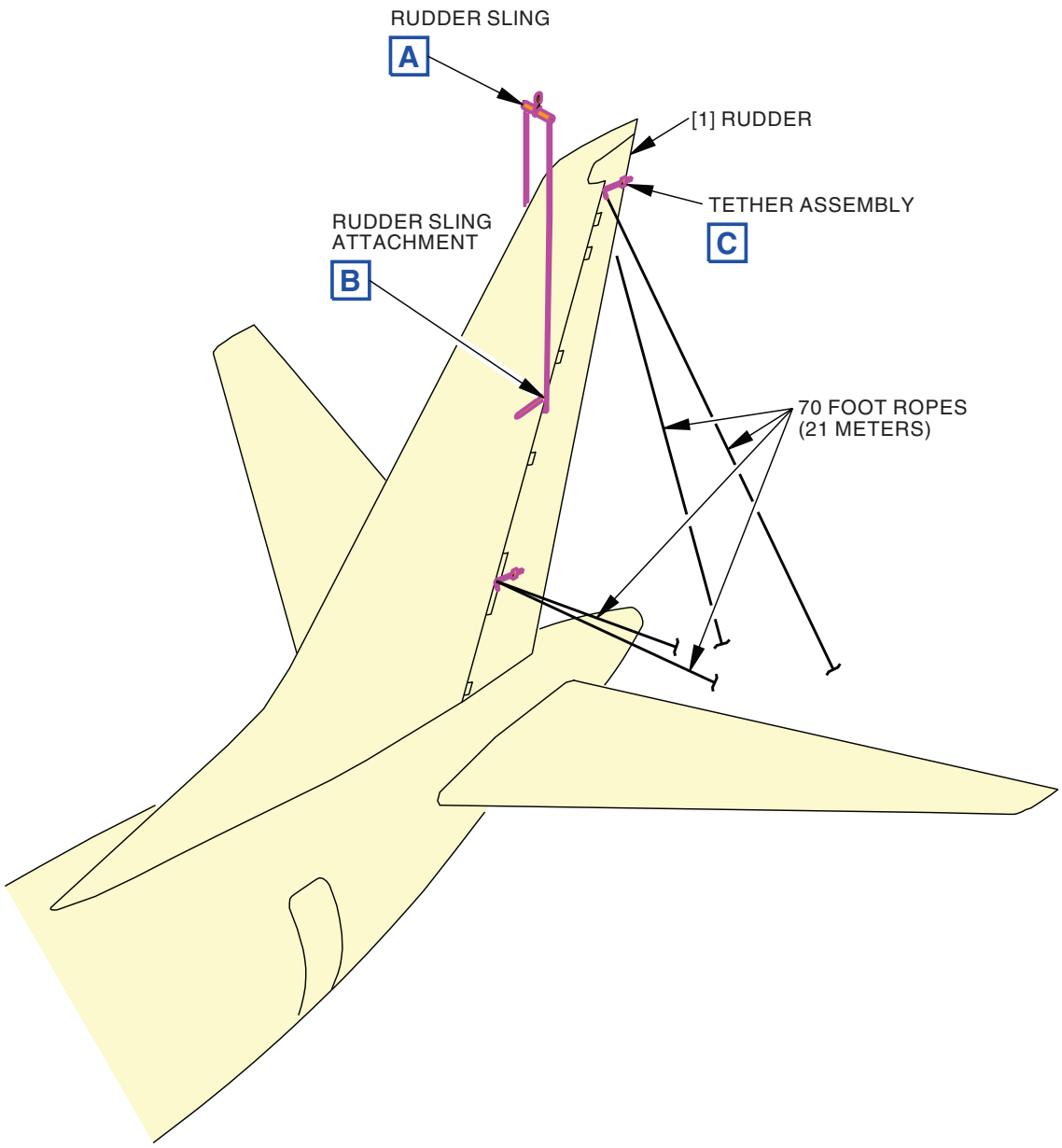
(9) Do this step to remove the rudder [1]:

(a) Use the load cell to make sure that the force used to lift the rudder [1] is not more than 257 lb (117 kg).

(b) Move the rudder [1] up and aft from the vertical fin and lower to the ground.

——— END OF TASK ———

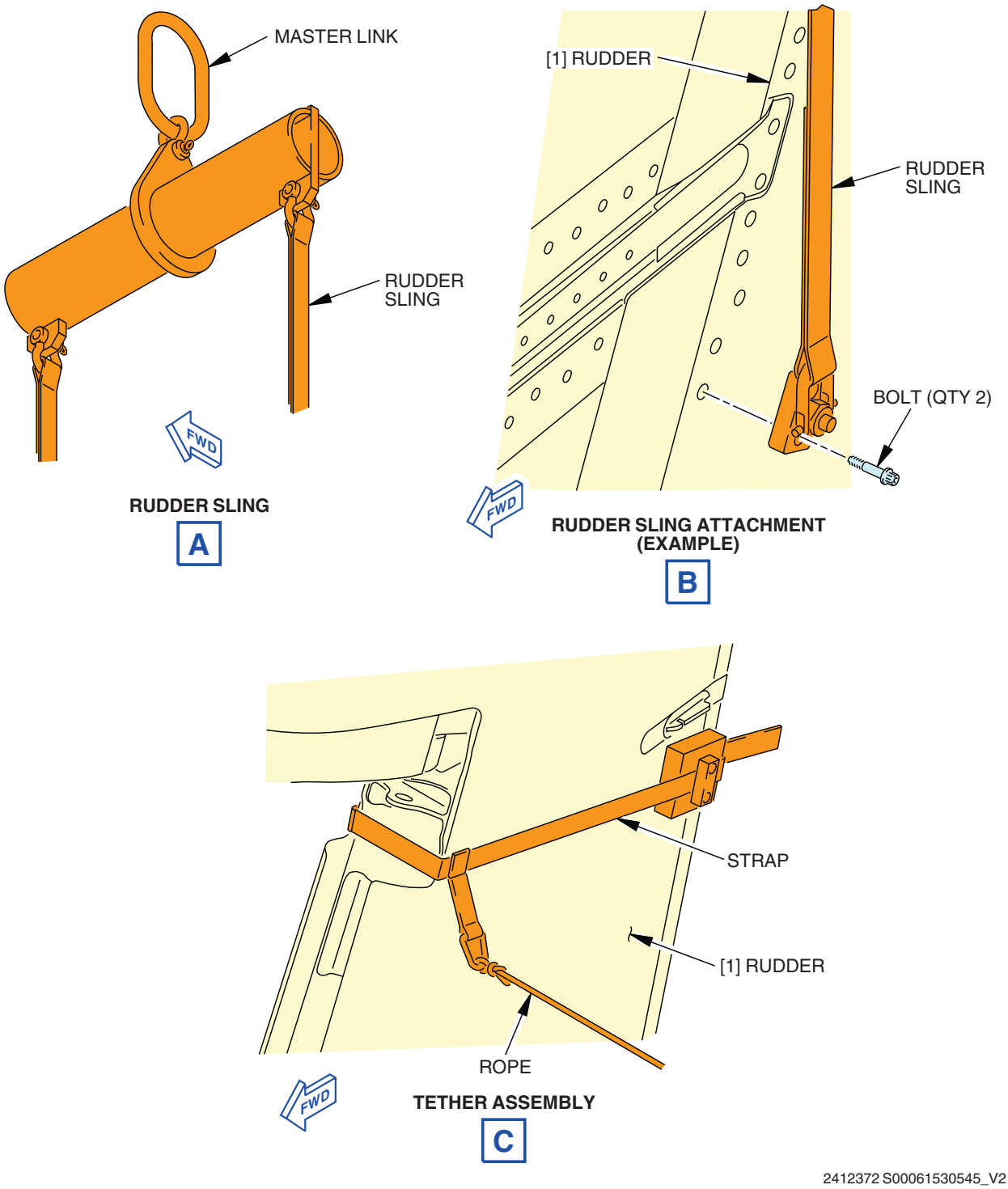
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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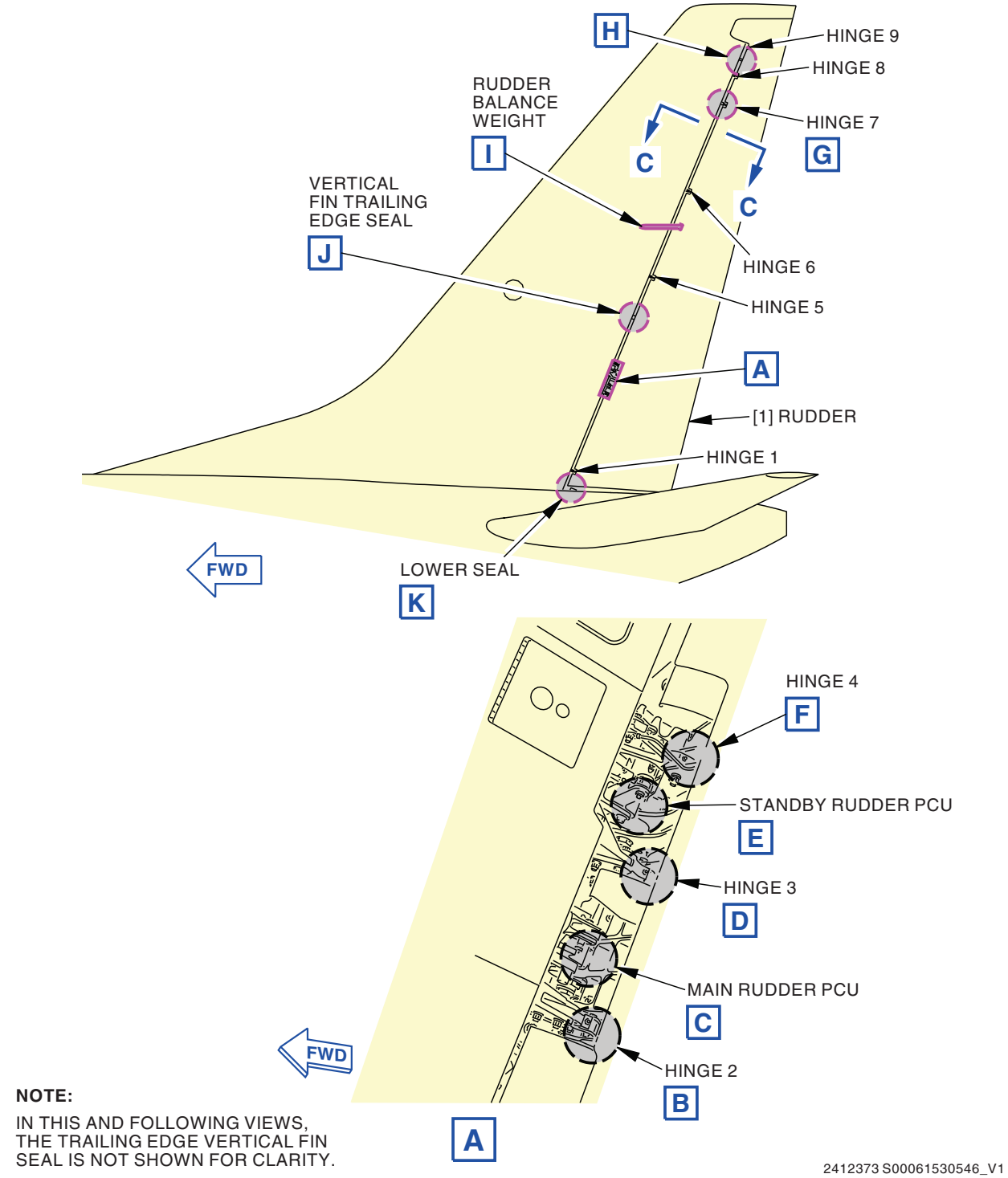
Rudder Sling Installation
Figure 401/27-21-11-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Sling Installation
Figure 401/27-21-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



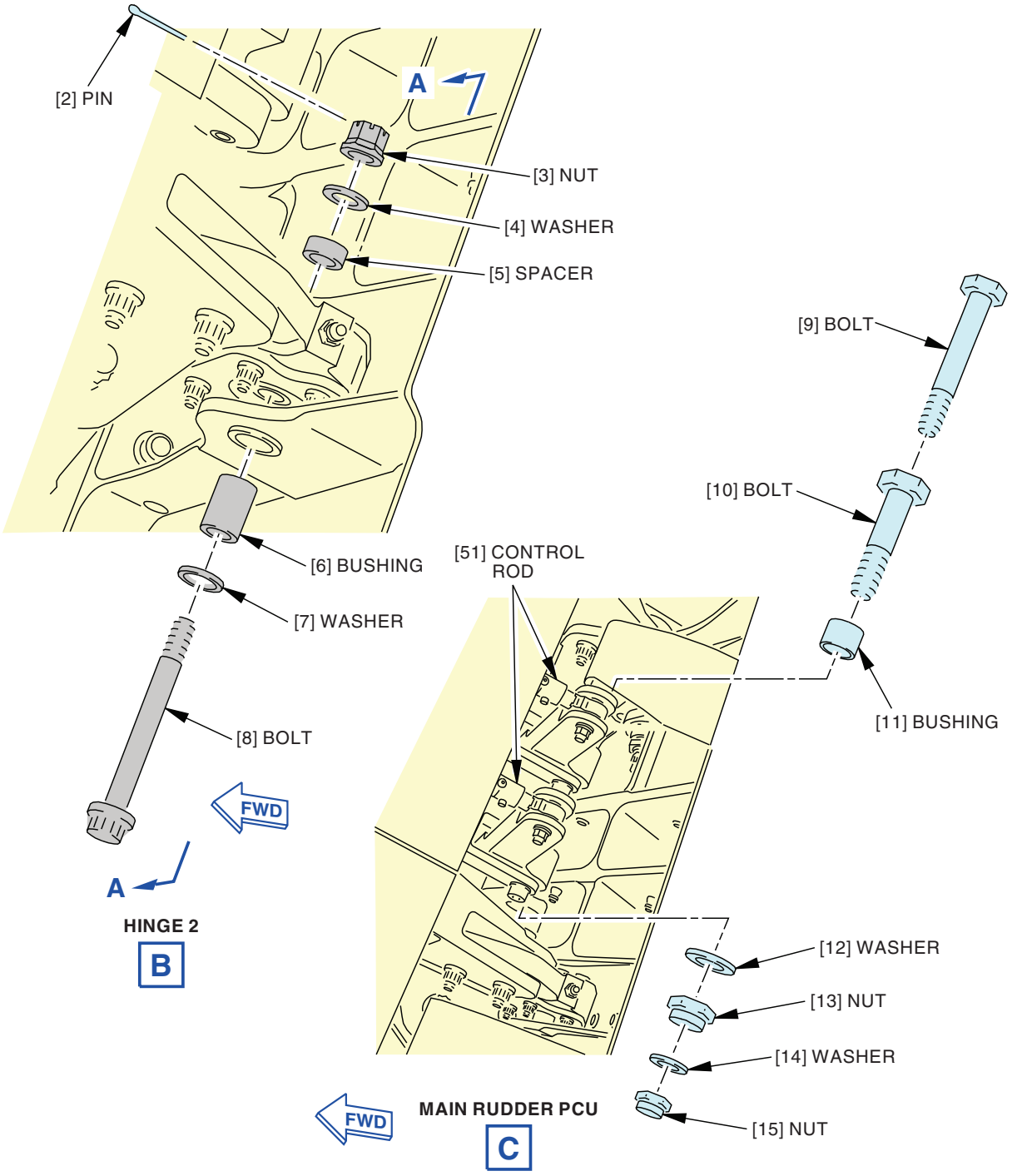
NOTE:

IN THIS AND FOLLOWING VIEWS, THE TRAILING EDGE VERTICAL FIN SEAL IS NOT SHOWN FOR CLARITY.

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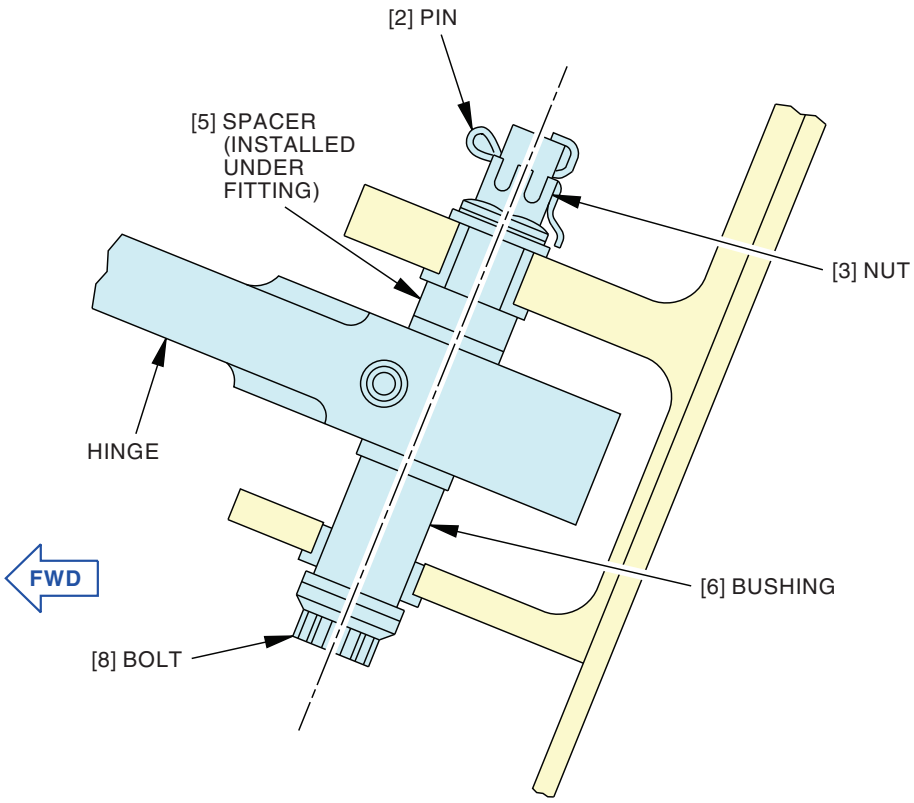
Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 1 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

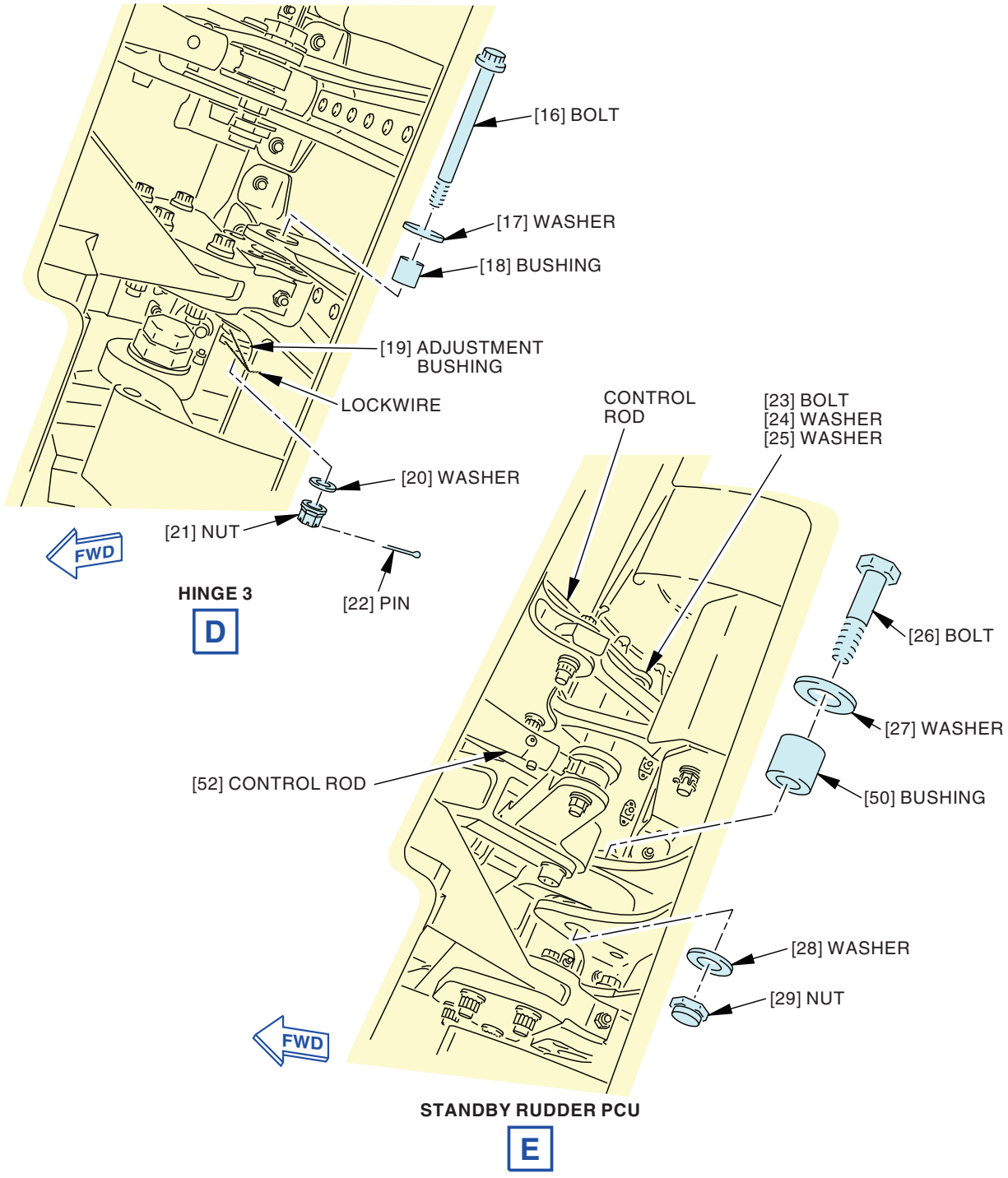


HINGE 2
A-A

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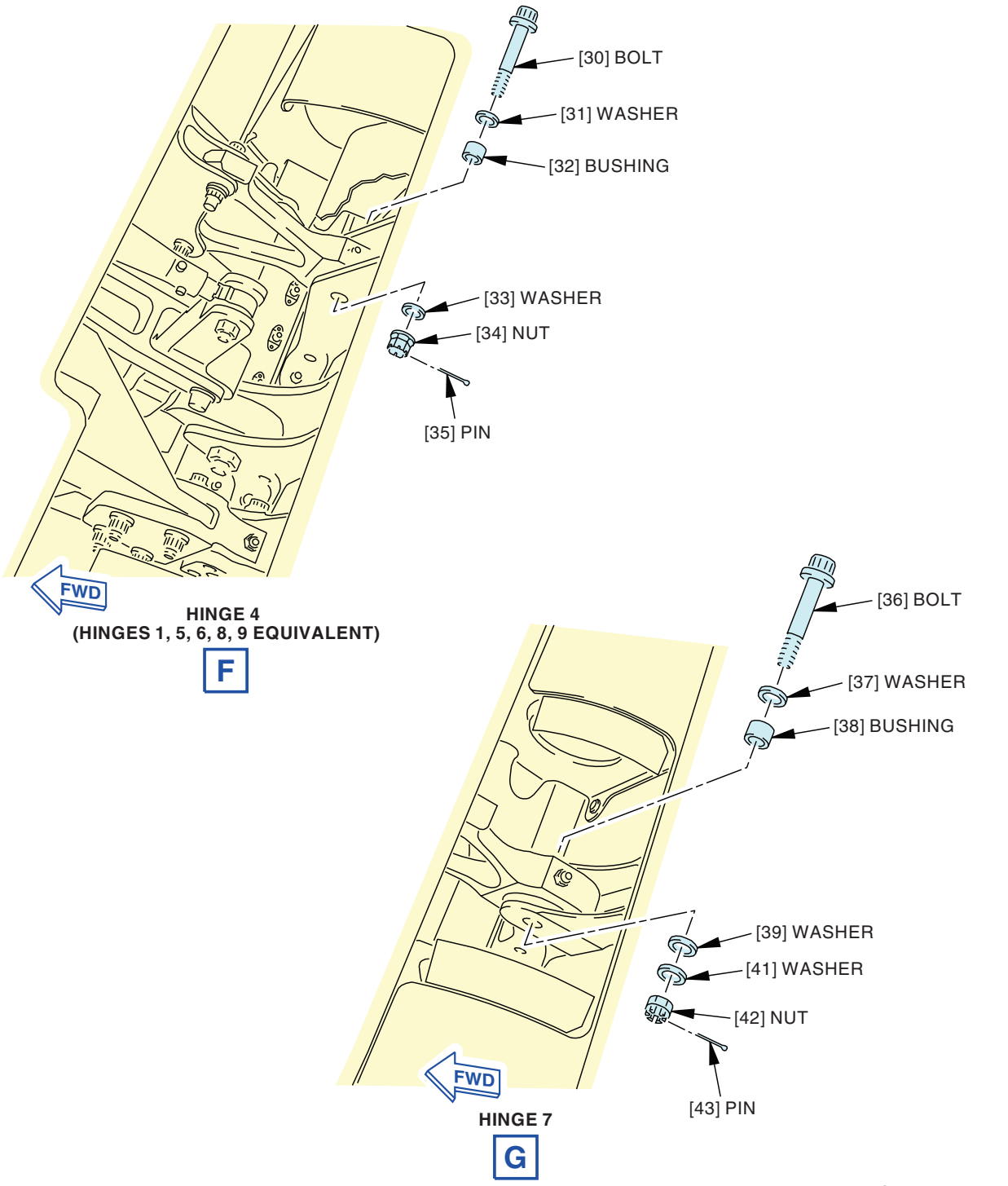
Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



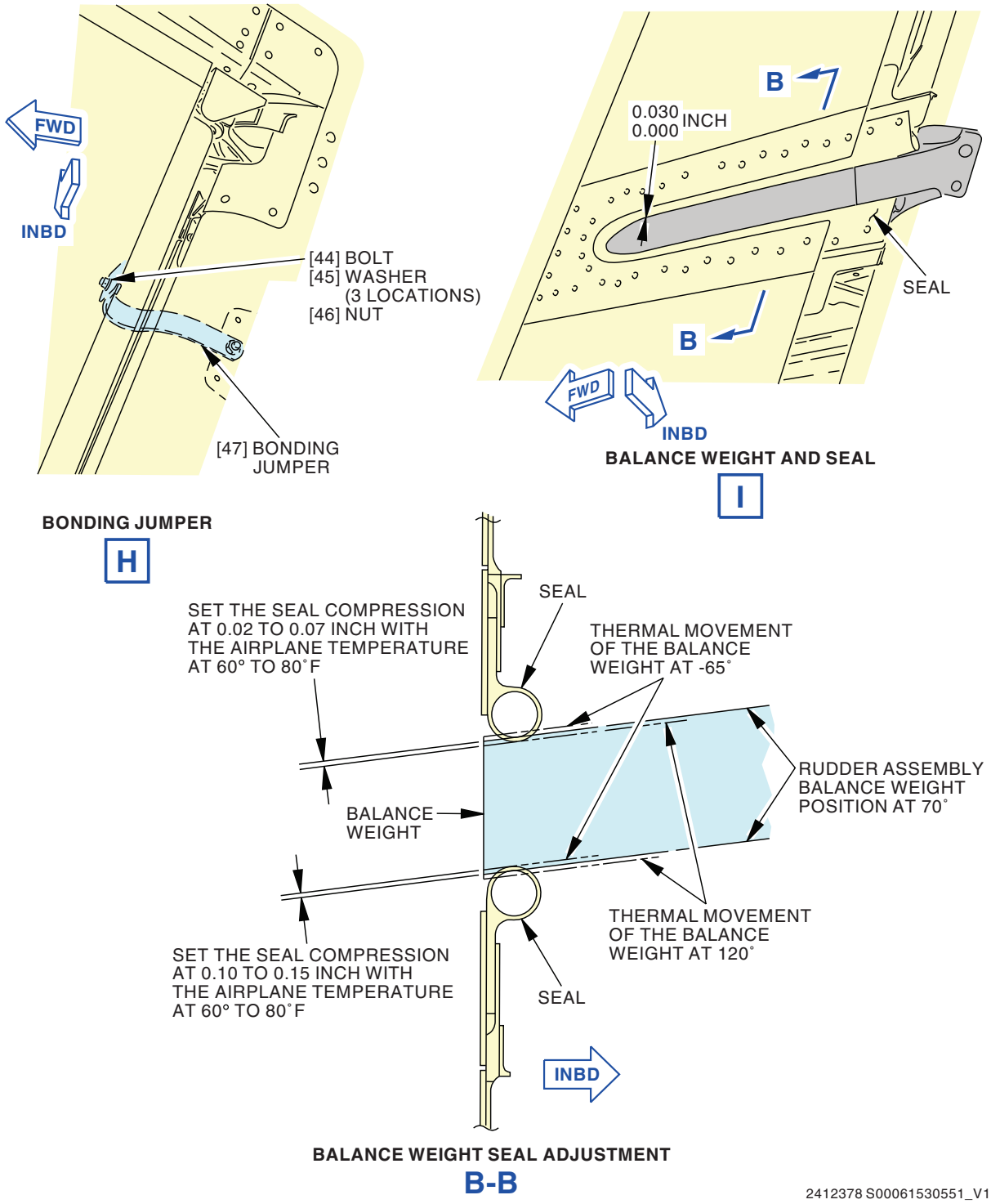
Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



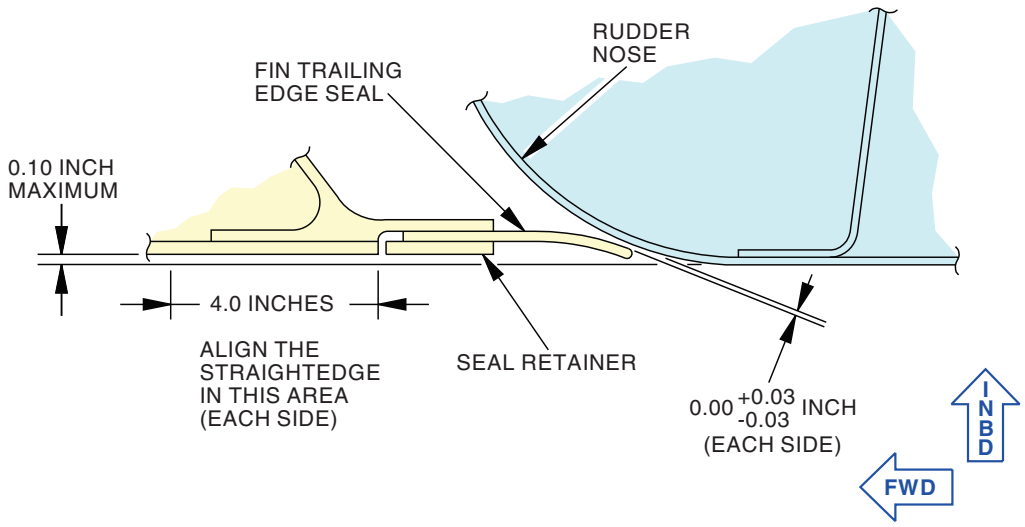
Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



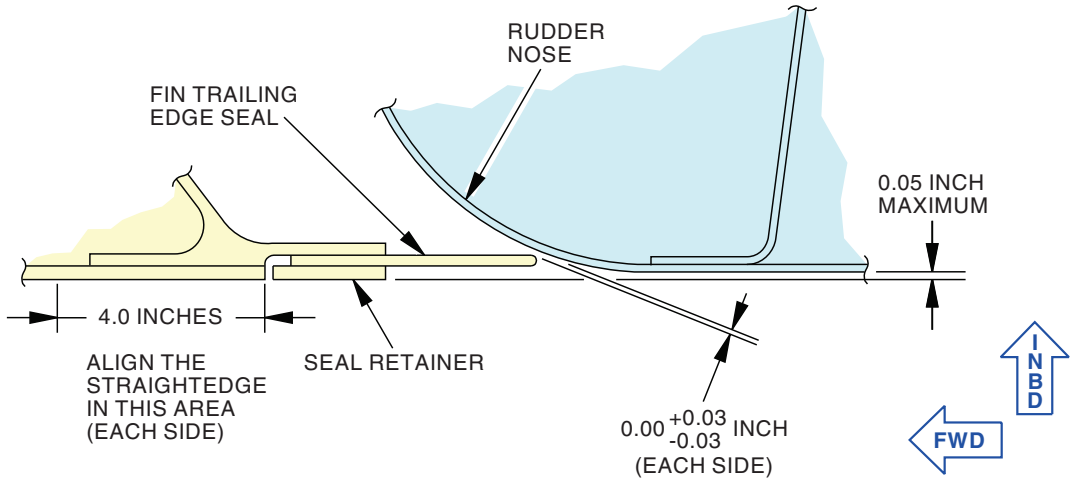
Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POSITIVE MISMATCH

C-C



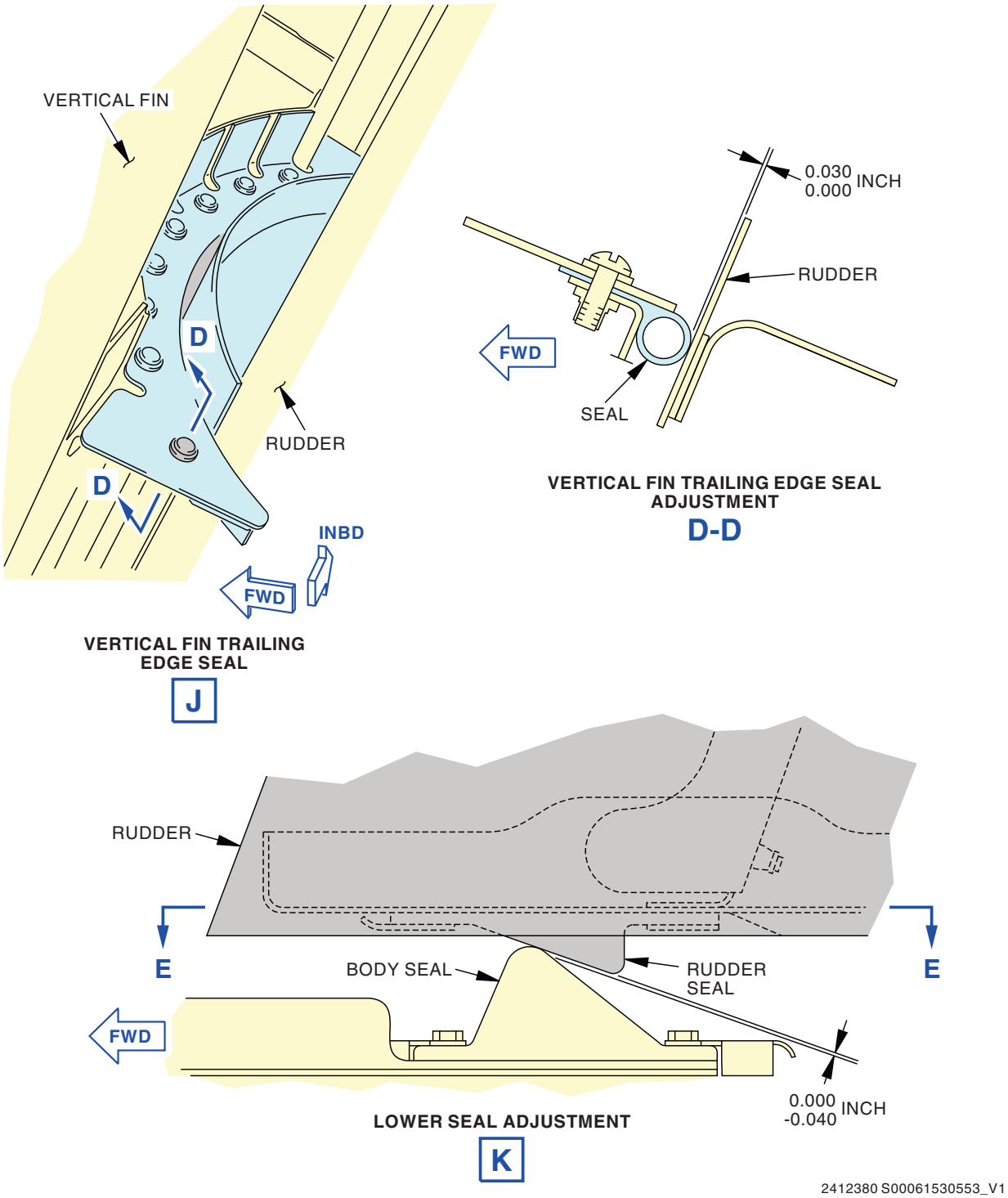
NEGATIVE MISMATCH

C-C

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Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Installation
Figure 402/27-21-11-990-802 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details